THE TWO-MINUS ADJOINT-CHARACTER CASE OF GINIBRE'S Q_3 CONDITION
FORMAL DETAILED SUPPLEMENT TO PARTS I–II

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Abstract. Let G be a compact connected Lie group with simple Lie algebra, let χ_{ad} be its adjoint character, and define

$$Q_3^G(n) = \int_{G\times G} (\chi_{\mathrm{ad}}(g) - \chi_{\mathrm{ad}}(h))^2 (\chi_{\mathrm{ad}}(g) + \chi_{\mathrm{ad}}(h))^n \, d\,\mathrm{Haar}(g)\, d\,\mathrm{Haar}(h).$$

We prove $Q_3^G(n) \geq 0$ for every such G and every integer $n \geq 0$. This is the specialization of Ginibre's condition (Q3) in which every function is χ_{ad}, exactly two factors carry a minus sign, and the other n factors carry plus signs; it is not a proof of the full cone-valued Q3 condition. The moment formula turns the odd-exponent problem into inequalities for invariant multiplicities in tensor powers of the adjoint representation. The proof follows the classification: type A uses transition ratios; types B, C, D use stable moments, finite correction prefixes, and uniform trace tails; and the exceptional types use exact character-ring prefixes and rectangular analytic tails. All finite decisions are made with exact integer or rational arithmetic or with outward-rounded intervals. The computer-assisted steps are stated as finite mathematical contracts, with their active domains and proof dependencies separated from certificate generation and transcript authentication.

Contents

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Remark 0.1 (Companion architecture)**.** This formal detailed supplement expands the B/C Pieri-correction development and prints the detailed certificate, transcript, and row ledgers. It is a load-bearing component of the proof of the residual B/C closure stated compactly in `paper.pdf`. The journal-facing Parts I–II remain consolidated in `paper.pdf`; the present supplement supplies the numbered proofs of the correction-prefix contracts consumed there. The compact contract table identifies every active result by label and proves that offsets beyond 27 are not consumed. Conditional and superseded development routes printed later in this expanded source are archival material, not part of that indexed proof. Part III is the separate full signed-product extension `full_q3_extension.pdf`.

Remark 0.2 (Publication proof spine and author self-audits)**.** The load-bearing dependency spine for Parts I–III and their detailed supplement is listed in `PUBLICATION_PROOF_SPINE.md` and, for the present theorem, in the final-facing dependency map below. The supplement's numbered correction-prefix proofs are load-bearing; explicitly marked diagnostic and superseded routes remain provenance and are not consumed. The archived historical report and the programs under `referee_audits/` are author-controlled self-audits. They provide redundant checks and source mappings; they are not independent peer review and are not premises of any theorem.

1. Introduction

Ginibre's condition (Q3) asks that, for a probability measure σ and every finite family $f_1, \dots, f_r$ in a specified multiplicative convex cone, each signed product integral

$$\iint \prod_{i=1}^{r} \big(f_i(x) \pm f_i(y)\big)\, d\sigma(x)\, d\sigma(y)$$

be nonnegative; see [2, §1, condition (Q3), equation (1.2)]. The present theorem addresses one precise noncommutative specialization. We take normalized Haar measure on a compact connected group with simple Lie algebra, set every f_i equal to the adjoint character, choose exactly two minus signs, and choose n plus signs. The resulting integral is $Q_3^G(n)$. We do not assert Q3 for a cone of functions, for unequal characters, or for four or more minus factors.

Even n is immediate from pointwise nonnegativity. Odd n is not: the adjoint character takes negative values, and expansion produces differences of large invariant-tensor multiplicities. The moment formula in lemma 2.3 converts the problem into exact inequalities for $m_k^G = \mathrm{mult}(\mathbf{1}, \mathrm{ad}^{\otimes k})$. The proof then separates into the classification families. Type A is governed by stable derangement moments and finite-rank transition ratios. The classical types use the exact stable trace law of Diaconis–Shahshahani, orthogonal and symplectic identities of Rains, finite correction bridges, and uniform trace tails. The exceptional types combine finite character-ring intervals with analytic rectangular tails.

Computer assistance enters only after the mathematical reduction has specified a finite integer, rational, or interval inequality. A source table is not accepted merely because its hash matches. For every moment ledger passed to an active source-replay stage, the clean replay regenerates the claimed exceptional or low classical moments from the Cartan datum by lemmas 2.4 and 2.5; in particular, the E_8 values through m_{100} require the decomposition only through $\mathrm{ad}^{\otimes 50}$. The ancillary E_8 values of Bourjaily–Plesser–Vergu [14, ancillary file `adjoint_tensor_ranks.tex`] are therefore comparison data, not an unverified premise. Likewise, every stable classical moment used in a finite correction is regenerated from the exact recurrence of lemma 4.67; the archived OEIS row is checked against that regeneration and is not used as a mathematical premise. Exact discrete calculations use GMP integers or rationals, and analytic comparisons use outward-rounded MPFR intervals.

Parts I–II are consolidated in this manuscript. The final-facing reductions and family closures consume a compact finite-correction contract and the accepted certificate ledger below. The formal detailed supplement prints the line-by-line correction derivations and historical ledgers; its numbered correction-prefix proofs are incoming nodes for the compact residual closure, while explicitly diagnostic and superseded material adds no hypothesis or incoming edge. The replay contract states which inputs are regenerated, which legacy certificates are audited without a full generator rerun, and which archived files are retained only as diagnostic history.

The choice of the adjoint character is not merely a source of numerical examples. Its powers are the characters of tensor powers of the adjoint module, so every Haar moment is an invariant-tensor multiplicity and can be attacked uniformly through the classification. Part III takes the smallest uniformly closed multiplicative convex cone generated by this character. By Ginibre's Proposition 1, that cone is exactly the largest function class that follows formally from single-generator signed-product inequalities; enlarging it requires genuinely mixed-character input. The resulting theorem is thus a complete statement about the adjoint tensor category, while remaining strictly narrower than a theorem for all central positive-definite functions.

Contribution and relation to prior results. Ginibre's Proposition 1 supplies a promotion mechanism once the required signed products are known on a self-conjugate generating set; it does not establish the repeated-adjoint inequality proved here. The exact stable trace laws of Diaconis–Shahshahani, the power-map identities of Rains, and the character bounds of Garibaldi–Guralnick–Rains provide inputs to individual family arguments, but none of those results compares the two moment diagonals in lemma 2.3 for every exponent and every simple type. Likewise, the adjoint tensor-rank calculations of Bourjaily–Plesser–Vergu provide comparison data for exceptional moments, not the sign theorem.

The contribution of Parts I–II is the uniform classification theorem and the mechanism joining three regimes that do not overlap automatically: stable invariant-tensor moments, finite-rank corrections, and analytic tails. The repeated-adjoint and exactly-two-minus restrictions are essential parts of the statement, not abbreviations for a result about arbitrary characters. In particular, this paper makes no claim for unequal irreducible characters or for all positive-definite class functions. The separate Part III uses Ginibre's promotion only after importing the theorem proved here; it is not used to justify the present theorem or its novelty.

The computer calculations are proof devices rather than the mathematical claim. Each active calculation has a numbered finite contract, and the compact residual B/C contract table below records the exact theorem labels, rank ranges, and moment offsets supplied by the detailed proof.

2. Statement and General Reductions

Definition 2.1 (Adjoint Q_3 functional)**.** For a compact connected Lie group G with simple Lie algebra and normalized Haar probability measure, set

$$Q_3^G(n) = \int_{G\times G} (\chi_{\mathrm{ad}}(g) - \chi_{\mathrm{ad}}(h))^2 (\chi_{\mathrm{ad}}(g) + \chi_{\mathrm{ad}}(h))^n \, d\,\mathrm{Haar}(g)\, d\,\mathrm{Haar}(h)$$

for $n \in \mathbb{Z}_{\geq 0}$, where χ_{ad} is the adjoint character.

Theorem 2.2 (Two-minus adjoint-character Q3 nonnegativity)**.** *For every compact connected Lie group G with simple Lie algebra and every integer $n \geq 0$,*

$$Q_3^G(n) \geq 0.$$

Lemma 2.3 (Moment formula)**.** *Let*

$$m_k^G = \int_G \chi_{\mathrm{ad}}(g)^k \, d\,\mathrm{Haar}(g) = \mathrm{mult}(\mathbf{1}, \mathrm{ad}^{\otimes k}).$$

Then

$$Q_3^G(n) = 2\sum_{k=0}^{n}\binom{n}{k}\left(m_{k+2}^G m_{n-k}^G - m_{k+1}^G m_{n-k+1}^G\right).$$

Proof. Expand $(\chi_{\mathrm{ad}}(g)+\chi_{\mathrm{ad}}(h))^n$ by the binomial theorem, multiply by $(\chi_{\mathrm{ad}}(g)-\chi_{\mathrm{ad}}(h))^2$, and integrate term by term over $G\times G$. Schur orthogonality identifies $\int_G \chi_{\mathrm{ad}}(g)^k\, d\,\mathrm{Haar}(g)$ with the invariant multiplicity $\mathrm{mult}(\mathbf{1},\mathrm{ad}^{\otimes k})$. □

Lemma 2.4 (Racah–Speiser adjoint tensor step)**.** *Let V_λ be an irreducible representation of a compact connected semisimple group, let ρ be the half-sum of the positive roots, and let $a(\mu)$ be the multiplicity of the weight μ in the adjoint representation. For every adjoint weight μ, proceed as follows. If $\lambda+\mu+\rho$ lies on a Weyl wall, assign zero. Otherwise let w be the unique Weyl element for which $w(\lambda+\mu+\rho)$ is strictly dominant, put*

$$\nu = w(\lambda+\mu+\rho)-\rho,$$

and assign $\mathrm{sgn}(w)a(\mu)$ to V_ν. After equal dominant weights are collected, the resulting signed sum is the irreducible decomposition of $V_\lambda\otimes\mathrm{ad}$.

Proof. For a weight η write $A_\eta = \sum_{w\in W}\mathrm{sgn}(w)e^{w\eta}$. The Weyl character formula gives $\chi_\lambda = A_{\lambda+\rho}/A_\rho$. Since the adjoint weight multiset is Weyl invariant,

$$A_{\lambda+\rho}\chi_{\mathrm{ad}} = \sum_\mu a(\mu)A_{\lambda+\mu+\rho}.$$

An alternant vanishes when its index is singular. If the index is regular, moving it to the strict dominant chamber changes the alternant by $\mathrm{sgn}(w)$, and division by A_ρ gives the character specified in the statement. Collecting equal dominant indices proves the formula. This is the Racah–Speiser procedure; the same shifted-wall convention is described in [14, Appendix B, especially (B.1)–(B.6)]. □

Lemma 2.5 (Character-pairing reconstruction of moments)**.** *Write the exact irreducible decomposition*

$$\chi_{\mathrm{ad}}^j = \sum_{\lambda\in P_+} c_j(\lambda)\chi_\lambda.$$

Then, for every $j\ge 0$,

$$m_{2j}^G = \sum_{\lambda\in P_+} c_j(\lambda)^2, \qquad m_{2j+1}^G = \sum_{\lambda\in P_+} c_j(\lambda)c_{j+1}(\lambda).$$

Consequently all moments through degree M can be reconstructed after iterating lemma 2.4 only through tensor power $\lceil M/2\rceil$.

Proof. The adjoint character is real, so

$$m_{a+b}^G = \int_G \chi_{\mathrm{ad}}(g)^a\,\overline{\chi_{\mathrm{ad}}(g)^b}\, d\,\mathrm{Haar}(g) = \left\langle \chi_{\mathrm{ad}}^a, \chi_{\mathrm{ad}}^b\right\rangle_{L^2(G)}.$$

Irreducible-character orthonormality makes the last expression $\sum_\lambda c_a(\lambda)c_b(\lambda)$. Taking $(a,b)=(j,j)$ and $(j,j+1)$ proves the two identities. □

Lemma 2.6 (Positivity of adjoint moments)**.** *The adjoint character is real-valued, and*

$$m_0^G = 1, \qquad m_1^G = 0, \qquad m_2^G = 1,$$

while

$$m_k^G = \dim\big((\mathfrak{g}_{\mathbf{C}}^{\otimes k})^G\big) \quad \textit{is a positive integer for every } k\ge 2.$$

In particular, every transition ratio used below has a nonzero positive denominator.

Proof. The adjoint action preserves a positive-definite real inner product B on the compact real form $\mathfrak{g}$, so its complexification is self-dual and its character is real. The compact-semisimple/complex-semisimple correspondence and decomposition into simple root systems in [1, Chapter V, §5, pp. 209–215, especially (5.5) and Theorem (5.8)] show that $\mathfrak{g}_{\mathbf{C}}$ is a complex simple Lie algebra. If a complex subspace of $\mathfrak{g}_{\mathbf{C}}$ is invariant under G, connectedness of G and differentiation make it invariant under $\mathrm{ad}(\mathfrak{g}_{\mathbf{C}})$; it is therefore an ideal. Thus the complexified adjoint representation is irreducible.

Character orthogonality now gives the displayed invariant-space formula and, because the character is real,

$$m_2^G = \int_G |\chi_{\mathrm{ad}}(g)|^2 \, d\,\mathrm{Haar}(g) = 1.$$

Degree zero consists of the scalars. A degree-one invariant is a vector fixed by the adjoint action, hence lies in the center of $\mathfrak{g}_{\mathbf{C}}$; simplicity gives $m_1^G = 0$. The tensor B is a nonzero invariant tensor of degree two. Simplicity and non-abelianness imply that the Cartan tensor

$$\omega(X, Y, Z) = B(X, [Y, Z])$$

is a nonzero invariant tensor of degree three. Every integer $k \geq 2$ has the form $2a + 3b$ with $a, b \geq 0$; tensoring a copies of B and b copies of ω, and using B to identify $\mathfrak{g}$ with its dual, gives a nonzero invariant tensor in degree k. Hence $m_k^G \geq 1$. □

Lemma 2.7 (Central quotient invariance)**.** *If $\pi : \widetilde{G} \to G$ is a finite central cover of compact connected Lie groups with simple Lie algebra, then*

$$Q_3^{\widetilde{G}}(n) = Q_3^G(n)$$

for every $n \geq 0$.

Proof. The adjoint characters satisfy $\chi_{\mathrm{ad}}^{\widetilde{G}} = \chi_{\mathrm{ad}}^G \circ \pi$. The push-forward of normalized Haar measure under π is normalized Haar measure. Apply this in the two variables of the defining integral. □

Lemma 2.8 (Reduction to the simple Lie algebra)**.** *Let G be a compact connected Lie group with simple Lie algebra $\mathfrak{g}$. There is a compact simply connected Lie group G_{sc} with Lie algebra $\mathfrak{g}$ and a finite central covering $G_{\mathrm{sc}} \to G$. Consequently, if G_1 and G_2 are compact connected Lie groups with isomorphic simple Lie algebras, then*

$$Q_3^{G_1}(n) = Q_3^{G_2}(n)$$

for every $n \geq 0$.

Proof. A simple Lie algebra is semisimple. For a compact connected semisimple group, the universal covering group is compact and its kernel is finite and central; moreover, the simply connected compact group is determined by its Lie algebra. These statements are recorded in [1, Chapter V, Remark (7.13) and §8, p. 232]. Thus $G_{\mathrm{sc}} \to G$ has the asserted properties. Isomorphic simple Lie algebras give isomorphic simply connected compact covering groups, so two applications of lemma 2.7 prove the displayed equality. □

Lemma 2.9 (Even exponents)**.** *For every G and every even $n \geq 0$, $Q_3^G(n) \geq 0$.*

Proof. For even n, both $(\chi_{\mathrm{ad}}(g) - \chi_{\mathrm{ad}}(h))^2$ and $(\chi_{\mathrm{ad}}(g) + \chi_{\mathrm{ad}}(h))^n$ are pointwise nonnegative. □

Definition 2.10 (Chain difference)**.** For odd exponents define

$$D_G(n) = Q_3^G(n+2) - 4Q_3^G(n).$$

Equivalently, with $n = 2m + 1$,

$$D_G(2m+1) = Q_3^G(2m+3) - 4Q_3^G(2m+1).$$

Lemma 2.11 (Chain propagation)**.** *Let $0 \le m_0 \le m_1$. Assume $D_G(2m+1) \ge 0$ for $m = m_0, m_0+1, \ldots, m_1$ and $Q_3^G(2m_0+1) \ge 0$. Then*

$$Q_3^G(2m+1) \ge 0$$

for every $m_0 \le m \le m_1 + 1$.

Proof. The recurrence

$$Q_3^G(2m+3) = 4Q_3^G(2m+1) + D_G(2m+1)$$

propagates nonnegativity one odd exponent at a time. □

3. Type A: The Adjoint Representation of SU(N)

This section proves the type A branch. Since finite central quotients do not change Q_3 by lemma 2.8, it suffices to use $\mathrm{SU}(N)$ as the representative of type A_{N-1}.

Theorem 3.1 (Stable-rank type A branch)**.** *For every $N \ge 2$ and every $n \ge 0$ satisfying $N \ge n+2$,*

$$Q_3^{\mathrm{SU}(N),\mathrm{ad}}(n) \ge 0.$$

Proof. Write $X = |\operatorname{tr} U|^2$ for U Haar-distributed in $\mathrm{SU}(N)$. For $0 \le k \le N$, scalar invariance identifies the $\mathrm{SU}(N)$ integral of X^k with the $\mathrm{U}(N)$ integral. Schur expansion and character orthogonality give

$$\mathbf{E}X^k = \sum_{\lambda \vdash k,\ \ell(\lambda) \le N} (\dim \mathrm{Sp}_\lambda)^2.$$

When $k \le N$ the length condition is void, and the regular representation of the symmetric group gives $\mathbf{E}X^k = k!$. Since $\chi_{\mathrm{ad}}(U) = X - 1$, this gives

$$m_r^{\mathrm{SU}(N)} = \sum_{j=0}^{r} (-1)^{r-j} \binom{r}{j} j! = !r \qquad (0 \le r \le N),$$

where $!r$ is the derangement number. If $N \ge n+2$, all moments appearing in lemma 2.3 are in this stable range.

Let Y have the exponential distribution of mean 1, and set $A = Y - 1$. The exponential generating function $M(z) = \mathbf{E}e^{zA} = e^{-z}(1-z)^{-1}$ satisfies

$$M''(z)M(z) - M'(z)^2 = M(z)^2(\log M)''(z) = \frac{e^{-2z}}{(1-z)^4}.$$

Thus $Q_3^{\mathrm{SU}(N)}(n)/2$ is the nth exponential coefficient of $e^{-2z}(1-z)^{-4}$. If these coefficients are denoted q_n, then $q_0 = 1$, $q_1 = 2$, and differentiating gives

$$q_{n+1} = (n+2)q_n + 2nq_{n-1} \qquad (n \ge 1).$$

Hence $q_n > 0$ for all $n \ge 0$. □

Definition 3.2 (Type A transition ratios)**.** For $N \ge 2$, put

$$m_k^{(N)} = \int_{\mathrm{SU}(N)} \chi_{\mathrm{ad}}(U)^k \, dU, \qquad \rho_k^{(N)} = \frac{m_{k+1}^{(N)}}{m_k^{(N)}} \quad (k \ge 3).$$

The transition-ratio inequality at (N,k) is

$$R_{N,k} := m_{k+2}^{(N)} m_k^{(N)} - (m_{k+1}^{(N)})^2 \ge 0,$$

equivalently $\rho_k^{(N)} \le \rho_{k+1}^{(N)}$.

Lemma 3.3 (Boundary reduction for type A)**.** *Fix $N \geq 4$. If*

$$\rho_3^{(N)} = \frac{9}{2}, \qquad \frac{9}{2} \leq \rho_k^{(N)} \leq \rho_{k+1}^{(N)} \quad (k \geq 3),$$

then $Q_3^{\mathrm{SU}(N)}(n) > 0$ for every odd $n \geq 5$.

Proof. Use the symmetric form

$$\frac{1}{2}Q_3(n) = \frac{1}{2}\sum_{j=0}^{n}\binom{n}{j}\big(m_{j+2}m_{n-j} + m_j m_{n+2-j} - 2m_{j+1}m_{n+1-j}\big).$$

For $3 \leq j \leq n-3$, division by $m_{j+1}m_{n+1-j}$ gives

$$\frac{\rho_{j+1}}{\rho_{n-j}} + \frac{\rho_{n+1-j}}{\rho_j} - 2 \geq 0$$

by monotonicity and AM-GM. The remaining depth-two boundary contribution

$$L_2 = B_0 + nB_1 + \binom{n}{2}B_2$$

uses only $m_0 = 1, m_1 = 0, m_2 = 1, m_3 = 2, m_4 = 9$ and the four ratios $R = \rho_{n-2}$, $S = \rho_{n-1}$, $T = \rho_n$, $U = \rho_{n+1}$. Substitution and $U \geq T$ give

$$\frac{L_2}{m_{n-2}} \geq RS\left(T^2 + \frac{n^2 - 5n + 2}{2}\right) - 2n(n-2)R + \frac{9}{2}n(n-1).$$

Since $R, S, T \geq 9/2$, the right hand side is at least

$$\frac{9(10n^2 - 66n + 765)}{16} > 0.$$

Thus the boundary contribution is positive and all interior contributions are nonnegative. □

Lemma 3.4 (Exact formal derivation of the low-rank type A recurrences)**.** *Put*

$$E_k^{(N)} = \mathbf{E}_{\mathrm{U}(N)}|\operatorname{tr} U|^{2k}, \qquad m_k^{(N)} = \sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}E_j^{(N)}.$$

Then the following recurrences hold:

N	*recurrence*
3	$(k+3)^2 m_{k+1} = k(7k+11)m_k + 8k(k+1)m_{k-1}, \quad k \geq 1,$
4	$0 = (k^3 + 22k^2 + 161k + 392)m_{k+4}$ $- (16k^3 + 230k^2 + 1054k + 1524)m_{k+3}$ $+ (10k^3 - 340k - 750)m_{k+2}$ $+ (72k^3 + 522k^2 + 1242k + 972)m_{k+1}$ $+ (45k^3 + 270k^2 + 495k + 270)m_k, \quad k \geq 0,$
5	$0 = 192(k+1)(k+2)(k+3)(k+6)m_k$ $+ 16(k+2)(k+3)(22k^2 + 184k + 309)m_{k+1}$ $+ (k+3)(129k^3 + 1245k^2 + 2570k - 1112)m_{k+2}$ $- (k+3)(30k^3 + 642k^2 + 4487k + 10196)m_{k+3}$ $+ (k+8)^2(k+10)^2 m_{k+4}, \quad k \geq 0.$

Proof. Gessel's determinant [3, §7, Theorem 16 and the formula following (26)] gives

$$F_N(t) := \sum_{k\geq 0} E_k^{(N)} \frac{t^k}{(k!)^2} = \det\bigl(I_{|i-j|}(2\sqrt{t})\bigr)_{1\leq i,j\leq N}.$$

Put $y = I_0(2\sqrt{t})$ and $z = y'$. The Bessel recurrence and $ty'' + y' - y = 0$ give

$$I_1(2\sqrt{t}) = \sqrt{t}\, z, \qquad I_{r+1}(2\sqrt{t}) = I_{r-1}(2\sqrt{t}) - \frac{r}{\sqrt{t}} I_r(2\sqrt{t}), \qquad z' = \frac{y-z}{t}.$$

Expanding the three determinants, with no series truncation, gives

$$\begin{aligned} F_3 &= z(2y^2 - yz - 2tz^2), \\ F_4 &= -\frac{1}{t}\bigl(4t^2z^4 - 8ty^2z^2 + 8tyz^3 - tz^4 + 4y^4 - 8y^3z + 4y^2z^2\bigr), \\ F_5 &= -\frac{4}{t^2}\bigl(2y^2 - yz - (2t+1)z^2\bigr) \\ &\qquad\qquad \cdot \bigl(4y^3 - 5y^2z + yz^2 - 4tyz^2 + 3tz^3\bigr). \end{aligned} \tag{3.4.1}$$

Here is a finite differential certificate for the coefficient recurrences. On $\mathbb{Q}[t, t^{-1}, y, z]$, let ∂ be the derivation

$$\partial t = 1, \qquad \partial y = z, \qquad \partial z = (y-z)/t,$$

and put

$$\vartheta = t\partial, \qquad \mathcal{S} = (\vartheta + 1)\partial.$$

If $F = \sum_{k\geq 0} a_k t^k/(k!)^2$, then

$$\mathcal{S}F = \sum_{k\geq 0} a_{k+1} \frac{t^k}{(k!)^2}, \qquad q(\vartheta)F = \sum_{k\geq 0} q(k) a_k \frac{t^k}{(k!)^2}. \tag{3.4.2}$$

For the following explicitly displayed polynomials, direct calculation in this Laurent-polynomial differential ring gives

$$\sum_j q_{N,j}(\vartheta)\mathcal{S}^j F_N = 0 \qquad (N = 3, 4, 5). \tag{3.4.3}$$

They are

$$\begin{array}{c|l}
N & (q_{N,0}(x), q_{N,1}(x), \ldots) \\
\hline
3 & \begin{array}{l}(9(x+1)^2,\ -(19x^2+78x+77),\\ \ 11x^2+70x+109,\ -(x+5)^2);\end{array} \\
4 & \begin{array}{l}(-64(x+1)^2(x+2),\\ \ 2(106x^3+763x^2+1791x+1366),\\ \ -(253x^3+2662x^2+9223x+10482),\\ \ 127x^3+1795x^2+8378x+12884,\\ \ -(23x^3+428x^2+2625x+5312),\\ \ (x+8)^2(x+9));\end{array} \\
5 & \begin{array}{l}(225(x+1)^2(x+2)^2,\\ \ -(x+2)^2(934x^2+6226x+10317),\\ \ 1487x^4+22936x^3+130422x^2+324256x+297420,\\ \ -2(554x^4+11290x^3+85109x^2+281496x+344526),\\ \ 367x^4+9494x^3+91097x^2+384340x+601200,\\ \ -(38x^4+1282x^3+15993x^2+87412x+176688),\\ \ (x+10)^2(x+12)^2).\end{array}
\end{array} \tag{3.4.4}$$

Thus (3.4).2–.3 give the $E_k^{(N)}$-recurrences with coefficients $q_{N,j}(k)$.

For completeness, we verify that these are exactly the binomial transforms of the three recurrences in the statement. Let

$$A_N(w) = \sum_{k\ge 0} E_k^{(N)} \frac{w^k}{k!}, \qquad M_N(w) = \sum_{k\ge 0} m_k^{(N)} \frac{w^k}{k!}.$$

The binomial-transform identity is $M_N(w) = e^{-w}A_N(w)$. If a proposed recurrence is written $\sum_r p_{N,r}(k)m_{k+r} = 0$, its exponential-generating-function operator is

$$\sum_r p_{N,r}(wD)D^r, \qquad D = \frac{d}{dw}.$$

Conjugating by e^{-w} replaces D by $D-1$ and wD by $wD-w$. Normal ordering with the single relation

$$Dw = wD + 1$$

and then taking coefficients gives the polynomials in (3.4).4, after index shifts $1, 1, 2$ for $N = 3, 4, 5$, respectively. The finitely many coefficients below those shifts vanish by the hook-length values for $E_k^{(N)}$.

Every algebraic operation just described is replayed without floating point or symbolic-library assumptions by

`character_ring_iter/verify_sun_recurrence_derivation_exact.py`.

It represents a monomial as $t^{a/2}y^bz^c$, expands the determinants over all permutations, normal-orders the Weyl algebra by the displayed commutation relation, applies $\partial z = (y-z)/t$, and requires the final Laurent polynomial to have zero terms. It also regenerates the initial $E_k^{(N)}$ from the hook-length formula and checks the omitted coefficients. The status-zero transcript reports zero reduced-identity terms and zero finite residuals for all three ranks. The source and transcript SHA-256 values are, respectively,

$$\begin{array}{l}\texttt{e08536805bd04122b6cf87ac1ba083bb}\\ \texttt{d5076cb2c1620baad6002ae3f42fb100}\end{array}, \qquad \begin{array}{l}\texttt{71423c65e6c02d1862e14d3008120816}\\ \texttt{e324d15a9642dd7af5143bf530c4a7ec}\end{array}.$$

This proves the three formal recurrences for every stated index. ☐

Theorem 3.5 (Low-rank type A ratio theorem)**.** *For $N = 3$ one has*

$$\rho_3^{(3)} = \rho_4^{(3)} = 4, \qquad 4 \le \rho_k^{(3)} \le \rho_{k+1}^{(3)} \quad (k \ge 3).$$

For $N = 4, 5$ one has

$$\rho_3^{(N)} = \frac{9}{2}, \qquad \frac{9}{2} \le \rho_k^{(N)} \le \rho_{k+1}^{(N)} \quad (k \ge 3).$$

Proof. Use the exact formal recurrences of lemma 3.4.

For $N = 3$, set $U_k = 8 - \rho_k$ and

$$G_k = \frac{32k + 100}{(k+4)^2}, \qquad H_k = \frac{32(k + 11/5)}{(k+3)^2}.$$

The first recurrence gives $U_{k+1} = \Phi_{k+1}(U_k)$, where

$$\Phi_k(u) = \frac{k^2 + 37k + 72 - 8k(k+1)/(8-u)}{(k+3)^2}.$$

This map is decreasing for $u < 8$. The exact base values are $U_5 = 111/32$, $G_5 = 260/81$, and $H_5 = 18/5$. The two induction margins are

$$\Phi_{k+1}(H_k) - G_{k+1} = \frac{4(34k^4 + 288k^3 + 683k^2 + 278k - 403)}{(k+4)^2(k+5)^2(5k^2 + 10k + 1)},$$

$$H_{k+1} - \Phi_{k+1}(G_k) = \frac{29k^2 + 91k + 54}{5(k+4)^2(2k^2 + 8k + 7)}.$$

They are positive for $k \ge 5$, so $G_k \le U_k \le H_k$. If r_k^* is the positive root of

$$(k+4)^2 r^2 - (k+1)(7k+18)r - 8(k+1)(k+2) = 0,$$

then, on writing $\epsilon_k^* = 8 - r_k^*$, the discriminant identity

$$(9k^2 + 103k + 238)^2 - 4(k+4)^2(288k + 864) = (9k^2 + 39k + 38)^2 + 16(k-3)(k+2)$$

gives $\epsilon_k^* \le G_k \le U_k$. Hence $\rho_{k+1} \ge \rho_k$ for $k \ge 5$; the exact initial ratios are $\rho_3 = \rho_4 = 4$ and $\rho_5 = 145/32$.

For $N = 4$, set $U_k = 15 - \rho_k$ and

$$G_k = \frac{1125}{10k + 71}, \qquad H_k = \frac{1125}{10k + 66}.$$

Solving the second recurrence for U_{k+3} gives a rational map increasing in U_k, U_{k+1}, U_{k+2} on the box $0 \le U_j \le H_j$ for $j \ge 7$. The exact base margins are

j	$U_j - G_j$	$H_j - U_j$
7	4847/73179	48089/211752
8	120402/1635179	286983/1581034
9	331053/4344746	103677/701636.

Clearing the positive denominators in the lower and upper induction margins gives, respectively,

$$\begin{aligned} P_-(s) &= 2432000s^5 + 100233600s^4 + 1609345435s^3 \\ &\quad + 12643460157s^2 + 48855354258s + 74691137880, \\ P_+(s) &= 128000s^5 + 24662400s^4 + 750722465s^3 \\ &\quad + 8917574583s^2 + 46522820852s + 89208809220, \end{aligned}$$

at $s = k - 7$. The denominators are

$$40(k+7)^2(k+8)(5k-2)(5k+3)(5k+8)(10k+101),$$
$$10(k+7)^2(k+8)(5k+48)(10k-9)(10k+1)(10k+11),$$

so the induction proves $G_k \le U_k \le H_k$ for $k \ge 7$. Since $H_{k+1} < G_k$, the ratios increase. Direct substitution of $m_3 = 2, m_4 = 9, m_5 = 43, m_6 = 245, m_7 = 1557, m_8 = 10829$ checks the preceding indices and the lower bound $\rho_k \ge 9/2$.

For $N = 5$, set $U_k = 24 - \rho_k$ and

$$G_k = \frac{2880}{10k+109}, \qquad H_k = \frac{288}{k+10}.$$

The third recurrence similarly gives an increasing three-variable map for $k \ge 6$. Its exact base margins are

j	$U_j - G_j$	$H_j - U_j$
6	92/1859	10/11
7	19597/163248	11345/15504
8	3424/21315	4272/7105.

After clearing the positive denominators, the two induction numerators at $s = k - 6$ are

$$\begin{aligned} R_-(s) &= 3750000s^6 + 174475000s^5 + 3214281000s^4 \\ &\quad + 30095015750s^3 + 153613752915s^2 \\ &\quad + 416461589985s + 482743428300, \\ R_+(s) &= 1900s^5 + 76662s^4 + 1135088s^3 + 7550562s^2 \\ &\quad + 22473804s + 23833584, \end{aligned}$$

with denominators

$$2(k+8)^2(k+10)^2(10k-11)(10k-1)(10k+9)(10k+139),$$
$$k(k-2)(k-1)(k+8)^2(k+10)^2(k+13).$$

Thus $G_k \le U_k \le H_k$ for $k \ge 6$, and again $H_{k+1} < G_k$. The exact initial ratios $9/2, 44/9, 6, 76/11$ complete the $N = 5$ proof. □

Lemma 3.6 (The $SU(3)$ boundary)**.** *The ratio inequalities in theorem 3.5 imply* $Q_3^{\mathrm{SU}(3)}(n) > 0$ *for every odd* $n \ge 11$.

Proof. The interior terms in the symmetric moment formula are nonnegative by the same AM–GM argument as in lemma 3.3. For the depth-two boundary put

$$\rho_{n-2} = 4 + s, \qquad \rho_{n-1} = 4 + s + t, \qquad s, t \ge 0.$$

Substitution of the $N = 3$ recurrence from theorem 3.5 gives

$$L_2 = \frac{m_{n-2}}{2(n+3)^2(n+4)^2} P(n, s, t),$$

where

$$\begin{aligned} P &= n^6(s^2 + st + 4s + 4t + 8) + n^5(9s^2 + 9st + 24s + 36t + 56) \\ &\quad + n^4(119s^2 + 119st + 884s + 476t + 2104) \\ &\quad + n^3(479s^2 + 479st + 4256s + 1916t + 10120) \\ &\quad + n^2(716s^2 + 716st + 7184s + 2864t + 17088) \\ &\quad + n(636s^2 + 636st + 6528s + 2544t + 14784) \\ &\quad + 576s^2 + 576st + 4608s + 2304t + 9216. \end{aligned}$$

Every coefficient is nonnegative and the constant term is positive. Thus $L_2 > 0$, while every interior term is nonnegative. $\square$

Theorem 3.7 (The $SU(2)$ row)**.** *For every $n \geq 0$, $Q_3^{\mathrm{SU}(2)}(n) \geq 0$.*

Proof. Parametrize $SU(2)$ conjugacy classes by $\alpha \in [0, \pi]$. The class density is $(2/\pi)\sin^2\alpha\, d\alpha$, and $\chi_{\mathrm{ad}}(\alpha) = 1 + 2\cos(2\alpha)$. For independent angles α, β, put $\theta = \alpha + \beta$ and $\phi = \alpha - \beta$. Folding the resulting diamond uses the identities

$$\chi_{\mathrm{ad}}(\alpha) + \chi_{\mathrm{ad}}(\beta) = 2 + 4\cos\theta\cos\phi,$$
$$(\chi_{\mathrm{ad}}(\alpha) - \chi_{\mathrm{ad}}(\beta))^2 = 16\sin^2\theta\sin^2\phi,$$
$$4\sin^2\alpha\sin^2\beta = (\cos\theta - \cos\phi)^2, \qquad d\alpha\, d\beta = \tfrac{1}{2}d\theta\, d\phi.$$

Thus folding by the C_2 Weyl symmetries sends

$$\frac{1}{2}(\chi_{\mathrm{ad}}(\alpha) - \chi_{\mathrm{ad}}(\beta))^2\, d\mu(\alpha)d\mu(\beta)$$

to the normalized USp(4) Weyl density

$$\frac{8}{\pi^2}(\cos\theta - \cos\phi)^2\sin^2\theta\sin^2\phi\, d\theta d\phi \quad (0 \leq \theta, \phi \leq \pi)$$

from [1, Chapter IV, (1.11)]. The normalization follows at $n = 0$ from Schur orthogonality. If V is the defining representation of USp(4), then

$$\chi_{\mathrm{ad}}(\alpha) + \chi_{\mathrm{ad}}(\beta) = 2 + 4\cos\theta\cos\phi = \chi_{\Lambda^2 V}(\theta, \phi).$$

Consequently

$$\frac{1}{2}Q_3^{\mathrm{SU}(2)}(n) = \dim\big(((\Lambda^2 V)^{\otimes n})^{\mathrm{USp}(4)}\big) \geq 0.$$

$\square$

Lemma 3.8 (Derangement-ratio gap)**.** *If $D_r = !r$, then for every $r \geq 3$,*

$$\frac{D_{r+2}}{D_{r+1}} - \frac{D_{r+1}}{D_r} \geq \frac{7}{18}.$$

Proof. The recurrence $D_{r+1} = (r+1)D_r + (-1)^{r+1}$ gives

$$\frac{D_{r+2}}{D_{r+1}} - \frac{D_{r+1}}{D_r} = 1 + \frac{(-1)^{r+2}}{D_{r+1}} - \frac{(-1)^{r+1}}{D_r}.$$

For even r this is greater than 1. For odd $r \geq 3$, use $D_r \geq D_3 = 2$ and $D_{r+1} \geq D_4 = 9$ to obtain $1 - 1/9 - 1/2 = 7/18$. $\square$

Lemma 3.9 (Uniform type A transition strip)**.** *For every $N \geq 6$ and every integer k with*

$$N - 1 \leq k \leq N + 33,$$

one has

$$R_{N,k} = m_{k+2}^{(N)} m_k^{(N)} - (m_{k+1}^{(N)})^2 > 0.$$

Proof. Let $D_r = !r$. By Schur–Weyl duality and RSK,

$$a! - E_a^{(N)} = \#\{\pi \in S_a : \mathrm{lds}(\pi) > N\}.$$

If the longest decreasing subsequence has length greater than N, some fixed set of $N+1$ positions appears in decreasing relative order. The union bound therefore gives

$$0 \leq a! - E_a^{(N)} \leq a! \frac{\binom{a}{N+1}}{(N+1)!}.$$

Fix k in the displayed strip and put $K = k+2$. Comparing the binomial transforms for $m_r^{(N)}$ and D_r, for $r \le K$ one obtains

$$\left|m_r^{(N)} - D_r\right| \le 3r!\frac{\binom{K}{N+1}}{(N+1)!} \le \tau D_r, \qquad \tau := 9\frac{\binom{K}{N+1}}{(N+1)!},$$

where $\sum_{j\ge 0} 1/j! < 3$ and $D_r \ge r!/3$ for $r \ge 2$. The latter bound follows directly from $D_r/r! = \sum_{j=0}^{r}(-1)^j/j!$: it is an alternating sum, with the minimum $1/3$ attained at $r=3$.

By lemma 3.8,

$$D_{k+2}D_k - D_{k+1}^2 \ge \frac{7}{18}D_kD_{k+1}.$$

Writing $m_r^{(N)} = D_r + e_r$, expanding $R_{N,k}$, and using $|e_r| \le \tau D_r$ together with

$$D_{k+1} \le (k+2)D_k, \qquad D_{k+2} \le (k+3)D_{k+1},$$

gives

$$R_{N,k} \ge D_kD_{k+1}\left[\frac{7}{18} - \tau(4k+10) - \tau^2(2k+5)\right]. \tag{$*$}$$

For $6 \le N \le 18$, the exact C++/GMP hook-length replay evaluates all 35 values $N-1 \le k \le N+33$ and finds every integer $R_{N,k}$ strictly positive. Now assume $N \ge 19$ and write $k = N+d$, $-1 \le d \le 33$. The bracket in $(*)$ is smallest at $d = 33$. In that case

$$\tau_N = 9\frac{\binom{N+35}{N+1}}{(N+1)!}, \qquad \frac{\tau_{N+1}}{\tau_N} = \frac{N+36}{(N+2)^2}.$$

Both error terms decrease for $N \ge 19$. Indeed, after putting $s = N-19$, the cleared numerator for the first ratio comparison is

$$4s^3 + 382s^2 + 10478s + 83928,$$

and for the second it is

$$2s^5 + 277s^4 + 14446s^3 + 362171s^2 + 4408498s + 20862654.$$

All coefficients are positive. At the initial value $N = 19, d = 33$, exact rational arithmetic gives

$$\tau_{19} = \frac{3737279683}{3143467008000},$$

and the bracket in $(*)$ equals

$$\frac{1280170982560916731574699}{9881384830384472064000000} > \frac{1}{10}.$$

Thus $(*)$ is strictly positive for every $N \ge 19$ and every allowed d. The GMP checker independently reconstructs the finite moments, the base rational margin, and the two shifted coefficient lists. $\square$

Lemma 3.10 (Fixed-band determinant criterion)**.** *Let μ be a probability measure on $[-1, A]$, where $A > c > u > b > 1$, and put $m_j = \int x^j\, d\mu(x)$. If*

$$p_B = \mu([b,u]), \qquad p_C = \mu([c,A])$$

satisfy, for an odd integer k,

$$p_Bp_Cb^k(c-u)^2 > 2(c+1)^2, \tag{3.10.1}$$

then $m_{k+2}m_k - m_{k+1}^2 > 0$.

Proof. For independent X, Y with law μ,

$$m_{k+2}m_k - m_{k+1}^2 = \frac{1}{2}\mathbf{E}\big[X^kY^k(X-Y)^2\big].$$

The two symmetric rectangles $[b,u]\times[c,A]$ contribute at least $P = p_Bp_Cb^kc^k(c-u)^2$. The negative contribution for which the positive variable x lies in $[c,A]$ has absolute value at most $H = \int_{[c,A]} x^k(x+1)^2\,d\mu(x)$. For $x \geq c$, the quotient $(x+1)/(x-u)$ is decreasing. Since $p_C \leq 1$, (3.10).1 gives

$$p_Bb^k(x-u)^2 \geq p_Bb^k(c-u)^2\frac{(x+1)^2}{(c+1)^2} > 2(x+1)^2.$$

Integrating first over $[b,u]$ and then over $[c,A]$ shows $P > 2H$. The remaining negative contribution, with positive variable in $(0,c)$, is at most $L = (c+1)^2c^k$, while (3.10).1 gives $P/2 > L$. All same-sign contributions are nonnegative, proving the claim. □

Lemma 3.11 (Quadratic type-A transition window)**.** *For $N \geq 25$,*

$$R_{N,k} > 0 \qquad \left(N+16 \leq k \leq \left\lfloor \frac{N^2}{11}\right\rfloor - 2\right).$$

Proof. Put $K = \lfloor N^2/11\rfloor$, $\theta = 121/176$, and $D_r =!r$. The RSK formula in the proof of lemma 3.9 gives, for $r \leq K$,

$$|m_r^{(N)} - D_r| \leq 3r!\frac{\binom{K}{N+1}}{(N+1)!}.$$

Since $K < (N+1)^2/11$, $(N+1)! \geq ((N+1)/e)^{N+1}$, and $e < 11/4$,

$$\frac{\binom{K}{N+1}}{(N+1)!} \leq \theta^{N+1}.$$

Also $D_r \geq r!/3$ for $r \geq 2$, so $|m_r^{(N)} - D_r| \leq \tau D_r$ with $\tau = 9\theta^{N+1}$. Expanding $R_{N,k}$ about the derangement determinant as in lemma 3.9 yields

$$R_{N,k} \geq D_kD_{k+1}\left[\frac{7}{18} - \tau(4k+10) - \tau^2(2k+5)\right]. \tag{3.11.1}$$

For $k \leq K$ the error terms are bounded by

$$9\theta^{N+1}\left(\frac{4N^2}{11}+10\right), \qquad 81\theta^{2N+2}\left(\frac{2N^2}{11}+5\right).$$

They decrease for $N \geq 25$: after clearing denominators, the two successive ratio inequalities reduce respectively to

$$220N^2 - 968N + 5566 > 0, \qquad 32670N^2 - 58564N + 869143 > 0.$$

At $N = 25$ their exact rational sum is less than $129/1000 < 7/18$. Thus the bracket in (3.11).1 is positive. The C++/GMP replay independently checks the two cleared polynomials and the exact $N = 25$ comparison. □

Proposition 3.12 (Uniform type-A endpoint overlaps)**.** *The transition determinant satisfies $R_{N,k} > 0$ in both ranges*

N	*odd* k
$N \geq 24$	$k \geq 71,$
$6 \leq N \leq 23$	$k \geq 53.$

Proof. Let U be Haar-distributed in $\mathrm{SU}(N)$, put $Y = |\operatorname{tr} U|^2$, $X = Y - 1$, and let μ_N be the law of X. Then $X \in [-1, N^2 - 1]$. The $\mathrm{SU}(N)$ and $\mathrm{U}(N)$ moments of Y agree, and

$$\mathbf{E}Y^r = \sum_{\lambda \vdash r,\ \ell(\lambda) \le N} (f^\lambda)^2 \le r!, \tag{3.12.1}$$

with equality for $r \le N$.

First suppose $N \ge 24$. Set $z = y - 7/2$ and

$$\begin{aligned} P(z) = 925503648 &+ 348262482z - 142832255z^2 - 40966535z^3 \\ &+ 8723312z^4 + 880405z^5 - 242416z^6 + 14686z^7 - 274z^8, \end{aligned}$$

and $T = 1467426013$. On $y \in [5/2, 9/2]$, $|z| \le 1$, $0 \le (y - 5/2)(9/2 - y) \le 1$, and $|P(z)| \le T$; outside that interval the quadratic factor is nonpositive. Consequently

$$\mathbf{1}_{[5/2,9/2]}(y) \ge \frac{(y - 5/2)(9/2 - y)P(y - 7/2)^2}{T^2}.$$

Using the factorial moments through degree 18 gives

$$\mu_N([3/2, 7/2]) \ge \eta_B := \frac{12292322913754866540 7}{65536 \cdot 1467426013^2}.$$

Paley–Zygmund applied to Y^{12}, using the moments $12!$ and $24!$, gives

$$\mu_N([4, N^2 - 1]) \ge \eta_C := \frac{88255484124721}{99271744277318310297 6}.$$

Exact rational arithmetic gives $\eta_B \eta_C (3/2)^{71} > 200$. Thus lemma 3.10 applies with $(b, u, c) = (3/2, 7/2, 4)$ for every odd $k \ge 71$.

Now let $6 \le N \le 23$. Put $z = y - 43/10$ and

$$Q(z) = 959324176457 + 276017837985z - 58083597219z^2 - 11664458272z^3 + 1230335979z^4,$$

and $S = 935204207737834/625$. On $[29/10, 57/10]$ one has $|z| \le 7/5$, $0 \le (y - 29/10)(57/10 - y) \le 49/25$, and $|Q(z)| \le S$. The same signed-polynomial minorant gives

$$\mu_N([19/10, 47/10]) \ge \delta_B := \frac{44525786465344542780431067563007}{4388427632471750532372897337983385 6}.$$

For $N \ge 10$ this uses the stable moments through degree 10; for $6 \le N \le 9$ the exact hook calculation has nonnegative excess, whose minimum is 1513726621221888441 at $N = 9$.

Paley–Zygmund applied to Y^{15} gives

$$\mu_N([5, N^2 - 1]) \ge \delta_C := \frac{471756427}{178410035553750000}.$$

For $N \ge 15$, this follows from $\mathbf{E}Y^{15} = 15!$ and $\mathbf{E}Y^{30} \le 30!$. For $6 \le N \le 14$, the exact hook calculation checks both $\mathbf{E}Y^{15} > 6^{15}$ and

$$\left(1 - \frac{6^{15}}{\mathbf{E}Y^{15}}\right)^2 \frac{(\mathbf{E}Y^{15})^2}{\mathbf{E}Y^{30}} \ge \delta_C;$$

the smallest excess occurs at $N = 14$ and is strictly positive. Finally, exact rational arithmetic gives

$$\delta_B \delta_C (19/10)^{52} (3/10)^2 > 72.$$

For every odd $k \ge 53$, lemma 3.10 applies with $(b, u, c) = (19/10, 47/10, 5)$. The C++/GMP replay reconstructs all finite hook moments, polynomial expectations, Paley–Zygmund domain checks, and the two terminal rational inequalities used above. $\square$

Proposition 3.13 (Finite type-A overlap rows). *One has $R_{N,k} > 0$ for the following odd transition indices:*

N	k	N	k
24	$57, 59, \ldots, 69$	6	$39, 41, \ldots, 51$
25	$59, 61, \ldots, 69$	$7, 8$	$41, 43, \ldots, 51$
26	$61, 63, \ldots, 69$	$9, 10$	$43, 45, \ldots, 51$
27	$65, 67, 69$	$11, 12$	$45, 47, 49, 51$
		$13, 14$	$47, 49, 51$
		$15, 16$	$49, 51$
		$17, 18$	$51.$

Proof. For $24 \le N \le 27$, put $K = k + 2$ and $\tau = 9\binom{K}{N+1}/(N+1)!$. The RSK union bound and determinant expansion in lemma 3.9 give

$$R_{N,k} \ge D_k D_{k+1} \left[\frac{7}{18} - \tau(4k+10) - \tau^2(2k+5) \right].$$

Exact rational evaluation over the displayed finite rows gives a bracket greater than $3/10$ in every case.

For $6 \le N \le 18$, the exact hook formula in (3.12).1 supplies $\mathbf{E}Y^r$ through $r = 53$; binomial transformation gives the adjoint moments, and direct integer evaluation gives $R_{N,k} > 0$ at every displayed pair. The C++/GMP checker recomputes these partitions and hook dimensions, requires exact division at every step, matches a hard-coded positive row minimum for each N, and rejects any missing transition index. ☐

Theorem 3.14 (Finite type A ratio theorem). *For every $N \ge 6$,*

$$\rho_3^{(N)} = \frac{9}{2}, \qquad \frac{9}{2} \le \rho_k^{(N)} \le \rho_{k+1}^{(N)} \quad (k \ge 3).$$

Proof. The stable part $3 \le k \le N - 2$ follows from the derangement moments above. Indeed $m_r^{(N)} = !r$ for $r \le N$, and lemma 3.8 gives both monotonicity and the lower bound $\rho_k^{(N)} \ge \rho_3^{(N)} = 9/2$.

For even $k = 2q$, Cauchy–Schwarz applied to the real functions χ_{ad}^q and χ_{ad}^{q+1} gives

$$(m_{k+1}^{(N)})^2 \le m_k^{(N)} m_{k+2}^{(N)}.$$

The denominators are positive by lemma 2.6, so this proves the even transition inequalities.

The uniform strip lemma 3.9 covers $N - 1 \le k \le N + 33$. The remaining odd transition range is covered by the following proved overlaps:

range	certificate
$N - 1 \le k \le N + 33$	*lemma* 3.9
k even	Cauchy–Schwarz, as above
$6 \le N \le 18,\ k \le 51$	*proposition* 3.13
$6 \le N \le 23,\ k \ge 53$	*proposition* 3.12
$24 \le N \le 27,\ k < 71$	*lemma* 3.11*and proposition* 3.13
$N \ge 24,\ k \ge 71$	*proposition* 3.12
$N \ge 28,\ k \le 69$	*lemma* 3.11.

These alternatives cover all odd $k \ge N - 1$: the first line covers through $N + 33$. For $6 \le N \le 18$, the finite rows meet the endpoint range $k \ge 53$; for $19 \le N \le 23$, that endpoint already begins before the first remaining odd index. For $24 \le N \le 27$, the finite and quadratic rows meet $k \ge 71$. If $N \ge 28$ and a remaining odd index is at most 69, then it is at least $N + 34 > N + 16$, while $\lfloor N^2/11 \rfloor - 2 \ge 69$, so lemma 3.11 applies. Every later odd index is covered by proposition 3.12.

The certificate output for the finite-overlap replay is:

uniform strip $N-1 \le k \le N+33$	ok,
endpoint constants	ok,
$N = 6, \dots, 18$	ok,
final line	`All SU(N) repair certificates verified.`

All arithmetic in the finite replay is integer or rational arithmetic; no floating-point comparison is used in a sign decision. The infinite parts are the displayed inequalities in lemmas 3.9 and 3.11 and proposition 3.12. □

Corollary 3.15 (Finite-rank type A branch)**.** *For every $N \ge 2$ and every $n \ge 0$ satisfying $N < n+2$,*

$$Q_3^{\mathrm{SU}(N),\mathrm{ad}}(n) \ge 0.$$

Proof. The $N = 2$ row is theorem 3.7. For $N = 3$, theorem 3.5 and lemma 3.6 cover every odd $n \ge 11$; the hook-length formula gives the exact positive values

n	1	3	5	7	9
$Q_3(n)/2$	2	28	654	21312	902860.

For $N = 4, 5$, theorem 3.5 and lemma 3.3 give positivity for odd $n \ge 5$. For $N \ge 6$, theorem 3.14 and lemma 3.3 do the same. The remaining odd cases $n = 1, 3$ follow from the first adjoint moments:

N	m_0	m_1	m_2	m_3	m_5	$Q_3(1)$	$Q_3(3)$
2	1	0	1	1	6	2	8
3	1	0	1	2	32	4	56
4	1	0	1	2	43	4	78
≥ 5	1	0	1	2	44	4	80.

Here $Q_3(1) = 2m_3$ and $Q_3(3) = 2(m_5 - 2m_3)$. Even n is lemma 2.9. □

Theorem 3.16 (Type A closure)**.** *For every $N \ge 2$ and every $n \ge 0$,*

$$Q_3^{\mathrm{SU}(N),\mathrm{ad}}(n) \ge 0.$$

Proof. If $N \ge n+2$, use theorem 3.1. If $N < n+2$, use corollary 3.15. □

4. The Non-A Classical Families

The non-A classical Lie algebras are represented by the standard B_b, C_b, and D_b families. The proof uses exact adjoint moments, Rains stabilization, and a final finite certificate. We use the irreducible-root-system rank conventions

$$B_b\ (b \ge 2), \qquad C_b\ (b \ge 2), \qquad D_b\ (b \ge 4).$$

The low-rank coincidences $B_1 = C_1 = A_1$ and $D_3 = A_3$ are already covered by theorem 3.16; $D_2 = A_1 \times A_1$ is not simple. The coincidence $B_2 = C_2$ causes harmless overlap between the two finite classical rows.

Definition 4.1 (Classical constants and first hits)**.** Let

$$C_G = \max_{g \in G}(-\chi_{\mathrm{ad}}(g)).$$

Define

$$s(G) = \begin{cases} 2b-1, & G = B_b, C_b, \\ \text{largest odd integer } \le b-3, & G = D_b, \end{cases}$$

the largest odd exponent covered by the stable moment comparison. After the row-gated bridge through $m = 29$ the first possible unbridged odd exponent is

$$N_{\mathrm{hit}}^{(29)}(G) = \max(63, s(G) + 2).$$

Equivalently,

$$\begin{aligned} &B_b, C_b: \quad N_{\mathrm{hit}}^{(29)} = \max(63, 2b+1), \\ &D_b: \qquad N_{\mathrm{hit}}^{(29)} = \max(63, b-1) \text{ for even } b, \quad \max(63, b-2) \text{ for odd } b. \end{aligned}$$

Lemma 4.2 (Classical negative endpoint)**.** *For the standard B_b, C_b, D_b representatives,*

$$C_{B_b} = b, \qquad C_{C_b} = b, \qquad C_{D_b} = \begin{cases} b, & b \text{ even}, \\ b-2, & b \text{ odd}. \end{cases}$$

Proof. Write the defining eigenvalue pairs as z_j, z_j^{-1} and put $x_j = (z_j + z_j^{-1})/2 \in [-1, 1]$ and $S = \sum_j x_j$, $Q = \sum_j x_j^2$. For B_b the defining trace is $T_1 = 1 + 2S$ and $T_2 = 1 + 4Q - 2b$. Since the adjoint representation is Λ^2 of the defining representation,

$$\chi_{\mathrm{ad}} = \frac{T_1^2 - T_2}{2} = b + 2S + 2(S^2 - Q).$$

This expression is affine in each x_j separately, so its minimum occurs at a vertex. At a vertex $Q = b$ and $S = b - 2k$, hence $\chi_{\mathrm{ad}} = 2S^2 + 2S - b$; the minimum is $-b$.

For D_b, $T_1 = 2S$ and $T_2 = 4Q - 2b$, again with $\chi_{\mathrm{ad}} = (T_1^2 - T_2)/2$. Thus

$$\chi_{\mathrm{ad}} = b + 2(S^2 - Q).$$

At a vertex this is $2S^2 - b$. If b is even, $S = 0$ is allowed and the minimum is $-b$; if b is odd, the nearest allowed values are $S = \pm 1$ and the minimum is $2 - b$.

For C_b, the adjoint representation is Sym^2 of the defining representation, so

$$\chi_{\mathrm{ad}} = \frac{T_1^2 + T_2}{2} = 2S^2 + 2Q - b.$$

This is bounded below by $-b$, and equality is attained at $x_1 = \cdots = x_b = 0$. ☐

Lemma 4.3 (Stable classical edge)**.** *For the classical families the stable comparison proves the following Chain differences:*

$$D_G(2r+1) = Q_3^G(2r+3) - 4Q_3^G(2r+1) > 0$$

whenever

G	*stable range*
B_b, C_b	$b \geq r + 2$
D_b	$b \geq 2r + 6.$

Equivalently, the stable edge covers all odd exponents at most $s(G)$, where $s(G)$ is defined in definition 4.1.

Proof. Write $T_j(U) = \mathrm{Tr}(U^j)$ in the defining representation. The adjoint characters are

$$\chi_{\mathrm{ad}}^{\mathrm{SO}} = (T_1^2 - T_2)/2, \qquad \chi_{\mathrm{ad}}^{\mathrm{Sp}} = (T_1^2 + T_2)/2.$$

Diaconis–Shahshahani [4, Theorem 4, p. 56, and Theorem 6, p. 59] give the exact stable joint trace moments

$$(T_1, T_2)_{\mathrm{SO}} \longmapsto (Z_1, 1 + \sqrt{2} Z_2), \qquad (T_1, T_2)_{\mathrm{Sp}} \longmapsto (Z_1, -1 + \sqrt{2} Z_2),$$

where Z_1, Z_2 are independent standard normals. Hence the stable orthogonal and symplectic adjoint variables are

$$X_{\mathrm{SO}} = \frac{Z_1^2 - 1 - \sqrt{2} Z_2}{2}, \qquad X_{\mathrm{Sp}} = \frac{Z_1^2 - 1 + \sqrt{2} Z_2}{2},$$

which have the same distribution. Their common moment generating function is

$$Z(t) = \frac{\exp(t^2/4 - t/2)}{\sqrt{1-t}}.$$

It remains to justify the stated finite-rank ranges. For B_b, the adjoint character is the Schur function $s_{(1,1)}$. Expand $s^s_{(1,1)} = \sum_\lambda c_\lambda s_\lambda$. Rains' orthogonal Schur integral criterion [5, §3, paragraph preceding Lemma 3.1] says that only partitions with even row lengths contribute to the $O(2b+1)$ integral. Such a partition has size $2s$ and at most s rows; thus $2b+1 \geq s$ removes the length cutoff. The integrand is unchanged by $U \mapsto -U$, so the $SO(2b+1)$ and $O(2b+1)$ integrals agree.

For C_b, the adjoint character is $s_{(2)}$. Every partition in the Pieri expansion of $s^s_{(2)}$ has at most s rows. Rains' symplectic criterion in the same source paragraph retains precisely the partitions in which every row length has even multiplicity. Their number of rows is even and at most $2\lfloor s/2 \rfloor$, so $b \geq \lfloor s/2 \rfloor$ removes the symplectic length cutoff.

For D_b, expand χ^s_{ad} into monomials $T_1^{2j} T_2^{s-j}$, each of weighted trace degree $2s$. If $b \geq s+1$, then $2s < 2b$, so Diaconis–Shahshahani's exact orthogonal trace-moment identity applies. In this strict degree range the $SO(2b)$ integral equals the $O(2b)$ integral: the difference is the determinant-isotypic integral, and the determinant representation first occurs in matrix-entry degree $2b$. This proves the finite-rank stabilization for every adjoint moment m_s in the three stated ranges.

For independent copies of the common stable variable, the exponential generating function of Q_3 is

$$F(t) = \mathbf{E}[(X-Y)^2 e^{t(X+Y)}] = \big((1-t)^{-3} + (1-t)^{-1}\big) e^{-t+t^2/2}.$$

Hence

$$F''(t) - 4F(t) = P(t)G(t),$$

where

$$G(t) = e^{-t+t^2/2}, \qquad P(t) = 12(1-t)^{-5} - 7(1-t)^{-3} - 4(1-t)^{-1} + (1-t).$$

Write $P(t) = \sum p_k t^k$ and $G(t) = \sum (-1)^k h_k t^k$. The coefficients satisfy

$$p_{k+1} > p_k, \qquad h_k \geq h_{k+1} > 0 \qquad (k \geq 0),$$

because

$$p_k = 12\binom{k+4}{4} - 7\binom{k+2}{2} - 4 + [k=0] - [k=1],$$

and

$$(k+1)h_{k+1} = h_k + h_{k-1}, \qquad h_{-1} = 0, \quad h_0 = 1.$$

Indeed, $p_0 = 2$, while $p_1 - p_0 = 32$, $p_2 - p_1 = 100$, and for $k \geq 2$,

$$p_{k+1} - p_k = 12\binom{k+4}{3} - 7(k+2) = (k+2)\big(2(k+4)(k+3) - 7\big) > 0.$$

The recurrence gives $h_1 = 1$ and positivity of every h_k by induction. For $k \geq 1$, its shifted form

$$kh_k = h_{k-1} + h_{k-2} \geq h_{k-1}$$

then implies

$$h_{k+1} = \frac{h_k + h_{k-1}}{k+1} \leq h_k.$$

For $N = 2r+1$,

$$D_{\mathrm{st}}(N) = N! \sum_{j=0}^{(N-1)/2} \big(p_{2j+1} h_{N-2j-1} - p_{2j} h_{N-2j}\big) > 0.$$

Here every summand is strictly positive: with $a = N - 2j - 1$, the coefficient inequalities give

$$p_{2j+1} h_a - p_{2j} h_{a+1} = (p_{2j+1} - p_{2j}) h_a + p_{2j}(h_a - h_{a+1}) > 0.$$

The Chain difference at $2r+1$ uses only $m_0^G, \dots, m_{2r+5}^G$. For B_b, C_b, the condition $b \ge r+2$ implies $b \ge \lfloor (2r+5)/2 \rfloor$; for D_b, the condition $b \ge 2r+6$ is exactly $b \ge (2r+5)+1$. The preceding stabilization therefore identifies every required finite-rank moment with the stable one. Substitution gives the displayed inequality. $\square$

Definition 4.4 (Classical row gate)**.** For an odd Chain target $2m+3$ and a finite row G with $C_G < 296$, the row is active in the exact middle bridge if

$$0 \le m \le 29, \qquad s(G) < 2m+3 \le 61.$$

The stable edge lemma 4.3 covers odd exponents $\le s(G)$, while the post-bridge hit begins at $N_{\mathrm{hit}}^{(29)}(G) = \max(63, s(G)+2)$. Thus the row gate is exactly the finite interval between stable coverage and the post-$m=29$ tail.

Lemma 4.5 (Pushforward tail criterion)**.** *Let $X = \chi_{\mathrm{ad}}(g)$ and $Y = \chi_{\mathrm{ad}}(h)$ for independent Haar variables, and let ψ_G be the positive finite measure on $\mathbb{R}$ defined by*

$$\int f(u)\, d\psi_G(u) = \mathbf{E}\big[(X-Y)^2 f(X+Y)\big].$$

Then $\psi_G(\mathbb{R}) = 2$ and

$$Q_3^G(n) = \int u^n \, d\psi_G(u).$$

For $c > 2C_G$ and $P_G(c) = \psi_G([c, 2\dim G])$,

$$Q_3^G(n) \ge c^n P_G(c) - 2(2C_G)^n$$

for every odd n. More generally, for $0 < a < 2C_G < c$ and $N_G(a) = \psi_G([-2C_G, -a])$,

$$Q_3^G(n) \ge c^n P_G(c) - (2C_G)^n N_G(a) - 2a^n.$$

Therefore the inequality

$$P_G(c) \ge N_G(a) \left(\frac{2C_G}{c}\right)^n + 2\left(\frac{a}{c}\right)^n$$

implies $Q_3^G(n) \ge 0$, and if it holds at an odd n_0 it holds for all later odd exponents.

Proof. The first identity is the definition of ψ_G with $f(u) = u^n$. By lemma 2.6, $\mathbf{E}X = 0$ and $\mathbf{E}X^2 = 1$, hence $\psi_G(\mathbb{R}) = \mathbf{E}(X-Y)^2 = 2$. For odd n, split the integral into $u \ge c$, $0 \le u < c$, and $u < 0$. The middle piece is nonnegative. Since $\chi_{\mathrm{ad}} \ge -C_G$, the negative support is contained in $[-2C_G, 0]$. On that interval $u^n \ge -(2C_G)^n$, and its ψ_G-mass is at most $\psi_G(\mathbb{R}) = 2$, giving the first lower bound. For the second, on $[-2C_G, -a]$ one has $u^n \ge -(2C_G)^n$, while on $[-a, 0]$ one has $u^n \ge -a^n$. Therefore

$$\int_{-2C_G}^{0} u^n \, d\psi_G(u) \ge -(2C_G)^n N_G(a) - a^n \psi_G([-a,0]) \ge -(2C_G)^n N_G(a) - 2a^n.$$

After division by c^n, the two terms in the sufficient condition are nonnegative multiples of $(2C_G/c)^n$ and $(a/c)^n$. Both bases lie in $(0,1)$, proving the monotonicity assertion. $\square$

Lemma 4.6 (Weyl caps and elementary pushforward conversion)**.** *Let $G = D_b = \mathrm{SO}(2b)$, let $r = b$, $d = b(2b-1)$, and let $d_1, \dots, d_r$ be the degrees of its Weyl group W. Use the type-D_b roots $e_i \pm e_j$, all of squared length 2, and write*

$$\sum_{\alpha > 0} \alpha(\theta)^2 = \kappa |\theta|^2.$$

Suppose $R > 0$ is small enough that the ball $\kappa|\theta|^2 \le R$ lies in one torus eigenangle chart, and suppose

$$\prod_{\alpha > 0} \left(\frac{\sin(\alpha(\theta)/2)}{\alpha(\theta)/2}\right)^2 \ge \eta_R > 0 \tag{4.6.1}$$

throughout that ball. If $X = \chi_{\mathrm{ad}}(g)$ for Haar g, then

$$\Pr\{X \ge d - R\} \ge \frac{\eta_R}{|W|} \frac{\prod_{i=1}^r d_i!}{(2\pi)^{r/2}\Gamma(d/2+1)} \left(\frac{R}{2\kappa}\right)^{d/2}. \tag{4.6.2}$$

Write $C = C_G$, put $x_0 = d - R$, and let P_G be the pushforward mass of lemma 4.5. If $x_0 - C > 2C$ and $x_0 \ge 1$, then

$$P_G(x_0 - C) \ge \Pr\{X \ge x_0\} \frac{(x_0 - 1)^2}{C + 1}. \tag{4.6.3}$$

If $t > 1$ and $x_0 - t > 2C$, then

$$P_G(x_0 - t) \ge \Pr\{X \ge x_0\}(x_0 - t)^2(1 - t^{-2}). \tag{4.6.4}$$

Proof. For $g = \exp(\theta)$ in the chosen torus chart,

$$d - \chi_{\mathrm{ad}}(g) = 2\sum_{\alpha>0}(1 - \cos\alpha(\theta)) \le \sum_{\alpha>0} \alpha(\theta)^2 = \kappa|\theta|^2.$$

Weyl integration, (4.6).1, and the Macdonald–Mehta integral therefore reduce the cap mass to

$$\frac{\eta_R}{|W|(2\pi)^r} \int_{\kappa|\theta|^2 \le R} \prod_{\alpha>0} \alpha(\theta)^2 \, d\theta.$$

The root product is homogeneous of degree $d - r$. Taking the radial part of the Macdonald–Mehta Gaussian integral gives

$$\int_{\kappa|\theta|^2 \le R} \prod_{\alpha>0} \alpha(\theta)^2 \, d\theta = \frac{(2\pi)^{r/2} \prod_i d_i!}{\Gamma(d/2+1)} \left(\frac{R}{2\kappa}\right)^{d/2},$$

which proves (4.6).2.

Let Y be an independent copy of X. Since $Y \ge -C$ and $\mathbf{E}Y = 0$, if $p = \Pr\{Y > 1\}$ then $0 = \mathbf{E}Y \ge p - C(1-p)$; hence $\Pr\{Y \le 1\} \ge 1/(C+1)$. On $\{X \ge x_0, Y \le 1\}$ one has $X + Y \ge x_0 - C$ and $(X-Y)^2 \ge (x_0-1)^2$. Independence proves (4.6).3. Also $\mathbf{E}Y^2 = 1$ by lemma 2.6, so Chebyshev gives $\Pr\{|Y| \le t\} \ge 1 - t^{-2}$. On $\{X \ge x_0, |Y| \le t\}$, both $X + Y$ and $X - Y$ are at least $x_0 - t$. This proves (4.6).4. □

Lemma 4.7 (Root-normalized type-B Weyl caps)**.** *Let $G = B_b = \mathrm{SO}(2b+1)$, let $r = b$, $d = b(2b+1)$, and let $d_i = 2i$ for $1 \le i \le b$. Use the actual type-B_b roots*

$$e_i \pm e_j \quad (i < j), \qquad e_i \quad (1 \le i \le b),$$

and put $\kappa = 2b - 1$. Suppose the ball $\kappa|\theta|^2 \le R$ lies in one torus eigenangle chart and

$$\prod_{\alpha>0} \left(\frac{\sin(\alpha(\theta)/2)}{\alpha(\theta)/2}\right)^2 \ge \eta_R > 0 \qquad (\kappa|\theta|^2 \le R). \tag{4.7.1}$$

If $X = \chi_{\mathrm{ad}}(g)$ for Haar $g \in G$, then

$$\Pr\{X \ge d - R\} \ge \frac{2^{-b}\eta_R}{|W|} \frac{\prod_{i=1}^b d_i!}{(2\pi)^{b/2}\Gamma(d/2+1)} \left(\frac{R}{2\kappa}\right)^{d/2}. \tag{4.7.2}$$

With $C = C_G$, $x_0 = d - R$, and P_G as in lemma 4.5, the conversion bounds (4.6).3 and (4.6).4 also hold.

Proof. For the displayed positive roots,

$$\sum_{\alpha>0} \alpha(\theta)^2 = \sum_{i<j} \big((\theta_i + \theta_j)^2 + (\theta_i - \theta_j)^2\big) + \sum_i \theta_i^2 = (2b-1)|\theta|^2.$$

The Macdonald–Mehta normalization used with the degrees $2, 4, \dots, 2b$ replaces each short root e_i by the squared-length-2 normal $\sqrt{2}e_i$, while leaving the roots $e_i \pm e_j$ unchanged. Hence the square of the actual Weyl discriminant is 2^{-b} times the Macdonald–Mehta discriminant square. Repeating the radial calculation in the proof of lemma 4.6 with this factor and with $\kappa = 2b-1$ gives (4.7).2. The proofs of (4.6).3 and (4.6).4 use only the lower support bound for the adjoint character, $\mathbf{E}X = 0$, $\mathbf{E}X^2 = 1$, and independence, so the same calculation applies to B_b. □

Lemma 4.8 (One-sided moment pushforward criterion)**.** *In the setting of lemma 4.5, put*

$$K_j^G = \int_{\mathbb{R}} u^j \, d\psi_G(u) = Q_3^G(j).$$

Let k, m be nonnegative integers, let $0 < a < 2C_G < c$, and suppose $2k + m \le n$. Define

$$H_{k,m}^G(a) = \sum_{j=0}^{2m} (-1)^j \binom{2m}{j} a^{-j} K_{2k+j}^G.$$

Then

$$\int_{-a}^{0} |u|^n \, d\psi_G(u) \le \frac{a^{n-2k}}{4^m} H_{k,m}^G(a).$$

Consequently,

$$P_G(c) \ge N_G(a) \left(\frac{2C_G}{c}\right)^n + \frac{a^{n-2k}}{4^m c^n} H_{k,m}^G(a) \tag{4.8.1}$$

implies $Q_3^G(n) \ge 0$. If the quantities on the right are fixed and $a, 2C_G < c$, validity at one odd exponent implies validity at every later odd exponent.

Proof. For $-a \le u \le 0$, write $x = -u/a \in [0,1]$. The arithmetic–geometric mean inequality gives

$$\frac{(1+x)^2}{4} \ge x.$$

Since $n - 2k \ge m$, it follows that

$$|u|^n = a^n x^n \le \frac{a^{n-2k}}{4^m} u^{2k} (1 - u/a)^{2m}.$$

The polynomial on the right is nonnegative on all of $\mathbb{R}$. Integration and binomial expansion give the first assertion. Splitting the negative part of the integral defining $Q_3^G(n)$ at $-a$ gives

$$Q_3^G(n) \ge c^n P_G(c) - (2C_G)^n N_G(a) - \frac{a^{n-2k}}{4^m} H_{k,m}^G(a),$$

which proves the criterion. The two ratios in (4.8).1 are strictly below one, proving the last assertion. □

Lemma 4.9 (Exact push moments from adjoint moments)**.** *Let X, Y be independent copies of the adjoint character of a compact group, and write $m_j = \mathbf{E}[X^j]$. Then the pushforward moments in lemma 4.8 satisfy, for every $q \ge 0$,*

$$K_q^G = 2 \sum_{i=0}^{q} \binom{q}{i} \big(m_{i+2} m_{q-i} - m_{i+1} m_{q-i+1}\big). \tag{4.9.1}$$

Consequently $K_0^G, \ldots, K_q^G$ are determined exactly by $m_0, \ldots, m_{q+2}$.

Proof. Expand $(X-Y)^2 (X+Y)^q$ and use independence. The two pure-square sums are equal after replacing i by $q - i$; collecting them with the mixed term gives (4.9).1. □

Lemma 4.10 (Stable finite-rank push moments). *Define the integers*

$$K_j = j![t^j]\Big(\big((1-t)^{-3}+(1-t)^{-1}\big)e^{-t+t^2/2}\Big).$$

For $G = B_b$ or C_b and every $0 \le j \le 2b-2$, one has

$$K_j^G = K_j.$$

For $G = D_b$ the same identity holds for every $0 \le j \le b-3$. In particular, if $2k+2m \le 2b-2$ in types B, C, or if $2k+2m \le b-3$ in type D, then

$$H^G_{k,m}(a) = \sum_{j=0}^{2m}(-1)^j\binom{2m}{j}a^{-j}K_{2k+j}.$$

Proof. The proof of lemma 4.3 computes the exponential generating function

$$\mathbf{E}[(X-Y)^2e^{t(X+Y)}] = \big((1-t)^{-3}+(1-t)^{-1}\big)e^{-t+t^2/2}$$

for the stable orthogonal/symplectic adjoint variable. The same proof invokes Rains' Schur-integral stabilization to identify the finite B_b, C_b adjoint moments through degree $2r+5$ whenever $b \ge r+2$. Take $r = b-2$. Since K_j^G uses adjoint moments only through degree $j+2 \le 2b$, the finite and stable values agree in types B, C. In type D_b, the exact decomposition of lemma 4.45 has $A_q(b) = B_q(b) = 0$ for $q < b$. Thus the finite adjoint moments used by K_j^G agree with their stable values whenever $j+2 < b$, equivalently $j \le b-3$. The final formula is the binomial expansion in lemma 4.8. □

Lemma 4.11 (Trace-tail reduction). *Let U be Haar-distributed in the defining representation of B_b, C_b, D_b, and put*

$$T_1(U) = \mathrm{Tr}(U), \qquad \tau_G(U) = \begin{cases}\mathrm{Tr}(U^2), & G = B_b, D_b,\\ -\mathrm{Tr}(U^2), & G = C_b.\end{cases}$$

For $1 < a < 2C_G < c$, define

$$p_1(G;c,a) = \Pr\{|T_1(U)| \ge \sqrt{2c+3a}\}, \qquad p_2(G;a) = \Pr\{\tau_G(U) \ge a\}.$$

Then

$$P_G(c) \ge c^2(1-a^{-2})(p_1(G;c,a)-p_2(G;a)), \qquad N_G(a) \le 2(C_G^2+1)p_2(G;a).$$

Proof. The defining-character identities are

$$\chi_{\mathrm{ad}}(U) = \frac{T_1(U)^2 - T_2(U)}{2} \quad (B_b, D_b), \qquad \chi_{\mathrm{ad}}(U) = \frac{T_1(U)^2 + T_2(U)}{2} \quad (C_b).$$

On the event $|T_1(U)| \ge \sqrt{2c+3a}$ and $\tau_G(U) \le a$ one has $\chi_{\mathrm{ad}}(U) \ge c+a$. If V is an independent Haar element with $|\chi_{\mathrm{ad}}(V)| \le a$, then $\chi_{\mathrm{ad}}(U)+\chi_{\mathrm{ad}}(V) \ge c$ and $|\chi_{\mathrm{ad}}(U)-\chi_{\mathrm{ad}}(V)| \ge c$. Chebyshev and Schur orthogonality give $\Pr\{|\chi_{\mathrm{ad}}(V)| \le a\} \ge 1-a^{-2}$, proving the lower bound for $P_G(c)$. For the negative tail, $\chi_{\mathrm{ad}}(U)+\chi_{\mathrm{ad}}(V) \le -a$ implies one of the two characters is at most $-a/2$. Put $X = \chi_{\mathrm{ad}}(U)$, $Y = \chi_{\mathrm{ad}}(V)$, and $A = \{X \le -a/2\}$. Symmetry, followed by independence and lemma 2.6, gives

$$\begin{aligned} N_G(a) &\le 2\,\mathbf{E}\big[(X-Y)^2\mathbf{1}_A\big] \\ &= 2\left(\mathbf{E}[X^2\mathbf{1}_A] - 2\mathbf{E}[X\mathbf{1}_A]\mathbf{E}[Y] + \Pr(A)\mathbf{E}[Y^2]\right) \\ &\le 2(C_G^2+1)\Pr(A). \end{aligned}$$

The displayed adjoint formulas imply $\{\chi_{\mathrm{ad}}(U) \le -a/2\} \subseteq \{\tau_G(U) \ge a\}$. □

Lemma 4.12 (Decoupled trace-tail reduction and excess-square negative tail)**.** *Use the notation of lemma 4.11. Let*

$$1 < a < 2C_G < c, \qquad d > 1, \qquad q > 1,$$

and put

$$p_1(G; c, d, q) = \Pr\{|T_1(U)| \geq \sqrt{2c + 2d + q}\}, \qquad p_2(G; q) = \Pr\{\tau_G(U) \geq q\}.$$

Then

$$P_G(c) \geq c^2(1 - d^{-2})\big(p_1(G; c, d, q) - p_2(G; q)\big). \tag{4.12.1}$$

Moreover,

$$N_G(a) \leq 2\mathbf{E}\big[(\tau_G(U) - a)_+^2\big]\,. \tag{4.12.2}$$

For every $v > 0$, this implies

$$N_G(a) \leq \frac{8e^{-2}}{v^2} e^{-va}\mathbf{E}e^{v\tau_G(U)}. \tag{4.12.3}$$

Proof. On the event $|T_1(U)| \geq \sqrt{2c + 2d + q}$ and $\tau_G(U) \leq q$, the two defining-character identities in lemma 4.11 give $\chi_{\mathrm{ad}}(U) \geq c + d$. For an independent Haar element V, the event $|\chi_{\mathrm{ad}}(V)| \leq d$ therefore gives both $\chi_{\mathrm{ad}}(U) + \chi_{\mathrm{ad}}(V) \geq c$ and $|\chi_{\mathrm{ad}}(U) - \chi_{\mathrm{ad}}(V)| \geq c$. Schur orthogonality and Chebyshev's inequality give probability at least $1 - d^{-2}$ for the latter event, proving (4.12).1.

Write $X = \chi_{\mathrm{ad}}(U)$ and $Y = \chi_{\mathrm{ad}}(V)$. Partition the event $X + Y \leq -a$ into $X \leq Y$ and $Y < X$. The two contributions are equal, and the diagonal has zero $(X - Y)^2$ weight. On the first part,

$$0 \leq Y - X \leq -a - 2X.$$

For all three classical families the defining-character identity is $X = (T_1(U)^2 - \tau_G(U))/2$, so $-2X \leq \tau_G(U)$. Hence

$$(X - Y)^2 \leq (\tau_G(U) - a)^2, \qquad \tau_G(U) \geq a,$$

which proves (4.12).2. Finally, for $z \geq a$, the maximum of $(z - a)^2 e^{-vz}$ is $4e^{-2}v^{-2}e^{-va}$, attained at $z = a + 2/v$. Applying this pointwise and then (4.12).2 proves (4.12).3. ☐

Lemma 4.13 (Rains-square high-rank trace certificate)**.** *Set*

$$r = \frac{2000001}{1000000}, \qquad \eta = \frac{359}{2000}, \qquad A = \frac{4571}{2000}, \qquad L = 2r + 3\eta.$$

For every B_b, C_b, D_b row with $C_G \geq 296$, the hypotheses of lemma 4.11 hold at $a = \eta C_G$ and $c = rC_G$, and the comparison in lemma 4.5 holds at the stable edge.

Proof. The positive T_1 tail is supplied by the one-trace density estimate of Courteaut–Johansson–Lambert, Theorem 1.4, combined with the Mills lower bound, giving

$$\Pr\{|T_1| \geq \sqrt{LC_G}\} \geq \exp(-AC_G)$$

throughout the checked range. The determinant and parity shifts in the orthogonal cases change T_1 by at most a deterministic unit, and the one-trace L^1 error is absorbed by the replayed inequality $E_1(C_G) \leq \exp(-AC_G)$.

For the τ_G tail, use the power-map decomposition at power 2. Rains' Theorem 1.5, equations (22)–(23), and Theorem 1.7 give the following centered case split. Here $X_N^U := 2\operatorname{Re}\operatorname{Tr} V$ for Haar $V \in U(N)$; thus it is the trace of Rains' real-projection eigenvalue multiset and is centered at mean 0, and $X_N^{\pm}$ is the centered trace in the nonfixed N-pair part of the orthogonal determinant component $O^{\pm}$. The deterministic fixed-trace contribution deleted by Rains' convention is

$$\kappa(O^+(2N)) = \kappa(O^-(2N)) = 0, \qquad \kappa(O^+(2N+1)) = 1, \qquad \kappa(O^-(2N+1)) = -1.$$

The full square-trace means are $\mathbf{E}\,\mathrm{Tr}(U^2) = 1$ for B_b, D_b and $\mathbf{E}\,\mathrm{Tr}(U^2) = -1$ for C_b, by the adjoint character identities in lemma 4.11 and Schur orthogonality. Thus centering τ_G at $\mathbf{E}\tau_G = 1$ gives:

G	$\tau_G - 1$	component pair-count lower bound
B_b	X_b^U	$b = C_G,$
D_b	$X^+_{\lceil b/2\rceil} + X^-_{\lceil (b-1)/2\rceil}$	$\lfloor b/2\rfloor \geq \lfloor C_G/2\rfloor,$
C_b	$-X^-_{\lceil (b-1)/2\rceil} - X^+_{\lceil b/2\rceil}$	$\lfloor b/2\rfloor = \lfloor C_G/2\rfloor.$

The deterministic shifts in this table are componentwise as follows. For B_b, equation (23) with $p = 2$ gives $O^+(2b+1)^2 \sim \mathrm{Re}\,U(b)$ after the source-side fixed $+1$ eigenvalue is omitted. Restoring that source contribution gives

$$\mathrm{Tr}(U^2) = 1 + X_b^U.$$

For D_b, equation (22) with $p = 2$ gives, with $n_0 = \lceil b/2\rceil$ and $n_1 = \lceil (b-1)/2\rceil$,

$$O^+(2b)^2 \sim O^+(2n_0) \oplus O^-(2n_1 + 1),$$

where the source component has fixed contribution 0, the $O^+(2n_0)$ component has fixed contribution 0, and the $O^-(2n_1+1)$ component has a deleted fixed contribution -1. Since $X^-_{n_1}$ denotes only the centered nonfixed trace, the centered full source trace is

$$\mathrm{Tr}(U^2) - 1 \sim X^+_{\lceil b/2\rceil} + X^-_{\lceil (b-1)/2\rceil}.$$

For C_b, Rains' equation (36) identifies $\mathrm{Sp}(2b)$ with $O^-(2b+2)$ after omitting fixed ± 1 eigenvalues, whose trace contribution is $+1 - 1 = 0$. Applying equation (22) with $n = b + 1$ gives

$$O^-(2b+2)^2 \sim O^-(2\lceil (b+1)/2\rceil) \oplus O^+(2\lceil b/2\rceil + 1).$$

The first component again has deleted fixed contribution 0, while the second has deleted fixed contribution $+1$. With the centered nonfixed traces denoted by $X^-_{\lceil (b-1)/2\rceil}$ and $X^+_{\lceil b/2\rceil}$, and with $\mathbf{E}\,\mathrm{Tr}(U^2) = -1$, the convention $\tau_G = -\,\mathrm{Tr}(U^2)$ gives

$$-\,\mathrm{Tr}(U^2) - 1 \sim -X^-_{\lceil (b-1)/2\rceil} - X^+_{\lceil b/2\rceil}.$$

Thus every non-unitary component has at least $N(C_G) = \lfloor C_G/2\rfloor$ nontrivial conjugate pairs. In the B_b row, Courteaut–Johansson–Lambert, Theorem 1.1, applies to the unitary trace and projection cannot increase total variation; in the C_b, D_b rows, Courteaut–Johansson–Lambert, Theorem 1.4, applies componentwise. The replay checks that the B_b unitary-projection error is bounded by the same two-component envelope used below. For $N(C) = \lfloor C/2\rfloor$, Theorem 1.4 gives the component bound

$$E_1(N) = \frac{5(\log(2N))^{1/4}}{\sqrt{\Gamma(2N+1)}}, \qquad N(C) = \lfloor C/2\rfloor.$$

The L^1 distance of the convolution from the corresponding Gaussian convolution is at most the sum of the two component errors. The Gaussian convolution has variance 2, so the one-sided Gaussian tail gives

$$\Pr\{\tau_G \geq \eta C_G\} \leq \exp\left(-\frac{(\eta C_G - 1)^2}{4}\right) + 2E_1(N(C_G)).$$

For completeness, we print the scalar certificate rather than leaving it as an unnamed replay. Put

$$\widehat{E}(C) = 5(\log(2C))^{1/4} \exp\{-C(\log(2C) - 1)\}, \qquad E_U(C) = \frac{19C^{1/4}(\log C)^{1/2}}{\Gamma(C+2)},$$

$$T(C) = \exp\left(-\frac{(\eta C - 1)^2}{4}\right) + 2\widehat{E}(\lfloor C/2\rfloor), qquadd(C) = 1 - \frac{1}{\eta^2 C^2}.$$

The elementary bound $(2C)! \geq (2C/e)^{2C}$ gives $E_1(C) \leq \widehat{E}(C)$. With the rational constants r, η, A, L displayed in the statement, define

$$\begin{aligned}
G(C) &= (A - L/2)C + \tfrac{1}{2}\log(LC) - \log(LC+1) - \tfrac{3}{2}\log 2,\\
R_0(C) &= e^{AC}T(C),\\
R_1(C) &= \frac{r(1+C^{-2})}{2d(C)} e^{AC}T(C)\left(\frac{2}{r}\right)^C,\\
R_2(C) &= \frac{4r}{\eta^3 C^2 d(C)} \exp\bigl(-\bigl(\log(r/\eta) - A\bigr)C\bigr).
\end{aligned} \tag{4.13.1}$$

The finite scalar certificate consists exactly of

$$G(C) \geq 0, \quad \widehat{E}(C)e^{AC} \leq 1, \quad E_U(C) \leq \widehat{E}(\lfloor C/2 \rfloor), \quad R_i(C) \leq \frac{1}{2}\ (i = 0, 1, 2). \tag{4.13.2}$$

The first two comparisons control the Gaussian lower tail and its total variation error, the third absorbs the type-B unitary projection error, and the last three are precisely the positive-tail/negative-tail ratios left by lemmas 4.5 and 4.11. The directed-MPFR replay evaluates (4.13).2 for every integer $296 \leq C \leq 10000$. At $C = 296$ it also checks, with outward rounding, the sign conditions

$$\begin{gathered}
A - L/2 - (2C)^{-1} > 0, quad A - \log(2C) + (4C\log(2C))^{-1} < 0,\\
A - \eta(\eta C - 1)/2 < 0, quad A - \eta(\eta C - 1)/2 + \log(2/r) < 0, quad A - \log(r/\eta) < 0.
\end{gathered} \tag{4.13.3}$$

These are the derivative upper bounds used for the five monotone factors in (4.13).2; their left sides only improve with C. Consequently the endpoint check propagates beyond 10000. This lists every scalar predicate decided by the high-rank replay. □

Proposition 4.14 (Stable and high-rank classical tail). *For the classical families B_b, C_b, D_b, all rows with the classical size parameter $C_G \geq 296$ satisfy*

$$Q_3^G(n) \geq 0$$

for every odd exponent $n \geq s(G) + 2$.

Proof. Rains moment stabilization identifies the adjoint moment prefix with the stable orthogonal/symplectic moment sequence through the required degree, and the stable Chain differences are exact positive integers. The finite-rank correction is controlled by lemma 4.13: after lemma 4.11 reduces the correction to the two defining-trace probabilities, the exact one-variable replay verifies the pushforward comparison in lemma 4.5 at the stable edge for every $C_G \geq 296$. The monotonicity clause of lemma 4.5 then gives all later odd exponents. □

Proposition 4.15 (Source-generated B/C row-gated bridge). *For $G = B_b$ or C_b, $2 \leq b \leq 30$, and every row-gated index $b - 1 \leq m \leq 29$, one has*

$$D_G(2m+1) > 0.$$

All 870 inequalities are regenerated from the Pieri rule and checked with exact integer arithmetic.

Proof. Write

$$e_2^j = \sum_{\lambda \vdash 2j} a_j(\lambda) s_\lambda, \qquad a_j(\lambda) \in \mathbb{Z}_{\geq 0}. \tag{4.15.1}$$

The orthogonal and symplectic Schur-integral criteria used in proposition 4.62 give

$$m_j^{B_b} = \sum_{\substack{\lambda:\ \lambda_i \text{ even} \\ \ell(\lambda) \leq 2b+1}} a_j(\lambda), \qquad m_j^{C_b} = \sum_{\substack{\lambda:\ \lambda_i \text{ even} \\ \lambda_1 \leq 2b}} a_j(\lambda). \tag{4.15.2}$$

Indeed, the first identity is the finite odd-orthogonal criterion. For the second, apply the standard involution ω: it sends h_2^j to e_2^j and s_α to $s_{\alpha'}$. A partition α has its row lengths in equal pairs precisely when every row length of α' is even, while $\ell(\alpha) \le 2b$ is equivalent to $(\alpha')_1 \le 2b$. This proves (4.15).2.

Pieri's rule gives the exact recurrence

$$a_{j+1}(\mu) = \sum_{\lambda\colon\ \mu/\lambda \text{ is a vertical two-strip}} a_j(\lambda), \qquad a_0(\varnothing) = 1. \tag{4.15.3}$$

For a band with largest active rank r, the verifier retains exactly the union

$$\ell(\lambda) \le 2r+1 \quad \text{or} \quad \lambda_1 \le 2r. \tag{4.15.4}$$

A discarded partition has both height and width beyond these bounds. Adding boxes cannot decrease either quantity, so none of its descendants can enter either sum in (4.15).2. Thus this truncation is exact. A retained partition is stored in ordinary coordinates when its height is bounded and in conjugate coordinates otherwise; in the latter coordinates (4.15).3 becomes horizontal two-strip Pieri. This is a bijective change of representation, not a further truncation.

The three simultaneous bands and their exact moment windows are

band	b	finite moments generated through
low	$2, 3, 4$	61
middle	$5, 6, 7, 8$	$47, 45, 43, 43$
high	$9, 10, \ldots, 19$	$41.$

(4.15.5)

The recurrence is evaluated modulo the first five, four, or three primes, respectively, from

$$\begin{gathered}(2305843009213694963, 2305843009214695031, 2305843009215694967,\\ 2305843009216695041, 2305843009217694971).\end{gathered} \tag{4.15.6}$$

The exact inversions in each CRT reconstruction verify that the moduli are pairwise coprime; primality is not required. The relevant products are

$$\begin{aligned}
P_3 &= 12259964326943078111827907482335947634252293072155482851 > s_{41},\\
P_4 &= 28269553036527760477584768395351886838659733871027006344612486973772241891 > s_{47},\\
P_5 &= 65185151242986397660135212167169058667750503816445442673844289788983521680612515978066230161 > s_{61}.
\end{aligned} \tag{4.15.7}$$

Each sum in (4.15).2 lies in $[0, s_j]$ by nonnegativity and the stable Schur sum. Hence its modular residues have a unique representative in that interval, and the GMP Chinese-remainder reconstruction is the exact finite moment. The code substitutes the result back into every prime and also verifies stabilization before $j = 2b + 2$.

Put $m_j^G = s_j + \delta_j^G$. Outside the generated windows, proposition 4.62 and nonnegativity give the proved box

$$-s_j \le \delta_j^G \le 0. \tag{4.15.8}$$

For each row the verifier expands, over $\mathbb{Z}$,

$$D_G(2m+1) = D_{\mathrm{st}}(2m+1) + \sum_j L_{m,j}\delta_j^G + \sum_{i \le j} Q_{m,i,j}\delta_i^G\delta_j^G. \tag{4.15.9}$$

An exact correction is substituted. For an interval correction, each linear term is minimized at the appropriate endpoint and each quadratic term at all four endpoint products; for a diagonal term the value at zero is included. These are the extrema because every interval in (4.15).8 is one-sided. Thus the resulting integer is a rigorous lower bound for (4.15).9.

There are

$$2\sum_{b=2}^{30}(31 - b) = 870$$

row-gated checks. The replay generates 832 correction values, evaluates 416 steps exactly and 454 by the stated interval substitution, and reports no failure. Its least lower bound is the exact value 1046 at $B_2, m = 1$. The source and the deterministic machine-C output have SHA-256 hashes

$$\begin{array}{l}\texttt{fbd5565215a7b976b67a4529ad21daa2}\\ \texttt{f235f2988da463c502370e3dab3ad797}\end{array}, \qquad \begin{array}{l}\texttt{d019c183f7dedff1049199b31a50d681}\\ \texttt{3e0c17a0c565df1e845119e2a9fba054}\end{array}. \tag{4.15.10}$$

This proves every asserted strict inequality. ☐

Proposition 4.16 (Correction-aware classical row-gated bridge)**.** *The classical rows in the finite bridge through $m = 29$ satisfy*

$$D_G(2m+1) \geq 0$$

for every Chain step $m = 0, \ldots, 29$ whose target exponent $2m+3$ is covered by the row gate.

Proof. The B_b, C_b rows are exactly proposition 4.15. For type D, the ranks $4 \leq b \leq 24$ are proposition 4.54, the ranks $25 \leq b \leq 52$ lie in the correction-aware interval range proved in proposition 4.51, and the ranks $b \geq 53$ are proposition 4.48. These ranges exhaust every classical row admitted by definition 4.4; all inequalities are strict, hence in particular nonnegative. ☐

Proposition 4.17 (First-hit trace rows)**.** *The 57 top finite rows at the first post-bridge hit satisfy*

$$Q_3^G(n) \geq 0$$

for every later odd exponent.

Proof. At the first hit

$$N_{\text{hit}}^{(29)}(G) = \max(63, s(G) + 2),$$

the 384-bit directed-rounding C++/MPFR certificate checks the one-trace lower bound against the negative correction terms for exactly 57 rows:

$$B : 276, 278, \ldots, 295, \quad C : 276, 278, \ldots, 295, \quad D : 276, 278, 280, \ldots, 295, 297.$$

Every interval lower endpoint is positive. The replay output is:

rows interval-certified	57,
minimum row	D_{281},
minimum log-margin interval	[0.66444449761076900114840811 26,
	0.6644444976107690011484081126].

The checker uses outward rounding for every rational operation, logarithm, exponential, and logarithmic subtraction. Its archived output has SHA-256

$$\begin{array}{l}\texttt{ca3a938c9d45d65eb892f6ba637ed8c3}\\ \texttt{a7b1d32a571a5be52e2f5efb80477176}\end{array}.$$

The monotonicity clause of lemma 4.5 then carries the inequality to all later odd exponents. ☐

Proposition 4.18 (Direct post-$m = 29$ low-rank tails)**.** *The following low-rank post-$m = 29$ rows satisfy*

$$Q_3^G(n) \geq 0$$

for every odd $n \geq N_{\text{hit}}^{(29)}(G)$:

family	*ranks*
B	$2 \leq b \leq 13$
C	$2 \leq b \leq 19$
D	$4 \leq b \leq 24$.

Proof. The exact box integrals for B_2, C_2, B_3, C_3 are replayed by the flags

`-b2-c2-post-m29-tail-certificate`,
`-b3-c3-post-m29-tail-certificate`.

For B_4, B_5, B_6, B_8, the fixed rational cap replays apply lemma 4.7; in particular they include the factor 2^{-b} absent from an unnormalized non-simply-laced Macdonald–Mehta product. Their cap data are

G	R	x_0	independent-variable event	c
B_4	9	27	$Y \le 1$	23
B_5	9	46	$Y \le 1$	41
B_6	1053/50	2847/50	$\|Y\| \le 3/2$	1386/25
B_8	1609/25	1791/25	$\|Y\| \le 3/2$	3507/50.

The corrected exact-rational margins are positive in all four rows.

The rows

$$B_7, B_9, B_{10}, B_{11}, B_{12}, B_{13}, \qquad C_4, \dots, C_{19}$$

use the exact determinant tails of proposition 4.41. Since $b \le 19$ here, definition 4.1 gives $N_{\mathrm{hit}}^{(29)}(G) = 63$. For B_{11}, B_{12}, B_{13} and C_{13}, C_{14}, C_{17}, the first-hit certificate proposition 4.44 supplies $Q_3^G(63) \ge 0$ and proposition 4.42 supplies every odd exponent before the exact-MGF onset. For C_{16}, the same first-hit certificate supplies the exponent 63 and the exact-MGF tail starts at 65. Every other row in the display has exact-MGF onset 63. Thus these certificates cover every odd exponent at least 63 in all B/C rows listed in the proposition.

For $D_4, \dots, D_9$, the fixed rational ball-cap and product-sine replays

`-d4-post-m29-tail-certificate`, `-d5-post-m29-tail-certificate`,
`-d6-b6-post-m29-tail-certificate`, `-d7-b7-c7-post-m29-tail-certificate`,
`-d8-post-m29-tail-certificate`, `-b8-c9-d9-post-m29-tail-certificate`

instantiate lemma 4.6. Their type-D cap and central-variable data are

b	x_0	independent-variable event	c
4	26	$Y \le 1$	22
5	41	$Y \le 1$	38
6	949/20	$\|Y\| \le 7/5$	921/20
7	73	$\|Y\| \le 8/5$	357/5
8	84	$\|Y\| \le 8/5$	412/5
9	767/10	$\|Y\| \le 8/5$	751/10.

For D_4, D_5, the replay uses the per-root bound $\bigl(\sin(\alpha/2)/(\alpha/2)\bigr)^2 \ge 1/4$. For D_6, D_7 it uses

$$\prod_{\alpha>0} \left(\frac{\sin(\alpha/2)}{\alpha/2}\right)^2 \ge \left(1 - \frac{R}{24}\right)^2,$$

which follows from $\sin z/z \ge 1 - z^2/6$ and $\prod_i (1-u_i) \ge 1 - \sum_i u_i$. For D_8, the Euler product and $\sum_{\alpha>0} (\alpha/2)^2 \le 9$ give the lower bound $(\sin 3/3)^2$; the replay inserts the rational inequality $\sin 3 \ge 15/106 - (15/106)^3/6$. For D_9, put $L = |\alpha|^2 = 2$. Concavity of $\log(1-u)$ and $\alpha(\theta)^2 \le LR/\kappa$ give

$$\prod_{\alpha>0} \left(1 - \frac{\alpha(\theta)^2}{24}\right)^2 \ge \left(1 - \frac{LR}{24\kappa}\right)^{2\kappa/L}.$$

The exact rational replays substitute these bounds in (4.6).2–(4.6).4 and prove the pushforward-tail inequality from odd exponent 63. For D_{10} and $D_{12}, \dots, D_{24}$, the exact finite-rank certificates in proposition 4.55 also start at odd exponent 63. The row D_{11} instead has an exact correction-aware Chain bridge through odd exponent 75 and an exact-MGF tail from exponent 75 onward. Finally, proposition 4.54 gives every row-gated Chain difference through $m = 29$, so the stable edge and lemma 2.11 reach odd exponent 61 in every row. These three cases cover every odd exponent at least 63. □

Proposition 4.19 (Post-$m = 29$ high-edge Chernoff trace tails). *For $G = B_b$ or $G = C_b$ with $218 \le b \le 275$, or $b = 277$, one has*

$$Q_3^G(n) \ge 0$$

for every odd $n \ge N_{\text{hit}}^{(29)}(G)$.

Proof. This is Proposition 24.17E in the proof ledger. The active replay uses 384-bit directed rounding in `character_ring_iter/trace_square_cutoff_mpfr.cpp`. Its source-supported CJL/Rains Chernoff estimate uses

$$(r, \eta) \in \{(2001/1000, 9/20), (10001/5000, 56/125), (99/50, 56/125)\}.$$

It checks the one-trace source range, the Chernoff admissibility window, and the trace-pushforward inequality of lemma 4.5 at $N_{\text{hit}}^{(29)}(G)$, with outward rounding for every rational operation, logarithm, exponential, and logarithmic subtraction. The local log

`certificates/classical_trace/first_hit_trace_replay.log`

exits with status 0 and has SHA-256 hash

`ca3a938c9d45d65eb892f6ba637ed8c3`
`a7b1d32a571a5be52e2f5efb80477176`.

All 118 interval lower endpoints are positive. The smallest is attained at B_{219} and C_{219}, and is enclosed by

$$[0.307429969543813258390342958\,8, 0.3074299695438132583903429588].$$

□

Lemma 4.20 (Type B unitary square-trace tail). *Let O be Haar-distributed in the defining representation of B_b, and put $\tau_B(O) = \operatorname{Tr}(O^2)$. For every real $a > 1$,*

$$\Pr\{\tau_B(O) \ge a\} \le 4 \exp\left(-\frac{(a-1)^2}{4}\right).$$

Proof. Use the standard representative $O \in \mathrm{SO}(2b+1)$. Rains [6, Theorem 1.5, equation (23)], specialized to power $p = 2$, identifies the nonfixed eigenvalues of O^2 with those of $\operatorname{Re} U(b)$. Thus, if V is Haar-distributed in $U(b)$, restoring the fixed $+1$ eigenvalue gives the distributional identity

$$\tau_B(O) \overset{d}{=} 1 + 2 \operatorname{Re} \operatorname{Tr} V.$$

Courteaut–Johansson–Lambert [11, Lemma 2.2, with $m = 1$] gives, for every $L > 0$,

$$\Pr\big\{(\operatorname{Re}\operatorname{Tr} V, \operatorname{Im}\operatorname{Tr} V) \notin [-L/2, L/2]^2\big\} < 4e^{-L^2/4}.$$

For every $0 < L < a - 1$, the event $\{2 \operatorname{Re}\operatorname{Tr} V \ge a - 1\}$ is contained in the displayed complement of the square. Hence its probability is at most $4e^{-L^2/4}$. Letting $L \uparrow a - 1$ proves the claim. □

Proposition 4.21 (Post-$m = 29$ middle B tails from the unitary square trace). *For every $124 \le b \le 217$ and every odd $n \ge 2b + 1$,*

$$Q_3^{B_b}(n) \ge 0.$$

Proof. Set

$$r = \frac{2001}{1000}, \qquad \eta = \frac{7}{20}, \qquad c = rb, \qquad a = \eta b, \qquad L = 2r + 3\eta = \frac{1263}{250}.$$

Then $1 < a < 2b < c$. In the notation of lemma 4.11, put

$$p_1 = \Pr\{|T_1(O)| \geq \sqrt{Lb}\}, \qquad p_2 = \Pr\{\tau_B(O) \geq \eta b\}.$$

The defining trace has mean zero. Courteaut–Johansson–Lambert [11, Theorem 1.4], which applies because $b \geq 124$, and the two-sided Mills lower bound with $\pi < 4$ give

$$p_1 \geq G_b - E_b,$$

where

$$G_b = \frac{e^{-Lb/2}\sqrt{Lb}}{\sqrt{2}(Lb+1)}, \qquad E_b = \frac{5(\log(2b))^{1/4}}{\sqrt{(2b)!}}.$$

By lemma 4.20,

$$p_2 \leq U_b := 4\exp\left(-\frac{(\eta b - 1)^2}{4}\right).$$

The directed-rounding MPFR replay

`certificates/post_m29/ post_m29_b_unitary_square_trace_mpfr_certificate.log`

checks, for every integer $124 \leq b \leq 217$, the four inequalities

$$E_b \leq \frac{1}{4}G_b, \qquad U_b \leq \frac{1}{4}G_b,$$

$$(rb)^2\big(1-(\eta b)^{-2}\big)\frac{G_b}{2} \geq 2\left(2(b^2+1)U_b(2/r)^{2b+1}\right),$$

and

$$(rb)^2\big(1-(\eta b)^{-2}\big)\frac{G_b}{2} \geq 2\left(2(\eta/r)^{2b+1}\right).$$

It uses 384-bit MPFR intervals, rounds every lower bound downward and every upper bound upward, exits with status 0, and reports `rows_checked=94` and `failures=0`. The four smallest natural-log margins are all attained at B_{124} and are respectively

$$241.28842853377649\ldots, \qquad 129.87520977348063\ldots,$$
$$130.69322950465274\ldots, \qquad 126.27950899088545\ldots.$$

The log has SHA-256

`7780bfe3253bf9e030a9a544dcd7327f`
`cd905aed8a2890ded191dc38efae8230`,

and the C++ source has SHA-256

`d0d503ffc4bd6cb0485a97a4cef2178a`
`930f3f00c38ffd4486cb66aa01652aa7`.

The first two inequalities imply $p_1 - p_2 \geq G_b/2$. Hence lemma 4.11 gives

$$P_{B_b}(c) \geq (rb)^2\big(1-(\eta b)^{-2}\big)\frac{G_b}{2}, \qquad N_{B_b}(a) \leq 2(b^2+1)U_b.$$

The final two checked inequalities imply

$$P_{B_b}(c) \geq N_{B_b}(a)(2/r)^{2b+1} + 2(\eta/r)^{2b+1}.$$

Thus lemma 4.5 applies at the odd exponent $2b+1$ and, by its monotonicity clause, at every later odd exponent. $\square$

Definition 4.22 (CJL Fredholm remainder)**.** For an integer $d \geq 1$ and $0 \leq v < d+1$, set

$$\mathfrak{b}_d(v) = \frac{\exp\big(v^2/(d+1)\big)}{1 - v^2/(d+1)^2}\,\frac{v^d}{d!}.$$

Definition 4.23 (CJL trace-norm envelope). For an integer $d \geq 1$ and $0 \leq v < d+1$, set

$$\mathfrak{a}_d(v) = \frac{\exp(v^2/(d+1))v^d}{d!} \frac{1}{1 - v/(d+1)} \frac{1}{\sqrt{1 - v^2/(d+1)^2}}.$$

Lemma 4.24 (Extended small-Fredholm trace MGF bounds). *Let Z be a centered full defining trace in one of the four orthogonal Jacobi components of Courteaut–Johansson–Lambert, or in the corresponding symplectic component, and let d be the total matrix dimension attached to that component. If*

$$0 \leq v < d+1, \qquad \mathfrak{a}_d(v) < 1,$$

then $\mathfrak{b}_d(v) < 1$ and

$$e^{v^2/2}(1 - \mathfrak{b}_d(v)) \leq \mathbf{E}e^{\pm vZ} \leq \frac{e^{v^2/2}}{1 - \mathfrak{b}_d(v)}. \tag{4.24.1}$$

Proof. Courteaut–Johansson–Lambert [11, Proposition 3.3] writes the centered transform as

$$\mathbf{E}e^{zZ} = e^{z^2/2} \det(I + QK_zQ).$$

The proof of their Lemma 3.6 starts from the trace-norm inequality in their equation (3.4), inserts their Bessel bound (2.16), and sums two geometric series. If the least Bessel index is denoted by the total dimension d, the calculation before their equation (3.5) gives, without using the later sufficient restriction $|z| < 2e^{-5/4}n$,

$$\|QK_zQ\|_{\mathcal{J}_1} \leq \frac{\exp(|z|^2/(d+1))|z|^d}{d!} \frac{1}{1 - |z|/(d+1)} \frac{1}{\sqrt{1 - |z|^2/(d+1)^2}} = \mathfrak{a}_d(|z|)$$

whenever $|z| < d+1$. The four kernels listed in Proposition 3.3 have least indices $2n, 2n+1, 2n+1, 2n+2$, respectively, exactly their corresponding total dimensions, so the same displayed calculation applies to every case.

The proof of their Lemma 3.7, in particular equation (3.6), also gives directly

$$\sum_{j \geq 1} \frac{1}{j} \left| \mathrm{Tr}(QK_zQ)^j \right| \leq -\log(1 - \mathfrak{b}_d(|z|)).$$

Here $\mathfrak{b}_d(v) < \mathfrak{a}_d(v) < 1$. Thus the trace-norm hypothesis justifies Plemelj's series, and the last display bounds the absolute value of the logarithm of the determinant. For real $z = \pm v$, the determinant is positive because it equals $e^{-v^2/2}\mathbf{E}e^{\pm vZ}$. Exponentiation proves (4.24).1. ☐

Lemma 4.25 (Layered exponential-tilt lower tail). *Let Z satisfy (4.24).1 at every argument used below, with the same dimension d. Fix $s, l > 0$, put $t = s - l$, and choose $\lambda_- > 0$ and pairs (u_i, λ_i), $i = 0, \dots, L-1$, such that*

$$0 < u_0 < \cdots < u_{L-1}.$$

Write $D(v) = 1 - \mathfrak{b}_d(v)$ and define

$$A_- = \frac{\exp(\lambda_-^2/2 - \lambda_- l)}{D(s)D(s - \lambda_-)}, \qquad A_i^+ = \frac{\exp(\lambda_i^2/2 - \lambda_i u_i)}{D(s)D(s + \lambda_i)},$$

$$g_i = 1 - A_- - A_i^+.$$

Suppose

$$0 < g_0 \leq g_1 \leq \cdots \leq g_{L-1}.$$

Then, with $w_i = e^{-su_i}$,

$$\Pr\{Z \geq t\} \geq V_d(s, l) := D(s)e^{-s^2/2} \left(w_{L-1}g_{L-1} + \sum_{i=0}^{L-2} (w_i - w_{i+1})g_i \right). \tag{4.25.1}$$

If the same MGF bounds hold after replacing Z by $-Z$, then

$$\Pr\{|Z| \geq t\} \geq 2V_d(s,l). \tag{4.25.2}$$

Proof. Tilt the law by $d\mathbf{P}_s = M(s)^{-1}e^{sZ}\,d\mathbf{P}$, where $M(s) = \mathbf{E}e^{sZ}$. Exponential Markov at the independently chosen parameters λ_- and λ_i, followed by lemma 4.24, gives

$$\mathbf{P}_s\{Z - s \leq -l\} \leq A_-, \qquad \mathbf{P}_s\{Z - s \geq u_i\} \leq A_i^+.$$

Consequently

$$\mathbf{P}_s\{t < Z \leq s + u_i\} \geq g_i.$$

For a decreasing function, the least integral compatible with these nested cumulative-mass lower bounds places each required mass increment at the right endpoint. Hence

$$\mathbf{E}_s[e^{-s(Z-s)}\mathbf{1}_{\{Z>t\}}] \geq w_{L-1}g_{L-1} + \sum_{i=0}^{L-2}(w_i - w_{i+1})g_i.$$

Multiplying by $M(s)e^{-s^2}$ and using $M(s) \geq e^{s^2/2}D(s)$ proves (4.25).1. Applying the same argument to $-Z$ and adding the two disjoint tails proves (4.25).2. $\square$

Lemma 4.26 (Arbitrary-parameter $B/C/D$ square-trace MGF bounds)**.** *For B_b, let $\tau_B = \mathrm{Tr}(O^2)$ and, for $x > 1$, $v > 0$, put*

$$U_B(x,v) = \exp\bigl(-v(x-1) + v^2\bigr).$$

Then

$$e^{-vx}\mathbf{E}e^{v\tau_B} \leq U_B(x,v). \tag{4.26.1}$$

For C_b, use the integers d_0, d_1 of lemma 4.32 and suppose $\mathfrak{a}_{d_i}(v) < 1$ for $i = 0,1$. Put

$$U_C(x,v) = \frac{\exp(-v(x-1) + v^2)}{(1 - \mathfrak{b}_{d_0}(v))(1 - \mathfrak{b}_{d_1}(v))}.$$

Then

$$e^{-vx}\mathbf{E}e^{v\tau_C} \leq U_C(x,v). \tag{4.26.2}$$

For D_b, put

$$m_0 = \left\lceil \frac{b}{2} \right\rceil, \qquad m_1 = \left\lceil \frac{b-1}{2} \right\rceil, \qquad d_0 = 2m_0, \qquad d_1 = 2m_1 + 1,$$

and suppose $\mathfrak{a}_{d_i}(v) < 1$ for $i = 0,1$. With $\tau_D = \mathrm{Tr}(O^2)$, put

$$U_D(x,v) = \frac{\exp(-v(x-1) + v^2)}{(1 - \mathfrak{b}_{d_0}(v))(1 - \mathfrak{b}_{d_1}(v))}.$$

Then

$$e^{-vx}\mathbf{E}e^{v\tau_D} \leq U_D(x,v). \tag{4.26.3}$$

Proof. For B_b, the Rains identity used in lemma 4.20 gives $\tau_B \stackrel{d}{=} 1 + 2\,\mathrm{Re}\,\mathrm{Tr}\,V$ with V Haar in $U(b)$. Courteaut–Johansson–Lambert [11, Lemma 2.12], applied to the real function $f(\theta) = 2v\cos\theta$, gives

$$\mathbf{E}e^{2v\,\mathrm{Re}\,\mathrm{Tr}\,V} \leq e^{v^2}.$$

This proves (4.26).1. For C_b, the independent Rains components Y_0, Y_1 in the proof of lemma 4.32 satisfy $\tau_C - 1 \stackrel{d}{=} -(Y_0 + Y_1)$. Apply the upper bound in lemma 4.24 to each centered component at $-v$, multiply the two bounds, and include the deterministic $+1$. This gives (4.26).2.

For D_b, Rains' power-two decomposition [6, Theorem 1.5, equation (22)], with the fixed eigenvalues restored as in the proof of lemma 4.13, gives independent centered full orthogonal traces Y_0, Y_1 of total dimensions d_0, d_1 such that

$$\tau_D - 1 \stackrel{d}{=} Y_0 + Y_1.$$

Apply the upper bound of lemma 4.24 at v to both components, multiply, and include the deterministic $+1$. This proves (4.26).3. □

Lemma 4.27 (Type B defining-trace MGF sandwich)**.** *Let O be Haar-distributed in* $\mathrm{SO}(2b+1)$*, put $T_b = \operatorname{Tr} O$, and, for $v \geq 0$, define*

$$\beta_b(v) = \mathfrak{b}_{2b+1}(v).$$

If $0 \leq v < 2e^{-5/4}b$, then $\beta_b(v) < 1$ and

$$e^{v^2/2}\big(1 - \beta_b(v)\big) \leq \mathbf{E}e^{vT_b} \leq \frac{e^{v^2/2}}{1 - \beta_b(v)}.$$

Proof. In the notation of Courteaut–Johansson–Lambert, the b nonfixed eigenvalue pairs of $O^+(2b+1)$ have Jacobi parameters $(1/2, -1/2)$. Their trace variable in Section 3 omits the deterministic $+1$ eigenvalue and has mean -1. Thus the linear factor in [11, Proposition 3.3] cancels when that eigenvalue is restored. Specializing the proposition at $\xi = -iv$ gives

$$\mathbf{E}e^{vT_b} = e^{v^2/2} \det\big(I + Q_b K_v^{+-} Q_b\big)\,.$$

For $v < 2e^{-5/4}b$, Lemmas 3.6–3.7 of the same source, with total matrix size $d = 2b+1$, give

$$\big|\log\det\big(I + Q_b K_v^{+-} Q_b\big)\big| \leq -\log\big(1 - \beta_b(v)\big).$$

The determinant is positive because it is $e^{-v^2/2}\mathbf{E}e^{vT_b}$. Exponentiating the last display proves both bounds. □

Lemma 4.28 (Exponentially tilted type B defining-trace tail)**.** *In the setting of lemma 4.27, let $t, h > 0$, put $s = t + h$, and suppose $t + 2h < 2e^{-5/4}b$. Define*

$$\rho_b(t,h) = 1 - \frac{e^{-h^2/2}}{1 - \beta_b(s)}\left(\frac{1}{1 - \beta_b(t)} + \frac{1}{1 - \beta_b(t+2h)}\right).$$

If $\rho_b(t,h) > 0$, then

$$\Pr\{T_b \geq t\} \geq V_b(t,h) := \big(1 - \beta_b(s)\big)\rho_b(t,h)\exp\left(-\frac{s^2}{2} - sh\right).$$

Proof. Write $M(v) = \mathbf{E}e^{vT_b}$ and tilt Haar probability by

$$d\mathbf{P}_s = M(s)^{-1}e^{sT_b}\,d\mathbf{P}.$$

Exponential Markov and lemma 4.27 give

$$\mathbf{P}_s\{T_b - s \geq h\} \leq \frac{e^{-h^2/2}}{(1 - \beta_b(s))(1 - \beta_b(s+h))}$$

and

$$\mathbf{P}_s\{T_b - s \leq -h\} \leq \frac{e^{-h^2/2}}{(1 - \beta_b(s))(1 - \beta_b(s-h))}.$$

Consequently $\mathbf{P}_s\{s - h < T_b < s + h\} \geq \rho_b(t,h)$. Since $s - h = t$,

$$\begin{aligned}\Pr\{T_b \geq t\} &\geq M(s)\,\mathbf{E}_s\big[e^{-sT_b}\mathbf{1}_{\{s-h<T_b<s+h\}}\big]\\ &\geq M(s)e^{-s(s+h)}\rho_b(t,h)\\ &\geq (1 - \beta_b(s))\rho_b(t,h)e^{-s^2/2-sh},\end{aligned}$$

as claimed. □

Proposition 4.29 (Post-$m = 29$ middle B tails from an MGF tilt)**.** *For every $62 \leq b \leq 123$ and every odd $n \geq 2b + 29$,*

$$Q_3^{B_b}(n) \geq 0.$$

Proof. Set

$$r=\frac{20001}{10000},\qquad \alpha=\frac{979}{250},\qquad h=\frac{601}{500},\qquad c_b=rb,\qquad a_b=\alpha\sqrt{b},$$

and put

$$t_b=\sqrt{2c_b+3a_b},\qquad s_b=t_b+h,\qquad V_b=V_b(t_b,h),$$

where the final quantity is defined in lemma 4.28. One has $1<a_b<2b<c_b$. In the notation of lemma 4.11,

$$p_1\ge \Pr\{T_b\ge t_b\}\ge V_b,\qquad p_2\le U_b:=4\exp\left(-\frac{(a_b-1)^2}{4}\right),$$

where the second inequality is lemma 4.20.

The 384-bit directed-rounding MPFR replay

`certificates/post_m29/ post_m29_bc_tilted_mgf_mpfr_certificate.log`

checks, for every integer $62\le b\le 123$, that

$$t_b+2h<2e^{-5/4}b,\qquad \rho_b(t_b,h)>0,\qquad V_b\ge 2U_b,$$

and that

$$c_b^2(1-a_b^{-2})\frac{V_b}{2}\ge 2\left(2(b^2+1)U_b(2/r)^{2b+29}\right),$$
$$c_b^2(1-a_b^{-2})\frac{V_b}{2}\ge 2\left(2(a_b/c_b)^{2b+29}\right).$$

Every lower quantity is rounded downward and every upper quantity upward. The combined B/C log exits with status 0, reports `B_rows_checked=62` and `failures=0`, and places all five B-side minima at B_{62}. The minimum domain and tilted-mass lower bounds are respectively

$$14.669512561242725\ldots,\qquad 0.028831039797142247\ldots,$$

and the three minimum natural-log comparison margins are

$$0.11377835996790686\ldots,\qquad 0.12021572562760810\ldots,\qquad 0.15645659337912115\ldots.$$

The log has SHA-256

`ca6860204466e4a6b38d9c5597511cd1`
`1cfebb4b1b65cab88db6ee45f9477111`,

and the C++ source has SHA-256

`0f5ad1037bcf738ab4ddd9c5b12f151a`
`adc1bff2b5c4726c60622f608ef9e0e7`.

The first comparison gives $p_1-p_2\ge V_b/2$. Hence lemma 4.11 gives

$$P_{B_b}(c_b)\ge c_b^2(1-a_b^{-2})\frac{V_b}{2},\qquad N_{B_b}(a_b)\le 2(b^2+1)U_b.$$

The final two checked comparisons imply the hypothesis of lemma 4.5 at the odd exponent $2b+29$. Its monotonicity clause proves every later odd exponent. ☐

Lemma 4.30 (Type C defining-trace MGF sandwich). *Let U be Haar-distributed in* $\mathrm{Sp}(2b)$*, put* $S_b=\operatorname{Tr}U$*, and define*

$$\gamma_b(v)=\mathfrak{b}_{2b}(v).$$

If $0\le v<2e^{-5/4}b$*, then* $\gamma_b(v)<1$ *and*

$$e^{v^2/2}\big(1-\gamma_b(v)\big)\le \mathbf{E}e^{vS_b}\le \frac{e^{v^2/2}}{1-\gamma_b(v)}.$$

Proof. For $\mathrm{Sp}(2b)$, the Jacobi parameters in [11, Section 3, equation (3.1)] are $(1/2, 1/2)$, b nontrivial conjugate pairs, and total matrix dimension $d = 2b$. There are no deterministic eigenvalues and the linear factor in [11, Proposition 3.3] vanishes. Substitution of $\xi = -iv$ in that proposition therefore gives

$$\mathbf{E}e^{vS_b} = e^{v^2/2} \det\bigl(I + Q_b K_v^{++} Q_b\bigr) .$$

For $v < 2e^{-5/4}b$, Lemmas 3.6–3.7 of the same source, with $d = 2b$, give

$$\bigl|\log \det\bigl(I + Q_b K_v^{++} Q_b\bigr)\bigr| \le -\log\bigl(1 - \mathfrak{b}_{2b}(v)\bigr).$$

The determinant is positive because it equals $e^{-v^2/2}\mathbf{E}e^{vS_b}$. Exponentiation proves both bounds. $\square$

Lemma 4.31 (Exponentially tilted type C defining-trace tail)**.** *In the setting of lemma 4.30, let $t, h > 0$, put $s = t + h$, and suppose $t + 2h < 2e^{-5/4}b$. Define*

$$\rho_b^C(t,h) = 1 - \frac{e^{-h^2/2}}{1-\gamma_b(s)} \left(\frac{1}{1-\gamma_b(t)} + \frac{1}{1-\gamma_b(t+2h)} \right).$$

If $\rho_b^C(t,h) > 0$, then

$$\Pr\{S_b \ge t\} \ge W_b(t,h) := \bigl(1-\gamma_b(s)\bigr)\rho_b^C(t,h) \exp\left(-\frac{s^2}{2} - sh \right).$$

Proof. The proof of lemma 4.28 uses only the two displayed MGF bounds in lemma 4.27. Apply its exponential tilt at $s = t + h$ with S_b in place of T_b and with γ_b in place of β_b. The two exponential-Markov estimates become

$$\mathbf{P}_s\{S_b - s \ge h\} \le \frac{e^{-h^2/2}}{(1-\gamma_b(s))(1-\gamma_b(s+h))},$$

$$\mathbf{P}_s\{S_b - s \le -h\} \le \frac{e^{-h^2/2}}{(1-\gamma_b(s))(1-\gamma_b(s-h))}.$$

Their sum leaves interval mass at least $\rho_b^C(t,h)$. On $s - h < S_b < s + h$, one has $e^{-sS_b} \ge e^{-s(s+h)}$; the lower MGF bound of lemma 4.30 then gives exactly the displayed $W_b(t,h)$. $\square$

Lemma 4.32 (Type C square-trace MGF tail)**.** *Let U be Haar-distributed in $\mathrm{Sp}(2b)$, where $b \ge 2$, and put $\tau_C(U) = -\operatorname{Tr}(U^2)$. Define*

$$m_0 = \left\lceil \frac{b+1}{2} \right\rceil, \quad m_1 = \left\lceil \frac{b}{2} \right\rceil, \quad n_0 = m_0 - 1, \quad n_1 = m_1,$$

$$d_0 = 2m_0, \qquad d_1 = 2m_1 + 1.$$

For $a > 1$, put $v = (a-1)/2$. If

$$v < 2e^{-5/4} \min(n_0, n_1),$$

then $\mathfrak{b}_{d_i}(v) < 1$ for $i = 0, 1$ and

$$\Pr\{\tau_C(U) \ge a\} \le \frac{\exp\bigl(-(a-1)^2/4\bigr)}{(1-\mathfrak{b}_{d_0}(v))(1-\mathfrak{b}_{d_1}(v))}.$$

Proof. Rains [6, Theorem 1.5 and the subsequent remark, equation (36)] identifies the nonfixed eigenvalues of $\mathrm{Sp}(2b)$ with those of $O^-(2b+2)$. The omitted eigenvalues are $+1, -1$, so after squaring their trace contribution is 2. Equation (22) of the same theorem at $p = 2$ decomposes the remaining eigenvalue distribution into independent components

$$O^-(2m_0) \oplus O^+(2m_1 + 1).$$

Let Y_0, Y_1 be the full defining traces of these two components. The first component has fixed eigenvalues $+1, -1$, of total trace 0; the second has a fixed $+1$. Restoring these fixed eigenvalues in Rains' identity therefore gives

$$\mathrm{Tr}(U^2) \stackrel{d}{=} Y_0 + Y_1 - 1, \qquad \tau_C(U) - 1 \stackrel{d}{=} -(Y_0 + Y_1).$$

By Courteaut–Johansson–Lambert [11, Section 3, equation (3.1)], the first component has Jacobi parameters $(1/2, 1/2)$ and n_0 nontrivial pairs, while the second has parameters $(1/2, -1/2)$ and n_1 nontrivial pairs. Proposition 3.3 shows that the nonfixed trace means are respectively 0 and -1; after the fixed traces 0 and $+1$ are restored, both Y_0, Y_1 are centered. Lemmas 3.6–3.7, at the argument $-v$, now give

$$\mathbf{E}e^{-vY_i} \le \frac{e^{v^2/2}}{1 - \mathfrak{b}_{d_i}(v)} \qquad (i = 0, 1).$$

Independence and exponential Markov yield

$$\Pr\{-(Y_0 + Y_1) \ge a - 1\} \le \frac{e^{-v(a-1)+v^2}}{(1 - \mathfrak{b}_{d_0}(v))(1 - \mathfrak{b}_{d_1}(v))}.$$

Since $v = (a-1)/2$, the exponent is $-(a-1)^2/4$. $\square$

Proposition 4.33 (Post-$m = 29$ middle C tails from MGF tilts)**.** *For every* $62 \le b \le 217$ *and every odd* $n \ge 2b + 29$*,*

$$Q_3^{C_b}(n) \ge 0.$$

Proof. Set

$$r = \frac{20001}{10000}, \qquad \alpha = \frac{979}{250}, \qquad h = \frac{601}{500}, \qquad c_b = rb, \qquad a_b = \alpha\sqrt{b},$$

and put

$$t_b = \sqrt{2c_b + 3a_b}, \qquad s_b = t_b + h, \qquad W_b = W_b(t_b, h),$$

where the final quantity is defined in lemma 4.31. For the square trace, use the integers m_i, n_i, d_i of lemma 4.32, put $v_b = (a_b - 1)/2$, and define

$$U_b^C = \frac{\exp\bigl(-(a_b - 1)^2/4\bigr)}{(1 - \mathfrak{b}_{d_0}(v_b))(1 - \mathfrak{b}_{d_1}(v_b))}.$$

One has $1 < a_b < 2b < c_b$. In the notation of lemma 4.11,

$$p_1 \ge \Pr\{S_b \ge t_b\} \ge W_b, \qquad p_2 \le U_b^C,$$

by lemmas 4.31 and 4.32.

The 384-bit directed-rounding MPFR replay

`certificates/post_m29/ post_m29_bc_tilted_mgf_mpfr_certificate.log`

checks, for every integer $62 \le b \le 217$, that

$$t_b + 2h < 2e^{-5/4}b, \qquad v_b < 2e^{-5/4}\min(n_0, n_1), \qquad \rho_b^C(t_b, h) > 0, \qquad W_b \ge 2U_b^C,$$

and that

$$c_b^2(1 - a_b^{-2})\frac{W_b}{2} \ge 2\left(2(b^2+1)U_b^C(2/r)^{2b+29}\right),$$

$$c_b^2(1 - a_b^{-2})\frac{W_b}{2} \ge 2\left(2(a_b/c_b)^{2b+29}\right).$$

Every lower quantity is rounded downward and every upper quantity upward. The replay exits with status 0, reports `C_rows_checked=156` and `failures=0`, and gives the minimum defining-trace domain, square-trace domain, and tilted-mass lower bounds

$$14.669512561242725\ldots, \qquad 2.7221542041383810\ldots, \qquad 0.028831039797142247\ldots.$$

The three minimum natural-log comparison margins are

$$1.5000727210859520\ldots,\qquad 1.5065100867456532\ldots,\qquad 0.15645659337912115\ldots.$$

The log has SHA-256

$$\begin{array}{l}\texttt{ca6860204466e4a6b38d9c5597511cd1}\\ \texttt{1cfebb4b1b65cab88db6ee45f9477111}\end{array},$$

and the C++ source has SHA-256

$$\begin{array}{l}\texttt{0f5ad1037bcf738ab4ddd9c5b12f151a}\\ \texttt{adc1bff2b5c4726c60622f608ef9e0e7}\end{array}.$$

The comparison $W_b \geq 2U_b^C$ gives $p_1 - p_2 \geq W_b/2$. Hence lemma 4.11 gives

$$P_{C_b}(c_b) \geq c_b^2(1 - a_b^{-2})\frac{W_b}{2}, \qquad N_{C_b}(a_b) \leq 2(b^2+1)U_b^C.$$

The final two checked comparisons imply the hypothesis of lemma 4.5 at the odd exponent $2b+29$. Its monotonicity clause proves every later odd exponent. □

Lemma 4.34 (Exact odd-orthogonal defining-trace MGFs)**.** *Let O be Haar-distributed in* $\mathrm{SO}(2b+1)$*, put $Z_b = \operatorname{Tr} O$, and, for $v > 0$ and $\epsilon \in \{+1, -1\}$, set $M_{b,\epsilon}(v) = \mathbf{E}e^{\epsilon v Z_b}$. Then*

$$M_{b,+}(v) = e^{v}\det\left(I_{|j-k|}(2v) - I_{j+k+1}(2v)\right)_{0\leq j,k<b}, \tag{4.34.1}$$

$$M_{b,-}(v) = e^{-v}\det\left(I_{|j-k|}(2v) + I_{j+k+1}(2v)\right)_{0\leq j,k<b}, \tag{4.34.2}$$

where I_m is the modified Bessel function. Both displayed matrices are positive definite.

Proof. For $O^+(2b+1)$, Courteaut–Johansson–Lambert [11, Section 3, equation (3.1)] identify the b nontrivial eigenvalue pairs with Jacobi parameters $(1/2, -1/2)$. Their exact Toeplitz–Hankel identity [11, Lemma 3.1, the E_n^{+-} formula], applied to $\psi(x) = e^{2vx}$, has Fourier coefficients $\widehat{\phi}_m = I_m(2v)$. Restoring the deterministic $+1$ eigenvalue gives (4.34).1.

For the argument $-v$, use $I_m(-2v) = (-1)^m I_m(2v)$. Conjugating the resulting matrix by $\mathrm{diag}(1, -1, 1, -1, \ldots)$ changes it to the plus-Hankel matrix in (4.34).2 and does not change its determinant. The two matrices are moment Gram matrices for the strictly positive Jacobi weights occurring in Lemma 3.1 of the cited source, so they are positive definite. □

Lemma 4.35 (Exact even-orthogonal defining-trace MGFs)**.** *For $b \geq 2$, let $Z_b^+ = \operatorname{Tr} O$ with O Haar-distributed in* $\mathrm{SO}(2b) = O^+(2b)$*. For $r \geq 1$, let $Z_r^- = \operatorname{Tr} O$ with O Haar-distributed in the determinant-minus component $O^-(2r+2)$. For $v > 0$,*

$$\mathbf{E}e^{vZ_b^+} = \frac{1}{2}\det\left(I_{|j-k|}(2v) + I_{j+k}(2v)\right)_{0\leq j,k<b}, \tag{4.35.1}$$

and

$$\mathbf{E}e^{vZ_r^-} = \det\left(I_{|j-k|}(2v) - I_{j+k+2}(2v)\right)_{0\leq j,k<r}. \tag{4.35.2}$$

Both displayed matrices are positive definite.

Proof. Courteaut–Johansson–Lambert [11, Section 3, equation (3.1)] identify $O^+(2b)$ with the Jacobi parameters $(-1/2, -1/2)$ and $O^-(2r+2)$ with the parameters $(1/2, 1/2)$. Apply their exact Toeplitz–Hankel identities [11, Lemma 3.1, the E_b^{--} and E_r^{++} formulas] to $\psi(x) = e^{2vx}$, whose Fourier coefficients are $I_j(2v)$.

For completeness, the factor $1/2$ in (4.35).1 is forced by the normalized $(-1/2, -1/2)$ Jacobi measure. In the orthonormal cosine basis $1, \sqrt{2}T_1, \sqrt{2}T_2, \ldots$, Andreief's identity gives a Gram determinant. The unscaled Toeplitz–Hankel matrix in the display is obtained by multiplying the first Gram row and column by $\sqrt{2}$, so its determinant is twice the normalized expectation. In particular it equals 2 at $v = 0$. The $(1/2, 1/2)$ formula is already normalized at $v = 0$. Finally, both matrices are Gram matrices, up to this positive diagonal rescaling, for their strictly positive Jacobi weights. Hence every principal pivot is positive. □

Lemma 4.36 (Exact low-rank trace supports). *For $O \in \mathrm{SO}(2b+1)$,*

$$-(2b-1) \le \operatorname{Tr} O \le 2b+1. \tag{4.36.1}$$

For $O \in O^-(2r+2)$,

$$|\operatorname{Tr} O| \le 2r. \tag{4.36.2}$$

For $U \in \mathrm{Sp}(2b)$ and $\tau_C = -\operatorname{Tr}(U^2)$,

$$|\operatorname{Tr} U| \le 2b, \qquad |\tau_C| \le 2b. \tag{4.36.3}$$

Proof. The nonreal eigenvalues occur in conjugate unit-circle pairs and each pair contributes a number in $[-2,2]$ to the trace. An element of $\mathrm{SO}(2b+1)$ has a forced $+1$ eigenvalue, giving the upper endpoint $2b+1$ and the lower endpoint $1-2b$. An element of $O^-(2r+2)$ has forced eigenvalues $+1,-1$, whose trace contributions cancel, leaving r conjugate pairs. A symplectic matrix has b such pairs and no forced real eigenvalue; squaring preserves the unit-circle pair structure. These observations give all three displays. □

Lemma 4.37 (Exact type-C defining-trace MGF). *Let U be Haar-distributed in $\mathrm{Sp}(2b)$, $b \ge 1$, and put $S_b = \operatorname{Tr} U$. For every $v > 0$,*

$$\mathbf{E} e^{vS_b} = \mathbf{E} e^{-vS_b} = \det\left(I_{|j-k|}(2v) - I_{j+k+2}(2v)\right)_{0 \le j,k < b}. \tag{4.37.1}$$

The displayed matrix is positive definite.

Proof. The remark following Rains' Theorem 1.5 [6, equation (36)] identifies the nonfixed eigenvalues of $\mathrm{Sp}(2b)$ with those of $O^-(2b+2)$. The latter component has fixed eigenvalues $+1,-1$, of total trace zero, so the full traces have the same distribution. Equation (4.35).2 with $r = b$ gives the determinant in (4.37).1 and its positive definiteness. Multiplication of U by the central element $-I$ preserves Haar measure and changes S_b to $-S_b$, proving the equality of the two MGFs. □

Lemma 4.38 (Exact type-C square-trace MGF). *Let U be Haar-distributed in $\mathrm{Sp}(2b)$, $b \ge 2$, and put $\tau_C = -\operatorname{Tr}(U^2)$. Define*

$$m_0 = \left\lceil \frac{b+1}{2} \right\rceil, \qquad m_1 = \left\lceil \frac{b}{2} \right\rceil,$$

and

$$H_m(v) = \det\left(I_{|j-k|}(2v) - I_{j+k+2}(2v)\right)_{0 \le j,k < m-1}.$$

With $M_{m_1,-}$ as in (4.34).2, one has, for every $v > 0$,

$$\mathbf{E} e^{v\tau_C} = e^v H_{m_0}(v) M_{m_1,-}(v). \tag{4.38.1}$$

Both determinant matrices on the right are positive definite.

Proof. Use Rains' identification $\mathrm{Sp}(2b) \sim O^-(2b+2)$ after omitting the fixed eigenvalues $+1,-1$ [6, remark following Theorem 1.5, equation (36)]. Theorem 1.5 at $p = 2$, specifically equation (22), decomposes the nonfixed eigenvalues after squaring into independent components

$$O^-(2m_0) \oplus O^+(2m_1+1).$$

If Y_0, Y_1 are their full traces, then the first component's fixed eigenvalues have total trace zero and the second component has a fixed $+1$. Restoring these eigenvalues gives

$$\tau_C - 1 \overset{d}{=} -(Y_0 + Y_1).$$

The law of Y_0 is symmetric because multiplication by $-I$ preserves the determinant-minus component in even dimension. Thus $\mathbf{E} e^{-vY_0} = H_{m_0}(v)$ by (4.35).2. By definition, $\mathbf{E} e^{-vY_1} = M_{m_1,-}(v)$. Independence proves (4.38).1. Positive definiteness is supplied by lemmas 4.34 and 4.35. □

Lemma 4.39 (Exact type-D square-trace MGF)**.** *Let O be Haar-distributed in* $\mathrm{SO}(2b)$ *and put* $\tau_D = \mathrm{Tr}(O^2)$. *Define*

$$m_0 = \left\lceil \frac{b}{2} \right\rceil, \qquad m_1 = \left\lceil \frac{b-1}{2} \right\rceil.$$

If $G_{m,+}(v) = \mathbf{E}e^{vZ_m^+}$ denotes the even-orthogonal MGF in (4.35).1 and $M_{m,-}(v)$ denotes the odd-orthogonal MGF in (4.34).2, then, for $v > 0$,

$$\mathbf{E}e^{v\tau_D} = e^v G_{m_0,+}(v) M_{m_1,-}(v). \tag{4.39.1}$$

Proof. Rains' power-two decomposition [6, Theorem 1.5, equation (22)], with the fixed eigenvalues restored, gives independent full traces Y_0, Y_1 on $O^+(2m_0)$ and $O^-(2m_1+1)$. The latter component has a forced eigenvalue -1, which is omitted in Rains' convention, and therefore $\tau_D - 1 \overset{d}{=} Y_0 + Y_1$; this is the type-$D$ identity used in the proof of lemma 4.26. Independence and lemmas 4.34 and 4.35 give (4.39).1. □

Lemma 4.40 (Layered tilt with exact two-sided MGFs)**.** *Let Z be a real random variable and write $M_\epsilon(v) = \mathbf{E}e^{\epsilon v Z}$ for $\epsilon \in \{+1, -1\}$. Fix $s > 0$ and $t > 0$, and choose $\lambda_- > 0$ and pairs (x_i, λ_i), $i = 0, \dots, L-1$, with $t < x_0 < \cdots < x_{L-1}$ and $\lambda_i > 0$. Assume the displayed MGFs are finite. For each sign define*

$$A_{\epsilon,-} = e^{\lambda_- t} \frac{M_\epsilon(s - \lambda_-)}{M_\epsilon(s)}, \qquad A_{\epsilon,i}^+ = e^{-\lambda_i x_i} \frac{M_\epsilon(s + \lambda_i)}{M_\epsilon(s)},$$

$$g_{\epsilon,i} = 1 - A_{\epsilon,-} - A_{\epsilon,i}^+, \qquad w_i = e^{-s x_i}.$$

If $\epsilon Z \le x_i$ almost surely, one may instead set $A_{\epsilon,i}^+ = 0$. If $0 < g_{\epsilon,0} \le \cdots \le g_{\epsilon,L-1}$, then

$$\Pr\{\epsilon Z \ge t\} \ge V_\epsilon := M_\epsilon(s) \left(w_{L-1} g_{\epsilon,L-1} + \sum_{i=0}^{L-2} (w_i - w_{i+1}) g_{\epsilon,i} \right). \tag{4.40.1}$$

Consequently $\Pr\{|Z| \ge t\} \ge V_+ + V_-$.

Proof. For fixed ϵ, tilt the law by $d\mathbf{P}_{\epsilon,s} = M_\epsilon(s)^{-1} e^{\epsilon s Z}\, d\mathbf{P}$. Exponential Markov gives

$$\mathbf{P}_{\epsilon,s}\{\epsilon Z < t\} \le A_{\epsilon,-}, \qquad \mathbf{P}_{\epsilon,s}\{\epsilon Z > x_i\} \le A_{\epsilon,i}^+.$$

The second inequality is replaced by the deterministic zero bound when $\epsilon Z \le x_i$ almost surely. Thus the tilted mass of $t \le \epsilon Z \le x_i$ is at least $g_{\epsilon,i}$. The same summation-by-parts argument as in lemma 4.25 bounds the integral of the decreasing weight $e^{-s\epsilon Z}$ by the bracket in (4.40).1. Since

$$\Pr\{\epsilon Z \ge t\} = M_\epsilon(s) \mathbf{E}_{\epsilon,s}[e^{-s\epsilon Z} \mathbf{1}_{\{\epsilon Z \ge t\}}],$$

multiplication by $M_\epsilon(s)$ proves the displayed inequality. The positive and negative tail events are disjoint for $t > 0$, so their lower bounds add. □

Proposition 4.41 (Low-rank B/C tails from exact determinant MGFs)**.** *For the rows and onsets in the following table,*

$$Q_3^G(n) \ge 0 \qquad \textit{for every odd } n \ge N_G.$$

G	B_7	B_9	B_{10}	B_{11}	B_{12}	B_{13}
N_G	63	63	63	75	81	83

G	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}
N_G	63	63	63	63	63	63	63	63

G	C_{12}	C_{13}	C_{14}	C_{15}	C_{16}	C_{17}	C_{18}	C_{19}
N_G	63	71	75	63	65	73	63	63

Proof. For each row put

$$c = rb, \qquad q = \alpha_q\sqrt{b}, \qquad a = \alpha_a\sqrt{b}, \qquad t = \sqrt{2c+2d+q}, \qquad s = t + l,$$
$$u_i = u_0 + i\Delta \quad (0 \le i \le 7), \qquad x_i = t + u_i, \qquad \lambda_- = f_- l, \qquad \lambda_i = f_+ u_i.$$

The fixed type-B parameters are

G	r	α_q	α_a	d	l	u_0	f_-	f_+	(k,m)	N_G
B_7	45/7	7	35/10	14/10	10	25/10	1/10	33/100	(3,3)	63
B_9	60/9	9	4333/1000	14/10	5	25/10	1/10	1/2	(2,6)	63
B_{10}	75/10	7	4250/1000	14/10	4	25/10	1/10	1/2	(2,7)	63
B_{11}	80/11	11	4523/1000	14/10	3	3	5/100	1/2	(3,7)	75
B_{12}	90/12	12	4619/1000	14/10	3	3	5/100	1/2	(3,8)	81
B_{13}	100/13	13	4715/1000	14/10	2	3	5/100	75/100	(4,8)	83

(4.41.1)

Here $\Delta = 1/10$, $f_q = 1$, and $f_a = 509/500$ in every type-B row. The fixed type-C parameters are

G	r	α_q	α_a	d	l	u_0	f_-	f_+	(k,m)	N_G
C_4	32/10	401/100	35/10	14/10	35/10	10	5/100	15/10	(1,2)	63
C_5	4	5	35/10	14/10	3	100	5/100	15/10	(2,2)	63
C_6	32/6	6	35/10	14/10	5	15/10	1/5	15/10	(2,3)	63
C_7	6	7	35/10	14/10	5	125/100	1/5	2	(3,3)	63
C_8	60/8	8	46/10	14/10	7	15/10	5/100	2	(3,4)	63
C_9	60/9	9	42/10	14/10	5	175/100	1/10	2	(4,4)	63
C_{10}	6	7	4	14/10	3	25/10	5/100	2	(2,7)	63
C_{11}	70/11	7	45/10	14/10	1	25/10	5/100	15/10	(3,7)	63
C_{12}	80/12	7	45/10	14/10	2	2	1/2	1/2	(3,8)	63
C_{13}	80/13	8	45/10	14/10	1/2	3	1/10	15/10	(3,9)	71
C_{14}	20/3	8	45/10	14/10	2	3	1/2	1	(3,10)	75
C_{15}	76/15	8	44/10	14/10	1	25/10	3/4	3/4	(5,9)	63
C_{16}	75/16	7	45/10	14/10	75/100	250/100	75/100	75/100	(4,11)	65
C_{17}	5	9	45/10	14/10	1	25/10	3/4	1/2	(3,13)	73
C_{18}	$226172/10^5$	$460576/10^5$	$438971/10^5$	$138552/10^5$	$105713/10^5$	$240817/10^5$	1	56/100	(5,8)	63
C_{19}	$226172/10^5$	$460576/10^5$	$438971/10^5$	$138552/10^5$	$105713/10^5$	$240817/10^5$	1	56/100	(5,8)	63

(4.41.2)

Here $f_q = 75841/10^5$ and $f_a = 79538/10^5$. Also $\Delta = 1/10$ except in the last two rows, where $\Delta = 30177/10^6$.

For B_b, apply lemma 4.40 separately to $\epsilon = +1, -1$, with the exact MGFs of lemma 4.34, and call the sum of the resulting lower bounds L_G. For C_b, use the symmetry and exact determinant in lemma 4.37; twice the positive-sign lower bound is L_G. Whenever an endpoint reaches the deterministic support, use the zero upper-tail clause justified by lemma 4.36.

Put $v_q = f_q(q-1)/2$ and $v_a = f_a(a-1)/2$. For type B, set

$$U_B(z,v) = \exp\{-v(z-1) + v^2\}.$$

For type C, let

$$U_C(z,v) = e^{-v(z-1)} H_{m_0}(v) M_{m_1,-}(v),$$

with m_0, m_1, H as in lemma 4.38. Chernoff's inequality and lemmas 4.26 and 4.38 give the required square-trace upper bounds; if $z > 2b$, the bound is zero by (4.36).3. Define

$$P_G^* = c^2(1-d^{-2})\big(L_G - U_G(q,v_q)\big), \qquad N_G^* = \frac{8e^{-2}}{v_a^2} U_G(a,v_a),$$

and

$$H_{k,m}(a) = \sum_{j=0}^{2m} (-1)^j \binom{2m}{j} a^{-j} K_{2k+j}.$$

The fixed 384-bit directed-rounding replay

`certificates/post_m29/ post_m29_low_bc_exact_mgf_mpfr_certificate.log` checks every geometry condition, determinant pivot, tilted-layer mass, layer monotonicity, positive difference $L_G - U_G(q, v_q)$, and polynomial expectation. At the displayed onset it checks

$$P_G^* \geq N_G^* \left(\frac{2b}{c}\right)^{N_G} + \frac{a^{N_G-2k}}{4^m c^{N_G}} H_{k,m}(a). \tag{4.41.3}$$

The replay constructs every K_j as an exact GMP integer, encloses every Bessel series remainder, and evaluates each determinant by interval Cholesky decomposition. It checks 22 rows, reports no failure, and has minimum final natural-log margin

$$0.07437283746004294208266355122 > 0$$

at C_{15}. Its log and source SHA-256 hashes are respectively

`5bf958c5ce217dfc64efc08c1ef72bd6`
`98aab30f1259eb3d93c9bcb0c8e0f3d2`,

`71cc2293c4ad5ba50d216c8f939cd1ec`
`604ce7b86d6a19e9be1fcb27adc421fe`.

The degree check $2k + 2m \leq 2b - 2$ holds in every row, so the stable moments used in (4.41).3 are supplied by lemma 4.10. Thus lemma 4.8 proves the claim at N_G. Since $2b/c < 1$ and $a/c < 1$ in every checked row, the monotonicity clause of that lemma proves every later odd exponent. ☐

Proposition 4.42 (Finite Chain bridges to the low-rank exact-MGF onsets)**.** *The Chain differences are nonnegative in the following ranges:*

G	$\{m : D_G(2m+1) \geq 0\}$
B_{11}	$31 \leq m \leq 35$
B_{12}	$31 \leq m \leq 38$
B_{13}	$31 \leq m \leq 39$
C_{13}	$31 \leq m \leq 33$
C_{14}	$31 \leq m \leq 35$
C_{17}	$31 \leq m \leq 34.$

(4.42.1)

Consequently, if $Q_3^G(63) \geq 0$ in one of these six rows, then $Q_3^G(n) \geq 0$ for every odd $63 \leq n < N_G$, where N_G is the onset in proposition 4.41.

Proof. Write the finite-row moment as $m_j^G = m_j^{\mathrm{st}} + \delta_j^G$. Expanding the Chain difference gives the exact identity

$$D_G(2m+1) = D_{\mathrm{st}}(2m+1) + \sum_j L_{m,j}\delta_j^G + \sum_{i \leq j} Q_{m,i,j}\delta_i^G\delta_j^G. \tag{4.42.2}$$

The authenticated source logs supply consecutive exact corrections through the following degrees:

G	B_{11}	B_{12}	B_{13}	C_{13}	C_{14}	C_{17}
last degree	51	53	57	43	51	45.

Every supplied correction is nonpositive. By proposition 4.62, every missing higher correction is also nonpositive; nonnegativity of the actual moment therefore gives $-m_j^{\mathrm{st}} \leq \delta_j^G \leq 0$. The exact-integer replay retains every known linear and quadratic term in (4.42).2, bounds a missing linear term below by $-L_{m,j} m_j^{\mathrm{st}}$ when $L_{m,j} > 0$, bounds a missing negative quadratic term using $|\delta_i^G \delta_j^G| \leq m_i^{\mathrm{st}} m_j^{\mathrm{st}}$, and drops only nonnegative terms.

The replay

`certificates/post_m29/ post_m29_low_bc_exact_onset_bridge_certificate.log`

checks all 34 differences in (4.42).1. Its smallest exact integer lower bound is

$$64688975233529723032612107788184688805229314688529030019703101781365210144991608409847821836 > 0$$

at $C_{13}, m = 31$. It records status zero and has SHA-256

$$\texttt{c156d1296b8f74afc24f9f48ee615f3a}$$
$$\texttt{ba57e5e9d87f671258139226f8f57d28}.$$

This proves (4.42).1. Starting from odd exponent 63, repeated use of lemma 2.11 gives the final assertion. □

Proposition 4.43 (Residual B/C tails from layered MGFs and one-sided moments)**.** *For every B_b with $14 \le b \le 61$ and every C_b with $20 \le b \le 61$, one has*

$$Q_3^G(n) \ge 0$$

for every odd $n \ge 2b + 29$.

Proof. The following table gives four exact rational parameter sets. In each row put

$$c = rb, \quad q = \alpha_q\sqrt{b}, \quad a = \alpha_a\sqrt{b}, \quad t = \sqrt{2c + 2d + q}, \quad s = t + l,$$
$$u_i = u_0 + i\Delta \quad (0 \le i \le 7), \qquad \lambda_- = l, \qquad \lambda_i = f_+ u_i,$$
$$v_q = f_q(q-1)/2, \qquad v_a = f_a(a-1)/2.$$

G	r	α_q	α_a	d	l	u_0	Δ	f_+	f_q	f_a	(k,m)
B_{14}–B_{15}	$\frac{16}{5}$	$\frac{529}{100}$	$\frac{213}{50}$	$\frac{143}{100}$	$\frac{109}{100}$	$\frac{13}{10}$	$\frac{31}{1000}$	1	1	$\frac{509}{500}$	$(5, b-6)$
B_{16}	$\frac{200001}{10^5}$	$\frac{45546}{10^4}$	$\frac{4284}{10^3}$	$\frac{1277}{10^3}$	$\frac{856}{10^3}$	$\frac{1847}{10^3}$	$\frac{34}{10^3}$	$\frac{567}{10^3}$	1	$\frac{101331}{10^5}$	$(5,10)$
B_{17}	$\frac{20446}{10^4}$	$\frac{44739}{10^4}$	$\frac{42902}{10^4}$	$\frac{1341}{10^3}$	$\frac{972}{10^3}$	$\frac{15366}{10^4}$	$\frac{332}{10^4}$	$\frac{7291}{10^4}$	1	$\frac{101331}{10^5}$	$(5,10)$
B_{18}–B_{61}	$\frac{20674}{10^4}$	$\frac{441721}{10^5}$	$\frac{429465}{10^5}$	$\frac{13501}{10^4}$	$\frac{103311}{10^5}$	$\frac{139822}{10^5}$	$\frac{31205}{10^6}$	1	1	$\frac{101331}{10^5}$	$(5,8)$
C_{20}–C_{61}	$\frac{226172}{10^5}$	$\frac{460576}{10^5}$	$\frac{438971}{10^5}$	$\frac{138552}{10^5}$	$\frac{105713}{10^5}$	$\frac{135104}{10^5}$	$\frac{30177}{10^6}$	1	$\frac{75841}{10^5}$	$\frac{79538}{10^5}$	$(5,8)$

For B_{14}, B_{15}, let $V_{b,+}, V_{b,-}$ be the exact-MGF lower bounds in lemma 4.40, using lemma 4.34, $L = 8$, $\lambda_- = l$, and $x_i = s + u_i$, $\lambda_i = u_i$, and put $L_b = V_{b,+} + V_{b,-}$. In the other four rows, let V_b be the lower bound in lemma 4.25 with the displayed parameters and put $L_b = 2V_b$. For $G = B_b, C_b$, respectively, write U_G for the corresponding function in lemma 4.26. Define

$$P_b^* = c^2(1 - d^{-2})\big(L_b - U_G(q, v_q)\big), \qquad N_b^* = \frac{8e^{-2}}{v_a^2} U_G(a, v_a),$$

and, using the stable push moments of lemma 4.10,

$$H_{k,m}(a) = \sum_{j=0}^{2m} (-1)^j \binom{2m}{j} a^{-j} K_{2k+j}.$$

The 384-bit directed-rounding MPFR replay

`certificates/post_m29/ post_m29_bc_layered_mgf_mpfr_certificate.log`

checks every intermediate domain and sign condition and, at $n_b = 2b + 29$, the final comparison

$$P_b^* \ge N_b^* \left(\frac{2b}{c}\right)^{n_b} + \frac{a^{n_b - 2k}}{4^m c^{n_b}} H_{k,m}(a). \tag{4.43.1}$$

For B_{14}, B_{15} it also evaluates

$$I_\nu(2v) = \sum_{r=0}^{180} \frac{v^{2r+\nu}}{r!(r+\nu)!} + R_{\nu,v}, \qquad 0 < R_{\nu,v} \le \frac{v^{362+\nu}}{181!(181+\nu)!} \frac{1}{1 - v^2/(181(181+\nu))},$$

after checking that the final ratio is below one for every argument and $0 \le \nu \le 29$. It evaluates both exact Toeplitz–Hankel determinants by interval Cholesky decomposition and checks every pivot strictly positive. It constructs the integers K_j exactly in GMP from their generating function, converts them to outward-rounded MPFR intervals, and uses outward rounding for every subsequent arithmetic operation. The combined replay checks 333 rows: 48 type-B, 42 type-C, and the 243 type-D rows used later in proposition 4.49. It exits with status 0 and reports `failures=0`. Its B/C lower margins include

condition	minimum directed lower bound
$c-2b,\ 2b-a,\ a-1,\ d-1$	0.00016
$1-\mathfrak{a}_d$	0.24176531795234819...
exact Cholesky pivot	0.85461901487040600...
g_0	0.0028157129657487152...
$\min_i(g_{i+1}-g_i)$	0.008012263371171611...
$\log P_b^* - \log(\text{right side of }(4.43).1)$	1.358823017167476129...

The two exact-determinant rows have final natural-log margins

$$6.045003937030598629\ldots \quad \text{and} \quad 7.590922674135796189\ldots.$$

Among the B/C rows the final minimum is attained at B_{16}. The combined-log minimum is the positive D_{220} margin quoted in proposition 4.49. The log has SHA-256

`3e4900e5e2de2ec1a7fb04830ac59dea`
`7369f8eebb1932ec08ac9ef73ba11654`,

and the C++ source has SHA-256

`71cc2293c4ad5ba50d216c8f939cd1ec`
`604ce7b86d6a19e9be1fcb27adc421fe`.

By lemmas 4.25 and 4.40, as applicable, the defining-trace event in lemma 4.12 has probability at least L_b. Equations (4.26).1–(4.26).2 bound the positive square-trace loss by $U_G(q, v_q)$. Therefore (4.12).1 gives $P_G(c) \geq P_b^*$. Equations (4.12).2–(4.12).3 give $N_G(a) \leq N_b^*$. The degree conditions for lemma 4.10 are $2k+2m = 2b-2$ for B_{14}, B_{15}, $2k+2m = 30 = 2(16)-2$ for B_{16}, B_{17}, and $2k+2m = 26 \leq 2b-2$ in the uniform rows. Thus (4.43).1 and lemma 4.8 prove the claim at n_b. Both ratios $2b/c$ and a/c are strictly below one, so the monotonicity clause of that lemma proves every later odd exponent. □

Proposition 4.44 (Post-$m=29$ first-hit lower bound)**.** *For the remaining post-$m=29$ finite residue rows*

$$B_{11},\ldots,B_{275},B_{277}, \qquad C_{13},\ldots,C_{275},C_{277}, \qquad D_{25},\ldots,D_{275},D_{277},D_{279},$$

one has

$$D_G\big(N_{\text{hit}}^{(29)}(G)-2\big) \geq 0$$

where

$$N_{\text{hit}}^{(29)}(G) = \max(63, s(G)+2).$$

Consequently $Q_3^G(N_{\text{hit}}^{(29)}(G)) \geq 0$ for each of these rows.

Proof. The 783 rows split into five arithmetic buckets:

bucket	rows	minimum positive margin row
BC exact $m=30$ window	14	B_{11}
BC stable-tail $m=30$	26	B_{19}
BC stable-tail sloped	490	B_{32}
D exact two-sided interval	28	D_{25}
D A-free frontier	225	D_{53}

For a row whose first hit is $n = 2m+3$, let

$$\Delta_{\text{st}}(m) = D_{\text{st}}(2m+1)$$

be the stable Chain difference, and write the finite-row moment correction as $m_j^G = m_j^{\mathrm{st}} + \delta_j^G$. Expanding definition 2.10 in the variables δ_j^G gives

$$D_G(2m+1) = \Delta_{\mathrm{st}}(m) + \sum_j L_{m,j}\delta_j^G + \sum_{i\le j} Q_{m,i,j}\delta_i^G\delta_j^G$$

with integer coefficients $L_{m,j}$ and $Q_{m,i,j}$ determined from lemma 2.3. The B/C checker computes these coefficients exactly and proves a positive integer lower bound in each of its 530 rows. Those rows use the proved nonpositive boundary correction, so the positive linear tail is compared with the negative quadratic terms. The accepted replay is restricted to B/C and does not parse a determinant-only type-D row. It reports

rows checked	530,
stable moments generated	$m_0, \dots, m_{557}$,
scope	one Chain step at $N_{\mathrm{hit}}^{(29)}(G)$,
exit status	0.

Its three minimum margins have the following decimal sizes:

bucket	minimum row	$\log_{10}$(minimum margin)
BC exact $m = 30$ window	B_{11}	85.79
BC stable-tail $m = 30$	B_{19}	85.79
BC stable-tail sloped	B_{32}	93.85

The replay log has SHA-256

`21642c87b90affa8e618bb9f97bf6ad3`
`46289380832690546488c6dcc7a7300e`.

For $D_{25}, \dots, D_{52}$, the exact two-sided interval replay of proposition 4.51 checks every Chain step from the stable edge through its direct-tail onset, hence in particular the displayed first-hit step. Its 707 exact checks have minimum lower endpoint $4695986280539928492451072 > 0$. For every displayed D_b with $b \ge 53$, the short-frontier replay of proposition 4.49 checks the required moment ceiling $N_D(b) + 2 \le 2b$ and every Chain step from $m = 30$ to the direct-tail onset using lemma 4.46.3. Its worst positive logarithmic lower bound is $167.515421966088354\ldots$ at $D_{53}, m = 50$. The two status-0 logs have SHA-256 hashes

`e47196c8097fb2a087b4673d47a5e34c`
`aae1cd1a316058759e17b13708e2d888`,
`876d3f752d03b8f35183f0b28af7c15e`
`9e28d104396fc65c215e5d854d8ae31f`.

Every one of the five buckets therefore has a strictly positive lower margin. Thus $D_G(N_{\mathrm{hit}}^{(29)}(G) - 2) \ge 0$ in every displayed row. The preceding odd exponent is either covered by the stable edge or by the row-gated bridge, so lemma 2.11 gives $Q_3^G(N_{\mathrm{hit}}^{(29)}(G)) \ge 0$ in every displayed row. The B/C first-hit replay supplies only this one Chain step and is not used as an all-later tail. The all-later claims are proved separately below by the interval, layered-MGF, tilted-MGF, and correction-bridge propositions. □

Lemma 4.45 (Exact finite-D moment decomposition)**.** *For $D_b = \mathrm{SO}(2b)$ write*

$$e_2^j = \sum_{\lambda \vdash 2j} c_j(\lambda) s_\lambda,$$

where the nonnegative integers $c_j(\lambda)$ are given by the iterated vertical-two-strip Pieri rule. Put

$$A_j(b) = \sum_{\substack{\nu \vdash j \\ \ell(\nu) > 2b}} c_j(2\nu_1, 2\nu_2, \dots)$$

and

$$B_j(b) = \begin{cases} 0, & j < b, \\ \sum_{\substack{\nu \vdash j-b \\ \ell(\nu) \le 2b}} c_j(1+2\nu_1, \dots, 1+2\nu_{\ell(\nu)}, 1^{2b-\ell(\nu)}), & j \ge b. \end{cases}$$

If $s_j = m_j^{\mathrm{st}}$, then the exact correction is

$$m_j^{D_b} - s_j = B_j(b) - A_j(b).$$

In particular, the determinant-isotypic sum $B_j(b)$ alone is not the finite-rank correction once $A_j(b)$ is nonzero.

Proof. The adjoint character of $\mathrm{SO}(2b)$ is e_2 in the defining eigenvalues. Normalized Haar measure on $\mathrm{SO}(2b)$ is obtained from normalized Haar measure on $O(2b)$ by multiplication by $1+\det$. Expand e_2^j by Pieri's rule. Rains' orthogonal Schur-integral criterion [5, §3] says that the untwisted $O(2b)$ integral retains exactly the Schur summands with even row lengths and at most $2b$ rows. The stable sum retains all even-row summands, so the omitted part is exactly $A_j(b)$.

For the determinant-twisted integral, use $\det s_\lambda = s_{\lambda+(1^{2b})}$ on $O(2b)$. The same criterion is nonzero precisely when λ has exactly $2b$ rows, all odd. Such a partition has the unique displayed form with $\nu \vdash j-b$; none exists for $j < b$. Its total contribution is $B_j(b)$. Adding the untwisted and determinant-twisted integrals gives $m_j^{D_b} = s_j - A_j(b) + B_j(b)$. □

Lemma 4.46 (Type-D determinant component and the A-free envelope)**.** *Let $b \ge 4$, and use the notation of lemma 4.45. Put*

$$p_j(b) = s_j - A_j(b).$$

Then

$$p_j(b) - B_j(b) = \int_{\mathrm{Sp}(2b-2)} \chi_{\omega_2}(V)^j \, dV, \tag{4.46.1}$$

where ω_2 is the second fundamental representation of $\mathrm{Sp}(2b-2)$*. Consequently, for every $j \ge 0$,*

$$0 \le B_j(b) \le p_j(b) \le s_j. \tag{4.46.2}$$

Moreover $A_j(b) = 0$ for $0 \le j \le 2b$. Hence throughout this window the actual type-D correction satisfies

$$0 \le m_j^{D_b} - s_j = B_j(b) \le s_j. \tag{4.46.3}$$

Proof. Normalized Haar measure on $O(2b)$ is the half-sum of normalized Haar measure on its determinant components. Since

$$p_j(b) = \int_{O(2b)} e_2(U)^j \, dU, \qquad B_j(b) = \int_{O(2b)} \det(U) e_2(U)^j \, dU,$$

the normalized $O^-(2b)$ component moment is $p_j(b) - B_j(b)$.

Rains [6, remark following Theorem 1.5, equation (36)] identifies the nonfixed eigenvalue distribution of $O^-(2b)$ with that of $\mathrm{Sp}(2b-2)$. Restoring the fixed eigenvalues $+1, -1$, let V denote the defining $2b-2$ dimensional symplectic representation. On eigenvalue multisets,

$$e_2(V \oplus \mathbf{1} \oplus (-\mathbf{1})) = e_2(V) - 1.$$

The symplectic decomposition $\Lambda^2 V \cong \mathbf{1} \oplus V_{\omega_2}$ therefore turns the last character into χ_{ω_2}. This proves (4.46).1. Its right side is the multiplicity of the trivial representation in $V_{\omega_2}^{\otimes j}$, so it is a nonnegative integer. Since $A_j(b), B_j(b) \ge 0$, this proves (4.46).2.

Finally, a partition ν with more than $2b$ nonzero parts has $|\nu| \ge 2b+1$. The defining sum for $A_j(b)$ is therefore empty when $j \le 2b$. Substitution in lemma 4.45 proves (4.46).3. □

Lemma 4.47 (Two-sided type-D correction box)**.** *Use the notation of lemma 4.45 and put* $\delta_j(b) = m_j^{D_b} - s_j$ *and* $U_j = s_j$*. Then*

$$\delta_j(b) = 0 \quad (0 \le j < b), \tag{4.47.1}$$

$$0 \le \delta_j(b) \le U_j \quad (b \le j \le 2b), \tag{4.47.2}$$

and

$$-s_j \le \delta_j(b) \le U_j \quad (j > 2b). \tag{4.47.3}$$

Proof. For $j < b$, both sums $A_j(b), B_j(b)$ vanish. For $j \le 2b$, the sum $A_j(b)$ vanishes, so lemma 4.46 gives $\delta_j(b) = B_j(b) \ge 0$. For all j, the same lemma gives $0 \le B_j(b) \le s_j$ and the definition gives $0 \le A_j(b) \le s_j$; hence $\delta_j(b) \ge -s_j$. The same inequalities give $\delta_j(b) = B_j(b) - A_j(b) \le B_j(b) \le s_j = U_j$ for every parity. Together with the preceding vanishing and nonnegativity statements, this proves all three displays. □

Proposition 4.48 (Unconditional A-free type-D row-gated bridge)**.** *For every* $b \ge 53$*, the type-*D_b *rows in the finite bridge through* $m = 29$ *satisfy*

$$D_{D_b}(2m+1) \ge 0$$

at every row-gated Chain step.

Proof. At a row-gated step $m \le 29$, the two diagonals in the exact moment formula use moment indices only through $2m + 5 \le 63$. Since $63 \le 2b$ for $b \ge 53$, lemma 4.47 supplies

$$\delta_j^{D_b} = 0 \quad (j < b), \qquad 0 \le \delta_j^{D_b} \le s_j \quad (b \le j \le 63).$$

The exact GMP/OpenMP verifier expands each Chain difference as in (4.15).9 and minimizes every linear and quadratic monomial over these intervals. For $53 \le b \le 63$ it checks the 36 row-gated pairs

$$\left\lfloor \frac{b-4}{2} \right\rfloor \le m \le 29.$$

Every lower bound is positive; the least is

$$1295951275047754385311107869606245254765288973412864475150697991241728 > 0$$

at $D_{53}, m = 24$. For $b \ge 64$ the row-gated range through $m = 29$ is empty. The verifier source and deterministic machine-C output have SHA-256 hashes

```
1e574e0c53e624df30e9c69bc24280b
8e6c56a099923428f0a27036b74a7454b,

a84df851316bf376f00933ff958b7ad0d
ac444acfa8f45d05d4221f17ddadd12 .
```

Thus every nonvacuous row in the proposition has a source-generated exact integer certificate. □

Proposition 4.49 (Unconditional short-onset type-D tails)**.** *For every* $53 \le b \le 295$ *and every odd integer*

$$n \ge N_{\mathrm{hit}}^{(29)}(D_b),$$

one has

$$Q_3^{D_b}(n) \ge 0.$$

Proof. For each rank, let $N_D(b)$ be the odd onset in

`certificates/post_m29/ post_m29_bc_layered_mgf_mpfr_certificate.log`.

The directed-rounding verifier uses

$$c_b = \frac{20674}{10000}b, \quad q_b = \frac{441721}{100000}\sqrt{b}, \quad a_b = \frac{429465}{100000}\sqrt{b}, \quad d = \frac{13501}{10000},$$
$$l = \frac{103311}{100000}, \quad u_i = \frac{139822}{100000} + i\frac{31205}{1000000} \quad (0 \le i < 8), \quad k = 5, \quad m = 8.$$

The lower and upper defining-trace tilt fractions and the positive square-trace tilt fraction are 1; the negative square-trace fraction is 101331/100000. The full defining trace of $\mathrm{SO}(2b)$ is centered, and lemmas 4.24 and 4.25 give the checked two-sided defining-trace lower bound. The type-D clause (4.26).3 gives both checked square-trace upper bounds. Since $2k + 2m = 26 \le b - 3$, the type-D clause of lemma 4.10 supplies the exact polynomial $H_{5,8}^{D_b}(a_b)$.

For each b, the verifier starts at $N_{\text{hit}}^{(29)}(D_b)$ and increases the odd exponent until the strict comparison

$$P_{D_b}(c_b) > N_{D_b}(a_b)\left(\frac{2b}{c_b}\right)^{N_D(b)} + \frac{a_b^{N_D(b)-10}}{4^8 c_b^{N_D(b)}} H_{5,8}^{D_b}(a_b) \tag{4.49.1}$$

holds. For odd b, $C_{D_b} = b - 2$, so the first ratio in lemma 4.8 is no larger than the conservative $2b/c_b$ used in (4.49).1. All geometry, trace-norm, determinant, nested-mass, polynomial, and final comparisons are outward-rounded at 384-bit MPFR precision. The machine-C run checks all 243 ranks, reports

`SUMMARY rows_checked=333 B_rows_checked=48 C_rows_checked=42 D_rows_checked=243 failures=0`,

and has minimum final natural-log margin

$$0.0078804615513483267599166931 34 > 0$$

at D_{220}. Its log and C++ source have respective SHA-256 hashes

`3e4900e5e2de2ec1a7fb04830ac59dea`
`7369f8eebb1932ec08ac9ef73ba11654`,

`71cc2293c4ad5ba50d216c8f939cd1ec`
`604ce7b86d6a19e9be1fcb27adc421fe`.

The verifier also checks $N_D(b) + 2 \le 2b$ in every row, with minimum slack 1. Thus lemma 4.8 proves the assertion for every odd $n \ge N_D(b)$.

It remains to reach these onsets. The exact GMP/OpenMP replay

`certificates/post_m29/ post_m29_d53_295_short_frontier_gmp_machine_c_certificate.log`

reads each onset exactly once, rejects an even or missing onset, and independently checks $N_D(b) + 2 \le 2b$. It expands every Chain difference in the stable moments and the correction variables and uses lemma 4.56 and corollary 4.57 together with lemma 4.46.3. It verifies every active step $m = 30, \ldots, 163$ for all $D_{53}, \ldots, D_{295}$, reports `failures=0`, and records minimum A-free slack 1. The worst positive frontier is $m = 50$, odd target 103, frontier D_{53}, with logarithmic lower bound

$$167.515421966088354.$$

The log and source hashes are

`876d3f752d03b8f35183f0b28af7c15e`
`9e28d104396fc65c215e5d854d8ae31f`,

`444cf4dbe59d5000a3315f31054ef07e`
`c1cf710edf91a7c685f973644d65ad74`.

The stable edge and proposition 4.48 give $Q_3^{D_b}(61) > 0$. The replayed Chain differences and lemma 2.11 reach $N_D(b)$, after which the monotone direct tail just proved supplies every later odd exponent. This proves the proposition. ∎

Proposition 4.50 (Unconditional D_{19} closure)**.** *For every odd integer $n \geq s(D_{19}) + 2 = 17$, one has*

$$Q_3^{D_{19}}(n) \geq 0.$$

Proof. For the monotone direct tail, use the rational geometry and layered tilts in the proof of proposition 4.49, but take polynomial degrees $(k, m) = (6, 10)$, square-trace parameter $\alpha = 9/2$, and square-tilt fraction 1. The required push moments have degrees at most $2k + 2m = 32$. The exact C++/GMP source replay computes the finite-D_{19} adjoint moments $m_{19}, \ldots, m_{34}$ from the Pieri sums in lemma 4.45; the lower moments are stable by the same lemma. Formula (4.9).1 then gives every required $K_q^{D_{19}}$ exactly. The sixteen status-0 machine-C source logs and the two final certificates are authenticated by

$$\texttt{certificates/post\_m29/d19\_exact\_mgf\_machine\_c.sha256}, \qquad \mathrm{SHA256} = \begin{matrix}\texttt{ead8966c1769cbaaaf6d7a6928df831d} \\ \texttt{65887f5286d76c0fa18a19fcceb05d5b}\end{matrix}. \tag{4.50.1}$$

The exact-moment generator source has SHA-256

$$\begin{matrix}\texttt{bc22efcafafe2d40b5f5430cb8f9cb04} \\ \texttt{6bc6b9b188da684e61dc6fdfaa27077e}\end{matrix}.$$

The directed 384-bit C++/MPFR verifier applies lemmas 4.8, 4.35, 4.39 and 4.40. At odd exponent 63 it reports exact defining- and square-determinant pivot lower bounds $0.8516811782\ldots$ and $0.8449575175\ldots$ and the strict natural-log margin

$$6.7370668175795237151\ldots > 0.$$

Its log and source hashes are

$$\begin{matrix}\texttt{42e987f476bacc0e7422adb382556222} \\ \texttt{7aa0511a58a6984f61adcf4c08b3e4b5}\end{matrix}, \qquad \begin{matrix}\texttt{1f296ce3a32756fbee27bdc44eb9749d} \\ \texttt{03dc88e74107eb7f15d91f8fda3a15e5}\end{matrix}. \tag{4.50.2}$$

Thus the monotonicity clause of lemma 4.8 supplies every odd $n \geq 63$.

It remains to reach 63. For $19 \leq j \leq 38$, the accepted determinant rows are the full corrections because $A_j(19) = 0$ by lemma 4.46. At $j = 39$ the machine-C source replay instead records the full identity

$$B_{39}(19) - A_{39}(19) = 284926397477977670115230306761403414 7. \tag{4.50.3}$$

The exact-GMP verifier parses legacy determinant rows only through $j = 38$, parses (4.50).3 as a full correction, and uses the proved two-sided box of lemma 4.47 at every other index. It checks all 24 Chain steps $m = 7, \ldots, 30$, reports no failure, and has minimum exact lower bound

$$36778114931645730 > 0.$$

The bridge log and source hashes are

$$\begin{matrix}\texttt{a0a46e0274a2fa4ded548edfbec2f2d8} \\ \texttt{3cc7d5cecc13be155aa37d77fe9e11a0}\end{matrix}, \qquad \begin{matrix}\texttt{edec7e651f112a342909230068651c9a} \\ \texttt{59dc44a58bffcb5fa9e0420bf9f7ad8b}\end{matrix}. \tag{4.50.4}$$

The stable edge supplies the odd exponents through $s(D_{19}) = 15$; these positive Chain differences and lemma 2.11 reach 63, where the monotone direct tail begins. □

Proposition 4.51 (Unconditional D_{25}–D_{52} closure)**.** *For every $25 \leq b \leq 52$ and every odd integer $n \geq s(D_b) + 2$, one has*

$$Q_3^{D_b}(n) \geq 0.$$

Proof. First consider the monotone direct tails. Use the rational parameters and tilt fractions in the proof of proposition 4.49; only the MGF supplier and the following polynomial pairs (k_b, m_b) change:

$$(5,6),\ (4,7),\ (5,7),\ (5,7) \quad \text{at } b = 25, 26, 27, 28, \qquad (k_b, m_b) = (5, 8) \quad (29 \leq b \leq 52). \tag{4.51.1}$$

Thus $2k_b + 2m_b \leq b - 3$ in every row, and lemma 4.10 supplies the exact polynomial $H_{k_b, m_b}^{D_b}(a_b)$.

For $25 \leq b \leq 28$, the defining-trace lower bound is obtained from lemmas 4.35 and 4.40; the trace law is symmetric because $-I \in \mathrm{SO}(2b)$. For $29 \leq b \leq 52$, it is obtained from lemmas 4.24 and 4.25. In every row, lemma 4.39 supplies both square-trace upper bounds. The directed 384-bit MPFR verifier evaluates each Bessel series through term 180, adds the positive geometric remainder bound displayed in the proof of proposition 4.43, and uses interval Cholesky decomposition for every exact determinant. It verifies all geometry, pivot, Fredholm, nested-mass, polynomial, and final comparison inequalities.

Let T_b be the resulting odd onset. The exact list is

$$\begin{aligned}(T_{25},\dots,T_{38}) &= (71,73,71,73,73,73,75,77,79,79,81,83,83,85),\\ (T_{39},\dots,T_{52}) &= (85,87,89,89,91,93,93,95,97,97,99,99,101,103).\end{aligned} \tag{4.51.2}$$

At T_b the verifier proves the strict inequality

$$P_{D_b}(c_b) > N_{D_b}(a_b)\left(\frac{2b}{c_b}\right)^{T_b} + \frac{a_b^{T_b-2k_b}}{4^{m_b}c_b^{T_b}} H^{D_b}_{k_b,m_b}(a_b). \tag{4.51.3}$$

The conservative factor $2b/c_b$ dominates the actual endpoint factor also when b is odd. Hence lemma 4.8 proves every odd exponent at least T_b. The machine-C log

`certificates/post_m29/ post_m29_d25_52_exact_mgf_mpfr_machine_c_certificate.log`

checks 28 rows with no failures. Its minimum final natural-log margin is

$$0.01375109087492612411266244772 > 0$$

at D_{39}. The exact defining- and square-determinant pivot minima are, respectively, $0.8522815995\dots$ and $0.8481405784\dots$. The log and source hashes are respectively

$$\begin{gathered}\texttt{f87ab4bce0b189878ef4c3528ce34687}\\ \texttt{1721bff05cb00ccb553a6c4e84e1f049},\\[4pt] \texttt{71cc2293c4ad5ba50d216c8f939cd1ec}\\ \texttt{604ce7b86d6a19e9be1fcb27adc421fe}.\end{gathered} \tag{4.51.4}$$

It remains to connect the stable edge to T_b. Expand each Chain difference in the correction variables as in lemma 4.56. The exact C++/GMP/OpenMP verifier substitutes the boxes of lemma 4.47; for every bilinear term it takes the minimum over the four interval endpoints, and for a diagonal term it also accounts for an interval containing zero. It checks every Chain index

$$\left\lfloor \frac{b-4}{2} \right\rfloor \le m \le \frac{T_b-3}{2} \tag{4.51.5}$$

for every $25 \le b \le 52$. These are exactly the steps whose target exponents run from $s(D_b)+2$ through T_b. The replay performs 707 exact integer checks and reports no failure. Its smallest exact lower bound is

$$4695986280539928492451072 > 0$$

at $D_{25}, m = 10$. The machine-C log

`certificates/post_m29/ post_m29_d25_52_correction_interval_gmp_machine_c_certificate.log`

and its C++ source have respective SHA-256 hashes

$$\begin{gathered}\texttt{e47196c8097fb2a087b4673d47a5e34c}\\ \texttt{aae1cd1a316058759e17b13708e2d888},\\[4pt] \texttt{9a6cf4da5b866d3b8fc2ce6d1d1a442f}\\ \texttt{d3b419702fda2296da83d2fb788379e4}.\end{gathered} \tag{4.51.6}$$

The stable edge, these positive Chain differences, and lemma 2.11 reach T_b; the direct tail then supplies every later odd exponent. ☐

Definition 4.52 (Legacy finite type-D envelope)**.** For $4 \le b \le 52$, let T_b be the first odd exponent of the monotone pushforward tail in the accepted row certificate cited in proposition 4.18. Let DEnv denote the following finite statement: for $4 \le b \le 52$ and every $b \le j \le T_b + 2$, the quantities of lemma 4.45 satisfy

$$0 \le B_j(b) - A_j(b) \le s_j.$$

It contains only finitely many integer inequalities and no asymptotic assertion.

Proposition 4.53 (The legacy envelope is false)**.** *The statement* DEnv *of definition 4.52 is false. Specifically, at* $b = 4$ *and* $j = 18$*,*

$$\begin{gathered}s_{18} = 658906772228668, \qquad A_{18}(4) = 369382858189470,\\ B_{18}(4) = 288601773705246, \qquad B_{18}(4) - A_{18}(4) = -80781084484224 < 0.\end{gathered}$$

Proof. The exact C++/GMP Pieri replay on machine C reports

$$m_{18}^{D_4} = 578125687744444 = s_{18} - 369382858189470 + 288601773705246.$$

The archived log

`certificates/post_m29/ d4_j18_denv_counterexample_gmp_machine_c.log`

has SHA-256

$$\texttt{463e3c9632bb92936620fa65c03d3ff5}\\ \texttt{abdb86940624f1621ffa976156d7d02f},$$

and the C++ source has SHA-256

$$\texttt{bc22efcafafe2d40b5f5430cb8f9cb04}\\ \texttt{6bc6b9b188da684e61dc6fdfaa27077e}.$$

All displayed identities are exact integer identities. The negative correction contradicts the lower inequality in DEnv. □

Proposition 4.54 (Exact D_4–D_{24} row-gated bridge)**.** *For every* $4 \le b \le 24$*, the type-*D_b *row satisfies*

$$D_{D_b}(2m+1) \ge 0$$

at every row-gated Chain step $0 \le m \le 29$*.*

Proof. The exact signed adjoint-moment sources are generated from the two Pieri sums in lemma 4.45. They supply the following full moment prefixes:

b	4	5	6, 7	8, …, 17	18	19	20, …, 24
exact through m_j	59	44	40	38	38	34	30.

(4.54.1)

At D_4, five distinct deterministic-primality-checked 63-bit moduli have product greater than 28^{59}. The Chinese remainder reconstruction is therefore the unique integer in the character bound $0 \le m_j^{D_4} \le 28^j$ for every $0 \le j \le 59$. It agrees with two independent archived reconstruction summaries through m_{59} and with the full Pieri source at all 37 overlapping indices $m_4, \ldots, m_{40}$.

For every index after the prefix in (4.54).1, the verifier uses the proved intervals of lemma 4.47. It expands each Chain difference exactly as an integer polynomial in the corrections; a bilinear term is minimized over all four interval endpoints, and a diagonal term also tests zero when its interval contains zero. The fixed-mode GMP/OpenMP replay

`certificates/post_m29/ post_m29_d4_24_exact_prefix_interval_gmp_certificate.log`

checks all 530 row-gated steps and reports `failures=0`. Its smallest exact lower endpoint is $34 > 0$, at $D_5, m = 0$. The separate D4 CRT replay reports five primes, 60 reconstructed moments, 37 full-source cross-checks, and no failure. Every signed source log records exactly one zero status, machine-C host `nb1cb2f`, and source SHA-256

$$\texttt{bc22efcafafe2d40b5f5430cb8f9cb04}\\ \texttt{6bc6b9b188da684e61dc6fdfaa27077e}.$$

Before a correction is used, the C++ loader verifies both $O_j^{\mathrm{even}} + O_j^{\det} = \delta_j$ and $s_j + \delta_j = m_j$. The parent accepted-artifact manifest authenticates both replay outputs and all source logs. Thus every exact or interval-substituted integer lower endpoint used in the stated rows is positive. □

Proposition 4.55 (Exact D_{10}–D_{24} terminal tails)**.** *For every* $10 \le b \le 24$ *and every odd integer* $n \ge 63$*,*

$$Q_3^{D_b}(n) \ge 0.$$

Proof. First consider D_{10}. Put $d_{10} = \dim D_{10}$ and use the following rational cap and squared-Chebyshev data:

$$\begin{array}{c|cccccccccc} b & d_b & C_{D_b} & R_b & \kappa_b & M_b & \ell_b & S_b & a_b & J_b \\ \hline 10 & 190 & 10 & 107 & 18 & 12400 & 9 & 4650 & 14 & 38. \end{array} \tag{4.55.1}$$

Here $\sum_{\alpha>0} \alpha(\theta)^2 = (2b-2)|\theta|^2$, which gives the displayed values of κ_b. More explicitly, if $p_{2j} = \sum_i \theta_i^{2j}$, expansion over the positive roots $e_i \pm e_j$ gives

$$\sum_{\alpha>0} \alpha(\theta)^4 = 6p_2^2 + (2b-8)p_4 \le \kappa_b p_2^2$$

and

$$\sum_{\alpha>0} \alpha(\theta)^6 = 30p_2p_4 + (2b-32)p_6 \le \kappa_b p_2^3.$$

For the second inequality, after putting $y_i = \theta_i^2/p_2$, the difference between the right and left sides, divided by p_2^3, is

$$\sum_i y_i(1-y_i)\big((2b-2) + (2b-32)y_i\big) \ge 0$$

for $b = 10$. On the cap $Q(\theta) = \kappa_b|\theta|^2 \le R_b$, these formulas also give $|\alpha(\theta)|^2 \le 2R_b/\kappa_b$.

The cap width is $w = 6/5$ and the positive threshold is $c_{10} = d_{10} - R_{10} - w = 409/5$. Weyl integration in exponential coordinates [1, Chapter IV, (1.11)], the Macdonald–Mehta normalization [12, Theorem 3.1(i)], and

$$\prod_{\alpha>0} \left(\frac{\sin(\alpha(\theta)/2)}{\alpha(\theta)/2}\right)^2 \ge M_b^{-1}$$

give an exact rational lower bound for $P_{D_b}(c_b)$. The checker proves the sine-product inequality by the rational exponential majorant

$$\frac{R_b}{12} + \frac{R_b^2}{1440\kappa_b} + \frac{R_b^3}{90720\kappa_b^2(1 - 2R_b/(39\kappa_b))},$$

using 90 Taylor terms. All remaining cap, quartic-tail, monotonicity, Weyl-order, and factorial comparisons are exact integer or rational inequalities.

For completeness, the displayed sine exponent follows directly from the Euler product for $\sin z/z$. The quadratic, quartic, and sextic terms are bounded by

$$\frac{R_b}{12}, \qquad \frac{R_b^2}{1440\kappa_b}, \qquad \frac{R_b^3}{90720\kappa_b^2},$$

respectively. Since $\pi^2 > 39/4$, each remaining Euler-product term is at most $2R_b/(39\kappa_b)$ times its predecessor; the geometric sum is the third term in the stated majorant. Also $R_b < \kappa_b\pi^2$, so the cap is contained in one eigenangle chart. Writing $(d_i) = (2, 4, \ldots, 2b-2, b)$, its resulting probability lower bound is

$$\frac{1}{|W(D_b)|M_b} \frac{\prod_i d_i!}{(2\pi)^{b/2}\Gamma(d_b/2+1)} \left(\frac{R_b}{2\kappa_b}\right)^{d_b/2}. \tag{4.55.2}$$

The verifier replaces the powers of π by an explicit rational lower bound. Namely, $\pi < 22/7$ gives

$$\frac{1}{(2\pi)^5\Gamma(96)} > \frac{1}{(44/7)^5\, 95!} \tag{4.55.2a}$$

for D_{10}. The checker uses this bound only in the direction decreasing the cap probability.

Let Y be an independent adjoint character. By lemma 4.45, its first four moments are stable; since $\mathbf{E}Y = 0$, $\mathbf{E}Y^2 = \mathbf{E}Y^3 = 1$, and $\mathbf{E}Y^4 = 6$, for $r = 1/2$, $s = 15/4$ one has

$$\mathbf{E}(Y-r)^2(Y-s)^2 = 6 - 2(r+s) + r^2 + s^2 + 4rs + r^2s^2 =: V_4.$$

If $y < -w$ and $x \geq d_b - R_b \geq r$, then

$$\frac{x-y}{x+w} \leq \frac{(r-y)(s-y)}{(r+w)(s+w)}.$$

Thus the lost $(X-Y)^2$-weight is at most $V_4(x+w)^2/((r+w)^2(s+w)^2)$. The verifier checks that the resulting quadratic function of x is positive and increasing from $x = d_b - R_b$ onward. This proves that the cap contribution is a lower bound for $P_{D_b}(c_b)$, rather than merely for unweighted Haar mass.

For a real variable u, define

$$R_b^{\mathrm{Ch}}(u) = T_{\ell_b}\left(1 - \frac{4d_b}{S_b}u + \frac{2}{S_b}u^2\right)^2.$$

For $u \leq -a_b$, the quantity $u(u-2d_b)$ is at least $a_b(a_b+2d_b)$; since $T_{\ell_b}(z)^2$ increases for $z \geq 1$, this polynomial is at least $R_b^{\mathrm{Ch}}(-a_b) > 0$. Hence

$$N_{D_b}(a_b) \leq \frac{\int R_b^{\mathrm{Ch}}(u)\, d\psi_{D_b}(u)}{R_b^{\mathrm{Ch}}(-a_b)}. \tag{4.55.3}$$

The numerator has degree $4\ell_b$ and is an exact linear combination of $K_0^{D_b}, \ldots, K_{4\ell_b}^{D_b}$. By lemma 4.9, these are reconstructed from the full adjoint moments through $m_{J_b}^{D_b}$. The machine-C C++/GMP sources provide the gap-free range $m_{10}, \ldots, m_{38}$ and verify, row by row,

$$m_j^{D_b} = s_j + O_j^{\mathrm{even}}(D_b) + O_j^{\mathrm{det}}(D_b). \tag{4.55.4}$$

The exact-rational C++/GMP replay then proves

$$P_{D_b}(c_b) > N_{D_b}(a_b)\left(\frac{2C_{D_b}}{c_b}\right)^{63} + 2\left(\frac{a_b}{c_b}\right)^{63}. \tag{4.55.5}$$

Its output is

`certificates/post_m29/ post_m29_d10_exact_tail_gmp_certificate.log`.

For the exact-MGF rows $11 \leq b \leq 24$, use the common parameters

$$\begin{gathered} c_b = \frac{20674}{10000}b, \quad a_b = \frac{429465}{100000}\sqrt{b}, \quad d_0 = \frac{13501}{10000}, \quad l = \frac{103311}{100000}, \\ u_i = \frac{139822}{100000} + i\frac{31205}{1000000} \quad (0 \leq i < 8), \qquad f_- = f_i = f_{\mathrm{sq}} = 1, \qquad f_{\mathrm{neg}} = \frac{101331}{100000}, \end{gathered} \tag{4.55.6}$$

Here the four f's are respectively the lower and upper defining-trace tilt fractions and the positive and negative square-trace tilt fractions. Use the following rank-dependent polynomial and square-trace parameters:

b	(k_b, m_b)	$q_b/\sqrt{b}$
11	$(3,7)$	$24/5$
$12, 13, 14$	$(4,6)$	$19/4$
$15, 16$	$(4,6)$	$9/2$
17	$(5,5)$	$441721/100000$
18	$(4,7)$	$441721/100000$
19	$(5,6)$	$441721/100000$
$20, 21$	$(5,7)$	$441721/100000$
22	$(5,8)$	$441721/100000$
23	$(5,8)$	$17/4$
24	$(5,9)$	$441721/100000.$

(4.55.7)

The source loader first requires one status-zero machine-C provenance line, the pinned C++/GMP generator hash, and both exact identities in (4.55).4 for every consumed moment. It then uses lemma 4.9 to reconstruct the finite-rank $K_j^{D_b}$ needed by $H^{D_b}_{k_b,m_b}(a_b)$.

Put $n_{11} = 75$ and $n_b = 63$ for $12 \le b \le 24$. The exact even-orthogonal defining-trace determinant of lemma 4.35, the exact square-trace determinant of lemma 4.39, and the layered tilt of lemma 4.40 supply the positive trace tail. The same square-trace determinant and (4.12).3 supply $N_{D_b}(a_b)$. With directed 384-bit MPFR intervals, the verifier proves at $n = n_b$

$$P_{D_b}(c_b) > N_{D_b}(a_b)\left(\frac{2b}{c_b}\right)^{n_b} + \frac{a_b^{n_b-2k_b}}{4^{m_b}c_b^{n_b}}H^{D_b}_{k_b,m_b}(a_b). \tag{4.55.8}$$

The factor $2b/c_b$ conservatively dominates $2C_{D_b}/c_b$. For D_{11}, the machine-C replay reports defining- and square-MGF pivot lower bounds $0.8426844977700063107\ldots$ and $0.9814906225170877019\ldots$, and final natural-log margin $0.09113279249064900 2559\ldots > 0$. Its output is

`certificates/post_m29/ post_m29_d11_exact_mgf_mpfr_machine_c_certificate.log`.

The separate exact C++/GMP/OpenMP bridge consumes the gap-free machine-C corrections through $m_{46}^{D_{11}}$, checks all 34 Chain steps from the stable edge through odd exponent 75, and reports no failure. Its minimum exact lower endpoint is $11976538 > 0$. Its output is

`certificates/post_m29/ post_m29_d11_onset75_bridge_gmp_machine_c_certificate.log`.

By the stable edge and lemma 2.11, the bridge proves the required odd exponents through 75; the exact-MGF comparison and lemma 4.8 prove every odd exponent from 75 onward.

For $D_{12}, \ldots, D_{24}$, the replay checks all 13 rows with no failure. Its minimum defining- and square-MGF Cholesky pivots are respectively $0.8380062150\ldots$ and $0.8377469953\ldots$, and its minimum final natural-log margin is

$$0.0920085575002693989685\ldots > 0$$

at D_{24}. The output is

`certificates/post_m29/ post_m29_d12_24_exact_mgf_mpfr_certificate.log`.

Finally, lemma 4.5 propagates (4.55).5 for D_{10}, while lemma 4.8 propagates (4.55).8 for $D_{11}, \ldots, D_{24}$. $\square$

Lemma 4.56 (D stable-envelope frontier monotonicity)**.** *Fix $m \ge 31$. Let $s_j = m_j^{\mathrm{st}}$ be the stable D-row moments, let $\Delta_{\mathrm{st}}(m)$ be the stable Chain difference, and let $L_{m,j}$ and $Q_{m,i,j}$ be the integer coefficients in the expansion*

$$D_G(2m+1) = \Delta_{\mathrm{st}}(m) + \sum_j L_{m,j}\delta_j^G + \sum_{i \le j} Q_{m,i,j}\delta_i^G\delta_j^G,$$

where $m_j^G = s_j + \delta_j^G$. For an integer rank boundary r, define

$$F_m(r) = \Delta_{\mathrm{st}}(m) + \sum_{\substack{j \ge r \\ L_{m,j} < 0}} L_{m,j}s_j + \sum_{\substack{i \le j,\ i,j \ge r \\ Q_{m,i,j} < 0}} Q_{m,i,j}s_is_j.$$

Then $F_m(r') \ge F_m(r)$ whenever $r' \ge r$. Moreover, if a type D_b row has determinant corrections satisfying

$$\delta_j^{D_b} = 0 \quad (j < b), \qquad 0 \le \delta_j^{D_b} \le s_j \quad (j \ge b),$$

then

$$D_{D_b}(2m+1) \ge F_m(b).$$

Proof. The difference $F_m(r') - F_m(r)$ is the negative of the sum of those displayed linear and quadratic summands whose indices lie in the set removed when the boundary is raised from r to r'. Every removed displayed summand is nonpositive because $s_j \geq 0$ and only negative coefficients are displayed; hence the difference is nonnegative.

For the row bound, expand $D_{D_b}(2m+1)$ in the corrections $\delta_j^{D_b}$. Terms with positive coefficient are nonnegative and may be discarded in a lower bound. If a negative linear coefficient occurs, then $0 \leq \delta_j^{D_b} \leq s_j$ gives $L_{m,j}\delta_j^{D_b} \geq L_{m,j}s_j$. If a negative quadratic coefficient occurs, then the product vanishes unless both indices are at least b, and $0 \leq \delta_i^{D_b}\delta_j^{D_b} \leq s_i s_j$ gives $Q_{m,i,j}\delta_i^{D_b}\delta_j^{D_b} \geq Q_{m,i,j}s_i s_j$. Adding these lower bounds gives $D_{D_b}(2m+1) \geq F_m(b)$. □

Corollary 4.57 (D frontier bridge reduction)**.** *Let $\mathcal{I}$ be a finite interval of type D ranks for which the stable-envelope hypotheses of lemma 4.56 hold. For a fixed m, let r_m be the smallest rank in $\mathcal{I}$ whose finite bridge still includes the Chain step $2m+1$. If $F_m(r_m) > 0$, then*

$$D_{D_b}(2m+1) > 0$$

for every active rank $b \in \mathcal{I}$ at that Chain step.

Proof. Every active rank satisfies $b \geq r_m$. By lemma 4.56, $D_{D_b}(2m+1) \geq F_m(b) \geq F_m(r_m) > 0$. □

Proposition 4.58 (Post-$m = 29$ D-interval finite tails)**.** *For $G = D_b$ with $98 \leq b \leq 275$ or $b = 277, 279$, one has*

$$Q_3^G(n) \geq 0$$

for every odd $n \geq N_{\text{hit}}^{(29)}(G)$.

Proof. Every displayed rank lies in $53 \leq b \leq 295$, so this is the corresponding subrange of proposition 4.49. The former $R = 38b$ onset and long frontier artifacts are retained only as superseded provenance; no theorem now consumes their false stable-envelope hypothesis. □

Proposition 4.59 (Post-$m = 29$ B/C interval direct tails)**.** *For $G = B_b$ with $14 \leq b \leq 217$ or $G = C_b$ with $20 \leq b \leq 217$, let $N_{\text{dir}}^{BC}(G)$ be the odd onset recorded for the row in*

certificates/post_m29/bc_interval_formula_mpfr_certificate.log.

Then

$$Q_3^G(n) \geq 0$$

for every odd $n \geq N_{\text{dir}}^{BC}(G)$.

Proof. The accepted C++/Optimus verifier

`certificates/post_m29/bc_cheb80_negative_tail_gmp_certificate.log`

first supplies the common negative-tail input used by the interval direct-tail template. It uses exact GMP rational arithmetic with the degree-8 Chebyshev majorant, scale 12000, and cutoff 80. The log exits with status 0, reports `failures=0`, and has SHA-256

`80fce36378ec2192dca2c0b616c083b4`
`6168cf5134b66e73775156b995ea303b`.

Its worst row is B_{217}, with exact upper mass log $-15.1757329227092583 < -15$.

The directed-rounding MPFR verifier

`certificates/post_m29/bc_interval_formula_mpfr_certificate.log`

then checks the hypothesis of lemma 4.5 at the logged row onset for each of the 402 rows. It exits with status 0, reports `failures=0`, and has SHA-256

`53530cfe3d4b1947a77bf45c5c5aaa56`
`a052ec1cde93c0704b68e004f22f2cb5`.

The largest logged onset is $N^{BC}_{\mathrm{dir}}(B_{217}) = 30897$. The smallest directed lower log margin at an onset, attained at B_{20}, is

$$0.00613017280715829168757 91 > 0.$$

By the monotonicity clause of lemma 4.5, the same inequality holds for every later odd exponent in the row. ☐

Corollary 4.60 (Uniform residual B/C direct-tail onset bound)**.** *For every row $G = B_b$ with $14 \le b \le 217$ or $G = C_b$ with $20 \le b \le 217$,*

$$N^{BC}_{\mathrm{dir}}(G) \le 30897.$$

Proof. This is the largest-onset clause in the directed-rounding MPFR verifier used in proposition 4.59: the largest logged onset is $N^{BC}_{\mathrm{dir}}(B_{217}) = 30897$. ☐

Corollary 4.61 (Initial residual B direct-tail onset values)**.** *The directed-rounding MPFR verifier used in proposition 4.59 records*

$$N^{BC}_{\mathrm{dir}}(B_{14}) = 925, \qquad N^{BC}_{\mathrm{dir}}(B_{15}) = 415, \qquad N^{BC}_{\mathrm{dir}}(B_{16}) = 329, \qquad N^{BC}_{\mathrm{dir}}(B_{17}) = 303.$$

Proof. These are the row entries for B_{14}, B_{15}, B_{16}, and B_{17} in the status-0 directed-rounding MPFR verifier log

`certificates/post_m29/bc_interval_formula_mpfr_certificate.log`,

whose SHA-256 hash and failure-free summary are cited in proposition 4.59. ☐

Proposition 4.62 (B/C Pieri correction sign)**.** *For $G = B_b$ or $G = C_b$, put $s_j = m^{\mathrm{st}}_j$ and $m^G_j = s_j + \delta^G_j$ in the first nonstable moment range $j \ge 2b + 2$. Then*

$$\delta^G_j \le 0.$$

More precisely, writing

$$e_2^j = \sum_\lambda c_j(\lambda) s_\lambda, \qquad h_2^j = \sum_\lambda d_j(\lambda) s_\lambda,$$

the coefficients $c_j(\lambda)$ and $d_j(\lambda)$ are nonnegative integers, and

$$\delta^{B_b}_j = -\sum_{\substack{\nu \vdash j \\ \ell(\nu) > 2b+1}} c_j(2\nu_1, 2\nu_2, \ldots),$$

while

$$\delta^{C_b}_j = -\sum_{\substack{\mu \vdash j \\ \ell(\mu) > b}} d_j(\mu_1, \mu_1, \ldots, \mu_{\ell(\mu)}, \mu_{\ell(\mu)}).$$

Proof. For B_b, the adjoint character is $e_2 = s_{(1,1)}$ in the defining eigenvalues. Pieri's rule expands e_2^j by adding vertical two-strips, so each $c_j(\lambda)$ is a nonnegative integer. Rains' orthogonal Schur-integral criterion [5, §3] says that the stable orthogonal integral keeps exactly the Schur summands whose row lengths are even, while the finite $\mathrm{SO}(2b{+}1)$ integral keeps exactly those among them with at most $2b{+}1$ nonzero rows. The omitted stable summands are precisely the partitions $(2\nu_1, 2\nu_2, \ldots)$ with $\ell(\nu) > 2b + 1$, giving the displayed negative sum.

For C_b, the adjoint character is $h_2 = s_{(2)}$. Pieri's rule expands h_2^j by adding horizontal two-strips, so each $d_j(\lambda)$ is nonnegative. Rains' symplectic Schur-integral criterion [5, §3] says that the stable symplectic integral keeps exactly the Schur summands whose row lengths occur with even multiplicity, while the finite $\mathrm{Sp}(2b)$ integral keeps exactly those among them with at most $2b$ nonzero rows. The omitted summands are exactly $(\mu_1, \mu_1, \ldots, \mu_{\ell(\mu)}, \mu_{\ell(\mu)})$ with $\ell(\mu) > b$, giving the second negative sum. ☐

Corollary 4.63 (Residual window-size bounds from the direct-tail table). *In the residual B/C windows of remark 4.839, with $q = b+1$, the quantity $j-q$ satisfies the following bounds:*

range of q	*upper bound for $j-q$*
$q = 15$	912
$16 \le q \le 17$	401
$q = 18$	287
$19 \le q \le 20$	283
$21 \le q \le 22$	303
$23 \le q \le 24$	337
$25 \le q \le 28$	423
$q \ge 29$	30870.

Proof. The residual window ends at $j = N_{\mathrm{dir}}^{BC}(G)+2$. For the finite small-$q$ ranges, the displayed maxima of $N_{\mathrm{dir}}^{BC}(G)+2-q$ are read directly from the status-0 directed-rounding MPFR verifier table cited in proposition 4.59; the attaining rows are respectively $B_{14}, B_{15}, B_{17}, B_{19}, B_{21}, B_{23}$, and B_{27}, with the C rows in the same q-ranges no larger. For $q \ge 29$, corollary 4.60 gives

$$j - q \le 30899 - q \le 30870.$$

□

Lemma 4.64 (Rightmost even-column wall removal). *Let $q \ge 2$, and let λ be a partition such that λ' is even and $\lambda_1 \ge q$. Put $L = \lambda_1$, and define a new partition μ by its column lengths:*

$$\mu'_c = \begin{cases} \lambda'_c, & 1 \le c \le L-q, \\ \lambda'_c - 2, & L-q < c \le L, \end{cases}$$

omitting zero columns at the right. Then μ' is even and $|\mu| = |\lambda| - 2q$. Moreover, the skew shape λ/μ admits an iterated Pieri filling with content (2^q) whose labels each occupy a horizontal two-strip. Consequently s_λ occurs with positive coefficient in $s_\mu h_2^q$; in particular the width-overflow Schur sum over column-even λ is Schur-supported inside

$$h_2^q \sum_{\nu' : \,\mathrm{even}} s_\nu$$

after this rightmost-column deletion. Applying the involution ω gives the corresponding row-even vertical-strip statement with e_2^q.

Proof. Since λ' is even, every nonzero column length λ'_c is a positive even integer. Thus the rightmost q nonzero columns all have length at least 2, so the displayed definition is nonnegative. The sequence (μ'_c) is weakly decreasing: the unchanged part is weakly decreasing, the rightmost part remains weakly decreasing after subtracting 2 from each of its entries, and at the boundary

$$\mu'_{L-q} = \lambda'_{L-q} \ge \lambda'_{L-q+1} \ge \lambda'_{L-q+1} - 2 = \mu'_{L-q+1}$$

when $L > q$. Hence μ is a partition. All nonzero μ'_c are even, and the total removed size is two boxes in each of q columns, giving $|\mu| = |\lambda| - 2q$.

List the removed columns from left to right as

$$c_d = L - q + d \qquad (1 \le d \le q).$$

The two boxes removed from column c_d lie in rows $\lambda'_{c_d} - 1$ and λ'_{c_d}. Let

$$a_i = \left\lceil \frac{i}{2} \right\rceil \qquad (1 \le i \le 2q).$$

Fill the upper removed box of column c_d by a_d and the lower removed box by a_{q+d}. The word

$$a_1, \ldots, a_{2q} = 1, 1, 2, 2, \ldots, q, q$$

has content (2^q). Since $q \geq 2$, one has $a_d < a_{q+d}$ for every d, so columns are strictly increasing.

It remains to check rows. If two upper removed boxes lie in the same row, their columns have the same even height and their labels are a_d in increasing column order, hence weakly increase. The same argument applies to lower removed boxes, using the weakly increasing labels a_{q+d}. An upper removed row has odd index, while a lower removed row has even index, because all column heights are even; hence an upper removed box and a lower removed box cannot lie in the same row. Thus the filling is semistandard.

For each label r, the two boxes carrying r lie in distinct columns, because columns are strictly increasing. Therefore the boxes of each label form a horizontal two-strip. Reading labels $1, \ldots, q$ gives an iterated Pieri chain from μ to λ, so Pieri's rule gives a positive coefficient of s_λ in $s_\mu h_2^q$. Summing over the possible column-even μ gives the stated Schur-support containment. Applying $\omega(s_\alpha) = s_{\alpha'}$ exchanges horizontal and vertical two-strips and gives the final row-even statement. □

Lemma 4.65 (Column-even Littlewood graph model)**.** *Let*

$$K(x) = \prod_{1 \leq a < b} (1 - x_a x_b)^{-1}.$$

For every $j \geq 0$, the stable B/C adjoint moment $s_j = m_j^{\mathrm{st}}$ appearing in proposition 4.62 is

$$s_j = [x_1^2 \cdots x_j^2]\, K(x_1, \ldots, x_j).$$

Equivalently, s_j is the number of loopless labelled multigraphs on $\{1, \ldots, j\}$ in which every vertex has degree 2, with parallel edges allowed.

Proof. Albion's classical Littlewood identity [7, Introduction, (1.1c)] gives

$$\sum_{\lambda' \colon \mathrm{even}} s_\lambda(x) = \prod_{1 \leq a < b} (1 - x_a x_b)^{-1},$$

where the paragraph following (1.1c) defines λ' as the conjugate partition and "even" as having only even parts. For the stable C-side, the Rains criterion used in proposition 4.62 gives

$$s_j = \left\langle h_2^j, \sum_{\lambda' \colon \mathrm{even}} s_\lambda \right\rangle.$$

For the stable B-side, the same criterion gives the Hall pairing of e_2^j with the row-even Schur sum; applying the standard involution $\omega(s_\lambda) = s_{\lambda'}$ sends this to the same displayed pairing with h_2^j and the column-even Schur sum. Since $h_2^j = h_{(2^j)}$ is Hall-dual to the monomial symmetric function $m_{(2^j)}$, this pairing is the coefficient of $m_{(2^j)}$ in K, hence the coefficient of the monomial $x_1^2 \cdots x_j^2$.

Finally, expanding

$$\prod_{1 \leq a < b \leq j} (1 - x_a x_b)^{-1} = \sum_{(m_{ab})_{a<b}} \prod_{a<b} (x_a x_b)^{m_{ab}}$$

chooses a nonnegative edge multiplicity m_{ab} for each unordered pair $a < b$. The coefficient of $x_1^2 \cdots x_j^2$ is therefore exactly the number of such multiplicity arrays with $\sum_{b \neq a} m_{\min(a,b),\max(a,b)} = 2$ for every a, i.e. the stated loopless labelled 2-regular multigraphs. □

Corollary 4.66 (Direct full-wall overflow deletion has a parity-zero obstruction)**.** *For every $q \geq 1$, the set of column-even semistandard tableaux of content (2^{2q+1}) whose terminal first row has length at least $2q$ is nonempty, whereas the stable one-label moment satisfies*

$$s_1 = 0.$$

Consequently no direct full-wall estimate which bounds every such width-overflow tableau at $j = 2q+1$ by a finite multiple of s_{j-2q} can be true.

Proof. Lemma 4.117 constructs, for each $q \geq 1$, a column-even semistandard tableau of content (2^{2q+1}) and shape $(2q+1, 2q+1)$. Its first row has length $2q+1$, hence at least $2q$, so the displayed width-overflow family is nonempty.

By lemma 4.65, s_1 counts loopless labelled 2-regular multigraphs on the one-label set. There is no unordered pair $a < b$ on one label, so the only loopless multigraph has degree 0 at its unique vertex. Hence there is no loopless 2-regular multigraph and $s_1 = 0$.

At $j = 2q+1$, deleting a full wall of $2q$ labels leaves the stable index $j - 2q = 1$. Any finite-multiple bound of the stated form therefore has right side 0, while the nonempty family above gives a positive left side. □

Lemma 4.67 (Stable Littlewood moments satisfy a three-term recurrence)**.** *Let s_n be the stable moment sequence of lemma 4.65. Put $s_{-1} = 0$, $s_0 = 1$, and $s_1 = 0$. For every $n \geq 1$,*

$$s_{n+1} = ns_n + ns_{n-1} - \binom{n}{2} s_{n-2}.$$

Proof. By lemma 4.65, s_n counts loopless labelled 2-regular multigraphs on an n-element label set. Such a graph is a set of connected components. A component of size 2 is the unique doubled edge on two labels. A component of size $r \geq 3$ is a simple unoriented cycle on its r labels, because any parallel edge would already use both degrees at its two endpoints.

Thus the exponential generating function

$$A(z) = \sum_{n \geq 0} s_n \frac{z^n}{n!}$$

is

$$A(z) = \exp\left(\frac{z^2}{2} + \sum_{r \geq 3} \frac{z^r}{2r}\right) = \exp\left(-\frac{z}{2} + \frac{z^2}{4}\right)(1-z)^{-1/2}.$$

Therefore

$$\frac{A'(z)}{A(z)} = -\frac{1}{2} + \frac{z}{2} + \frac{1}{2(1-z)} = \frac{z - \frac{1}{2}z^2}{1-z},$$

or equivalently

$$(1-z)A'(z) = \left(z - \frac{1}{2}z^2\right) A(z).$$

Comparing the coefficient of $z^n/n!$ gives

$$s_{n+1} - ns_n = ns_{n-1} - \binom{n}{2} s_{n-2},$$

which is the displayed recurrence. □

Lemma 4.68 (Cumulative deleted-only graph counts have a factorial bound)**.** *For the stable moment sequence s_n of lemma 4.65 and every $K \geq 0$,*

$$\sum_{d=0}^{K} (d+1)s_d \leq (K+2)!.$$

Proof. By lemma 4.65, s_d counts loopless labelled 2-regular multigraphs on a fixed d-element label set. Each connected component is either the doubled edge on two labels or a simple unoriented cycle on at least three labels. Encode such a graph as a permutation by writing each doubled edge as a transposition and each simple cycle as the cycle whose least label is written first and whose second label is smaller than its last label. The components are disjoint, so this gives a well-defined permutation, and the graph is recovered from the cycle decomposition of that permutation by forgetting the chosen orientation of cycles of length at least three and replacing every transposition by a doubled edge. Hence $s_d \le d!$.

Therefore
$$\sum_{d=0}^{K}(d+1)s_d \le \sum_{d=0}^{K}(d+1)d! = \sum_{r=1}^{K+1} r!.$$
The last sum is at most $(K+2)!$: there are $K+1$ summands, each at most $(K+1)!$, and $K+1 < K+2$. ☐

Lemma 4.69 (Loopless graph deficit repair)**.** *Let Γ be a loopless labelled multigraph on a finite vertex set V in which every vertex has degree 2, with parallel edges allowed. Let $S \subset V$, put $V_0 = V \setminus S$, and let Γ_0 be the graph obtained from Γ by deleting S and all incident edges. For $v \in V_0$ put*
$$c_v = 2 - \deg_{\Gamma_0}(v).$$
If the nonzero deficits are not concentrated as $c_v = 2$ at a single vertex, then there exists a loopless multigraph R on V_0 such that
$$\deg_R(v) = c_v \qquad (v \in V_0).$$
Consequently $\Gamma_0 \cup R$ is a loopless labelled 2-regular multigraph on V_0.

Proof. For each $v \in V_0$, the number c_v is the number of edges of Γ joining v to a vertex of S, counted with multiplicity. Thus $c_v \in \{0, 1, 2\}$. The sum
$$C = \sum_{v \in V_0} c_v$$
is the number of edges from S to V_0. Since the total degree of the vertices in S is $2|S|$ and each edge internal to S contributes 2 to that degree, C is even.

Let $A = \{v : c_v = 2\}$ and $B = \{v : c_v = 1\}$. Then $|B|$ is even. If $A = B = \varnothing$, take R empty. If $A = \varnothing$, pair the vertices of B arbitrarily and put one edge on each pair.

Assume next that A is nonempty. If $|A|$ is even, pair the vertices of A arbitrarily and put two parallel edges on each pair; also pair the vertices of B arbitrarily with one edge on each pair. If $|A|$ is odd, then the excluded concentration case is exactly $|A| = 1$ and $B = \varnothing$. When $|A| \ge 3$, choose three vertices of A and connect them by a triangle, then pair the remaining vertices of A with two parallel edges on each pair and pair B arbitrarily. When $|A| = 1$, the exclusion forces $|B| \ge 2$; connect the unique vertex of A to two vertices of B, and pair the remaining vertices of B arbitrarily.

In every case the constructed graph R has no loops and has degree c_v at each v. Adding it to Γ_0 preserves looplessness and gives total degree $\deg_{\Gamma_0}(v) + c_v = 2$ at every vertex of V_0. ☐

Lemma 4.70 (Canonical graph deficit repair away from the obstruction)**.** *In the setting of lemma 4.69, assume the label set V_0 is linearly ordered and that the nonzero deficits are not concentrated as $c_v = 2$ at a single vertex. Then there is a canonically chosen loopless multigraph*
$$R_{\mathrm{can}}(\Gamma, S)$$
on V_0, determined only by the ordered deficit word $(c_v)_{v \in V_0}$, such that
$$\deg_{R_{\mathrm{can}}(\Gamma,S)}(v) = c_v \qquad (v \in V_0).$$
Consequently
$$\Gamma_0 \cup R_{\mathrm{can}}(\Gamma, S)$$
is a loopless labelled 2-regular multigraph on V_0, and the repair graph adds exactly
$$\frac{1}{2}\sum_{v \in V_0} c_v$$

edges.

Proof. Let

$$A = \{v \in V_0 : c_v = 2\}, \qquad B = \{v \in V_0 : c_v = 1\},$$

each listed in the fixed increasing order. As in the proof of lemma 4.69, $|B|$ is even.

If $A = B = \varnothing$, set $R_{\mathrm{can}} = \varnothing$. If $A = \varnothing$, pair the ordered list B consecutively and put one edge on each pair.

Suppose $A \neq \varnothing$. If $|A|$ is even, pair the ordered list A consecutively and put two parallel edges on each pair; also pair the ordered list B consecutively and put one edge on each pair. If $|A|$ is odd and $|A| \geq 3$, take the first three vertices of A and put a triangle on them, then pair the remaining vertices of A consecutively with two parallel edges on each pair, and pair B consecutively with one edge on each pair. Finally, if $|A| = 1$, the excluded obstruction implies $|B| \geq 2$; connect the unique vertex of A to the first two vertices of B, and pair the remaining vertices of B consecutively.

All choices are made by increasing order, so the construction is determined by the ordered deficit word. No loop is introduced: every edge joins two distinct vertices in a pair, in a triangle, or between the unique vertex of A and a vertex of B. The degree check is case-by-case from the construction. Vertices of A receive degree 2, vertices of B receive degree 1, and all other vertices receive degree 0. This proves $\deg_{R_{\mathrm{can}}}(v) = c_v$.

Since the sum of degrees of R_{can} is $\sum_v c_v$, the number of added edges is half that sum. Adding R_{can} to Γ_0 gives degree $\deg_{\Gamma_0}(v) + c_v = 2$ at every retained vertex and preserves looplessness. ☐

Lemma 4.71 (Deleted graph attachments are two-port data)**.** *Let Γ be a loopless labelled 2-regular multigraph on a finite linearly ordered label set V, and let $S \subset V$. Put $V_0 = V \setminus S$. For each $s \in S$, introduce two ordered ports*

$$(s, 1), \qquad (s, 2).$$

The multiset of all edges of Γ with at least one endpoint in S is equivalently encoded by the following port datum:

(1) *a matching of some ports into unordered pairs*

$$\{(s, a), (t, b)\} \qquad (s, t \in S, \ s \neq t),$$

representing the edges internal to S; and

(2) *a map from every unmatched port to V_0, representing the cross-edge endpoint of that port.*

For each $v \in V_0$, the number of unmatched ports mapped to v is the deficit

$$c_v = 2 - \deg_{\Gamma_0}(v)$$

of lemma 4.69. Moreover this port datum is canonically determined by the edge multiset using the ambient label order, and it determines the edge multiset.

Proof. For each unordered pair of distinct labels $x < y$, let m_{xy} be the edge multiplicity and label the formal parallel copies by $1, \ldots, m_{xy}$. For every edge copy incident to S, record at each endpoint $s \in S$ the opposite endpoint and this copy index. Order the two records at a fixed s lexicographically by the opposite endpoint and then by the copy index, and assign them to the ports $(s, 1), (s, 2)$ in that order. If an edge copy has both endpoints in S, pair the two ports assigned to its two endpoint records. Looplessness gives distinct labels $s \neq t$ for such a pair. If an edge joins $s \in S$ to $v \in V_0$, leave the assigned port unmatched and map it to v. Since Γ has degree 2 at every $s \in S$, every port is used exactly once. The construction is canonical from the ordered labels and the edge-copy ordering.

Conversely, from such a port datum, each matched pair $\{(s, a), (t, b)\}$ gives one internal edge between s and t, and each unmatched port (s, a) mapped to v gives one cross edge between s and v. This reconstructs the multiset of all edges incident to S. Finally, the cross edges incident to a retained vertex v are exactly the edges lost from v when S is deleted, so their number is $2 - \deg_{\Gamma_0}(v) = c_v$. ☐

Proposition 4.72 (Canonical graph deficit splices are recovered from attachment data). *Work in the setting of lemma 4.70. Let*

$$\Gamma^* = \Gamma_0 \cup R_{\mathrm{can}}(\Gamma, S)$$

be the repaired loopless labelled 2-regular multigraph on $V_0 = V \setminus S$. Suppose a datum A determines the multiset of all edges of Γ with at least one endpoint in S, split as

$$E_S(\Gamma) = \{\textit{edges with both endpoints in } S\}, \qquad E_\times(\Gamma) = \{\textit{edges with one endpoint in } S \textit{ and one endpoint in } V_0\}.$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad A.$$

In particular, in the non-obstructed graph-splice route the remaining branch-data problem is exactly the recovery of the deleted internal edges and cross-attachments incident to S, equivalently the two-port datum of lemma 4.71; the canonical degree repair itself carries no additional choice.

Proof. From $E_\times(\Gamma)$ we compute, for every $v \in V_0$,

$$c_v = \#\{\text{edges in } E_\times(\Gamma) \text{ incident to } v\},$$

with multiplicity. Therefore lemma 4.70 reconstructs the same canonical repair multigraph $R_{\mathrm{can}}(\Gamma, S)$ from the ordered deficit word $(c_v)_{v \in V_0}$.

Since Γ^* was formed as the multigraph union $\Gamma_0 \cup R_{\mathrm{can}}(\Gamma, S)$, subtracting these known edge-multiplicities recovers Γ_0. The original graph Γ is the disjoint union of the retained part Γ_0, the deleted internal edge multiset $E_S(\Gamma)$, and the cross-edge multiset $E_\times(\Gamma)$. All three pieces are now known, so Γ is recovered. □

Lemma 4.73 (Canonical graph repair is not a witness-free supplier). *Let $q \geq 9$. Let*

$$S = \{s_1, \dots, s_q\}, \qquad V_0 = \{a_1, \dots, a_q\}$$

be disjoint ordered label sets. For each permutation $\pi \in \mathfrak{S}_q$, let Γ_π be the loopless labelled 2-regular multigraph on $S \cup V_0$ obtained by putting two parallel edges between s_i and $a_{\pi(i)}$ for every i.

Deleting S from every Γ_π gives the same retained graph $\Gamma_0 = \varnothing$ on V_0, the same deficit word $c_{a_r} = 2$ for $1 \leq r \leq q$, and hence the same canonical repaired graph of lemma 4.70. The family has cardinality $q!$, and

$$q! > 2^{2q}.$$

Consequently the canonical graph deficit repair, together with only $2q$ unrestricted branch bits, cannot be an injective recovery map on the unrestricted graph family. In particular, for the residual fixed-witness range $q \geq 15$, the target-size graph repair must use additional structure from the fixed witness crossings; the witness-free graph repair is not the final supplier.

Proof. Each Γ_π is loopless and every vertex has degree 2: each s_i is incident to the two parallel edges joining it to $a_{\pi(i)}$, and each a_r is hit by the unique s_i with $\pi(i) = r$, again with multiplicity two. After deleting S, no edge remains on V_0, so Γ_0 is empty and $c_{a_r} = 2$ for all r. Since $q \geq 9$, the deficits are not concentrated at a single vertex, and lemma 4.70 applies. Its input ordered deficit word is the same for every π, so its repaired graph is also the same for every π.

The graphs Γ_π are distinct because the neighbour of s_1 in V_0 determines $\pi(1)$, and similarly for the other labels. Hence the fiber over that common repaired graph has size $q!$.

It remains only to compare with the branch budget. One has

$$9! = 362880 > 262144 = 2^{18}.$$

If $q! > 2^{2q}$ and $q \geq 9$, then

$$(q+1)! = (q+1)q! > (q+1)2^{2q} > 4 \cdot 2^{2q} = 2^{2(q+1)}.$$

Thus induction gives $q! > 2^{2q}$ for every $q \geq 9$. Therefore $2q$ bits, which distinguish at most 2^{2q} cases, cannot separate this fiber. The final sentence is the specialization to residual $q \geq 15$. □

Lemma 4.74 (The single-deficit obstruction is component-local)**.** *In the setting of lemma 4.69, suppose that the only nonzero deficit is $c_v = 2$ at a single vertex $v \in V_0$. Put $U = V_0 \setminus \{v\}$. Then no edge of Γ joins a vertex of $S \cup \{v\}$ to a vertex of U. Consequently the induced graph on $S \cup \{v\}$ is a union of connected components of Γ, the induced graph on U is a loopless labelled 2-regular multigraph, and the component containing v has all two edges incident to v joining v to vertices of S, counted with multiplicity.*

Conversely, if for some $v \in V_0$ no edge joins $S \cup \{v\}$ to $V_0 \setminus \{v\}$ and v has no neighbour in V_0, then the deletion of S has exactly the single nonzero deficit $c_v = 2$.

Proof. For any $u \in U$, the equality $c_u = 0$ means that no edge of Γ joins u to S. The equality $c_v = 2$, together with $\deg_\Gamma(v) = 2$, means that both edges incident to v join v to S, counted with multiplicity; in particular no edge joins v to U. Thus no edge joins $S \cup \{v\}$ to U. All degrees inside the induced graph on U are therefore still 2, and all degrees inside the induced graph on $S \cup \{v\}$ are also still 2. This proves the component decomposition and the asserted description of the component containing v.

For the converse, the absence of edges from $S \cup \{v\}$ to $V_0 \setminus \{v\}$ gives $c_u = 0$ for every $u \in V_0 \setminus \{v\}$, while the absence of a neighbour of v in V_0 gives $\deg_{\Gamma_0}(v) = 0$ and hence $c_v = 2$. ☐

Lemma 4.75 (Single-deficit attachments are forced port matchings)**.** *Work in the setting of lemma 4.71. Suppose that the deletion of S has exactly one nonzero deficit, namely*

$$c_v = 2$$

for a retained vertex $v \in V_0$. Then the port datum of lemma 4.71 has the following form:

1. *exactly two ports are unmatched and mapped to v;*
2. *no port is mapped to any vertex of $V_0 \setminus \{v\}$; and*
3. *every remaining port is matched to a port of a distinct deleted label.*

This restricted port datum determines the induced graph on $S \cup \{v\}$, and the full graph is recovered from that induced graph together with the loopless 2-regular graph induced on $V_0 \setminus \{v\}$.

Proof. By lemma 4.71, the number of ports mapped to a retained vertex $u \in V_0$ is exactly the deficit c_u. Since the only nonzero deficit is $c_v = 2$, exactly two ports are mapped to v, and no port is mapped to any vertex of $V_0 \setminus \{v\}$. Every port is used exactly once in the port datum, so all remaining ports are matched internally. The definition of the port datum permits an internal matched pair only between distinct deleted labels, proving the third item.

The two ports mapped to v give the two edges from v to S, and each matched pair gives one internal edge among the deleted labels. Hence the restricted port datum determines precisely the induced graph on $S \cup \{v\}$. By lemma 4.74, no edge joins $S \cup \{v\}$ to $V_0 \setminus \{v\}$, and the induced graph on $V_0 \setminus \{v\}$ is already loopless and 2-regular. The original graph is therefore the disjoint union of these two induced graphs. ☐

Lemma 4.76 (Single-deficit port codes split by connected component)**.** *Work in the setting of lemma 4.75, and let H be the loopless labelled multigraph on $S \cup \{v\}$ determined by the restricted port datum. Let H_v be the connected component of H containing v, and put*

$$S_v = V(H_v) \cap S.$$

Then H_v is a loopless connected labelled 2-regular multigraph on $S_v \cup \{v\}$, and its two edges incident to v are exactly the two ports mapped to v. Every other connected component of H is a loopless connected labelled 2-regular multigraph on a subset of S and is encoded only by internally matched deleted-label ports. The restricted port datum decomposes uniquely as the v-component subdatum together with these deleted-only component subdata.

Proof. By lemma 4.75, the restricted port datum has exactly two unmatched ports, both mapped to v, no port mapped to any other retained label, and all remaining ports internally matched between distinct deleted labels. Hence the graph H determined by the datum is loopless. The two unmatched ports give the two edges incident to v, and the two ports at each deleted label are each used once, so every vertex of H has degree 2.

Connected components of a loopless labelled 2-regular multigraph are again loopless labelled connected 2-regular multigraphs. The component containing v therefore has vertex set $S_v \cup \{v\}$ and contains the two edges coming from the two ports mapped to v. Since no port is mapped to a retained label other than v, every component not containing v has all vertices in S and all of its edges come from internally matched deleted-label ports.

Finally, connected components partition the edge multiset of H. Restricting the port datum to the ports whose endpoints lie in one component gives the corresponding component subdatum, and the union of these subdata reconstructs the original restricted port datum. The partition into connected components is unique, proving the claimed decomposition. □

Lemma 4.77 (The single-deficit component is an anchored cycle)**.** *Work in the setting of lemma 4.76. If $|S_v| = 1$, then H_v consists of the unique deleted label in S_v, the retained vertex v, and two parallel edges between them. If $|S_v| \geq 2$, then H_v has no parallel edges and its underlying simple graph is a cycle on $S_v \cup \{v\}$. Equivalently, deleting v and its two incident edges leaves a simple path through all labels of S_v, whose endpoints are the two deleted neighbours of v.*

The same dichotomy holds for every deleted-only component in lemma 4.76: it is either two deleted labels joined by two parallel edges, or a simple cycle on at least three deleted labels.

Proof. Let C be any connected component of the loopless labelled 2-regular multigraph H. If C has a pair of parallel edges between two labels a and b, those two edge copies already contribute degree 2 at both a and b. Since every vertex of C has degree 2 and C is connected, no further vertex or edge can lie in C. Thus a component with parallel edges is exactly the two-vertex double-edge component.

If C has no parallel edges, it is a finite connected simple graph in which every vertex has degree 2. Start at any vertex and follow an edge without immediately backtracking. Finiteness gives a first repeated vertex, and the degree-two condition forces the resulting cycle to contain every vertex of C; otherwise a vertex on the cycle would need an additional edge to the remaining connected part, giving degree at least 3. Hence C is a simple cycle.

Apply this classification to the component H_v. Since v has degree 2 in H_v, the two-vertex case with v is exactly the double edge from v to the unique deleted label in S_v. In the simple-cycle case, removing v and its two incident edges breaks the cycle into the asserted simple path through all deleted labels of S_v. Applying the same component classification to the components not containing v gives the deleted-only statement. □

Lemma 4.78 (Counting single-deficit anchored components)**.** *Let S be a finite label set with $|S| = q$, and let $v \notin S$. For a loopless labelled 2-regular multigraph H on $S \cup \{v\}$, let S_v be the set of labels of S lying in the connected component of v. Fix $T \subset S$ with $|T| = h$. The number of such graphs with $S_v = T$ is*

$$a_h\, s_{q-h}, \qquad a_1 = 1, \qquad a_h = \frac{h!}{2} \quad (h \geq 2),$$

and there are no graphs with $h = 0$. Consequently the number of such graphs whose v-component has exactly h deleted labels is

$$\binom{q}{h} a_h\, s_{q-h} \qquad (1 \leq h \leq q).$$

Proof. The case $h = 0$ is impossible because v would have degree 0 in a graph in which every vertex has degree 2 and loops are forbidden. Fix $T \subset S$ with $|T| = h \geq 1$.

The component classification proved in lemma 4.77 applies to every connected component of a loopless labelled 2-regular multigraph. Hence the component containing v is determined as follows.

If $h = 1$, it is forced to be the two parallel edges between v and the unique label of T. If $h \geq 2$, deleting v leaves a simple path through all labels of T, and adjoining v to the two endpoints recovers the simple cycle. The number of unoriented simple paths through the h labelled vertices of T is $h!/2$, since a linear order and its reversal give the same path and these are the only repetitions.

The components not containing v form an arbitrary loopless labelled 2-regular multigraph on $S \setminus T$, independently of the anchored component. By lemma 4.65, the number of such deleted-only graphs is s_{q-h}. This gives $a_h s_{q-h}$ for fixed T. There are $\binom{q}{h}$ choices for T, giving the displayed component-size count. □

Lemma 4.79 (Anchored single-deficit components have a support-path code)**.** *Let S be a finite label set and let $v \notin S$. A loopless labelled 2-regular multigraph H on $S \cup \{v\}$ is recovered from the following data:*

1. *the support $T \subset S$ of the connected component containing v;*
2. *if $|T| \geq 2$, an ordering $(t_1, \ldots, t_{|T|})$ of the elements of T, taken modulo reversal; and*
3. *the loopless labelled 2-regular multigraph induced on $S \setminus T$.*

The recovery rule is canonical: if $|T| = 1$, put two parallel edges between v and the unique label of T; if $|T| \geq 2$, put the cycle

$$v - t_1 - t_2 - \cdots - t_{|T|} - v$$

and then take the disjoint union with the graph on $S \setminus T$. Conversely, the graph H determines exactly these data.

Proof. For a given graph H, let T be the set of labels of S in the connected component of v. If this component contains a pair of parallel edges between two labels, then those two edge copies already give degree 2 at both endpoints; connectedness and the degree-two condition force the component to be exactly that double-edge component. Thus, when $|T| = 1$, the component of v is the forced double edge from v to the unique label of T.

If there are no parallel edges in the component of v, it is a finite connected simple graph in which every vertex has degree 2, hence a simple cycle. For $|T| \geq 2$, deleting v from this cycle leaves a simple path through all labels of T. Reading that path from one endpoint to the other gives an ordering of T, and reversing the reading gives the only other ordering describing the same path. The remaining connected components are precisely the induced loopless labelled 2-regular multigraph on $S \setminus T$.

Conversely, the displayed rule constructs a loopless labelled 2-regular multigraph on $T \cup \{v\}$ whose component containing v has support T. The reversal of $(t_1, \ldots, t_{|T|})$ gives the same unoriented cycle, so the rule is well defined on reversal classes. Taking the disjoint union with the given graph on $S \setminus T$ gives a loopless labelled 2-regular multigraph on $S \cup \{v\}$. The two constructions are inverse by the component classification in the preceding paragraphs. □

Lemma 4.80 (A single retained deficit is repaired by inserting into one retained edge)**.** *Let U be a finite linearly ordered label set with $|U| \geq 2$, let $v \notin U$, and let Γ_U be a loopless labelled 2-regular multigraph on U. Choose the lexicographically least edge ab of Γ_U, counting parallel copies by their multiplicity order. Let*

$$\mathcal{I}_v(\Gamma_U)$$

be the multigraph on $U \cup \{v\}$ obtained by deleting that one copy of ab and adding the two edges va and vb.

Then $\mathcal{I}_v(\Gamma_U)$ is a loopless labelled 2-regular multigraph on $U \cup \{v\}$. Moreover Γ_U is recovered from $\mathcal{I}_v(\Gamma_U)$ and v by deleting the two edges incident to v and restoring one edge between their other endpoints.

Proof. Since Γ_U is loopless and every vertex has degree 2, it has at least one edge; the chosen edge joins two distinct labels $a, b \in U$. Deleting one copy of ab lowers the degrees of a and b by one.

Adding va and vb restores the degrees of a and b and gives degree 2 at v. Every other vertex has unchanged degree. No loop is introduced because $v \notin U$ and $a \neq b$.

In $\mathcal{I}_v(\Gamma_U)$ the vertex v is incident exactly to the two edges va and vb. Deleting those two edges lowers the degrees of a and b by one and removes v; adding one copy of ab restores exactly the edge deleted in the construction. Thus the stated inverse recovers Γ_U. □

Proposition 4.81 (Single-deficit graphs have a stable retained repair)**.** *Work in the setting of lemma 4.74. Suppose that the deletion of S has exactly one nonzero deficit $c_v = 2$ and that $U = V_0 \setminus \{v\}$ has at least two vertices. Let Γ_U be the graph induced by Γ on U, and set*

$$\Gamma^* = \mathcal{I}_v(\Gamma_U),$$

using lemma 4.80. Then Γ^* is a loopless labelled 2-regular multigraph on V_0. Furthermore, the original graph Γ is recovered from

$$S, \quad v, \quad \Gamma^*, \quad \text{and the support-path code of lemma 4.79.}$$

Thus, in the single-deficit case with $|V_0 \setminus \{v\}| \geq 2$, the stable retained output carries no additional choice; the remaining branch-data problem is the recovery of the retained vertex v and the support-path code, including the deleted-only graph on the complement of the component containing v.

Proof. By lemma 4.74, no edge joins $S \cup \{v\}$ to U, and the graph induced on U is loopless and 2-regular. Since $|U| \geq 2$, lemma 4.80 applies and gives a loopless labelled 2-regular graph Γ^* on $U \cup \{v\} = V_0$.

The same lemma recovers Γ_U from Γ^* and v. The support-path code of lemma 4.79 recovers the induced graph on $S \cup \{v\}$. Again by lemma 4.74, the original graph is the disjoint union of these two induced graphs. Hence the displayed data recover Γ. □

Lemma 4.82 (Deleted-only single-deficit components are independent data)**.** *Let S contain five distinct labels*

$$s_0, d_1, d_2, d_3, d_4,$$

let $v \notin S$, and let $U = \{u_1, u_2\}$ be disjoint from $S \cup \{v\}$. There exist two distinct loopless labelled 2-regular multigraphs on $S \cup U \cup \{v\}$ such that

(1) *deleting S gives the same single retained deficit $c_v = 2$;*
(2) *the stable retained repair of proposition 4.81 is the same graph Γ^*;*
(3) *the component containing v has the same support $\{s_0\}$ and the same two parallel edges from v to s_0; and*
(4) *the two graphs differ only in the deleted-only component on $\{d_1, d_2, d_3, d_4\}$.*

Consequently a recovery statement for the whole original graph cannot omit the induced deleted-only graph on $S \setminus T$, where T is the support of the component containing v, unless an additional argument proves $T = S$ or supplies that deleted-only graph by some other target datum.

Proof. Let Γ_U be the graph with two parallel edges between u_1 and u_2. Let H_0 be the graph with two parallel edges between v and s_0. On $D = \{d_1, d_2, d_3, d_4\}$, take the two simple cycles

$$d_1 - d_2 - d_3 - d_4 - d_1 \qquad \text{and} \qquad d_1 - d_3 - d_2 - d_4 - d_1.$$

Let Γ and Γ' be the disjoint unions of Γ_U, H_0, and these two respective cycles on D. Both graphs are loopless and every vertex has degree 2.

After deleting S, both graphs leave Γ_U on U and the isolated vertex v, so the only nonzero retained deficit is $c_v = 2$. Since the retained complement graph Γ_U is the same in both cases, proposition 4.81 produces the same repaired retained graph $\Gamma^* = \mathcal{I}_v(\Gamma_U)$. The component containing v is H_0 in both cases, hence has support $\{s_0\}$ and the same anchored double edge. The two graphs are nevertheless distinct because the cycle on D is different. This proves the four displayed properties.

The final assertion follows because the displayed data S, v, Γ^* and the anchored support/path data are identical for Γ and Γ', while the original graphs are not identical. The missing datum is exactly the induced deleted-only graph on $S \setminus T$. ☐

Lemma 4.83 (Deleted-only complements exceed the retained-repair branch budget)**.** *Let $q \geq 15$. Let S be a linearly ordered q-element set, choose $s_0 \in S$, and put $D = S \setminus \{s_0\}$. Let $v \notin S$, and let U be a linearly ordered set disjoint from $S \cup \{v\}$ which carries a fixed loopless labelled 2-regular multigraph Γ_U. Assume $|U| \geq 2$, and put*

$$\Gamma^* = \mathcal{I}_v(\Gamma_U)$$

as in lemma 4.80.

There are more than 2^{2q} loopless labelled 2-regular multigraphs Γ on $S \cup U \cup \{v\}$ such that

1. *deleting S gives exactly one nonzero deficit, namely $c_v = 2$;*
2. *the stable retained repair of proposition 4.81 is the fixed graph Γ^*;*
3. *the component containing v has support $\{s_0\}$, so the anchored support is a fixed one-point witness interval and the adjacent-rank internal-edge condition is vacuous; and*
4. *the only varying datum is the deleted-only component on D.*

Consequently the retained-repair target Γ^*, even together with v, the anchored interval support, and an unrestricted $2q$-bit branch word, cannot recover this family. A single-deficit fixed-witness supplier must therefore either prove the deleted-only complement is absent, recover it from the actual target-size path, or encode it in a target object which is not just the stable retained repair.

Proof. For every unoriented simple cycle C through all labels of D, let Γ_C be the disjoint union of Γ_U, the two parallel edges between v and s_0, and the cycle C on D. This graph is loopless and every vertex has degree 2. Deleting S leaves Γ_U on U and isolates v, so the only nonzero retained deficit is $c_v = 2$. Since Γ_U is fixed, the stable retained repair is the same graph $\Gamma^* = \mathcal{I}_v(\Gamma_U)$ for every C.

The component containing v is always the double edge between v and s_0. Thus its support is the fixed one-point interval $\{s_0\}$, and there are no anchored internal edges. Different cycles C give different graphs Γ_C, and the only varying part is the deleted-only component on D.

The set D has $q-1$ labels. The number of unoriented simple cycles on a fixed $(q-1)$-element labelled set is

$$\frac{(q-2)!}{2}.$$

At $q = 15$ this is

$$\frac{13!}{2} = 3113510400 > 1073741824 = 2^{30} = 2^{2q}.$$

If $(q-2)!/2 > 2^{2q}$ and $q \geq 15$, then

$$\frac{(q-1)!}{2} = (q-1)\frac{(q-2)!}{2} > 4 \cdot 2^{2q} = 2^{2(q+1)}.$$

Induction gives $(q-2)!/2 > 2^{2q}$ for every $q \geq 15$.

For the displayed family, S, v, Γ^*, and the anchored interval support are fixed. A $2q$-bit word has only 2^{2q} values, fewer than the number of graphs in the family. Hence these data cannot be injective on the family. The final sentence is exactly this obstruction applied to any single-deficit supplier based only on the retained repair target. ☐

Lemma 4.84 (Unrestricted anchored path orders exceed the branch budget)**.** *Let $q \geq 10$. Let S be a q-element linearly ordered label set, let $v \notin S$, and let U be a linearly ordered label set disjoint from $S \cup \{v\}$ which carries a fixed loopless labelled 2-regular multigraph Γ_U. Assume $|U| \geq 2$, and put*

$$\Gamma^* = \mathcal{I}_v(\Gamma_U)$$

as in lemma 4.80.

There are $q!/2$ loopless labelled 2-regular multigraphs Γ on $S \cup U \cup \{v\}$ such that

(1) *deleting S gives exactly one nonzero deficit, namely $c_v = 2$;*
(2) *the stable retained repair of proposition 4.81 is the same graph Γ^* for every member of the family; and*
(3) *the support of the v-component is all of S, while the anchored path order varies over all orderings of S modulo reversal.*

Moreover

$$\frac{q!}{2} > 2^{2q}.$$

Consequently the repaired retained graph, the vertex v, and an unrestricted $2q$-bit branch word cannot by themselves recover the unrestricted anchored path order in the residual range $q \geq 15$.

Proof. For each ordering $(s_1, \ldots, s_q)$ of S, taken modulo reversal, let $H(s_1, \ldots, s_q)$ be the cycle

$$v - s_1 - s_2 - \cdots - s_q - v$$

on $S \cup \{v\}$. Let Γ be the disjoint union of this cycle with the fixed graph Γ_U on U. It is loopless and 2-regular on $S \cup U \cup \{v\}$. After deleting S, the vertex v is isolated, while the graph on U is still Γ_U; hence the only nonzero deficit is $c_v = 2$. Since the retained complement graph is the same fixed Γ_U in every case, proposition 4.81 produces the same repaired retained graph $\Gamma^* = \mathcal{I}_v(\Gamma_U)$ for every member of the family. Distinct orderings modulo reversal give distinct unoriented cycles through the labelled set $S \cup \{v\}$, so the family has cardinality $q!/2$.

For the inequality, the base case is

$$\frac{10!}{2} = 1814400 > 1048576 = 2^{20}.$$

If $q!/2 > 2^{2q}$ and $q \geq 10$, then

$$\frac{(q+1)!}{2} = (q+1)\frac{q!}{2} > (q+1)2^{2q} > 4 \cdot 2^{2q} = 2^{2(q+1)}.$$

Thus induction proves $q!/2 > 2^{2q}$ for every $q \geq 10$. A $2q$-bit word has only 2^{2q} possible values, so it cannot distinguish all members of this common repaired-output fiber. Since the residual B/C range has $q \geq 15$, the final sentence follows. □

Proposition 4.85 (Monotone anchored order leaves only support bits for the anchored component)**.** *Work in the setting of proposition 4.81. Assume that, in the support-path code of lemma 4.79, the anchored path order on the support $T \subset S$ is the increasing order of T in the ambient order on S, taken modulo reversal. Then the original graph Γ is recovered from*

$$S, \quad v, \quad \Gamma^*, \quad \Gamma_{S \setminus T}, \quad \mathbf{1}_T \in \{0,1\}^S,$$

where $\Gamma_{S \setminus T}$ is the loopless labelled 2-regular graph induced by Γ on $S \setminus T$. Consequently, under any fixed-witness crossing theorem which determines the retained deficit vertex v and forces this monotone anchored order, the single-deficit graph splice needs no separate path-order branch data for the anchored component. A full supplier must still either supply or absorb the deleted-only graph $\Gamma_{S \setminus T}$, or prove $T = S$.

Proof. The characteristic word $\mathbf{1}_T$ and the ordered set S recover the support T. If $|T| = 1$, lemma 4.79 prescribes the unique double edge from v to the single label of T. If $|T| \geq 2$, the monotonicity hypothesis says that the ordering of T modulo reversal is the increasing order, so the same lemma reconstructs the anchored cycle

$$v - t_1 - \cdots - t_{|T|} - v$$

where $t_1 < \cdots < t_{|T|}$ are the elements of T in the ambient order.

The graph Γ^* and the vertex v recover the retained complement graph Γ_U by lemma 4.80, as used in proposition 4.81. Taking the disjoint union of this recovered retained complement, the reconstructed anchored component on $T \cup \{v\}$, and the supplied deleted-only graph $\Gamma_{S\setminus T}$ gives Γ, by lemmas 4.74 and 4.76. The final sentence is the same recovery statement with the independent deleted-only datum separated, as required by lemma 4.82. □

Lemma 4.86 (Adjacent support edges force monotone anchored order)**.** *Let S be a finite linearly ordered label set, let $v \notin S$, and let $T \subset S$ with $|T| = h \geq 2$. Let P be a simple path through all labels of T. Suppose every edge of P joins two labels which are adjacent in the induced order on T. Then P is the monotone path through T, uniquely up to reversal.*

Proof. Write the elements of T in increasing order as

$$t_1 < t_2 < \cdots < t_h.$$

The unordered pairs of adjacent labels in the induced order are exactly

$$\{t_i, t_{i+1}\} \qquad (1 \leq i < h).$$

There are $h-1$ such pairs. A simple path through all h labels of T also has exactly $h-1$ edges. Since every edge of P is one of the displayed adjacent pairs and P is connected on all vertices of T, it must use all of them. Hence P is the path

$$t_1 - t_2 - \cdots - t_h,$$

read in one direction or the other. □

Proposition 4.87 (Adjacent support-edge recovery criterion)**.** *Work in the setting of proposition 4.81. Assume that, in the support-path code of lemma 4.79, every internal edge of the anchored path on T joins adjacent labels in the induced order on T. Then Γ is recovered from*

$$S, \quad v, \quad \Gamma^*, \quad \Gamma_{S\setminus T}, \quad \mathbf{1}_T \in \{0,1\}^S.$$

Consequently this leaf removes the anchored path-order choice once v, the support, and the deleted-only component have been supplied; it does not by itself encode the independent deleted-only graph on $S \setminus T$.

Proof. If $|T| = 1$, the support-path code has the forced double edge from v to the unique label of T, so there is no internal path-order choice. If $|T| \geq 2$, lemma 4.86 shows that the anchored path order on T is the increasing order, modulo reversal. Therefore proposition 4.85 applies and gives recovery from the displayed data. The final sentence is precisely this recovery criterion interpreted for the fixed-witness supplier. □

Lemma 4.88 (Consecutive frontier events give adjacent support labels)**.** *Let S be a finite linearly ordered label set and let $T \subset S$. Suppose each $t \in T$ is assigned a frontier event $e(t)$ in a linearly ordered set and that this assignment is order-preserving in the sense that, for $t, t' \in T$,*

$$t < t' \iff e(t) < e(t').$$

If $a, b \in T$ and no event $e(t)$ with $t \in T$ lies strictly between $e(a)$ and $e(b)$, then a and b are adjacent in the induced order on T.

Proof. Assume, after interchanging a and b if necessary, that $a < b$. The order-preserving hypothesis gives $e(a) < e(b)$. If a label $t \in T$ satisfied $a < t < b$, then the same hypothesis would give

$$e(a) < e(t) < e(b),$$

contradicting the assumption that no support event lies strictly between $e(a)$ and $e(b)$. Hence no element of T lies strictly between a and b, which is exactly adjacency in the induced order on T. □

Proposition 4.89 (Consecutive frontier-event edges give single-deficit recovery)**.** *Work in the setting of proposition 4.81. Let $T \subset S$ be the support of the anchored component in lemma 4.79. Suppose the labels of T are assigned order-preservingly to their fixed-witness frontier events, and every internal edge of the anchored path on T has endpoints whose assigned frontier events are consecutive among the events assigned to T. Then Γ is recovered from*

$$S, \quad v, \quad \Gamma^*, \quad \Gamma_{S\setminus T}, \quad \mathbf{1}_T \in \{0,1\}^S.$$

Consequently the frontier-event condition settles only the anchored path-order part of the single-deficit recovery; the deleted-only graph remains an explicit supplier input unless $T = S$.

Proof. By lemma 4.88, each internal edge of the anchored path joins adjacent labels in the induced order on T. Therefore proposition 4.87 applies and gives the displayed recovery. The final sentence is this criterion in the language of the fixed-witness crossing construction. □

Lemma 4.90 (Full frontier adjacency restricts to support adjacency)**.** *Let S be a finite linearly ordered label set, let $T \subset S$, and let $e : S \to E$ be a map into a linearly ordered set such that, for $s, s' \in S$,*

$$s < s' \iff e(s) < e(s').$$

If $a, b \in T$ and no event $e(s)$ with $s \in S$ lies strictly between $e(a)$ and $e(b)$, then a and b are adjacent in the induced order on T.

Proof. The restriction of e to T satisfies the same order equivalence. Since $T \subset S$, the absence of an event $e(s)$ with $s \in S$ strictly between $e(a)$ and $e(b)$ implies the same absence for events $e(t)$ with $t \in T$. Hence lemma 4.88 applies to the restricted map $T \to E$ and gives the asserted adjacency. □

Proposition 4.91 (Full frontier-adjacent edges give single-deficit recovery)**.** *Work in the setting of proposition 4.81. Let $T \subset S$ be the support of the anchored component in lemma 4.79. Suppose all labels of S are assigned to their fixed-witness frontier events by a map $e : S \to E$ satisfying*

$$s < s' \iff e(s) < e(s') \qquad (s, s' \in S).$$

If every internal edge of the anchored path on T has endpoints whose assigned frontier events are consecutive among the full event set assigned to S, then Γ is recovered from

$$S, \quad v, \quad \Gamma^*, \quad \Gamma_{S\setminus T}, \quad \mathbf{1}_T \in \{0,1\}^S.$$

Consequently full-frontier adjacency removes the path-order choice without pre-filtering by the unknown support. It still leaves the independent deleted-only component on $S\setminus T$ to be supplied, absorbed into the target, or ruled out by proving $T = S$.

Proof. Let ab be an internal edge of the anchored path on T. By hypothesis, no frontier event assigned to a label of S lies strictly between the events assigned to a and b. By lemma 4.90, the labels a and b are adjacent in the induced order on T. Thus every internal edge of the anchored path joins adjacent support labels. Applying proposition 4.87 gives the displayed recovery. The final sentence is the same criterion expressed in the full frontier-event list visible in the fixed-witness crossing construction. □

Lemma 4.92 (Single-deficit components factor from the retained graph)**.** *Let S be a finite label set, let $v \notin S$, and let U be a finite label set disjoint from $S \cup \{v\}$. Among loopless labelled 2-regular multigraphs on $S \cup \{v\} \cup U$, the graphs whose deletion of S has exactly one nonzero deficit, namely $c_v = 2$, are in bijection with pairs*

$$(\Gamma_1, \Gamma_2),$$

where Γ_1 is a loopless labelled 2-regular multigraph on $S \cup \{v\}$ and Γ_2 is a loopless labelled 2-regular multigraph on U. Consequently their number is

$$s_{|S|+1} s_{|U|}.$$

Proof. By lemma 4.74, a graph with exactly the deficit $c_v = 2$ has no edge joining $S \cup \{v\}$ to U, and the induced graphs on $S \cup \{v\}$ and on U are both loopless labelled 2-regular multigraphs. Thus restriction to the two induced subgraphs gives the displayed pair.

Conversely, the disjoint union of any such pair is loopless and has degree 2 at every vertex. After deleting S, every vertex of U still has degree 2, while v has no neighbour in U and hence has degree 0 in the retained graph. Therefore the unique nonzero deficit is $c_v = 2$. The two constructions are inverse to one another. The product count follows from lemma 4.65, which identifies s_n with the number of loopless labelled 2-regular multigraphs on any fixed n-element label set. ☐

Corollary 4.93 (The residual-floor all-graph single-deficit family exceeds the fixed-fiber budget)**.** *Let $|S| = 15$, fix $v \notin S$, and let U be a 14-element label set disjoint from $S \cup \{v\}$. The all-graph single-deficit family of lemma 4.92 has cardinality*

$$s_{16}s_{14} = 23184038368042864094640,$$

whereas a target consisting of an arbitrary stable retained graph on 15 labels together with a 30-bit branch word has size

$$2^{30}s_{15} = 158126931013997166592.$$

In particular,

$$s_{16}s_{14} > 2^{30}s_{15}.$$

Consequently a supplier which only compresses the witness-free all-graph single-deficit family to a stable retained graph/path on $j - q = 15$ labels and $2q$ branch bits cannot be injective at the first residual boundary $q = 15$, $j = 30$.

Proof. By lemma 4.92, the stated family has cardinality $s_{|S|+1}s_{|U|} = s_{16}s_{14}$. Iterating lemma 4.67 gives

$$s_{14} = 10157945044, \qquad s_{15} = 147267180508, \qquad s_{16} = 2282355168060.$$

Hence

$$s_{16}s_{14} = 23184038368042864094640$$

and

$$2^{30}s_{15} = 158126931013997166592.$$

The displayed strict inequality follows. The final sentence is then the pigeonhole principle applied to a domain larger than its proposed target set. ☐

Lemma 4.94 (B/C corrections as Littlewood bad-shape sums)**.** *Let $a_j(\lambda)$ be the Schur coefficient*

$$h_2^j = \sum_\lambda a_j(\lambda)s_\lambda.$$

For $b \geq 1$ and $j \geq 0$,

$$|\delta_j^{B_b}| = \sum_{\substack{\lambda'\colon \text{even} \\ \lambda_1 \geq 2b+2}} a_j(\lambda),$$

and

$$|\delta_j^{C_b}| = \sum_{\substack{\lambda'\colon \text{even} \\ \ell(\lambda) \geq 2b+2}} a_j(\lambda).$$

Proof. For B_b, proposition 4.62 gives

$$|\delta_j^{B_b}| = \sum_{\substack{\nu \vdash j \\ \ell(\nu) > 2b+1}} c_j(2\nu_1, 2\nu_2, \ldots),$$

where $e_2^j = \sum_\lambda c_j(\lambda)s_\lambda$. Applying the Hall involution ω gives

$$c_j(2\nu_1, 2\nu_2, \ldots) = a_j\big((2\nu_1, 2\nu_2, \ldots)'\big),$$

because $\omega(e_2) = h_2$ and $\omega(s_\lambda) = s_{\lambda'}$. The map

$$\nu \longmapsto \lambda = (2\nu_1, 2\nu_2, \ldots)'$$

is a bijection from partitions with $\ell(\nu) \geq 2b+2$ to partitions λ whose conjugate has only even parts and whose width satisfies $\lambda_1 \geq 2b+2$. This proves the B_b identity.

For C_b, proposition 4.62 gives

$$|\delta_j^{C_b}| = \sum_{\substack{\mu \vdash j \\ \ell(\mu) > b}} a_j(\mu_1, \mu_1, \ldots, \mu_{\ell(\mu)}, \mu_{\ell(\mu)}).$$

The partitions of the displayed paired-row form are exactly the partitions λ for which λ' has only even parts. The condition $\ell(\mu) > b$ is equivalent to $\ell(\lambda) = 2\ell(\mu) \geq 2b+2$. This proves the C_b identity. ☐

Lemma 4.95 (Canonical witnesses for residual bad shapes)**.** *Let $a_j(\lambda)$ be as in lemma 4.94. Equivalently, $a_j(\lambda)$ counts semistandard tableaux of shape λ and content (2^j). Fix $q \geq 1$.*

If λ' is even and $\lambda_1 \geq 2q$, then every such tableau T has a canonical q-element witness

$$W_q(T) = \{T(1,1), T(1,3), \ldots, T(1, 2q-1)\} \subset \{1, \ldots, j\}.$$

If λ' is even and $\ell(\lambda) \geq 2q$, then every such tableau T has a canonical q-element witness

$$H_q(T) = \{T(1,1), T(3,1), \ldots, T(2q-1, 1)\} \subset \{1, \ldots, j\}.$$

Proof. The coefficient statement is the standard complete-symmetric expansion: $h_2^j = h_{(2^j)}$ is the generating function for semistandard tableaux with content (2^j).

For the width witness, the first row of T is weakly increasing and each label occurs exactly twice in the whole tableau. If two entries among $T(1,1), T(1,3), \ldots, T(1, 2q-1)$ were equal, then weak increase would force at least three copies of that label in the first row between the two odd positions. This contradicts content (2^j). Hence $W_q(T)$ has cardinality q.

For the height witness, the first column of T is strictly increasing. Therefore the entries $T(1,1), T(3,1), \ldots, T(2q-1, 1)$ are distinct, and $H_q(T)$ has cardinality q. ☐

Definition 4.96 (Pieri two-strip path)**.** A Pieri two-strip path of length j is a sequence of partitions

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$

such that each skew difference $\lambda^{(t)}/\lambda^{(t-1)}$ is a horizontal strip of size 2. The path is column-even if $(\lambda^{(j)})'$ has only even parts. For $q \geq 1$ define

$$C_q^w(\lambda^\bullet) = \{t : \min(q, \lceil \lambda_1^{(t)}/2 \rceil) - \min(q, \lceil \lambda_1^{(t-1)}/2 \rceil) = 1\}$$

and

$$C_q^h(\lambda^\bullet) = \{t : \min(q, \lceil \ell(\lambda^{(t)})/2 \rceil) - \min(q, \lceil \ell(\lambda^{(t-1)})/2 \rceil) = 1\}.$$

Lemma 4.97 (Pieri paths encode the coefficient tableaux)**.** *For every $j \geq 0$, semistandard tableaux of content (2^j) are in bijection with Pieri two-strip paths of length j. Under this bijection, the tableau shape is $\lambda^{(j)}$.*

Proof. Given a tableau T, let $\lambda^{(t)}$ be the shape of the subtableau formed by entries at most t. The two boxes labelled t are not in the same column, because columns are strictly increasing, so $\lambda^{(t)}/\lambda^{(t-1)}$ is a horizontal strip of size 2.

Conversely, given such a path, fill every box of $\lambda^{(t)}/\lambda^{(t-1)}$ with the label t. Along each row, boxes are added only at the right end as t increases, so rows are weakly increasing. If a box in row $r > 1$ and column c is added at time t, then the box directly above it belongs to $\lambda^{(t)}$ by the partition condition; the horizontal-strip condition says that this upper box was not added at the same time, so it was added earlier and has smaller label. Thus columns are strictly increasing. The two constructions are inverse to each other. ☐

Lemma 4.98 (Witnesses as odd-frontier crossings)**.** *Let T be a semistandard tableau of content (2^j), and let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)} = \lambda(T)$$

be the Pieri shape path in which $\lambda^{(t)}$ is the shape of the subtableau of entries at most t. Put

$$\omega_q(t) = \min\{q, \lceil \lambda_1^{(t)}/2 \rceil\}, \qquad \eta_q(t) = \min\{q, \lceil \ell(\lambda^{(t)})/2 \rceil\}.$$

If $\lambda_1(T) \geq 2q$, then

$$W_q(T) = \{t : \omega_q(t) - \omega_q(t-1) = 1\}.$$

If $\ell(\lambda(T)) \geq 2q$, then

$$H_q(T) = \{t : \eta_q(t) - \eta_q(t-1) = 1\}.$$

Proof. The boxes labelled t form the skew shape $\lambda^{(t)}/\lambda^{(t-1)}$. Since columns in T are strictly increasing and the label t occurs exactly twice, this skew shape is a horizontal two-strip of size 2. Hence the first-row length can increase by at most 2, and the height can increase by at most 1.

For the width statement, the box $(1, c)$ has label t exactly when

$$\lambda_1^{(t-1)} < c \leq \lambda_1^{(t)}.$$

Taking $c = 2r - 1$ with $1 \leq r \leq q$, this is equivalent, because the odd frontier levels are spaced by 2 and the first row grows by at most 2 at one step, to

$$\omega_q(t) - \omega_q(t-1) = 1 \quad \text{and} \quad \omega_q(t) = r.$$

Thus the labels at $(1, 1), (1, 3), \ldots, (1, 2q - 1)$ are precisely the labels whose steps increase ω_q.

For the height statement, the box $(r, 1)$ has label t exactly when

$$\ell(\lambda^{(t-1)}) < r \leq \ell(\lambda^{(t)}).$$

Taking $r = 2a - 1$ with $1 \leq a \leq q$, and using that a horizontal strip can create at most one new row, gives the analogous equivalence with $\eta_q(t) - \eta_q(t-1) = 1$. Hence the first-column witnesses are exactly the steps increasing η_q. ☐

Lemma 4.99 (Terminal threshold after all frontier crossings)**.** *Let $\lambda^\bullet$ be a column-even Pieri two-strip path of length j, and fix $q \geq 1$. If $|C_q^w(\lambda^\bullet)| = q$, then $\lambda_1^{(j)} \geq 2q - 1$, and the width bad-shape condition is exactly the exclusion $\lambda_1^{(j)} \neq 2q - 1$. If $|C_q^h(\lambda^\bullet)| = q$, then automatically $\ell(\lambda^{(j)}) \geq 2q$.*

Proof. For the width statistic, the quantity

$$\min(q, \lceil \lambda_1^{(t)}/2 \rceil)$$

starts at 0, is nondecreasing, and can increase by at most 1 at each Pieri step, because a horizontal two-strip adds at most two boxes to the first row. Therefore $|C_q^w(\lambda^\bullet)| = q$ implies

$$\min(q, \lceil \lambda_1^{(j)}/2 \rceil) = q,$$

hence $\lambda_1^{(j)} \geq 2q - 1$. Under this conclusion, the inequality $\lambda_1^{(j)} \geq 2q$ is equivalent to $\lambda_1^{(j)} \neq 2q - 1$.

The height statistic is analogous. A horizontal strip creates at most one new row, so

$$\min(q, \lceil \ell(\lambda^{(t)})/2 \rceil)$$

also starts at 0, is nondecreasing, and increases by at most 1 at each step. Thus $|C_q^h(\lambda^\bullet)| = q$ implies $\ell(\lambda^{(j)}) \geq 2q - 1$. Since $\lambda^\bullet$ is column-even, the first column length $\ell(\lambda^{(j)})$ is even whenever it is nonzero. Here it is nonzero because $q \geq 1$, so $\ell(\lambda^{(j)}) \geq 2q$. ☐

Lemma 4.100 (Local frontier states at crossing steps)**.** *Let $\lambda^\bullet$ be a Pieri two-strip path, and list $C_q^w(\lambda^\bullet) = \{s_1 < \cdots < s_m\}$. For each $1 \le r \le m$,*

$$\lambda_1^{(s_r)} \in \{2r-1, 2r\},$$

and

$$\lambda_1^{(s_r-1)} = \begin{cases} 0, & r = 1, \\ 2r-3 \text{ or } 2r-2, & r > 1. \end{cases}$$

Similarly, if $C_q^h(\lambda^\bullet) = \{u_1 < \cdots < u_m\}$, then for each $1 \le r \le m$,

$$\ell(\lambda^{(u_r-1)}) = 2r-2, \qquad \ell(\lambda^{(u_r)}) = 2r-1.$$

Proof. For the width statistic, put

$$\omega_q(t) = \min(q, \lceil \lambda_1^{(t)}/2 \rceil).$$

The sequence $\omega_q(t)$ starts at 0, is nondecreasing, and increases by at most 1 at each step, since a horizontal two-strip adds at most two boxes to the first row. Therefore, at the r-th time s_r where it increases,

$$\omega_q(s_r - 1) = r - 1, \qquad \omega_q(s_r) = r.$$

For $r = 1$, the first equality gives $\lambda_1^{(s_1-1)} = 0$. For $r > 1$ it gives $\lambda_1^{(s_r-1)} \in \{2r-3, 2r-2\}$. The second equality gives $\lambda_1^{(s_r)} \ge 2r-1$, while the horizontal two-strip bound gives $\lambda_1^{(s_r)} \le \lambda_1^{(s_r-1)} + 2 \le 2r$, proving the width claims.

For the height statistic, put

$$\eta_q(t) = \min(q, \lceil \ell(\lambda^{(t)})/2 \rceil).$$

At the r-th time u_r where it increases, $\eta_q(u_r - 1) = r - 1$ and $\eta_q(u_r) = r$. A horizontal strip creates at most one new row. If $\ell(\lambda^{(u_r-1)}) \le 2r-3$, then $\ell(\lambda^{(u_r)}) \le 2r-2$, so η_q would not increase to r. Thus $\ell(\lambda^{(u_r-1)}) = 2r-2$, and the one-row bound forces $\ell(\lambda^{(u_r)}) = 2r-1$. □

Lemma 4.101 (Local crossing strips)**.** *Let $\lambda^\bullet$ be a Pieri two-strip path. If s_r is the r-th element of $C_q^w(\lambda^\bullet)$, then the horizontal strip*

$$\lambda^{(s_r)}/\lambda^{(s_r-1)}$$

contains the frontier box $(1, 2r-1)$, and exactly one of the following three alternatives holds:

(i) $\lambda_1^{(s_r-1)} = 2r-3, \quad \lambda_1^{(s_r)} = 2r-1,$
and the two new first-row boxes are $(1, 2r-2), (1, 2r-1)$;

(ii) $\lambda_1^{(s_r-1)} = 2r-2, \quad \lambda_1^{(s_r)} = 2r-1,$
and the second new box is not in the first row and not in column $2r-1$;

(iii) $\lambda_1^{(s_r-1)} = 2r-2, \quad \lambda_1^{(s_r)} = 2r,$
and the two new first-row boxes are $(1, 2r-1), (1, 2r)$.

For $r = 1$, only alternative (iii) *can occur.*

If u_r is the r-th element of $C_q^h(\lambda^\bullet)$, then $\lambda^{(u_r)}/\lambda^{(u_r-1)}$ contains the frontier box $(2r-1, 1)$, and its other box is not in column 1.

Proof. By lemma 4.100, a width crossing has

$$\lambda_1^{(s_r)} \in \{2r-1, 2r\}.$$

Since $\lambda_1^{(s_r-1)} < 2r-1 \le \lambda_1^{(s_r)}$, the strip contains the box $(1, 2r-1)$. The same lemma gives $\lambda_1^{(s_r-1)} = 0$ for $r = 1$ and $\lambda_1^{(s_r-1)} \in \{2r-3, 2r-2\}$ for $r > 1$. Combining these possibilities with the fact that the strip has size 2 gives exactly the three listed alternatives. In alternative (ii), the second new box cannot lie in the first row because the first-row length increases by only one, and it cannot lie in column $2r-1$ because the strip is horizontal. For $r=1$, the path starts with first-row length $0=2r-2$, and a horizontal strip of size 2 from the empty partition gives alternative (iii).

For height, lemma 4.100 gives

$$\ell(\lambda^{(u_r-1)}) = 2r-2, \qquad \ell(\lambda^{(u_r)}) = 2r-1.$$

Thus the strip contains the new first-column box $(2r-1,1)$. Since the strip is horizontal, its other box cannot also lie in column 1. $\square$

Lemma 4.102 (Odd alternating height crossings create every row). *Let $q \geq 1$ and let*

$$\lambda^\bullet : \quad \varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q)}$$

be a column-even Pieri two-strip path of length $2q$. If

$$C_q^h(\lambda^\bullet) = \{1,3,\ldots,2q-1\},$$

then

$$\ell(\lambda^{(t)}) = t \qquad (0 \leq t \leq 2q).$$

Consequently every two-strip step creates exactly one new row; at time t the strip $\lambda^{(t)}/\lambda^{(t-1)}$ contains the new first-column box $(t,1)$, and its other box is not in column 1.

If instead

$$C_q^h(\lambda^\bullet) = \{2,4,\ldots,2q\},$$

then no such path exists.

Proof. Assume first that $C_q^h(\lambda^\bullet) = \{1,3,\ldots,2q-1\}$. For $1 \leq r \leq q$, lemma 4.100 applied to the r-th height crossing $u_r = 2r-1$ gives

$$\ell(\lambda^{(2r-2)}) = 2r-2, \qquad \ell(\lambda^{(2r-1)}) = 2r-1.$$

This proves the asserted length formula at all odd times and at all even times up to $2q-2$. The terminal even time follows from lemma 4.99, which gives $\ell(\lambda^{(2q)}) \geq 2q$, while a horizontal two-strip can create at most one new row from $\lambda^{(2q-1)}$, whose length is $2q-1$. Hence $\ell(\lambda^{(2q)}) = 2q$.

Thus every step raises the length by exactly one. The new row at time t is row t, so the strip contains the box $(t,1)$. Since the strip is horizontal, its other box cannot lie in the same column.

For the even alternating set, the first Pieri step starts from the empty partition and adds a horizontal strip of size 2. Such a strip must create the first row, so

$$\ell(\lambda^{(0)}) = 0, \qquad \ell(\lambda^{(1)}) = 1.$$

Therefore $\min(q, \lceil \ell(\lambda^{(1)})/2 \rceil) - \min(q, \lceil \ell(\lambda^{(0)})/2 \rceil) = 1$, so $1 \in C_q^h(\lambda^\bullet)$. This contradicts $C_q^h(\lambda^\bullet) = \{2,4,\ldots,2q\}$. $\square$

Lemma 4.103 (Off-column boxes encode odd alternating height paths). *Let $\lambda^\bullet$ be as in lemma 4.102 with*

$$C_q^h(\lambda^\bullet) = \{1,3,\ldots,2q-1\}.$$

For each $1 \leq t \leq 2q$, let

$$b_t = (r_t, c_t)$$

be the unique box of $\lambda^{(t)}/\lambda^{(t-1)}$ different from $(t,1)$. Then $c_t \geq 2$, and the sequence

$$(b_1, \ldots, b_{2q})$$

uniquely determines the whole Pieri path $\lambda^\bullet$.

Moreover, the terminal shape $\lambda^{(2q)}$ is column-even if and only if, for every $c \geq 2$, the number of indices t with $c_t = c$ is even. In that case the off-column boxes have a canonical same-column pairing, obtained by listing the times with fixed column c increasingly and pairing consecutive times.

Proof. By lemma 4.102, the t-th two-strip contains $(t,1)$ and exactly one further box outside column 1; hence $c_t \geq 2$. Starting from $\lambda^{(0)} = \varnothing$, the recursion

$$\lambda^{(t)} = \lambda^{(t-1)} \cup \{(t,1), b_t\} \qquad (1 \leq t \leq 2q)$$

recovers every intermediate partition from the displayed sequence of boxes. Thus the sequence determines the path.

At the terminal time, column 1 has exactly the boxes $(1,1), \ldots, (2q,1)$, so its length is $2q$. Every terminal box outside column 1 is one of the off-column boxes b_t, because each step adds only the first-column box and b_t, and boxes are never removed. Therefore, for each $c \geq 2$, the length of column c in $\lambda^{(2q)}$ is precisely

$$\#\{t : c_t = c\}.$$

The terminal shape is column-even exactly when all these numbers are even. When they are even, increasing order of the finitely many times in each fixed column gives a canonical pairing of those times, and hence of the corresponding off-column boxes. □

Lemma 4.104 (Row choices encode the off-column boxes)**.** *In the setting of lemma 4.103, the row indices*

$$r_1, \ldots, r_{2q}$$

determine the off-column boxes $b_t = (r_t, c_t)$ and hence determine the whole Pieri path. More explicitly, for each t one has $1 \leq r_t \leq t$, and

$$c_t = \begin{cases} \lambda^{(t-1)}_{r_t} + 1, & r_t < t, \\ 2, & r_t = t. \end{cases}$$

Thus, recursively from $\lambda^{(0)} = \varnothing$, the row-choice sequence determines

$$\lambda^{(t)} = \lambda^{(t-1)} \cup \{(t,1), (r_t, c_t)\}.$$

Proof. By lemma 4.102, $\ell(\lambda^{(t)}) = t$ and $\ell(\lambda^{(t-1)}) = t-1$. Hence the off-column box added at time t lies in one of the rows $1, \ldots, t$, proving $1 \leq r_t \leq t$.

If $r_t < t$, then row r_t already exists in $\lambda^{(t-1)}$. Since $\lambda^{(t)}/\lambda^{(t-1)}$ is obtained by adding boxes to a Young diagram, the new box in row r_t must be the first box after the previous row end, namely at column $\lambda^{(t-1)}_{r_t} + 1$.

If $r_t = t$, then the row did not exist in $\lambda^{(t-1)}$. The step already adds $(t,1)$, and the off-column box is in the same new row. A Young diagram cannot contain (t,c) with $c > 2$ unless it also contains $(t,2)$, but only two boxes are added at this step. Since the off-column box is not $(t,1)$, it must be $(t,2)$.

Thus c_t is determined by the previous partition and r_t in the displayed way. Starting from $\lambda^{(0)} = \varnothing$, this recursion determines all b_t and all $\lambda^{(t)}$. The final claim also follows from lemma 4.103. □

Lemma 4.105 (Odd alternating height paths are even-column standard tableaux)**.** *Let $q \geq 1$. The map that sends a column-even Pieri two-strip path*

$$\lambda^{\bullet} : \quad \varnothing = \lambda^{(0)} \subset \cdots \subset \lambda^{(2q)}$$

with

$$C^h_q(\lambda^{\bullet}) = \{1, 3, \ldots, 2q-1\}$$

to its shifted off-column recording tableau is a bijection onto standard Young tableaux of size $2q$ whose columns all have even length.

Explicitly, if $b_t = (r_t, c_t)$ is the off-column box of the t-th strip, put the entry t in the shifted box $(r_t, c_t - 1)$. The resulting tableau is standard and has only even columns. Conversely, given a *standard Young tableau U of size $2q$ with even columns, let $\nu^{(t)}$ be the shape of the boxes of U with entries at most t, and define*

$$\lambda^{(t)} = \{(i,1) : 1 \le i \le t\} \cup \{(i,c+1) : (i,c) \in \nu^{(t)}\}.$$

Then $\lambda^\bullet$ is a column-even Pieri two-strip path with the displayed odd alternating height-crossing set, and the two constructions are inverse.

Proof. Start with an odd alternating height path. By lemma 4.102, each step adds the first-column box $(t,1)$ and one off-column box $b_t = (r_t, c_t)$. Shift all off-column boxes one column left. At time t, the shifted shape is obtained from the previous shifted shape by adding the single box $(r_t, c_t - 1)$, because the unshifted off-column boxes are exactly the boxes outside column 1 of $\lambda^{(t)}$. These shifted shapes are therefore a saturated Young-lattice chain from $\varnothing$ to a partition of size $2q$. Filling the box added at time t with t gives a standard Young tableau. The terminal shifted shape has column lengths equal to the terminal column lengths of $\lambda^{(2q)}$ in columns $2, 3, \ldots$, so its columns are even by column-evenness of $\lambda^{(2q)}$.

Conversely, let U be a standard Young tableau of size $2q$ with even-column shape, and let $\nu^{(t)}$ be its standard growth chain. The displayed formula for $\lambda^{(t)}$ adds a first column of height t and shifts $\nu^{(t)}$ one column to the right. Since $\nu^{(t)}$ is a partition, the result is a partition. From $t-1$ to t, the first-column box $(t,1)$ is added, and the unique box of $\nu^{(t)}/\nu^{(t-1)}$ is added one column to the right. Thus $\lambda^{(t)}/\lambda^{(t-1)}$ is a horizontal strip of size 2: the two new boxes lie in columns 1 and at least 2.

The terminal first column has length $2q$, and every other terminal column of $\lambda^{(2q)}$ is a column of the even-column shape of U. Hence $\lambda^\bullet$ is column-even. Also $\ell(\lambda^{(t)}) = t$ for every t, so

$$\min(q, \lceil t/2 \rceil) - \min(q, \lceil (t-1)/2 \rceil)$$

is 1 exactly when t is odd and $1 \le t \le 2q-1$. Therefore $C_q^h(\lambda^\bullet) = \{1, 3, \ldots, 2q-1\}$.

The two constructions are inverse because one deletes the forced first column and shifts the remaining boxes left, while the other restores that first column and shifts the standard growth chain right. □

Lemma 4.106 (Schensted contraction of the odd alternating height model)**.** *Let $q \ge 1$ and let $\lambda^\bullet$ be a column-even Pieri two-strip path of length $2q$ with*

$$C_q^h(\lambda^\bullet) = \{1, 3, \ldots, 2q-1\}.$$

Let U be the even-column standard tableau associated to $\lambda^\bullet$ by lemma 4.105, and let π be the fixed-point-free involution on $\{1, \ldots, 2q\}$ whose Schensted insertion tableau is U. Contract the adjacent pairs

$$\{1,2\}, \{3,4\}, \ldots, \{2q-1, 2q\}$$

to vertices $1, \ldots, q$. Each transposition of π then gives an edge between the corresponding vertices, with a loop exactly when the transposition is one of the adjacent pairs $\{2v-1, 2v\}$.

The resulting contracted object is a labelled degree-two multigraph $\Gamma^\circ(\lambda^\bullet)$ on $\{1, \ldots, q\}$, with parallel edges and loops allowed. Moreover there is a word

$$\epsilon(\lambda^\bullet) \in \{0,1\}^{2q}$$

such that the pair $(\Gamma^\circ(\lambda^\bullet), \epsilon(\lambda^\bullet))$ recovers $\lambda^\bullet$.

If $\Gamma^\circ(\lambda^\bullet)$ has no loops, then it is a loopless labelled 2-regular multigraph counted by the stable moment s_q in lemma 4.65. Thus the loopless-contraction subfamily of the odd alternating height fiber satisfies the required $2^{2q} s_q$ bound. The only obstruction to this direct stable-graph target is the presence of adjacent transpositions $\{2v-1, 2v\}$ in π.

Proof. For a standard tableau U, the inverse Robinson–Schensted correspondence with equal insertion and recording tableaux gives a unique involution π with insertion tableau U. Rains [5, Lemma 3.3] records the classical Schensted fact that the number of fixed points of this involution is the number of odd columns of U. Since U has only even columns, π has no fixed points; hence it is a fixed-point-free involution, i.e. a perfect matching of $\{1,\dots,2q\}$.

After contracting the adjacent pairs to vertices $1,\dots,q$, every vertex has two incident ports, namely $2v-1$ and $2v$, and each port is matched exactly once by π. Therefore the contracted multigraph has degree two at every vertex, counting a loop as degree two. A transposition contributes a loop precisely when both of its endpoints lie in the same adjacent pair, which is exactly the displayed adjacent-pair case.

It remains to record why $2q$ bits are enough to recover the original matching from the contracted multigraph. For each vertex v, order its two incident edge-slots canonically from the contracted multigraph: first by the other endpoint label, then by the multiplicity number of the edge, and for a loop by its two loop-slots. Let the two bits attached to the ports $2v-1$ and $2v$ record which of these two ordered slots receives each port. The word $\epsilon(\lambda^\bullet)$ is the collection of these $2q$ slot choices. The contracted multigraph together with this slot assignment reconstructs every matched pair of ports, hence reconstructs π.

The Schensted map recovers U from π, and lemma 4.105 then recovers $\lambda^\bullet$ from U. Hence the contracted graph and the word $\epsilon(\lambda^\bullet)$ recover the original path.

If there are no loops, then $\Gamma^\circ(\lambda^\bullet)$ is loopless and degree two at every vertex, with parallel edges allowed. By lemma 4.65, the number of possible loopless contracted graphs is s_q, and the number of possible bit words is at most 2^{2q}. Hence the loopless-contraction subfamily has size at most $2^{2q}s_q$. The preceding loop criterion gives the final sentence. □

Lemma 4.107 (Loop repair for contracted degree-two graphs)**.** *Let $q \geq 2$, and let Γ° be a labelled degree-two multigraph on $\{1,\dots,q\}$, with parallel edges and loops allowed, where each loop contributes degree 2. Let*

$$L = \{v : \Gamma^\circ \text{ has a loop at } v\}.$$

Then there is a canonically chosen loopless labelled degree-two multigraph

$$\mathcal{R}(\Gamma^\circ)$$

on $\{1,\dots,q\}$, determined by Γ° and the usual order on $\{1,\dots,q\}$, such that Γ° is recoverable from $\mathcal{R}(\Gamma^\circ)$ and L.

Proof. Because a loop contributes degree 2, a vertex in L has exactly one loop and no other incident edge. Hence the subgraph on the complement L^c is already loopless and degree two at every vertex of L^c.

If $L = \varnothing$, set $\mathcal{R}(\Gamma^\circ) = \Gamma^\circ$. Suppose next that $|L|$ is even. Delete the loops on L, list the vertices of L in increasing order, pair them consecutively, and put two parallel edges on each pair. This gives a loopless degree-two graph on L, disjoint from the unchanged graph on L^c.

If $|L|$ is odd and at least 3, delete the loops on L, put a triangle on the first three vertices of L, and put two parallel edges on each consecutive pair of the remaining vertices of L. Again this gives a loopless degree-two graph on L, disjoint from the unchanged graph on L^c.

It remains to handle $|L| = 1$, say $L = \{v\}$. If L^c had only one vertex, then that vertex would have degree two in the loopless graph on L^c, which is impossible. Thus L^c has at least two vertices. Its loopless degree-two graph therefore has at least one non-loop edge. Choose the lexicographically least edge ab in L^c, counting parallel edges by their multiplicity order. Delete one copy of ab, delete the loop at v, and add the two edges va and vb. The resulting graph is loopless and degree two at every vertex.

The recovery from $\mathcal{R}(\Gamma^\circ)$ and L is inverse to these constructions. If $L = \varnothing$, do nothing. If $|L|$ is even or at least 3, the repair edges are exactly the edges whose two endpoints lie in L; delete them and restore one loop at every vertex of L. If $L = \{v\}$, the two edges incident to v in $\mathcal{R}(\Gamma^\circ)$ have other endpoints a and b with $a \neq b$; delete those two edges, restore the loop at v, and add one edge ab. This recovers the original graph in every case. □

Proposition 4.108 (Odd alternating height full-zone decoder). *For every $q \geq 2$, the height fixed-witness fiber with*

$$j = 2q, \qquad S = \{1, 3, \dots, 2q-1\}$$

satisfies

$$\#\mathcal{F}^h(S) \leq 2^{2q} s_q.$$

Equivalently, the odd alternating height full-zone case of proposition 4.530 has the required decoder.

Proof. By corollary 4.114, this fixed-witness fiber is equinumerous with the column-even Pieri paths of length $2q$ with $C_q^h = \{1, 3, \dots, 2q-1\}$. For such a path $\lambda^\bullet$, apply lemma 4.105 and then lemma 4.106. This gives a contracted degree-two graph $\Gamma^\circ(\lambda^\bullet)$ with loops allowed. Apply lemma 4.107 to get the loopless graph

$$H(\lambda^\bullet) = \mathcal{R}(\Gamma^\circ(\lambda^\bullet)).$$

Fix once and for all the lexicographic order on loopless labelled degree-two multigraphs and the lexicographic order on the finite set $\mathcal{P}_q^{\text{even}}$ of two-step growth paths. Since lemmas 4.65 and 4.118 identify both sets as having cardinality s_q, these orders give a fixed bijection between loopless graphs and paths in $\mathcal{P}_q^{\text{even}}$. Let $P^*(\lambda^\bullet)$ be the path corresponding to $H(\lambda^\bullet)$.

The branch word has two parts. The first q bits are the characteristic function of the loop set

$$L(\lambda^\bullet) = \{v : \Gamma^\circ(\lambda^\bullet) \text{ has a loop at } v\}.$$

Given $P^*(\lambda^\bullet)$, the fixed inverse bijection recovers $H(\lambda^\bullet)$. After $H(\lambda^\bullet)$ and this loop set are known, lemma 4.107 recovers $\Gamma^\circ(\lambda^\bullet)$. The remaining q bits record, for each vertex v, which of the two canonically ordered incident edge-slots of $\Gamma^\circ(\lambda^\bullet)$ receives the port $2v-1$; the port $2v$ then receives the other slot. This recovers the fixed-point-free involution π, hence its Schensted tableau U, and finally recovers $\lambda^\bullet$ by lemma 4.105.

Thus the assignment

$$\lambda^\bullet \longmapsto \big(P^*(\lambda^\bullet), \text{the } 2q\text{-bit branch word}\big)$$

is injective, with a decoder depending only on the alternating set S. Since $|\mathcal{P}_q^{\text{even}}| = s_q$ and there are 2^{2q} possible branch words, this gives the claimed bound. □

Proposition 4.109 (Even alternating height full-zone fiber is empty). *For every $q \geq 1$, the height fixed-witness fiber with*

$$j = 2q, \qquad S = \{2, 4, \dots, 2q\}$$

is empty. In particular it satisfies

$$\#\mathcal{F}^h(S) \leq 2^{2q} s_q.$$

Proof. By corollary 4.114, this fixed-witness fiber is equinumerous with the column-even Pieri paths of length $2q$ with $C_q^h = \{2, 4, \dots, 2q\}$. The second half of lemma 4.102 proves that no such path exists. The cardinality is therefore zero, giving the displayed bound. □

Corollary 4.110 (Height alternating full-zone decoder). *For every $q \geq 2$ and*

$$S \in \big\{\{1, 3, \dots, 2q-1\}, \{2, 4, \dots, 2q\}\big\},$$

the height fixed-witness fiber $\mathcal{F}^h(S)$ in the alternating full-zone case of proposition 4.530 satisfies

$$\#\mathcal{F}^h(S) \leq 2^{2q} s_q.$$

Proof. For $S = \{1, 3, \dots, 2q-1\}$ this is proposition 4.108. For $S = \{2, 4, \dots, 2q\}$ it is proposition 4.109. □

Lemma 4.111 (Odd alternating width full-zone path is forced)**.** *Let $q \geq 1$, and let $\lambda^\bullet$ be a column-even Pieri two-strip path of length $2q$ such that*

$$C_q^w(\lambda^\bullet) = \{1, 3, \ldots, 2q-1\}, \qquad \lambda_1^{(2q)} \geq 2q.$$

Then, for $0 \leq r \leq q$,

$$\lambda^{(2r)} = (2r, 2r),$$

with the convention $\lambda^{(0)} = \varnothing$, and for $1 \leq r \leq q$,

$$\lambda^{(2r-1)} = (2r, 2r-2).$$

In particular there is exactly one such path.

Proof. The terminal partition has size $4q$. Since it is column-even, every column has even positive height, hence height at least 2. The hypothesis $\lambda_1^{(2q)} \geq 2q$ therefore gives

$$4q = |\lambda^{(2q)}| \geq 2\lambda_1^{(2q)} \geq 4q.$$

Thus $\lambda_1^{(2q)} = 2q$, every column has height exactly 2, and

$$\lambda^{(2q)} = (2q, 2q).$$

All intermediate partitions are contained in this rectangle, so they have at most two rows.

Fix $1 \leq r \leq q$. The time $2r-1$ is the r-th width crossing. By lemma 4.100,

$$\lambda_1^{(2r-1)} \in \{2r-1, 2r\}.$$

If $\lambda_1^{(2r-1)} = 2r-1$, then the size $|\lambda^{(2r-1)}| = 4r-2$ and the two-row bound force

$$\lambda^{(2r-1)} = (2r-1, 2r-1).$$

For $r < q$, the next time $2r$ is not a width crossing, so $\lambda_1^{(2r)} \leq 2r$. From $(2r-1, 2r-1)$, a horizontal two-strip inside the terminal two-row rectangle and with first-row length at most $2r$ cannot be added: the only possible first-row addition is $(1, 2r)$, while the matching second-row addition $(2, 2r)$ lies in the same column. For $r = q$, the same obstruction prevents the final step from reaching $(2q, 2q)$. Hence this case is impossible, and

$$\lambda_1^{(2r-1)} = 2r.$$

The size $4r-2$ and the two-row bound then force

$$\lambda^{(2r-1)} = (2r, 2r-2).$$

For $r < q$, the time $2r$ is not a width crossing, so $\lambda_1^{(2r)} \leq 2r$. Since $|\lambda^{(2r)}| = 4r$ and $\lambda^{(2r)}$ has at most two rows, this forces $\lambda^{(2r)} = (2r, 2r)$. The case $r = q$ is the terminal rectangle already proved. This gives the displayed path at every time, and uniqueness follows. ☐

Proposition 4.112 (Width alternating full-zone decoder)**.** *For every $q \geq 2$ and*

$$S \in \big\{\{1, 3, \ldots, 2q-1\}, \{2, 4, \ldots, 2q\}\big\},$$

the width fixed-witness fiber $\mathcal{F}^w(S)$ in the alternating full-zone case of proposition 4.530 satisfies

$$\#\mathcal{F}^w(S) \leq 2^{2q} s_q.$$

Proof. If $S = \{2, 4, \ldots, 2q\}$, the fiber is empty. Indeed, the first Pieri step from $\varnothing$ is necessarily the horizontal strip giving the partition (2), so $1 \in C_q^w(\lambda^\bullet)$ for every path. By corollary 4.114, no width fixed-witness path has the even alternating crossing set.

It remains to consider $S = \{1, 3, \ldots, 2q-1\}$. By corollary 4.114, this fiber is equinumerous with column-even Pieri paths of length $2q$ with this crossing set and terminal first-row length at least $2q$. By lemma 4.111, there is exactly one such path.

For $q \geq 2$ the stable graph model is nonempty: for $q = 2$ take the graph with two parallel edges between the two vertices, and for $q \geq 3$ take the labelled cycle on $\{1, \ldots, q\}$. Hence $s_q \geq 1$ by lemma 4.65. Therefore the singleton odd fiber, and the empty even fiber, both have cardinality at most $2^{2q} s_q$. ☐

Corollary 4.113 (Alternating full-zone decoder is closed)**.** *For every $q \geq 2$, for both alternating sets*

$$S \in \big\{\{1, 3, \ldots, 2q-1\}, \{2, 4, \ldots, 2q\}\big\},$$

and for either fixed-witness fiber $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$ in proposition 4.530, the fixed fiber has cardinality at most $2^{2q} s_q$.

Proof. The height fibers are handled by corollary 4.110; the width fibers are handled by proposition 4.112. ☐

Corollary 4.114 (Fixed fibers as Pieri crossing paths)**.** *Fix $q \geq 1$, $j \geq q$, and $S \subset \{1, \ldots, j\}$ with $|S| = q$. The width fiber*

$$\{T : \lambda(T)' \text{ even},\ \lambda_1(T) \geq 2q,\ W_q(T) = S\}$$

is equinumerous with the column-even Pieri two-strip paths $\lambda^\bullet$ of length j such that $C_q^w(\lambda^\bullet) = S$ and $\lambda_1^{(j)} \geq 2q$. The height fiber

$$\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\}$$

is equinumerous with the column-even Pieri two-strip paths $\lambda^\bullet$ of length j such that $C_q^h(\lambda^\bullet) = S$.

Proof. Use the bijection of lemma 4.97. The terminal column-even condition is exactly $\lambda(T)'$ even. By lemma 4.98, the width witness is the crossing set $C_q^w(\lambda^\bullet)$ whenever $\lambda_1(T) \geq 2q$, and the height witness is $C_q^h(\lambda^\bullet)$ whenever $\ell(\lambda(T)) \geq 2q$. In the height case the latter terminal condition is automatic from lemma 4.99, because $|S| = q$. ☐

Lemma 4.115 (Bounded column defect after label deletion)**.** *Let T be a semistandard tableau of shape λ and content (2^j), with λ' even. For a subset $S \subset \{1, \ldots, j\}$, let $T \setminus S$ be the labelled retained subdiagram obtained by deleting from T the two boxes carrying each label in S, while keeping the original row and column coordinates of all remaining boxes. Then $T \setminus S$ has content $(2^{j-|S|})$ on the complementary labels, its remaining entries are weakly increasing along each retained row and strictly increasing along each retained column, and at most $2|S|$ columns contain an odd number of remaining boxes.*

Proof. The row weak-increase and column strict-increase conditions are inherited by subsets of boxes in each row and column. The content statement is immediate from the definition of content (2^j). For a column c, let d_c be the number of deleted boxes in that column. Since λ' is even, column c of λ contains an even number of boxes. Hence the parity of the number of boxes remaining in column c of $T \setminus S$ is the parity of d_c. Therefore a column of $T \setminus S$ can be odd only if it contains at least one deleted box. The total number of deleted boxes is $2|S|$, so at most $2|S|$ columns can be odd. ☐

Lemma 4.116 (Label deletion need not give a skew shape)**.** *The retained subdiagram in lemma 4.115 is not, in general, a skew Young diagram. This remains true for column-even semistandard tableaux of content (2^j).*

Proof. Take the semistandard tableau of shape $(4, 4)$

$$\begin{matrix} 1 & 1 & 2 & 2 \\ 3 & 3 & 4 & 4 \end{matrix}$$

and delete the two boxes labelled 3. The original shape has four columns, each of height 2, hence λ' is even, and the content is (2^4). After deletion, the second row has boxes only in columns 3 and 4, while the first row has boxes in columns $1, 2, 3, 4$. If this retained set were α/β for partitions $\beta \subset \alpha$, then the number of boxes deleted from the left of the first two rows would be $\beta_1 = 0$ and $\beta_2 = 2$, contradicting $\beta_1 \geq \beta_2$. Thus the retained set is not a skew Young diagram. ☐

Lemma 4.117 (Row packing does not repair witness deletion)**.** *For every $q \geq 1$ there is a column-even semistandard tableau T of content (2^{2q+1}) with $\lambda_1(T) \geq 2q$ such that, for its width witness $S = W_q(T)$, deleting all labels in S and then left-packing each retained row does not produce a Young diagram.*

Proof. Let T have shape $(2q + 1, 2q + 1)$, with first row

$$1, 1, 2, 2, \ldots, q, q, q + 1$$

and second row

$$q + 1, q + 2, q + 2, \ldots, 2q + 1, 2q + 1.$$

Rows are weakly increasing, columns are strictly increasing, and each label $1, \ldots, 2q + 1$ occurs exactly twice. The shape has two boxes in every column, so it is column-even. The width witness is $W_q(T) = \{1, \ldots, q\}$, read from the first-row odd positions $1, 3, \ldots, 2q - 1$. Deleting the two copies of every label in $W_q(T)$ leaves one box in the first row, labelled $q + 1$, and all $2q + 1$ boxes in the second row. Left-packing rows therefore gives row lengths $(1, 2q + 1)$, which are not weakly decreasing. Hence the packed rows do not form a Young diagram. ☐

Lemma 4.118 (Column-even tableaux as two-step growth paths)**.** *Let T be a semistandard tableau of content (2^j) whose shape has only even column lengths, and let k be any integer such that every column of T has length at most $2k$. Krattenthaler's semistandard growth-diagram bijection* [8, Theorem 4] *encodes T by a sequence of partitions*

$$\varnothing = \rho_0, \rho_1, \ldots, \rho_{2j} = \varnothing$$

such that every ρ_t has at most k columns, each adjacent pair differs by a vertical strip, and for every $i = 1, \ldots, j$ the two-step block

$$\rho_{2i-2} \supseteq \rho_{2i-1} \subseteq \rho_{2i}$$

corresponding, under Krattenthaler's index convention, to tableau label $j - i + 1$ satisfies

$$|\rho_{2i-2}| - 2|\rho_{2i-1}| + |\rho_{2i}| = 2.$$

Conversely, such sequences are counted by the same tableaux.

Proof. Krattenthaler's Theorem 4 gives a bijective equality between semistandard tableaux with prescribed content $j_1, \ldots, j_n$, m odd columns, and all columns of length at most $2k$, and sequences

$$\varnothing = \rho_0, \rho_1, \ldots, \rho_{2n} = (1^m)$$

with at most k columns, vertical-strip adjacent differences, and

$$|\rho_{2i-2}| - 2|\rho_{2i-1}| + |\rho_{2i}| = j_{n-i+1}.$$

Apply this with $n = j$, $m = 0$, and $j_1 = \cdots = j_j = 2$. Since all prescribed contents are equal to 2, the reversal of the index in the displayed source formula does not change the two-step condition; it only identifies source block i with tableau label $j - i + 1$, or equivalently label ℓ with block $j - \ell + 1$. ☐

Lemma 4.119 (Knuth growth-diagram cells are locally reversible)**.** *In the first arbitrary-filling variant of Krattenthaler's growth diagrams,* [9, §4.1, Theorem 7]*, a cell filled by a nonnegative integer m with corner labels*

$$\begin{matrix} \nu & \lambda \\ \rho & \mu \end{matrix}$$

is governed by mutually inverse forward and backward algorithms. The forward algorithm determines λ from (m, ρ, μ, ν) when $\rho \subseteq \mu$ and $\rho \subseteq \nu$ differ by horizontal strips. The backward algorithm *determines (m, ρ) from (λ, μ, ν) when $\mu \subseteq \lambda$ and $\nu \subseteq \lambda$ differ by horizontal strips. Consequently, on any finite Ferrers cell arrangement in this variant, the left and bottom boundary labels together with the cell entries determine the top and right boundary labels, and the top and right boundary labels determine the cell entries and the remaining corner labels.*

Proof. Krattenthaler [9, §4.1] constructs arbitrary-filling growth diagrams by separating each cell entry into a 0–1 blow-up, applying the ordinary Fomin local rules, and shrinking back to the original diagram. Immediately after Figure 9, the source records the equivalent direct cell algorithms: the forward algorithm computes the northeast corner λ from the cell entry and the other three corners, while the backward algorithm computes the southwest corner ρ and the cell entry from the other three corners. The theorem following those algorithms states the corresponding boundary bijection for Ferrers shapes. Iterating the forward cell rule across the cells gives the first global determination statement, and iterating the backward cell rule from the top-right boundary gives the second. □

Corollary 4.120 (Parsed boundaries recover Knuth subfillings)**.** *Let F be a finite Ferrers cell arrangement equipped with the first arbitrary-filling growth diagram of lemma 4.119, and let $E \subseteq F$ be a Ferrers subarrangement. Suppose that a parser has recovered the partition labels on the top/right boundary of E. Then the shape of E and those boundary labels determine, by a deterministic backward sweep, every cell entry of E and every corner label on or inside E.*

Consequently, in the right-boundary auxiliary setting of proposition 4.253, any auxiliary datum which is a fixed function of such a recovered subfilling is already a fixed function of the parsed outside growth-boundary data and requires no additional branch word.

Proof. The subarrangement E is itself a finite Ferrers cell arrangement in the orientation used by Krattenthaler's local rules. By lemma 4.119, the backward cell algorithm determines the southwest corner and the cell entry of a cell from its northeast, southeast, and northwest corner labels. Starting from the known top/right boundary of E and sweeping cells leftward and downward therefore recovers every cell entry and every remaining corner label in E.

If the right-boundary auxiliary datum is defined from one of these recovered subfillings, then the preceding deterministic recovery expresses that datum as a fixed function of the already parsed outside boundary labels. Thus it is an auxiliary input of the type allowed in proposition 4.253, not an extra encoded choice. □

Lemma 4.121 (Dual RSK' growth-diagram cells are locally reversible)**.** *In the fourth arbitrary-filling variant of Krattenthaler's growth diagrams,* [9, §4.4, Theorem 11]*, a cell filled by a nonnegative integer m with corner labels*

$$\begin{matrix} \nu & \lambda \\ \rho & \mu \end{matrix}$$

is governed by mutually inverse forward and backward algorithms with vertical strip adjacent differences. The forward algorithm determines λ from (m, ρ, μ, ν) when $\rho \subseteq \mu$ and $\rho \subseteq \nu$ differ by vertical strips. The backward algorithm determines (m, ρ) from (λ, μ, ν) when $\mu \subseteq \lambda$ and $\nu \subseteq \lambda$ differ by vertical strips.

Consequently, on any finite Ferrers cell arrangement in this variant, the left and bottom boundary labels together with the cell entries determine the top and right boundary labels, and the top and right boundary labels determine the cell entries and the remaining corner labels.

Proof. Krattenthaler [9, §4.4] recalls the fourth arbitrary-filling growth-diagram variant, denoted dual RSK'. Immediately before Theorem 11 the source gives the direct cell algorithms (F4) and (B4): (F4) computes the northeast corner λ from the cell entry and the other three corners, while (B4) computes the southwest corner ρ and the cell entry from the other three corners. The hypotheses in those algorithms are exactly the vertical-strip containment conditions displayed above. Theorem 11 then states the corresponding boundary bijection for Ferrers shapes. Iterating (F4) across the cells gives the forward determination, and iterating (B4) from the top/right boundary gives the backward determination. ☐

Lemma 4.122 (Dual-RSK' cell size conservation)**.** *In the one-cell fourth arbitrary-filling growth diagram of lemma 4.121, let the cell entry be m and write the corner labels as*

$$\begin{matrix} \nu & \lambda \\ \rho & \mu. \end{matrix}$$

Then

$$|\lambda| - |\mu| - |\nu| + |\rho| = m.$$

Proof. Use Krattenthaler's forward algorithm (F4) from [9, §4.4]. Extend all partitions by zero parts. Let c_i be the carry entering row i, with $c_1 = m$. Put

$$a_i = \chi(\rho_i = \mu_i = \nu_i), \qquad d_i = \min(\mu_i, \nu_i) - \rho_i.$$

Because μ/ρ and ν/ρ are vertical strips, d_i is either 0 or 1. The forward rule sets

$$b_i = \min(a_i, c_i), \qquad \lambda_i = \max(\mu_i, \nu_i) + b_i,$$

and updates

$$c_{i+1} = c_i - b_i + d_i.$$

For all sufficiently large i, the partition parts are zero and the carry has fallen to zero, so the following finite sum is legitimate. Since

$$\max(\mu_i, \nu_i) = \mu_i + \nu_i - \min(\mu_i, \nu_i) = \mu_i + \nu_i - \rho_i - d_i,$$

we have

$$\lambda_i - \mu_i - \nu_i + \rho_i = b_i - d_i = c_i - c_{i+1}.$$

Summing over i telescopes to

$$|\lambda| - |\mu| - |\nu| + |\rho| = c_1 - \lim_i c_i = m.$$

☐

Corollary 4.123 (Diagonal-two cells have padded carrier valleys)**.** *In the setting of lemma 4.122, suppose that*

$$|\lambda| - |\rho| = 2.$$

Then the left/bottom valley satisfies

$$|\nu| - 2|\rho| + |\mu| = 2 - m.$$

Proof. By lemma 4.122,

$$m = |\lambda| - |\mu| - |\nu| + |\rho|.$$

Using $|\lambda| - |\rho| = 2$ gives

$$|\nu| - 2|\rho| + |\mu| = |\mu| + |\nu| - |\rho| - |\lambda| + 2 = 2 - m.$$

☐

Corollary 4.124 (Parsed dual-RSK' boundaries recover subfillings)**.** *Let F be a finite Ferrers cell arrangement equipped with the fourth arbitrary-filling growth diagram of lemma 4.121, and let $E \subseteq F$ be a Ferrers subarrangement. Suppose that a parser has recovered the partition labels on the top/right boundary of E. Then the shape of E and those boundary labels determine, by a deterministic backward sweep, every cell entry of E and every corner label on or inside E.*

Proof. The proof is the vertical-strip analogue of corollary 4.120. By lemma 4.121, the backward cell algorithm of the fourth variant determines the southwest corner and the cell entry from the northeast, southeast, and northwest corner labels. Sweeping the cells of E leftward and downward from the recovered top/right boundary therefore determines every entry and every remaining corner label. □

Lemma 4.125 (Krattenthaler's semistandard path is a dual-RSK' boundary)**.** *Use the concrete growth-diagram construction in the proof sketch of* [8, Theorem 4, Steps 2–3]. *For a semistandard tableau T with content $(j_1, \ldots, j_n)$ and even-column endpoint parameter $m = 0$, the partition sequence counted in lemma 4.118 is read from the top/right boundary of the Step 3 half-square growth diagram. That Step 3 diagram is the fourth arbitrary-filling, dual-RSK' variant of lemma 4.121.*

Consequently, if a later decoder identifies a Ferrers subarrangement of this Step 3 diagram whose top/right boundary lies in a recovered segment of

$$\rho_0, \rho_1, \ldots, \rho_{2j},$$

then corollary 4.124 recovers the corresponding subfilling and all its internal corner labels without additional branch choices.

Proof. In the proof sketch of [8, Theorem 4], Step 2 applies a Knuth-type inverse growth-diagram construction to obtain a symmetric matrix with nonnegative integer entries. Step 3 then applies the Knuth-type extension cited there from Krattenthaler's fourth variant, and the output sequence is obtained by reading the partitions along the top/right boundary of the resulting half-square cell arrangement. Krattenthaler's [9, §4.4, Theorem 11] is precisely that fourth arbitrary-filling, dual-RSK' boundary bijection. Applying this description in the $m = 0$ and $j_1 = \cdots = j_j = 2$ specialization used in lemma 4.118 gives the first assertion. The final assertion is then exactly corollary 4.124 applied to the identified subarrangement. □

Lemma 4.126 (Full Step-3 half-square entries determine the semistandard path)**.** *Use the Step 3 dual-RSK' half-square in the $m = 0$, content (2^j) specialization of lemma 4.118. Any datum which determines every cell entry of this full Step 3 half-square determines the whole semistandard boundary sequence*

$$\rho_0, \rho_1, \ldots, \rho_{2j}.$$

Proof. By lemma 4.125, the sequence $\rho_0, \rho_1, \ldots, \rho_{2j}$ is the top/right boundary of the Step 3 half-square growth diagram. In Krattenthaler's Step 3 construction the left and bottom boundary labels of the full half-square are the fixed empty boundary labels of the growth diagram. Hence the known cell entries together with this fixed left/bottom boundary determine the top/right boundary by the forward part of lemma 4.121. This top/right boundary is exactly the displayed ρ-sequence. □

Lemma 4.127 (Step-3 cells are Step-2 half-matrix entries)**.** *Use the construction in* [8, Theorem 4, Steps 2–3] *in the $m = 0$ specialization of lemma 4.118. Let M be the symmetric nonnegative integer matrix produced by the Step 2 first-variant inverse growth diagram. The Step 3 dual-RSK' half-square is filled by one off-diagonal half of M: each Step 3 cell entry is the corresponding entry of M, and the transposed Step 2 entry is the same integer by symmetry. Consequently, a recovered Step 3 subfilling determines the entries of every off-diagonal Step 2 cell whose cell or transposed cell lies in that recovered half-square; the Step 2 diagonal entries are known to be zero.*

Proof. In [8, Theorem 4], Step 2 is explicitly the Knuth-type inverse growth-diagram construction of [9, §4.1]; in the semistandard extension it outputs a symmetric matrix with nonnegative integer entries and zeros on the diagonal. Step 3 then replaces the ordinary final growth-diagram pass by the Knuth-type extension of [9, §4.4]: one half of the symmetric Step 2 matrix is used as the filling of the half-square, and the partitions are read along the top/right boundary after applying the dual-RSK' local rules. In the $m = 0$ case there is no extra-letter region to delete; the only discarded cells are the diagonal and the opposite half of the symmetric matrix. Thus each Step 3 cell entry is literally the corresponding off-diagonal entry of M, and the opposite off-diagonal entry is equal to it because M is symmetric. □

Lemma 4.128 (Step-2 matrix degrees are the doubled content)**.** *Use the construction in* [8, Theorem 4, Step 2] *and* [9, §4.1, Theorem 7] *for a semistandard tableau of content* (2^j)*. Let*

$$M = (M_{ab})_{1\le a,b\le j}$$

be the Step 2 first-variant Knuth matrix. Then M is symmetric, has zero diagonal, and satisfies

$$\sum_{b=1}^{j} M_{ab} = 2 = \sum_{b=1}^{j} M_{ba} \qquad (1 \le a \le j).$$

Proof. The cited Step 2 construction applies the arbitrary-filling Knuth correspondence of [9, §4.1, Theorem 7]. In the rectangular form of that correspondence, an entry M_{ab} contributes M_{ab} copies of the biletter with row label a and column label b; the row sums give the weight of one semistandard tableau and the column sums give the weight of the other. In Krattenthaler's Theorem 4 construction these two tableaux are the equal insertion/recording pair obtained from the jeu-de-taquin-preprocessed tableau, and the source records that the resulting matrix is symmetric with zero diagonal. Specializing to content (2^j) means that each ordinary label a occurs exactly twice. Hence both the row and column sums of the Step 2 matrix are equal to 2 for every label. □

Lemma 4.129 (HKKO even-bound up-down boundary events)**.** *In the notation of Huh–Kim–Krattenthaler–Okada, let* $\mathrm{MUD}^*_n(w;\mu\to\nu)$ *be the marked up-down class used for the even-bound Littlewood identities. Its marked set is a subset of*

$$E_w(T) = \{j : \ell(T^{2j-2}) < \ell(T^{2j-1}) = \ell(T^{2j}) = w\}.$$

After the coefficient extraction from Theorem 7.15 to Theorem 8.7 of [10]*, an index $j \in E_w(T)$ becomes an eligible boundary-up step in the corresponding marked vacillating class* $\mathrm{MVT}^*_n(w;\mu\to\nu)$*; that is, the corresponding marked vacillating index is a $+\epsilon_w$ step whose starting point has w-th coordinate* 0*.*

Proof. Definition 7.12 of [10] defines $E_w(T)$ by the displayed formula and defines $\mathrm{MUD}^*_n(w;\mu\to\nu)$ by requiring the marking set to be a subset of $E_w(T)$. Thus a marked up-down event first raises the length from below w to w at time $2j-1$ and keeps length w at time $2j$.

Theorem 8.7 is obtained from the even-bound up-down identities of Theorem 7.15 by coefficient extraction into marked vacillating tableaux. Definition 8.6 then states the admissibility condition for the resulting marks: a marked index in MVT^* must be a $+\epsilon_w$ step whose starting point lies on $x_w = 0$. Hence the marked events coming from $E_w(T)$ are exactly eligible boundary-up steps in the vacillating model, as recorded in lemma 4.134. □

Lemma 4.130 (HKKO boundary-event splice)**.** *In the even-bound proof of* [10, Theorem 7.15]*, let Z_0 be the fixed-point class appearing in the proof of equation (7.32). Thus an element $(T,S)\in Z_0$ satisfies*

$$S \neq \varnothing, \qquad j\in S \Rightarrow \ell(T^{2j-1}) = w-1, \qquad j\notin S \Rightarrow 2j-1 \text{ is not a length peak of } T.$$

For every $j \in S$, the local operation ψ of Figure 7 changes the configuration at the j-th up-down pair, removes the mark, and produces an unmarked up-down tableau T'. Applied for all $j\in S$, this operation is a bijection from Z_0 onto the class Y in the proof of (7.32). It preserves the global endpoints μ,ν and the length $2n$, and it is reversible from T' and the selected set S.

Proof. In the proof of [10, Theorem 7.15], after equation (7.36), the authors define the fixed-point set Z_0 by exactly the three displayed conditions. They then define $\psi(T,S) = (T',\varnothing)$ by applying the operation shown in Figure 7 for each $j \in S$. The paragraph following the definition states that

ψ is a bijection from Z_0 to the class Y. Both Z_0 and Y are classes of w-up-down tableaux from the same μ to the same ν and of the same length $2n$, so the operation preserves these global data. Bijection gives the asserted reversibility from the output together with the selected set. ☐

Lemma 4.131 (Explicit Z_0 landing criterion). *Let*

$$U = (U^0, U^1, \ldots, U^{2N})$$

be a w-up-down tableau from μ to ν, and let $M \subseteq [N]$. Suppose first that $M = \varnothing$. Then the identity operation gives an unmarked up-down tableau with the same endpoints and length.

Suppose instead that $M \neq \varnothing$ and the following three conditions hold:

$$\ell(U^r) < w \qquad (0 \le r \le 2N),$$

$$m \in M \quad \Longrightarrow \quad \ell(U^{2m-1}) = w - 1,$$

and

$$m \notin M \quad \Longrightarrow \quad \neg\big(\ell(U^{2m-2}) < \ell(U^{2m-1}) > \ell(U^{2m})\big).$$

Then (U, M) lies in the Z_0 domain of lemma 4.130. Consequently the HKKO boundary-event splice produces an unmarked w-up-down tableau with endpoints μ, ν and length $2N$, and the operation is reversible from the output and M.

Proof. If $M = \varnothing$, there are no selected boundary events to splice, and the identity operation has the asserted properties.

Assume $M \neq \varnothing$. The first displayed condition says precisely that (U, M) lies in the class $\mathrm{MUD}_N^{<}(w; \mu \to \nu)$ of Definition 7.12 of [10]. The second and third displayed conditions are the two fixed-point conditions defining Z_0 in the proof of [10, Theorem 7.15, equation (7.32)]: selected indices have midpoint length $w - 1$, and unselected odd midpoints are not length peaks. Hence $(U, M) \in Z_0$. Applying lemma 4.130 gives the unmarked output with the same endpoints and length and gives reversibility from that output together with M. ☐

Lemma 4.132 (Penultimate-row creations give the selected Z_0 midpoint). *Let*

$$U = (U^0, \ldots, U^{2N})$$

be a w-up-down tableau, and let $M \subseteq [N]$. Suppose that, for every $m \in M$, the partition U^{2m-2} has at most $w - 2$ nonzero parts and U^{2m-1} is obtained from U^{2m-2} by adding one box in row $w - 1$. Then

$$m \in M \quad \Longrightarrow \quad \ell(U^{2m-1}) = w - 1.$$

Consequently, if the same U also satisfies the below-wall and unselected non-peak conditions of lemma 4.131, then the selected midpoint condition in that criterion is automatic.

Proof. Fix $m \in M$. Since U^{2m-2} has at most $w-2$ nonzero parts, its $(w-1)$-st part is zero. Adding one box in row $w-1$ makes the $(w-1)$-st part of U^{2m-1} positive. As U^{2m-1} is a partition, all parts below row $w - 1$ are zero. Hence its number of nonzero parts is exactly $w - 1$. The final sentence is immediate after substituting this implication into the hypotheses of lemma 4.131. ☐

Lemma 4.133 (Z_0 landing is stable under even-time concatenation). *Let*

$$U = (U^0, \ldots, U^{2N}) \quad \text{and} \quad V = (V^0, \ldots, V^{2M})$$

be w-up-down tableaux with $U^{2N} = V^0$. Let $A \subseteq [N]$ and $B \subseteq [M]$. Suppose that U satisfies

$$\ell(U^r) < w \quad (0 \le r \le 2N), \qquad a \in A \Rightarrow \ell(U^{2a-1}) = w - 1,$$

and

$$a \notin A \Rightarrow \neg\big(\ell(U^{2a-2}) < \ell(U^{2a-1}) > \ell(U^{2a})\big),$$

and that V satisfies the analogous three conditions with B in place of A. Let W be the even-time concatenation

$$W = (U^0, \dots, U^{2N}, V^1, \dots, V^{2M})$$

and put

$$C = A \cup \{N + b : b \in B\} \subseteq [N + M].$$

Then W satisfies the same three Z_0 landing conditions with selected set C.

Proof. Every state of W is either a state of U or a state of V, with the common endpoint counted once. Hence $\ell(W^r) < w$ for all $0 \le r \le 2(N + M)$.

If $c \in C$, then either $c = a \in A$, in which case $W^{2c-1} = U^{2a-1}$, or $c = N + b$ with $b \in B$, in which case $W^{2c-1} = V^{2b-1}$. The selected midpoint condition follows from the corresponding condition for U or V.

Finally let $c \notin C$. If $c \le N$, then $c \notin A$ and the three states $W^{2c-2}, W^{2c-1}, W^{2c}$ are the corresponding three states of U. If $c > N$, write $c = N + b$ with $1 \le b \le M$; then $b \notin B$ and the same three states are the shifted corresponding states of V. In both cases the non-peak condition follows from the appropriate segment. □

Lemma 4.134 (HKKO even-bound marked boundary model)**.** *Let $\mathrm{SYT}_{n,2w}[r]$ be the set of standard Young tableaux of size n, width at most $2w$, and exactly r odd columns. In the notation of Huh–Kim–Krattenthaler–Okada, for $0 \le k \le w - 1$,*

$$|\mathrm{SYT}_{n,2w}[k]| = |\mathrm{MVT}^0_n(w; (1^k, 0^{w-k}) \to (0^w))|$$

and

$$|\mathrm{SYT}_{n,2w}[2w - k]| = |\mathrm{MVT}^1_n(w; (1^k, 0^{w-k}) \to (0^w))|.$$

Moreover

$$|\mathrm{SYT}_{n,2w}[k]| + |\mathrm{SYT}_{n,2w}[2w - k]| = |\mathrm{MVT}^*_n(w; (1^k, 0^{w-k}) \to (0^w))|,$$

and for $k = w$,

$$|\mathrm{SYT}_{n,2w}[w]| = |\mathrm{MVT}^*_n(w; (1^w) \to (0^w))|.$$

In MVT^*_n, every marked index is a nonzero vacillating step $+\epsilon_w$ whose starting point lies on the boundary hyperplane $x_w = 0$, and the underlying vacillating path has no zero steps. For $0 \le k \le w-1$, the two refinements MVT^0_n and MVT^1_n are separated by whether the first $+\epsilon_w$ step is unmarked or marked.

Proof. Definitions 8.1, 8.5, and 8.6 of [10] define $\mathrm{SYT}_{n,w}[r]$, marked vacillating tableaux, the subset MVT^*_n, and the refinements $\mathrm{MVT}^0_n, \mathrm{MVT}^1_n$. In Definition 8.6, a mark is allowed only on a step ϵ_w whose starting point is on $x_w = 0$, and the same definition requires the underlying path to have no zero step. The 0/1 split is exactly the unmarked/marked state of the first such step. The four displayed cardinality identities are precisely Theorem 8.7 of [10]. □

Lemma 4.135 (Marked boundary branch count)**.** *Let $T = (T^0, \dots, T^n)$ be a w-vacillating tableau with no zero steps, and write it as a walk in*

$$\{x_1 \ge \cdots \ge x_w \ge 0\}.$$

Let

$$B(T) = \{r : T^r - T^{r-1} = +\epsilon_w \text{ and } T^{r-1}_w = 0\}$$

be the set of eligible boundary-up steps. For this fixed underlying vacillating path T, the elements of $\mathrm{MVT}^*_n(w; \mu \to \nu)$ with underlying path T are exactly the choices of a subset of $B(T)$. In particular, if a repair construction designates a set $B_0 \subseteq B(T)$ of size at most $2h$ and leaves all eligible steps outside B_0 unmarked, then the number of possible marking choices is at most 2^{2h}.

Proof. By Definition 8.5 of [10], a marked vacillating tableau is a pair (T, S) with $S \subseteq [n]$. Definition 8.6 says that membership in MVT_n^* requires the underlying path to have no zero step and requires every marked index to be a $+\epsilon_w$ step whose starting point lies on $x_w = 0$. For fixed T, this condition is exactly $S \subseteq B(T)$. Thus the possible markings are precisely the subsets of $B(T)$. The final claim is the same statement restricted to the designated set B_0, whose number of subsets is at most 2^{2h}. □

Lemma 4.136 (Eligible boundary-up steps are last-row creations). *Let*

$$V = (V^0, \dots, V^N)$$

be a w-vacillating tableau, viewed as a walk in the chamber

$$x_1 \geq x_2 \geq \cdots \geq x_w \geq 0.$$

For a time t, the step t belongs to the eligible boundary-up set

$$B(V) = \{t : V^t - V^{t-1} = +\epsilon_w \text{ and } V_w^{t-1} = 0\}$$

if and only if V^{t-1} has at most $w-1$ nonzero parts and V^t is obtained from V^{t-1} by adding one box in row w.

Proof. In the chamber notation of Definition 8.2 of [10], recalled in lemma 4.134, the vector ϵ_w adds one to the last coordinate and leaves all other coordinates unchanged. The condition $V_w^{t-1} = 0$ is exactly that the starting partition has no nonzero w-th part. Thus the step is $+\epsilon_w$ from the boundary hyperplane precisely when it creates a single box in row w. □

Lemma 4.137 (Odd-column removal for standard tableaux). *Let U be a standard Young tableau of size n with m odd-length columns and with all column lengths at most $2k+1$. Krattenthaler's reversible odd-column removal in* [8, §4, item (2)] *gives a bijection between such tableaux U and pairs (U^{ev}, A), where U^{ev} is a standard Young tableau of size $n-m$ with only even columns and all column lengths at most $2k$, and $A \subset \{1, \dots, n\}$ has cardinality m.*

Proof. Krattenthaler [8, §4, item (2)] treats standard Young tableaux whose columns have length at most $2k+1$. Starting with the last entry in the rightmost odd column and proceeding right-to-left through the currently rightmost odd column, he applies the inverse Robinson–Schensted insertion until all odd columns have disappeared. The cited paragraph states that the output is a standard Young tableau of size $n-m$ with only even columns, all of length at most $2k$, together with a subset of $\{1, \dots, n\}$ of cardinality m, and that all steps are reversible. This is exactly the displayed bijection. □

Lemma 4.138 (Odd-column removal is not the final fixed-witness map). *Let U be a standard Young tableau of size $2(j-q)$ with m odd-length columns. If $m > 0$, the output tableau U^{ev} of lemma 4.137 has size strictly smaller than $2(j-q)$. Consequently, applying odd-column removal as the final operation after witness-label deletion cannot by itself produce an element of $\mathcal{P}_{j-q}^{\mathrm{even}}$, unless the retained shape is already column-even.*

Proof. Lemma 4.137 sends a standard tableau of size n with m odd columns to a column-even standard tableau of size $n-m$. Substituting $n = 2(j-q)$ gives output size $2(j-q)-m$, which is strictly smaller than $2(j-q)$ when $m > 0$. But the paths in $\mathcal{P}_{j-q}^{\mathrm{even}}$ correspond to semistandard tableaux of content (2^{j-q}) and therefore have total size $2(j-q)$. Hence the odd-column-removal operation can only serve as an internal reversible splice or boundary-mark mechanism; it is already target-size preserving only in the case $m = 0$. □

Lemma 4.139 (Standardization of doubled-content tableaux). *Let T be a semistandard tableau of shape λ and content (2^j). For each label $a \in \{1, \dots, j\}$, replace the leftmost box carrying a by $2a-1$ and the rightmost box carrying a by $2a$. The resulting filling* $\mathrm{std}(T)$ *is a standard Young tableau of* *shape λ. The map $T \mapsto \operatorname{std}(T)$ is injective. It preserves the shape, and hence preserves the column-even condition, the width and height thresholds, and the canonical witness boxes of lemma 4.95. Deleting the two boxes carrying a semistandard label a in T is the same box deletion as deleting the standard adjacent pair $\{2a-1, 2a\}$ in $\operatorname{std}(T)$.*

Proof. The two boxes carrying a fixed label a cannot lie in the same column, because columns in T are strictly increasing. Thus the leftmost and rightmost boxes are well-defined.

Consider two boxes in the same row, with the left box carrying label a and the right box carrying label b. Since rows of T are weakly increasing, $a \le b$. If $a < b$, then every standardized value assigned to the left box is at most $2a < 2b-1$, which is at most the standardized value assigned to the right box. If $a = b$, then the two boxes are the leftmost and rightmost occurrences of a in that row and receive $2a-1$ and $2a$. Hence rows of $\operatorname{std}(T)$ are strictly increasing.

Now consider two boxes in the same column, with the upper box carrying a and the lower box carrying b. Since columns of T are strictly increasing, $a < b$. The standardized value in the upper box is at most $2a$, while the standardized value in the lower box is at least $2b-1$, and $2a < 2b-1$. Hence columns of $\operatorname{std}(T)$ are strictly increasing. The entries of $\operatorname{std}(T)$ are exactly $1, 2, \ldots, 2j$, so it is standard.

Injectivity follows by replacing each pair $\{2a-1, 2a\}$ in $\operatorname{std}(T)$ by a. The remaining assertions are immediate because the operation changes only labels, not boxes. □

Lemma 4.140 (Standardized witness-pair local states)**.** *Let T be a semistandard tableau of content (2^j), and let $\operatorname{std}(T)$ be the standard tableau of lemma 4.139.*

If $\lambda_1(T) \ge 2q$ and $s_r = T(1, 2r-1)$ for $1 \le r \le q$, then the two boxes deleted by the width witness label s_r are exactly the boxes carrying the adjacent standard pair $\{2s_r - 1, 2s_r\}$ in $\operatorname{std}(T)$. The frontier box $(1, 2r-1)$ carries one of these two entries. Thus the bit

$$\epsilon_r^w(T) = \begin{cases} 0, & \operatorname{std}(T)(1, 2r-1) = 2s_r - 1, \\ 1, & \operatorname{std}(T)(1, 2r-1) = 2s_r \end{cases}$$

records which member of the adjacent pair lies at the width frontier. If the other copy of s_r lies in the first row, then it is in column $2r$ when $\epsilon_r^w(T) = 0$, and in column $2r-2$ when $\epsilon_r^w(T) = 1$.

If $\ell(\lambda(T)) \ge 2q$ and $t_r = T(2r-1, 1)$ for $1 \le r \le q$, then the two boxes deleted by the height witness label t_r are exactly the boxes carrying the adjacent standard pair $\{2t_r - 1, 2t_r\}$ in $\operatorname{std}(T)$, and the frontier box $(2r-1, 1)$ carries $2t_r - 1$.

Proof. For every label a, lemma 4.139 replaces the two boxes carrying a by the adjacent standard entries $2a-1$ and $2a$. This proves the adjacent-pair statement for both width and height witnesses.

For the width statement, the frontier box $(1, 2r-1)$ is one of the two boxes carrying s_r, hence in the standardized tableau it carries either $2s_r - 1$ or $2s_r$, giving the displayed bit. If the other copy lies in the first row, lemma 4.549 places it in column $2r-2$ or $2r$. The standardization assigns $2s_r - 1$ to the leftmost copy and $2s_r$ to the rightmost copy, so the other first-row copy is in column $2r$ in the first case and in column $2r-2$ in the second.

For the height statement, lemma 4.550 says that the second copy of t_r is not in the first column. Since the frontier copy is in column 1, it is the leftmost copy and therefore receives the standardized label $2t_r - 1$. □

Lemma 4.141 (Frontier crossing time inside a deleted standard pair)**.** *Let T be a semistandard tableau of content (2^j), let $U = \operatorname{std}(T)$, and let*

$$\varnothing = \sigma^{(0)} \subset \sigma^{(1)} \subset \cdots \subset \sigma^{(2j)} = \lambda(T)$$

be the standard shape chain of U.

If $\lambda_1(T) \ge 2q$ and $s_r = T(1, 2r-1)$ for $1 \le r \le q$, then the standard time

$$\kappa_r^w = 2s_r - 1 + \epsilon_r^w(T)$$

is the unique time in the adjacent pair $\{2s_r-1, 2s_r\}$ *at which the box* $(1, 2r-1)$ *is added. Equivalently,*

$$\sigma_1^{(\kappa_r^w-1)} < 2r-1 \leq \sigma_1^{(\kappa_r^w)}.$$

If $\ell(\lambda(T)) \geq 2q$ *and* $t_r = T(2r-1, 1)$ *for* $1 \leq r \leq q$*, then the standard time*

$$\kappa_r^h = 2t_r - 1$$

is the unique time in the adjacent pair $\{2t_r-1, 2t_r\}$ *at which the box* $(2r-1, 1)$ *is added. Equivalently,*

$$\ell(\sigma^{(\kappa_r^h-1)}) < 2r-1 \leq \ell(\sigma^{(\kappa_r^h)}).$$

Proof. In a standard tableau, the box carrying entry m is added exactly when the standard shape chain passes from $\sigma^{(m-1)}$ to $\sigma^{(m)}$.

For the width statement, lemma 4.140 says that the frontier box $(1, 2r-1)$ carries $2s_r - 1$ when $\epsilon_r^w(T) = 0$ and carries $2s_r$ when $\epsilon_r^w(T) = 1$. Thus it is added at κ_r^w. Before that time the first row does not contain column $2r-1$; after that time it does. This is exactly the displayed inequality. The time is unique because one standard entry labels one box.

For the height statement, lemma 4.140 says that the frontier box $(2r-1, 1)$ carries $2t_r - 1$. Therefore it is added at $\kappa_r^h = 2t_r - 1$. Before that time the first column has length below $2r-1$; after that time it has length at least $2r-1$. Again uniqueness is the uniqueness of the standard entry in that box. ☐

Lemma 4.142 (Frontier crossing times are ordered deleted times)**.** *Let* T *be a semistandard tableau of content* (2^j)*, let* $U = \mathrm{std}(T)$*, and let*

$$\varnothing = \sigma^{(0)} \subset \sigma^{(1)} \subset \cdots \subset \sigma^{(2j)} = \lambda(T)$$

be the standard shape chain of U*.*

Assume $\lambda_1(T) \geq 2q$*. Put* $s_r = T(1, 2r-1)$*,* $S = W_q(T)$*, and*

$$D(S) = \bigcup_{s\in S}\{2s-1, 2s\}, \qquad \kappa_r^w = 2s_r - 1 + \epsilon_r^w(T) \quad (1 \leq r \leq q).$$

Then

$$2s_r - 1 \leq \kappa_r^w \leq 2s_r, \qquad \kappa_r^w \in D(S), \qquad \kappa_1^w < \kappa_2^w < \cdots < \kappa_q^w.$$

Moreover,

$$\{m : \min(q, \lceil\sigma_1^{(m)}/2\rceil) - \min(q, \lceil\sigma_1^{(m-1)}/2\rceil) = 1\} = \{\kappa_r^w : 1 \leq r \leq q\}.$$

Assume instead that $\ell(\lambda(T)) \geq 2q$*. Put* $t_r = T(2r-1, 1)$*,* $S = H_q(T)$*, and*

$$D(S) = \bigcup_{t\in S}\{2t-1, 2t\}, \qquad \kappa_r^h = 2t_r - 1 \quad (1 \leq r \leq q).$$

Then

$$2t_r - 1 \leq \kappa_r^h \leq 2t_r, \qquad \kappa_r^h \in D(S), \qquad \kappa_1^h < \kappa_2^h < \cdots < \kappa_q^h.$$

Moreover,

$$\{m : \min(q, \lceil\ell(\sigma^{(m)})/2\rceil) - \min(q, \lceil\ell(\sigma^{(m-1)})/2\rceil) = 1\} = \{\kappa_r^h : 1 \leq r \leq q\}.$$

Proof. For the width case, $\epsilon_r^w(T) \in \{0, 1\}$ by definition, so $2s_r - 1 \leq \kappa_r^w \leq 2s_r$ and $\kappa_r^w \in D(S)$. The entries $s_1, \ldots, s_q$ occur in weakly increasing order along the first row, and they are distinct by lemma 4.95; hence $s_1 < \cdots < s_q$. Therefore

$$\kappa_r^w \leq 2s_r < 2s_{r+1} - 1 \leq \kappa_{r+1}^w$$

for $1 \leq r < q$.

By lemma 4.141, κ_r^w is exactly the standard time at which the box $(1, 2r-1)$ is added. At that time the first row length passes from $2r-2$ to $2r-1$, so the displayed minimum increases by one. Conversely, if the displayed minimum increases at time m, then the single box added from $\sigma^{(m-1)}$ to

$\sigma^{(m)}$ must be the first-row box $(1, 2r-1)$ for a unique $1 \le r \le q$. The same cited lemma identifies its standard time with κ_r^w. This proves the width equality of crossing sets.

For the height case, the inequalities $2t_r - 1 \le \kappa_r^h \le 2t_r$ and $\kappa_r^h \in D(S)$ are immediate from $\kappa_r^h = 2t_r - 1$. The first-column entries are strictly increasing, so $t_1 < \cdots < t_q$, and hence $\kappa_1^h < \cdots < \kappa_q^h$.

By lemma 4.141, κ_r^h is exactly the standard time at which the box $(2r-1, 1)$ is added. At that time the first-column length passes from $2r-2$ to $2r-1$, so the displayed height minimum increases by one. Conversely, if that minimum increases at time m, then the added box is the first-column box $(2r-1, 1)$ for a unique $1 \le r \le q$, and lemma 4.141 again gives $m = \kappa_r^h$. □

Lemma 4.143 (Fixed-witness frontier assignments are strictly ordered)**.** *In the width setting of lemma 4.142, the assignment*

$$e^w : S = W_q(T) \longrightarrow D(S), \qquad e^w(s_r) = \kappa_r^w$$

satisfies

$$s < s' \quad \Longleftrightarrow \quad e^w(s) < e^w(s') \qquad (s, s' \in S).$$

In the height setting, the assignment

$$e^h : S = H_q(T) \longrightarrow D(S), \qquad e^h(t_r) = \kappa_r^h$$

satisfies the same equivalence.

Proof. We prove the width case; the height case is identical. By lemma 4.95, the labels

$$s_r = T(1, 2r-1) \qquad (1 \le r \le q)$$

are distinct, and they occur weakly increasingly along the first row. Hence

$$s_1 < s_2 < \cdots < s_q.$$

Lemma 4.142 gives

$$\kappa_1^w < \kappa_2^w < \cdots < \kappa_q^w.$$

Thus, for $s = s_r$ and $s' = s_{r'}$, one has $s < s'$ exactly when $r < r'$, and this is exactly when $e^w(s) = \kappa_r^w < \kappa_{r'}^w = e^w(s')$. □

Corollary 4.144 (Actual frontier-consecutive edges give single-deficit recovery)**.** *Work in the setting of proposition 4.81, and let $A \subset S$ be the support of the anchored component in lemma 4.79.*

In the width setting of lemma 4.142, suppose $S = W_q(T)$, write $S = \{s_1 < \cdots < s_q\}$ as there, and assign $s_r \mapsto \kappa_r^w$. If every internal edge of the anchored path on A has endpoints whose assigned times are consecutive in the full ordered list

$$\kappa_1^w < \kappa_2^w < \cdots < \kappa_q^w,$$

then Γ is recovered from

$$S, \quad v, \quad \Gamma^*, \quad \Gamma_{S \setminus A}, \quad \mathbf{1}_A \in \{0,1\}^S,$$

where $\Gamma_{S\setminus A}$ is the graph induced by Γ on $S \setminus A$.

In the height setting, the same assertion holds with $S = H_q(T)$, $S = \{t_1 < \cdots < t_q\}$, and the full ordered list

$$\kappa_1^h < \kappa_2^h < \cdots < \kappa_q^h.$$

Thus, in the actual fixed-witness fibers, the single-deficit crossing leaf is the identification of v together with this full-frontier consecutiveness of the anchored internal edges, plus a separate supplier for the deleted-only graph unless $A = S$.

Proof. In the width case, lemma 4.143 shows that the assignment $s_r \mapsto \kappa_r^w$ satisfies the order equivalence required in proposition 4.91. The displayed consecutiveness hypothesis is exactly consecutiveness among the full event set assigned to S. That proposition therefore gives the stated recovery. The height case is identical, using the assignment $t_r \mapsto \kappa_r^h$. The final sentence just records the two cases in the language of the fixed-witness fibers. □

Lemma 4.145 (Residual windows have a q-bit retained-vertex code)**.** *Let $G = B_b$ with $14 \le b \le 217$ or $G = C_b$ with $20 \le b \le 217$, and put $q = b + 1$. Let j be in the residual window of remark 4.839, and let $S \subset \{1, \dots, j\}$ have cardinality q. Then the ordered retained set*

$$V_0 = \{1, \dots, j\} \setminus S$$

has cardinality at most 2^q. Consequently any retained vertex $v \in V_0$ is encoded by a word in $\{0,1\}^q$, using its rank in the increasing order of V_0.

Proof. For all residual rows, $q \ge 15$. By corollary 4.60, and since j lies in the residual window of remark 4.839,

$$j \le N_{\mathrm{dir}}^{BC}(G) + 2 \le 30899.$$

Therefore

$$|V_0| = j - q \le 30899 - 15 = 30884 < 32768 = 2^{15} \le 2^q.$$

Listing V_0 increasingly identifies its elements with $1, \dots, |V_0|$, and the displayed inequality embeds this rank set into the 2^q binary words of length q. □

Proposition 4.146 (q spare bits remove v-identification from the single-deficit leaf)**.** *Work in one residual row and moment of remark 4.839. Put $q = b+1$, let $S \subset \{1, \dots, j\}$ have cardinality q, and put $V_0 = \{1, \dots, j\} \setminus S$. In the setting of proposition 4.81, let $A \subset S$ be the support of the anchored component.*

Assume that the actual width fixed-witness fiber has $S = W_q(T)$ and every internal edge of the anchored path on A has endpoints whose assigned times are consecutive in the full ordered list $\kappa_1^w < \cdots < \kappa_q^w$. Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \Gamma_{S \setminus A}, \quad \mathbf{1}_A \in \{0,1\}^S, \quad \rho(v) \in \{0,1\}^q,$$

where $\rho(v)$ is the q-bit rank code of lemma 4.145. The same assertion holds in the height fixed-witness fiber with $S = H_q(T)$ and $\kappa_1^h < \cdots < \kappa_q^h$.

Consequently, on the single-deficit leaf, the remaining fixed-witness crossing theorem no longer has to identify v: it only has to prove the displayed full-κ consecutiveness of anchored internal edges. The support word and the retained-vertex code together use at most $2q$ branch bits, while the deleted-only graph $\Gamma_{S \setminus A}$ remains an explicit supplier obligation unless it is absorbed into the target or absent.

Proof. Lemma 4.145 recovers v from S and the word $\rho(v)$. In the width case, after v is recovered, corollary 4.144 applies to the displayed full-κ^w consecutiveness hypothesis and recovers Γ from S, v, Γ^*, $\Gamma_{S \setminus A}$, and $\mathbf{1}_A$. The height case is identical with κ^h in place of κ^w. The last sentence counts the q support bits and the q retained-vertex bits, and records the separate deleted-only input. □

Lemma 4.147 (Consecutive frontier times are adjacent witness ranks)**.** *Let*

$$\kappa_1 < \kappa_2 < \cdots < \kappa_q$$

be a strictly ordered list. If $1 \le r, r' \le q$ and no κ_i lies strictly between κ_r and $\kappa_{r'}$, then $|r - r'| = 1$ unless $r = r'$. Conversely, if $|r - r'| = 1$, then no κ_i lies strictly between κ_r and $\kappa_{r'}$.

Proof. Assume $r < r'$. If $r' - r \ge 2$, then κ_{r+1} lies strictly between κ_r and $\kappa_{r'}$, contradiction. Hence $r' = r + 1$. The case $r' > r$ after interchanging r, r' is the same. Conversely, if $r' = r + 1$, the strict ordering leaves no index strictly between r and r', and hence no listed value lies strictly between κ_r and $\kappa_{r'}$. The reversed adjacent case is identical. □

Proposition 4.148 (Witness-rank adjacent edges give residual single-deficit recovery)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \dots, j\}$ have cardinality q, and use the notation of proposition 4.146.*

In the width fixed-witness fiber, write

$$S = W_q(T) = \{s_1 < \cdots < s_q\}.$$

If every internal edge of the anchored path on A joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, then Γ is recovered from

$$S, \quad \Gamma^*, \quad \Gamma_{S\setminus A}, \quad \mathbf{1}_A \in \{0,1\}^S, \quad \rho(v) \in \{0,1\}^q.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$ and the condition that every internal anchored edge joins $t_r, t_{r'}$ with $|r - r'| = 1$.

Consequently the remaining single-deficit crossing theorem may be stated without frontier-time notation: after the repaired retained graph is fixed, the geometric condition for the anchored component is that anchored internal edges join adjacent witness ranks; the deleted-only graph remains separate unless it is otherwise supplied.

Proof. In the width case, lemma 4.142 gives the strictly ordered frontier list

$$\kappa_1^w < \kappa_2^w < \cdots < \kappa_q^w.$$

If an internal anchored edge joins s_r to $s_{r'}$ with $|r - r'| = 1$, then lemma 4.147 shows that the corresponding frontier times κ_r^w and $\kappa_{r'}^w$ are consecutive in the full ordered list. Thus the hypotheses of proposition 4.146 hold, and that proposition gives the displayed recovery. The height case is identical, using

$$\kappa_1^h < \kappa_2^h < \cdots < \kappa_q^h.$$

□

Lemma 4.149 (Full-rank adjacent anchored paths have interval support)**.** *Let*

$$S = \{s_1 < \cdots < s_q\}$$

be a linearly ordered witness set, let $T \subset S$, and let P be a simple path through all labels of T. Suppose every edge of P joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$ in the full ordered set S. Then T is a consecutive interval of S: either $T = \varnothing$, or

$$T = \{s_a, s_{a+1}, \dots, s_b\}$$

for uniquely determined $1 \le a \le b \le q$. If $|T| \ge 2$, then P is the monotone path through this interval, uniquely up to reversal.

Proof. If $T = \varnothing$ there is nothing to prove. Put

$$I = \{r : s_r \in T\} \subset \{1, \dots, q\}.$$

The path P gives a connected graph on the vertex set I. Every edge joins indices differing by one. If I had a gap, say $r < r'$ in I with no element of I strictly between them, then no path using only unit-difference edges could connect the part of I at or below r to the part at or above r'. This contradicts connectedness. Hence I is an interval $\{a, \dots, b\}$, proving the support statement and uniqueness of a, b.

When $|T| \ge 2$, the path P has $|T| - 1 = b - a$ edges. The only possible edges inside the interval are

$$\{s_i, s_{i+1}\} \qquad (a \le i < b),$$

also $b - a$ edges. Since P is connected on every vertex of the interval and uses only these possible edges, it uses all of them. Thus it is the monotone path through $s_a, s_{a+1}, \dots, s_b$, read in one direction or the other. □

Corollary 4.150 (Adjacent-rank single-deficit recovery has interval support code)**.** *Work in one residual row and moment of remark 4.839. Put $q = b+1$, let $S \subset \{1, \dots, j\}$ have cardinality q, and use the notation of proposition 4.148. Suppose the width fixed-witness fiber has*

$$S = W_q(T) = \{s_1 < \cdots < s_q\}$$

and every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$. Then the anchored support is a consecutive witness interval, and Γ is recovered from

$$S, \quad \Gamma^*, \quad \Gamma_{S \setminus A}, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

where $\rho(v)$ is the retained-vertex rank code of lemma 4.145 and $\iota(A)$ is the lexicographic rank code of the support interval $\{s_a, \dots, s_b\}$ among all nonempty intervals of S. The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Thus, under the adjacent-rank crossing condition, the support data in the single-deficit leaf are interval endpoints rather than an arbitrary subset of the witness set; the retained vertex and interval support still use at most the allowed $2q$ branch bits in the residual range. This does not encode the deleted-only graph on the complement of the interval.

Proof. Apply lemma 4.149 to the anchored path on its support. It shows that the support is a consecutive interval of the full witness list and that the anchored path is the monotone path through that interval up to reversal. The number of nonempty intervals in a q-element ordered set is

$$\frac{q(q+1)}{2}.$$

In the residual range $q \geq 15$, this number is at most 2^q, so the lexicographic rank of the interval is encoded by a q-bit word: at $q = 15$ one has $15 \cdot 16/2 = 120 < 2^{15}$, and the implication from q to $q+1$ follows because

$$\frac{(q+1)(q+2)}{q(q+1)} = \frac{q+2}{q} \leq 2.$$

The interval code recovers the support characteristic vector $\mathbf{1}_A \in \{0,1\}^S$. Therefore proposition 4.148 applies and recovers Γ from S, Γ^*, $\Gamma_{S \setminus A}$, the recovered support vector, and the retained-vertex code $\rho(v)$. The height case is identical with the ordered list $t_1 < \cdots < t_q$. □

Corollary 4.151 (Full-support adjacent-rank single-deficit recovery)**.** *Work in one residual row and moment of remark 4.839. Put $q = b+1$, let $S \subset \{1, \dots, j\}$ have cardinality q, and use the notation of proposition 4.148. Suppose the width fixed-witness fiber has*

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support is all of S, and every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$. Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q,$$

where $\rho(v)$ is the retained-vertex rank code of lemma 4.145. The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, in the adjacent-rank single-deficit leaf, the deleted-only component obstruction is absent once the anchored support is the whole witness set. The remaining proper-support case is exactly the case in which the deleted-only graph on $S \setminus A$ has to be supplied, absorbed into the target, or ruled out.

Proof. If $A = S$, then the support interval in corollary 4.150 is the whole ordered witness set, so its interval code is determined by S. The graph $\Gamma_{S \setminus A}$ is the unique empty graph on the empty label set. Thus the data required by corollary 4.150 are fixed functions of

$$S, \quad \Gamma^*, \quad \rho(v).$$

That corollary recovers Γ in the width case, and the height case is identical. The final sentence is the contrapositive localization of the only datum removed in this full-support case. $\square$

Corollary 4.152 (Proper-support adjacent-rank leaves reduce to deleted-only readout)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and use the notation of corollary 4.150. Suppose the width fixed-witness fiber has*

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support A is a proper nonempty interval of S, and every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$. Let Δ_A be any datum, determined without using an additional branch word, which determines the deleted-only graph

$$\Gamma_{S \setminus A}.$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q, \quad \Delta_A.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved, the proper-support single-deficit leaf has no remaining path-order, support-set, or retained-vertex ambiguity. Its remaining supplier obligation is exactly a branch-free target-path readout of $\Gamma_{S \setminus A}$, or a theorem excluding proper support.

Proof. The interval code $\iota(A)$ recovers the proper support interval A. Together with the retained-vertex code $\rho(v)$, the datum Δ_A therefore supplies every input listed in corollary 4.150: the deleted-only graph is the graph recovered from Δ_A. Applying that corollary recovers Γ in the width case, and the height case is identical.

The final sentence is just the same recovery statement read against the branch accounting in corollary 4.150: the allowed $2q$ branch bits are already the q-bit retained-vertex code and the q-bit interval code, so the remaining deleted-only graph must be supplied by the target-path/parser data or eliminated by proving $A = S$. $\square$

Lemma 4.153 (Line-adjacent deleted-only complements are forced)**.** *Let*

$$S = \{s_1 < \cdots < s_q\}$$

be a finite linearly ordered label set, and let

$$A = \{s_a, s_{a+1}, \ldots, s_b\} \qquad (1 \le a \le b \le q)$$

be a nonempty interval. Put $D = S \setminus A$. Let Γ_D be a loopless labelled 2-regular multigraph on D, with parallel edges allowed. Suppose every edge of Γ_D joins two labels $s_r, s_{r'}$ with

$$|r - r'| = 1.$$

Then the following dichotomy holds.

1. *If either $a - 1$ or $q - b$ is odd, no such graph Γ_D exists.*
2. *If both $a - 1$ and $q - b$ are even, then Γ_D is uniquely determined by S and A: it consists of two parallel edges between s_{2u-1} and s_{2u} for $1 \le u \le (a-1)/2$, and two parallel edges between s_{b+2u-1} and s_{b+2u} for $1 \le u \le (q-b)/2$.*

Proof. The deleted set D is the disjoint union of the left interval $\{s_1, \ldots, s_{a-1}\}$ and the right interval $\{s_{b+1}, \ldots, s_q\}$. Since A is nonempty, no label in the left interval is adjacent in full witness rank to a label in the right interval. Hence the adjacency hypothesis makes Γ_D the disjoint union of its restrictions to these two intervals.

It remains to analyze one contiguous interval

$$I = \{s_\ell, s_{\ell+1}, \ldots, s_r\}$$

of deleted labels. Write $d = r - \ell + 1$. If $d = 0$, there is nothing to prove. If $d \geq 1$, let m_i be the multiplicity of the edge between s_i and s_{i+1} for $\ell \leq i < r$. The adjacency hypothesis and looplessness allow no other edges inside I. The endpoint s_ℓ has degree m_ℓ, so $m_\ell = 2$ when $d \geq 2$, while for $d = 1$ degree 2 is impossible. For an interior vertex s_i one has

$$m_{i-1} + m_i = 2.$$

Thus the multiplicities are forced alternately as

$$m_\ell = 2, \quad m_{\ell+1} = 0, \quad m_{\ell+2} = 2, \quad \ldots.$$

At the right endpoint the degree condition is $m_{r-1} = 2$. This holds exactly when d is even. Therefore an odd-length interval is impossible, while an even-length interval has the unique doubled-adjacent-pair graph

$$(s_\ell, s_{\ell+1}),\ (s_{\ell+2}, s_{\ell+3}), \ldots,\ (s_{r-1}, s_r),$$

with two parallel copies of each listed edge.

Applying this interval analysis to the left and right components of D gives the stated parity obstruction and the displayed unique graph. □

Corollary 4.154 (Line-adjacent proper-support leaves need no deleted-only readout)**.** *Work in one residual row and moment of remark 4.839. Put $q = b+1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and use the notation of corollary 4.150. Suppose the width fixed-witness fiber has*

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support is a proper nonempty interval $A = \{s_a, \ldots, s_b\}$, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and every edge of the deleted-only graph $\Gamma_{S\setminus A}$ also joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$ in the full witness list. If either $a - 1$ or $q - b$ is odd, these hypotheses are inconsistent. If both are even, then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved for both the anchored component and the deleted-only complement, the proper-support single-deficit leaf has no deleted-only readout obligation.

Proof. The interval code $\iota(A)$ recovers A, hence the integers a and b. If one of $a - 1$ and $q - b$ is odd, then lemma 4.153 says that no deleted-only graph satisfying the full-rank adjacency hypothesis exists. This gives the inconsistency statement.

If both integers are even, the same lemma recovers the deleted-only graph $\Gamma_{S\setminus A}$ from S and A alone. Use this forced graph as the branch-free datum Δ_A in corollary 4.152. That corollary recovers Γ in the width case. The height case is identical, with the ordered witness list $t_1 < \cdots < t_q$ replacing $s_1 < \cdots < s_q$. □

Corollary 4.155 (Actual frontier-consecutive deleted-only complements need no readout)**.** *Work in one residual row and moment of remark 4.839. Put $q = b+1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and use the notation of corollary 4.150. Suppose the width fixed-witness fiber has*

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

and that the anchored support is a proper nonempty interval $A = \{s_a, \ldots, s_b\}$. Assume:

1. *every internal edge of the anchored path on A has endpoints whose assigned times are consecutive in the full ordered list*

$$\kappa_1^w < \cdots < \kappa_q^w;$$

2. *every edge of the deleted-only graph $\Gamma_{S\setminus A}$ has endpoints whose assigned times are consecutive in the same full ordered list.*

If either $a-1$ or $q-b$ is odd, these hypotheses are inconsistent. If both are even, then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$ and $\kappa_1^h < \cdots < \kappa_q^h$.

Consequently the proper-support single-deficit leaf is closed once the actual fixed-witness parser proves full-frontier consecutiveness for the anchored internal edges and the deleted-only complement edges.

Proof. For the width case, lemma 4.147 turns item (1) into the adjacent-rank condition for every anchored internal edge. The same lemma turns item (2) into the condition that every deleted-only edge joins labels $s_r, s_{r'}$ with $|r - r'| = 1$ in the full witness list. Therefore the hypotheses of corollary 4.154 are satisfied, and that corollary gives the inconsistency/recovery dichotomy.

The height case is identical, using the ordered list $\kappa_1^h < \cdots < \kappa_q^h$ and the height witness labels $t_1 < \cdots < t_q$. □

Proposition 4.156 (Actual frontier-consecutive single-deficit graphs use only the interval code)**.** *Work in one residual row and moment of remark 4.839. Put $q = b+1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and work in the setting of proposition 4.81. Let $A \subset S$ be the support of the anchored component in lemma 4.79.*

In the width fixed-witness fiber suppose

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

with full frontier list

$$\kappa_1^w < \cdots < \kappa_q^w.$$

Assume that every internal edge of the anchored path on A has endpoints whose assigned times are consecutive in this full list, and that every edge of the deleted-only graph $\Gamma_{S \setminus A}$ also has endpoints whose assigned times are consecutive in this full list. Then A is a nonempty interval in the ordered witness set, and Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$ and $\kappa_1^h < \cdots < \kappa_q^h$.

Consequently the single-deficit graph leaf, under actual full-frontier consecutiveness for both the anchored component and the deleted-only complement, consumes only the already budgeted retained-vertex rank code and support-interval code.

Proof. Consider the width case. The component containing v has nonempty support A, because the single retained deficit has the two ports at v attached to deleted labels by lemma 4.75. By lemma 4.147, the full-list consecutiveness of the anchored internal edges implies that every such edge joins adjacent witness ranks. Then lemma 4.149 shows that A is a nonempty interval.

If $A = S$, then corollary 4.151 recovers Γ from S, Γ^*, and $\rho(v)$, so the displayed data recover Γ as well. If A is a proper interval, then corollary 4.155 applies to the same full-list consecutiveness hypotheses. Its inconsistent parity alternative cannot occur for the actual graph under discussion, and its recovery alternative gives Γ from the displayed data.

The height case is identical with the ordered height witness list and $\kappa_1^h < \cdots < \kappa_q^h$ replacing the width data. □

Lemma 4.157 (Small deleted-only complements are forced)**.** *Let D be a finite label set, and let Γ_D be a loopless labelled 2-regular multigraph on D, with parallel edges allowed. If $|D| = 0$, then Γ_D is the unique empty graph. If $|D| = 1$, no such graph exists. If $|D| = 2$, then Γ_D is the unique doubled edge on the two labels. If $|D| = 3$, then Γ_D is the unique simple triangle on the three labels.*

Proof. For $|D| = 0$, the empty edge multiset is the only possible graph and the degree condition is vacuous. For $|D| = 1$, looplessness forbids an edge, so the unique vertex would have degree 0, not degree 2.

For $|D| = 2$, each edge must join the two distinct labels. Degree 2 at either label therefore forces exactly two parallel copies of that edge.

For $|D| = 3$, a pair of parallel edges between two labels would already use degree 2 at those two labels and would leave the third label with degree 0, so no parallel edge can occur. Thus the graph is simple. The only simple graph on three labels in which every vertex has degree 2 is the triangle. ☐

Corollary 4.158 (Near-full proper-support leaves need no deleted-only readout)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and use the notation of corollary 4.152. Suppose the width fixed-witness fiber has*

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support A is a proper nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and

$$|S \setminus A| \leq 3.$$

If $|S \setminus A| = 1$, then these hypotheses are inconsistent. If $|S \setminus A| = 2$ or $|S \setminus A| = 3$, then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved, the only proper-support deleted-only readout still not covered by forced small complements alone is the case

$$|S \setminus A| \geq 4.$$

Proof. Put $D = S \setminus A$. By lemma 4.76, the deleted-only part induced on D is a loopless labelled 2-regular multigraph. If $|D| = 1$, this contradicts lemma 4.157, so the hypotheses are inconsistent.

If $|D| = 2$ or $|D| = 3$, then lemma 4.157 says that the deleted-only graph on D is uniquely determined by the label set D, hence by S and $\iota(A)$. Use this forced graph as the branch-free datum Δ_A in corollary 4.152. That corollary recovers Γ in the width case, and the height case is identical.

The final sentence combines these two cases with corollary 4.151, which already covers $D = \varnothing$. ☐

Corollary 4.159 (Small deleted-only complements fit in the support code)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and use the notation of corollary 4.150. Suppose the width fixed-witness fiber has*

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and

$$|S \setminus A| \leq 7.$$

Then there is a q-bit code

$$\zeta(A, \Gamma_{S \setminus A}) \in \{0,1\}^q,$$

depending only on the interval A and on the deleted-only graph induced on $S \setminus A$, such that Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \zeta(A, \Gamma_{S \setminus A}) \in \{0,1\}^q.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved, the proper-support deleted-only readout that is not already absorbed by the existing support-code slot begins only in the case

$$|S \setminus A| \geq 8.$$

Proof. For a fixed ordered q-element witness set S, the number of nonempty intervals $A \subseteq S$ with $|S \setminus A| = d$ is $d+1$. For such an interval, lemma 4.65 gives exactly s_d possible loopless labelled 2-regular graphs on $S \setminus A$, where s_d is the stable moment sequence of lemma 4.67. Iterating that recurrence gives

$$s_0 = 1, \quad s_1 = 0, \quad s_2 = 1, \quad s_3 = 1, \quad s_4 = 6, \quad s_5 = 22, \quad s_6 = 130, \quad s_7 = 822.$$

Therefore the number of pairs $(A, \Gamma_{S\setminus A})$ with $|S \setminus A| \leq 7$ is

$$\sum_{d=0}^{7} (d+1)s_d = 7656 < 32768 = 2^{15} \leq 2^q,$$

because every residual row has $q \geq 15$. Listing these pairs in lexicographic order gives the advertised q-bit code $\zeta(A, \Gamma_{S\setminus A})$.

The code recovers both A and the deleted-only graph $\Gamma_{S\setminus A}$. Applying corollary 4.150 with these recovered data recovers Γ in the width case, and the height case is identical. The final sentence is the same branch accounting: the q retained-vertex bits and the q packed support/deleted-only bits are the whole allowed $2q$ branch budget. □

Corollary 4.160 (Residual joint branch code absorbs complements of size nine)**.** *Work in one residual row and moment of remark 4.839. Put $q = b+1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and use the notation of corollary 4.150. Suppose the width fixed-witness fiber has*

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and

$$|S \setminus A| \leq 9.$$

Then there is a $2q$-bit code

$$\xi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q},$$

depending only on the retained deficit vertex v, the interval A, and the deleted-only graph induced on $S \setminus A$, such that Γ is recovered from

$$S, \quad \Gamma^*, \quad \xi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q}.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved, the proper-support deleted-only readout that is not already absorbed by the total $2q$ branch budget begins only in the case

$$|S \setminus A| \geq 10.$$

Proof. For fixed S and a fixed complement size d, there are $d+1$ nonempty intervals $A \subseteq S$ with $|S \setminus A| = d$. For each such interval, lemma 4.65 gives exactly s_d possible deleted-only graphs on $S \setminus A$. Iterating lemma 4.67 one more step beyond corollary 4.159 gives

$$s_8 = 6202, \qquad s_9 = 52552,$$

and hence

$$M_9 := \sum_{d=0}^{9} (d+1)s_d = 588994.$$

Thus, for fixed S, the number of triples

$$(v, A, \Gamma_{S\setminus A}) \qquad \text{with} \qquad |S \setminus A| \le 9$$

is at most

$$(j-q)M_9.$$

Since j lies in the residual window of remark 4.839, one has $j \le N_{\mathrm{dir}}^{BC}(G) + 2$.

For the first three residual B rows, corollary 4.61 gives the exact onset values. Therefore

q	$j-q \le$	$(j-q)M_9$	2^{2q}
15	912	537162528	1073741824
16	401	236186594	4294967296
17	314	184944116	17179869184.

In each of these rows, $(j-q)M_9 < 2^{2q}$. For all remaining residual rows one has $q \ge 18$, and corollary 4.60 gives

$$j - q \le 30899 - q \le 30881.$$

Consequently

$$(j-q)M_9 \le 30881 \cdot 588994 = 18188723714 < 68719476736 = 2^{36} \le 2^{2q}.$$

Listing all triples $(v, A, \Gamma_{S\setminus A})$ in lexicographic order gives the advertised $2q$-bit code ξ. The code recovers v, A, and the deleted-only graph. Applying corollary 4.150 with these recovered inputs recovers Γ in the width case, and the height case is identical. The final sentence is exactly the displayed packing statement applied to proper support. $\square$

Corollary 4.161 (Stratified residual joint branch code)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and use the notation of corollary 4.150. Define*

$$K(q) = \begin{cases} 9, & q = 15, \\ 10, & 16 \le q \le 17, \\ 11, & q = 18, \\ 12, & 19 \le q \le 20, \\ 13, & 21 \le q \le 22, \\ 14, & 23 \le q \le 24, \\ 15, & q \ge 25. \end{cases}$$

Suppose the width fixed-witness fiber has

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and

$$|S \setminus A| \le K(q).$$

Then there is a $2q$-bit code

$$\Xi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q},$$

depending only on the retained deficit vertex v, the interval A, and the deleted-only graph induced on $S \setminus A$, such that Γ is recovered from

$$S, \quad \Gamma^*, \quad \Xi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q}.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved, the unabsorbed proper-support deleted-only readout is restricted to the cases

$$|S \setminus A| \ge K(q) + 1.$$

Proof. Let
$$M_K = \sum_{d=0}^{K} (d+1) s_d.$$
As in the proof of corollary 4.160, for fixed S the number of triples $(v, A, \Gamma_{S \setminus A})$ with $|S \setminus A| \leq K$ is at most $(j-q)M_K$. Iterating lemma 4.67 gives

K	M_K
9	588994
10	6080128
11	68940568
12	851837540
13	11388066236
14	163757241896
15	2520032130024.

Combining these values with the residual window bounds of corollary 4.63 gives

range of q	$K(q)$	$(j-q)M_{K(q)} \leq$	2^{2q} lower bound
$q = 15$	9	537162528	1073741824
$16 \leq q \leq 17$	10	2438131328	4294967296
$q = 18$	11	19785943016	68719476736
$19 \leq q \leq 20$	12	241070023820	274877906944
$21 \leq q \leq 22$	13	3450584069508	4398046511104
$23 \leq q \leq 24$	14	55186190518952	70368744177664
$25 \leq q \leq 28$	15	1065973591000152	1125899906842624
$q \geq 29$	15	77793391853840880	288230376151711744.

In every row of the table the third entry is strictly smaller than the fourth. Thus the triples $(v, A, \Gamma_{S \setminus A})$ with $|S \setminus A| \leq K(q)$ inject into the 2^{2q} possible branch words. Listing them in lexicographic order gives the code Ξ.

The code recovers v, A, and the deleted-only graph. Applying corollary 4.150 with these recovered inputs recovers Γ in the width case; the height case is identical. The final sentence is the complement of the displayed packing range. □

Corollary 4.162 (High-rank linear joint branch code)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, assume $q \geq 45$, let $S \subset \{1, \dots, j\}$ have cardinality q, and use the notation of corollary 4.150. Set*
$$L(q) = \left\lfloor \frac{q}{4} \right\rfloor + 10.$$
Suppose the width fixed-witness fiber has
$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$
the anchored support A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and
$$|S \setminus A| \leq L(q).$$
Then there is a $2q$-bit code
$$\Theta(v, A, \Gamma_{S \setminus A}) \in \{0,1\}^{2q},$$
depending only on the retained deficit vertex v, the interval A, and the deleted-only graph induced on $S \setminus A$, such that Γ is recovered from
$$S, \quad \Gamma^*, \quad \Theta(v, A, \Gamma_{S \setminus A}) \in \{0,1\}^{2q}.$$
The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, for residual rows with $q \geq 45$, after the adjacent-rank crossing condition has been proved, the unabsorbed proper-support deleted-only readout is restricted to

$$|S \setminus A| \geq \left\lfloor \frac{q}{4} \right\rfloor + 11.$$

Proof. By corollary 4.60, every residual row has $j \leq 30899$. Hence the retained deficit vertex has at most $j - q \leq 30899$ choices. Let $L = L(q)$. As in the proof of corollary 4.161, for fixed S the number of triples

$$(v, A, \Gamma_{S\setminus A}) \qquad \text{with} \qquad |S \setminus A| \leq L$$

is at most

$$30899 \sum_{d=0}^{L} (d+1) s_d.$$

By lemma 4.68, this is at most

$$30899\,(L+2)!.$$

We claim that

$$30899\,(L(q)+2)! < 2^{2q} \qquad (45 \leq q \leq 218).$$

At $q = 45$, $L(q) = 21$ and the exact integer inequality is

$$30899 \cdot 23! = 798801465214806893199360000 < 1237940039285380274899124224 = 2^{90}.$$

When q first reaches 48, $L(q)$ increases from 21 to 22; the left side is multiplied by 24, while the right side from $q = 45$ to $q = 48$ is multiplied by $2^6 = 64$. After that, each further increase of $L(q)$ by one requires four increases of q, so the right side is multiplied by $2^8 = 256$. Throughout the residual range $q \leq 218$, one has $L(q) \leq 64$, so the left side is multiplied by at most 66 at each such step. Hence the displayed inequality holds for every $45 \leq q \leq 218$.

Thus the triples $(v, A, \Gamma_{S\setminus A})$ with $|S \setminus A| \leq L(q)$ inject into the 2^{2q} possible branch words. Listing them in lexicographic order gives the code Θ. The code recovers v, A, and the deleted-only graph. Applying corollary 4.150 with these recovered inputs recovers Γ in the width case; the height case is identical. The final sentence is the complement of the displayed packing range. ☐

Corollary 4.163 (Residual one-third joint branch code)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and use the notation of corollary 4.150. Set*

$$H(q) = \left\lceil \frac{q}{3} \right\rceil = \left\lfloor \frac{q+2}{3} \right\rfloor.$$

Suppose the width fixed-witness fiber has

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and

$$|S \setminus A| \leq H(q).$$

Then there is a $2q$-bit code

$$\Upsilon(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q},$$

depending only on the retained deficit vertex v, the interval A, and the deleted-only graph induced on $S \setminus A$, such that Γ is recovered from

$$S, \quad \Gamma^*, \quad \Upsilon(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q}.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved, the unabsorbed proper-support deleted-only readout is restricted to

$$|S \setminus A| \geq \left\lceil \frac{q}{3} \right\rceil + 1.$$

Proof. As in corollary 4.162, it is enough to prove

$$30899\,(H(q)+2)! < 2^{2q} \qquad (15 \leq q \leq 218).$$

For $H(q) = 5$, the only residual case is $q = 15$, and

$$30899 \cdot 7! = 155730960 < 1073741824 = 2^{30}.$$

For $H(q) = 6$, the worst residual case is $q = 16$, and

$$30899 \cdot 8! = 1245847680 < 4294967296 = 2^{32}.$$

It remains to handle $7 \leq h := H(q) \leq 73$. For fixed h, the smallest residual q with $H(q) = h$ is $q = 3h - 2$, so it suffices to prove

$$30899\,(h+2)! < 2^{6h-4}.$$

At $h = 7$ one has the stronger margin

$$8 \cdot 30899 \cdot 9! = 89701032960 < 274877906944 = 2^{38}.$$

For $7 \leq h \leq 61$, increasing h by one multiplies the left side by $h + 3 \leq 64$ and the right side by $2^6 = 64$, so this factor-8 margin is preserved through $h = 61$. For $62 \leq h \leq 73$, the total possible loss of margin from $h = 61$ to $h = 73$ is at most

$$\left(\frac{75}{64}\right)^{12} < 8.$$

Thus the strict inequality still holds for every $62 \leq h \leq 73$. Since $q \leq 218$ gives $H(q) \leq 73$, the displayed bound holds in the whole residual range.

By lemma 4.68, the number of triples $(v, A, \Gamma_{S\setminus A})$ with $|S\setminus A| \leq H(q)$ is then strictly less than 2^{2q}. Listing those triples lexicographically gives the code Υ, and the recovery follows from corollary 4.150 exactly as in corollary 4.162. The height case is identical. □

Corollary 4.164 (High-rank one-third-plus-eight joint branch code)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, assume $q \geq 80$, let $S \subset \{1, \dots, j\}$ have cardinality q, and use the notation of corollary 4.150. Set*

$$P(q) = \left\lceil \frac{q}{3} \right\rceil + 8.$$

Suppose the width fixed-witness fiber has

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and

$$|S \setminus A| \leq P(q).$$

Then there is a $2q$-bit code

$$\Pi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q},$$

depending only on the retained deficit vertex v, the interval A, and the deleted-only graph induced on $S \setminus A$, such that Γ is recovered from

$$S, \quad \Gamma^*, \quad \Pi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q}.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, for residual rows with $q \geq 80$, after the adjacent-rank crossing condition has been proved, the unabsorbed proper-support deleted-only readout is restricted to

$$|S \setminus A| \geq \left\lceil \frac{q}{3} \right\rceil + 9.$$

Proof. As in corollary 4.162, it is enough to prove

$$30899\,(P(q)+2)! < 2^{2q} \qquad (80 \leq q \leq 218).$$

Put $h = P(q)$. Then $35 \leq h \leq 81$. For $h = 35$, the worst residual case is $q = 80$, and the exact integer inequality is

$$30899 \cdot 37! < 2^{160}.$$

For $h = 36$, the worst residual case is $q = 82$, and

$$30899 \cdot 38! < 2^{164}.$$

For $36 \leq h \leq 61$, increasing h by one increases the worst-case q by three, so the right side is multiplied by $2^6 = 64$, while the left side is multiplied by $h+3 \leq 64$. Thus the displayed inequality propagates from $h = 36$ through $h = 61$.

At $h = 61$, we have the stronger margin

$$256 \cdot 30899 \cdot 63! < 2^{314}.$$

For $62 \leq h \leq 81$, each further step multiplies the left side by at most 84 and the right side by 64. The total possible loss of margin is at most

$$\left(\frac{84}{64}\right)^{20} = \left(\frac{21}{16}\right)^{20} < 256,$$

because $(21/16)^4 < 3$. Hence the strict inequality remains valid through $h = 81$, which covers the residual endpoint $q = 218$.

By lemma 4.68, the number of triples $(v, A, \Gamma_{S\setminus A})$ with $|S\setminus A| \leq P(q)$ is then strictly less than 2^{2q}. Listing those triples lexicographically gives the code Π, and the recovery follows from corollary 4.150 exactly as in corollary 4.162. The height case is identical. ☐

Corollary 4.165 (Exact high-rank one-third-plus-ten joint branch code)**.** *Work in one residual row and moment of remark 4.839. Put $q = b+1$, assume $q \geq 86$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and use the notation of corollary 4.150. Set*

$$Q(q) = \left\lceil \frac{q}{3} \right\rceil + 10.$$

Suppose the width fixed-witness fiber has

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and

$$|S \setminus A| \leq Q(q).$$

Then there is a $2q$-bit code

$$\Psi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q},$$

depending only on the retained deficit vertex v, the interval A, and the deleted-only graph induced on $S \setminus A$, such that Γ is recovered from

$$S, \quad \Gamma^*, \quad \Psi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q}.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, for residual rows with $q \geq 86$, after the adjacent-rank crossing condition has been proved, the unabsorbed proper-support deleted-only readout is restricted to

$$|S \setminus A| \geq \left\lceil \frac{q}{3} \right\rceil + 11.$$

Proof. Let

$$M_K = \sum_{d=0}^{K} (d+1) s_d$$

for the stable moment sequence of lemma 4.67. As in corollary 4.162, the number of triples $(v, A, \Gamma_{S\setminus A})$ with $|S \setminus A| \leq Q(q)$ is at most

$$30899\, M_{Q(q)}$$

because $j - q \leq 30899$ in the residual window. It remains to check

$$30899\, M_{\lceil q/3 \rceil + 10} < 2^{2q} \qquad (86 \leq q \leq 218).$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_complement_packing_gmp_certificate.log`

computes the s_d from lemma 4.67, forms the cumulative sums M_K, and checks the displayed integer inequality for all 133 integers q in the range $86 \leq q \leq 218$. The replay was run on `optimus`; it exits with status 0, reports `failures=0`, `q_checked=133`, `k_min=39`, `k_max=83`, and has SHA-256

`bbbfd5c129c4f8d70e10d38dd1dbc5cf`
`24ce4a4cd3491c5e579aeaeb90253125`.

The log reports its closest approximate ratio at $q = 88$, $K = 40$, with

$$\log_{10}\left(30899\, M_{40}/2^{176}\right) \approx -0.115519519802795401 < 0.$$

Therefore the triples with $|S \setminus A| \leq Q(q)$ inject into the 2^{2q} possible branch words. Listing them lexicographically gives the code Ψ, and the recovery follows from corollary 4.150 exactly as in corollary 4.162. The height case is identical. □

Definition 4.166 (Exact residual complement frontier)**.** For $15 \leq q \leq 218$, let

$$M_K = \sum_{d=0}^{K} (d+1) s_d$$

for the stable moment sequence of lemma 4.67, and let $W(q)$ be the residual window bound for $j - q$:

$$W(q) = \begin{cases} 912, & q = 15, \\ 401, & 16 \leq q \leq 17, \\ 287, & q = 18, \\ 283, & 19 \leq q \leq 20, \\ 303, & 21 \leq q \leq 22, \\ 337, & 23 \leq q \leq 24, \\ 423, & 25 \leq q \leq 28, \\ 30899 - q, & q \geq 29. \end{cases}$$

Define $F(q)$ to be the largest integer K with $0 \leq K \leq q - 2$ such that

$$W(q) M_K < 2^{2q}.$$

Corollary 4.167 (Exact residual complement frontier joint branch code)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \dots, j\}$ have cardinality q, and use the notation of corollary 4.150. Let $F(q)$ and $W(q)$ be as in definition 4.166. The exact C++/GMP/OpenMP replay*

certificates/post_m29/bc_complement_frontier_gmp_certificate.log

computes these frontiers for every $15 \le q \le 218$ from the recurrence of lemma 4.67; it exits with status 0, reports failures=0, q_checked=204, min_frontier=9, max_frontier=83, and max_frontier_minus_q=-6. Its SHA-256 is

09ae87166af7f93ae6ef4df0322fcacd
ec86fc6ae5701ce0117851a655643b70.

Its closest accepted ratio is at $q = 62$, $F(q) = 30$, with

$$\log_{10}\bigl(W(62)M_{30}/2^{124}\bigr) = -0.0065440917006611698 5 < 0.$$

Suppose the width fixed-witness fiber has

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the anchored support A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and

$$|S \setminus A| \le F(q).$$

Then there is a $2q$-bit code

$$\Phi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q},$$

depending only on the retained deficit vertex v, the interval A, and the deleted-only graph induced on $S \setminus A$, such that Γ is recovered from

$$S, \quad \Gamma^*, \quad \Phi(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q}.$$

The same assertion holds in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved, the unabsorbed proper-support deleted-only readout is restricted to

$$|S \setminus A| \ge F(q) + 1.$$

Proof. The residual window bound is corollary 4.63; for $q \ge 29$ the final line is the sharper bound $j - q \le 30899 - q$ from corollary 4.60. Hence the retained deficit vertex has at most $W(q)$ possible ranks.

For fixed S, the number of triples

$$(v, A, \Gamma_{S\setminus A}) \qquad \text{with} \qquad |S \setminus A| \le F(q)$$

is at most $W(q)M_{F(q)}$, since for each complement size d there are $d + 1$ interval supports A and s_d loopless 2-regular deleted-only graphs on $S \setminus A$. By the defining inequality for $F(q)$, this number is strictly less than 2^{2q}. Listing those triples lexicographically gives the code Φ.

The code recovers v, A, and the deleted-only graph. Applying corollary 4.150 with these recovered inputs recovers Γ in the width case; the height case is identical. The final sentence is the complement of the exact packed range. ∎

Corollary 4.168 (Combined residual complement threshold)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \dots, j\}$ have cardinality q, and use the notation of* *corollary 4.150. Define*

$$K_0(q)=\begin{cases}9, & q=15,\\ 10, & 16\le q\le 17,\\ 11, & q=18,\\ 12, & 19\le q\le 20,\\ 13, & 21\le q\le 22,\\ 14, & 23\le q\le 24,\\ 15, & q\ge 25,\end{cases}$$

$$L_0(q)=\begin{cases}\lfloor q/4\rfloor+10, & q\ge 45,\\ 0, & 15\le q\le 44,\end{cases}\qquad H_0(q)=\lceil q/3\rceil,$$

$$P_0(q)=\begin{cases}\lceil q/3\rceil+8, & q\ge 80,\\ 0, & 15\le q\le 79,\end{cases}$$

$$Q_0(q)=\begin{cases}\lceil q/3\rceil+10, & q\ge 86,\\ 0, & 15\le q\le 85,\end{cases}$$

let $F_0(q)=F(q)$ be the exact complement frontier of definition 4.166, and

$$R(q)=\max\{K_0(q),L_0(q),H_0(q),P_0(q),Q_0(q),F_0(q)\}.$$

Suppose the width fixed-witness fiber has

$$S=W_q(T)=\{s_1<\cdots<s_q\},$$

the anchored support A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r-r'|=1$, and

$$|S\setminus A|\le R(q).$$

Then there is a $2q$-bit code

$$\Omega(v,A,\Gamma_{S\setminus A})\in\{0,1\}^{2q},$$

depending only on the retained deficit vertex v, the interval A, and the deleted-only graph induced on $S\setminus A$, such that Γ is recovered from

$$S,\quad \Gamma^*,\quad \Omega(v,A,\Gamma_{S\setminus A})\in\{0,1\}^{2q}.$$

The same assertion holds in the height fixed-witness fiber with $S=H_q(T)=\{t_1<\cdots<t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved, the unabsorbed proper-support deleted-only readout is restricted to

$$|S\setminus A|\ge R(q)+1.$$

Proof. Choose one of $K_0(q)$, $L_0(q)$, $H_0(q)$, $P_0(q)$, $Q_0(q)$, or $F_0(q)$ attaining the maximum $R(q)$. If $K_0(q)$ attains the maximum, the assertion is exactly corollary 4.161. If $H_0(q)$ attains the maximum, it is exactly corollary 4.163. If $L_0(q)$ attains the maximum, then $q\ge 45$ and the assertion is exactly corollary 4.162. If $P_0(q)$ attains the maximum, then $q\ge 80$ and the assertion is exactly corollary 4.164. If $Q_0(q)$ attains the maximum, then $q\ge 86$ and the assertion is exactly corollary 4.165. If $F_0(q)$ attains the maximum, the assertion is exactly corollary 4.167. The final sentence is the complement of the combined packing range. ∎

Lemma 4.169 (Witness deletion is consecutive standard-time deletion)**.** *Let T be a semistandard tableau of content (2^j), let $U = \mathrm{std}(T)$, and let*

$$\varnothing = \sigma^{(0)} \subset \sigma^{(1)} \subset \cdots \subset \sigma^{(2j)} = \lambda(T)$$

be the standard shape chain of U, where $\sigma^{(r)}$ is the shape formed by the entries at most r. Let

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)} = \lambda(T)$$

be the Pieri two-strip path of lemma 4.97. Then

$$\sigma^{(2a)} = \lambda^{(a)} \quad \text{for } 0 \le a \le j,$$

and each pair

$$\sigma^{(2a-2)} \subset \sigma^{(2a-1)} \subset \sigma^{(2a)}$$

adds exactly the two boxes of semistandard label a.

Consequently, for $S \subset \{1, \dots, j\}$, deleting all semistandard labels in S is the same box deletion as deleting the consecutive standard-time pairs

$$D(S) = \bigcup_{s \in S} \{2s - 1, 2s\}$$

from U. The set $D(S)$ is a union of at most $|S|$ maximal consecutive intervals of standard times, and these intervals are exactly the doubled standard-time form of the deletion joints in definition 4.249.

Proof. By lemma 4.139, the boxes carrying semistandard label a in T are precisely the boxes carrying standard entries $2a - 1$ and $2a$ in U. Therefore the shape formed by entries at most $2a$ in U is exactly the shape formed by semistandard labels at most a in T, which is $\lambda^{(a)}$ by lemma 4.97. This proves $\sigma^{(2a)} = \lambda^{(a)}$ and identifies the two single-box additions between times $2a - 2$ and $2a$ with the two boxes of label a.

The deletion statement follows from the same identification of boxes. Since each deleted semistandard label contributes the adjacent standard pair $\{2s - 1, 2s\}$, $D(S)$ is a union of at most $|S|$ maximal consecutive intervals. Under the label-to-block conversion used in definition 4.249, maximal consecutive deleted labels are the same deleted objects as maximal consecutive deleted two-step blocks, with the order reversed by lemma 4.118. Reversing order preserves the number and endpoint nature of the intervals, so these are exactly the same deletion joints in standard-time coordinates. □

Lemma 4.170 (Low-label boxes determine the standard even readout)**.** *Let T be a semistandard tableau of content (2^j), let $U = \mathrm{std}(T)$, and let*

$$\varnothing = \sigma^{(0)} \subset \sigma^{(1)} \subset \cdots \subset \sigma^{(2j)} = \lambda(T)$$

be the standard shape chain of U. Fix $M \le j$. Any datum which determines, for each $1 \le a \le M$, the two boxes of T carrying semistandard label a, determines the even standard-time endpoints

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \dots, \sigma^{(2M)}.$$

The readout is

$$\sigma^{(2a)} = \bigcup_{1 \le u \le a} \{\text{boxes of } T \text{ carrying label } u\} \qquad (0 \le a \le M),$$

with the empty union for $a = 0$.

Proof. By lemma 4.139, the boxes carrying semistandard label u in T are exactly the boxes carrying the adjacent standard pair $\{2u - 1, 2u\}$ in U. Therefore the boxes of U with entries at most $2a$ are precisely the union of the boxes of T carrying labels $1, \dots, a$. Since $\sigma^{(2a)}$ is the shape formed by entries at most $2a$ in U, the displayed formula follows for every $a \le M$. □

Lemma 4.171 (Step-2 Knuth prefixes determine low-label boxes)**.** *Use the Step 2 first arbitrary-filling Knuth growth diagram in the proof of* [8, Theorem 4] *for a semistandard tableau T of content* (2^j)*. Let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$

be the Pieri two-strip path of T. Fix $M \le j$. Any datum which determines the Step 2 boundary prefix

$$\lambda^{(0)}, \lambda^{(1)}, \dots, \lambda^{(M)}$$

determines, for each $1 \le a \le M$, the two boxes of T carrying semistandard label a.

Moreover, the same conclusion holds if the datum determines a Ferrers subarrangement C of the Step 2 first-variant growth diagram, the cell entries of C, and the left/bottom boundary labels of C, and the forward boundary sweep on C contains the displayed prefix.

Proof. By lemma 4.97, the two boxes carrying label a are exactly the skew difference

$$\lambda^{(a)} \setminus \lambda^{(a-1)}.$$

Thus the displayed boundary prefix determines those two-box sets for every $a \le M$.

For the final assertion, apply the forward part of lemma 4.119 to the known cell entries and known left/bottom boundary labels of C. Sweeping across C determines its top/right boundary labels. If that boundary contains the displayed prefix, the first paragraph recovers the low-label boxes. □

Lemma 4.172 (Step-2 Knuth intervals determine label boxes)**.** *Use the Step 2 first arbitrary-filling Knuth growth diagram in the proof of* [8, Theorem 4] *for a semistandard tableau T of content* (2^j)*. Let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$

be the Pieri two-strip path of T. Fix $1 \le a \le b \le j$. Any datum which determines the Step 2 boundary interval

$$\lambda^{(a-1)}, \lambda^{(a)}, \dots, \lambda^{(b)}$$

determines, for every $a \le s \le b$, the two boxes of T carrying semistandard label s.

Moreover, the same conclusion holds if the datum determines a Ferrers subarrangement C of the Step 2 first-variant growth diagram, the cell entries of C, and the left/bottom boundary labels of C, and the forward boundary sweep on C contains the displayed boundary interval.

Proof. By lemma 4.97, the two boxes carrying label s are exactly the skew difference

$$\lambda^{(s)} \setminus \lambda^{(s-1)}.$$

The displayed boundary interval contains both endpoints of this difference for every $a \le s \le b$, so it determines all listed two-box label sets.

For the final assertion, apply the forward part of lemma 4.119 to the known cell entries and known left/bottom boundary labels of C. Sweeping across C determines its top/right boundary labels. If that boundary contains the displayed interval, the first paragraph recovers the label boxes. □

Lemma 4.173 (Step-2 Knuth continuations determine continuation boxes)**.** *Use the Step 2 first arbitrary-filling Knuth growth diagram in the proof of* [8, Theorem 4] *for a semistandard tableau T of content* (2^j)*. Let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$

be the Pieri two-strip path of T. Fix $M \le j$. Any datum which determines the Step 2 boundary continuation

$$\lambda^{(M)}, \lambda^{(M+1)}, \dots, \lambda^{(j)}$$

determines, for each $M < t \le j$, the two boxes of T carrying semistandard label t.

Moreover, the same conclusion holds if the datum determines a Ferrers subarrangement C of the Step 2 first-variant growth diagram, the cell entries of C, and the left/bottom boundary labels of C, and the forward boundary sweep on C contains the displayed continuation segment.

Proof. By lemma 4.97, the two boxes carrying label t are exactly the skew difference

$$\lambda^{(t)} \setminus \lambda^{(t-1)}.$$

Thus the displayed boundary continuation determines those two-box sets for every $M < t \leq j$.

For the final assertion, apply the forward part of lemma 4.119 to the known cell entries and known left/bottom boundary labels of C. Sweeping across C determines its top/right boundary labels. If that boundary contains the displayed continuation segment, the first paragraph recovers the labelled continuation boxes. □

Lemma 4.174 (Step-2 Knuth matrix is unique from the labelled tableau)**.** *Use the Step 2 first arbitrary-filling Knuth growth diagram in the proof of* [8, Theorem 4]*. For a fixed semistandard tableau T in that construction, the Step 2 symmetric nonnegative integer matrix is uniquely determined by T.*

Proof. In [8, Theorem 4, Step 2], the first arbitrary-filling Knuth construction of [9, §4.1, Theorem 7] is applied in the inverse direction to a pair of equal semistandard boundary tableaux. The equality of the two tableaux is exactly the symmetry statement recorded in the source for the resulting matrix. Hence the labelled tableau T determines both the P- and Q-side top/right boundary paths of the full Step 2 rectangular growth diagram.

By lemma 4.119, the backward local algorithm for this first arbitrary-filling variant determines the southwest corner and the cell entry of each cell from the other three corner labels. Starting from the full top/right boundary determined by T and sweeping backward through the whole rectangle therefore recovers every Step 2 cell entry. This recovered filling is the unique Step 2 matrix associated to T. □

Lemma 4.175 (Step-2 Q-prefix strips determine boundary prefixes)**.** *Use the Step 2 first arbitrary-filling Knuth growth diagram in the proof of* [8, Theorem 4]*, and orient it as the rectangular first-variant RSK diagram of* [9, §4.1, Theorem 7]*. Let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$

be the Q-side semistandard boundary path of this diagram; equivalently, $\lambda^{(a)}/\lambda^{(a-1)}$ is the set of boxes of the recording tableau carrying label a. For $M \leq j$, let C_M be the Step 2 rectangular subarrangement consisting of the first M cells in the Q-coordinate direction and all cells in the opposite coordinate direction. If the entries of C_M are known, then

$$\lambda^{(0)}, \lambda^{(1)}, \ldots, \lambda^{(M)}$$

are determined.

Proof. In the rectangular case, [9, §4.1, Theorem 7] identifies the first arbitrary-filling growth diagram with RSK. The first half of the top/right boundary sequence is the recording tableau Q: the boxes added in the difference $\lambda^{(a)}/\lambda^{(a-1)}$ are precisely the boxes carrying the label a in Q. The subarrangement C_M is anchored on the global empty left/bottom boundary, so those boundary labels are all $\varnothing$. Applying the forward local rules of lemma 4.119 to the known entries of C_M therefore determines the corresponding top/right boundary labels, and these include exactly $\lambda^{(0)}, \ldots, \lambda^{(M)}$. □

Lemma 4.176 (Step-2 Q-suffix strips determine boundary continuations)**.** *Use the Step 2 first arbitrary-filling Knuth growth diagram and notation of lemma 4.175. For $M \leq j$, let C_M^+ be the rectangular subarrangement consisting of the cells whose Q-coordinate label lies in*

$$\{M+1, \ldots, j\}$$

and whose opposite coordinate is arbitrary. Suppose a datum determines the cell entries of C_M^+ and the partition labels on the left/bottom boundary of C_M^+. Then the same datum determines the Q-side *continuation boundary*

$$\lambda^{(M)}, \lambda^{(M+1)}, \dots, \lambda^{(j)}.$$

Proof. The subarrangement C_M^+ is a finite rectangular Ferrers cell arrangement in the first arbitrary-filling Knuth growth diagram. Applying the forward local rules of lemma 4.119 to the known entries and known left/bottom boundary labels determines the top/right boundary labels of C_M^+. In the orientation of lemma 4.175, the part of this boundary in the Q-coordinate direction is exactly the segment of the recording boundary after the first M labels, namely

$$\lambda^{(M)}, \lambda^{(M+1)}, \dots, \lambda^{(j)}.$$

□

Lemma 4.177 (Only the internal Q-cut of a suffix strip is non-global)**.** *Use the Step 2 first arbitrary-filling Knuth growth diagram and notation of lemma 4.176. The left/bottom boundary of the suffix strip C_M^+ is the union of:*

(1) *the part lying on the global empty left/bottom boundary of the full Step 2 rectangle; and*
(2) *the internal separating boundary between the prefix strip C_M and the suffix strip C_M^+.*

The labels on the first part are all $\varnothing$. Consequently any datum which determines the ordered partition labels on the internal separating boundary determines the full left/bottom boundary of C_M^+.

Proof. The strips C_M and C_M^+ are obtained by cutting the full Step 2 rectangle between the Q-coordinate labels M and $M+1$. The suffix strip still contains every cell in the opposite coordinate direction. Hence the incoming boundary for a forward sweep on C_M^+ has only two kinds of edges: those which already lie on the global incoming boundary of the full rectangle, and those created by the cut from the removed prefix strip. The first arbitrary filling RSK rectangle is anchored on the global empty left/bottom boundary, so the labels of the first kind are all $\varnothing$. The labels of the second kind are exactly the ordered internal cut labels. Knowing those ordered cut labels therefore supplies every non-fixed label on the left/bottom boundary of C_M^+. □

Lemma 4.178 (Q-prefix entries determine the internal Q-cut)**.** *Use the Step 2 first arbitrary-filling Knuth growth diagram and notation of lemmas 4.175 and 4.177. If a datum determines all cell entries of the Q-prefix strip C_M, then it determines the ordered partition labels on the internal separating boundary between C_M and C_M^+.*

Proof. The prefix strip C_M is anchored on the global empty left/bottom boundary of the full Step 2 rectangle. Applying the forward local rules of lemma 4.119 to the known entries of C_M therefore determines every boundary label produced by the forward sweep on C_M. The boundary component of C_M created by cutting the full rectangle between the Q-coordinate labels M and $M+1$ is exactly the internal separating boundary between C_M and C_M^+, so its ordered labels are determined. □

Lemma 4.179 (Low-label incidence cells cover the Step-2 Q-prefix strip)**.** *Use the notation of lemma 4.175 and lemma 4.127. For $M \le j$, let H_M be the Step 3 half-square cell set consisting of the cells corresponding to unordered off-diagonal Step 2 label pairs*

$$\{a,b\} \subset \{1,\dots,j\}, \qquad a \neq b, \qquad \min(a,b) \le M.$$

Then every non-diagonal cell of the Step 2 Q-prefix strip C_M has its cell or its transpose represented in H_M. Consequently, recovering the Step 3 subfilling on any carrier containing H_M determines all non-diagonal entries of C_M.

Proof. The strip C_M consists of the Step 2 cells whose Q-coordinate label lies in $\{1,\dots,M\}$. A non-diagonal cell of this strip therefore has ordered label coordinates (a,b) with $a \le M$ and $a \neq b$. The Step 2 matrix is symmetric, and Step 3 keeps exactly one off-diagonal representative of the unordered pair $\{a,b\}$. This representative belongs to H_M because $\min(a,b) \le a \le M$. The last sentence is then lemma 4.127. □

Corollary 4.180 (Low-label incidence entries determine the internal Q-cut). *Use the Step 2 and Step 3 notation of lemma 4.179. If a datum determines the Step 3 cell entries on every cell of H_M, then it determines the ordered partition labels on the internal separating boundary between the Step 2 Q-prefix strip C_M and the suffix strip C_M^+.*

Proof. By lemma 4.179, the recovered entries on H_M determine every non-diagonal entry of the Step 2 Q-prefix strip C_M. The diagonal entries of the Step 2 matrix are zero by lemma 4.127. Hence all cell entries of C_M are determined. Applying lemma 4.178 gives the ordered internal cut labels. □

Lemma 4.181 (Continuation incidence cells cover the Step-2 Q-suffix strip). *Use the notation of lemma 4.176 and lemma 4.127. For $M \le j$, let J_M be the Step 3 half-square cell set corresponding to unordered off-diagonal Step 2 label pairs*

$$\{a, b\} \subset \{1, \dots, j\}, \qquad a \neq b, \qquad \max(a, b) > M.$$

Then every non-diagonal cell of the Step 2 Q-suffix strip C_M^+ has its cell or its transpose represented in J_M. Consequently, recovering the Step 3 subfilling on any carrier containing J_M determines all non-diagonal entries of C_M^+; the diagonal entries of C_M^+ are zero.

Moreover, in the Step 3 label order $j, j-1, \dots, 1$, the set J_M is exactly the first $j - M$ strict-left columns, namely the columns incident to the continuation labels $M + 1, \dots, j$.

Proof. A non-diagonal cell of the Step 2 Q-suffix strip has ordered label coordinates (a, b) with $a > M$ and $a \neq b$. Since the Step 2 matrix is symmetric and Step 3 retains one representative of each unordered off-diagonal pair, that cell or its transpose is represented by the Step 3 cell for the unordered pair

$$\{a, b\}.$$

This pair belongs to J_M because $\max(a, b) \ge a > M$. The entries agree by lemma 4.127, and the diagonal entries of the Step 2 matrix are zero by the same construction.

In the Step 3 half-square the labels are ordered as $j, j-1, \dots, 1$. A cell whose unordered pair has maximum label greater than M is incident to one of the labels $M + 1, \dots, j$, and in the retained strict half-square it lies in the column of such a continuation label. These are precisely the first $j - M$ strict-left columns. □

Lemma 4.182 (Low-label incidence splits into cross and low-low cells). *Use the Step 3 half-square labelling of lemma 4.179. For $M \le j$, let X_M be the cell set corresponding to unordered pairs*

$$\{a, b\}, \qquad \min(a, b) \le M < \max(a, b),$$

and let L_M be the cell set corresponding to unordered pairs

$$\{a, b\}, \qquad 1 \le a < b \le M.$$

Then

$$H_M = X_M \sqcup L_M.$$

Moreover, if the Step 3 half-square is ordered by labels $j, j-1, \dots, 1$ as in [8, Theorem 4]*, then X_M lies in the first $j - M$ strict-left columns, namely the columns incident to the continuation labels $M + 1, \dots, j$.*

Proof. The definition of H_M says that an unordered off-diagonal pair belongs to it exactly when at least one of its two labels is at most M. Such a pair has either both labels at most M, which is precisely L_M, or exactly one label at most M, which is precisely X_M. These alternatives are disjoint and exhaust H_M.

In Krattenthaler's Step 3 half-square the labels are ordered $j, j-1, \dots, 1$. A cell of X_M has one incident label in $M + 1, \dots, j$; the half-square representative of that unordered pair appears in the strict-left column indexed by that continuation label. Hence every cell of X_M lies in the first $j - M$ strict-left columns. □

Lemma 4.183 (Low-low cells are a terminal dual-RSK' corner)**.** *Use the Step 3 dual-RSK' half-square of lemma 4.125 and the notation of lemma 4.182. For $M \le j$, the cell set L_M is the terminal triangular Ferrers subarrangement on the labels $M, M-1, \dots, 1$ in the Step 3 label order $j, j-1, \dots, 1$. Its top/right boundary in the full Step 3 diagram is the terminal segment*

$$\rho_{2(j-M)}, \rho_{2(j-M)+1}, \dots, \rho_{2j}$$

of the semistandard growth path. Consequently, any data determining this boundary segment determine every Step 3 entry on L_M and every corner label on or inside L_M.

Proof. The Step 3 half-square is indexed by the ordered labels $j, j-1, \dots, 1$; its cells are the off-diagonal unordered pairs of distinct labels, represented in one triangular half. The subset with both incident labels at most M is therefore the principal triangular corner cut out by the terminal ordered label list $M, M-1, \dots, 1$. This is exactly the cell set L_M from lemma 4.182.

By lemma 4.125, the sequence $\rho_0, \rho_1, \dots, \rho_{2j}$ is read along the top/right boundary of this Step 3 diagram. The labels $j, j-1, \dots, M+1$ occupy the first $j-M$ label positions, each contributing two boundary steps in the content (2^j) specialization of lemma 4.118. Hence the boundary of the remaining terminal M-label corner begins at $\rho_{2(j-M)}$ and ends at ρ_{2j}, giving the displayed segment. Applying corollary 4.124 to this terminal Ferrers subarrangement recovers its cell entries and internal corner labels from that top/right boundary. □

Corollary 4.184 (The terminal low-low carrier is exactly the terminal-boundary readout)**.** *Use the notation of lemma 4.183. Let $\mathfrak{A}$ be any datum. The datum $\mathfrak{A}$ determines the terminal boundary segment*

$$\rho_{2(j-M)}, \rho_{2(j-M)+1}, \dots, \rho_{2j}$$

if and only if $\mathfrak{A}$ identifies the terminal Ferrers subarrangement L_M and determines the partition labels on its top/right boundary. Thus the direct low-low carrier route with $E = L_M$ is precisely the terminal-boundary readout. A larger parsed carrier can be a source only by recovering this terminal boundary after the backward sweep.

Proof. Lemma 4.183 identifies L_M as the terminal triangular Ferrers subarrangement and identifies its top/right boundary with the displayed boundary segment. Therefore a datum determines that boundary segment exactly when it determines the top/right boundary labels of this particular Ferrers subarrangement. The final sentence is this equivalence applied to the carrier choice $E = L_M$. □

Corollary 4.185 (Any parsed carrier containing the low-low corner recovers the terminal boundary)**.** *Use the notation of lemma 4.183. Let E be a Ferrers subarrangement of the Step 3 dual-RSK' half-square which contains the terminal low-low subarrangement L_M. If a datum determines E and the partition labels on the top/right boundary of E, then it determines the terminal boundary segment*

$$\rho_{2(j-M)}, \rho_{2(j-M)+1}, \dots, \rho_{2j}.$$

Consequently every direct carrier source for L_M is, after the dual-RSK' backward sweep, a terminal-boundary source.

Proof. Applying corollary 4.124 to the determined carrier E recovers every corner label on or inside E. Since $L_M \subseteq E$, the top/right boundary labels of the subarrangement L_M are among these recovered corner labels. By lemma 4.183, that top/right boundary is exactly the displayed terminal segment of the Step 3 semistandard path. □

Lemma 4.186 (Consecutive Step–3 label intervals have boundary segments)**.** *Use the Step–3 dual-RSK' half-square of lemma 4.125. Let*

$$1 \le a \le b \le j$$

and let $E_{a,b}$ be the Ferrers subarrangement on the consecutive Step–3 labels

$$b, b-1, \dots, a$$

inside the global label order $j, j-1, \dots, 1$. Then the top/right boundary of $E_{a,b}$ in the full Step–3 diagram is the boundary segment

$$\rho_{2(j-b)}, \rho_{2(j-b)+1}, \dots, \rho_{2(j-a+1)}.$$

Consequently, if a parser determines the partition labels on this boundary, then it determines every Step–3 cell entry and every corner label on or inside $E_{a,b}$.

Proof. In the Step–3 half-square, the boundary labels are ordered by $j, j-1, \dots, 1$, and in the content (2^j) specialization each label contributes two boundary steps to the path

$$\rho_0, \rho_1, \dots, \rho_{2j}.$$

The labels preceding the interval $b, b-1, \dots, a$ are $j, j-1, \dots, b+1$, of which there are $j-b$. Hence the interval begins at the boundary label $\rho_{2(j-b)}$. The interval itself has $b-a+1$ labels, so its boundary segment ends after $2(b-a+1)$ additional steps, namely at

$$\rho_{2(j-b)+2(b-a+1)} = \rho_{2(j-a+1)}.$$

This proves the displayed segment.

The final statement is exactly corollary 4.124 applied to the consecutive-label Ferrers subarrangement $E_{a,b}$. ☐

Corollary 4.187 (Consecutive Step–3 carrier data recover interval boundaries)**.** *In the setting of lemma 4.186, suppose a parser determines the cell entries of $E_{a,b}$ and the partition labels on the left/bottom boundary of $E_{a,b}$. Then the same data determine the boundary segment*

$$\rho_{2(j-b)}, \rho_{2(j-b)+1}, \dots, \rho_{2(j-a+1)}.$$

Proof. Apply the forward determination in lemma 4.121 to the finite subarrangement $E_{a,b}$. Its cell entries together with its left/bottom boundary labels determine its top/right boundary labels. By lemma 4.186, that top/right boundary is exactly the displayed segment. ☐

Lemma 4.188 (Deleted intervals carry ordered frontier events)**.** *Let T be a semistandard tableau of content (2^j), let $U = \mathrm{std}(T)$, and let*

$$\varnothing = \sigma^{(0)} \subset \sigma^{(1)} \subset \cdots \subset \sigma^{(2j)} = \lambda(T)$$

be the standard shape chain of U.

In the width case assume $\lambda_1(T) \geq 2q$, put $s_r = T(1, 2r-1)$ and $S = W_q(T)$. Write the maximal consecutive label intervals in S as

$$J_\alpha = \{a_\alpha, a_\alpha + 1, \dots, b_\alpha\}, \qquad 1 \leq \alpha \leq L,$$

with $b_\alpha + 1 < a_{\alpha+1}$ for $\alpha < L$. Then the corresponding maximal standard-time components of $D(S)$ are

$$\widetilde{J}_\alpha = \{2a_\alpha - 1, 2a_\alpha, \dots, 2b_\alpha\}.$$

For

$$E^w_\alpha = \{\kappa^w_r : s_r \in J_\alpha\},$$

one has

$$E^w_\alpha \subset \widetilde{J}_\alpha, \qquad |E^w_\alpha| = |J_\alpha|,$$

and the elements of E^w_α occur in the same order as the labels in J_α. If $\alpha < \beta$, then every element of E^w_α is smaller than every element of E^w_β. Under the raw block deletion definition 4.249, the same interval J_α becomes the deleted block interval

$$\{j - b_\alpha + 1, \dots, j - a_\alpha + 1\}.$$

The same conclusions hold in the height case, with $t_r = T(2r-1, 1)$*,* $S = H_q(T)$*, and* $E^h_\alpha = \{\kappa^h_r : t_r \in J_\alpha\}$*.*

Proof. The maximal consecutive intervals in S are disjoint, ordered by label, and their union is S. Since

$$D(S) = \bigcup_{s \in S} \{2s-1, 2s\},$$

the labels in $J_\alpha = \{a_\alpha, \dots, b_\alpha\}$ contribute exactly $\{2a_\alpha - 1, 2a_\alpha, \dots, 2b_\alpha\}$. For $\alpha < L$, the gap $b_\alpha + 1 < a_{\alpha+1}$ leaves at least the missing pair $\{2b_\alpha + 1, 2b_\alpha + 2\}$ before the next component. These are therefore precisely the maximal standard-time components of $D(S)$.

For the width case, lemma 4.142 gives $\kappa^w_r \in \{2s_r - 1, 2s_r\} \subset D(S)$ and $\kappa^w_1 < \cdots < \kappa^w_q$. Hence the events whose labels lie in J_α lie in $\widetilde{J}_\alpha$, one event for each label, and their order is the label order. The separation of the $\widetilde{J}_\alpha$ gives the stated ordering between different intervals. The height case is identical, using the height half of lemma 4.142.

Finally, the raw block deletion uses the label-to-block conversion $s \mapsto j - s + 1$ from definition 4.249. Thus a consecutive label interval $\{a_\alpha, \dots, b_\alpha\}$ maps to the consecutive block interval $\{j - b_\alpha + 1, \dots, j - a_\alpha + 1\}$, with order reversed. □

Lemma 4.189 (Consecutive witness labels have consecutive frontier ranks)**.** *In the width case of lemma 4.188, write*

$$W_q(T) = \{s_1 < s_2 < \cdots < s_q\}.$$

Let $J_\alpha = \{a_\alpha, \dots, b_\alpha\}$ *be a maximal consecutive label interval in* $W_q(T)$*, put* $h_\alpha = |J_\alpha|$*, and let* p_α *be the unique index with* $s_{p_\alpha} = a_\alpha$*. Then*

$$s_{p_\alpha + i - 1} = a_\alpha + i - 1 \qquad (1 \le i \le h_\alpha),$$

and the ordered frontier-event list at J_α *is*

$$E^w_\alpha = \{\kappa^w_{p_\alpha}, \kappa^w_{p_\alpha+1}, \dots, \kappa^w_{p_\alpha+h_\alpha-1}\}.$$

Moreover, for $1 \le i \le h_\alpha$*,*

$$\sigma_1^{(\kappa^w_{p_\alpha+i-1} - 1)} < 2(p_\alpha + i - 1) - 1 \le \sigma_1^{(\kappa^w_{p_\alpha+i-1})}.$$

In the height case, with

$$H_q(T) = \{t_1 < t_2 < \cdots < t_q\},$$

the analogous assertion holds: if $t_{p_\alpha} = a_\alpha$*, then* $t_{p_\alpha+i-1} = a_\alpha + i - 1$*, the ordered event list is*

$$E^h_\alpha = \{\kappa^h_{p_\alpha}, \kappa^h_{p_\alpha+1}, \dots, \kappa^h_{p_\alpha+h_\alpha-1}\},$$

and

$$\ell(\sigma^{(\kappa^h_{p_\alpha+i-1} - 1)}) < 2(p_\alpha + i - 1) - 1 \le \ell(\sigma^{(\kappa^h_{p_\alpha+i-1})})$$

for $1 \le i \le h_\alpha$*.*

Proof. We prove the width statement; the height statement is identical with t_r, κ^h_r in place of s_r, κ^w_r. The labels $s_1 < \cdots < s_q$ list $W_q(T)$ in increasing order. Since $J_\alpha = \{a_\alpha, \dots, b_\alpha\}$ is an interval contained in $W_q(T)$, the labels of $W_q(T)$ that belong to J_α are exactly

$$a_\alpha, a_\alpha + 1, \dots, b_\alpha.$$

They occur in the same increasing order in the list $s_1, \dots, s_q$. Therefore, starting from the unique index p_α with $s_{p_\alpha} = a_\alpha$, one has $s_{p_\alpha+i-1} = a_\alpha + i - 1$ for $1 \le i \le h_\alpha$.

By definition in lemma 4.188, $E^w_\alpha = \{\kappa^w_r : s_r \in J_\alpha\}$, so the preceding identification of the ranks gives the displayed event list. Finally, lemma 4.141 gives the displayed first-row crossing inequality for each rank $p_\alpha + i - 1$. □

Corollary 4.190 (Joint frontier-slot concatenation is the full frontier list)**.** *In the width case of lemma 4.188, write the maximal consecutive label intervals in $S = W_q(T)$ as*

$$J_\alpha = \{a_\alpha, \dots, b_\alpha\} \qquad (1 \le \alpha \le L),$$

and write

$$E^w_\alpha = \{e^w_{\alpha,1} < \cdots < e^w_{\alpha,h_\alpha}\}$$

for the ordered frontier-event set at J_α. Then the concatenated list

$$e^w_{1,1}, \dots, e^w_{1,h_1}, e^w_{2,1}, \dots, e^w_{2,h_2}, \dots, e^w_{L,1}, \dots, e^w_{L,h_L}$$

is exactly

$$\kappa^w_1 < \kappa^w_2 < \cdots < \kappa^w_q.$$

Consequently two entries adjacent in this concatenated joint-slot list are consecutive entries in the full width frontier list.

The same assertion holds in the height case, with $S = H_q(T)$ and the ordered frontier list $\kappa^h_1 < \cdots < \kappa^h_q$.

Proof. The intervals $J_1, \dots, J_L$ are the maximal consecutive components of S, so they are disjoint and their union is S. By lemma 4.188, each E^w_α consists exactly of the frontier events attached to labels in J_α, has cardinality $h_\alpha = |J_\alpha|$, and every event in E^w_α is smaller than every event in E^w_β when $\alpha < \beta$. Therefore the displayed concatenation is strictly increasing and contains precisely one frontier event for each label in S.

Lemma 4.142 says that the full width frontier events attached to the labels of $S = W_q(T)$ are precisely $\kappa^w_1 < \cdots < \kappa^w_q$ in increasing order. The concatenated list and the full frontier list are increasing lists with the same elements, hence they are equal term by term. Adjacent entries of the concatenated list are therefore adjacent entries of the full frontier list. The height case uses the height part of lemma 4.188 and lemma 4.142 in the same way. ☐

Proposition 4.191 (Joint-slot adjacent graph edges close the single-deficit graph leaf)**.** *Work in one residual row and moment of remark 4.839. Put $q = b+1$ and let $S \subset \{1, \dots, j\}$ have cardinality q. Work in the setting of proposition 4.81, and let $A \subset S$ be the support of the anchored component in lemma 4.79.*

In the width fixed-witness fiber, suppose $S = W_q(T)$ and form the concatenated joint-slot list

$$e^w_{1,1}, \dots, e^w_{1,h_1}, e^w_{2,1}, \dots, e^w_{2,h_2}, \dots, e^w_{L,1}, \dots, e^w_{L,h_L}$$

from the ordered frontier-event sets of the maximal witness-label intervals, as in corollary 4.190. Assume that every internal edge of the anchored path on A and every edge of the deleted-only graph $\Gamma_{S\setminus A}$ has endpoints whose assigned frontier events are adjacent entries of this concatenated joint-slot list. Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q.$$

The same assertion holds in the height fixed-witness fiber, with $S = H_q(T)$ and the height joint-slot list. Consequently the graph part of the repaired fixed-witness supplier is reduced to proving this joint-slot adjacency condition for the actual target-size parser.

Proof. In the width case, corollary 4.190 identifies the concatenated joint-slot list with the full ordered frontier list

$$\kappa^w_1 < \cdots < \kappa^w_q.$$

Thus the displayed joint-slot adjacency hypothesis is exactly the full-frontier consecutiveness hypothesis for the anchored internal edges and the deleted-only complement edges in proposition 4.156. That proposition recovers Γ from the displayed data.

The height case is identical, using the height half of corollary 4.190 and the list $\kappa^h_1 < \cdots < \kappa^h_q$. ☐

Proposition 4.192 (Combined complement packing leaves only large joint-slot adjacency)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and let $R(q)$ be the complement threshold of corollary 4.168. Work in the single-deficit setting of proposition 4.81, and let $A \subset S$ be the support of the anchored component in lemma 4.79.*

In the width fixed-witness fiber with

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

the graph recovery has the following two supplier alternatives.

(1) *If A is a nonempty interval of S, every internal edge of the anchored path joins two labels $s_r, s_{r'}$ with $|r - r'| = 1$, and*
$$|S \setminus A| \leq R(q),$$
then Γ is recovered from
$$S, \quad \Gamma^*, \quad \Omega(v, A, \Gamma_{S\setminus A}) \in \{0,1\}^{2q}$$
for the combined complement code of corollary 4.168.

(2) *If $|S \setminus A| \geq R(q)+1$ and every internal edge of the anchored path on A and every edge of the deleted-only graph $\Gamma_{S\setminus A}$ has endpoints whose assigned frontier events are adjacent entries of the concatenated joint-slot list of corollary 4.190, then Γ is recovered from*
$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q.$$

The same two alternatives hold in the height fixed-witness fiber with $S = H_q(T) = \{t_1 < \cdots < t_q\}$.

Consequently, after the adjacent-rank crossing condition has been proved and the combined complement code has been used, the graph-side source-extraction leaf for the repaired fixed-witness supplier is precisely the actual target-size parser theorem asserting the joint-slot adjacency in item (2) for the complementary range $|S \setminus A| \geq R(q) + 1$.

Proof. The first width alternative is exactly corollary 4.168. The second width alternative is exactly proposition 4.191. The height alternatives are the height halves of the same two results. The final sentence is the partition of the proper-support deleted-only complement into $|S \setminus A| \leq R(q)$ and $|S \setminus A| \geq R(q) + 1$. □

Lemma 4.193 (The all-but-one complement is beyond the combined threshold)**.** *For every residual rank parameter $15 \leq q \leq 218$, the combined threshold $R(q)$ of corollary 4.168 satisfies*

$$R(q) \leq q - 2.$$

Consequently, if A is a one-point subset of a q-element witness set S, then

$$|S \setminus A| = q - 1 \geq R(q) + 1.$$

Proof. Use the definitions of $K_0, L_0, H_0, P_0, Q_0, F_0$ in corollary 4.168. The exact frontier verifier cited in corollary 4.167 reports

$$F_0(q) - q \leq -6$$

throughout the residual range, so $F_0(q) \leq q - 2$. For $15 \leq q \leq 24$, the displayed values of K_0 give

q	15	16, 17	18	19, 20	21, 22	23, 24
$K_0(q)$	9	10	11	12	13	14,

and the functions L_0, H_0, P_0, Q_0 are no larger than these values or zero in this range, while each listed value is at most $q-2$. For $25 \leq q \leq 44$, one has $K_0(q) = 15$, $L_0(q) = P_0(q) = Q_0(q) = 0$, and $H_0(q) = \lceil q/3 \rceil \leq 15$, so $R(q) \leq 15 \leq q - 2$. For $45 \leq q \leq 79$,

$$L_0(q) = \lfloor q/4 \rfloor + 10 \leq q - 2, \qquad H_0(q) = \lceil q/3 \rceil \leq q - 2,$$

and $K_0(q) = 15 \leq q - 2$, while $P_0(q) = Q_0(q) = 0$. For $80 \leq q \leq 85$, the only new threshold is $P_0(q) = \lceil q/3 \rceil + 8 \leq q - 2$, and the earlier bounds remain valid. For $86 \leq q \leq 218$, the largest possible new threshold is $Q_0(q) = \lceil q/3 \rceil + 10 \le q-2$, and the earlier bounds remain valid. Together with the bound for F_0, every entry in the maximum defining $R(q)$ is at most $q-2$. $\square$

Proposition 4.194 (Large-complement adjacency is not graph-formal)**.** *Let $15 \le q \le 218$ and let*

$$S = \{s_1 < s_2 < \cdots < s_q\}$$

be a linearly ordered witness set. There is a loopless labelled 2-regular multigraph Γ in the single-deficit retained-repair setting of proposition 4.81 with anchored support

$$A = \{s_1\},$$

such that

$$|S \setminus A| \ge R(q) + 1,$$

but the deleted-only graph $\Gamma_{S\setminus A}$ has an edge whose endpoints are not adjacent entries of the full ordered frontier list, equivalently not adjacent entries of the concatenated joint-slot list of corollary 4.190.

Consequently the large-complement joint-slot adjacency hypothesis in proposition 4.192 is not a formal consequence of the retained-repair graph data. A source-extraction proof of that hypothesis must use the actual target-size parser to exclude these deleted-only graph choices, or must replace adjacency by an explicit target-window datum that recovers the deleted-only graph.

Proof. Let $U = \{u_1, u_2\}$ be disjoint from $S \cup \{v\}$ and put a double edge between u_1 and u_2. Put

$$D = S \setminus \{s_1\} = \{d_1 < d_2 < \cdots < d_{q-1}\}.$$

Since $q \ge 15$, the set D has at least four elements. On D, take the simple cycle

$$d_1 - d_3 - d_2 - d_4 - d_5 - \cdots - d_{q-1} - d_1.$$

Let Γ be the disjoint union of this cycle, the double edge between u_1 and u_2, and the double edge between v and s_1. This graph is loopless and every vertex has degree 2. Deleting S leaves the fixed retained double edge on U and an isolated retained vertex v, so the only nonzero retained deficit is $c_v = 2$; hence this is the single-deficit setting of proposition 4.81. The component containing v has support $A = \{s_1\}$.

By lemma 4.193,

$$|S \setminus A| = q - 1 \ge R(q) + 1.$$

The deleted-only graph contains the edge $d_1 d_3$. In the full ordered list of S, the label d_2 lies strictly between d_1 and d_3, so these endpoints are not adjacent witness ranks. By corollary 4.190, the concatenated joint-slot list is the same full ordered frontier list, so the same edge is not joint-slot adjacent. This proves the example.

The final assertion follows because all retained-repair graph data in the construction are valid before any target-size parser restriction is imposed, while the displayed deleted-only edge violates joint-slot adjacency. $\square$

Lemma 4.195 (Step–2 submatrices recover induced Littlewood graphs)**.** *Let*

$$M = (M_{ab})_{1 \le a,b \le j}$$

be a symmetric nonnegative integer matrix with zero diagonal and row sums equal to 2. Let Γ_M be the loopless labelled multigraph on $\{1, \ldots, j\}$ with M_{ab} parallel edges between a and b for $a < b$. Then Γ_M is a loopless labelled 2-regular multigraph. For every subset $D \subset \{1, \ldots, j\}$, the entries

$$M_{ab} \qquad (a, b \in D,\ a < b)$$

determine the induced deleted-only graph $(\Gamma_M)_D$.

In the content (2^j) Step–2/Step–3 construction of lemmas 4.127 and 4.128, any recovered Step–3 subfilling containing the cell for each unordered pair $\{a, b\} \subset D$ determines this same induced graph.

Proof. The zero diagonal forbids loops, and the symmetry lets M_{ab} be interpreted as the multiplicity of the unordered edge $\{a, b\}$. The degree of vertex a in Γ_M is

$$\sum_{b<a} M_{ba} + \sum_{a<b} M_{ab} = \sum_{b=1}^{j} M_{ab} = 2,$$

so Γ_M is loopless and 2-regular. The induced graph on D keeps exactly the edges whose two endpoints lie in D, hence exactly the displayed entries.

For a semistandard tableau of content (2^j), the Step–2 matrix has the displayed symmetry, zero diagonal, and degree condition by lemma 4.128. By lemma 4.127, each recovered Step–3 cell entry is the corresponding off-diagonal entry of this Step–2 matrix, up to the symmetric transpose. Therefore recovering all Step–3 cells indexed by unordered pairs inside D recovers all entries M_{ab} with $a, b \in D$, $a < b$, and hence recovers $(\Gamma_M)_D$. □

Proposition 4.196 (Target-window submatrices recover the large deleted-only graph)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and work in the single-deficit setting of proposition 4.81. Let $A \subset S$ be the support of the anchored component and put*

$$D = S \setminus A.$$

Assume that A is a nonempty proper interval in the ordered witness set, that every internal edge of the anchored path joins adjacent witness ranks, and that the Littlewood graph Γ is the graph Γ_M associated by lemma 4.195 to the Step–2 matrix of the original tableau.

Suppose that a datum recovered from the target-size parser, without using an additional branch word, determines the Step–3 cell entry for every unordered pair $\{a, b\} \subset D$. Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0, 1\}^q, \quad \iota(A) \in \{0, 1\}^q$$

and that target-window datum. The same assertion holds in the height fixed-witness fiber.

Consequently, in the large-complement branch $|S \setminus A| \geq R(q) + 1$, one sufficient replacement for the false graph-formal adjacency theorem is a target-size parser/carrier theorem recovering the Step–3 subfilling on all unordered pairs inside $S \setminus A$.

Proof. The target-window datum recovers the Step–3 entries on all unordered pairs in D. By lemma 4.195, these entries determine the induced deleted-only graph $\Gamma_D = \Gamma_{S \setminus A}$. Use this recovered graph as the datum Δ_A in corollary 4.152. The retained-vertex code $\rho(v)$ and the interval code $\iota(A)$ supply the remaining inputs of that corollary, while the adjacent-rank hypothesis on the anchored internal edges is one of its hypotheses. Therefore Γ is recovered. The height case is the height half of the same corollary.

The final sentence is this recovery statement specialized to the complementary range left by proposition 4.192. □

Proposition 4.197 (Ferrers carrier covers recover the large deleted-only graph)**.** *Work in one residual row and moment of remark 4.839. Put $q = b + 1$, let $S \subset \{1, \ldots, j\}$ have cardinality q, and work in the single-deficit setting of proposition 4.81. Let $A \subset S$ be the support of the anchored component and put*

$$D = S \setminus A.$$

Assume that A is a nonempty proper interval in the ordered witness set, that every internal edge of the anchored path joins adjacent witness ranks, and that the Littlewood graph Γ is the graph Γ_M associated by lemma 4.195 to the Step–2 matrix of the original tableau.

Suppose the Step–3 half-square admits a finite family of Ferrers subarrangements

$$E_1, \ldots, E_N$$

such that the target-size parser determines, without an additional branch word, the partition labels on the top/right boundary of each E_r, and such that every Step–3 cell corresponding to an unordered pair

$$\{a,b\} \subset D, \qquad a<b,$$

belongs to at least one E_r. Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q$$

and those carrier boundaries. The same assertion holds in the height fixed-witness fiber.

Consequently, in the large-complement branch $|S\setminus A| \geq R(q)+1$, it is enough to prove a target-size carrier-cover theorem: the parsed Step–3 Ferrers carriers cover every unordered pair inside $S \setminus A$.

Proof. Apply corollary 4.124 to each Ferrers subarrangement E_r. The recovered top/right boundary of E_r determines every Step–3 cell entry in E_r. Since the family covers every cell indexed by an unordered pair inside D, the same parser datum determines the Step–3 cell entry for every unordered pair $\{a,b\} \subset D$.

The hypotheses of proposition 4.196 are therefore satisfied, with the recovered carrier boundaries supplying the target-window datum. That proposition recovers Γ from the displayed data. The height case is identical, using the height half of the same proposition. ☐

Corollary 4.198 (Interval-hull carrier boundaries recover the large deleted-only graph)**.** *Work in the setting of proposition 4.197. Write the maximal consecutive label intervals of*

$$D = S \setminus A$$

as

$$D = \bigsqcup_{\alpha=1}^{L} \{a_\alpha, a_\alpha+1, \dots, b_\alpha\}, \qquad b_\alpha + 1 < a_{\alpha+1}.$$

Suppose that for every $1 \leq \alpha \leq \beta \leq L$ the target-size parser determines, without an additional branch word, the boundary segment

$$\rho_{2(j-b_\beta)}, \rho_{2(j-b_\beta)+1}, \dots, \rho_{2(j-a_\alpha+1)}$$

of the Step–3 semistandard path. Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q$$

and those boundary segments. The same assertion holds in the height fixed-witness fiber.

Proof. For each pair $\alpha \leq \beta$, let $E_{\alpha,\beta}$ be the consecutive Step–3 label-interval subarrangement

$$E_{a_\alpha, b_\beta}$$

of lemma 4.186. That lemma identifies the top/right boundary of $E_{\alpha,\beta}$ with the displayed segment. Hence the assumed parser data recover the top/right boundary of each $E_{\alpha,\beta}$.

Let $\{x,y\} \subset D$ with $x<y$. Then x lies in a unique interval $\{a_\alpha, \dots, b_\alpha\}$ and y lies in a unique interval $\{a_\beta, \dots, b_\beta\}$ with $\alpha \leq \beta$. Therefore both labels belong to the consecutive interval

$$\{a_\alpha, a_\alpha+1, \dots, b_\beta\},$$

so the Step–3 cell for $\{x,y\}$ lies in $E_{\alpha,\beta}$. Thus the family $\{E_{\alpha,\beta}\}_{1\leq\alpha\leq\beta\leq L}$ covers every unordered pair inside D. Applying proposition 4.197 recovers Γ. The height case is identical. ☐

Corollary 4.199 (Terminal boundaries above the deleted maximum recover the large deleted-only graph)**.** *Work in the setting of proposition 4.196, and assume $D = S \setminus A$ is nonempty. Let*

$$M_D = \max D.$$

Let M be any integer with

$$M_D \leq M \leq j.$$

Suppose the target-size parser determines, without an additional branch word, the Step–3 terminal boundary segment

$$\rho_{2(j-M)}, \rho_{2(j-M)+1}, \dots, \rho_{2j}.$$

Then Γ *is recovered from*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q$$

and that terminal boundary segment. The same assertion holds in the height fixed-witness fiber.

Proof. Every unordered pair $\{a,b\} \subset D$ has $a, b \le M_D \le M$. Hence its Step–3 cell lies in the low-low terminal cell set L_{M_D} of lemma 4.182 and also in L_M. By lemma 4.183, the displayed terminal boundary segment determines every Step–3 entry on L_M. In particular it determines every Step–3 cell entry indexed by an unordered pair inside D. Applying proposition 4.196 recovers Γ. The height case is the height half of that proposition. $\square$

Proposition 4.200 (Continuation columns plus terminal boundaries recover the large deleted-only graph)**.** *Work in the setting of proposition 4.196, and assume* $D = S \setminus A$ *is nonempty. Let* M *be an integer with*

$$1 \le M \le j.$$

Suppose the target-size parser determines, without an additional branch word, the following two pieces of Step–3 data:

(1) *every Step–3 cell entry corresponding to an unordered pair* $\{a,b\} \subset \{1,\dots,j\}$ *with* $\max(a,b) > M$*;*
(2) *the terminal boundary segment*

$$\rho_{2(j-M)}, \rho_{2(j-M)+1}, \dots, \rho_{2j}.$$

Then Γ *is recovered from*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q$$

and those parser data. The same assertion holds in the height fixed-witness fiber.

Proof. Let $\{a,b\} \subset D$ with $a < b$. If $b > M$, then the corresponding Step–3 cell entry is determined by item (1). If $b \le M$, then both labels are at most M, so the cell lies in the low-low terminal cell set L_M of lemma 4.182. By lemma 4.183, the boundary segment in item (2) determines every Step–3 entry on L_M, and hence determines this cell entry.

Thus the parser data determine the Step–3 entry for every unordered pair inside D. Applying proposition 4.196 recovers Γ. The height case is the height half of that proposition. $\square$

Corollary 4.201 (Strict-left data plus a terminal boundary recover the large deleted-only graph)**.** *Work in the setting of proposition 4.196, and also in the right-boundary auxiliary setting of proposition 4.253. Assume the final zone is in the non-full tight-height setting of lemma 4.339. Put*

$$M = 2m, \qquad Z_L = \{j - 2m + 1, \dots, j\},$$

and assume $D = S \setminus A$ *is nonempty. Suppose the outside datum* $\mathcal{D}_{\mathrm{out}}(P)$ *has been recovered as in proposition 4.253, from the retained blocks and the earlier-zone inverses using only the branch words already present in that recovery. Suppose further that the remaining target-window data determine, without an additional branch word, the terminal Step–3 boundary segment*

$$\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j}.$$

Then Γ *is recovered from*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q$$

together with the recovered outside datum and this terminal boundary segment.

Proof. By lemma 4.341, the outside datum $\mathcal{D}_{\mathrm{out}}(P)$ determines every Step–3 entry in the continuation column subarrangement J_M, whose columns are incident to the labels $M+1,\dots,j$. Hence it determines every Step–3 cell entry indexed by an unordered pair $\{a,b\}$ with $\max(a,b) > M$.

The displayed terminal boundary segment is the segment required in proposition 4.200 for this value of M. Applying that proposition gives the claimed recovery of Γ. $\square$

Corollary 4.202 (Full-budget terminal boundary lists recover the large deleted-only graph)**.** *Work in the setting of proposition 4.196 and in the right-boundary auxiliary setting of proposition 4.253. Assume the final zone is in the non-full tight-height setting of lemma 4.339. Put*

$$M = 2m, \qquad Z_L = \{j-2m+1,\dots,j\},$$

and assume $D = S \setminus A$ is nonempty and $m \geq 1$. Suppose the outside datum $\mathcal{D}_{\mathrm{out}}(P)$ has been recovered as in proposition 4.253, using only the branch words already present in that recovery. Suppose further that the remaining target-window data determine a canonically ordered finite set

$$\mathcal{B}_{\mathrm{tw}}$$

of possible terminal Step–3 boundary segments

$$\tau = (\tau_{2(j-2m)}, \tau_{2(j-2m)+1}, \dots, \tau_{2j}),$$

that the true segment

$$(\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j})$$

belongs to $\mathcal{B}_{\mathrm{tw}}$, and that

$$|\mathcal{B}_{\mathrm{tw}}| \leq 2^{2m}.$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, $\mathcal{B}_{\mathrm{tw}}$, and the already budgeted final-zone word $\beta_L \in \{0,1\}^{2m}$.

Proof. Encode the rank of the true terminal boundary segment in the canonical order on $\mathcal{B}_{\mathrm{tw}}$ by the word β_L. Since $|\mathcal{B}_{\mathrm{tw}}| \leq 2^{2m}$, this uses no branch data beyond the already budgeted final-zone word and selects the true segment from $\mathcal{B}_{\mathrm{tw}}$.

With that selected segment in hand, the hypotheses of corollary 4.201 are satisfied. That corollary recovers Γ from the displayed data. $\square$

Corollary 4.203 (Full-budget low-low candidate lists recover the large deleted-only graph)**.** *Work in the setting of proposition 4.196 and in the right-boundary auxiliary setting of proposition 4.253. Assume the final zone is in the non-full tight-height setting of lemma 4.339. Put*

$$M = 2m, \qquad Z_L = \{j-2m+1,\dots,j\},$$

and let L_{2m} be the low-low cell set of lemma 4.182. Assume $D = S \setminus A$ is nonempty and $m \geq 1$. Suppose the outside datum $\mathcal{D}_{\mathrm{out}}(P)$ has been recovered as in proposition 4.253, using only the branch words already present in that recovery. Suppose further that the remaining target-window data determine a canonically ordered finite set

$$\mathcal{L}_{\mathrm{tw}}$$

of possible Step–3 entry assignments on L_{2m}, that the true assignment belongs to $\mathcal{L}_{\mathrm{tw}}$, and that

$$|\mathcal{L}_{\mathrm{tw}}| \leq 2^{2m}.$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, $\mathcal{L}_{\mathrm{tw}}$, and the already budgeted final-zone word $\beta_L \in \{0,1\}^{2m}$.

Proof. Use the $2m$ bits of β_L to encode the rank of the true low-low assignment in the canonical order on $\mathcal{L}_{\mathrm{tw}}$. Since $|\mathcal{L}_{\mathrm{tw}}| \le 2^{2m}$, this selects the true Step–3 entries on L_{2m} without any branch data beyond the already budgeted final-zone word.

By lemma 4.341, the recovered outside datum determines every Step–3 entry in the continuation column subarrangement J_M, whose columns are incident to the labels $M+1, \ldots, j$. Let $\{a, b\} \subset D$ with $a < b$. If $b > M$, then the corresponding Step–3 entry is determined by the recovered continuation columns. If $b \le M$, then the cell lies in the low-low cell set $L_M = L_{2m}$, and the selected low-low assignment determines it.

Thus every Step–3 cell indexed by an unordered pair inside D is recovered. By lemma 4.195, these entries determine the induced deleted-only graph $\Gamma_{S\setminus A}$. Using this graph as the datum Δ_A in corollary 4.152 recovers Γ from the displayed data. □

Corollary 4.204 (Full-budget odd-packet classes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203, and assume $m \ge 9$. Suppose the remaining target-window data determine*

$$\lambda^{(2m)}, \qquad E_1, \ldots, E_m,$$

the labelled continuation boxes, and a canonically ordered set

$$\Omega_{\mathrm{tw}} \subseteq \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

which contains the true deleted odd-packet assignment and satisfies

$$|\Omega_{\mathrm{tw}}| \le 2^{2m}.$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, Ω_{tw}, and the already budgeted final-zone word $\beta_L \in \{0,1\}^{2m}$.

Proof. Let $\mathcal{L}_{\mathrm{tw}}$ be the low-low candidate set constructed in corollary 4.463, with the displayed set Ω_{tw} in place of the half-budget class there. That construction is a fixed function of the target-window data. Its membership and injectivity proof does not use the half-budget cardinality assumption: the true low-low assignment belongs to $\mathcal{L}_{\mathrm{tw}}$, and the map from $\mathcal{L}_{\mathrm{tw}}$ to the corresponding deleted odd-packet assignment is injective into Ω_{tw}. Hence

$$|\mathcal{L}_{\mathrm{tw}}| \le |\Omega_{\mathrm{tw}}| \le 2^{2m}.$$

The hypotheses of corollary 4.203 are therefore satisfied with this $\mathcal{L}_{\mathrm{tw}}$, and that corollary recovers Γ. □

Corollary 4.205 (Bounded signature lists recover the large deleted-only graph)**.** *Work in the setting of corollary 4.204, and assume $m \ge 9$. Suppose the remaining target-window data determine*

$$\lambda^{(2m)}, \qquad E_1, \ldots, E_m,$$

the labelled continuation boxes, and a canonically ordered finite set

$$\Sigma_{\mathrm{tw}}$$

of commuting normal-form signature words such that the true fixed-even standard tableau has signature in Σ_{tw} and

$$|\Sigma_{\mathrm{tw}}| \le 2^m.$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, these target-window data, and the already budgeted final-zone word $\beta_L \in \{0,1\}^{2m}$.

Proof. Corollary 4.430 constructs from the displayed target-window data a canonically ordered set
$$\Omega_{\mathrm{tw}}(\Sigma_{\mathrm{tw}}) \subseteq \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$
which contains the true deleted odd-packet assignment and has cardinality at most 2^{2m}. Applying corollary 4.204 to this full-budget odd-packet class recovers Γ. □

Corollary 4.206 (Nine-to-twelve broad odd-packet classes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203. Suppose*
$$9 \leq m \leq 12$$
and suppose the remaining target-window data determine
$$\lambda^{(2m)}, \qquad E_1, \dots, E_m,$$
and the labelled continuation boxes. Then Γ is recovered from the data
$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$
the recovered outside datum, these target-window data, and the already budgeted final-zone word
$$\beta_L \in \{0,1\}^{2m}.$$

Proof. Let
$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$
with the lexicographic order of definition 4.354. The displayed target-window data determine this set. By lemma 4.355, it contains the true deleted odd-packet assignment. By lemma 4.360,
$$|\Omega_{\mathrm{tw}}| \leq 2(m-2)!.$$
For $9 \leq m \leq 12$,
$$2(m-2)! \leq 2^{2m};$$
the four cases give
$$10080 \leq 262144, \quad 80640 \leq 1048576, \quad 725760 \leq 4194304, \quad 7257600 \leq 16777216.$$
Thus the hypotheses of corollary 4.204 hold, and that corollary recovers Γ. □

Corollary 4.207 (Thirteen-to-twenty-eight broad odd-packet classes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203. Suppose*
$$13 \leq m \leq 28$$
and suppose the remaining target-window data determine
$$\lambda^{(2m)}, \qquad E_1, \dots, E_m,$$
and the labelled continuation boxes. Then Γ is recovered from the data
$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$
the recovered outside datum, these target-window data, and the already budgeted final-zone word
$$\beta_L \in \{0,1\}^{2m}.$$

Proof. Let
$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$
with the lexicographic order of definition 4.354. The displayed target-window data determine this set, and lemma 4.355 puts the true deleted odd-packet assignment in it.

Lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with the cardinality of the associated fixed-even standard-tableau fiber
$$\mathcal{S}(\gamma; e_1, \dots, e_{m-1}).$$

By lemma 4.389,

$$|\Omega_{\mathrm{tw}}| = |\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| \le W(\gamma).$$

Since $13 \le m \le 28$, proposition 4.390 gives

$$W(\gamma) \le 2^{2m}.$$

Thus the hypotheses of corollary 4.204 hold with the unrefined compatibility set Ω_{tw}, and that corollary recovers Γ. □

Corollary 4.208 (Top-fixed envelope non-frontier broad odd-packet classes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203. Suppose $m \ge 13$, and suppose the remaining target-window data determine*

$$\lambda^{(2m)}, \qquad E_1, \ldots, E_m,$$

and the labelled continuation boxes. Let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

with the lexicographic order of definition 4.354. If $\Omega_{\mathrm{tw}} \neq \varnothing$, let $(\gamma; e_1, \ldots, e_{m-1})$ be the associated fixed-even datum of definition 4.379, and assume

$$(\gamma, e_{m-1}) \notin \mathcal{P}_m^{\mathrm{env}}.$$

Then Γ is recovered from the data

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, these target-window data, and the already budgeted final-zone word

$$\beta_L \in \{0,1\}^{2m}.$$

Proof. The displayed target-window data determine Ω_{tw}, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\mathrm{tw}} = \varnothing$, there is no true assignment in the present fiber, so the recovery assertion is vacuous.

Assume $\Omega_{\mathrm{tw}} \neq \varnothing$. As in the proof of corollary 4.207, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})|.$$

By lemma 4.392,

$$|\Omega_{\mathrm{tw}}| \le W^{\mathrm{top}}(\gamma; e_{m-1}).$$

Since $(\gamma, e_{m-1}) \notin \mathcal{P}_m^{\mathrm{env}}$, definition 4.391 gives

$$W^{\mathrm{top}}(\gamma; e_{m-1}) \le 2^{2m}.$$

Thus the hypotheses of corollary 4.204 hold with the unrefined compatibility set Ω_{tw}, and that corollary recovers Γ. □

Corollary 4.209 (Top-two envelope non-frontier broad odd-packet classes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203. Suppose $m \ge 13$, and suppose the remaining target-window data determine*

$$\lambda^{(2m)}, \qquad E_1, \ldots, E_m,$$

and the labelled continuation boxes. Let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

with the lexicographic order of definition 4.354. If $\Omega_{\mathrm{tw}} \neq \varnothing$, let $(\gamma; e_1, \ldots, e_{m-1})$ be the associated fixed-even datum of definition 4.379, and assume

$$(\gamma, e_{m-1}, e_{m-2}) \notin \mathcal{P}_m^{\mathrm{env},2}.$$

Then Γ *is recovered from the data*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, these target-window data, and the already budgeted final-zone word

$$\beta_L \in \{0,1\}^{2m}.$$

Proof. The displayed target-window data determine Ω_{tw}, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\mathrm{tw}} = \varnothing$, there is no true assignment in the present fiber, so the recovery assertion is vacuous.

Assume $\Omega_{\mathrm{tw}} \neq \varnothing$. As in the proof of corollary 4.208, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})|.$$

By lemma 4.393,

$$|\Omega_{\mathrm{tw}}| \leq W^{\mathrm{top},2}(\gamma; e_{m-1}, e_{m-2}).$$

Since $(\gamma, e_{m-1}, e_{m-2}) \notin \mathcal{P}_m^{\mathrm{env},2}$, definition 4.391 gives

$$W^{\mathrm{top},2}(\gamma; e_{m-1}, e_{m-2}) \leq 2^{2m}.$$

Thus the hypotheses of corollary 4.204 hold with the unrefined compatibility set Ω_{tw}, and that corollary recovers Γ. ☐

Corollary 4.210 (Fixed-depth envelope non-frontier broad odd-packet classes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203. Let* $d \geq 3$*, and suppose* $m \geq d+1$*. Suppose the remaining target-window data determine*

$$\lambda^{(2m)}, \qquad E_1, \dots, E_m,$$

and the labelled continuation boxes. Let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

with the lexicographic order of definition 4.354. If $\Omega_{\mathrm{tw}} \neq \varnothing$*, let* $(\gamma; e_1, \dots, e_{m-1})$ *be the associated fixed-even datum of definition 4.379, and assume*

$$(\gamma, e_{m-1}, e_{m-2}, \dots, e_{m-d}) \notin \mathcal{P}_m^{\mathrm{env},d}.$$

Then Γ *is recovered from the data*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, these target-window data, and the already budgeted final-zone word

$$\beta_L \in \{0,1\}^{2m}.$$

Proof. The displayed target-window data determine Ω_{tw}, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\mathrm{tw}} = \varnothing$, there is no true assignment in the present fiber, so the recovery assertion is vacuous.

Assume $\Omega_{\mathrm{tw}} \neq \varnothing$. As in the proof of corollary 4.209, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})|.$$

If $d = 3$, then lemma 4.394 gives

$$|\Omega_{\mathrm{tw}}| \leq W^{\mathrm{top},3}(\gamma; e_{m-1}, e_{m-2}, e_{m-3}).$$

If $d \geq 4$, then lemma 4.395 gives

$$|\Omega_{\mathrm{tw}}| \leq W^{\mathrm{top},d}(\gamma; e_{m-1}, e_{m-2}, \dots, e_{m-d}).$$

In both cases, the displayed non-frontier hypothesis and definition 4.391 give

$$|\Omega_{\mathrm{tw}}| \le 2^{2m}.$$

Thus the hypotheses of corollary 4.204 hold with the unrefined compatibility set Ω_{tw}, and that corollary recovers Γ. $\square$

Corollary 4.211 (Twenty-nine top-three envelope broad odd-packet classes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203. Suppose $m = 29$, and suppose the remaining target-window data determine*

$$\lambda^{(58)}, \qquad E_1, \ldots, E_{29},$$

and the labelled continuation boxes. Then Γ is recovered from the data

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, these target-window data, and the already budgeted final-zone word

$$\beta_L \in \{0,1\}^{58}.$$

Proof. Let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(58)}; E_1, \ldots, E_{29})$$

with the lexicographic order of definition 4.354. The displayed target-window data determine this set, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\mathrm{tw}} = \varnothing$, the assertion is vacuous.

Assume $\Omega_{\mathrm{tw}} \ne \varnothing$, and let $(\gamma; e_1, \ldots, e_{28})$ be the associated fixed-even datum. As in the proof of corollary 4.207, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify the cardinality of Ω_{tw} with

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{28})|.$$

By lemma 4.394,

$$|\Omega_{\mathrm{tw}}| \le W^{\mathrm{top},3}(\gamma; e_{28}, e_{27}, e_{26}).$$

Proposition 4.398 gives

$$\mathcal{P}_{29}^{\mathrm{env},3} = \varnothing,$$

so the right side is at most 2^{58}. Thus the hypotheses of corollary 4.204 hold with the unrefined compatibility set Ω_{tw}, and that corollary recovers Γ. $\square$

Corollary 4.212 (Thirty fixed-depth envelope broad odd-packet classes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203. Suppose $m = 30$, and suppose the remaining target-window data determine*

$$\lambda^{(60)}, \qquad E_1, \ldots, E_{30},$$

and the labelled continuation boxes. Then Γ is recovered from the data

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, these target-window data, and the already budgeted final-zone word

$$\beta_L \in \{0,1\}^{60}.$$

Proof. Let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(60)}; E_1, \ldots, E_{30})$$

with the lexicographic order of definition 4.354. The displayed target-window data determine this set, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\mathrm{tw}} = \varnothing$, the assertion is vacuous.

Assume $\Omega_{\mathrm{tw}} \ne \varnothing$, and let $(\gamma; e_1, \ldots, e_{29})$ be the associated fixed-even datum. As in the proof of corollary 4.207, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify the cardinality of Ω_{tw} with

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{29})|.$$

By lemma 4.395,

$$|\Omega_{\mathrm{tw}}| \leq W^{\mathrm{top},7}(\gamma; e_{29}, e_{28}, e_{27}, e_{26}, e_{25}, e_{24}, e_{23}).$$

Proposition 4.399 gives

$$\mathcal{P}_{30}^{\mathrm{env},7} = \varnothing,$$

so the right side is at most 2^{60}. Thus the hypotheses of corollary 4.204 hold with the unrefined compatibility set Ω_{tw}, and that corollary recovers Γ. □

Corollary 4.213 (Thirty-one fixed-depth envelope broad odd-packet classes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203. Suppose $m = 31$, and suppose the remaining target-window data determine*

$$\lambda^{(62)}, \qquad E_1, \ldots, E_{31},$$

and the labelled continuation boxes. Then Γ is recovered from the data

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, these target-window data, and the already budgeted final-zone word

$$\beta_L \in \{0,1\}^{62}.$$

Proof. Let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(62)}; E_1, \ldots, E_{31})$$

with the lexicographic order of definition 4.354. The displayed target-window data determine this set, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\mathrm{tw}} = \varnothing$, the assertion is vacuous.

Assume $\Omega_{\mathrm{tw}} \neq \varnothing$, and let $(\gamma; e_1, \ldots, e_{30})$ be the associated fixed-even datum. As in the proof of corollary 4.207, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify the cardinality of Ω_{tw} with

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{30})|.$$

By lemma 4.395,

$$|\Omega_{\mathrm{tw}}| \leq W^{\mathrm{top},10}(\gamma; e_{30}, e_{29}, e_{28}, e_{27}, e_{26}, e_{25}, e_{24}, e_{23}, e_{22}, e_{21}).$$

Proposition 4.400 gives

$$\mathcal{P}_{31}^{\mathrm{env},10} = \varnothing,$$

so the right side is at most 2^{62}. Thus the hypotheses of corollary 4.204 hold with the unrefined compatibility set Ω_{tw}, and that corollary recovers Γ. □

Corollary 4.214 (Column-even tight prefixes recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203 and in the setting of corollary 4.328. Let*

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}$$

be the true tight-prefix Pieri path. Assume $m \geq 2$ and that $\lambda^{(2m)}$ is column-even. Suppose the final-zone replacement window V_L^* and branch word

$$\beta_L \in \{0,1\}^{2m}$$

are chosen by the fixed odd-alternating height full-zone decoder of proposition 4.108 applied to this tight prefix. Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, V_L^*, and β_L.

Proof. By corollary 4.328, the displayed prefix is an odd alternating height full-zone path in the domain of proposition 4.108. The decoder therefore recovers the whole prefix path

$$\lambda^{(0)}, \lambda^{(1)}, \ldots, \lambda^{(2m)}$$

from V_L^* and β_L.

The recovered prefix path determines the boxes carrying the semistandard labels $1, \ldots, 2m$ by lemma 4.97. Hence it determines the prefix labelled tableau. By lemma 4.174, that labelled tableau determines the Step–2 symmetric matrix entries whose two labels are at most $2m$; by lemma 4.127, these are exactly the Step–3 entries on the low-low cell set L_{2m}.

The recovered outside datum determines the continuation-column entries J_{2m} by lemma 4.341. Let $\{a, b\} \subset D$ with $a < b$. If $b > 2m$, the corresponding Step–3 entry is recovered from J_{2m}; if $b \leq 2m$, it is recovered from L_{2m}. Thus every Step–3 cell indexed by an unordered pair inside D is recovered. As in proposition 4.196, lemma 4.195 and corollary 4.152 recover Γ from these entries and the displayed graph data. □

Corollary 4.215 (Strict-left low-low carriers recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203. Suppose that, from*

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \ldots, \rho_{2(j-2m)},$$

a deterministic rule identifies a Ferrers subarrangement E of the Step–3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$L_{2m} \subseteq E.$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, the carrier E, its recovered top/right boundary labels, and the verified inclusion $L_{2m} \subseteq E$.

Proof. By lemma 4.339, the recovered outside datum determines

$$\rho_0, \rho_1, \ldots, \rho_{2(j-2m)}.$$

Thus the displayed deterministic rule is applied to data already recovered from the target-window and outside data. It identifies the carrier E, its top/right boundary labels, and the inclusion $L_{2m} \subseteq E$.

The recovered top/right boundary labels determine every Step–3 entry on E by corollary 4.124. Hence they determine the true Step–3 assignment on L_{2m}.

By lemma 4.341, the recovered outside datum determines every Step–3 entry in the continuation column subarrangement J_{2m}. Let $\{a, b\} \subset D$ with $a < b$. If $b > 2m$, then the corresponding Step–3 entry is determined by J_{2m}; if $b \leq 2m$, then the corresponding cell lies in L_{2m} and is determined by the recovered carrier subfilling. Hence every Step–3 cell indexed by an unordered pair inside D is recovered. As in proposition 4.196, lemma 4.195 and corollary 4.152 recover Γ from these entries and the displayed graph data. □

Corollary 4.216 (Strict-left incidence-cover carriers recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203 and in the setting of definition 4.440, and also in the non-full tight-height setting of lemma 4.339. Put*

$$M = 2m, \qquad Z_L = \{j - 2m + 1, \ldots, j\},$$

and assume $m \geq 29$. Let $A(m)$ and $B(m)$ be the frontier functions of proposition 4.452. Let d_a be the residual low-low degrees of definition 4.440. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \ldots, \rho_{2(j-2m)},$$

a deterministic rule identifies a target-window-determined set

$$U_{\mathrm{tw}} \subseteq \{1 \le a \le 2m : d_a > 0\},$$

identifies a Ferrers subarrangement E of the Step–3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$\{\{a,b\} : 1 \le a < b \le 2m,\ \{a,b\} \cap U_{\mathrm{tw}} \neq \varnothing\} \subseteq E.$$

Assume moreover that one positive-residual incidence-cover threshold holds. Namely, put

$$r_i = \#\{1 \le a \le 2m : d_a = i\} \quad (i = 1,2), \qquad D = \sum_{a=1}^{2m} d_a, \qquad D(U_{\mathrm{tw}}) = \sum_{a \in U_{\mathrm{tw}}} d_a.$$

Assume one of

$$|U_{\mathrm{tw}}| > r_1 + r_2 - A(m) \qquad \text{or} \qquad D(U_{\mathrm{tw}}) > D - B(m).$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, the carrier E, its recovered top/right boundary labels, the incidence-cover data, and the already budgeted final-zone word $\beta_L \in \{0,1\}^{2m}$.

Proof. By lemma 4.339, the recovered outside datum determines

$$\rho_0, \rho_1, \ldots, \rho_{2(j-2m)}.$$

Thus the displayed deterministic rule is applied to data already recovered from the target-window and outside data. It identifies U_{tw}, the carrier E, its top/right boundary labels, and the displayed incidence inclusion.

The recovered top/right boundary labels determine every Step–3 entry on E by corollary 4.124. Hence the target-window data determine the true Step–3 entries on every low-low cell incident to U_{tw}. If

$$|U_{\mathrm{tw}}| > r_1 + r_2 - A(m),$$

then the positivity of U_{tw} gives

$$\#\{a \notin U_{\mathrm{tw}} : d_a > 0\} = r_1 + r_2 - |U_{\mathrm{tw}}| < A(m),$$

so corollary 4.450 gives a canonical full-budget low-low candidate set containing the true assignment. If instead $D(U_{\mathrm{tw}}) > D - B(m)$, then corollary 4.451 gives the same candidate set. The hypotheses of corollary 4.203 are therefore satisfied with this candidate set, and that corollary recovers Γ. ☐

Corollary 4.217 (Hall-augmented non-column-even carriers recover the large deleted-only graph)**.** *Work in the non-full tight-height setting of corollary 4.327 and lemma 4.339 and in the setting of corollary 4.215. Let*

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}$$

be the true tight-prefix Pieri path, and assume that $\lambda^{(2m)}$ is not column-even. Suppose that the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the paired endpoint-defect set of lemma 4.331, the continuation-incidence sets

$$N(a) = \{t \in \{2m+1, \ldots, j\} : C_t \cap P_a \neq \varnothing\},$$

where P_a are the adjacent endpoint-defect pairs and C_t is the two-column set hit by continuation label t, as in lemmas 4.334 and 4.336, and the lexicographically least canonical Hall charge of lemma 4.335. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \ldots, \rho_{2(j-2m)},$$

together with $\lambda^{(2m)}$, the endpoint-defect pairs, the continuation incidences, the canonical continuation Hall charge, and the Hall-charged continuation blocks recovered by corollary 4.516, a deterministic rule identifies a Ferrers subarrangement E of the Step–3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$L_{2m} \subseteq E.$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum and these Hall-augmented carrier data.

Proof. The first recovery hypothesis makes $\lambda^{(2m)}$, the endpoint-defect pairs, the continuation-incidence sets, and the canonical Hall charge fixed functions of the displayed target-window data. By lemma 4.339, the recovered outside datum determines

$$\rho_0, \rho_1, \dots, \rho_{2(j-2m)}.$$

Therefore the displayed data determine the Hall-charged continuation blocks by corollary 4.516, and hence every input to the deterministic Hall-augmented carrier rule. Thus they determine E, its top/right boundary labels, and the inclusion $L_{2m} \subseteq E$. Applying corollary 4.215 recovers Γ. ☐

Corollary 4.218 (Hall-augmented incidence-cover carriers recover the large deleted-only graph)**.** *Work in the non-full tight-height setting of corollary 4.327 and lemma 4.339 and in the setting of corollary 4.216 and definition 4.440. Let*

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}$$

be the true tight-prefix Pieri path, assume $m \geq 29$, and assume that $\lambda^{(2m)}$ is not column-even. Suppose that the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the paired endpoint-defect set of lemma 4.331, the continuation-incidence sets

$$N(a) = \{t \in \{2m+1, \dots, j\} : C_t \cap P_a \neq \varnothing\},$$

where P_a are the adjacent endpoint-defect pairs and C_t is the two-column set hit by continuation label t, as in lemmas 4.334 and 4.336, and the lexicographically least canonical Hall charge of lemma 4.335. Let $A(m)$ and $B(m)$ be the frontier functions of proposition 4.452. Let d_a be the residual low-low degrees of definition 4.440. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

together with $\lambda^{(2m)}$, the endpoint-defect pairs, the continuation incidences, the canonical continuation Hall charge, and the Hall-charged continuation blocks recovered by corollary 4.516, a deterministic rule identifies a target-window-determined set

$$U_{\text{tw}} \subseteq \{1 \leq a \leq 2m : d_a > 0\},$$

identifies a Ferrers subarrangement E of the Step–3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$\{\{a,b\} : 1 \leq a < b \leq 2m, \ \{a,b\} \cap U_{\text{tw}} \neq \varnothing\} \subseteq E.$$

Assume moreover that one positive-residual incidence-cover threshold holds. Namely, put

$$r_i = \#\{1 \leq a \leq 2m : d_a = i\} \quad (i = 1,2), \qquad D = \sum_{a=1}^{2m} d_a, \qquad D(U_{\text{tw}}) = \sum_{a \in U_{\text{tw}}} d_a.$$

Assume one of

$$|U_{\text{tw}}| > r_1 + r_2 - A(m) \qquad \text{or} \qquad D(U_{\text{tw}}) > D - B(m).$$

Then Γ *is recovered from*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum and these Hall-augmented incidence-cover carrier data.

Proof. The first recovery hypothesis makes $\lambda^{(2m)}$, the endpoint-defect pairs, the continuation-incidence sets, and the canonical Hall charge fixed functions of the displayed target-window data. By lemma 4.339, the recovered outside datum determines

$$\rho_0, \rho_1, \ldots, \rho_{2(j-2m)}.$$

Therefore the displayed data determine the Hall-charged continuation blocks by corollary 4.516, and hence every input to the deterministic Hall-augmented incidence-cover carrier rule. Thus they determine U_{tw}, E, its top/right boundary labels, the incidence inclusion, and the displayed positive-residual threshold. Applying corollary 4.216 recovers Γ. $\square$

Corollary 4.219 (Continuation segments supply the Hall-augmented graph carrier)**.** *Work in the setting of corollary 4.217. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the Pieri continuation segment

$$\lambda^{(2m)} \subset \lambda^{(2m+1)} \subset \cdots \subset \lambda^{(j)}.$$

Using this recovered segment, form the endpoint-defect pairs, the continuation incidences, and the canonical continuation Hall charge of lemma 4.513, and recover the Hall-charged continuation blocks of corollary 4.515. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \ldots, \rho_{2(j-2m)},$$

together with the recovered continuation segment, its canonical Hall data, and the Hall-charged continuation blocks, a deterministic rule identifies a Ferrers subarrangement E of the Step–3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$L_{2m} \subseteq E.$$

Then Γ *is recovered from*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, the recovered continuation segment, and this carrier rule.

Proof. The first partition in the recovered continuation segment is $\lambda^{(2m)}$. For each continuation label $t = 2m+1, \ldots, j$, the difference

$$\lambda^{(t)} \setminus \lambda^{(t-1)}$$

is exactly the pair of boxes carrying label t by lemma 4.97. Thus the recovered continuation segment supplies the prefix endpoint and the continuation label boxes.

By lemma 4.513, the same segment determines the endpoint-defect pairs, the continuation incidences, and the canonical Hall charge. Moreover, corollaries 4.515 and 4.516 identify the Hall-charged continuation blocks as fixed functions of the canonical Hall charge and the recovered outside datum. Therefore the deterministic carrier rule in the present statement supplies the Hall-augmented carrier rule required by corollary 4.217. Applying that corollary recovers Γ. $\square$

Corollary 4.220 (Internal Q-cuts supply the Hall-augmented graph carrier)**.** *Work in the setting of corollary 4.219. Let C_{2m}^+ be the Step 2 Q-suffix strip of lemma 4.176, and let C_{2m} be the corresponding Step 2 Q-prefix strip. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the ordered partition labels on the internal separating boundary between the Step 2 Q-prefix strip C_{2m} and the suffix strip C_{2m}^{+}. Use corollary 4.342 to recover the Pieri continuation segment and then form its canonical Hall data and Hall-charged continuation blocks. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

together with the recovered continuation segment, its canonical Hall data, and the Hall-charged continuation blocks, a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$L_{2m} \subseteq E.$$

Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, the recovered internal Q-cut, and this Hall-augmented carrier rule.

Proof. Corollary 4.342 recovers the Pieri continuation segment from the displayed target-window data and the internal cut. The remaining hypotheses are then exactly the continuation-segment Hall-carrier hypotheses of corollary 4.219. Applying that corollary recovers Γ. □

Lemma 4.221 (Column-parity defects are local to deletion intervals)**.** *Let T be a semistandard tableau of content (2^j) whose shape has only even column lengths. Let $S \subset \{1, \dots, j\}$, and write the maximal consecutive intervals in S as*

$$J_\alpha = \{a_\alpha, a_\alpha + 1, \dots, b_\alpha\}, \qquad 1 \le \alpha \le L.$$

For each interval put

$$O_\alpha(T) = \{c : \#\{\textit{boxes of } T \textit{ in column } c \textit{ whose label lies in } J_\alpha\} \textit{ is odd}\}.$$

Then

$$|O_\alpha(T)| \le 2|J_\alpha|$$

for every α. If

$$O_{\mathrm{ret}}(T,S) = \{c : \#\{\textit{boxes of } T \setminus S \textit{ in column } c\} \textit{ is odd}\},$$

then

$$O_{\mathrm{ret}}(T,S) = O_1(T) \triangle O_2(T) \triangle \cdots \triangle O_L(T),$$

where $\triangle$ denotes symmetric difference. Consequently, every odd column of the retained diagram $T \setminus S$ lies in at least one $O_\alpha(T)$, and the final column-parity defect is controlled by at most

$$\sum_{\alpha=1}^{L} 2|J_\alpha| = 2|S|$$

joint-local column candidates.

Proof. For a fixed α, the interval J_α contains $|J_\alpha|$ labels, and each label occurs exactly twice in T. Hence the total number of boxes whose label lies in J_α is $2|J_\alpha|$. A column can belong to $O_\alpha(T)$ only if it contains at least one of these boxes, so $|O_\alpha(T)| \le 2|J_\alpha|$.

Let $d_{\alpha,c}$ be the number of boxes in column c whose label lies in J_α. Since the original shape of T has even column lengths, the parity of column c after deleting all labels in S is

$$\sum_{\alpha=1}^{L} d_{\alpha,c} \pmod 2.$$

If this parity is odd, then at least one summand $d_{\alpha,c}$ is odd, and therefore $c \in O_\alpha(T)$ for that α. More precisely, the parity is odd exactly when c belongs to an odd number of the sets $O_\alpha(T)$, which is the stated symmetric-difference identity. Summing the preceding size bounds over α gives the final estimate. □

Lemma 4.222 (Joint endpoints determine the parity defects)**.** *Let T be a semistandard tableau of content (2^j), and let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)} = \lambda(T)$$

be its Pieri two-strip path. Let $U = \mathrm{std}(T)$, and let

$$\varnothing = \sigma^{(0)} \subset \sigma^{(1)} \subset \cdots \subset \sigma^{(2j)} = \lambda(T)$$

be the standard shape chain of U. Let $J_\alpha = \{a_\alpha, a_\alpha + 1, \ldots, b_\alpha\}$ be one of the maximal consecutive intervals in a subset $S \subset \{1, \ldots, j\}$, and define $O_\alpha(T)$ as in lemma 4.221. Then

$$O_\alpha(T) = \{c \geq 1 : (\lambda^{(b_\alpha)})'_c - (\lambda^{(a_\alpha-1)})'_c \equiv 1 \pmod 2\}.$$

Equivalently,

$$O_\alpha(T) = \{c \geq 1 : (\sigma^{(2b_\alpha)})'_c - (\sigma^{(2a_\alpha-2)})'_c \equiv 1 \pmod 2\}.$$

Here missing column lengths are interpreted as 0. In particular, the local parity-defect candidates at the joint are determined by the two endpoint partitions of the deleted standard-time component

$$\{2a_\alpha - 1, 2a_\alpha, \ldots, 2b_\alpha\}.$$

Moreover $|O_\alpha(T)|$ is even.

Proof. By lemma 4.97, the boxes whose labels lie in J_α are exactly the skew difference $\lambda^{(b_\alpha)}/\lambda^{(a_\alpha-1)}$. Hence, for each column c, the number of boxes in column c whose labels lie in J_α is

$$(\lambda^{(b_\alpha)})'_c - (\lambda^{(a_\alpha-1)})'_c.$$

Taking this equality modulo 2 gives the displayed description of $O_\alpha(T)$. By lemma 4.169,

$$\sigma^{(2b_\alpha)} = \lambda^{(b_\alpha)} \quad \text{and} \quad \sigma^{(2a_\alpha-2)} = \lambda^{(a_\alpha-1)}.$$

Substitution gives the equivalent standard-time formula.

The total number of boxes in the deleted interval is $2|J_\alpha|$, which is even. This total is the sum, over all columns, of the displayed column differences. Therefore the number of columns for which the difference is odd is even. ☐

Corollary 4.223 (Joint parity defects have canonical pairs)**.** *In the setting of lemma 4.222, write*

$$O_\alpha(T) = \{o_1 < o_2 < \cdots < o_{2r_\alpha}\}.$$

Then $r_\alpha \leq |J_\alpha|$, and the pairs

$$\{o_1, o_2\},\ \{o_3, o_4\},\ \ldots,\ \{o_{2r_\alpha-1}, o_{2r_\alpha}\}$$

are determined by the two endpoint partitions $\sigma^{(2a_\alpha-2)}$ and $\sigma^{(2b_\alpha)}$ of the deleted standard-time component.

Proof. Lemma 4.222 proves that $|O_\alpha(T)|$ is even, so the displayed increasing enumeration and adjacent pairing are defined. The same lemma identifies $O_\alpha(T)$ from the two standard-time endpoint partitions, hence the increasing order and the adjacent pairing are also determined by those endpoints. Finally, lemma 4.221 gives $|O_\alpha(T)| \leq 2|J_\alpha|$, so $r_\alpha \leq |J_\alpha|$. ☐

Corollary 4.224 (Frontier events cover paired joint defects)**.** *In either the width or height setting of lemma 4.188, let*

$$E_\alpha(T) = \{e_1 < e_2 < \cdots < e_{h_\alpha}\}$$

be the ordered frontier-event set at J_α, and let

$$\mathcal{O}_\alpha(T) = \{\{o_1, o_2\}, \ldots, \{o_{2r_\alpha-1}, o_{2r_\alpha}\}\}$$

be the paired defect set of corollary 4.223. Then $h_\alpha = |J_\alpha|$ and $r_\alpha \leq h_\alpha$. Hence the rule

$$\{o_{2u-1}, o_{2u}\} \longmapsto e_u \qquad (1 \leq u \leq r_\alpha)$$

is an order-preserving injection from the paired defect slots at the joint into the frontier-event slots at the same joint.

Proof. Lemma 4.188 gives $|E_\alpha(T)| = |J_\alpha|$, so $h_\alpha = |J_\alpha|$. Corollary 4.223 gives $r_\alpha \le |J_\alpha|$. Therefore $r_\alpha \le h_\alpha$, and the displayed assignment is well defined. It is order-preserving by construction from the increasing enumerations. □

Lemma 4.225 (Local branch words encode orientation and selected defect slots)**.** *In either the width or height setting of lemma 4.188, fix one deletion interval J_α and put $h_\alpha = |J_\alpha|$. Let*

$$E_\alpha(T) = \{e_1 < e_2 < \cdots < e_{h_\alpha}\}$$

be its ordered frontier-event set, and let $\mathcal{O}_\alpha(T)$ be the paired defect set of corollary 4.223, matched to the first r_α event slots by corollary 4.224.

In the width case, let θ_i be the bit $\epsilon^w_r(T)$ attached by lemma 4.140 to the unique witness label whose frontier event is e_i. In the height case, put $\theta_i = 0$ for every i. Then, for every subset $M_\alpha \subseteq \{1, \dots, r_\alpha\}$ of paired defect slots, the data

$$\big((\theta_1, \dots, \theta_{h_\alpha}), M_\alpha\big)$$

inject into $\{0,1\}^{2h_\alpha}$ by using the first h_α bits for the word θ and the last h_α bits for the characteristic function of M_α inside the matched event slots.

Proof. The ordered event list has length h_α by lemma 4.188. In the width case each event belongs to a unique witness label in J_α, so lemma 4.140 supplies one bit θ_i for that event. In the height case the same lemma forces the frontier copy to be the lower member of the adjacent standard pair, so the all-zero word records the orientation data.

By corollary 4.224, the paired defect slots are matched injectively to the first $r_\alpha \le h_\alpha$ ordered event slots. Thus the characteristic function of any subset M_α extends uniquely by zeros to a word of length h_α. Concatenating this selection word with θ gives an element of $\{0,1\}^{2h_\alpha}$. The two halves of the concatenated word recover θ and M_α, so the encoding is injective. □

Lemma 4.226 (Branch words recover the local repair data)**.** *In the setting of lemma 4.225, write $J_\alpha = \{a_\alpha, a_\alpha + 1, \dots, b_\alpha\}$ and $h_\alpha = |J_\alpha|$. Let*

$$\beta_\alpha = (\beta_1, \dots, \beta_{2h_\alpha}) \in \{0,1\}^{2h_\alpha}$$

be the branch word produced there from the orientation word and a subset M_α of paired defect slots. Put

$$\sigma^- = \sigma^{(2a_\alpha - 2)}, \qquad \sigma^+ = \sigma^{(2b_\alpha)}.$$

Then J_α, σ^-, σ^+, and β_α determine the following local data:

1. *the ordered frontier-event list*
$$E_\alpha(T) = \{e_1 < \cdots < e_{h_\alpha}\},$$
with
$$e_i = 2(a_\alpha + i - 1) - 1 + \beta_i \quad \textit{in the width case,}$$
and
$$e_i = 2(a_\alpha + i - 1) - 1 \quad \textit{in the height case;}$$
2. *the paired local defect set, obtained by listing increasingly the columns*
$$O_\alpha(T) = \{c \ge 1 : (\sigma^+)'_c - (\sigma^-)'_c \equiv 1 \pmod 2\}$$
and pairing adjacent entries;
3. *the selected subset M_α, by reading the last h_α bits of β_α on the matched defect slots of corollary 4.224.*

Consequently, after the branch word is fixed, the only missing input for lemma 4.227 is an actual zero-free vacillating segment and an injection of the selected paired defect slots into its eligible boundary-up steps.

Proof. The labels in the interval J_α occur, in increasing order, as $a_\alpha, a_\alpha+1, \dots, b_\alpha$. In the width case the first h_α bits of β_α are the orientation bits θ_i of lemma 4.225. The event attached to the label $a_\alpha+i-1$ is therefore $2(a_\alpha+i-1)-1+\theta_i$ by lemma 4.141. Since $\theta_i=\beta_i$, this gives the displayed width formula. In the height case lemma 4.141 gives the lower time $2(a_\alpha+i-1)-1$, and the first half of the branch word is the all-zero word by lemma 4.225. The ordering and the fact that these are exactly the events of $E_\alpha(T)$ are supplied by lemma 4.188.

Lemma 4.222 identifies $O_\alpha(T)$ from the two endpoint partitions σ^- and σ^+. Then corollary 4.223 determines the adjacent pairing. Finally, lemma 4.225 encodes M_α in the second half of β_α after the matched event-slot injection of corollary 4.224; reading those bits recovers exactly M_α. The last sentence is just the remaining hypothesis in lemma 4.227 after the recovered data have been fixed. □

Lemma 4.227 (Local branch words give admissible HKKO markings)**.** *In the setting of lemma 4.225, suppose a joint repair construction also supplies a zero-free w-vacillating path*

$$V=(V^0,\dots,V^N)$$

and an injection ι_α from the paired defect slots $\{1,\dots,r_\alpha\}$ into the eligible boundary-up set

$$B(V)=\{t : V^t-V^{t-1}=+\epsilon_w \text{ and } V_w^{t-1}=0\}.$$

Then, for every subset $M_\alpha\subseteq\{1,\dots,r_\alpha\}$, the pair

$$\big(V,\iota_\alpha(M_\alpha)\big)$$

is an admissible element of the HKKO marked class $\mathrm{MVT}_N^*(w;V^0\to V^N)$. For fixed V and ι_α, different subsets M_α give different markings, and the second half of the local branch word of lemma 4.225 recovers M_α.

Proof. By hypothesis V has no zero steps, and every element in the image of ι_α is a step $+\epsilon_w$ whose starting point has w-th coordinate 0. Therefore Definition 8.6 of [10], recorded in lemma 4.134, says that $(V,\iota_\alpha(M_\alpha))$ belongs to $\mathrm{MVT}_N^*(w;V^0\to V^N)$.

Since ι_α is injective, distinct subsets of $\{1,\dots,r_\alpha\}$ have distinct images. Finally, lemma 4.225 encodes M_α as the last h_α bits of the branch word, extended by zeros outside the matched defect slots. Hence that half of the branch word recovers M_α. □

Lemma 4.228 (Frontier-event eligibility supplies the local marks)**.** *In the setting of lemma 4.226, suppose a joint repair construction supplies a zero-free w-vacillating segment*

$$V=(V^0,\dots,V^N)$$

and an order-preserving assignment of the recovered frontier-event slots $e_1<\dots<e_{h_\alpha}$ to distinct local times

$$\tau_1<\dots<\tau_{h_\alpha} \quad \text{in } \{1,\dots,N\}.$$

Let r_α be the number of paired defect slots. If

$$\tau_u\in B(V) \qquad (1\le u\le r_\alpha),$$

then the rule $u\mapsto\tau_u$ is the required injection of paired defect slots into eligible boundary-up steps in lemma 4.227. Consequently every selected subset M_α encoded in the second half of the branch word gives an admissible HKKO marking, and M_α is recovered from that half of the branch word.

Proof. The times $\tau_1, \dots, \tau_{h_\alpha}$ are distinct by hypothesis. Since $r_\alpha \le h_\alpha$ by corollary 4.224, the rule $u \mapsto \tau_u$ is an injection on $\{1, \dots, r_\alpha\}$. The displayed eligibility condition says that its image lies in $B(V)$. Therefore the hypotheses of lemma 4.227 hold with $\iota_\alpha(u) = \tau_u$. That lemma gives the admissible marking statement and recovery of the selected subset from the second half of the branch word. ☐

Corollary 4.229 (Fixed-witness defect slots use exactly the local branch budget)**.** *Let S be a fixed witness set and decompose it into maximal consecutive label intervals*

$$J_1, \dots, J_L.$$

For each J_α, let $h_\alpha = |J_\alpha|$ and let

$$\beta_\alpha \in \{0,1\}^{2h_\alpha}$$

be the local branch word of lemma 4.225. Then the endpoint partitions at the deletion joint and the word β_α determine the paired local odd-column defect slots and the selected subset of those slots. The number of paired slots at J_α is at most h_α, and the selected subset uses only the last h_α bits of β_α.

Moreover, if the local repair segment realizes the recovered ordered frontier-event list by distinct local times whose first paired-defect slots are eligible boundary-up steps as in lemma 4.228, then the selected subset gives admissible HKKO marks with no branch data beyond β_α. Across all deletion joints the total branch budget is

$$\prod_{\alpha=1}^{L} 2^{2h_\alpha} = 2^{2|S|}.$$

Proof. For a fixed joint, lemma 4.226 recovers the ordered frontier-event list, the paired local defect set, and the selected subset from the joint endpoints and β_α. The same lemma obtains the paired defect set by applying lemma 4.222 and then corollary 4.223; corollary 4.224 gives at most h_α paired slots and matches them to the first $r_\alpha \le h_\alpha$ ordered frontier events. The construction of lemma 4.225 stores the selected subset in the last h_α bits, so no other branch source is used for the defect selection.

Under the stated realization hypothesis, lemma 4.228 turns those selected slots into admissible HKKO markings and recovers the selected subset from the same last half of β_α. Finally, the maximal intervals J_α partition S, so

$$\sum_{\alpha=1}^{L} h_\alpha = |S|.$$

Multiplying the local branch counts gives the displayed total budget. ☐

Lemma 4.230 (Decoded branch data suffice for the local Z_0 splice)**.** *In the setting of lemma 4.226, let*

$$\beta_\alpha \in \{0,1\}^{2h_\alpha}$$

be a local branch word, and let $M_\alpha \subseteq \{1, \dots, r_\alpha\}$ be the selected paired-defect subset recovered from its second half. Suppose a joint repair construction supplies a w-up-down tableau

$$U = (U^0, U^1, \dots, U^{2N})$$

from μ to ν, together with distinct indices

$$\tau_1, \dots, \tau_{h_\alpha} \in [N]$$

assigned to the ordered frontier-event slots. Put

$$M_\alpha^\tau = \{\tau_u : u \in M_\alpha\}.$$

If $M_\alpha^\tau = \varnothing$, set $U^* = U$. If $M_\alpha^\tau \ne \varnothing$, assume

$$\ell(U^r) < w \qquad (0 \le r \le 2N),$$

$$u \in M_\alpha \quad \Longrightarrow \quad \ell(U^{2\tau_u - 1}) = w - 1,$$

and

$$m \notin M_\alpha^\tau \quad \Longrightarrow \quad \neg\big(\ell(U^{2m-2}) < \ell(U^{2m-1}) > \ell(U^{2m})\big).$$

Then the HKKO boundary-event splice produces an unmarked w-up-down tableau U^* with endpoints μ, ν and length $2N$. Moreover the operation is reversible from U^*, the branch word β_α, and the assigned indices $\tau_1, \ldots, \tau_{h_\alpha}$.

Proof. By lemma 4.226, the branch word β_α recovers the selected subset M_α. Since the assigned indices $\tau_1, \ldots, \tau_{h_\alpha}$ are distinct, the set $M_\alpha^\tau = \{\tau_u : u \in M_\alpha\}$ is then also recovered from β_α and the assignment τ.

If $M_\alpha^\tau = \varnothing$, the identity operation has the asserted endpoint, length, and reversibility properties. If $M_\alpha^\tau \neq \varnothing$, the three displayed hypotheses are exactly the conditions of lemma 4.131 with $M = M_\alpha^\tau$. That lemma applies lemma 4.130 and gives an unmarked output U^* with the same endpoints and length, reversible from U^* and M_α^τ. Since M_α^τ is recovered from β_α and the assigned indices, the claimed reversibility follows. □

Lemma 4.231 (Deleted intervals have canonical HKKO event indices)**.** *In the setting of lemma 4.226, write $J_\alpha = \{a_\alpha, \ldots, b_\alpha\}$ and $h_\alpha = |J_\alpha|$. Normalize the deleted standard-time component*

$$\{2a_\alpha - 1, 2a_\alpha, \ldots, 2b_\alpha\}$$

to the local interval $\{1, \ldots, 2h_\alpha\}$ by subtracting $2a_\alpha - 2$. Then the adjacent standard pair attached to the label $a_\alpha + i - 1$ is the local pair

$$\{2i - 1, 2i\} \qquad (1 \le i \le h_\alpha).$$

Thus, for a local HKKO segment whose h_α up-down pairs are indexed by these deleted adjacent pairs in label order, the canonical event-index assignment is

$$\tau_i = i \qquad (1 \le i \le h_\alpha).$$

This assignment is determined by J_α alone. Consequently, in lemma 4.230 for such a local segment one has $M_\alpha^\tau = M_\alpha$, and the HKKO splice is reversible from the output, the fixed interval J_α, and the branch word β_α.

Proof. The standard-time component associated with J_α is $\{2a_\alpha - 1, \ldots, 2b_\alpha\}$ by lemma 4.188. For the i-th label in the interval, namely $a_\alpha + i - 1$, the adjacent standardized pair is

$$\{2(a_\alpha + i - 1) - 1,\ 2(a_\alpha + i - 1)\}.$$

Subtracting $2a_\alpha - 2$ gives $\{2i - 1, 2i\}$. Hence the natural local up-down pair index of that event slot is i, giving $\tau_i = i$. Since J_α gives a_α and h_α, no additional data are needed to recover this assignment.

With $\tau_i = i$, the selected HKKO index set $M_\alpha^\tau = \{\tau_u : u \in M_\alpha\}$ equals M_α. The final reversibility assertion is therefore the reversibility assertion of lemma 4.230, with the assigned indices recovered from the fixed interval J_α. □

Lemma 4.232 (Top-wall peak-loop segment)**.** *Let $w \ge 3$, let $N \ge 0$, and let τ be a partition with exactly $w - 2$ nonzero parts. For a subset $M \subseteq [N]$, define a sequence*

$$U = (U^0, U^1, \ldots, U^{2N})$$

by setting $U^{2m} = \tau$ for every $0 \le m \le N$ and, for $1 \le m \le N$, setting U^{2m-1} to be obtained from τ by adding one box in row $w - 1$ if $m \in M$, and by adding one box in row 1 if $m \notin M$. Then U is a w-up-down tableau from τ to τ,

$$\ell(U^r) < w \qquad (0 \le r \le 2N),$$

and its odd length-peak set is exactly M:

$$\{m \in [N] : \ell(U^{2m-2}) < \ell(U^{2m-1}) > \ell(U^{2m})\} = M.$$

Moreover, every selected midpoint has length $w-1$ and is obtained by adding one box in row $w-1$ from a partition with exactly $w-2$ nonzero parts.

Proof. Since τ has exactly $w-2$ nonzero parts, adding one box in row $w-1$ creates a partition of length $w-1$, and adding one box in row 1 creates a partition still having length $w-2$. In each two-step block the first step adds one box to τ and the second removes that same box, so the sequence is a w-up-down tableau from τ back to τ.

All even states have length $w-2$. If $m \in M$, then the odd state U^{2m-1} has length $w-1$, so

$$\ell(U^{2m-2}) < \ell(U^{2m-1}) > \ell(U^{2m}).$$

If $m \notin M$, then the odd state has length $w-2$, equal to the adjacent even-state lengths, so there is no length peak at $2m-1$. This proves the displayed equality for the odd length-peak set. The below-wall bound follows from the lengths $w-2$ and $w-1$, both strictly less than w. The final assertion is the construction in the selected case. $\square$

Lemma 4.233 (Below-wall peakless connectors)**.** *Let $w \geq 1$, and let μ, ν be partitions with*

$$\ell(\mu) < w, \qquad \ell(\nu) < w.$$

Then there is a w-up-down tableau

$$C = (C^0, \dots, C^{2R})$$

from μ to ν such that

$$\ell(C^r) < w \qquad (0 \leq r \leq 2R)$$

and C has no odd length peaks:

$$\neg\big(\ell(C^{2a-2}) < \ell(C^{2a-1}) > \ell(C^{2a})\big) \qquad (1 \leq a \leq R).$$

Moreover C can be chosen from μ and ν alone.

Proof. Put $\zeta = \mu \vee \nu$, the componentwise maximum of μ and ν. Then ζ is a partition and $\ell(\zeta) < w$. Starting from μ, add the boxes of ζ/μ one at a time, row by row from top to bottom, to obtain a chain of partitions from μ to ζ. Starting from ζ, remove the boxes of ζ/ν one at a time, row by row from bottom to top, to obtain a chain from ζ to ν. Concatenating these two chains gives partitions

$$\gamma_0 = \mu, \gamma_1, \dots, \gamma_R = \nu$$

all below the wall, such that each adjacent pair is comparable and differs by one box.

If $\gamma_{a-1} \subseteq \gamma_a$, set

$$(C^{2a-2}, C^{2a-1}, C^{2a}) = (\gamma_{a-1}, \gamma_a, \gamma_a).$$

If $\gamma_a \subseteq \gamma_{a-1}$, set

$$(C^{2a-2}, C^{2a-1}, C^{2a}) = (\gamma_{a-1}, \gamma_{a-1}, \gamma_a).$$

The overlaps agree at the even times, so this defines a w-up-down tableau from μ to ν. Each nontrivial adjacent difference is a single box and hence a vertical strip; the other adjacent difference in the same two-step block is empty. Since every displayed state is one of the γ_a, all states remain below the wall.

In an increasing two-step block the midpoint equals the right endpoint, so its length cannot be strictly larger than both endpoint lengths. In a decreasing two-step block the midpoint equals the left endpoint, with the same conclusion. Thus no odd midpoint is a length peak. The construction used only μ and ν. $\square$

Lemma 4.234 (Peakless connectors attach the top-wall loop)**.** *Let $w \geq 3$. Let*

$$A = (A^0, \dots, A^{2R}), \qquad L = (L^0, \dots, L^{2N}), \qquad B = (B^0, \dots, B^{2S})$$

be w-up-down tableaux with

$$A^{2R} = L^0, \qquad L^{2N} = B^0.$$

Assume all three segments stay below the wall:

$$\ell(A^r) < w, \qquad \ell(L^r) < w, \qquad \ell(B^r) < w$$

for all admissible r. Assume also that A and B have no odd length peaks:

$$\neg\big(\ell(A^{2a-2}) < \ell(A^{2a-1}) > \ell(A^{2a})\big) \quad (1 \leq a \leq R),$$

and

$$\neg\big(\ell(B^{2b-2}) < \ell(B^{2b-1}) > \ell(B^{2b})\big) \quad (1 \leq b \leq S).$$

Let

$$M_L = \{\ell_0 \in [N] : \ell(L^{2\ell_0-2}) < \ell(L^{2\ell_0-1}) > \ell(L^{2\ell_0})\}$$

be the odd peak set of L. Form the even-time concatenation

$$W = (A^0, \dots, A^{2R}, L^1, \dots, L^{2N}, B^1, \dots, B^{2S}).$$

Then W is a w-up-down tableau from A^0 to B^{2S}, it stays below the wall, and its odd length-peak set is exactly

$$\{R + \ell_0 : \ell_0 \in M_L\}.$$

Proof. Even-time concatenation of up-down tableaux with matching endpoints is again an up-down tableau, with start A^0 and end B^{2S}. Every state of W is a state of one of the three input segments, so the below-wall condition is inherited.

For $1 \leq m \leq R$, the three states $W^{2m-2}, W^{2m-1}, W^{2m}$ are the corresponding three states of A, so there is no odd peak by hypothesis on A. For $R < m \leq R + N$, write $m = R + \ell_0$; the three states are the shifted corresponding states of L, so m is an odd peak of W exactly when $\ell_0 \in M_L$. For $R + N < m \leq R + N + S$, write $m = R + N + b$; the three states are the shifted corresponding states of B, so there is no odd peak by hypothesis on B. These three cases prove the displayed peak set. ◻

Corollary 4.235 (Top-wall loops attach to arbitrary below-wall endpoints)**.** *Let $w \geq 3$, let μ, ν be partitions with*

$$\ell(\mu) < w, \qquad \ell(\nu) < w,$$

and let τ be a partition with exactly $w - 2$ nonzero parts. For every $N \geq 0$ and every $M \subseteq [N]$, there are integers $R, S \geq 0$ and a w-up-down tableau

$$W = (W^0, \dots, W^{2(R+N+S)})$$

from μ to ν, staying below the wall, whose odd length-peak set is

$$\{R + m : m \in M\}.$$

The connector parts of W may be chosen from (μ, τ) and (τ, ν) alone.

Proof. Apply lemma 4.233 to obtain a peakless below-wall connector A from μ to τ and a peakless below-wall connector B from τ to ν. Apply lemma 4.232 to τ and M to obtain a below-wall loop L from τ to τ whose odd length peaks are exactly M. The concatenation statement is then exactly lemma 4.234. The final assertion follows from the canonical connector construction in lemma 4.233. ◻

Lemma 4.236 (Peak-supported branch selections land in Z_0)**.** *In the setting of lemma 4.226, suppose a local repair construction supplies a w-up-down tableau*

$$U = (U^0, U^1, \ldots, U^{2N})$$

and distinct assigned event indices

$$\tau_1, \ldots, \tau_{h_\alpha} \in [N].$$

Let

$$P(U) = \{m \in [N] : \ell(U^{2m-2}) < \ell(U^{2m-1}) > \ell(U^{2m})\}$$

be the set of odd length peaks. Assume

$$\ell(U^r) < w \qquad (0 \le r \le 2N),$$

$$P(U) \subseteq \{\tau_u : 1 \le u \le r_\alpha\},$$

and

$$m \in P(U) \quad \Longrightarrow \quad \ell(U^{2m-1}) = w - 1.$$

Define

$$M_\alpha = \{u \in \{1, \ldots, r_\alpha\} : \tau_u \in P(U)\}.$$

If β_α is the local branch word whose second half encodes this M_α, then the HKKO boundary-event splice of lemma 4.230 applies to U and is reversible from the unmarked output, β_α, and the assigned indices. In particular, the unselected non-peak condition in lemma 4.131 is automatic from the peak-support hypothesis.

Proof. The set $M_\alpha^\tau = \{\tau_u : u \in M_\alpha\}$ of lemma 4.230 is exactly $P(U)$ by the definition of M_α and the inclusion $P(U) \subseteq \{\tau_u : 1 \le u \le r_\alpha\}$. Therefore the displayed top-wall condition on $P(U)$ gives

$$u \in M_\alpha \quad \Longrightarrow \quad \ell(U^{2\tau_u - 1}) = w - 1.$$

If $m \notin M_\alpha^\tau$, then $m \notin P(U)$, which is precisely the negation of the odd length-peak condition at m. The below-wall condition is one of the hypotheses. Thus all hypotheses of lemma 4.230 hold. That lemma gives the splice and the stated reversibility. The last sentence is the same implication $m \notin M_\alpha^\tau \Rightarrow m \notin P(U)$. ◻

Lemma 4.237 (Canonical loop segments align the local event indices)**.** *Let $w \ge 3$, let $h \ge 0$, and let μ, ν be partitions with*

$$\ell(\mu) < w, \qquad \ell(\nu) < w.$$

Let $\eta_w = (1^{w-2})$, the partition with $w-2$ parts all equal to 1. For $M \subseteq [h]$, choose the connector A from μ to η_w by the canonical construction in lemma 4.233, choose the loop L from η_w to itself by lemma 4.232, and choose the connector B from η_w to ν by the same canonical connector construction. Let R be the half-length of A, and concatenate A, L, B as in lemma 4.234. If

$$\vartheta_i = R + i \qquad (1 \le i \le h),$$

then the resulting segment W is a below-wall w-up-down tableau from μ to ν with odd length-peak set

$$P(W) = \{\vartheta_i : i \in M\}.$$

Every midpoint indexed by $P(W)$ has length $w - 1$. The indices $\vartheta_1, \ldots, \vartheta_h$ are distinct and are determined by w, h, and the left endpoint μ; in particular they carry no branch data.

Consequently, in the setting of lemma 4.226, if $h = h_\alpha$, $M = M_\alpha \subseteq \{1, \ldots, r_\alpha\}$, and the local repair segment is chosen as above, then the hypotheses of lemma 4.236 hold with assigned indices $\tau_i = \vartheta_i$. Thus the HKKO splice applies with no additional event-index storage.

Proof. The partition η_w has exactly $w-2$ nonzero parts, so lemma 4.232 applies to η_w and M. The two connector segments are below-wall and have no odd length peaks by lemma 4.233. Therefore lemma 4.234 gives a below-wall segment W from μ to ν whose odd peak set is the loop peak set shifted by the half-length R of the left connector. Since the loop peak set is M, this is exactly

$$P(W) = \{R + i : i \in M\} = \{\vartheta_i : i \in M\}.$$

The selected midpoints are the selected midpoints of the loop, so lemma 4.232 gives length $w-1$ at each of them.

The connector construction of lemma 4.233 is canonical once its two endpoints are fixed: it passes through the componentwise maximum and adds or removes boxes in the prescribed row order. Hence the half-length R of the connector from μ to η_w is determined by w and μ, and the displayed indices are determined by w, h, μ. Their distinctness is immediate from $\vartheta_i = R + i$.

For the final assertion, the displayed peak identity gives

$$P(W) \subseteq \{\vartheta_i : 1 \le i \le r_\alpha\}$$

because $M_\alpha \subseteq \{1, \dots, r_\alpha\}$. The below-wall condition and the length-$w-1$ condition at peaks have just been proved. Thus lemma 4.236 applies with $\tau_i = \vartheta_i$. Since the assigned indices are determined by the local endpoint data and h_α, they require no additional storage. □

Corollary 4.238 (Canonical loop splice is locally invertible)**.** *In the setting of lemma 4.226, suppose $h = h_\alpha$, the local endpoints μ, ν satisfy*

$$\ell(\mu) < w, \qquad \ell(\nu) < w,$$

and the selected subset $M_\alpha \subseteq \{1, \dots, r_\alpha\}$ is the subset recovered from the second half of the local branch word β_α. Construct the local segment W from μ to ν by lemma 4.237, and let W^* be its HKKO boundary-event splice. Then W^* is an unmarked w-up-down tableau from μ to ν, and the marked preimage segment W is recoverable from W^*, β_α, w, h_α, and the local endpoint μ. Thus the canonical loop splice introduces no hidden local data beyond the branch word.

Proof. Lemma 4.237 supplies assigned indices

$$\tau_i = \vartheta_i = R + i \qquad (1 \le i \le h_\alpha)$$

and proves that they are determined by w, h_α, μ. The same lemma checks the hypotheses of lemma 4.236; hence lemma 4.230 applies to W and produces an unmarked segment W^* with the same endpoints μ, ν. Its reversibility statement recovers W from W^*, β_α, and the assigned indices. Since the assigned indices are themselves recovered from w, h_α, μ, the displayed data recover W. No further local datum is used. □

Lemma 4.239 (Local endpoints canonically choose a below-wall parameter)**.** *Let μ and ν be partitions. Define*

$$w(\mu, \nu) = \max\{3, \ell(\mu) + 1, \ell(\nu) + 1\}.$$

Then $w(\mu, \nu) \ge 3$,

$$\ell(\mu) < w(\mu, \nu), \qquad \ell(\nu) < w(\mu, \nu),$$

and the integer $w(\mu, \nu)$ is a fixed function of the endpoint pair (μ, ν).

Proof. The inequalities $w(\mu, \nu) \ge \ell(\mu) + 1$ and $w(\mu, \nu) \ge \ell(\nu) + 1$ are immediate from the definition of the maximum, and they imply the two strict below-wall inequalities. The same definition also gives $w(\mu, \nu) \ge 3$ and shows that no additional choice is involved. □

Proposition 4.240 (Below-wall endpoints supply the local HKKO splice package)**.** *Work in the setting of lemma 4.226. Put $h = h_\alpha$, and let μ, ν be the local endpoint partitions for a deletion joint. Suppose that*

$$\ell(\mu) < w, \qquad \ell(\nu) < w.$$

Then, for every local branch word

$$\beta_\alpha \in \{0,1\}^{2h_\alpha},$$

there is a canonical assigned index list

$$\tau_1 < \cdots < \tau_{h_\alpha}$$

and a canonical w-up-down segment W from μ to ν whose selected indices are exactly the subset M_α recovered from the second half of β_α. The HKKO boundary-event splice produces an unmarked w-up-down segment W^* from μ to ν, and the marked preimage W is recoverable from

$$W^*, \quad \beta_\alpha, \quad w, \quad h_\alpha, \quad \mu.$$

No event-index choice or marking choice beyond the branch word is introduced.

Proof. By lemma 4.226, the branch word recovers the selected subset $M_\alpha \subseteq \{1,\dots,r_\alpha\}$ from its second half. Apply lemma 4.237 to the endpoints μ, ν, the integer $h = h_\alpha$, and this subset. It gives a canonical below-wall w-up-down segment W from μ to ν and a canonical assigned index list $\tau_i = \vartheta_i$ determined by w, h_α, μ; its peak set is exactly the image of M_α under that list.

The same lemma checks the hypotheses of lemma 4.236. Hence lemma 4.230 applies and the HKKO boundary-event splice produces an unmarked segment W^* with the same endpoints. Its reversibility statement recovers W from W^*, β_α, and the assigned indices. Since the assigned indices are themselves fixed functions of w, h_α, μ, the displayed data recover W. □

Lemma 4.241 (Canonical below-wall packet outputs are image-compatible)**.** *Work in the setting of lemma 4.226. Put $h = h_\alpha$, let μ, ν be endpoint partitions for a local packet, and put*

$$w = w(\mu, \nu)$$

as in lemma 4.239. For every local branch word

$$\beta_\alpha \in \{0,1\}^{2h},$$

there are canonically chosen segments

$$W(\mu,\nu,\beta_\alpha), \qquad W^*(\mu,\nu,\beta_\alpha),$$

such that $W^*(\mu,\nu,\beta_\alpha)$ is an unmarked HKKO packet output from μ to ν in the image of the canonical below-wall splice package of proposition 4.240, and the marked preimage $W(\mu,\nu,\beta_\alpha)$ is recoverable from

$$W^*(\mu,\nu,\beta_\alpha), \quad \beta_\alpha, \quad w, \quad h, \quad \mu.$$

No packet-output image choice is introduced beyond the endpoint pair and the branch word.

Proof. Lemma 4.239 gives

$$\ell(\mu) < w, \qquad \ell(\nu) < w.$$

Therefore proposition 4.240 applies to the endpoint pair μ, ν, the canonical wall w, and the branch word β_α. In the proof of that proposition the endpoint pair and branch word canonically choose the assigned index list, the below-wall marked segment W, and its unmarked HKKO splice output W^*. The same proposition states that W is recovered from

$$W^*, \quad \beta_\alpha, \quad w, \quad h, \quad \mu.$$

Since W^* is produced by that splice, it lies in the image of the canonical below-wall splice package. All choices used in its construction are fixed functions of μ, ν and β_α. □

Lemma 4.242 (Connector insertion is not target-size preserving)**.** *Let P be a two-step path with j labelled blocks, and let $S \subset \{1,\dots,j\}$ have cardinality q. Suppose one forms a new windowed object* *by keeping every block whose label is not in S and, for the maximal deleted intervals, inserting local two-step windows of half-lengths $N_1, \dots, N_L$. Then the resulting windowed object has*

$$(j-q) + \sum_{\alpha=1}^{L} N_\alpha$$

two-step windows. Hence it belongs to $\mathcal{P}_{j-q}^{\text{even}}$ by block count only if every inserted window has half-length 0.

Consequently the connector-loop segments of corollaries 4.235 and 4.238 and lemma 4.237 cannot be the final fixed-witness map by raw insertion when any connector-loop window has positive half-length. They may only be used inside a target-size-preserving local splice or deterministic parser whose output still has exactly $j-q$ two-step blocks.

Proof. Deleting the q labelled blocks leaves exactly $j-q$ retained blocks. Raw insertion of local windows adds N_α two-step windows at the α-th deleted interval and does not remove any retained block. Summing over the maximal deleted intervals gives the displayed count. The paths in $\mathcal{P}_{j-q}^{\text{even}}$ have, by definition, exactly $j-q$ two-step blocks, so equality of block counts forces $\sum_\alpha N_\alpha = 0$. Since each N_α is nonnegative, this is equivalent to $N_\alpha = 0$ for all α.

The connector-loop segments have positive half-length whenever they contain a nonempty connector or loop. Therefore raw insertion of such a segment cannot be the final target-size map. A valid use must occur inside a construction whose output block count is restored to $j-q$, which is precisely the target-size-preserving splice or parser alternative stated in the last sentence. □

Lemma 4.243 (Unchanged retained windows leave no target-size slots). *Let P be a two-step path with j labelled blocks, let $S \subset \{1, \dots, j\}$ have cardinality q, and put $I(S) = \{j-s+1 : s \in S\}$. Let P^* be a two-step path with exactly $j-q$ two-step windows. Suppose a deterministic parser of P^* returns pairwise disjoint output windows consisting of:*

(1) *one retained window for each $i \notin I(S)$, equal to the original block $B_i(P)$; and*
(2) *local output windows U_α^* attached to the maximal deleted intervals of $I(S)$,*

and suppose these returned windows exhaust the $j-q$ two-step windows of P^*. Then every local output window U_α^* has half-length 0.

Consequently, a target-size fixed-witness repair cannot both keep every nondeleted block as a full unchanged output window and also insert any positive-length local HKKO or connector segment. Any positive-length local repair must replace or merge some neighbouring retained windows, or the local inverse must use a zero-window slot together with the endpoint and branch data.

Proof. The set $I(S)$ has q elements, so there are exactly $j-q$ indices $i \notin I(S)$. The parser therefore already returns $j-q$ retained two-step windows in item (1). Since these windows are pairwise disjoint and the whole target path P^* has exactly $j-q$ two-step windows, no additional returned window can have positive half-length. Hence each U_α^* has half-length 0.

The consequence is the same counting statement applied to any proposed repair whose output lies in the target set $\mathcal{P}_{j-q}^{\text{even}}$: a positive-length local segment would add at least one extra two-step window unless it replaces or merges windows that were otherwise counted among the unchanged retained blocks. If no such replacement or merger occurs, the local parser has only a zero-window slot at that joint, so inversion at the joint can use only the endpoint data and the branch word. □

Lemma 4.244 (Zero-window unchanged repairs force endpoint matching). *Let*

$$P = (\rho_0, \rho_1, \dots, \rho_{2j})$$

be a two-step path with j labelled blocks, let $S \subset \{1, \dots, j\}$, and put $I(S) = \{j-s+1 : s \in S\}$. Write the maximal consecutive intervals of $I(S)$ as

$$K_\alpha = \{u_\alpha, u_\alpha + 1, \dots, v_\alpha\}.$$

Suppose a target two-step path P^* is parsed as the unchanged retained blocks $B_i(P)$, $i \notin I(S)$, in their original order, with no positive-length local output window at any deleted interval. Then each deleted interval satisfies the corresponding endpoint matching condition:

$$\rho_{2u_\alpha - 2} = \rho_{2v_\alpha} \qquad \text{if } 1 < u_\alpha \le v_\alpha < j,$$
$$\rho_{2v_\alpha} = \varnothing \qquad \text{if } u_\alpha = 1 \le v_\alpha < j,$$

and

$$\rho_{2u_\alpha - 2} = \varnothing \qquad \text{if } 1 < u_\alpha \le v_\alpha = j.$$

There is no endpoint condition when $K_\alpha = \{1, \ldots, j\}$.

Consequently, an unchanged-retained zero-window repair can apply only to deletion joints whose boundary partitions already match the start/end requirements above. At any nonmatching joint, a target-size repair must change or merge neighbouring retained windows, or else use a local mechanism that is not parsed as an unchanged zero-window deletion.

Proof. Consider first an internal interval $1 < u_\alpha \le v_\alpha < j$. Since K_α is maximal, the neighbouring block indices $u_\alpha - 1$ and $v_\alpha + 1$ are retained. With no positive local output window between them, these two retained output blocks are adjacent in the parsed target path. The right endpoint of $B_{u_\alpha - 1}(P)$ is $\rho_{2u_\alpha - 2}$, and the left endpoint of $B_{v_\alpha + 1}(P)$ is ρ_{2v_α}. Adjacent two-step windows in a single path share their common endpoint, so $\rho_{2u_\alpha - 2} = \rho_{2v_\alpha}$.

If $u_\alpha = 1$ and $v_\alpha < j$, then the first retained block is $B_{v_\alpha + 1}(P)$ and, with no preceding local window, it must start at the initial partition of P^*, namely $\varnothing$. Its left endpoint is ρ_{2v_α}, proving the second displayed equality. The terminal case $1 < u_\alpha \le v_\alpha = j$ is the same argument at the right end: the last retained block before the deleted interval has right endpoint $\rho_{2u_\alpha - 2}$, and the target path ends at $\varnothing$. If all blocks are deleted, there are no retained neighbouring windows and hence no endpoint equality is forced.

The consequence is the contrapositive of these necessary conditions. □

Lemma 4.245 (Endpoint matching is sufficient for raw retained concatenation)**.** *Let*

$$P = (\rho_0, \rho_1, \ldots, \rho_{2j})$$

be a two-step path with j labelled blocks, let $S \subset \{1, \ldots, j\}$, and put $I(S) = \{j - s + 1 : s \in S\}$. Suppose every maximal deleted interval $K_\alpha = \{u_\alpha, \ldots, v_\alpha\}$ of $I(S)$ satisfies the matching condition displayed in lemma 4.244. Then concatenating the retained blocks

$$B_i(P) = (\rho_{2i-2}, \rho_{2i-1}, \rho_{2i}), \qquad i \notin I(S),$$

in their original order gives a two-step path with exactly $j - |S|$ blocks. Each retained block keeps its original vertical-strip adjacent differences and its original weight.

Proof. Order the retained indices increasingly. If two consecutive retained indices $i < i'$ satisfy $i' = i + 1$, then the right endpoint of $B_i(P)$ is ρ_{2i} and the left endpoint of $B_{i'}(P)$ is $\rho_{2i'-2} = \rho_{2i}$, so the blocks concatenate.

If $i' > i + 1$, then the missing indices between them form a maximal deleted interval $K_\alpha = \{i + 1, \ldots, i' - 1\}$. This is an internal interval, so the matching condition gives

$$\rho_{2i} = \rho_{2(i+1)-2} = \rho_{2(i'-1)} = \rho_{2i'-2},$$

which is again exactly the endpoint equality needed to concatenate $B_i(P)$ and $B_{i'}(P)$.

If the first retained index is $i > 1$, then the deleted prefix $\{1, \ldots, i - 1\}$ is a maximal deleted interval and the boundary matching condition gives $\rho_{2i-2} = \varnothing$, so the first retained block starts at the required initial partition. If there is no deleted prefix, then $i = 1$ and $\rho_0 = \varnothing$ already. The terminal suffix is identical: if the last retained index is $i < j$, the matching condition gives $\rho_{2i} = \varnothing$, and otherwise $i = j$ gives the original terminal partition $\rho_{2j} = \varnothing$.

Thus the retained blocks concatenate to a single two-step path. The number of retained indices is $j-|S|$, so this path has exactly $j-|S|$ blocks. The vertical-strip and weight assertions are inherited from the original blocks. □

Lemma 4.246 (Endpoint-matched zero-window codes are not universal). *Let $h \geq 1$ and let*

$$\mu = (5, 4, 3, 2, 1).$$

There are at least 5^h local two-step segments

$$U = (U^0, U^1, \ldots, U^{2h})$$

from μ to μ such that every adjacent difference is a vertical strip and every two-step block has weight 2:

$$|U^{2a-2}| - 2|U^{2a-1}| + |U^{2a}| = 2 \qquad (1 \leq a \leq h).$$

Consequently no zero-window local inverse that is given only the endpoint μ, the length h, and a branch word in $\{0,1\}^{2h}$ can recover all endpoint-matched local segments of this type.

Thus a matched zero-window step in the fixed-witness repair cannot use a universal endpoint-only inverse for arbitrary local growth segments; it must use the fixed witness-crossing restrictions, or else the target path must carry additional replacement or merger data.

Proof. The partition $\mu = (5, 4, 3, 2, 1)$ has five removable corner boxes, one at the end of each row. For each word

$$c = (c_1, \ldots, c_h) \in \{1, 2, 3, 4, 5\}^h,$$

define a local segment by

$$U^{2a-2} = \mu, \qquad U^{2a} = \mu \quad (1 \leq a \leq h),$$

and let U^{2a-1} be the partition obtained from μ by deleting the corner box in row c_a. Removing one corner box and adding it back are both vertical-strip moves. Moreover

$$|U^{2a-2}| - 2|U^{2a-1}| + |U^{2a}| = |\mu| - 2(|\mu| - 1) + |\mu| = 2.$$

Different words c give different midpoint sequences, hence different local segments. Therefore there are at least 5^h such segments.

The number of branch words of length $2h$ is $2^{2h} = 4^h$, and $5^h > 4^h$. If the local output window has length zero, all these segments have the same output and the same endpoint data. A map using only those data and a $2h$-bit branch word therefore has fewer possible codes than local segments, so it cannot be injective on this family. The final sentence is this counting obstruction applied to any proposed zero-window local inverse. □

Corollary 4.247 (Singleton zero-window packet inverses are not universal). *There is no fixed inverse which recovers every one-block local segment*

$$U = (U^0, U^1, U^2)$$

from only the common endpoint pair

$$U^0 = U^2 = (5, 4, 3, 2, 1),$$

the known singleton interval, and a branch word in $\{0,1\}^2$, when the local output window has half-length zero and the segment is required only to have vertical-strip adjacent differences and weight

$$|U^0| - 2|U^1| + |U^2| = 2.$$

Consequently, a height singleton packet cannot be discharged by a universal endpoint-plus-branch zero-window inverse in the ambient growth-path category. Any singleton-packet supplier must use the fixed witness-crossing restrictions or carry replacement/merged target data that recover the midpoint.

Proof. This is the case $h = 1$ of lemma 4.246. There are at least five such segments with the displayed common endpoints, obtained by deleting and restoring one of the five removable corner boxes of $(5, 4, 3, 2, 1)$. A zero-window code with the displayed endpoint pair and a two-bit branch word has only four possible values. Therefore it cannot be injective on this five-element family, so no fixed inverse using only those data can recover all segments. The final sentence is the same obstruction applied to a proposed singleton-packet zero-window decoder before any fixed-witness or merged-target information has been used. □

Corollary 4.248 (Path formulation of the repaired deletion problem). *To prove the fixed-witness estimates in proposition 4.563, it is enough to construct, for the growth paths of lemma 4.118 attached to a fixed witness set S, an injection into*

$$\{0,1\}^{2q} \times \mathcal{P}^{\mathrm{even}}_{j-q},$$

where $\mathcal{P}^{\mathrm{even}}_{j-q}$ denotes the two-step growth paths corresponding to column-even semistandard tableaux of content (2^{j-q}). The path operation must remove the q two-step blocks whose tableau labels lie in S–equivalently, blocks $j - s + 1$ for $s \in S$–repair the resulting joints, and recover the original path from the repaired path, S, and the $2q$ branch bits. The source-backed wall-event model available for this repair is the up-down boundary-event model of lemma 4.129, the reversible boundary-event splice of lemma 4.130, and the marked boundary-step specialization in lemma 4.134; the standard-tableau odd-column removal available after a joint has been converted into a genuine standard tableau is lemma 4.137, but lemma 4.138 shows that this removal cannot be the final fixed-witness map unless no odd columns remain; the semistandard-to-standard bridge is the doubled-label standardization of lemma 4.139, with the local adjacent-pair frontier states of lemma 4.140 and the crossing times of lemma 4.141, ordered inside the deleted pairs by lemma 4.142, distributed across deletion joints by lemma 4.188, and assigned their consecutive local frontier ranks by lemma 4.189; the matching column-parity defects are localized joint-by-joint by lemma 4.221 and are determined by the adjacent endpoint partitions by lemma 4.222, with canonical paired defect slots from corollary 4.223 assigned to ordered frontier-event slots by corollary 4.224; the allowed local branch word is made explicit in lemma 4.225, recovers the local event/defect data by lemma 4.226, and becomes an admissible HKKO marking once the selected slots are realized as eligible boundary-up steps by lemmas 4.227 and 4.228. By lemma 4.169, it remains to connect the deleted consecutive standard-time pairs, equivalently the deleted two-step blocks, and their ordered per-joint frontier times to the penultimate-row creations of lemma 4.132 in the up-down splice segment and to the marked boundary-up steps of lemma 4.136 after coefficient extraction, to place the resulting local segment in the domain of the HKKO splice lemma 4.130 using the explicit criterion lemma 4.131 and the decoded-splice package lemma 4.230, with the canonical local event indices of lemma 4.231, the explicit top-wall peak-loop model of lemma 4.232, the peakless connector attachment of lemma 4.234, the arbitrary-endpoint connector and index-alignment leaves corollary 4.235 and lemma 4.237, the local splice inversion of corollary 4.238, the below-wall package proposition 4.240, and the peak-supported landing criterion of lemma 4.236, while respecting the target-size guard of lemma 4.242 and the no-spare-window constraint of lemma 4.243, as well as the endpoint-matching obstruction of lemma 4.244. Conversely, lemma 4.245 says that matched zero-window joints are exactly the cases where the retained windows already concatenate as a target-size path, while lemma 4.246 rules out a universal endpoint-only inverse for all matched local loops with only the $2h$ branch bits. Thus the remaining construction must output an object of $\mathcal{P}^{\mathrm{even}}_{j-q}$ through a target-size-preserving splice or parser, cannot simply append positive-length local windows while keeping all retained windows unchanged, cannot simply skip nonmatching deleted intervals, cannot recover arbitrary matched local loops from only endpoint data and the branch word, and must recover the original path from that repaired path and the decoded branch words. The replacement-zone recovery criterion of proposition 4.252 is the remaining positive parser shape: a local repair may absorb neighbouring retained blocks only if the parsed replacement *zone outputs exactly the number of retained blocks in that zone and has a local inverse. The canonical halo zones of lemma 4.254 provide a deterministic one-block-neighbour replacement-zone choice from S with the exact global block-count identity needed by that parser. The branch-accounting form proposition 4.541 shows that two branch bits per deleted witness block inside each replacement zone are exactly sufficient for the required $2^{2q}s_{j-q}$ fixed-fiber bound. lemma 4.257 shows that the retained material inside one zone consists only of endpoint halo blocks and one- or two-block gaps between deleted packets. lemma 4.258 bounds the replacement output in each zone by twice the number of localized deleted blocks. lemma 4.259 gives the exact retained-minus-deleted balance of a zone from its endpoint halo bits, internal retained gap lengths, and deleted packet lengths. corollary 4.318 sharpens this on the height side in the residual range: height witness labels are separated, so the canonical halo zones contain only singleton deleted packets and have no negative local balance. corollary 4.319 further classifies the tight height zones: they are exactly right-boundary alternating singleton strings; every other height zone has positive surplus. corollary 4.320 translates such a tight zone back to labels: the deleted labels are precisely $\{1, 3, \ldots, 2m-1\}$ and the retained labels are $\{2, 4, \ldots, 2m\}$. corollary 4.328 gives the exact remaining gate: if $m \geq 2$ and the prefix endpoint at label $2m$ is already column-even, the closed odd alternating height decoder applies; otherwise the missing datum is the singleton local case or the endpoint/continuation parity repair. corollary 4.330 removes the only combinatorial freedom in that singleton case: the deleted label-1 strip is the forced first-row domino, so the unresolved singleton work is only the target-size merge with the retained label-2 block and the surrounding endpoint/continuation recovery. lemma 4.331 makes the continuation datum explicit: the columns in which the later labels must contribute odd parity are exactly the odd columns of the tight prefix endpoint, with canonical adjacent pairing. lemma 4.332 also shows that this full endpoint-defect set cannot simply be routed through the tight zone's own m frontier slots: a concrete tight $m = 2$ prefix has three paired endpoint defects. lemma 4.334 gives the replacement structure: those paired endpoint defects are nevertheless chargeable injectively to distinct continuation labels whose strips hit the corresponding defect pair. corollary 4.326 gives the parser-facing geometry of every non-full tight height zone: the block immediately to its left is outside the one-block halo and is therefore returned unchanged, so the left endpoint of the tight zone is visible without branch data. corollary 4.327 packages this for the right-boundary auxiliary criterion: the parsed separator block and the terminal endpoint $\varnothing$ determine the endpoint pair of the tight zone. lemma 4.514 then locates the Hall-matched continuation labels on the parser side: their corresponding blocks are never inside the tight zone, but lie either in the unchanged complement or in strictly earlier halo-zone windows. lemma 4.335 removes the remaining hidden choice in this continuation charge: the matching is the lexicographically least Hall injection determined by the recovered incidence relation. lemma 4.336 identifies that incidence relation in semistandard coordinates: a continuation label is incident to a defect pair exactly when one of its two tableau boxes lies in one of the paired columns. lemma 4.337 records the exact conversion from labelled boxes to the path segment: the prefix endpoint plus the two boxes of every continuation label determine $\lambda^{(2m)} \subset \cdots \subset \lambda^{(j)}$. lemma 4.513 packages the same information at the path level: recovering the Pieri continuation segment from label $2m$ onward determines the paired endpoint defects, all continuation incidences, and the canonical Hall charge. proposition 4.338 combines these pieces into the tight-zone auxiliary criterion: outside recovery of the prefix endpoint and labelled continuation boxes supplies the tight-zone endpoint pair, the incidence graph, and the canonical Hall charge with no new branch data. proposition 4.495 uses the smaller direct Hall-data packaging for the current carrier route: it is enough for the target-window data to recover the non-column-even prefix endpoint, paired endpoint defects, continuation incidences, canonical Hall charge, and a Hall-augmented Step 3 carrier containing L_{2m}. Corollary 4.516 proves that the Hall-charged continuation blocks selected by this direct Hall charge are already part of the recovered outside datum, without first recovering the full Pieri continuation segment. proposition 4.497 remains one source for this direct interface: a recovered Pieri continuation segment determines the Hall data, and corollary 4.515 recovers the selected outside blocks from that segment. lemmas 4.176* *and 4.181 and corollary 4.342 make the continuation-segment half smaller: the outside strict-left subfilling already supplies the Step 2 Q-suffix entries, so the target window only has to recover the internal separating boundary between the Step 2 Q-prefix strip and that suffix strip; the remaining incoming suffix boundary is the fixed global empty boundary by lemma 4.177. Lemma 4.178 and corollaries 4.180 and 4.343 identify this cut from the opposite side: the Q-prefix entries determine the cut, the low-label incidence entries determine those prefix entries, and in the non-full tight setting the recovered strict-left columns reduce that readout to the low-low cell set* L_{2m}*. lemma 4.339 identifies the growth-path data already supplied by the right-boundary outside parser: it recovers the strict-left initial segment of the Krattenthaler growth path, the segment corresponding to continuation labels* $2m+1, \ldots, j$*. proposition 4.340 names the remaining bridge in a closure-ready form: if this strict-left segment decodes the Pieri prefix endpoint and continuation boxes, then all tight-zone auxiliary data are parser-supplied and branch-free. lemma 4.344 shows that this decoding is not a consequence of the strict-left segment alone: a tight-zone inverse must also use the constraints or branch data native to the final right-boundary zone. lemma 4.347 localizes the missing information to the endpoint-defect data of a deleted singleton packet, so the next inverse target is recovery of that packet datum from the final-zone constraints or right-boundary auxiliary data before applying the packet-local branch decoder. lemma 4.348 rules out the next tempting shortcut: retained boxes together with their smallest column-even closure can miss an all-deleted even-column pair. Thus the final tight-zone inverse must recover the full tight-prefix packet data, not only the endpoint-defect parity. lemma 4.349 sharpens this once more: even the prefix endpoint, the retained even-label boxes, and the continuation boxes can agree while two odd deleted packets are interchanged. Thus the residual odd-packet matching itself must be recovered from the repaired target window or from the already budgeted branch word. lemma 4.352 identifies the available bit source: in height packets the first half of the local branch word is unused by the local recovery formulas. Proposition 4.353 therefore reduces the residual odd-packet matching to a finite ambiguity bound* $|\Omega| \leq 2^m$ *for the parser-visible tight-prefix datum. Definition 4.354, lemma 4.355, and corollary 4.365 make this* Ω *explicit from the tight-prefix endpoint and retained even strips, and give a first sufficient rank leaf: a uniform cardinality bound for this endpoint/even-strip compatibility set. Definition 4.368, lemma 4.369, and corollary 4.370 replace the recursive Pieri-path formulation of* Ω *by an equivalent fixed-cell semistandard completion problem with ordinary row and column inequalities. Lemma 4.371, definition 4.372, proposition 4.373, and corollary 4.374 strip off the now forced first column and reduce the remaining cardinality leaf to a fixed-even growth-chain count in the non-first-column diagram. Lemma 4.375 records that every nonterminal reduced even strip is a singleton; only the terminal even strip can have two boxes. Definition 4.376, lemma 4.377, and corollary 4.378 then reverse that count into an exact corner-removal tree; the unresolved estimate is the uniform leaf bound for this tree. Definition 4.379, lemma 4.380, and corollary 4.381 remove the terminal fixed strip and identify this tree with an odd-size standard-tableau fiber in which all even entries are fixed. Definition 4.382, lemma 4.383, and corollary 4.384 then identify the same fiber with interval-constrained linear extensions of the odd-cell Young subposet. Definition 4.385, lemma 4.386, and corollary 4.387 give the exact terminal recurrence and show that a non-removable final fixed even cell has only two possible final odd blockers. Lemma 4.507 shows that the full fixed-even* 2^m *bound is false in this breadth: a compatible* $m = 26$ *datum has* $254531860 > 2^{26}$ *fixed-even tableaux. Thus the residual supplier cannot be the uniform fixed-even estimate alone; it must use the extra parser-visible outside data of the canonical halo/right-boundary construction. Lemma 4.357 and corollary 4.358 and lemmas 4.359 and 4.360 and corollary 4.362 and propositions 4.363 and 4.364 remove every case* $m \leq 8$ *from that live ambiguity: when* $m = 1$ *the odd packet is forced, and when* $2 \leq m \leq 6$ *the binary first-two-coordinate prefix and a residual permutation bound already fit inside the* 2^m *branch budget; the* $m = 7, 8$ *cases are closed by exact finite enumeration of the tight data. Corollary 4.366 records the corrected live leaf: construct, from those parser-visible outside data, a canonically ordered refined subset of the broad compatibility set that contains the true odd-packet assignment and has size at most*

2^m. *Definition 4.401, proposition 4.426, and corollary 4.427 then isolate the exact remaining overflow datum: after the branch-free parser-visible right-boundary data fix the commuting normal-form signature, every remaining signature class is binary at each non-commuting stage and hence has size at most* 2^m. *Lemma 4.402 gives the first local decoder rule for that signature: the symbol is* $*$ *exactly when the fixed even cell is blocked, and otherwise it is recovered from the lower shape after deleting the top odd/even pair. Definition 4.403, proposition 4.404, and corollary 4.405 iterate this rule: the whole commuting signature is determined by the associated fixed-even datum and its odd sublevel shape chain. Proposition 4.406 and corollary 4.424 translate this shape chain back to the reduced half-step diagrams already present in the fixed-cell growth-chain model. Lemma 4.407 and corollary 4.425 show that the reduced fixed-even datum itself is not a separate mystery once the tight-prefix endpoint and retained even strips are visible; the live remaining decoder is the true reduced half-step chain. Lemma 4.408 and corollary 4.409 then identify those half-steps as first-column deletions of the true odd-time tight-prefix Pieri shapes, so a decoder for the true tight-prefix shape chain also supplies the signature input. Corollary 4.410 translates the same target into doubled-standardization coordinates: it is enough to recover the even standard-time endpoints through time* $4m$. *Corollary 4.411 then gives the growth-diagram form of the same task: a parsed dual-RSK' carrier subarrangement, together with a deterministic standardization readout from the recovered subfilling and internal corner labels, supplies those endpoints by a backward sweep. Corollary 4.412 makes that readout concrete: it is enough for the recovered carrier data to determine the two boxes carrying each semistandard label* $1, \dots, 2m$. *Lemma 4.356 also identifies the same readout with the tight-prefix packet datum* $(\lambda^{(2m)}; E_1, \dots, E_m; x_1, \dots, x_m)$ *used in the finite odd-packet rank criterion. Lemma 4.171 and corollary 4.413 then restate the readout in the actual two-stage Krattenthaler construction: it is enough for the recovered Step 3 carrier cells to determine a Step 2 Knuth subarrangement whose forward boundary contains the prefix* $\lambda^{(0)}, \dots, \lambda^{(2m)}$. *Lemma 4.127 and corollary 4.414 remove the cell-entry part of that readout: since Step 3 is run on one off-diagonal half of the symmetric Step 2 matrix, a recovered Step 3 carrier already supplies the entries of any Step 2 cells it contains, up to transpose. Lemma 4.175 and corollary 4.415 then remove the left/bottom-boundary part for the anchored Q-prefix strip: the strip starts on the global empty boundary, so its entries alone recover* $\lambda^{(0)}, \dots, \lambda^{(2m)}$ *by the first-variant RSK boundary description. Lemma 4.179 and corollary 4.416 make the required Step 3 carrier explicit: it is enough to recover the half-square cells representing unordered label pairs with at least one label in* $\{1, \dots, 2m\}$. *Lemmas 4.182 and 4.341 then split this target: the cross cells with one continuation label are already recovered from the strict-left boundary segment, so the live target is the low-low triangle* L_{2m}. *Corollaries 4.417 and 4.418 record the corresponding entry-level criteria. Lemma 4.183 and corollary 4.419 sharpen this once more:* L_{2m} *is the terminal low-label dual-RSK' corner, so it is enough to recover its boundary segment* $\rho_{2(j-2m)}, \dots, \rho_{2j}$ *from the target-window data. Corollary 4.420 translates that boundary target back into replacement-zone language: a branch-free decoder for the original terminal right-boundary zone subpath is sufficient, but stronger than the rank-budget route requires. Corollary 4.428 records the smaller target compatible with the existing branch accounting: the target-window data may instead determine a canonical at-most-*2^m *refined ambiguity class* Ω_{tw} *for the odd packets. Corollary 4.429 identifies one such class: the fiber of the commuting normal-form signature inside the broad compatibility set. Corollary 4.432 rewrites the base datum in parser-facing terms: the retained strips are just the two-box readouts for labels* $2, 4, \dots, 2m$. *Corollary 4.433 rewrites this once more in growth-chain terms: the even standard-time endpoint chain, together with the continuation boxes, supplies the base datum and the signature fiber. Corollary 4.434 reduces the same criterion to the low-label box readout: recovering the two boxes carrying each label* $1, \dots, 2m$, *together with the labelled continuation boxes, already recovers that even chain and therefore the signature-fiber ambiguity class. Corollary 4.435 records the sharper conclusion used by the current route: the same low-label readout determines the actual deleted odd-packet assignment, so the refined ambiguity class may be chosen to be a singleton. Corollary 4.436 replaces the abstract continuation-box hypothesis by a Step 2 continuation sweep: once a recovered Knuth* *subarrangement contains the boundary segment $\lambda^{(2m)}, \ldots, \lambda^{(j)}$, the labelled continuation boxes are read off from consecutive differences. Corollary 4.437 then pulls this back to the existing low-low triangle target: the strict-left segment recovers every Step 3 cell incident to a continuation label, so the remaining low-low entries recover the whole Step 2 boundary path and close the singleton rank route. Corollary 4.438 is the branch-budget version of the same target: exact recovery of L_{2m} is not necessary if the target-window data determine a canonical at-most-2^m list of possible low-low subfillings containing the true one. Definition 4.440 and proposition 4.441 make this row-free: because the Step 2 matrix has doubled-content row sums and the strict-left segment already recovers every continuation-column entry, the true low-low subfilling must lie in an explicitly determined finite degree-constrained ambient set. The remaining supplier is therefore a uniform target-window refinement of that ambient set, not a row-by-row residual computation. Proposition 4.442 closes the sparse residual-degree part of this ambient set under the relaxed final-zone budget: if the total residual low-low degree $D = \sum_a d_a$ satisfies $(D-1)!! \le 2^{2m}$, then the whole degree-constrained ambient set already fits in the full branch word. In particular it also closes the active-label case $(2r-1)!! \le 2^{2m}$, where $r = \#\{a : d_a > 0\}$. Thus the live low-low source extraction starts only after this double-factorial bound fails. Proposition 4.443 shrinks that remaining branch further: if $r_2 = \#\{a : d_a = 2\}$, then each low-low assignment has at least $2^{\lceil r_2/2 \rceil}$ labelled-port realizations, so the whole degree-pruned ambient set still fits the full final-zone budget whenever*

$$(D-1)!! \le 2^{2m+\lceil r_2/2 \rceil}.$$

Definition 4.444 and proposition 4.446 replace these sufficient double-factorial tests by the exact labelled max-degree-two profile count $\mathsf{N}(r_1, r_2)$, where $r_i = \#\{a : d_a = i\}$. Only the exact degree-profile dense cases $\mathsf{N}(r_1, r_2) > 2^{2m}$ still need an additional target-window low-low candidate source. The certificate-backed proposition 4.452 further shows that every such dense case has at least $A(m)$ positive residual labels, where $A(m)$ is the exact active-support dense-profile frontier defined there, and total residual low-low degree at least $B(m)$, where $B(m)$ is the corresponding exact total-degree frontier. Equivalently, corollary 4.453 closes the continuation-heavy complement: if $r_0 = \#\{a : d_a = 0\}$ and $C = \sum_a c_a$, the live dense branch has

$$r_0 \le 2m - A(m), \qquad C \le 4m - B(m).$$

Proposition 4.461 records the matching obstruction to a degree-only refinement: residual degrees $d_a = 1$ already give $(2m-1)!!$ low-low assignments, exceeding the 2^m branch budget for $m \ge 3$. Thus the refinement has to use the target-window signature or equivalent semistandard data, not only the recovered residual degrees. Lemma 4.345 gives the concrete four-label version of the same obstruction inside the Step 3 construction: two examples have the same strict-left continuation subfilling and the same residual low-label degrees, but different L_4 fillings. Corollary 4.463 gives the current refinement mechanism: any parser-visible refined odd-packet class Ω_{tw} of size at most 2^m induces, by inverse Step 2 uniqueness, a low-low candidate class of the same size bound. Thus the live target can be phrased either as a bounded refined odd-packet class or as the induced bounded low-low class. Corollary 4.466 eliminates the separate refined-class construction from this live target: once the target-window data recover the base tight-prefix endpoint, retained even strips, labelled continuation boxes, and the true commuting signature κ, the bounded low-low class is the induced image of the signature fiber $\Omega_{\mathrm{tw}}(\kappa)$. Corollary 4.430 records the relaxed full-branch version: exact recovery of κ is stronger than necessary if the target-window data instead recover an at-most-2^m list of possible commuting signatures containing the true one. Lemma 4.126 and corollary 4.439 then plug this branch-budgeted low-low recovery into the actual local inverse: after the first m branch bits select the true low-low filling, the strict-left continuation columns complete the Step 3 half-square, whose forward boundary is the original terminal ρ-subpath. Corollary 4.467 packages the resulting final-facing sufficient condition: base data plus κ from the target window give the tight right-boundary inverse. Proposition 4.468 then inserts this inverse into the global right-boundary auxiliary criterion: with

$A(P) = \mathcal{D}_{\mathrm{out}}(P)$, the final tight-zone inverse is a fixed function of the allowed parser data and the already budgeted branch word. Corollary 4.503 and proposition 4.504 supplies the matching right-boundary inverse for the already closed small leaves $m \le 8$ and inserts it into the global auxiliary criterion: in those cases the full tight compatibility set is itself small enough, so no target-window signature is needed. Proposition 4.505 packages the split condition actually needed by the global supplier: base target-window data suffice for $m \le 8$ and, by the full-branch broad-Ω bound, also for $9 \le m \le 28$; only the range $m \ge 29$ still requires the commuting normal-form signature κ. Proposition 4.469 removes κ as an independent large-zone readout target: in the remaining supplier range $m \ge 29$ (the statement is available already for $m \ge 9$), it is enough that the same target-window tuple recover the even standard-time endpoint chain through time $4m$ and the labelled continuation boxes. Proposition 4.470 pushes this to the concrete box readout: recovering the two boxes carrying each label $1, \dots, 2m$, together with the labelled continuation boxes, supplies the large final-zone inverse. Proposition 4.472 pushes the same large-zone criterion to the Step 3 data already split by the strict-left machinery: it is enough to recover the low-low triangle entries L_{2m}, since the strict-left outside datum supplies the continuation columns. Proposition 4.473 records the branch-free boundary version of this criterion: recovering the terminal low-label boundary segment $\rho_{2(j-2m)}, \dots, \rho_{2j}$ recovers L_{2m} and hence supplies the large final-zone inverse. Proposition 4.474 uses the whole final-zone branch word directly: a canonical at-most-2^{2m} list of possible terminal boundary segments containing the true one is already enough to recover the original final-zone subpath. Proposition 4.476 is the half-budget boundary version used by the refined-rank route: a canonical at-most-2^m list of possible terminal boundary segments is enough, because each candidate has a deterministic low-low subfilling. Proposition 4.477 records the corresponding half-budget low-low form. Proposition 4.478 is the direct final-zone version: if the target-window data determine a canonical at-most-2^{2m} list of possible L_{2m} fillings containing the true one, all $2m$ branch bits may select that filling and the full Step 3 half-square then recovers the final-zone subpath. Proposition 4.479 gives the current supplier-side form: it is enough for the target-window data to recover a bounded refined odd-packet class Ω_{tw}, since its induced low-low candidates inject into it. Proposition 4.480 records the relaxed full-branch version: an at-most-2^{2m} odd-packet class also suffices when all final-zone branch bits are used directly for the right-boundary inverse. Corollary 4.464 records the same source in the low-low language: the induced low-low candidates still inject into the full-budget odd-packet class. Proposition 4.481 is the corresponding parser-facing signature-list supplier: an at-most-2^m list of commuting signatures gives an at-most-2^{2m} odd-packet class by the binary signature-fiber theorem, and corollary 4.465 converts it to the same full-budget low-low candidate source. Proposition 4.482 uses the existing binary-prefix factorial bound to close the unrefined broad compatibility set for $9 \le m \le 12$ under this relaxed budget; those cases no longer require the commuting-signature refinement once the base target-window data are recovered. Proposition 4.483 uses the exact recursive fixed-even envelope certificate to close the unrefined broad compatibility set for $13 \le m \le 28$ under the same relaxed budget. Proposition 4.398 verifies exactly that the same recursive envelope already exceeds the full branch budget at 144 non-column-even frontier shapes for $m = 29$, and that all other $m = 29$ shapes remain below budget. It also verifies exactly 346 over-budget top-fixed pairs (γ, e), exactly 62 over-budget top-two triples, and no over-budget top-three quadruple at $m = 29$. Thus proposition 4.484 closes every broad-Ω case off the top-fixed envelope frontier $\mathcal{P}_m^{\mathrm{env}}$, while proposition 4.487 closes the remaining $m = 29$ frontier by using the next two retained even cells. Propositions 4.399 and 4.488 closes the $m = 30$ frontier at fixed depth seven, and propositions 4.400 and 4.489 closes the $m = 31$ frontier at fixed depth ten. Propositions 4.485 and 4.486 then close every $m \ge 32$ case outside some target-window-determined fixed-depth envelope frontier. The live broad-Ω branch therefore starts at $m \ge 32$ and must remain over budget at every fixed depth. Lemma 4.490 records the matching threshold check: the same factorial estimate exceeds the full branch budget from $m = 13$ onward; the certified recursive envelope supplies the replacement bound through $m = 28$. Proposition 4.491 gives the concrete carrier form: a target-window/strict-left rule that identifies a Step 3 carrier

containing H_{2m} supplies the needed low-low entries and hence the large final-zone inverse. Proposition 4.492 sharpens the live carrier target: because the strict-left outside datum already supplies the cross-incidence part, the carrier for the remaining large range only has to contain the terminal low-low cell set L_{2m}. Corollary 4.421 inserts the already recovered strict-left ρ-segment into this criterion, leaving only the carrier-identification, Step 2 subarrangement, and left/bottom boundary readout step from that segment together with the target window and endpoint pair. Corollary 4.422 records the corresponding anchored-strip version: it is enough to identify a Step 3 carrier whose recovered half-matrix contains the Step 2 Q-prefix strip C_{2m} up to transpose. Corollary 4.423 is the same statement in incidence form: the carrier must contain the cell set H_{2m}. Proposition 4.506 makes the data source explicit: it is enough for the right-boundary replacement window V_L^*, together with the already recovered outside datum, to determine that signature. Definition 4.508 and proposition 4.509 give a sufficient binary-branching criterion for restricted instances of this count, while lemma 4.510 shows that this criterion is too strong even before the broader counterexample above. Lemma 4.511 gives the local branch-control dichotomy: each odd corner removal either commutes past the fixed even removal, or is immediately blocked by a fixed even box to its east or south. propositions 4.253 and 4.542 record the resulting recovery criterion: a right-boundary tight zone may use only such already recovered outside data as an auxiliary input, with no new branch words and with the same $2^{2q}s_{j-q}$ fixed-fiber count. Proposition 4.815 inserts the split target-window inverse into this final supplier: base target-window data handle the right-boundary C-side alternative through $m = 28$, through the certified $m = 29, 30, 31$ envelope frontiers, and through every fixed-depth non-frontier case. Only the $m \geq 32$ branch that remains in every fixed-depth envelope frontier still needs the commuting-signature or bounded signature-list readout in this split form. Proposition 4.816 gives the terminal-source version of those last readouts: for $m \geq 29$ in the non-column-even final-zone branch, the degree-pruned ambient set already closes the sparse residual-degree cases of proposition 4.442 and the degree-profile cases of proposition 4.443, and then the exact degree-profile cases of proposition 4.446; only the cases with $\mathsf{N}(r_1, r_2) > 2^{2m}$, hence at least $A(m)$ positive residual labels and total residual degree at least $B(m)$ by proposition 4.452, still need a source beyond the strict-left degree equations. The supplier now accepts either a partial low-low readout whose residual profile falls back under the exact full-budget frontier, by proposition 4.447, a full-budget odd-packet class, a bounded commuting-signature list, or a canonical full-budget list of possible L_{2m} low-low fillings containing the true one; the odd-packet and signature-list alternatives now both feed this last artifact through corollaries 4.464 and 4.465. Corollary 4.449 gives a checkable sufficient form of the partial-readout alternative: enough recovered low-low edge mass forces the residual total degree below $B(m)$. Corollary 4.450 gives the parser-facing sufficient form of this partial-readout alternative: an incidence readout on enough low labels closes the source as soon as the unexposed complement is below the certified active-support or degree frontier. In the same branch, corollary 4.451 records the corresponding cardinality and covered-degree tests for this complement condition, and corollary 4.456 normalizes those tests to the positive-residual part of the cover. Finally, corollary 4.453 gives the parser-side equivalent bounds $r_0 \leq 2m - A(m)$ and $C \leq 4m - B(m)$ for the fully continued labels and total low-continuation incidence. A full-budget list of possible terminal Step 3 boundary segments $\rho_{2(j-2m)}, \ldots, \rho_{2j}$, and in particular exact recovery of that segment, induces such a low-low candidate list by corollary 4.475. The earlier carrier and Hall-data formulations have all been reduced to this single terminal artifact. Proposition 4.494 removes one large subcase from that terminal target: if the tight prefix endpoint is already column-even, the closed odd-alternating full-zone decoder supplies the final right-boundary inverse. The live $m \geq 29$ carrier problem is therefore the non-column-even tight-prefix endpoint case. In that remaining case the useful auxiliary object is not a row-indexed table: it is the direct endpoint-defect/continuation incidence structure, the canonical Hall charge, and the Hall-selected continuation blocks recovered from that Hall charge and the outside datum, as isolated in propositions 4.495 and 4.496 and corollary 4.516. Corollary 4.501 records the corresponding shared terminal-source form: the same Hall-augmented incidence-cover carrier also *supplies the large-complement deleted-only graph branch. Corollary 4.498 narrows the continuation-segment part further to recovery of the internal Step 2 Q-cut between the prefix and suffix strips; the entries of that suffix are already supplied by the strict-left Step 3 columns, and the rest of the incoming suffix boundary is globally empty. Corollary 4.343 ties this boundary target back to the carrier target: once the target-window mechanism supplies the low-low entries on L_{2m}, the internal Q-cut and the continuation segment are also recovered. Thus a direct carrier containing L_{2m} closes the non-column-even branch without first recovering the continuation segment, and corollary 4.502 records that the same carrier also supplies the target-window signature input. Proposition 4.493 is the corresponding partial-carrier version: it is enough for the carrier to contain all low-low cells incident to a determined set of low labels, provided the unexposed complement is below the certified $A(m), B(m)$ frontier. The weaker full-budget candidate-list route closes it after the final-zone branch word selects the true low-low filling. Its graph counterpart, corollary 4.216, shows that this same partial carrier also recovers the large deleted-only graph; the Hall route remains available when the internal Q-cut or direct Hall data are recovered first. Corollary 4.350 shows the limitation of this auxiliary object: Hall data alone do not determine the residual odd-packet matching, so the remaining carrier rule must also use the actual final target-window or Step 3 boundary information. Corollary 4.351 shows that adding the Hall-selected continuation pieces does not remove this limitation; those pieces are an interface for a carrier or bounded-candidate source, not the source itself. lemma 4.125 identifies the semistandard growth path used here with the top/right boundary of the Step 3 dual-RSK' half-square in Krattenthaler's Theorem 4 construction. corollary 4.124 is the matching parser-to-filling bridge for that boundary: once the relevant top/right boundary of an outside dual-RSK' subdiagram is parsed, Krattenthaler's backward local rules recover the subfilling without further choices. lemma 4.517 assigns the ordered frontier-event packets to those same canonical zones, so the remaining local construction can be made zone-by-zone. The packet branch data are still available inside a zone by the deterministic splitting of lemma 4.518. If the zone replacement window exposes the retained halo blocks, then lemma 4.519 recovers the endpoint pairs needed by the packet-local HKKO inverses. lemma 4.520 sharpens this endpoint recovery: each packet endpoint needs only the adjacent retained halo block on that side, or the zone endpoint if the packet touches the zone boundary. lemma 4.521 records the limitation of this saving: if separate packet inverses are called for every deleted component, the adjacent endpoint blocks range over all retained halo indices. Thus a true target-size saving must come from a merged endpoint rule, a packet-compression rule, or a zero-window packet mechanism, not merely from packetwise endpoint lookup. proposition 4.522 then assembles those packet inverses and retained halo blocks into the full zone inverse. lemma 4.523 shows that positive packet outputs cannot be exposed as additional disjoint windows after all retained halo blocks have already been exposed unchanged. lemma 4.525 gives the sharper budget: disjoint positive packet output windows must be paid for by the same number of retained halo windows that are not exposed unchanged, and raw length-$|K|$ packet outputs cannot fit in zones with fewer retained than deleted blocks. corollary 4.526 refines this into an exact compression accounting rule: each unit by which packet outputs are shortened buys one more visible retained halo block, after the local surplus has been spent. corollary 4.527 makes this exact: raw packet outputs can expose at most the local surplus $|Z_\gamma \setminus I| - |Z_\gamma \cap I|$ retained blocks. corollary 4.528 records the tight case: when the raw packet lengths exactly fill the target-size window, no retained halo block is exposed unchanged, so packet endpoints must come from a merged decoder rather than from visible retained blocks. lemma 4.256 shows that the alternating witness sets at $j = 2q$ are exactly such single-zone tight cases. proposition 4.530 isolates the resulting first hard positive target: a full-zone alternating repair is precisely an injection into one target path of length q and one 2q-bit branch word. lemma 4.102 proves the first structural fact for that target: on the odd alternating height side, every step creates a new row, while the even alternating height side is empty. lemma 4.103 then shows that the off-column boxes determine the path and are canonically paired by column in the terminal column-even case. lemma 4.104 compresses those off-column boxes further to their row choices. lemma 4.105 identifies the resulting odd* *alternating height fiber with even-column standard Young tableaux of size $2q$. lemma 4.106 contracts the Schensted matching of such a tableau to the stable graph target whenever no adjacent-pair loop is present. lemma 4.107 repairs those adjacent-pair loops by a canonical loopless graph plus the loop-set bits, and proposition 4.108 closes the odd alternating height full-zone decoder within the same $2q$-bit branch budget. proposition 4.109 and corollary 4.110 then close the height side of the alternating full-zone obstruction; the lemma 4.111 and proposition 4.112 close the width side: the odd width case has a single forced rectangular path, and the even width case is empty. Thus corollary 4.113 closes the alternating full-zone obstruction. The remaining canonical-halo work is the non-alternating zone-local decoder. proposition 4.531 isolates the remaining count-compatible form: the target-size zone window must decode the retained halo blocks and the packet outputs from the same replacement blocks, either by merging them or by using zero-window packet data. proposition 4.535 then plugs the below-wall HKKO splice into this count-compatible form: once the target-size window recovers the retained halo blocks, the packet endpoints determined from those retained blocks and the existing packet branch words canonically construct the image-compatible packet outputs. Thus the parser no longer has to output those packet objects separately; it must supply the retained blocks and the fixed readouts from the resulting marked preimages. lemma 4.551 closes the bad-boundary part of those fixed readouts: width boundary second-copy positions are selected by the orientation half of the local branch word, and height boundary second copies never remain in the bad first column. lemma 4.255 removes the remaining internal zone-endpoint datum: unchanged complement blocks determine all canonical-zone endpoints, using the fixed initial and terminal endpoints of the growth-path model at boundary zones. proposition 4.538 promotes this zone-local inverse to the fixed-witness fiber estimate for any canonical halo parser with those below-wall packet-splice readouts.*

Proof. By lemma 4.118, column-even semistandard tableaux of content (2^j) are equivalently represented by two-step growth paths with one two-step block of total weight 2 for each label. Thus a fixed witness set S identifies q labelled two-step blocks, using the label-to-block reversal in lemma 4.118. If the displayed injection exists for either the width-witness or height-witness family, then the target set has cardinality $2^{2q}s_{j-q}$, because lemma 4.65 and lemma 4.118 count the same column-even tableaux of content (2^{j-q}). This is exactly the corresponding fiber estimate in proposition 4.563. ☐

Definition 4.249 (Raw block deletion datum)**.** Let

$$P = (\rho_0, \rho_1, \dots, \rho_{2j})$$

be a two-step growth path as in lemma 4.118, and let $S \subset \{1, \dots, j\}$. Put

$$I(S) = \{j - s + 1 : s \in S\}.$$

The raw block deletion datum $R(P, S)$ is the ordered list of triples

$$B_i(P) = (\rho_{2i-2}, \rho_{2i-1}, \rho_{2i}), \qquad i \notin I(S),$$

together with deletion-joint markers at the following places: before the first retained index i when $i > 1$, with boundaries $\varnothing$ and ρ_{2i-2}; between two consecutive retained indices $i < i'$ when $i' > i+1$, with boundaries ρ_{2i} and $\rho_{2i'-2}$; and after the last retained index i when $i < j$, with boundaries ρ_{2i} and $\varnothing$.

Lemma 4.250 (Raw deletion localizes the repair)**.** *Let P and S be as in definition 4.249. Every retained triple in $R(P, S)$ is a two-step block with vertical-strip adjacent differences and weight 2:*

$$|\rho_{2i-2}| - 2|\rho_{2i-1}| + |\rho_{2i}| = 2.$$

The only failures of $R(P, S)$ to be a single two-step growth path with $j - |S|$ blocks can occur at deletion joints, and the number of deletion joints is at most $|S|$.

Proof. For $i \notin I(S)$, the retained triple is one of the original two-step blocks of P, so the vertical-strip and weight-2 conditions are inherited from lemma 4.118. If two consecutive retained indices

$i < i'$ satisfy $i' = i+1$, then the right endpoint of $B_i(P)$ is $\rho_{2i} = \rho_{2i'-2}$, so the two blocks concatenate without a new condition. If the first retained index is 1, the raw datum begins at $\rho_0 = \varnothing$; otherwise the initial endpoint condition is exactly the first marked deletion joint. Similarly, if the last retained index is j, the raw datum ends at $\rho_{2j} = \varnothing$; otherwise the terminal endpoint condition is exactly the last marked deletion joint. Finally, if two consecutive retained indices satisfy $i' > i+1$, the blocks were separated by at least one deleted block, and the only missing internal path condition is the compatibility of the two boundary partitions ρ_{2i} and $\rho_{2i'-2}$ at that splice. Thus all repair data are concentrated at the marked deletion joints. These joints are in bijection with maximal consecutive intervals of deleted indices, so there are at most $|I(S)| = |S|$ of them. ☐

Proposition 4.251 (Local inverses give global fixed-witness recovery)**.** *Let*

$$P = (\rho_0, \rho_1, \ldots, \rho_{2j})$$

be a two-step growth path as in lemma 4.118, let $S \subset \{1, \ldots, j\}$ have cardinality q, and put $I(S) = \{j - s + 1 : s \in S\}$. Write the maximal consecutive intervals of $I(S)$ as

$$K_\alpha = \{u_\alpha, u_\alpha + 1, \ldots, v_\alpha\} \qquad (1 \le \alpha \le L)$$

in increasing block order. Suppose a repair assignment sends P to

$$\left(P^*, (\beta_\alpha)_{\alpha=1}^{L}\right)$$

and satisfies the following three properties.

(1) *There is a deterministic parser, depending only on S, which applied to P^* returns the retained windows outside the intervals K_α and local output windows U_α^* for the intervals K_α.*
(2) *The retained windows returned by the parser are exactly the original two-step blocks*

$$(\rho_{2i-2}, \rho_{2i-1}, \rho_{2i}) \qquad (i \notin I(S))$$

in their original order.

(3) *For each α, the parser returns the endpoint pair*

$$\mu_\alpha = \rho_{2u_\alpha - 2}, \qquad \nu_\alpha = \rho_{2v_\alpha},$$

and a fixed local inverse recovers the deleted subpath

$$(\rho_{2u_\alpha-2}, \rho_{2u_\alpha-1}, \ldots, \rho_{2v_\alpha})$$

from U_α^*, β_α, μ_α, ν_α, and K_α.

Then the original path P is uniquely recoverable from S, P^*, and the words $\beta_1, \ldots, \beta_L$. Consequently, after composing with the bijection lemma 4.97, the original semistandard tableau is uniquely recoverable from the same data.

Proof. The set S determines $I(S)$ and hence the ordered list of maximal intervals K_α. Applying the deterministic parser to P^* gives all retained windows and all local output windows U_α^* in this same order. By the second hypothesis, every original two-step block with index outside $I(S)$ is therefore recovered.

For each maximal deleted interval K_α, the third hypothesis recovers the whole deleted subpath from the corresponding local output window, branch word, endpoints, and the known interval K_α. The intervals K_α and the retained block indices are disjoint and together cover $\{1, \ldots, j\}$. Interleaving the recovered deleted subpaths with the recovered retained two-step blocks in block order reconstructs the full sequence

$$\rho_0, \rho_1, \ldots, \rho_{2j}.$$

The endpoint overlaps agree because the recovered deleted subpaths and retained windows are all subpaths of the same original path at their common boundary partitions. Hence P is uniquely recovered.

Finally, lemma 4.97 is a bijection between the semistandard tableaux of content (2^j) and their Pieri two-strip paths, so recovering P recovers the original tableau. □

Proposition 4.252 (Replacement zones give global fixed-witness recovery)**.** *Let*

$$P = (\rho_0, \rho_1, \dots, \rho_{2j})$$

be a two-step growth path as in lemma 4.118, let $S \subset \{1, \dots, j\}$ have cardinality q, and put $I(S) = \{j - s + 1 : s \in S\}$. Let

$$Z_\alpha = \{a_\alpha, a_\alpha + 1, \dots, b_\alpha\} \qquad (1 \le \alpha \le L)$$

be disjoint intervals, in increasing order, such that every index in $I(S)$ belongs to exactly one Z_α. Put

$$R = \{1, \dots, j\} \setminus \bigcup_{\alpha=1}^{L} Z_\alpha.$$

Suppose a repair assignment sends P to

$$\left(P^*, (\beta_\alpha)_{\alpha=1}^{L}\right)$$

and satisfies the following properties.

1. *The intervals Z_α are determined by S.*
2. *P^* is a two-step path with exactly $j - q$ two-step windows, and a deterministic parser depending only on S decomposes those windows, in order, into unchanged retained blocks for indices $i \in R$ and local replacement windows V_α^* for the intervals Z_α.*
3. *For each α, the local replacement window V_α^* has exactly*
$$|Z_\alpha \setminus I(S)|$$
two-step blocks.
4. *The unchanged retained blocks returned by the parser are exactly*
$$(\rho_{2i-2}, \rho_{2i-1}, \rho_{2i}) \qquad (i \in R)$$
in their original order.
5. *For each α, the parser returns the endpoint pair*
$$\mu_\alpha = \rho_{2a_\alpha - 2}, \qquad \nu_\alpha = \rho_{2b_\alpha},$$
and a fixed local inverse recovers the original subpath
$$(\rho_{2a_\alpha-2}, \rho_{2a_\alpha-1}, \dots, \rho_{2b_\alpha})$$
from V_α^*, β_α, μ_α, ν_α, and Z_α.

Then the original path P is uniquely recoverable from S, P^*, and the words $\beta_1, \dots, \beta_L$.

Proof. The set S determines $I(S)$ and, by hypothesis, the intervals Z_α and the complement R. Applying the deterministic parser to P^* gives the unchanged retained windows for R and the replacement windows V_α^* in their original order. By the fourth hypothesis, all original blocks with indices in R are recovered.

For each replacement interval Z_α, the fifth hypothesis recovers the whole original subpath from the corresponding parsed replacement window, branch word, endpoints, and known interval. The intervals Z_α and the singleton indices in R partition $\{1, \dots, j\}$. Interleaving the recovered replacement-zone subpaths with the recovered retained blocks in block order reconstructs the full sequence

$$\rho_0, \rho_1, \dots, \rho_{2j}.$$

The endpoint overlaps agree because every recovered piece is recovered with the endpoint pair it had in the original path, and every unchanged retained block is the original block. Thus P is uniquely recovered. □

Proposition 4.253 (Right-boundary auxiliary replacement-zone recovery)**.** *Work in the setting and notation of proposition 4.252. Suppose that the zones are ordered increasingly and that the last one is a right-boundary zone,*

$$Z_L = \{a_L, \dots, j\}.$$

Assume properties (1)–(4) of proposition 4.252. Assume also that the parser returns the endpoint pair

$$\rho_{2a_\alpha - 2}, \qquad \rho_{2b_\alpha}$$

for every zone. For every $\alpha < L$, suppose a fixed local inverse recovers the original subpath on Z_α from V_α^*, β_α, that endpoint pair, and Z_α.

Let the recovered outside datum be

$$\mathcal{D}_{\text{out}}(P) = \Big((B_i(P))_{i \in R}, \big((\rho_{2a_\alpha - 2}, \dots, \rho_{2b_\alpha}) \big)_{\alpha < L} \Big).$$

Suppose that an auxiliary datum $A(P)$ for the right-boundary zone is a fixed function of S and $\mathcal{D}_{\text{out}}(P)$, and that a fixed inverse recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L - 2}, \rho_{2j}, \quad Z_L, \quad A(P).$$

Then the original path P is uniquely recoverable from S, P^*, and the words $\beta_1, \dots, \beta_L$.

Proof. Applying the deterministic parser to P^* gives the unchanged retained blocks for R, the replacement windows V_α^*, and the endpoint pair of every zone. By property (4) of proposition 4.252, the retained blocks in R are the original blocks.

For each $\alpha < L$, the assumed local inverse recovers the original subpath on Z_α. Therefore the datum $\mathcal{D}_{\text{out}}(P)$ is determined from S, P^*, and the branch words: its first component is supplied by the parsed retained blocks in R, and its second component is supplied by the subpaths just recovered for the zones strictly left of Z_L.

Since $A(P)$ is a fixed function of S and this recovered outside datum, $A(P)$ is also recovered. The right-boundary inverse then recovers the original subpath on Z_L. The singleton indices in R and the intervals $Z_1, \dots, Z_L$ partition $\{1, \dots, j\}$. Interleaving the recovered subpaths and retained blocks in block order reconstructs the full sequence

$$\rho_0, \rho_1, \dots, \rho_{2j}.$$

The endpoint overlaps agree for the same reason as in proposition 4.252: each recovered piece carries the endpoint pair it had in the original path. Hence P is uniquely recoverable. ☐

Lemma 4.254 (Canonical halo replacement zones)**.** *Let $j \geq 1$ and let $S \subset \{1, \dots, j\}$. Put*

$$I(S) = \{j - s + 1 : s \in S\}$$

and define the one-block halo of $I(S)$ by

$$H(S) = \{r \in \{1, \dots, j\} : \ |r - i| \leq 1 \text{ for some } i \in I(S)\}.$$

Write the maximal consecutive intervals of $H(S)$ as

$$Z_\alpha = \{a_\alpha, a_\alpha + 1, \dots, b_\alpha\}.$$

Then the intervals Z_α are determined by S, are pairwise disjoint, and cover every deleted block index in $I(S)$. If

$$R = \{1, \dots, j\} \setminus \bigcup_\alpha Z_\alpha,$$

then

$$|R| + \sum_\alpha |Z_\alpha \setminus I(S)| = j - |S|.$$

Moreover every internal boundary of a zone is separated from every other zone by at least one index not in $H(S)$.

Proof. The set $I(S)$ is determined by S, and $H(S)$ is determined by $I(S)$. The maximal consecutive intervals of a finite ordered set are uniquely defined, so the intervals Z_α are determined by S, pairwise disjoint, and their union is $H(S)$. Since every $i \in I(S)$ satisfies $|i - i| \le 1$, one has $I(S) \subseteq H(S)$, so the zones cover every deleted block index.

The sets R and the intervals Z_α partition $\{1, \dots, j\}$. Hence

$$j = |R| + \sum_\alpha |Z_\alpha|.$$

Because $I(S) \subseteq \bigcup_\alpha Z_\alpha$ and the zones are disjoint,

$$\sum_\alpha |Z_\alpha| = |I(S)| + \sum_\alpha |Z_\alpha \setminus I(S)|.$$

Substituting $|I(S)| = |S|$ gives the displayed count.

Finally, if two maximal intervals of $H(S)$ are distinct and ordered, then the index immediately after the first interval, if it exists before the second, does not belong to $H(S)$; otherwise the two intervals would be part of the same maximal consecutive interval. This is the stated separation. □

Lemma 4.255 (Complement blocks expose canonical halo zone endpoints)**.** *Let*

$$P = (\rho_0, \rho_1, \dots, \rho_{2j})$$

be a two-step growth path as in lemma 4.118, let $S \subset \{1, \dots, j\}$, and let $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$ be the canonical halo replacement zones of lemma 4.254. Let

$$R = \{1, \dots, j\} \setminus \bigcup_\gamma Z_\gamma.$$

Suppose one is given S and all complement blocks

$$B_r(P) = (\rho_{2r-2}, \rho_{2r-1}, \rho_{2r}) \qquad (r \in R),$$

and note that lemma 4.118 gives $\rho_0 = \rho_{2j} = \varnothing$. Then the endpoint pair

$$\rho_{2a_\gamma - 2}, \qquad \rho_{2b_\gamma}$$

is determined for every canonical halo zone.

Proof. The set S determines the canonical halo zones and the complement R by lemma 4.254. Fix one zone $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$.

For the left endpoint, if $a_\gamma = 1$, then $\rho_{2a_\gamma - 2} = \rho_0 = \varnothing$ is fixed. If $a_\gamma > 1$, then $a_\gamma - 1$ is not in the one-block halo; otherwise Z_γ would not be the maximal halo component beginning at a_γ. Hence $a_\gamma - 1 \in R$, and the recovered complement block $B_{a_\gamma - 1}(P)$ has right endpoint

$$\rho_{2(a_\gamma - 1)} = \rho_{2a_\gamma - 2}.$$

For the right endpoint, if $b_\gamma = j$, then $\rho_{2b_\gamma} = \rho_{2j} = \varnothing$ is fixed. If $b_\gamma < j$, then $b_\gamma + 1$ is not in the one-block halo, again by maximality of the halo component. Thus $b_\gamma + 1 \in R$, and the recovered complement block $B_{b_\gamma + 1}(P)$ has left endpoint

$$\rho_{2(b_\gamma + 1) - 2} = \rho_{2b_\gamma}.$$

These formulas determine both endpoints of every zone. □

Lemma 4.256 (Alternating witnesses form one tight halo zone)**.** *Let $q \ge 1$, put $j = 2q$, and let*

$$S_{\text{odd}} = \{1, 3, \dots, 2q - 1\}, \qquad S_{\text{even}} = \{2, 4, \dots, 2q\}.$$

For either choice $S \in \{S_{\text{odd}}, S_{\text{even}}\}$, the canonical halo construction of lemma 4.254 has a single replacement zone

$$Z = \{1, 2, \dots, 2q\}.$$

Moreover

$$|Z \cap I(S)| = |Z \setminus I(S)| = q,$$

and the components of $Z \cap I(S)$ are the q singleton deleted blocks of one parity.

Proof. Since $j = 2q$, the label-to-block map is $s \mapsto 2q - s + 1$. It sends S_{odd} to the even block indices

$$I(S_{\text{odd}}) = \{2, 4, \dots, 2q\},$$

and sends S_{even} to the odd block indices

$$I(S_{\text{even}}) = \{1, 3, \dots, 2q - 1\}.$$

Thus, in either case, $I(S)$ consists of all block indices of one parity and has cardinality q.

The one-block halo of either parity class is all of $\{1, \dots, 2q\}$: every index of the opposite parity is adjacent to an element of $I(S)$, with the endpoints covered by the endpoint parity element itself or by its adjacent neighbour. Hence the maximal consecutive intervals of the halo consist of the single interval $Z = \{1, \dots, 2q\}$.

Inside this zone, $Z \cap I(S) = I(S)$ has cardinality q, and $Z \setminus I(S)$ is the complementary parity class, also of cardinality q. Since no two elements of one parity class are consecutive, the maximal components of $Z \cap I(S)$ are exactly the q singleton deleted blocks. □

Lemma 4.257 (Canonical halo zones have short retained gaps)**.** *Let $j \geq 1$, let $S \subset \{1, \dots, j\}$, put $I = I(S)$, and let $Z_\alpha = \{a_\alpha, \dots, b_\alpha\}$ be one canonical halo zone of lemma 4.254. Write the maximal consecutive components of $Z_\alpha \cap I$ as*

$$K_1 = \{u_1, \dots, v_1\}, \dots, K_m = \{u_m, \dots, v_m\}, \qquad u_1 \leq v_1 < u_2 \leq v_2 < \cdots < u_m \leq v_m.$$

Then

$$a_\alpha = \max(1, u_1 - 1), \qquad b_\alpha = \min(j, v_m + 1),$$

and, for $1 \leq t < m$,

$$u_{t+1} - v_t \in \{2, 3\}.$$

Equivalently, inside a canonical halo zone the retained gaps between successive deleted components have length one or two, and the retained prefix and suffix have length at most one.

Proof. The full halo $H(S)$ is the union of the one-block halos of the maximal consecutive components of I. The zone Z_α is one maximal consecutive component of this union, and the intervals $K_1, \dots, K_m$ are exactly the components of I whose one-block halos lie in this component of $H(S)$.

The one-block halo of $K_t = \{u_t, \dots, v_t\}$ inside $\{1, \dots, j\}$ is

$$\{\max(1, u_t - 1), \dots, \min(j, v_t + 1)\}.$$

Since the halos of $K_1, \dots, K_m$ form one consecutive component, the left endpoint of their union is the left endpoint of the first halo and the right endpoint is the right endpoint of the last halo. This proves the displayed formulas for a_α and b_α.

For consecutive components of I, maximality gives $u_{t+1} \geq v_t + 2$. If instead $u_{t+1} \geq v_t + 4$, then the index $v_t + 2$ lies strictly between the two components and has distance at least two from every element of both K_t and K_{t+1}. It also has distance at least two from every other component of I, because the other components lie farther to the left or right. Hence $v_t + 2 \notin H(S)$, which would split the two halos into different components of $H(S)$, contradicting that both lie in Z_α. Thus $u_{t+1} \leq v_t + 3$, and $u_{t+1} - v_t \in \{2, 3\}$.

The final sentence is just a restatement: the retained gap between K_t and K_{t+1} is $\{v_t + 1, \dots, u_{t+1} - 1\}$, whose size is $u_{t+1} - v_t - 1 \in \{1, 2\}$, and the endpoint formulas show that at most one retained halo index can occur before K_1 or after K_m. □

Lemma 4.258 (Canonical halo zones have local retained budget)**.** *Let $j \ge 1$, let $S \subset \{1, \dots, j\}$, and let Z_α be the canonical halo replacement zones of lemma 4.254. Put $I = I(S)$. Then every nonempty zone satisfies*

$$|Z_\alpha \setminus I| \le 2\,|Z_\alpha \cap I|.$$

Consequently

$$\sum_\alpha |Z_\alpha \setminus I| \le 2|S|.$$

Proof. Fix a zone Z_α. Since Z_α is a maximal component of the one-block halo of I, it contains at least one element of I. Every retained index $r \in Z_\alpha \setminus I$ belongs to the halo, so there is an $i \in Z_\alpha \cap I$ with $|r - i| = 1$. Assign r to one such adjacent deleted index, choosing the smaller one if two are available. A fixed deleted index i can receive only the retained neighbours $i - 1$ and $i + 1$, so it receives at most two retained indices. This gives an injection from $Z_\alpha \setminus I$ into two labelled slots over $Z_\alpha \cap I$, proving the first displayed inequality.

The zones are disjoint and cover I, so summing the first inequality over α gives

$$\sum_\alpha |Z_\alpha \setminus I| \le 2 \sum_\alpha |Z_\alpha \cap I| = 2|I| = 2|S|.$$

$\square$

Lemma 4.259 (Canonical halo zone balance)**.** *Let $j \ge 1$, let $S \subset \{1, \dots, j\}$, put $I = I(S)$, and let $Z_\alpha = \{a_\alpha, \dots, b_\alpha\}$ be one canonical halo zone of lemma 4.254. Write the maximal consecutive components of $Z_\alpha \cap I$ as*

$$K_t = \{u_t, \dots, v_t\} \qquad (1 \le t \le m),$$

ordered increasingly. Put $h_t = |K_t|$, and for $1 \le t < m$ put

$$g_t = u_{t+1} - v_t - 1.$$

Finally set

$$\epsilon_L = \begin{cases} 1, & u_1 > 1, \\ 0, & u_1 = 1, \end{cases} \qquad \epsilon_R = \begin{cases} 1, & v_m < j, \\ 0, & v_m = j. \end{cases}$$

Then

$$|Z_\alpha \cap I| = \sum_{t=1}^{m} h_t, \qquad |Z_\alpha \setminus I| = \epsilon_L + \epsilon_R + \sum_{t=1}^{m-1} g_t.$$

Equivalently, the local retained-minus-deleted balance is

$$|Z_\alpha \setminus I| - |Z_\alpha \cap I| = \epsilon_L + \epsilon_R + \sum_{t=1}^{m-1} g_t - \sum_{t=1}^{m} h_t.$$

Proof. The first identity holds because the intervals K_t are precisely the connected components of $Z_\alpha \cap I$.

By lemma 4.257, the left endpoint of the zone is $a_\alpha = \max(1, u_1 - 1)$. Hence there is one retained prefix index before K_1 exactly when $u_1 > 1$, and none when $u_1 = 1$; this contributes ϵ_L. Similarly, the right endpoint $b_\alpha = \min(j, v_m + 1)$ gives one retained suffix index exactly when $v_m < j$, contributing ϵ_R.

Between K_t and K_{t+1} the retained indices are exactly

$$\{v_t + 1, \dots, u_{t+1} - 1\},$$

whose cardinality is g_t. The retained prefix, the internal retained gaps, and the retained suffix are disjoint and exhaust $Z_\alpha \setminus I$. Summing their sizes gives the second identity, and subtracting the first identity gives the balance formula. $\square$

Lemma 4.260 (Height witness labels are separated)**.** *Let T be a semistandard tableau of content (2^j) with $\ell(\lambda(T)) \geq 2q$, and put*

$$t_r = T(2r-1,1) \qquad (1 \leq r \leq q).$$

Then

$$t_r + 1 < t_{r+1} \qquad (1 \leq r < q).$$

Consequently the height witness set $H_q(T) = \{t_1, \ldots, t_q\}$ contains no two consecutive labels.

Proof. The first column of a semistandard tableau is strictly increasing. For $1 \leq r < q$ the entries in rows $2r-1$, $2r$, and $2r+1$ therefore satisfy

$$t_r = T(2r-1,1) < T(2r,1) < T(2r+1,1) = t_{r+1}.$$

All entries are integers, so $t_r + 1 < t_{r+1}$. This is exactly the assertion that no two labels in $H_q(T)$ are consecutive. □

Lemma 4.261 (Height witness blocks have retained collars)**.** *Let T be a semistandard tableau of content (2^j) with $\ell(\lambda(T)) \geq 2q$, put*

$$t_r = T(2r-1,1) \qquad (1 \leq r \leq q), \qquad S = H_q(T),$$

and fix $s = t_r \in S$. Let

$$u = j - s + 1$$

be the corresponding block index in the growth path of lemma 4.118. Then $u > 1$ and $u - 1 \notin I(S)$. Moreover, if $u < j$, then $u + 1 \notin I(S)$. Equivalently, the successor label $s+1$ is retained, and the predecessor label $s-1$ is retained whenever it exists.

Proof. Since $\ell(\lambda(T)) \geq 2q$, the box $(2r,1)$ belongs to T. Column strictness gives

$$s = T(2r-1,1) < T(2r,1) \leq j,$$

so $s < j$. Also $s+1 \notin S$: for $k \leq r$ one has $t_k \leq s$, while for $k > r$ the separation lemma gives $s + 1 < t_{r+1} \leq t_k$. Thus no t_k equals $s+1$. The inequality $s < j$ gives $u = j - s + 1 > 1$, and $s + 1 \notin S$ is equivalent, under $a \mapsto j - a + 1$, to $u - 1 \notin I(S)$.

It remains to prove the predecessor assertion. Assume $s > 1$. If $s - 1 \in S$, then $s - 1 = t_k$ for some k. Since $t_1 < \cdots < t_q$ and $s - 1 < s = t_r$, one has $k < r$. If $k < r$, then lemma 4.260, applied along the increasing list from k to r, gives in particular

$$t_k + 1 < t_{k+1} \leq t_r = s,$$

so $t_k \leq s-2$, contradicting $t_k = s-1$. Hence $s - 1 \notin S$. The condition $u < j$ is equivalent to $s > 1$, and $s - 1 \notin S$ is equivalent to $u + 1 \notin I(S)$. □

Lemma 4.262 (Height successor pairs have forced frontier form)**.** *Let T be a semistandard tableau of content (2^j) with $\ell(\lambda(T)) \geq 2q$, and let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)} = \lambda(T)$$

be its Pieri two-strip path. Put

$$t_r = T(2r-1,1) \qquad (1 \leq r \leq q), \qquad S = H_q(T).$$

For a fixed r, set $s = t_r$. Then $s < j$, $s + 1 \notin S$, and

$$\ell(\lambda^{(s-1)}) = 2r - 2, \qquad \ell(\lambda^{(s)}) = 2r - 1.$$

Moreover

$$\lambda^{(s)} \setminus \lambda^{(s-1)} = \{(2r-1,1), x_s\}$$

for a box x_s not in the first column, while

$$\ell(\lambda^{(s+1)}) \in \{2r-1, 2r\}.$$

If $\ell(\lambda^{(s+1)}) = 2r$, then the retained successor strip $\lambda^{(s+1)} \setminus \lambda^{(s)}$ contains the new first-column box $(2r, 1)$; otherwise it contains no first-column box below row $2r - 1$.

Proof. Since $\ell(\lambda(T)) \geq 2q$, the first-column box $(2r, 1)$ belongs to T. Column strictness gives

$$s = T(2r - 1, 1) < T(2r, 1) \leq j,$$

so $s < j$. The labels $t_1 < \cdots < t_q$ are strictly increasing, and lemma 4.260 gives $t_r + 1 < t_{r+1}$ for $r < q$. Hence $s + 1$ is not one of the labels $t_1, \ldots, t_q$, proving $s + 1 \notin S$.

The box $(2r - 1, 1)$ carries label s, so it is absent from $\lambda^{(s-1)}$ and present in $\lambda^{(s)}$. Thus

$$\ell(\lambda^{(s-1)}) \leq 2r - 2, \qquad \ell(\lambda^{(s)}) \geq 2r - 1.$$

For $r = 1$ the first inequality forces $\ell(\lambda^{(s-1)}) = 0$. For $r > 1$, the box $(2r - 2, 1)$ lies above $(2r-1, 1)$ in the first column, so its label is smaller than s and it is already present in $\lambda^{(s-1)}$. Hence $\ell(\lambda^{(s-1)}) = 2r - 2$ in every case.

The strip $\lambda^{(s)} \setminus \lambda^{(s-1)}$ is horizontal of size two by lemma 4.97. It contains $(2r-1, 1)$ and therefore cannot also contain $(2r, 1)$, because those two boxes are in the same column. Thus $\ell(\lambda^{(s)}) = 2r - 1$. The second box x_s carrying label s is therefore not in the first column.

The next strip $\lambda^{(s+1)} \setminus \lambda^{(s)}$ is again horizontal of size two, so it can create at most one new row. Since $\ell(\lambda^{(s)}) = 2r - 1$, this gives $\ell(\lambda^{(s+1)}) \in \{2r - 1, 2r\}$. If the length becomes $2r$, the new row is created precisely by the first-column box $(2r, 1)$, which must belong to the successor strip. If the length stays $2r - 1$, no first-column box below row $2r - 1$ is added at that step. $\square$

Corollary 4.263 (Height anchor pairs reduce to off-frontier data)**.** *Work in the height setting of corollary 4.322. Let $u \in Z_\gamma \cap I(S)$ be a deleted singleton block, and put*

$$s = j - u + 1.$$

Let r be determined by $s = t_r$ in the ordered height witness set $S = H_q(T)$. Then the retained left-neighbour block $u - 1$ corresponds to the successor label $s + 1$, and the two-label Pieri segment

$$\lambda^{(s-1)} \subset \lambda^{(s)} \subset \lambda^{(s+1)}$$

has the forced frontier form of lemma 4.262. Consequently, a target-window readout which determines $\lambda^{(s-1)}$, the off-first-column box x_s, and the retained successor strip

$$\lambda^{(s+1)} \setminus \lambda^{(s)}$$

determines this two-label Pieri segment. Equivalently, the height anchor-pair inverse need not encode the odd first-column witness box; the remaining Pieri-side input is off-frontier data together with the retained successor strip.

Proof. The label-to-block convention in lemma 4.118 sends a label a to the block $j - a + 1$. Thus the deleted singleton block u corresponds to the deleted height witness label $s = j - u + 1$, while the block $u - 1$ corresponds to $s + 1$. Since $u - 1 \in Z_\gamma \setminus I(S)$ by corollary 4.323, this is the retained left-neighbour block.

Applying lemma 4.262 to the witness label $s = t_r$ gives

$$\lambda^{(s)} = \lambda^{(s-1)} \cup \{(2r - 1, 1), x_s\}$$

with x_s off the first column, and gives the stated restriction on the successor strip. Therefore the data $\lambda^{(s-1)}$, x_s, and $\lambda^{(s+1)} \setminus \lambda^{(s)}$ reconstruct first $\lambda^{(s)}$ and then $\lambda^{(s+1)}$. The final sentence is just this reconstruction read as a restriction on what the merged height anchor-pair inverse has to supply. $\square$

Corollary 4.264 (Height anchor pairs have two-sided retained collar data)**.** *Work in the height setting of corollary 4.322. Let $u_t \in Z_\gamma \cap I(S)$ be a deleted singleton block, and put $s_t = j - u_t + 1$. Then the left-neighbour block $u_t - 1$ exists and is retained. If $u_t < j$, then the right-neighbour block*

$u_t + 1$ *also exists and is retained. Moreover, whenever these neighbours exist, they lie in the same canonical halo zone* Z_γ.

Consequently, a positive-height merged-window construction for the pair $(u_t - 1, u_t)$ *has deterministic access, from the canonical halo parser, to the retained left collar and, except when* $u_t = j$, *to the retained right collar. The exceptional case* $u_t = j$ *is precisely the terminal right boundary side, where the zone endpoint* ρ_{2j} *is part of the endpoint data.*

Proof. Apply lemma 4.261 to the height witness label s_t. It gives $u_t > 1$ and $u_t - 1 \notin I(S)$, so the left-neighbour block exists and is retained. If $u_t < j$, the same lemma gives $u_t + 1 \notin I(S)$, so the right-neighbour block exists and is retained.

The canonical halo zone Z_γ is a connected component of the one-block halo of $I(S)$. Hence every existing index at distance one from $u_t \in Z_\gamma \cap I(S)$ lies in that same component. Thus the retained neighbours just identified lie in Z_γ. If $u_t = j$, the right neighbour does not exist and the right side of the singleton block is the global endpoint ρ_{2j}, which is included as the right endpoint of any terminal zone containing u_t. □

Corollary 4.265 (Step–3 pair boundary is the anchor-pair segment)**.** *Work in the setting of corollary 4.263. Let* $P = (\rho_0, \rho_1, \dots, \rho_{2j})$ *be the Step–3 boundary growth path of lemma 4.118. Then the retained-successor/deleted-witness pair of labels* $(s+1, s)$ *is represented on the Step–3 top/right boundary by the consecutive two-block segment*

$$\rho_{2u-4}, \rho_{2u-3}, \rho_{2u-2}, \rho_{2u-1}, \rho_{2u}.$$

Consequently, any target-window readout which determines this Step–3 boundary segment determines the pair segment required by the local inverse hypothesis of proposition 4.532.

Proof. By lemma 4.118, tableau label ℓ corresponds to block $j-\ell+1$ of the Step–3 boundary growth path. The deleted witness label is $s = j - u + 1$, hence it corresponds to block u. The retained successor label $s + 1$ corresponds to block

$$j - (s + 1) + 1 = u - 1.$$

Thus the adjacent label pair $(s + 1, s)$ is exactly the adjacent block pair $(u - 1, u)$. The Step–3 boundary segment covered by these two blocks is

$$\rho_{2(u-1)-2}, \rho_{2(u-1)-1}, \rho_{2(u-1)}, \rho_{2u-1}, \rho_{2u},$$

which is the displayed segment. This is precisely the segment appearing in the pair-inverse hypothesis of proposition 4.532. □

Proposition 4.266 (Step–3 pair-boundary carriers supply anchor-pair inverses)**.** *Work in the height setting of proposition 4.532. Suppose that the parser there decomposes* V^*_γ *into the unchanged anchor blocks and one merged window* M_t *for each deleted singleton* u_t, *and suppose that for each* t *the data*

$$M_t, \quad \beta_t, \quad S, \quad Z_\gamma, \quad t, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine the Step–3 boundary segment

$$\rho_{2u_t-4}, \rho_{2u_t-3}, \rho_{2u_t-2}, \rho_{2u_t-1}, \rho_{2u_t}.$$

Then the pair-inverse hypothesis of proposition 4.532 holds for every t, *and hence the original height halo-zone subpath is recovered from* V^*_γ, β_γ, *the zone endpoint pair, and* Z_γ.

Proof. For each deleted singleton u_t, put $s_t = j - u_t + 1$. The pair $(u_t - 1, u_t)$ is the retained-successor/deleted-witness pair for labels $(s_t + 1, s_t)$ by corollary 4.263. Therefore corollary 4.265 identifies the displayed boundary segment with the consecutive original two-block segment required in the local pair-inverse hypothesis. The assumed deterministic readout from M_t, β_t, the zone data, and the endpoint pair is therefore a valid fixed pair inverse for this t. Applying this to every t supplies all pair inverses required by proposition 4.532, and that proposition recovers the full original zone subpath. □

Corollary 4.267 (Two-label Step–3 carrier data supply anchor-pair inverses)**.** *Work in the setting of proposition 4.266. For the deleted singleton u_t, put $s_t = j - u_t + 1$, and let E_{s_t,s_t+1} be the two-label Step–3 subarrangement of lemma 4.186. Suppose that, for each t, the data*

$$M_t, \quad \beta_t, \quad S, \quad Z_\gamma, \quad t, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine the cell entries of E_{s_t,s_t+1} and the partition labels on its left/bottom boundary. Then the hypotheses of proposition 4.266 hold, and hence the original height halo-zone subpath is recovered from V^*_γ, β_γ, the zone endpoint pair, and Z_γ.

Proof. Apply corollary 4.187 with $a = s_t$ and $b = s_t + 1$. Since $u_t = j - s_t + 1$, the boundary segment recovered there is

$$\rho_{2(j-(s_t+1))}, \dots, \rho_{2(j-s_t+1)} = \rho_{2u_t-4}, \dots, \rho_{2u_t}.$$

Thus each merged-window datum determines the Step–3 pair-boundary segment required in proposition 4.266. That proposition then gives the anchor-pair inverses and recovers the zone subpath. □

Lemma 4.268 (Two-label Step–3 carrier entries are three-valued)**.** *Use the Step–3 dual-RSK' half-square in the content (2^j) specialization of lemma 4.118. For $1 \le s < j$, the two-label subarrangement $E_{s,s+1}$ has a single cell, and the entry of that cell belongs to $\{0, 1, 2\}$.*

Proof. The Step–3 half-square has one off-diagonal cell for each unordered pair of distinct labels in the chosen half of the symmetric Step–2 matrix. Therefore the subarrangement on the two labels $s + 1, s$ has exactly one cell.

Let $N = (N_{ab})$ be the Step–2 matrix of lemma 4.128. By lemma 4.127, the unique Step–3 cell entry of $E_{s,s+1}$ is either $N_{s,s+1}$ or $N_{s+1,s}$, which are equal by symmetry. The entries of N are nonnegative integers and every row sum is 2 by lemma 4.128; hence $0 \le N_{s,s+1} \le 2$. Thus the unique cell entry lies in $\{0, 1, 2\}$. □

Lemma 4.269 (Two-label Step–3 carriers have diagonal jump two)**.** *Use the Step–3 dual-RSK' half-square in the content (2^j) specialization of lemma 4.118. For $1 \le s < j$, let*

$$\begin{matrix} \nu & \lambda \\ \rho & \mu \end{matrix}$$

be the four corner labels of the one-cell carrier $E_{s,s+1}$, with ρ the southwest corner and λ the northeast corner in lemma 4.121. Then

$$|\lambda| - |\rho| = 2.$$

Proof. In Krattenthaler's Step–3 half-square the labels are ordered $j, j-1, \dots, 1$, and the cell for the unordered adjacent pair $\{s, s+1\}$ lies next to the deleted diagonal. Consider the southwest hook of this cell in the retained strict half-square: the cell itself, the cells in the same retained row between it and the left boundary, and the cells in the same retained column between it and the bottom boundary. Because s and $s+1$ are adjacent in the label order, this hook is exactly the set of off-diagonal representatives incident to one of the two labels $s, s+1$; which one occurs depends only on which of the two symmetric halves was retained in Step 3.

Sum the one-cell conservation identity of lemma 4.122 over this hook. All internal edge labels cancel. The two exposed hook ends lie on the fixed empty left and bottom boundaries of the Step–3 construction, while the remaining outer diagonal corners are precisely ρ and λ. Hence

$$|\lambda| - |\rho|$$

is the sum of the Step–3 entries in the hook.

By lemma 4.127, those entries are exactly one off-diagonal half of the corresponding Step–2 symmetric matrix entries incident to the selected label. The diagonal Step–2 entry is zero, and the row and column sums of the Step–2 matrix are both 2 by lemma 4.128. Therefore the hook sum is 2, giving the displayed diagonal jump. □

Corollary 4.270 (Raw one-cell top/right corners are not target blocks). *Use the Step–3 dual-RSK' half-square in the content* (2^j) *specialization of lemma 4.118. Let*

$$\begin{matrix} \nu & \lambda \\ \rho & \mu \end{matrix}$$

be the four corner labels of a two-label one-cell carrier, and let e *be its cell entry. If the three non-southwest corners are read as a two-step growth block with middle label* λ*, then*

$$|\nu| - 2|\lambda| + |\mu| = -2 - e.$$

Consequently this triple is not a content-2 target block. Thus the one-cell corner readout in proposition 4.283 cannot be supplied by placing the raw top/right carrier corners as a single target block; it needs a content-normalized or genuinely collective grouped encoding.

Proof. By lemma 4.269,

$$|\lambda| - |\rho| = 2.$$

The one-cell conservation identity of lemma 4.122 gives

$$e = |\lambda| - |\mu| - |\nu| + |\rho|,$$

so

$$|\nu| + |\mu| = |\lambda| + |\rho| - e = 2|\lambda| - 2 - e.$$

Subtracting $2|\lambda|$ gives the displayed identity. Finally lemma 4.268 gives $e \in \{0, 1, 2\}$, so the displayed value is negative and is not 2. □

Proposition 4.271 (Two-bit cell-entry carriers reduce anchor pairs to boundary recovery). *Work in the setting of proposition 4.266. Fix once and for all an injection*

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

For the deleted singleton u_t*, put* $s_t = j - u_t + 1$ *and let* e_t *be the unique Step–3 cell entry of* E_{s_t,s_t+1}*. Suppose that, for each* t*, the data*

$$M_t, \quad S, \quad Z_\gamma, \quad t, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine the partition labels on the left/bottom boundary of E_{s_t,s_t+1}*, and suppose the two-bit branch word satisfies* $\beta_t = \kappa(e_t)$*. Then the hypotheses of corollary 4.267 hold. Consequently the original height halo-zone subpath is recovered from* V_γ^**,* β_γ*, the zone endpoint pair, and* Z_γ*.*

Proof. By lemma 4.268, $e_t \in \{0, 1, 2\}$, so the fixed injection κ lets the inverse read e_t from the two-bit word β_t. Since E_{s_t,s_t+1} has a single cell, this determines all cell entries of that two-label subarrangement. The left/bottom boundary labels are determined by hypothesis from M_t, the zone data, the endpoint pair, and t. Thus the data

$$M_t, \quad \beta_t, \quad S, \quad Z_\gamma, \quad t, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine both the cell entries and the left/bottom boundary labels of E_{s_t,s_t+1} for every t. Corollary 4.267 applies and gives the stated recovery of the original height halo-zone subpath. □

Proposition 4.272 (Two-label left/bottom carriers recover positive height zones). *Work in the height setting of proposition 4.532. Suppose that the parser there decomposes* V_γ^* *into the unchanged anchor blocks and one merged window* M_t *for each deleted singleton* u_t*, listed in increasing* t*. Fix an injection*

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

For each t*, put* $s_t = j - u_t + 1$*, let* E_{s_t,s_t+1} *be the corresponding two-label Step–3 carrier, and let* e_t *be its unique cell entry. Assume that the data*

$$M_t, \quad S, \quad Z_\gamma, \quad t, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine the ordered partition labels on the left/bottom boundary of E_{s_t,s_t+1}, and that the deterministic two-bit split of β_γ satisfies $\beta_t = \kappa(e_t)$ for every t. Then the local inverse hypothesis of proposition 4.532 holds for every deleted singleton in Z_γ, and the original height halo-zone subpath is recovered from V_γ^*, β_γ, the zone endpoint pair, and Z_γ.

Proof. The hypotheses for a fixed t are exactly the hypotheses of proposition 4.271: the merged window and endpoint/parser data determine the left/bottom boundary of E_{s_t,s_t+1}, while the two-bit word records the unique cell entry through the fixed injection κ. Therefore proposition 4.271 supplies the anchor-pair inverse for this t. Applying this argument to every deleted singleton supplies all local pair inverses required in proposition 4.532. That proposition then recovers the full original height halo-zone subpath. □

Corollary 4.273 (Raw carrier boundaries are not automatically target windows)**.** *The construction required in proposition 4.272 cannot be supplied, in the ambient one-cell dual-RSK' category, by simply declaring the left/bottom boundary of the two-label carrier $E_{s,s+1}$ to be a content-2 target two-step window. A target merged window must use the doubled-content restrictions, neighbouring parser data, or an explicit content-normalized encoding of that carrier boundary.*

Proof. Consider a single cell in the fourth arbitrary-filling dual-RSK' local rule of lemma 4.121. Put

$$m = 0, \qquad \rho = \varnothing, \qquad \mu = (1), \qquad \nu = \varnothing.$$

The hypotheses of the forward local rule hold: $\rho \subseteq \mu$ and $\rho \subseteq \nu$, and the two differences are vertical strips. Applying the forward rule gives $\lambda = (1)$, so this is a valid one-cell dual-RSK' local datum.

The natural left/bottom valley of this cell is

$$\nu \supseteq \rho \subseteq \mu \qquad \text{or, equivalently,} \qquad \mu \supseteq \rho \subseteq \nu.$$

In either order its two-step content weight is

$$|\nu| - 2|\rho| + |\mu| = 1.$$

But every target two-step block in the content (2^j) growth-path model has weight 2 by lemma 4.118. Thus the raw left/bottom boundary of a dual-RSK' carrier cell is not, by the local rules alone, a valid target two-step window. Any valid merged-window construction must therefore use additional structure beyond this raw boundary identification. □

Definition 4.274 (Bottom padding)**.** For a partition $\alpha = (\alpha_1, \dots, \alpha_\ell)$ and $e \in \{0, 1, 2\}$, define

$$\operatorname{pad}_e(\alpha) = (\alpha_1, \dots, \alpha_\ell, \underbrace{1, \dots, 1}_{e \text{ times}}),$$

with the convention that $\operatorname{pad}_0(\alpha) = \alpha$. For a valley of partitions $\nu \supseteq \rho \subseteq \mu$, put

$$\operatorname{Pad}_e(\nu, \rho, \mu) = \big(\operatorname{pad}_e(\nu), \rho, \mu\big).$$

Lemma 4.275 (Bottom padding normalizes a deficient carrier valley)**.** *Let*

$$\nu \supseteq \rho \subseteq \mu$$

be partitions such that ν/ρ and μ/ρ are vertical strips, and let $e \in \{0, 1, 2\}$. If

$$|\nu| - 2|\rho| + |\mu| = 2 - e,$$

then $\operatorname{Pad}_e(\nu, \rho, \mu)$ is a valid content-2 target two-step block. Moreover $\operatorname{Pad}_e(\nu, \rho, \mu)$ together with e determines (ν, ρ, μ).

Proof. Because ν/ρ is a vertical strip, for every row index i the difference $\nu_i - \rho_i$ is either 0 or 1. The partition $\operatorname{pad}_e(\nu)$ is obtained by appending e new bottom rows of length 1. Against the omitted zero rows of ρ, each appended row therefore contributes one new box, and the old rows still contribute the vertical strip ν/ρ. Hence $\operatorname{pad}_e(\nu)/\rho$ is a vertical strip. The strip μ/ρ is vertical by hypothesis.

The content weight of the padded block is

$$|\operatorname{pad}_e(\nu)| - 2|\rho| + |\mu| = |\nu| + e - 2|\rho| + |\mu| = 2.$$

Thus the displayed triple is a content-2 target two-step block in the sense of lemma 4.118. Finally, since e is known and the padding consists of the last e rows of $\operatorname{pad}_e(\nu)$, deleting those rows recovers ν; the middle and right entries of the triple are already ρ and μ. ☐

Proposition 4.276 (Padded carrier valleys recover positive height zones)**.** *Work in the height setting of proposition 4.532. Suppose that the parser there decomposes V_γ^* into the unchanged anchor blocks and one merged window M_t for each deleted singleton u_t, listed in increasing t. Fix an injection*

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2.$$

For each t, put $s_t = j - u_t + 1$. Let

$$\begin{matrix} \nu_t & \lambda_t \\ \rho_t & \mu_t \end{matrix}$$

be the corner labels of the one-cell Step–3 carrier E_{s_t,s_t+1}, and let e_t be that cell entry. Assume that the merged window is the padded carrier valley

$$M_t = \operatorname{Pad}_{e_t}(\nu_t, \rho_t, \mu_t),$$

and that the deterministic two-bit split of β_γ satisfies $\beta_t = \kappa(e_t)$ for every t. Then the original height halo-zone subpath is recovered from V_γ^*, β_γ, the zone endpoint pair, and Z_γ.

Proof. For each t, the word β_t determines e_t through the fixed injection κ. By lemma 4.269, the carrier satisfies $|\lambda_t| - |\rho_t| = 2$. Therefore corollary 4.123 gives

$$|\nu_t| - 2|\rho_t| + |\mu_t| = 2 - e_t.$$

Therefore lemma 4.275 shows that the target block M_t and this recovered value of e_t determine the left/bottom valley (ν_t, ρ_t, μ_t) of E_{s_t,s_t+1}. Since the carrier has one cell by lemma 4.268, this is exactly the left/bottom boundary and all cell entries required in corollary 4.267. Applying that corollary for every deleted singleton supplies the local pair inverses in proposition 4.532, and that proposition recovers the whole original zone subpath. ☐

Lemma 4.277 (Endpoint-compatible blocks form a replacement window)**.** *Let $S \subset \{1, \dots, j\}$, put $I = I(S)$, let $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$ be a replacement zone, and let*

$$h_1 < h_2 < \cdots < h_N$$

be the retained slot indices in $Z_\gamma \setminus I$. For each r let

$$C_r = (L_r, X_r, R_r)$$

be a two-step block with vertical-strip adjacent differences and content weight 2. Suppose that

$$L_{h_1} = \rho_{2a_\gamma - 2}, \qquad R_{h_N} = \rho_{2b_\gamma},$$

and

$$R_{h_i} = L_{h_{i+1}} \qquad (1 \le i < N).$$

Then the concatenation

$$L_{h_1}, X_{h_1}, R_{h_1} = L_{h_2}, X_{h_2}, \dots, R_{h_{N-1}} = L_{h_N}, X_{h_N}, R_{h_N}$$

is a two-step replacement window from $\rho_{2a_\gamma-2}$ to ρ_{2b_γ} with exactly $|Z_\gamma \setminus I|$ two-step blocks. The deterministic parser which cuts this concatenation after each block recovers the ordered blocks $C_{h_1}, \dots, C_{h_N}$.

Proof. The displayed endpoint equalities are exactly the compatibility conditions needed to concatenate the triples C_{h_i} into a single partition sequence. Each triple contributes one two-step block, and its vertical-strip and weight-2 conditions are part of the hypotheses. Since $N = |Z_\gamma \setminus I|$, the concatenated sequence has the required target size. Its first and last endpoints are the prescribed zone endpoints. The parser is fixed by the known ordered slot list $h_1 < \cdots < h_N$, so cutting the sequence into consecutive triples recovers each displayed block. □

Proposition 4.278 (Endpoint-compatible padded carriers recover positive height zones)**.** *Work in the height setting of proposition 4.532. Let*

$$I = I(S),$$

fix an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2,$$

and let

$$Z_\gamma \cap I = \{u_1 < \cdots < u_m\},$$

and let A_γ be the canonical right-anchor set. For each retained slot $r \in Z_\gamma \setminus I$, define a candidate target block C_r as follows. If $r \in A_\gamma$, let

$$C_r = B_r(P).$$

If $r = u_t - 1$ for some t, put $s_t = j - u_t + 1$, let

$$\begin{matrix} \nu_t & \lambda_t \\ \rho_t & \mu_t \end{matrix}$$

be the corner labels of E_{s_t,s_t+1}, let e_t be its cell entry, and set

$$C_r = \mathrm{Pad}_{e_t}(\nu_t, \rho_t, \mu_t).$$

Then each C_r is a valid content-2 two-step block. Assume also that these ordered blocks satisfy the endpoint equalities in lemma 4.277. If the two-bit split of the branch word satisfies $\beta_t = \kappa(e_t)$ for every t, then the original height halo-zone subpath is recovered from the resulting replacement window, β_γ, the zone endpoint pair, and Z_γ.

Proof. For $r \in A_\gamma$, the block $C_r = B_r(P)$ is an original block of a path in lemma 4.118, hence has vertical-strip adjacent differences and content weight 2. For $r = u_t - 1$, the diagonal-jump lemma lemma 4.269 and corollary 4.123 give

$$|\nu_t| - 2|\rho_t| + |\mu_t| = 2 - e_t.$$

Thus lemma 4.275 makes $C_r = \mathrm{Pad}_{e_t}(\nu_t, \rho_t, \mu_t)$ a valid content-2 two-step block.

By lemma 4.277, the ordered list of candidate blocks concatenates to a target-size replacement window V^*_γ, and the deterministic parser recovers the unchanged anchor blocks $C_r = B_r(P)$ and the merged padded-carrier windows C_{u_t-1} in increasing order. Therefore the hypotheses of proposition 4.276 are satisfied, and that proposition recovers the original height halo-zone subpath. □

Proposition 4.279 (Tagged collective padded splices recover positive height zones)**.** *Work in the height setting of proposition 4.532. Let*

$$Z_\gamma \cap I = \{u_1 < \cdots < u_m\},$$

and let A_γ be the canonical right-anchor set. Fix an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

For each deleted singleton u_t, put $s_t = j - u_t + 1$, let

$$\begin{matrix} \nu_t & \lambda_t \\ \rho_t & \mu_t \end{matrix}$$

be the corner labels of E_{s_t,s_t+1}, and let e_t be its cell entry.

Suppose that the replacement window is parsed, by a deterministic parser depending only on S, Z_γ, and the displayed replacement window, as an ordered tagged list

$$(\tau_1, C_1), \dots, (\tau_N, C_N), \qquad C_i = (L_i, X_i, R_i),$$

whose tag set is exactly the disjoint union

$$\{\mathrm{anc}(r) : r \in A_\gamma\} \sqcup \{\mathrm{car}(t) : 1 \le t \le m\}.$$

Assume that the blocks are defined by their tags as follows:

$$C_i = B_r(P) \quad \textit{if } \tau_i = \mathrm{anc}(r),$$

and

$$C_i = \mathrm{Pad}_{e_t}(\nu_t, \rho_t, \mu_t) \quad \textit{if } \tau_i = \mathrm{car}(t).$$

Assume also that the ordered blocks are endpoint-compatible:

$$L_1 = \rho_{2a_\gamma - 2}, \qquad R_N = \rho_{2b_\gamma}, \qquad R_i = L_{i+1} \quad (1 \le i < N),$$

and that the two-bit split of the branch word satisfies

$$\beta_t = \kappa(e_t) \qquad (1 \le t \le m).$$

Then the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath. No additional tag word is used.

Proof. First each tagged block is a valid content-2 block. Anchor blocks are original two-step blocks of the path in lemma 4.118. For carrier tags, lemma 4.269 and corollary 4.123 give

$$|\nu_t| - 2|\rho_t| + |\mu_t| = 2 - e_t,$$

so lemma 4.275 makes

$$\mathrm{Pad}_{e_t}(\nu_t, \rho_t, \mu_t)$$

a valid content-2 block.

The number of tags is

$$|A_\gamma| + m = |Z_\gamma \setminus I|$$

by corollary 4.322. The displayed endpoint equalities therefore concatenate the parsed blocks to a target-size replacement window from $\rho_{2a_\gamma - 2}$ to ρ_{2b_γ}.

The deterministic parser recovers the tagged list from the replacement window and the known zone data. For an anchor tag $\mathrm{anc}(r)$, this directly recovers the original block $B_r(P)$. For a carrier tag $\mathrm{car}(t)$, the branch word recovers e_t through κ, and then lemma 4.275 recovers (ν_t, ρ_t, μ_t) from the padded block and e_t. Since E_{s_t, s_t+1} has one cell with entry e_t, this gives its cell entry and left/bottom boundary. Applying corollary 4.187 with $a = s_t$ and $b = s_t + 1$ recovers the boundary segment

$$\rho_{2u_t - 4}, \rho_{2u_t - 3}, \rho_{2u_t - 2}, \rho_{2u_t - 1}, \rho_{2u_t}.$$

By corollary 4.265, this is the original two-block segment on $\{u_t - 1, u_t\}$.

Finally, corollary 4.323 says that the blocks of Z_γ are partitioned into the anchor singleton blocks $r \in A_\gamma$ and the paired intervals $\{u_t - 1, u_t\}$ for $1 \le t \le m$. Interleaving the recovered anchor blocks and recovered two-block segments in original block order reconstructs

$$\rho_{2a_\gamma - 2}, \rho_{2a_\gamma - 1}, \dots, \rho_{2b_\gamma}.$$

□

Corollary 4.280 (Tagged splices require parser-visible tags)**.** *In the use of proposition 4.279, the tag order and tag labels are part of the deterministic target-window parser. They cannot be supplied by an additional word, and they cannot be chosen after seeing the source path unless the chosen order is itself recoverable from the target window and the known zone data.*

Proof. The only branch data used in proposition 4.279 are the two-bit words $\beta_t = \kappa(e_t)$ for the deleted singletons. The proposition assumes that the replacement window is parsed as the ordered tagged list by a deterministic parser depending only on S, Z_γ, and the replacement window itself. If the tag order were supplied by an additional word, or chosen from the source path without being recovered by that parser, then the map would use extra information beyond the allowed target-size replacement window and the fixed two-bit singleton words. Such a map is not the hypothesis of the proposition and does not feed the fixed-fiber count. ☐

Proposition 4.281 (Grouped collective carrier splices recover positive height zones)**.** *Work in the height setting of proposition 4.532. Let*

$$Z_\gamma \cap I = \{u_1 < \cdots < u_m\},$$

and let A_γ be the canonical right-anchor set. Fix an injection

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2.$$

For each deleted singleton u_t, put $s_t = j - u_t + 1$, let

$$\begin{matrix} \nu_t & \lambda_t \\ \rho_t & \mu_t \end{matrix}$$

be the corner labels of E_{s_t,s_t+1}, and let e_t be its cell entry.

Suppose that the replacement window is parsed, by a deterministic parser depending only on S, Z_γ, and that replacement window, into ordered target subwindows

$$W_1, \dots, W_L$$

together with parser-visible tag sets

$$\Theta_\ell \subseteq \{\mathrm{anc}(r) : r \in A_\gamma\} \sqcup \{\mathrm{car}(t) : 1 \le t \le m\}.$$

Assume the tag sets are pairwise disjoint and their union is the displayed full tag set. Assume also that W_ℓ has exactly $|\Theta_\ell|$ two-step target blocks and that the ordered subwindows concatenate from $\rho_{2a_\gamma-2}$ to ρ_{2b_γ}.

Finally, assume that for each ℓ there is a fixed decoder which, from

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma},$$

and from the words β_t for tags $\mathrm{car}(t) \in \Theta_\ell$, recovers:

1. *the original block $B_r(P)$ for every $\mathrm{anc}(r) \in \Theta_\ell$;*
2. *the value e_t and the carrier valley (ν_t, ρ_t, μ_t) for every $\mathrm{car}(t) \in \Theta_\ell$.*

The carrier words are required to satisfy $\beta_t = \kappa(e_t)$. Then the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath. No additional tag or group word is used.

Proof. The parser-visible subwindows concatenate to a replacement window of total length

$$\sum_{\ell=1}^{L} |\Theta_\ell| = |A_\gamma| + m = |Z_\gamma \setminus I|,$$

where the last equality is corollary 4.322. Thus the parsed subwindows have the target size required by the canonical halo repair criterion.

Apply the fixed decoder to each group. Anchor tags directly recover the corresponding original blocks. For a carrier tag $\mathrm{car}(t)$, the recovered value e_t and valley (ν_t, ρ_t, μ_t) give the unique cell entry and left/bottom boundary of E_{s_t,s_t+1}. Hence corollary 4.187, applied with $a = s_t$ and $b = s_t + 1$, recovers

$$\rho_{2u_t-4}, \rho_{2u_t-3}, \rho_{2u_t-2}, \rho_{2u_t-1}, \rho_{2u_t}.$$

By corollary 4.265, this is the original two-block segment on $\{u_t - 1, u_t\}$.

The full tag set is recovered by the parser and is partitioned among the groups, so all anchor blocks and all paired two-block segments are recovered. Corollary 4.323 says that these pieces partition the original blocks of Z_γ. Interleaving them in original block order reconstructs

$$\rho_{2a_\gamma-2}, \rho_{2a_\gamma-1}, \ldots, \rho_{2b_\gamma}.$$

□

Proposition 4.282 (Step–3 boundary readouts supply grouped carrier decoders)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \ldots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Fix the injection

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block $B_r(P)$ for every* $\operatorname{anc}(r) \in \Theta_\ell$*;*
(2) *a Ferrers subarrangement E_ℓ of the Step–3 dual-RSK' half-square of lemma 4.125, together with the partition labels on the top/right boundary of E_ℓ;*
(3) *for every* $\operatorname{car}(t) \in \Theta_\ell$*, the inclusion of the one-cell carrier E_{s_t,s_t+1} in E_ℓ, where $s_t = j - u_t + 1$.*

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\operatorname{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L),$$

where e_t is the Step–3 cell entry of E_{s_t,s_t+1}. Then the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath. No additional Step–3 boundary word is used.

Proof. Fix a group ℓ. By item (2) and corollary 4.124, the displayed parser-visible data determine every cell entry and every corner label on or inside E_ℓ. Hence, for each carrier tag $\operatorname{car}(t) \in \Theta_\ell$, item (3) identifies within the recovered subfilling the one-cell carrier E_{s_t,s_t+1} and therefore recovers its entry e_t and its left/bottom valley

$$(\nu_t, \rho_t, \mu_t).$$

Item (1) recovers the anchor blocks for the anchor tags in the same group. Thus each group has the fixed decoder required in proposition 4.281. The displayed relation $\beta_t = \kappa(e_t)$ is exactly the carrier-word hypothesis there. Applying that proposition gives the claimed recovery of the original zone subpath.

The top/right boundary labels of E_ℓ are, by hypothesis, determined by the parsed subwindow and known zone data. They are not supplied by an additional branch word, so the procedure stays within the counted target-window data. □

Proposition 4.283 (One-cell Step–3 corner readouts supply grouped carrier decoders)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \ldots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block* $B_r(P)$ *for every* $\mathrm{anc}(r) \in \Theta_\ell$*;*
(2) *for every* $\mathrm{car}(t) \in \Theta_\ell$*, the three non-southwest corner labels*

$$\nu_t, \quad \lambda_t, \quad \mu_t$$

of the one-cell Step–3 carrier

$$E_{s_t,s_t+1}, \qquad s_t = j - u_t + 1,$$

where the corner notation is

$$\begin{matrix} \nu_t & \lambda_t \\ \rho_t & \mu_t. \end{matrix}$$

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\mathrm{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L),$$

where e_t *is the Step–3 cell entry of* E_{s_t,s_t+1}*. Then the replacement window,* β_γ*, the zone endpoint pair, and* Z_γ *recover the original height halo-zone subpath. No additional Step–3 boundary word is used.*

Proof. Fix a carrier tag $\mathrm{car}(t) \in \Theta_\ell$. By lemma 4.121, the one-cell backward rule (B4) for the fourth arbitrary-filling dual-RSK' diagram determines the southwest corner ρ_t and the cell entry e_t from the three other corner labels

$$\lambda_t, \quad \mu_t, \quad \nu_t.$$

Therefore item (2) recovers both e_t and the left/bottom carrier valley

$$(\nu_t, \rho_t, \mu_t)$$

for every carrier tag in the group. Item (1) recovers the anchor blocks. Thus the fixed decoders required in proposition 4.281 are available, and the displayed relation is exactly the carrier-word hypothesis there. Applying that proposition gives the original height halo-zone subpath.

The recovered corner labels are parser-visible functions of the group subwindow and known zone data by hypothesis. Hence this readout introduces no new branch word beyond the already counted β_γ. □

Proposition 4.284 (Padded valleys supply one-cell corner readouts)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \dots, W_L$$

with parser-visible tag sets Θ_ℓ *satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let*

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ*, the data*

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block* $B_r(P)$ *for every* $\mathrm{anc}(r) \in \Theta_\ell$*;*
(2) *for every* $\mathrm{car}(t) \in \Theta_\ell$*, the padded carrier valley*

$$\mathrm{Pad}_{e_t}(\nu_t, \rho_t, \mu_t)$$

of the one-cell Step–3 carrier E_{s_t,s_t+1}*, where* $s_t = j - u_t + 1$ *and the corner notation is*

$$\begin{matrix} \nu_t & \lambda_t \\ \rho_t & \mu_t. \end{matrix}$$

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\mathrm{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L).$$

Then the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath. No additional Step–3 boundary word or corner word is used.

Proof. Fix a carrier tag $\mathrm{car}(t) \in \Theta_\ell$. The word β_t determines e_t because κ is injective. By lemma 4.269, the carrier satisfies

$$|\lambda_t| - |\rho_t| = 2.$$

Hence corollary 4.123 gives

$$|\nu_t| - 2|\rho_t| + |\mu_t| = 2 - e_t.$$

Therefore lemma 4.275 recovers (ν_t, ρ_t, μ_t) from the parser-visible padded carrier valley and the decoded value of e_t. The forward half of lemma 4.121 then determines the northeast corner λ_t from

$$e_t, \quad \rho_t, \quad \mu_t, \quad \nu_t.$$

Thus the same data determine the three non-southwest corner labels

$$\nu_t, \quad \lambda_t, \quad \mu_t$$

for every carrier tag. Together with item (1), this supplies the hypotheses of proposition 4.283. Applying that proposition recovers the original height halo-zone subpath. □

Corollary 4.285 (Double-successor carriers are covered by the grouped padded-valley decoder)**.** *In the setting of proposition 4.284, no additional hypothesis $e_t \le 1$ is needed. In particular, if a carrier tag has $e_t = 2$, then the same parsed padded-valley data and the two-bit carrier word recover that carrier valley and hence feed the grouped positive-height decoder.*

Proof. The injection

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

is part of the grouped decoder data. Thus the carrier word $\beta_t = \kappa(e_t)$ recovers e_t also when $e_t = 2$. Lemma 4.275 then recovers the unpadded valley (ν_t, ρ_t, μ_t) from $\mathrm{Pad}_{e_t}(\nu_t, \rho_t, \mu_t)$ and this decoded value of e_t. The proof of proposition 4.284 uses no additional restriction on e_t beyond $e_t \in \{0,1,2\}$, so it applies to the double-successor case exactly as stated. □

Corollary 4.286 (Padded-valley endpoints do not determine the midpoint)**.** *In the ambient one-cell dual-RSK' local category, the data*

$$e, \qquad \mathrm{pad}_e(\nu), \qquad \mu$$

together with the diagonal-jump condition $|\lambda| - |\rho| = 2$ do not determine the middle partition ρ of the padded carrier valley

$$\mathrm{Pad}_e(\nu, \rho, \mu).$$

Consequently the grouped padded-valley supplier cannot be replaced by an endpoint-only readout of the padded left endpoint and right endpoint, unless additional Step–3 restrictions or parser-visible data determine ρ or an equivalent midpoint datum.

Proof. Take

$$e = 0, \qquad \nu = \mu = (2,1).$$

There are two possible southwest corners

$$\rho^{(1)} = (2), \qquad \rho^{(2)} = (1,1).$$

For each $a = 1, 2$, the differences $\nu/\rho^{(a)}$ and $\mu/\rho^{(a)}$ are vertical strips. Applying Krattenthaler's forward algorithm (F4), in the carry notation used in the proof of lemma 4.122, gives

$$\lambda^{(1)} = (2,1,1) \qquad \text{and} \qquad \lambda^{(2)} = (2,2).$$

Indeed, for $\rho^{(1)} = (2)$ the carry is created in the second row and placed in the third row, while for $\rho^{(2)} = (1,1)$ it is created in the first row and placed in the second row. Thus both are valid one-cell dual-RSK' data, and in both cases

$$|\lambda^{(a)}| - |\rho^{(a)}| = 2.$$

The endpoint data are identical:

$$\mathrm{pad}_0(\nu) = (2,1), \qquad \mu = (2,1),$$

but the midpoint is either (2) or $(1,1)$. Hence the displayed endpoint data and the diagonal-jump condition do not determine ρ. ☐

Corollary 4.287 (Padded-valley midpoint defect normal form). *Let ν, ρ, μ be partitions such that ν/ρ and μ/ρ are vertical strips. Put*

$$\eta_i = \min(\nu_i, \mu_i)$$

after extending all partitions by zero parts, and define the midpoint defect set

$$D(\nu,\rho,\mu) = \{\, i : \rho_i = \eta_i - 1 \,\}.$$

Then ρ is determined by ν, μ, and $D(\nu,\rho,\mu)$ through

$$\rho_i = \begin{cases} \eta_i - 1, & i \in D(\nu,\rho,\mu), \\ \eta_i, & i \notin D(\nu,\rho,\mu). \end{cases}$$

Moreover

$$|\nu| - 2|\rho| + |\mu| = \sum_i |\nu_i - \mu_i| + 2\,|D(\nu,\rho,\mu)|.$$

In particular, for an actual two-label Step–3 carrier with cell entry e,

$$|D(\nu,\rho,\mu)| = \frac{2 - e - \sum_i |\nu_i - \mu_i|}{2}.$$

Thus, once e, $\mathrm{pad}_e(\nu)$, and μ are known, recovering the padded carrier valley is equivalent to recovering this finite midpoint defect set.

Proof. Because ν/ρ and μ/ρ are vertical strips, one has

$$\nu_i - \rho_i,\ \mu_i - \rho_i \in \{0,1\}$$

for every row i. Hence

$$\eta_i - \rho_i = \min(\nu_i,\mu_i) - \rho_i \in \{0,1\},$$

which proves the displayed reconstruction formula for ρ from the set $D(\nu,\rho,\mu)$.

The size identity follows from

$$|\nu| - 2|\rho| + |\mu| = (|\nu| + |\mu| - 2|\eta|) + 2(|\eta| - |\rho|).$$

The first parenthesis is $\sum_i |\nu_i - \mu_i|$, and the second is $2|D(\nu,\rho,\mu)|$ by the reconstruction formula.

For an actual carrier, corollary 4.123 gives $|\nu| - 2|\rho| + |\mu| = 2 - e$, so the displayed cardinality formula follows. Finally, when e and $\mathrm{pad}_e(\nu)$ are known, the definition of bottom padding recovers ν by deleting the final e appended unit rows; the reconstruction formula then shows that the only remaining datum in the padded valley is $D(\nu,\rho,\mu)$. ☐

Corollary 4.288 (Two-label midpoint defects are singleton-sized). *For an actual two-label Step–3 carrier with cell entry e, put*

$$A(\nu,\mu) = \sum_i |\nu_i - \mu_i|.$$

Then

$$A(\nu,\mu) + 2\,|D(\nu,\rho,\mu)| = 2 - e.$$

Consequently $D(\nu,\rho,\mu)$ is empty except possibly when

$$e = 0, \qquad \nu = \mu.$$

In that exceptional case $D(\nu,\rho,\mu)$ is a singleton. Thus the only midpoint information not determined by e, $\mathrm{pad}_e(\nu)$, and μ is one defect row in the zero-entry equal-endpoint case.

Proof. The displayed identity is the size formula of corollary 4.287 combined with corollary 4.123. Since lemma 4.268 gives $e \in \{0,1,2\}$, the right side is at most 2. Hence $|D(\nu,\rho,\mu)| \le 1$.

If $D(\nu,\rho,\mu)$ is nonempty, then $|D| = 1$ and the displayed identity forces $2 - e - A(\nu,\mu) = 2$. Therefore $e = 0$ and $A(\nu,\mu) = 0$, which is equivalent to $\nu = \mu$. Conversely, if $e = 0$ and $\nu = \mu$, then $A(\nu,\mu) = 0$, so the displayed identity gives $|D| = 1$. In all remaining cases the defect set is empty. ☐

Corollary 4.289 (The singleton defect row is the predecessor of the northeast box)**.** *For an actual two-label Step–3 carrier, suppose*

$$e = 0, \qquad \nu = \mu,$$

and write

$$D(\nu,\rho,\mu) = \{d\}.$$

Then λ/ν consists of one box, and that box lies in row $d+1$. Consequently the singleton defect row d is determined by the row of the unique box of λ/ν.

Proof. Use the notation of the F4 carry computation in the proof of lemma 4.122. Since $e = 0$, the initial carry is $c_1 = 0$. Since $\nu = \mu$ and $D(\nu,\rho,\mu) = \{d\}$, one has

$$\min(\mu_i,\nu_i) - \rho_i = \begin{cases} 1, & i = d, \\ 0, & i \ne d. \end{cases}$$

For rows $i < d$ the carry is zero and the defect term is zero, so $\lambda_i = \nu_i$. At row d the equality $\rho_d = \nu_d - 1$ gives $a_d = 0$ in the F4 rule, so no box is placed in row d and the carry leaving that row is $c_{d+1} = 1$. At row $d+1$, the defect term is zero and $\rho_{d+1} = \nu_{d+1} = \mu_{d+1}$, so $a_{d+1} = 1$ and the incoming carry is placed there:

$$\lambda_{d+1} = \nu_{d+1} + 1, \qquad c_{d+2} = 0.$$

All later rows again have zero carry and zero defect term, hence $\lambda_i = \nu_i$ for $i \ne d+1$. Therefore λ/ν is the single box in row $d+1$. ☐

Corollary 4.290 (Exceptional northeast rows are removable-row successors)**.** *In the exceptional case of corollary 4.289, the defect row d is a removable-corner row of the common endpoint:*

$$\nu_d > \nu_{d+1}.$$

Conversely, let ν be any partition and let d be any row satisfying $\nu_d > \nu_{d+1}$. Put $\mu = \nu$, $e = 0$, and let ρ be obtained from ν by decreasing the dth row by one box. Then the one-cell F4 rule produces a valid northeast partition λ with

$$\lambda/\nu = \{\text{one box in row } d+1\}, \qquad |\lambda| - |\rho| = 2.$$

Thus, in the zero-entry equal-endpoint case, the northeast added-box row is exactly the successor of a removable row of the common endpoint.

Proof. Extend partitions by zero parts. If $D(\nu,\rho,\mu) = \{d\}$ and $\nu = \mu$, then $\rho_i = \nu_i$ for $i \ne d$ and $\rho_d = \nu_d - 1$. Since ρ is a partition,

$$\nu_d - 1 = \rho_d \ge \rho_{d+1} = \nu_{d+1},$$

which is equivalent to $\nu_d > \nu_{d+1}$.

Conversely suppose $\nu_d > \nu_{d+1}$ and define ρ by lowering row d. The displayed inequality gives $\rho_d = \nu_d - 1 \geq \nu_{d+1} = \rho_{d+1}$, and all other partition inequalities are unchanged, so ρ is a partition. Moreover ν/ρ and μ/ρ are vertical strips, each consisting of the same one box in row d.

Run the F4 carry rule with $e = 0$. The carry is zero before row d. At row d, the defect term is one and $a_d = 0$, so the carry leaving row d is one and no box is placed in row d. At row $d+1$, the defect term is zero and $a_{d+1} = 1$, so the incoming carry is placed there and then vanishes. Hence

$$\lambda_{d+1} = \nu_{d+1} + 1, \qquad \lambda_i = \nu_i \quad (i \neq d+1).$$

The inequality $\nu_d > \nu_{d+1}$ gives $\lambda_d = \nu_d \geq \nu_{d+1} + 1 = \lambda_{d+1}$, and the remaining partition inequalities are inherited from ν. Therefore λ is a partition and λ/ν is exactly one box in row $d+1$. Since $|\rho| = |\nu| - 1$ and $|\lambda| = |\nu| + 1$, one has $|\lambda| - |\rho| = 2$. ☐

Corollary 4.291 (Exceptional northeast row is not endpoint-forced)**.** *The row of the exceptional northeast box is not determined by the endpoint data*

$$e, \qquad \mathrm{pad}_e(\nu), \qquad \mu.$$

More precisely, with $e = 0$ and $\nu = \mu = (3,2,1)$, two valid choices of the middle partition give different northeast rows:

$$\rho = (2,2,1), \quad \lambda = (3,3,1), \qquad \textit{and} \qquad \rho' = (3,1,1), \quad \lambda' = (3,2,2).$$

The first has the unique box of λ/ν in row 2, while the second has the unique box of λ'/ν in row 3.

Proof. For $\nu = (3,2,1)$, rows 1 and 2 are both removable-corner rows. Applying corollary 4.290 with $d = 1$ gives $\rho = (2,2,1)$ and the northeast added box in row 2, hence $\lambda = (3,3,1)$. Applying the same corollary with $d = 2$ gives $\rho' = (3,1,1)$ and the northeast added box in row 3, hence $\lambda' = (3,2,2)$. In both cases $e = 0$ and the visible endpoint data are the same, because $\mathrm{pad}_0(\nu) = \nu$ and $\mu = \nu$. The two northeast rows are different. ☐

Corollary 4.292 (Exceptional removable-row ambiguity is unbounded)**.** *For every $k \geq 1$ there is a common endpoint partition $\nu = \mu$ for which the exceptional one-cell data*

$$e = 0, \qquad \mathrm{pad}_0(\nu) = \nu, \qquad \mu = \nu$$

admit at least k different removable-row choices in the sense of corollary 4.290. Consequently no fixed local word of length c bits, together with the endpoint data $(e, \mathrm{pad}_e(\nu), \mu)$, can recover the exceptional removable row for all ambient one-cell dual-RSK' data once $k > 2^c$.

Proof. Let

$$\nu = \mu = (k, k-1, \ldots, 2, 1).$$

For each $1 \leq d \leq k$ one has $\nu_d > \nu_{d+1}$, with the convention $\nu_{k+1} = 0$. Hence d is a removable-corner row of the common endpoint. By corollary 4.290, for each such d the partition $\rho^{(d)}$ obtained by decreasing the dth row of ν by one box, together with $e = 0$ and $\mu = \nu$, gives a valid one-cell F4 datum whose northeast corner $\lambda^{(d)}$ satisfies

$$\lambda^{(d)}/\nu = \{\text{one box in row } d+1\}, \qquad |\lambda^{(d)}| - |\rho^{(d)}| = 2.$$

The k choices have the same endpoint data $(e, \mathrm{pad}_e(\nu), \mu) = (0, \nu, \nu)$ but different removable rows. Thus a word of length c bits can distinguish at most 2^c of them. Choosing $k > 2^c$ proves the final assertion. ☐

Corollary 4.293 (Two-bit carrier words do not recover exceptional removable rows)**.** *In the ambient one-cell dual-RSK' local category, the exceptional removable row cannot be recovered for all data from*

$$e, \qquad \mathrm{pad}_e(\nu), \qquad \mu,$$

and one additional two-bit word. In particular, the counted carrier word $\beta = \kappa(e)$ in the grouped positive-height supplier cannot by itself serve as the removable-row readout in proposition 4.297.

Proof. Apply corollary 4.292 with $k = 5$. The five valid exceptional one-cell data have the same values of

$$e = 0, \qquad \mathrm{pad}_e(\nu) = \nu, \qquad \mu = \nu,$$

but have five different removable rows. A two-bit word has only four values, so two of these five data have the same displayed input and the same two-bit word. No decoder from those data can distinguish their removable rows.

If the word is the counted carrier word $\beta = \kappa(e)$, then in the exceptional case $e = 0$ it is fixed to the single value $\kappa(0)$, so it carries no removable-row choice at all. □

Proposition 4.294 (Midpoint-defect readouts supply grouped padded valleys)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \ldots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

1. *the original block $B_r(P)$ for every $\mathrm{anc}(r) \in \Theta_\ell$;*
2. *for every $\mathrm{car}(t) \in \Theta_\ell$, the padded left endpoint $\mathrm{pad}_{e_t}(\nu_t)$, the right endpoint μ_t, and the midpoint defect set*

$$D(\nu_t, \rho_t, \mu_t)$$

of the one-cell Step–3 carrier E_{s_t, s_t+1}, where $s_t = j - u_t + 1$.

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\mathrm{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L).$$

Then the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath. No additional midpoint word is used.

Proof. For a carrier tag $\mathrm{car}(t) \in \Theta_\ell$, the word β_t determines e_t through the fixed injection κ. From $\mathrm{pad}_{e_t}(\nu_t)$ and e_t one recovers ν_t by deleting the final e_t appended unit rows. Then corollary 4.287 reconstructs ρ_t from ν_t, μ_t, and the recovered defect set $D(\nu_t, \rho_t, \mu_t)$. Thus the same parser-visible data determine the padded carrier valley

$$\mathrm{Pad}_{e_t}(\nu_t, \rho_t, \mu_t)$$

for every carrier tag in the group.

Together with item (1), these are exactly the hypotheses of proposition 4.284. Applying that proposition recovers the original height halo-zone subpath. □

Proposition 4.295 (Singleton-defect row readouts supply grouped padded valleys)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \ldots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block* $B_r(P)$ *for every* $\mathrm{anc}(r) \in \Theta_\ell$*;*
(2) *for every* $\mathrm{car}(t) \in \Theta_\ell$*, the padded left endpoint* $\mathrm{pad}_{e_t}(\nu_t)$ *and the right endpoint* μ_t *of the one-cell Step–3 carrier* E_{s_t,s_t+1}*, where* $s_t = j - u_t + 1$*;*
(3) *for every* $\mathrm{car}(t) \in \Theta_\ell$ *with* $e_t = 0$ *and* $\nu_t = \mu_t$*, the unique row* d_t *of the singleton defect set* $D(\nu_t, \rho_t, \mu_t)$*.*

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\mathrm{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L).$$

Then the replacement window, β_γ*, the zone endpoint pair, and* Z_γ *recover the original height halo-zone subpath. No additional defect-set word is used.*

Proof. For a carrier tag $\mathrm{car}(t) \in \Theta_\ell$, the word β_t determines e_t through the fixed injection κ. From $\mathrm{pad}_{e_t}(\nu_t)$ and e_t one recovers ν_t by deleting the final e_t appended unit rows. The right endpoint μ_t is part of item (2).

By corollary 4.288, the defect set is empty unless $e_t = 0$ and $\nu_t = \mu_t$. In the exceptional case item (3) recovers its unique row d_t, and hence recovers

$$D(\nu_t, \rho_t, \mu_t) = \{d_t\}.$$

Thus the data determine the midpoint defect set for every carrier tag. Proposition 4.294 then recovers the original height halo-zone subpath. ☐

Proposition 4.296 (Northeast-box row readouts supply grouped padded valleys)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \dots, W_L$$

with parser-visible tag sets Θ_ℓ *satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let*

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ*, the data*

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block* $B_r(P)$ *for every* $\mathrm{anc}(r) \in \Theta_\ell$*;*
(2) *for every* $\mathrm{car}(t) \in \Theta_\ell$*, the padded left endpoint* $\mathrm{pad}_{e_t}(\nu_t)$ *and the right endpoint* μ_t *of the one-cell Step–3 carrier* E_{s_t,s_t+1}*, where* $s_t = j - u_t + 1$*;*
(3) *for every* $\mathrm{car}(t) \in \Theta_\ell$ *with* $e_t = 0$ *and* $\nu_t = \mu_t$*, the row* r_t *of the unique box in* λ_t/ν_t*.*

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\mathrm{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L).$$

Then the replacement window, β_γ*, the zone endpoint pair, and* Z_γ *recover the original height halo-zone subpath. No additional defect-row word is used.*

Proof. For each carrier tag, the word β_t determines e_t, and item (2) recovers ν_t from $\mathrm{pad}_{e_t}(\nu_t)$ and e_t. If the carrier is not in the exceptional case $e_t = 0$ and $\nu_t = \mu_t$, then corollary 4.288 makes the defect set empty.

In the exceptional case, item (3) gives the row r_t of the unique box of λ_t/ν_t. By corollary 4.289, the singleton defect row is $d_t = r_t - 1$. Hence

$$D(\nu_t, \rho_t, \mu_t) = \{r_t - 1\}.$$

Thus the hypotheses of proposition 4.295 are satisfied, and that proposition recovers the original height halo-zone subpath. ☐

Proposition 4.297 (Removable-row readouts supply grouped padded valleys)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \ldots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

1. *the original block $B_r(P)$ for every $\mathrm{anc}(r) \in \Theta_\ell$;*
2. *for every $\mathrm{car}(t) \in \Theta_\ell$, the padded left endpoint $\mathrm{pad}_{e_t}(\nu_t)$ and the right endpoint μ_t of the one-cell Step–3 carrier E_{s_t,s_t+1}, where $s_t = j - u_t + 1$;*
3. *for every $\mathrm{car}(t) \in \Theta_\ell$ with $e_t = 0$ and $\nu_t = \mu_t$, the removable-corner row d_t of ν_t whose successor row is the row of the unique box in λ_t/ν_t.*

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\mathrm{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L).$$

Then the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath. No northeast-corner partition readout is used.

Proof. For each carrier tag, the word β_t determines e_t, and item (2) recovers ν_t from $\mathrm{pad}_{e_t}(\nu_t)$ and e_t. If the carrier is not in the exceptional case $e_t = 0$ and $\nu_t = \mu_t$, then corollary 4.288 makes the defect set empty.

In the exceptional case, item (3) gives the removable row d_t. By corollary 4.290, the unique box of λ_t/ν_t lies in row $d_t + 1$. Thus the hypotheses of proposition 4.296 are satisfied, with $r_t = d_t + 1$, and that proposition recovers the original height halo-zone subpath. □

Corollary 4.298 (The removable-row exception gate is parser-visible)**.** *Work in the setting of proposition 4.297. For a carrier tag $\mathrm{car}(t) \in \Theta_\ell$, suppose the group data determine the padded left endpoint $\mathrm{pad}_{e_t}(\nu_t)$ and the right endpoint μ_t, and suppose the carrier word satisfies $\beta_t = \kappa(e_t)$. Then*

$$e_t = 0 \text{ and } \nu_t = \mu_t$$

if and only if

$$\beta_t = \kappa(0) \qquad \text{and} \qquad \mathrm{pad}_{e_t}(\nu_t) = \mu_t.$$

Consequently the set of carrier tags for which the removable-row readout in item (3) of proposition 4.297 is required is determined from the parser-visible group data and the already counted carrier words, before any removable row is read.

Proof. Since κ is injective and $\beta_t = \kappa(e_t)$, the equality $\beta_t = \kappa(0)$ is equivalent to $e_t = 0$. When $e_t = 0$, the bottom-padding convention in definition 4.274 gives

$$\mathrm{pad}_{e_t}(\nu_t) = \mathrm{pad}_0(\nu_t) = \nu_t.$$

Thus, under $\beta_t = \kappa(0)$, the equality $\mathrm{pad}_{e_t}(\nu_t) = \mu_t$ is equivalent to $\nu_t = \mu_t$. This proves both implications. The final sentence follows because item (2) of proposition 4.297 makes the two displayed endpoint partitions part of the group readout, while β_t is the already counted carrier word. □

Proposition 4.299 (Visible-gated northeast rows supply grouped padded valleys)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \ldots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block $B_r(P)$ for every $\operatorname{anc}(r) \in \Theta_\ell$;*
(2) *for every $\operatorname{car}(t) \in \Theta_\ell$, the padded left endpoint $\operatorname{pad}_{e_t}(\nu_t)$ and the right endpoint μ_t of the one-cell Step–3 carrier E_{s_t,s_t+1}, where $s_t = j - u_t + 1$;*
(3) *for every carrier tag satisfying the visible gate*

$$\beta_t = \kappa(0) \qquad \text{and} \qquad \operatorname{pad}_{e_t}(\nu_t) = \mu_t,$$

the row r_t of the unique box in λ_t/ν_t.

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\operatorname{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L).$$

Then the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath. No full visible-gated corner or boundary readout is used.

Proof. By corollary 4.298, the displayed visible gate is equivalent, for each carrier tag, to

$$e_t = 0, \qquad \nu_t = \mu_t.$$

Thus item (3) supplies the northeast-box row exactly for the carrier tags required in item (3) of proposition 4.296. Items (1) and (2) are the same as in that proposition, and the carrier-word relation is also the same. Applying proposition 4.296 therefore recovers the original height halo-zone subpath. □

Corollary 4.300 (Visible-gated northeast rows are target-block midpoints)**.** *Let*

$$\begin{matrix} \nu & \lambda \\ \rho & \mu \end{matrix}$$

be the corner labels of an actual two-label Step–3 carrier, and suppose that its entry is $e = 0$ and that $\nu = \mu$. Let r be the row of the unique box in λ/ν. Define X by lowering row $r-1$ of ν by one box and leaving all other rows unchanged. Then

$$X = \rho.$$

Consequently

$$(\nu, X, \nu)$$

is a valid content-2 target two-step block. Conversely, this equal-endpoint target block determines the northeast-box row r as one plus the unique row in which ν and X differ.

Proof. By corollary 4.288, the defect set $D(\nu,\rho,\mu)$ is a singleton. Write it as $\{d\}$. Since $\nu = \mu$, corollary 4.287 gives

$$\rho_i = \begin{cases} \nu_i - 1, & i = d, \\ \nu_i, & i \ne d. \end{cases}$$

By corollary 4.289, the unique box of λ/ν lies in row $d+1$. Hence $d = r-1$, and the displayed formula is exactly the definition of X. Thus $X = \rho$.

The differences ν/ρ and $\mu/\rho = \nu/\rho$ are vertical strips, and $|\rho| = |\nu| - 1$. Therefore

$$|\nu| - 2|\rho| + |\nu| = 2,$$

so $(\nu, X, \nu) = (\nu, \rho, \mu)$ is a valid content-2 target two-step block. The final assertion follows because this block has equal endpoints and its middle partition differs from that endpoint in the single row $d = r - 1$. $\square$

Proposition 4.301 (Visible-gated removable rows supply grouped padded valleys)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \dots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma - 2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block $B_r(P)$ for every $\mathrm{anc}(r) \in \Theta_\ell$;*
(2) *for every $\mathrm{car}(t) \in \Theta_\ell$, the padded left endpoint $\mathrm{pad}_{e_t}(\nu_t)$ and the right endpoint μ_t of the one-cell Step–3 carrier E_{s_t, s_t+1}, where $s_t = j - u_t + 1$.*

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\mathrm{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L),$$

and that, for every carrier tag satisfying the visible gate

$$\beta_t = \kappa(0) \qquad \text{and} \qquad \mathrm{pad}_{e_t}(\nu_t) = \mu_t,$$

the parser-visible group data determine the removable-corner row d_t of the common endpoint whose successor row is the row of the unique box in λ_t/ν_t. Then the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath. No hidden source-side exception predicate is used.

Proof. By corollary 4.298, the displayed visible gate is equivalent, for each carrier tag, to the source condition

$$e_t = 0, \qquad \nu_t = \mu_t.$$

Thus the removable rows supplied in the final hypothesis are supplied exactly for the carrier tags required in item (3) of proposition 4.297. Items (1) and (2) are the same as in that proposition, and the carrier-word relation is also the same. Therefore proposition 4.297 applies and recovers the original height halo-zone subpath. $\square$

Proposition 4.302 (Visible-gated midpoints supply removable rows)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \dots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma - 2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block $B_r(P)$ for every $\mathrm{anc}(r) \in \Theta_\ell$;*
(2) *for every $\mathrm{car}(t) \in \Theta_\ell$, the padded left endpoint $\mathrm{pad}_{e_t}(\nu_t)$ and the right endpoint μ_t of the one-cell Step–3 carrier E_{s_t, s_t+1}, where $s_t = j - u_t + 1$;*

(3) *for every carrier tag satisfying the visible gate*

$$\beta_t = \kappa(0) \qquad \textit{and} \qquad \mathrm{pad}_{e_t}(\nu_t) = \mu_t,$$

the southwest corner ρ_t of the same one-cell carrier.

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\mathrm{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L).$$

Then the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath. Equivalently, for flagged carrier tags, the remaining row datum in the visible-gated local inverse is supplied by the recovered middle partition.

Proof. Fix a flagged carrier tag. By corollary 4.298, the visible gate is equivalent to

$$e_t = 0, \qquad \nu_t = \mu_t.$$

Since $e_t = 0$, the padded left endpoint is ν_t itself. Item (3) gives ρ_t, and corollary 4.288 says that the defect set $D(\nu_t, \rho_t, \mu_t)$ is a singleton. Because $\nu_t = \mu_t$, the definition of the defect set reduces to

$$D(\nu_t, \rho_t, \mu_t) = \{\, i : \rho_{t,i} = \nu_{t,i} - 1 \,\}.$$

Let d_t be this unique row. Then corollary 4.290 says that d_t is the removable-corner row of the common endpoint whose successor row is the row of the unique box in λ_t/ν_t.

Thus item (3) of proposition 4.301 is satisfied for every flagged carrier tag, while items (1), (2), and the carrier-word relation are exactly the remaining hypotheses there. Applying that proposition recovers the original height halo-zone subpath. □

Proposition 4.303 (Visible-gated target blocks recover positive height zones)**.** *Work in the height setting of proposition 4.532. Let*

$$I = I(S),$$

fix an injection

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2,$$

and write

$$Z_\gamma \cap I = \{u_1 < \cdots < u_m\}.$$

Let β_t denote the deterministic two-bit subword of β_γ attached to the retained carrier slot u_t. Let A_γ be the canonical right-anchor set. For each retained slot $r \in Z_\gamma \setminus I$, suppose a candidate target block

$$C_r = (L_r, X_r, R_r)$$

is given as follows.

(1) *If $r \in A_\gamma$, then $C_r = B_r(P)$.*

(2) *If $r = u_t - 1$ for some t, put $s_t = j - u_t + 1$, let*

$$\begin{matrix} \nu_t & \lambda_t \\ \rho_t & \mu_t \end{matrix}$$

be the corner labels of E_{s_t, s_t+1}, and let e_t be its cell entry. Assume that C_r is a valid content-2 two-step block with

$$L_r = \mathrm{pad}_{e_t}(\nu_t), \qquad R_r = \mu_t.$$

If the visible gate

$$\beta_t = \kappa(0) \qquad \textit{and} \qquad L_r = R_r$$

holds, let r_t be the row of the unique box in λ_t/ν_t, and assume that X_r is obtained from the common endpoint $L_r = R_r$ by lowering row $r_t - 1$ by one box and leaving all other rows unchanged.

Assume that the ordered blocks C_r satisfy the endpoint equalities in lemma 4.277, and that the two-bit split of the branch word satisfies $\beta_t = \kappa(e_t)$ for every t. Then the original height halo-zone subpath is recovered from the resulting replacement window, β_γ, the zone endpoint pair, and Z_γ.

Proof. By lemma 4.277, the ordered candidate blocks concatenate to a target-size replacement window, and the deterministic parser recovers each block $C_r = (L_r, X_r, R_r)$ in slot order. Use singleton tag sets in that order: $\mathrm{anc}(r)$ for $r \in A_\gamma$ and $\mathrm{car}(t)$ for $r = u_t - 1$. These tags are determined by S and Z_γ, have total size $|Z_\gamma \setminus I|$, and partition the anchor and carrier tags required in proposition 4.281.

For an anchor singleton, the parsed block is $B_r(P)$ by item (1). For a carrier singleton, item (2) identifies the parsed left and right endpoints with

$$\mathrm{pad}_{e_t}(\nu_t), \qquad \mu_t.$$

Since $\beta_t = \kappa(e_t)$, the displayed gate $L_r = R_r$ is exactly

$$\mathrm{pad}_{e_t}(\nu_t) = \mu_t$$

and hence is the visible gate of proposition 4.302. When that gate holds, the equality $\beta_t = \kappa(0)$ and injectivity of κ give $e_t = 0$. The endpoint equality then gives

$$L_r = R_r = \nu_t = \mu_t.$$

By corollary 4.300, the row-encoded middle partition specified in item (2) is the southwest corner ρ_t. Thus the same parsed singleton block supplies the flagged southwest corner required there.

Thus the singleton parsing satisfies the hypotheses of proposition 4.302. Applying that proposition recovers the original height halo-zone subpath. □

Corollary 4.304 (Singleton target blocks do not replace the grouped source)**.** *The source theorem needed for the positive-height C-side supplier is not obtained by declaring the actual one-cell carrier valleys and unchanged anchors to be an automatically endpoint-compatible list of singleton target blocks. The criterion of proposition 4.303 is a sufficient local inverse once such blocks have been constructed; it does not construct them from the height halo hypotheses alone. In particular, the live source statement remains the grouped-window readout of proposition 4.302, or a collective target-window construction which supplies the same padded endpoints and visible-gated middle partitions.*

Proof. Proposition 4.303 assumes the endpoint equalities of lemma 4.277. Those equalities are extra source data: by corollary 4.312, the bottom-padded one-cell carrier valley is not forced by the Step–3 one-cell boundary convention to span the adjacent original two-block interval.

Nor can the endpoint problem be removed by a zero-padding or separate-buffer argument. Corollary 4.313 gives a valid positive-height zone with nonzero carrier padding, and with more nonzero padded carriers than surplus anchors. Corollary 4.314 shows that the full $e = 2$ carrier case occurs inside a positive-height zone. Thus the singleton target-block criterion is a valid sufficient inverse only after a source construction has supplied the endpoint-compatible blocks. The source obligation that remains attached to remark 4.839 is therefore the collective grouped target-window readout already isolated in proposition 4.302. □

Corollary 4.305 (One-cell Step–3 corners supply visible-gated midpoint readouts)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \dots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma - 2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block* $B_r(P)$ *for every* $\operatorname{anc}(r) \in \Theta_\ell$*;*
(2) *for every* $\operatorname{car}(t) \in \Theta_\ell$*, the three non-southwest corner labels*

$$\nu_t, \quad \lambda_t, \quad \mu_t$$

of the one-cell Step–3 carrier E_{s_t,s_t+1}*, where* $s_t = j - u_t + 1$*.*

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\operatorname{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L),$$

where e_t *is the Step–3 cell entry of* E_{s_t,s_t+1}*. Then the hypotheses of proposition 4.302 hold. Consequently the replacement window,* β_γ*, the zone endpoint pair, and* Z_γ *recover the original height halo-zone subpath.*

Proof. Fix a carrier tag $\operatorname{car}(t) \in \Theta_\ell$. By lemma 4.121, the one-cell backward rule recovers the southwest corner ρ_t and the cell entry e_t from the three displayed non-southwest corners. Hence the same parser-visible group data determine the padded left endpoint $\operatorname{pad}_{e_t}(\nu_t)$ and the right endpoint μ_t.

If the visible gate

$$\beta_t = \kappa(0) \qquad \text{and} \qquad \operatorname{pad}_{e_t}(\nu_t) = \mu_t$$

holds, then the already recovered ρ_t supplies the southwest corner required in item (3) of proposition 4.302. The anchor blocks are item (1), the padded endpoints are recovered as above, and the carrier-word relation is the displayed hypothesis. Thus all hypotheses of proposition 4.302 hold, and that proposition gives the asserted zone recovery. □

Corollary 4.306 (Step–3 boundary readouts supply visible-gated midpoint readouts)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \dots, W_L$$

with parser-visible tag sets Θ_ℓ *satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Fix the injection*

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

from that setting. Suppose that, for each ℓ*, the data*

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block* $B_r(P)$ *for every* $\operatorname{anc}(r) \in \Theta_\ell$*;*
(2) *a Ferrers subarrangement* E_ℓ *of the Step–3 dual-RSK' half-square of lemma 4.125, together with the partition labels on the top/right boundary of* E_ℓ*;*
(3) *for every* $\operatorname{car}(t) \in \Theta_\ell$*, the inclusion of the one-cell carrier* E_{s_t,s_t+1} *in* E_ℓ*, where* $s_t = j-u_t+1$*.*

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\operatorname{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L),$$

where e_t *is the Step–3 cell entry of* E_{s_t,s_t+1}*. Then the hypotheses of proposition 4.302 hold. Consequently the replacement window,* β_γ*, the zone endpoint pair, and* Z_γ *recover the original height halo-zone subpath.*

Proof. Fix a group ℓ. By item (2) and corollary 4.124, the parser-visible data recover every cell entry and every corner label on or inside E_ℓ. In particular, for each carrier tag $\operatorname{car}(t) \in \Theta_\ell$, item (3) identifies the one-cell carrier E_{s_t,s_t+1} inside this recovered subfilling, so the data recover its non-southwest corners

$$\nu_t, \quad \lambda_t, \quad \mu_t.$$

The anchor blocks are recovered by item (1). Thus the hypotheses of corollary 4.305 are satisfied for the same parser-visible grouped subwindows and the same carrier word relation. Applying that corollary gives the visible-gated midpoint readouts and hence the asserted zone recovery. □

Corollary 4.307 (Visible-gated one-cell corners are enough)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \dots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Fix the injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2$$

from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma - 2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block $B_r(P)$ for every $\operatorname{anc}(r) \in \Theta_\ell$;*
(2) *for every $\operatorname{car}(t) \in \Theta_\ell$, the padded left endpoint $\operatorname{pad}_{e_t}(\nu_t)$ and the right endpoint μ_t of the one-cell Step–3 carrier E_{s_t, s_t+1}, where $s_t = j - u_t + 1$;*
(3) *for every carrier tag satisfying the visible gate*

$$\beta_t = \kappa(0) \qquad \text{and} \qquad \operatorname{pad}_{e_t}(\nu_t) = \mu_t,$$

the three non-southwest corner labels

$$\nu_t, \quad \lambda_t, \quad \mu_t$$

of the same one-cell carrier.

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\operatorname{car}(t) \in \Theta_\ell, \ 1 \le \ell \le L).$$

Then the hypotheses of proposition 4.302 hold. Consequently the replacement window, β_γ, the zone endpoint pair, and Z_γ recover the original height halo-zone subpath.

Proof. Items (1) and (2) are exactly the anchor and endpoint readouts required in proposition 4.302. The displayed carrier-word relation is also the same.

It remains only to supply the southwest corner for the carrier tags whose visible gate holds. For such a tag, item (3) gives the three non-southwest corners of the one-cell carrier. By lemma 4.121, the one-cell backward rule recovers the southwest corner ρ_t from those three corners. Thus item (3) of proposition 4.302 is satisfied exactly on the visible-gated carrier tags, and that proposition gives the claimed recovery. □

Corollary 4.308 (Visible-gated Step–3 boundary readouts are enough)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \dots, W_L$$

with parser-visible tag sets Θ_ℓ satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Fix the injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2$$

from that setting. Suppose that, for each ℓ, the data

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma - 2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block $B_r(P)$ for every $\operatorname{anc}(r) \in \Theta_\ell$;*
(2) *for every $\operatorname{car}(t) \in \Theta_\ell$, the padded left endpoint $\operatorname{pad}_{e_t}(\nu_t)$ and the right endpoint μ_t of the one-cell Step–3 carrier E_{s_t, s_t+1}, where $s_t = j - u_t + 1$;*

(3) *a Ferrers subarrangement* E_ℓ^{vis} *of the Step–3 dual-RSK' half-square of lemma 4.125, together with the partition labels on the top/right boundary of* E_ℓ^{vis}*;*

(4) *for every carrier tag satisfying the visible gate*

$$\beta_t = \kappa(0) \qquad \textit{and} \qquad \mathrm{pad}_{e_t}(\nu_t) = \mu_t,$$

the inclusion of the one-cell carrier E_{s_t,s_t+1} *in* E_ℓ^{vis}*.*

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\mathrm{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L).$$

Then the hypotheses of proposition 4.302 hold. Consequently the replacement window, β_γ*, the zone endpoint pair, and* Z_γ *recover the original height halo-zone subpath.*

Proof. For a fixed group ℓ, item (3) and corollary 4.124 recover every cell entry and every corner label on or inside E_ℓ^{vis}. Hence, for every carrier tag whose visible gate holds, item (4) identifies the one-cell carrier inside this recovered subfilling and recovers its three non-southwest corners

$$\nu_t, \quad \lambda_t, \quad \mu_t.$$

Together with items (1) and (2), these are exactly the hypotheses of corollary 4.307. Applying that corollary gives the visible-gated midpoint readouts and hence the asserted zone recovery. □

Corollary 4.309 (Raw Step–3 corner triples are not grouped target windows)**.** *The source condition in corollary 4.306 is not discharged by taking, for each carrier tag, the three raw non-southwest corner labels of its one-cell Step–3 carrier as a singleton target subwindow of the grouped replacement window. Any construction that uses the Step–3 boundary criterion must recover those corner labels from a content-normalized or collective grouped target subwindow, rather than identify the raw corner triple itself with one content-*2 *target block.*

Proof. Fix a carrier tag and write the four corner labels of its one-cell carrier as

$$\begin{matrix} \nu & \lambda \\ \rho & \mu \end{matrix}$$

with e the cell entry. If the raw non-southwest triple (ν, λ, μ) were a singleton target subwindow in the grouped replacement window of proposition 4.281, then it would be a target two-step block in the content (2^j) growth-path model, hence a content-2 block by lemma 4.118. But corollary 4.270 gives

$$|\nu| - 2|\lambda| + |\mu| = -2 - e,$$

which is not 2. This contradicts the target-block weight required in the grouped replacement window. Hence the raw corner-triple identification cannot supply the source condition. □

Corollary 4.310 (One-cell Step–3 corners supply removable-row readouts)**.** *Work in the setting of proposition 4.281. Assume that the replacement window is parsed into ordered subwindows*

$$W_1, \dots, W_L$$

with parser-visible tag sets Θ_ℓ *satisfying the tag-partition, length, and concatenation hypotheses of that proposition. Let*

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2$$

be the fixed injection from that setting. Suppose that, for each ℓ*, the data*

$$W_\ell, \quad \Theta_\ell, \quad S, \quad Z_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}$$

determine:

(1) *the original block* $B_r(P)$ *for every* $\mathrm{anc}(r) \in \Theta_\ell$*;*

(2) *for every* $\operatorname{car}(t) \in \Theta_\ell$, *the three non-southwest corner labels*

$$\nu_t, \quad \lambda_t, \quad \mu_t$$

of the one-cell Step–3 carrier E_{s_t,s_t+1}, *where* $s_t = j - u_t + 1$.

Finally suppose that the carrier words satisfy

$$\beta_t = \kappa(e_t) \qquad (\operatorname{car}(t) \in \Theta_\ell,\ 1 \le \ell \le L),$$

where e_t *is the Step–3 cell entry of* E_{s_t,s_t+1}. *Then the replacement window,* β_γ, *the zone endpoint pair, and* Z_γ *recover the original height halo-zone subpath through the removable-row readout criterion of proposition 4.297.*

Proof. Fix a carrier tag $\operatorname{car}(t) \in \Theta_\ell$. By lemma 4.121, the one-cell backward rule recovers the southwest corner ρ_t and the cell entry e_t from the three displayed corner labels. From e_t and ν_t the padded left endpoint $\operatorname{pad}_{e_t}(\nu_t)$ is determined, while μ_t is already one of the displayed labels.

If $e_t = 0$ and $\nu_t = \mu_t$, then corollary 4.289 says that λ_t/ν_t consists of one box, and its row is the successor of the singleton defect row. Equivalently, by corollary 4.290, the predecessor row is a removable-corner row of ν_t. Since λ_t and ν_t are known, the row of the unique box in λ_t/ν_t is known, and hence so is the removable row required in item (3) of proposition 4.297.

Thus the hypotheses of proposition 4.297 are satisfied: the anchor blocks are item (1), the padded carrier endpoints are recovered from item (2) and e_t, and the exceptional removable rows are recovered as above. That proposition recovers the original height halo-zone subpath. □

Proposition 4.311 (Pair-endpoint padded carriers give endpoint-compatible placement)**.** *Work in the setting of proposition 4.278. For each deleted singleton* u_t, *with* $s_t = j - u_t + 1$, *let*

$$\begin{matrix} \nu_t & \lambda_t \\ \rho_t & \mu_t \end{matrix}$$

be the corner labels of E_{s_t,s_t+1} *and let* e_t *be its cell entry. Assume that the padded carrier has the pair endpoints*

$$\operatorname{pad}_{e_t}(\nu_t) = \rho_{2u_t-4}, \qquad \mu_t = \rho_{2u_t}.$$

If the two-bit split of the branch word satisfies $\beta_t = \kappa(e_t)$ *for every* t, *then the original height halo-zone subpath is recovered from the resulting replacement window,* β_γ, *the zone endpoint pair, and* Z_γ.

Proof. Define the candidate blocks C_r as in proposition 4.278. For anchors $r \in A_\gamma$, one has $C_r = B_r(P)$, so its endpoints are

$$\rho_{2r-2}, \qquad \rho_{2r}.$$

For the paired slot $r = u_t - 1$, the two displayed hypotheses say that C_r has endpoints

$$\rho_{2u_t-4}, \qquad \rho_{2u_t},$$

namely the endpoints of the original two-block interval $\{u_t - 1, u_t\}$.

By corollary 4.323, the zone indices are partitioned into the anchor singleton blocks $r \in A_\gamma$ and the adjacent pairs $\{u_t - 1, u_t\}$. Therefore the candidate blocks, ordered by their retained slots $r \in Z_\gamma \setminus I$, cover consecutive original index intervals whose union is Z_γ. Adjacent candidate blocks have matching endpoints because the right endpoint of one covered interval is the left endpoint of the next covered interval in the original path. The first and last candidate endpoints are exactly the zone endpoints $\rho_{2a_\gamma-2}$ and ρ_{2b_γ}.

Thus the endpoint equalities in lemma 4.277 hold. Proposition 4.278 then gives the claimed recovery of the original height halo-zone subpath. □

Corollary 4.312 (Pair-endpoint padding is not source-automatic)**.** *The pair-endpoint identities assumed in proposition 4.311 do not follow from the Step–3 one-cell boundary convention alone. In particular, the bottom-padded carrier valley cannot be declared to span the adjacent original two-block interval without an additional endpoint splice or an additional restriction beyond the ambient Step–3 construction.*

Proof. Specialize the Step–3 construction to content (2^2). The Step–2 matrix has zero diagonal and row sums equal to 2 by lemma 4.128, so its unique off-diagonal entry is 2. By lemma 4.127, the unique Step–3 carrier $E_{1,2}$ therefore has cell entry $e = 2$.

The full Step–3 left and bottom boundaries are labelled by empty partitions in Krattenthaler's construction, as used in lemma 4.126. Hence the one-cell carrier corners before applying the forward rule satisfy

$$\rho = \mu = \nu = \varnothing.$$

Running the carry formula in the proof of lemma 4.122 with $m = 2$ and these empty boundary partitions gives $b_1 = b_2 = 1$ and $b_i = 0$ for $i \geq 3$, hence the northeast corner is

$$\lambda = (1, 1).$$

Thus

$$\mathrm{pad}_e(\nu) = \mathrm{pad}_2(\varnothing) = (1, 1).$$

On the other hand, lemma 4.118 gives

$$\rho_0 = \rho_4 = \varnothing$$

for the full $m = 0$, content-(2^2) boundary path. Under the label-to-block convention, the adjacent label pair $(2, 1)$ is the full two-block interval, so the left endpoint required by the identity in proposition 4.311 is $\rho_0 = \varnothing$. Therefore

$$\mathrm{pad}_e(\nu) = (1, 1) \neq \varnothing = \rho_0.$$

This disproves only the source-automatic discharge of the pair-endpoint hypothesis; the conditional placement implication of proposition 4.311 remains valid. □

Corollary 4.313 (Positive height zones do not force zero carrier padding)**.** *The positive-height halo hypotheses do not force the two-label Step–3 carrier entry e_t to vanish. Moreover, nonzero carrier entries in one positive-height zone need not inject into the surplus-anchor set A_γ. Consequently, the replacement positive-height placement cannot be completed by proving that all positive-height padded carrier windows have zero padding, nor by assigning an independent surplus-anchor buffer to each nonzero padded carrier.*

Proof. Consider the semistandard tableau of content (2^5)

$$T = \begin{array}{cccc} 1 & 1 & 2 & 4 \\ 2 & 3 & 3 & 5 \\ 4 & & & \\ 5 & & & \end{array}.$$

Its column lengths are $4, 2, 2, 2$, so the shape is column-even. For $q = 2$ the height witness set is

$$H_2(T) = \{T(1, 1), T(3, 1)\} = \{1, 4\}.$$

Under the block convention $s \mapsto j - s + 1$, the deleted block set is

$$I(H_2(T)) = \{5, 2\}.$$

The one-block halo is the single zone $Z = \{1, 2, 3, 4, 5\}$, with three retained indices and two deleted indices, hence this is a positive-height zone.

The first-variant Knuth matrix associated to this tableau is

$$M = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}.$$

It is symmetric, has zero diagonal, and every row sum is 2. The row-word insertion check for the nonzero biletters

$$(1,2),(1,3),(2,1),(2,5),(3,1),(3,4),(4,3),(4,5),(5,2),(5,4)$$

gives insertion and recording tableaux both equal to the displayed T. Thus this is a valid Step–2 datum for the source construction of lemma 4.128.

For the positive-zone deleted witness label $s = 4$, the retained successor is $s + 1 = 5$. By lemma 4.127, the unique Step–3 cell entry of $E_{4,5}$ is the corresponding off-diagonal matrix entry

$$e = M_{4,5} = 1.$$

Thus a positive-height zone can have a genuinely nonzero padding parameter.

The same zone has deleted block list $u_1 = 2, u_2 = 5$. Its surplus-anchor set from corollary 4.322 is

$$A_\gamma = \{u_1 + 1\} = \{3\},$$

because $u_2 - u_1 = 3$ and $u_2 = j$. The other deleted witness label is $s = 1$, whose retained successor is 2, and the same matrix gives

$$M_{1,2} = 1.$$

Therefore both deleted singletons in this positive-height zone have nonzero adjacent carrier entry, while the zone has only one surplus anchor. No splice rule which requires a distinct surplus-anchor buffer for each nonzero carrier can cover this valid positive-height example. $\square$

Corollary 4.314 (Positive height zones may have double successor carriers)**.** *The positive-height carrier splice cannot be reduced to the case $e_t \in \{0,1\}$, nor to smoothing each deleted witness through two distinct retained graph neighbours. There is a column-even content-(2^7) tableau*

$$T = \begin{matrix} 1 & 1 & 2 & 2 \\ 3 & 3 & 4 & 7 \\ 4 & 5 & & \\ 5 & 6 & & \\ 6 & & & \\ 7 & & & \end{matrix}$$

whose height witness set for $q = 2$ is $H_2(T) = \{1,4\}$, whose canonical halo zone

$$Z = \{3,4,5,6,7\}$$

has positive surplus, and whose deleted witness label $s = 4$ has adjacent Step–3 carrier entry

$$e = M_{4,5} = 2.$$

Proof. The rows of T are weakly increasing and the columns are strictly increasing. Its row lengths are

$$(4,4,2,2,1,1),$$

so its column lengths are $(6,4,2,2)$, all even. Each label $1,\ldots,7$ appears exactly twice. The first-column entries in rows 1 and 3 are 1 and 4, so $H_2(T) = \{1,4\}$.

The first-variant Knuth matrix associated to this tableau is

$$M = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

It is symmetric, has zero diagonal, and every row sum is 2. The row-word insertion check for the nonzero biletters

$$(1,3),(1,6),(2,7),(2,7),(3,1),(3,6),(4,5),(4,5),$$
$$(5,4),(5,4),(6,1),(6,3),(7,2),(7,2)$$

gives insertion and recording tableaux both equal to the displayed T. Thus this is a valid Step–2 datum for the source construction of lemma 4.128.

Under the block convention $s \mapsto j - s + 1$, the witness set $\{1,4\}$ gives deleted block set

$$I = \{7,4\}.$$

The one-block halo of I is the single zone

$$Z = \{3,4,5,6,7\}.$$

It contains two deleted indices and three retained indices, so the zone has positive surplus. For the deleted block $u = 4$, the corresponding witness label is $s = 7 - u + 1 = 4$, and the retained successor label is 5. By lemma 4.127, the unique Step–3 cell entry of the adjacent two-label carrier $E_{4,5}$ is the off-diagonal matrix entry $M_{4,5} = 2$. Hence an actual positive-height zone can require the full two-unit padding case. □

Corollary 4.315 (Collar-only normalization is not universal)**.** *In the ambient growth-path category, the merged positive-height window in proposition 4.317 cannot be supplied by leaving the retained left collar unchanged and then using only the retained right collar, the outer endpoint pair, the known singleton position, and a two-bit word to recover the deleted singleton block. Equivalently, the actual merged window must carry deleted-sensitive target data, such as a content-normalized carrier-boundary readout, or use further fixed-height restrictions beyond the ambient two-step axioms.*

Proof. Let

$$\mu = (5,4,3,2,1),$$

and fix two removable corners c_L and c_R of μ. For every removable corner c of μ, form the three-block segment

$$U^0 = \mu, \quad U^1 = \mu \setminus \{c_L\}, \quad U^2 = \mu,$$
$$U^3 = \mu \setminus \{c\}, \quad U^4 = \mu,$$
$$U^5 = \mu \setminus \{c_R\}, \quad U^6 = \mu.$$

Each consecutive triple

$$(U^0, U^1, U^2), \qquad (U^2, U^3, U^4), \qquad (U^4, U^5, U^6)$$

has vertical-strip adjacent differences and content weight 2, because each middle partition is obtained from μ by deleting one removable corner. Thus these are valid ambient left-collar, deleted-singleton, and right-collar blocks.

The staircase partition μ has five removable corners, so the choice of c gives five distinct deleted singleton blocks. The retained left collar, the retained right collar, the outer endpoints $(U^0, U^6) = (\mu, \mu)$, and the known middle position are the same for all five choices. A two-bit word has only four values. Therefore no decoder using only those data and a two-bit word can recover the middle block for all five segments. □

Corollary 4.316 (Right collars are parser-local after left-collar readout)**.** *Work in the height setting of proposition 4.532. Suppose that the parser decomposes V_γ^* into unchanged anchor blocks indexed by A_γ and merged windows M_t indexed by the increasing deleted singletons*

$$Z_\gamma \cap I = \{u_1 < \cdots < u_m\}.$$

Assume that each M_t determines the original retained left-collar block

$$B_{u_t-1}(P) = (\rho_{2u_t-4}, \rho_{2u_t-3}, \rho_{2u_t-2}).$$

Then, for each t, the full parsed zone data determine the retained right-collar block $B_{u_t+1}(P)$ whenever $u_t < j$; if $u_t = j$, the right side is the endpoint ρ_{2j}.

Proof. By corollary 4.264, if $u_t < j$ then $u_t + 1$ is a retained index in Z_γ. If $u_t + 1 \in A_\gamma$, the parser returns $B_{u_t+1}(P)$ as an unchanged anchor block. If $u_t + 1 \notin A_\gamma$, then corollary 4.323 gives

$$u_t + 1 = u_k - 1$$

for a unique k. Since the deleted singleton list is increasing, this forces $k = t + 1$ and $u_{t+1} = u_t + 2$. The assumed left-collar readout from M_{t+1} therefore determines

$$B_{u_{t+1}-1}(P) = B_{u_t+1}(P).$$

Thus the right collar is determined in every nonterminal case. If $u_t = j$, there is no right-collar block and the right endpoint is the global terminal partition ρ_{2j}, which is part of the endpoint data for the terminal zone. □

Proposition 4.317 (Coupled collar carriers recover positive height zones)**.** *Work in the height setting of proposition 4.532. Suppose the parser decomposes V_γ^* into unchanged anchor blocks and merged windows $M_1, \ldots, M_m$ for the increasing deleted singleton list*

$$Z_\gamma \cap I = \{u_1 < \cdots < u_m\}.$$

Fix an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Assume the following.

1. *Each M_t determines the retained left-collar block $B_{u_t-1}(P)$.*
2. *After the right collars are recovered by corollary 4.316, the parsed zone data*

$$S, \quad Z_\gamma, \quad V_\gamma^*, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}, \quad t$$

   determine the left/bottom boundary of the two-label carrier E_{s_t,s_t+1}, where $s_t = j - u_t + 1$.
3. *If e_t is the unique cell entry of E_{s_t,s_t+1}, then the deterministic two-bit split of β_γ satisfies $\beta_t = \kappa(e_t)$ for every t.*

Then the original height halo-zone subpath is recovered from V_γ^*, β_γ, the zone endpoint pair, and Z_γ.

Proof. For each t, item (2) determines the left/bottom boundary of E_{s_t,s_t+1}. By lemma 4.268, its unique cell entry lies in $\{0, 1, 2\}$, and item (3) recovers that entry from β_t. Hence corollary 4.187, applied with $a = s_t$ and $b = s_t + 1$, recovers the top/right boundary segment

$$\rho_{2u_t-4}, \rho_{2u_t-3}, \rho_{2u_t-2}, \rho_{2u_t-1}, \rho_{2u_t}.$$

This is the original two-block segment on block indices $u_t - 1, u_t$, by corollary 4.265.

The anchor blocks indexed by A_γ are recovered unchanged by the parser. By corollary 4.323, the blocks in Z_γ are partitioned into those anchor singleton blocks and the pairs $\{u_t - 1, u_t\}$. Interleaving the recovered anchor blocks and recovered two-block segments in increasing block order reconstructs the whole subpath

$$\rho_{2a_\gamma-2}, \rho_{2a_\gamma-1}, \ldots, \rho_{2b_\gamma}.$$

□

Corollary 4.318 (Height halo zones have singleton nonnegative packets)**.** *Let T be a semistandard tableau of content (2^j) with $\ell(\lambda(T)) \geq 2q$, set $S = H_q(T)$, put $I = I(S) = \{j - s + 1 : s \in S\}$, and assume $j \geq 2q$. Let $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$ be one canonical halo zone of lemma 4.254, and write the maximal consecutive components of $Z_\gamma \cap I$ as*

$$K_t = \{u_t, \dots, v_t\} \qquad (1 \leq t \leq m).$$

Then every K_t is a singleton. Moreover the local balance

$$\Delta_\gamma = |Z_\gamma \setminus I| - |Z_\gamma \cap I|$$

is nonnegative.

Proof. If some K_t had more than one element, then it would contain two consecutive block indices r and $r+1$. The corresponding tableau labels $j - r + 1$ and $j - r$ would then be two consecutive elements of $S = H_q(T)$, contradicting lemma 4.260. Thus every K_t is a singleton.

Use the notation of lemma 4.259. Since all components are singletons, $h_t = 1$ for all t. If $\epsilon_L + \epsilon_R \geq 1$, then every internal gap has $g_t \geq 1$, so

$$\Delta_\gamma = \epsilon_L + \epsilon_R + \sum_{t=1}^{m-1} g_t - m \geq \epsilon_L + \epsilon_R + (m-1) - m \geq 0.$$

It remains to consider $\epsilon_L = \epsilon_R = 0$. Then the first deleted block in the zone is 1 and the last deleted block in the zone is j, so this zone contains all elements of I. Hence $|Z_\gamma \cap I| = q$ and $Z_\gamma = \{1, \dots, j\}$. Therefore

$$\Delta_\gamma = (j - q) - q = j - 2q \geq 0$$

by the hypothesis $j \geq 2q$. □

Corollary 4.319 (Height zero-surplus zones are right-boundary alternating)**.** *In the setting of corollary 4.318, write the singleton components of $Z_\gamma \cap I$ as*

$$K_t = \{u_t\} \qquad (1 \leq t \leq m), \qquad u_1 < \cdots < u_m.$$

Then $u_1 > 1$. Moreover, with

$$\epsilon_R = \begin{cases} 1, & u_m < j, \\ 0, & u_m = j, \end{cases}$$

one has the exact surplus formula

$$|Z_\gamma \setminus I| - |Z_\gamma \cap I| = \epsilon_R + \sum_{t=1}^{m-1} (u_{t+1} - u_t - 2).$$

Consequently the height zone has zero surplus if and only if $u_m = j$ and

$$u_{t+1} = u_t + 2 \qquad (1 \leq t < m).$$

Proof. Let $t_q = T(2q-1, 1)$ be the largest height witness label. Since $\ell(\lambda(T)) \geq 2q$, the first-column entry $T(2q, 1)$ exists. Column strictness gives

$$t_q < T(2q, 1) \leq j,$$

so $t_q < j$. Every label in $S = H_q(T)$ is at most t_q, and therefore every block index $j - s + 1$ in I is at least $j - t_q + 1 > 1$. In particular $u_1 > 1$.

By corollary 4.318, all deleted components in the zone are singletons. Thus, in the notation of lemma 4.259, one has $h_t = 1$ for all t and $\epsilon_L = 1$ because $u_1 > 1$. Also $g_t = u_{t+1} - u_t - 1$. Substituting these values into the balance formula gives

$$|Z_\gamma \setminus I| - |Z_\gamma \cap I| = 1 + \epsilon_R + \sum_{t=1}^{m-1} (u_{t+1} - u_t - 1) - m = \epsilon_R + \sum_{t=1}^{m-1} (u_{t+1} - u_t - 2).$$

The separated-label result lemma 4.260 gives $u_{t+1} \geq u_t + 2$, so every summand is nonnegative. Hence the surplus is zero exactly when $\epsilon_R = 0$ and every summand vanishes, which is the displayed right-boundary alternating condition. □

Corollary 4.320 (Height zero-surplus zones are alternating label prefixes). *In the setting of corollary 4.319, suppose the height zone has zero surplus and put $m = |Z_\gamma \cap I|$. Then*

$$Z_\gamma = \{j - 2m + 1, \dots, j\},$$

$$Z_\gamma \cap I = \{j - 2m + 2,\ j - 2m + 4, \dots, j\}, \qquad Z_\gamma \setminus I = \{j - 2m + 1,\ j - 2m + 3, \dots, j - 1\}.$$

Under the label-to-block correspondence $r \mapsto j - r + 1$, the deleted labels in this zone are exactly

$$\{1, 3, \dots, 2m - 1\},$$

and the retained labels in this zone are exactly

$$\{2, 4, \dots, 2m\}.$$

Thus every tight residual height zone is the terminal-block encoding of an odd alternating label prefix.

Proof. By corollary 4.319, zero surplus is equivalent to $u_m = j$ and $u_{t+1} = u_t + 2$ for $1 \leq t < m$. Hence

$$u_t = j - 2(m - t) \qquad (1 \leq t \leq m).$$

The deleted block set in the zone is therefore

$$Z_\gamma \cap I = \{u_1, \dots, u_m\} = \{j - 2m + 2,\ j - 2m + 4, \dots, j\}.$$

Since $u_1 > 1$ and $u_m = j$, the one-block halo component has left endpoint $u_1 - 1 = j - 2m + 1$ and right endpoint j, proving the formula for Z_γ. The retained blocks are the complementary alternating indices in this interval, namely

$$Z_\gamma \setminus I = \{j - 2m + 1,\ j - 2m + 3, \dots, j - 1\}.$$

Finally, a block index r corresponds to label $j - r + 1$. Applying this map to the displayed deleted and retained block sets gives the two displayed label sets. □

Corollary 4.321 (Positive height halo zones contain a retained anchor). *In the setting of corollary 4.319, suppose*

$$|Z_\gamma \setminus I| - |Z_\gamma \cap I| > 0.$$

Then there is an index

$$r \in Z_\gamma \setminus I$$

which is adjacent to exactly one deleted singleton block in $Z_\gamma \cap I$: exactly one of $r - 1$ and $r + 1$ belongs to $Z_\gamma \cap I$. More precisely, either $u_m < j$ and one may take $r = u_m + 1$, or there is an index $t < m$ with $u_{t+1} = u_t + 3$ and one may take $r = u_t + 1$.

Proof. By corollary 4.319,

$$|Z_\gamma \setminus I| - |Z_\gamma \cap I| = \epsilon_R + \sum_{t=1}^{m-1} (u_{t+1} - u_t - 2),$$

and every summand on the right is nonnegative. If the left side is positive, then either $\epsilon_R = 1$ or some summand is positive.

If $\epsilon_R = 1$, then $u_m < j$ and the right halo index $r = u_m + 1$ lies in $Z_\gamma \setminus I$. It is adjacent to the deleted singleton u_m, and it is not adjacent to any later deleted singleton because u_m is the last deleted index in the zone.

Otherwise, for some $t < m$ one has $u_{t+1} - u_t - 2 > 0$. Since lemma 4.257 gives $u_{t+1} - u_t \in \{2, 3\}$ for singleton components in one halo zone, this means $u_{t+1} = u_t + 3$. The index $r = u_t + 1$ is a retained index in the gap between the two deleted singletons. It is adjacent to u_t and has distance two from u_{t+1}; all other deleted singletons are farther away in block order. Thus it is adjacent to exactly one deleted singleton in $Z_\gamma \cap I$. $\square$

Corollary 4.322 (Height surplus is carried by canonical right anchors)**.** *In the setting of corollary 4.319, define*

$$A_\gamma = \{u_m + 1 : \ u_m < j\} \cup \{u_t + 1 : \ 1 \le t < m, \ u_{t+1} = u_t + 3\}.$$

Then

$$A_\gamma \subseteq Z_\gamma \setminus I,$$

every element of A_γ is adjacent to exactly one deleted singleton in $Z_\gamma \cap I$, and

$$|A_\gamma| = |Z_\gamma \setminus I| - |Z_\gamma \cap I|.$$

Thus the positive surplus of a height halo zone is represented by a canonical ordered list of one-sided retained anchors.

Proof. If $u_m < j$, then the right endpoint of the one-block halo component is $u_m + 1$, so $u_m + 1 \in Z_\gamma$. It is not in I, since otherwise it would be consecutive to the deleted singleton u_m. It is adjacent to u_m and to no later deleted singleton.

If $u_{t+1} = u_t + 3$, then the index $u_t + 1$ lies in the retained gap between the two deleted singletons u_t and u_{t+1}. It belongs to $Z_\gamma \setminus I$, is adjacent to u_t, and has distance two from u_{t+1}; the remaining deleted singletons are farther away. Hence every element of A_γ is a retained index adjacent to exactly one deleted singleton. The displayed set is canonical because it is defined from the increasing list of deleted singleton indices in the canonical halo zone.

It remains to count it. The first set in the union contributes ϵ_R, and the second contributes one element exactly for each gap with $u_{t+1} - u_t = 3$. By lemma 4.257, all gaps between singleton components inside a canonical halo zone have size 2 or 3. Therefore

$$|A_\gamma| = \epsilon_R + \sum_{t=1}^{m-1} (u_{t+1} - u_t - 2).$$

The right side is the surplus $|Z_\gamma \setminus I| - |Z_\gamma \cap I|$ by corollary 4.319. $\square$

Corollary 4.323 (Removing height surplus anchors leaves left-neighbour pairs)**.** *In the setting of corollary 4.322, one has*

$$(Z_\gamma \setminus I) \setminus A_\gamma = \{u_t - 1 : \ 1 \le t \le m\}.$$

Consequently, after deleting the canonical surplus anchors, the retained indices in the height zone are in canonical bijection with the deleted singletons by

$$u_t \longmapsto u_t - 1.$$

Proof. By corollary 4.319, one has $u_1 > 1$, so $u_1 - 1$ is the retained prefix index of the zone. This is the element $u_t - 1$ for $t = 1$.

Between consecutive deleted singletons $u_t < u_{t+1}$, the short-gap lemma lemma 4.257 gives $u_{t+1} - u_t \in \{2, 3\}$. If $u_{t+1} = u_t + 2$, the only retained index in the gap is

$$u_t + 1 = u_{t+1} - 1,$$

and it is not in A_γ. If $u_{t+1} = u_t + 3$, the retained gap indices are $u_t + 1$ and $u_t + 2$; the first belongs to A_γ by definition, while the second is $u_{t+1} - 1$ and is not in A_γ.

Finally, if $u_m < j$, the retained suffix index $u_m + 1$ belongs to A_γ; if $u_m = j$, there is no suffix retained index. Thus the retained indices not removed by A_γ are exactly the indices $u_t - 1$ for $1 \le t \le m$. These indices are distinct and each lies immediately to the left of the corresponding deleted singleton u_t, giving the stated bijection. $\square$

Corollary 4.324 (Only the terminal height halo zone can have zero surplus)**.** *In the setting of corollary 4.318, order the canonical halo zones increasingly by their block intervals. At most one height halo zone has zero surplus*

$$|Z_\gamma \setminus I| - |Z_\gamma \cap I| = 0.$$

If such a zone exists, it is the last zone, its right endpoint is j, and it has the alternating prefix form described in corollary 4.320. Every height halo zone strictly to its left has positive surplus.

Proof. Let Z_γ be a zero-surplus height halo zone, and write its singleton deleted components as in corollary 4.319. That corollary gives $u_m = j$. Therefore the canonical halo component containing this last deleted block has right endpoint j. Since the canonical halo zones are disjoint intervals ordered increasingly, a zone with right endpoint j is the last zone, and no other zone can also contain the block j. Hence there is at most one zero-surplus zone, and if it exists it is terminal.

Corollary 4.320 gives the displayed alternating prefix form for this terminal zone. Finally, corollary 4.318 says that every height halo zone has nonnegative surplus. Any zone strictly to the left of the terminal zero-surplus zone is not itself zero-surplus by uniqueness, and hence has positive surplus. ☐

Corollary 4.325 (Nonterminal height halo zones have retained anchors)**.** *In the setting of corollary 4.318, order the canonical halo zones increasingly by their block intervals. Every height halo zone except possibly the unique terminal zero-surplus zone of corollary 4.324 contains a retained anchor as in corollary 4.321.*

Proof. By corollary 4.324, every height halo zone which is not the unique terminal zero-surplus zone has positive surplus. The anchor assertion is then exactly corollary 4.321. ☐

Corollary 4.326 (Non-full tight height prefixes have a visible separator)**.** *In the setting of corollary 4.320, suppose $j > 2m$. Then*

$$2m + 1 \notin S, \qquad 2m + 2 \notin S \quad \text{if } 2m + 2 \leq j.$$

Moreover the block index $j - 2m$, corresponding to label $2m + 1$, lies outside the full one-block halo $H(S)$. Hence it belongs to the unchanged complement R of lemma 4.254, and its right endpoint is the left endpoint of the tight zone:

$$\rho_{2(j-2m)} = \rho_{2(j-2m+1)-2}.$$

Proof. By corollary 4.320, the tight zone is

$$Z_\gamma = \{j - 2m + 1, \dots, j\}.$$

If $2m+1 \in S$, then the corresponding deleted block is $j-2m$. Its one-block halo contains $j-2m+1$, so the maximal halo component containing Z_γ would extend left to include $j - 2m$, contradicting the displayed left endpoint of Z_γ.

If $2m + 2 \in S$, then the corresponding deleted block is $j - 2m - 1$. Its one-block halo contains $j - 2m$, and the tight zone contains $j - 2m + 1$. These two halo pieces are consecutive, so again they would form one larger halo component, contradicting the displayed left endpoint of Z_γ.

Now let $s \in S$. If $s \leq 2m - 1$, then its deleted block belongs to the tight zone and its halo is contained in Z_γ. If $s \geq 2m + 3$, then its deleted block index is at most $j - 2m - 2$, so its one-block halo ends at most at $j - 2m - 1$. The two excluded cases $s = 2m + 1, 2m + 2$ have just been ruled out. Thus no deleted block has one-block halo containing $j - 2m$, and so $j - 2m \notin H(S)$.

The complement assertion follows from the definition of R in lemma 4.254. Finally, the right endpoint of block $j - 2m$ is

$$\rho_{2(j-2m)},$$

while the left endpoint of the zone whose first block is $j - 2m + 1$ is

$$\rho_{2(j-2m+1)-2}.$$

These are the same index. ☐

Corollary 4.327 (Non-full tight height zone endpoints are parser-visible). *Work in the setting of corollary 4.326, and put*

$$a_\gamma = j - 2m + 1, \qquad Z_\gamma = \{a_\gamma, \dots, j\}.$$

Let

$$P = (\rho_0, \rho_1, \dots, \rho_{2j})$$

be a column-even two-step growth path for the tableau. In any replacement-zone parser that returns unchanged blocks for indices in the complement R of the canonical halo zones, the endpoint pair required for the right-boundary zone Z_γ,

$$\rho_{2a_\gamma - 2}, \qquad \rho_{2j},$$

is determined by the parsed unchanged separator block $B_{a_\gamma - 1}(P)$ and the known terminal endpoint $\rho_{2j} = \varnothing$.

Proof. Corollary 4.326 says that $a_\gamma - 1 = j - 2m$ belongs to the unchanged complement R. Hence the parser returns

$$B_{a_\gamma - 1}(P) = (\rho_{2a_\gamma - 4}, \rho_{2a_\gamma - 3}, \rho_{2a_\gamma - 2}).$$

Its right endpoint is exactly the left endpoint $\rho_{2a_\gamma - 2}$ of the zone. The right endpoint of this right-boundary zone is the terminal endpoint of the full two-step growth path, which is $\rho_{2j} = \varnothing$ by lemma 4.118. Thus both endpoints are determined from the parsed separator block and the fixed terminal endpoint. ☐

Corollary 4.328 (Column-even endpoint gate for tight height prefixes). *In the setting of corollary 4.320, let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$

be the Pieri two-strip path of T. Then

$$\ell(\lambda^{(t)}) = t \qquad (0 \le t \le 2m - 1),$$

and

$$\ell(\lambda^{(2r)}) = 2r \qquad (1 \le r < m).$$

Moreover

$$\ell(\lambda^{(2m)}) \in \{2m - 1, 2m\}.$$

If $m \ge 2$ and $\lambda^{(2m)}$ is column-even, then the prefix

$$\lambda^{(0)} \subset \cdots \subset \lambda^{(2m)}$$

is an odd alternating height full-zone path, with

$$C^h_m(\lambda^{(0)}, \dots, \lambda^{(2m)}) = \{1, 3, \dots, 2m - 1\},$$

and is in the domain of proposition 4.108. If $m = 1$, or if $\lambda^{(2m)}$ is not column-even, then the closed full-zone alternating decoder does not by itself supply this local zone. In the non-column-even case either column 1 has odd length $2m - 1$, or at least one off-first-column column of $\lambda^{(2m)}$ has odd length. Any target-size decoder for this tight zone must therefore use endpoint or continuation data beyond the closed full-zone alternating decoder.

Proof. By corollary 4.320, the first m height witness labels are

$$t_r = 2r - 1 \qquad (1 \le r \le m).$$

For each such r, lemma 4.100 gives

$$\ell(\lambda^{(2r-2)}) = 2r - 2, \qquad \ell(\lambda^{(2r-1)}) = 2r - 1.$$

This proves $\ell(\lambda^{(t)}) = t$ for all odd $t \le 2m-1$ and all even $t \le 2m-2$. For $1 \le r < m$, applying the same display to $r+1$ gives

$$\ell(\lambda^{(2r)}) = 2r.$$

Finally, the horizontal strip at time $2m$ creates at most one new row from length $2m-1$, so

$$\ell(\lambda^{(2m)}) \in \{2m-1, 2m\}.$$

If $m \ge 2$ and $\lambda^{(2m)}$ is column-even, its first column has even nonzero length. Since that length is one of $2m-1$ and $2m$, it must be $2m$. Therefore every step in the prefix creates exactly one row. The height statistic

$$\min(m, \lceil \ell(\lambda^{(t)})/2 \rceil)$$

then increases exactly at the odd times $1, 3, \ldots, 2m-1$. This proves the displayed crossing set, and the column-even endpoint hypothesis makes the prefix a column-even Pieri path of length $2m$. Hence it is exactly an odd alternating height full-zone path, so proposition 4.108 applies.

If $m = 1$, the proposition just cited is not available in this local size, so this singleton tight zone remains part of the endpoint/continuation local decoder. If $\lambda^{(2m)}$ is not column-even, then some column of $\lambda^{(2m)}$ has odd length. Column 1 has length either $2m-1$ or $2m$, so the odd column is either column 1 with length $2m-1$, or it is an off-first-column odd column. In either case the prefix endpoint is not a column-even endpoint of the type used in the closed full-zone alternating decoder, so a decoder for the tight zone must incorporate the endpoint or the later continuation parity data. □

Lemma 4.329 (The first semistandard label is forced)**.** *Let T be a semistandard tableau of content (2^j) with $j \ge 1$. Then the two boxes carrying label 1 are exactly*

$$(1,1) \qquad \textit{and} \qquad (1,2).$$

Equivalently, in the Pieri path of lemma 4.97, the first strip is the forced horizontal domino

$$\lambda^{(1)} = (2).$$

Proof. No box carrying label 1 can lie below the first row: if such a box were in row $r > 1$, then the box immediately above it would have a strictly smaller label by column strictness, impossible. Hence both copies of label 1 lie in the first row. Since rows are weakly increasing, the boxes carrying the smallest label are the leftmost boxes of that row. There are exactly two copies, so they occupy columns 1 and 2.

The Pieri-path statement is the same assertion under lemma 4.97: the skew difference $\lambda^{(1)}/\lambda^{(0)}$ consists exactly of the two boxes carrying label 1. □

Corollary 4.330 (Singleton tight height zones have forced deleted strip)**.** *In the setting of corollary 4.328, suppose $m = 1$. Then the tight height zone is*

$$Z_\gamma = \{j-1, j\}, \qquad Z_\gamma \cap I = \{j\}, \qquad Z_\gamma \setminus I = \{j-1\},$$

and its deleted tableau label is 1. The deleted Pieri strip is the forced domino

$$\lambda^{(1)}/\lambda^{(0)} = \{(1,1), (1,2)\}.$$

Thus the singleton tight C-side local problem has no remaining choice about the deleted semistandard boxes; the remaining local data are the target-size merge with the retained label-2 block and the endpoint/continuation parity needed to recover the surrounding growth-path window.

Proof. The displayed formulas for Z_γ, $Z_\gamma \cap I$, and $Z_\gamma \setminus I$ are the case $m = 1$ of corollary 4.320. The same corollary says that the deleted label set in the zone is $\{1\}$ and that the retained label set is $\{2\}$. The assertion about the deleted strip is exactly lemma 4.329.

Finally, this zone has one retained block and one deleted block, but proposition 4.540 allows exactly $|Z_\gamma \setminus I| = 1$ replacement block for the zone. Therefore there is no separate positive replacement window for the deleted strip in addition to the retained label-2 block; the remaining task is precisely the stated target-size merge and endpoint/continuation recovery. □

Lemma 4.331 (Continuation parity equals the tight-prefix endpoint defect)**.** *In the setting of corollary 4.328, put*

$$O_{\mathrm{pre}} = \{c \geq 1 : (\lambda^{(2m)})'_c \equiv 1 \pmod 2\}$$

and

$$O_{\mathrm{cont}} = \{c \geq 1 : (\lambda^{(j)})'_c - (\lambda^{(2m)})'_c \equiv 1 \pmod 2\}.$$

Then

$$O_{\mathrm{cont}} = O_{\mathrm{pre}}.$$

In particular, if the tight prefix endpoint is not column-even, the remaining labels $2m+1, \dots, j$ must add an odd number of boxes exactly in the columns of O_{pre}. The set O_{pre} has even cardinality, and its adjacent increasing pairing is determined by the prefix endpoint $\lambda^{(2m)}$.

Proof. The final shape $\lambda^{(j)}$ is column-even, so

$$(\lambda^{(j)})'_c \equiv 0 \pmod 2 \qquad (c \geq 1).$$

Therefore

$$(\lambda^{(j)})'_c - (\lambda^{(2m)})'_c \equiv -(\lambda^{(2m)})'_c \equiv (\lambda^{(2m)})'_c \pmod 2,$$

which proves $O_{\mathrm{cont}} = O_{\mathrm{pre}}$.

If the prefix endpoint is not column-even, then O_{pre} is nonempty, and the equality just proved says exactly that the continuation adds odd column parity in those and only those columns. The size $|\lambda^{(2m)}| = 4m$ is even, so the sum of the column lengths of $\lambda^{(2m)}$ is even; hence the number of odd column lengths is even. Thus O_{pre} has even cardinality. Listing its columns increasingly and pairing adjacent entries is therefore a canonical pairing determined by $\lambda^{(2m)}$. □

Lemma 4.332 (Tight-prefix endpoint defects need not fit the local slots)**.** *There is a column-even Pieri two-strip path*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(7)}$$

with $C_2^h(\lambda^\bullet) = \{1,3\}$ such that the tight height zone for $S = \{1,3\}$ has $m = 2$ deleted singleton blocks, but the endpoint defect set of the prefix through label 4 has three paired defects.

Consequently, the non-column-even tight-prefix continuation cannot in general be repaired by assigning every endpoint-defect pair to one of the m local frontier slots of the tight zone. A valid target-size decoder must either use the continuation jointly, discard/cancel some endpoint defects before the local marking step, or encode a smaller selected subset through the tight-zone branch word.

Proof. Consider the path

$$\begin{aligned}
\lambda^{(0)} &= \varnothing,\\
\lambda^{(1)} &= (2),\\
\lambda^{(2)} &= (3,1),\\
\lambda^{(3)} &= (4,1,1),\\
\lambda^{(4)} &= (6,1,1),\\
\lambda^{(5)} &= (6,3,1),\\
\lambda^{(6)} &= (6,5,1),\\
\lambda^{(7)} &= (6,6,1,1).
\end{aligned}$$

Each successive difference is a horizontal strip of size 2: the added boxes are, respectively,

$$(1,1),(1,2);\quad (2,1),(1,3);\quad (3,1),(1,4);\quad (1,5),(1,6);$$
$$(2,2),(2,3);\quad (2,4),(2,5);\quad (2,6),(4,1).$$

The terminal conjugate partition is

$$(\lambda^{(7)})' = (4,2,2,2,2,2),$$

so the terminal shape is column-even.

The lengths of the successive shapes are

$$0,1,2,3,3,3,3,4.$$

Thus

$$\min(2,\lceil \ell(\lambda^{(t)})/2\rceil)$$

increases exactly at $t=1$ and $t=3$, so $C_2^h(\lambda^\bullet)=\{1,3\}$. For $j=7$ and $S=\{1,3\}$, the label-to-block map gives

$$I(S)=\{7,5\}.$$

The one-block halo is the single zone $\{4,5,6,7\}$, with deleted blocks $\{5,7\}$ and retained blocks $\{4,6\}$; hence this is the tight height zone with $m=2$.

At the prefix endpoint one has

$$\lambda^{(4)}=(6,1,1),\qquad (\lambda^{(4)})'=(3,1,1,1,1,1).$$

Therefore

$$O_{\text{pre}}=\{1,2,3,4,5,6\},$$

which has three adjacent pairs. Since the tight zone has only $m=2$ local frontier slots, these three endpoint-defect pairs cannot all be assigned to distinct local slots of that tight zone. ☐

Lemma 4.333 (Finite Hall injection criterion)**.** *Let A and B be finite sets, and let $N(a)\subseteq B$ be a neighbour set for each $a\in A$. If*

$$\left|\bigcup_{a\in X} N(a)\right| \ge |X| \qquad \text{for every } X\subseteq A,$$

then there is an injection $\phi: A\to B$ with $\phi(a)\in N(a)$ for every $a\in A$.

Proof. We prove the assertion by induction on $|A|$. The cases $|A|=0,1$ are immediate. Suppose first that every nonempty proper subset $X\subsetneq A$ satisfies

$$\left|\bigcup_{a\in X} N(a)\right| \ge |X|+1.$$

Choose $a_0\in A$ and $b_0\in N(a_0)$. For any $Y\subseteq A\setminus\{a_0\}$, the neighbour set after removing b_0 has size at least $|Y|$, because the original neighbour set of Y has size at least $|Y|+1$ when Y is nonempty. The induction hypothesis gives an injection of $A\setminus\{a_0\}$ into $B\setminus\{b_0\}$ respecting neighbours. Sending a_0 to b_0 completes the matching.

It remains to consider the case where some nonempty proper subset $X\subsetneq A$ has equality:

$$\left|\bigcup_{a\in X} N(a)\right| = |X|.$$

Put $B_X=\bigcup_{a\in X} N(a)$. By induction, X injects into B_X. For $Y\subseteq A\setminus X$,

$$\left|\bigcup_{a\in Y} N(a)\setminus B_X\right| = \left|B_X\cup\bigcup_{a\in Y} N(a)\right| - |B_X| \ge |X\cup Y| - |X| = |Y|.$$

Thus the induction hypothesis also matches $A \setminus X$ into $B \setminus B_X$. Combining the two injections proves the claim. □

Lemma 4.334 (Continuation labels cover endpoint defects)**.** *In the setting of lemma 4.331, write*

$$O_{\mathrm{pre}} = \{o_1 < o_2 < \cdots < o_{2r}\}$$

and pair adjacent entries

$$P_a = \{o_{2a-1}, o_{2a}\} \qquad (1 \le a \le r).$$

For each continuation label $t > 2m$, let

$$C_t = \{c \ge 1 : (\lambda^{(t)})'_c - (\lambda^{(t-1)})'_c = 1\}$$

be the set of columns hit by the horizontal two-strip of label t. Then there is an injection

$$\phi : \{1, \dots, r\} \hookrightarrow \{2m+1, \dots, j\}$$

such that

$$C_{\phi(a)} \cap P_a \ne \varnothing \qquad (1 \le a \le r).$$

Thus the paired endpoint defects of a tight prefix can be charged to distinct continuation labels, although by lemma 4.332 they cannot in general all be charged to the tight zone's own frontier slots.

Proof. For a subset $A_0 \subseteq \{1, \dots, r\}$, put

$$P(A_0) = \bigcup_{a \in A_0} P_a.$$

Every column in $P(A_0)$ belongs to O_{pre}. By lemma 4.331, the continuation from $2m+1$ through j adds an odd, hence positive, number of boxes in each such column. Therefore the total number of incidences

$$(t, c) \quad \text{with} \quad t > 2m,\ c \in C_t \cap P(A_0)$$

is at least $|P(A_0)| = 2|A_0|$.

Each continuation label t contributes a horizontal two-strip, so $|C_t| = 2$ and it contributes at most two incidences to $P(A_0)$. Hence the number of continuation labels with $C_t \cap P(A_0) \ne \varnothing$ is at least $|A_0|$. This is exactly the Hall condition for the paired-defect set, with neighbours those continuation labels whose strips hit the pair. Applying lemma 4.333 gives the desired injection. □

Lemma 4.335 (The continuation Hall charge can be chosen canonically)**.** *In the setting of lemma 4.334, order the defect pairs by $1 < \cdots < r$ and the continuation labels by their usual order. For each pair set*

$$N(a) = \{t \in \{2m+1, \dots, j\} : C_t \cap P_a \ne \varnothing\}.$$

Among all injections

$$\phi : \{1, \dots, r\} \hookrightarrow \{2m+1, \dots, j\} \qquad \text{with} \qquad \phi(a) \in N(a),$$

there is a lexicographically least word

$$(\phi(1), \dots, \phi(r)).$$

This canonical injection is a fixed function of the incidence family $(N(a))_{a=1}^r$.

Consequently, if the incidence family is recovered from parser-visible outside growth-diagram subfillings by corollary 4.120 or corollary 4.124, then the chosen Hall charge and the charged continuation labels are admissible auxiliary data for proposition 4.253; no additional branch word is needed to select the matching.

Proof. Lemma 4.334 proves that at least one such injection exists. Since both the defect-pair set and the continuation-label set are finite and linearly ordered, the set of admissible image words $(\phi(1), \dots, \phi(r))$ is a nonempty finite set in lexicographic order. It therefore has a unique least element. The admissible words are determined exactly by the neighbour sets $N(a)$ together with the injectivity condition, so this least word is a fixed function of the incidence family.

If the incidence family is recovered from parsed outside subfillings in either listed growth-diagram variant, then the same fixed lexicographic rule computes the chosen matching from the already recovered outside data. Thus the matching is an auxiliary datum of the kind allowed in proposition 4.253, rather than a new encoded choice. □

Lemma 4.336 (Continuation incidence is semistandard-local)**.** *Let T be a semistandard tableau of content (2^j), and let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$

be its Pieri two-strip path under lemma 4.97. For a label t, put

$$C_t = \{c \geq 1 : (\lambda^{(t)})'_c - (\lambda^{(t-1)})'_c = 1\}.$$

Then C_t is exactly the set of columns occupied by the two boxes of T carrying label t. Consequently, in the setting of lemma 4.334, the incidence relation

$$C_t \cap P_a \neq \varnothing$$

is determined by the positions of the two semistandard boxes carrying the continuation label t.

Proof. By lemma 4.97, the skew difference $\lambda^{(t)}/\lambda^{(t-1)}$ is precisely the set of boxes of T carrying label t. Passing to conjugates records column lengths. Hence $(\lambda^{(t)})'_c - (\lambda^{(t-1)})'_c = 1$ exactly when this skew difference contains a box in column c. Since the difference is a horizontal strip, there is at most one such box in each column, and the displayed set is exactly the two columns occupied by label t. The final assertion is the definition of the incidence relation in lemma 4.334 read with this identification. □

Lemma 4.337 (Continuation boxes determine the Pieri continuation)**.** *Let T be a semistandard tableau of content (2^j), and let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$

be its Pieri two-strip path. Fix $a < j$. If one is given $\lambda^{(a)}$ and, for each label $t > a$, the two boxes of T carrying label t, then the whole Pieri continuation segment

$$\lambda^{(a)} \subset \lambda^{(a+1)} \subset \cdots \subset \lambda^{(j)}$$

is determined. Explicitly,

$$\lambda^{(t)} = \lambda^{(a)} \cup \bigcup_{u=a+1}^{t} \{\textit{boxes of } T \textit{ carrying label } u\} \qquad (a < t \leq j).$$

Consequently, in the tight-height setting with $a = 2m$, recovery of the prefix endpoint $\lambda^{(2m)}$ together with the labelled continuation boxes recovers the Hall-charge auxiliary data of lemma 4.513.

Proof. By lemma 4.97, $\lambda^{(t)}$ is the shape formed by all boxes whose labels are at most t. The boxes whose labels are at most a form $\lambda^{(a)}$, and for $u > a$ the two boxes labelled u are exactly the skew difference $\lambda^{(u)}/\lambda^{(u-1)}$. Thus adding, in label order, the recovered two-box sets for $u = a+1, \dots, t$ to $\lambda^{(a)}$ gives the displayed formula for every $t > a$.

The final sentence is the case $a = 2m$ followed by lemma 4.513. □

Proposition 4.338 (Recovered continuation boxes supply the tight auxiliary data)**.** *Work in the non-full tight-height setting of corollary 4.327, and let*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$

be the Pieri path of the tableau. Suppose the parser data outside the tight right-boundary zone determine the prefix endpoint $\lambda^{(2m)}$ and, for each continuation label $t = 2m+1, \dots, j$, the two boxes of the tableau carrying label t. Then those outside data determine:

1. *the growth-path endpoint pair*
$$\rho_{2(j-2m+1)-2}, \qquad \rho_{2j}$$
of the tight right-boundary zone;
2. *the endpoint-defect pairs of $\lambda^{(2m)}$;*
3. *every incidence $C_t \cap P_a \neq \varnothing$ between a continuation label and an endpoint-defect pair; and*
4. *the canonical continuation Hall charge of lemma 4.335.*

Thus, once a right-boundary tight-zone inverse is written using only these data as its auxiliary input, the auxiliary input is of the kind allowed by proposition 4.253.

Proof. The endpoint pair in item (1) is determined by corollary 4.327: the parsed separator block gives the left endpoint and the terminal endpoint is $\rho_{2j} = \varnothing$.

By lemma 4.337, the recovered prefix endpoint and recovered labelled continuation boxes determine the whole Pieri continuation segment

$$\lambda^{(2m)} \subset \lambda^{(2m+1)} \subset \cdots \subset \lambda^{(j)}.$$

Then lemma 4.513 determines the endpoint-defect pairs, the continuation incidence relation, and the canonical Hall charge from that segment. These are precisely items (2)–(4).

All data listed above are fixed functions of the outside parser data under the stated recovery hypothesis. Therefore they may be used as the auxiliary datum $A(P)$ in proposition 4.253 without adding branch choices. □

Lemma 4.339 (Outside data recover the strict-left growth segment)**.** *Work in the non-full tight-height setting of corollary 4.327. Put*

$$a_\gamma = j - 2m + 1, \qquad Z_\gamma = \{a_\gamma, \dots, j\},$$

and order the canonical halo zones increasingly, with Z_γ the final right-boundary zone. In the outside datum $\mathcal{D}_{\text{out}}(P)$ of proposition 4.253, the unchanged complement blocks and the recovered earlier zone subpaths determine the whole initial growth segment

$$\rho_0, \rho_1, \dots, \rho_{2a_\gamma - 2} = \rho_0, \rho_1, \dots, \rho_{2(j-2m)}.$$

Under the label-to-block convention of lemma 4.118, this is exactly the two-step growth-block segment corresponding to continuation labels

$$2m+1, \dots, j.$$

Proof. Since the canonical halo zones are disjoint intervals and $Z_\gamma = \{a_\gamma, \dots, j\}$ is the final one, every block index $i < a_\gamma$ lies either in the unchanged complement R or in a canonical halo zone strictly to the left of Z_γ. The outside datum in proposition 4.253 contains the original two-step block $B_i(P)$ for every such $i \in R$, and it contains the whole recovered subpath for each earlier halo zone. Interleaving these blocks and subpaths in increasing block order therefore reconstructs the initial segment through the right endpoint of block $a_\gamma - 1$, namely

$$\rho_0, \rho_1, \dots, \rho_{2(a_\gamma - 1)} = \rho_0, \rho_1, \dots, \rho_{2a_\gamma - 2}.$$

The formula $a_\gamma = j - 2m + 1$ gives the displayed equality with $\rho_{2(j-2m)}$.

Finally, lemma 4.118 identifies tableau label ℓ with block $j-\ell+1$. As ℓ runs from $2m+1$ to j, this block index runs from $j-2m$ down to 1, exactly the block interval $\{1,\dots,j-2m\}$ reconstructed above. □

Proposition 4.340 (Strict-left growth bridge supplies tight auxiliary data)**.** *Work in the non-full tight-height setting of lemma 4.339. Suppose that the strict-left Krattenthaler growth segment*

$$\rho_0, \rho_1, \dots, \rho_{2(j-2m)}$$

determines the prefix endpoint $\lambda^{(2m)}$ of the Pieri path and, for each continuation label $t = 2m+1,\dots,j$, the two boxes of the tableau carrying label t. Then the outside datum $\mathcal{D}_{\mathrm{out}}(P)$ determines all auxiliary data listed in proposition 4.338. Consequently, any tight right-boundary inverse that uses only those data as its auxiliary input satisfies the auxiliary-data hypothesis of proposition 4.253.

Proof. By lemma 4.339, the outside datum $\mathcal{D}_{\mathrm{out}}(P)$ determines the strict-left segment

$$\rho_0, \rho_1, \dots, \rho_{2(j-2m)}.$$

The hypothesis of the present proposition then determines $\lambda^{(2m)}$ and the two boxes carrying every continuation label $2m+1,\dots,j$. Applying proposition 4.338 to these recovered data gives the tight-zone endpoint pair, endpoint-defect pairs, continuation incidences, and canonical Hall charge. These are exactly the listed auxiliary data. Since all of them are fixed functions of $\mathcal{D}_{\mathrm{out}}(P)$, a right-boundary inverse using only these data has auxiliary input of the form allowed in proposition 4.253. □

Lemma 4.341 (Strict-left data recover continuation-column Step-3 entries)**.** *Work in the non-full tight-height setting of lemma 4.339. Put $M = 2m$. In the Step 3 dual-RSK' half-square of lemma 4.125, let J_M be the Ferrers subarrangement formed by the first $j-M$ strict-left columns, i.e. the columns incident to the continuation labels $M+1,\dots,j$ in the label order $j, j-1,\dots,1$. Then the outside datum $\mathcal{D}_{\mathrm{out}}(P)$ determines every cell entry of J_M and every corner label on or inside J_M.*

Proof. By lemma 4.339, the outside datum determines the strict-left growth segment

$$\rho_0, \rho_1, \dots, \rho_{2(j-M)}.$$

Under lemma 4.118, this is exactly the Step 3 top/right boundary segment for the continuation labels $M+1,\dots,j$. These labels are the first $j-M$ labels in Krattenthaler's Step 3 order $j, j-1,\dots,1$, so the displayed segment is the top/right boundary of J_M. Applying corollary 4.124 to this Ferrers subarrangement recovers its cell entries and internal corner labels. □

Corollary 4.342 (Strict-left data reduce continuation recovery to the internal Q-cut)**.** *Work in the non-full tight-height setting of lemma 4.339. Put $M = 2m$, and let C_M^+ be the Step 2 Q-suffix strip of lemma 4.176; let C_M be the corresponding Step 2 Q-prefix strip. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the ordered partition labels on the internal separating boundary between the Step 2 Q-prefix strip C_M and the suffix strip C_M^+. Then the same data determine the Pieri continuation segment

$$\lambda^{(2m)} \subset \lambda^{(2m+1)} \subset \cdots \subset \lambda^{(j)}.$$

Proof. By lemma 4.341, the outside datum $\mathcal{D}_{\mathrm{out}}(P)$ determines every Step 3 entry in the first $j-M$ strict-left columns. By lemma 4.181, those entries determine every non-diagonal entry of the Step 2 Q-suffix strip C_M^+, and the diagonal entries of that strip are zero. Hence the displayed target-window data determine all cell entries of C_M^+.

By the hypothesis and lemma 4.177, the same data determine the full left/bottom boundary of C_M^+. Together with those boundary labels, lemma 4.176 determines the Q-side continuation boundary

$$\lambda^{(M)}, \lambda^{(M+1)}, \dots, \lambda^{(j)}.$$

Since $M = 2m$, this is the asserted Pieri continuation segment. □

Corollary 4.343 (Strict-left plus low-low entries recover the internal Q-cut)**.** *Work in the non-full tight-height setting of lemma 4.339. Put $M = 2m$, and let L_M be the low-low Step 3 cell set of lemma 4.182. Let C_M and C_M^+ be the corresponding Step 2 Q-prefix and Q-suffix strips. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine every Step 3 cell entry on L_M. Then the same data determine the ordered partition labels on the internal separating boundary between the Step 2 Q-prefix strip C_M and the suffix strip C_M^+. Consequently they determine the Pieri continuation segment

$$\lambda^{(2m)} \subset \lambda^{(2m+1)} \subset \cdots \subset \lambda^{(j)}.$$

Proof. By lemma 4.341, the outside datum $\mathcal{D}_{\mathrm{out}}(P)$ determines every Step 3 entry in J_M, the first $j - M$ strict-left columns. By lemma 4.182, the cross part X_M of H_M lies in J_M, while the low-low part is L_M. The hypothesis supplies the entries on L_M, so the displayed target-window data determine every Step 3 entry on

$$H_M = X_M \sqcup L_M.$$

Corollary 4.180 then determines the ordered internal cut labels between C_M and C_M^+. Finally corollary 4.342 applies with this recovered internal cut and gives the displayed Pieri continuation segment. □

Lemma 4.344 (The strict-left dual-RSK' boundary alone does not determine the tight prefix)**.** *In the $m = 0$ specialization of Krattenthaler's semistandard construction, there exist two column-even semistandard tableaux of content (2^6) in the same tight height fixed-witness fiber*

$$H_2(T) = \{1, 3\}$$

for which the Step 3 dual-RSK' subfilling in the first two strict-left columns is identical, and hence its top/right boundary segment

$$\rho_0, \rho_1, \rho_2, \rho_3, \rho_4$$

is identical, but the Pieri prefix endpoints at label 4 are different and both prefix endpoints are non-column-even. Consequently, the bridge in proposition 4.340 cannot be proved from the recovered strict-left boundary segment together with the tight-witness condition alone; a successful tight-zone inverse must also use the extra constraints or data supplied by the tight right-boundary zone.

Proof. Consider the following two symmetric zero-diagonal matrices with rows and columns indexed by $1, \dots, 6$:

$$A_{14} = A_{41} = A_{16} = A_{61} = A_{23} = A_{32} = A_{25} = A_{52} = A_{35} = A_{53} = A_{46} = A_{64} = 1,$$

all other entries being zero, and

$$B_{13} = B_{31} = B_{16} = B_{61} = B_{24} = B_{42} = B_{25} = B_{52} = B_{35} = B_{53} = B_{46} = B_{64} = 1,$$

all other entries being zero. Every row sum of both matrices is 2. Applying ordinary RSK in the row-word convention used in lemma 4.119 gives, for A, the tableau

$$T_A = \begin{matrix} 1 & 1 & 3 & 4 \\ 2 & 2 & 5 & 6 \\ 3 & 5 & & \\ 4 & 6 & & \end{matrix}$$

and, for B, the tableau

$$T_B = \begin{matrix} 1 & 1 & 2 & 3 & 4 \\ 2 & 4 & 5 & 5 & 6 \\ 3 \\ 6 \end{matrix}.$$

Indeed, the bottom words inserted under top labels $1, \dots, 6$ are

$$4, 6, 3, 5, 2, 5, 1, 6, 2, 3, 1, 4$$

for A, and

$$3, 6, 4, 5, 1, 5, 2, 6, 2, 3, 1, 4$$

for B; direct row insertion gives the displayed insertion tableau in each case, and symmetry gives the same recording tableau. Both final shapes are column-even:

$$(4, 4, 2, 2)' = (4, 4, 2, 2), \qquad (5, 5, 1, 1)' = (4, 2, 2, 2, 2).$$

The heights of the Pieri prefixes for T_A are

$$1, 2, 3, 4, 4, 4,$$

and the heights of the Pieri prefixes for T_B are

$$1, 2, 3, 3, 3, 4.$$

Therefore, in both cases, the capped height statistic $\min(2, \lceil \ell(\lambda^{(t)})/2 \rceil)$ increases exactly at $t = 1$ and $t = 3$. Thus both tableaux lie in the tight height fixed-witness fiber $H_2(T) = \{1, 3\}$. However the Pieri prefix endpoints after labels at most 4 are

$$\lambda_A^{(4)} = (4, 2, 1, 1), \qquad \lambda_B^{(4)} = (5, 2, 1),$$

so they are different. Their conjugates are, respectively,

$$(4, 2, 1, 1)' = (4, 2, 1, 1), \qquad (5, 2, 1)' = (3, 2, 1, 1, 1),$$

so neither prefix endpoint is column-even.

Now order the rows and columns of the Step 2 symmetric matrix as in Krattenthaler's proof, namely $6, 5, 4, 3, 2, 1$, and pass to the Step 3 strict lower-triangular dual-RSK' half-square. The first two strict-left columns are exactly the entries incident to labels 6 and 5 on the strict-left side. For both A and B, these nonzero entries are the four unit cells joining

$$6 \text{ to } 1, \qquad 6 \text{ to } 4, \qquad 5 \text{ to } 2, \qquad 5 \text{ to } 3,$$

with all other entries in those two columns equal to zero. Therefore the Step 3 subfilling in the first two strict-left columns is the same for A and B. Since the left and bottom boundaries of this subarrangement are empty, the forward dual-RSK' local rules of lemma 4.121 give the same top/right boundary segment $\rho_0, \dots, \rho_4$.

Thus the same strict-left boundary segment and the same tight height witness condition are compatible with two different non-column-even Pieri prefix endpoints at label 4. The final sentence follows. □

Lemma 4.345 (Strict-left data alone do not determine the low-low triangle)**.** *In the two examples of lemma 4.344, put $M = 4$. The recovered Step 3 strict-left continuation-column subfilling, and hence the residual degree vector*

$$d_a = 2 - \sum_{t=5}^{6} u_{\{a,t\}} \qquad (1 \le a \le 4),$$

is the same for T_A and T_B. Nevertheless the Step 3 low-low assignments on L_4 are different:

$$T_A : \quad u_{\{1,4\}} = u_{\{2,3\}} = 1, \qquad T_B : \quad u_{\{1,3\}} = u_{\{2,4\}} = 1,$$

with all other low-low entries equal to 0 *in the two cases. Consequently, the low-low supplier in the large tight right-boundary zone cannot be derived from the strict-left continuation subfilling or its residual degrees alone; it must also use target-window, right-boundary, or equivalent semistandard data.*

Proof. The preceding proof shows that the first two strict-left columns of the Step 3 dual-RSK' half-square are identical for T_A and T_B. These are the columns incident to the continuation labels 6 and 5, so they are exactly the continuation-column subfilling for $M = 4$. Reading the displayed matrices, the entries incident from the low labels $1, 2, 3, 4$ to the continuation labels $5, 6$ are, in both examples,

$$\{1, 6\}, \qquad \{2, 5\}, \qquad \{3, 5\}, \qquad \{4, 6\},$$

each with multiplicity one and all other continuation incidences zero. Hence

$$\sum_{t=5}^{6} u_{\{a,t\}} = 1 \qquad (1 \leq a \leq 4),$$

and the residual degree vector is $d_1 = d_2 = d_3 = d_4 = 1$ for both examples.

By lemma 4.127, the Step 3 low-low entries on L_4 are exactly the off-diagonal Step 2 matrix entries whose two labels lie in $\{1, 2, 3, 4\}$. In the matrix A, the only such nonzero entries are $A_{14} = A_{41} = 1$ and $A_{23} = A_{32} = 1$. In the matrix B, the only such nonzero entries are $B_{13} = B_{31} = 1$ and $B_{24} = B_{42} = 1$. This gives the displayed two different low-low assignments with the same continuation subfilling and the same residual degrees. □

Corollary 4.346 (Strict-left data alone do not determine the terminal boundary)**.** *In the two examples of lemma 4.344, put* $M = 4$*. The recovered strict-left continuation-column subfilling and the residual degree vector are the same, but the terminal Step 3 boundary segment*

$$\rho_{2(j-M)}, \rho_{2(j-M)+1}, \ldots, \rho_{2j}$$

cannot be the same. Consequently, the terminal-boundary supplier for the large tight right-boundary zone cannot be a function only of the strict-left continuation subfilling or of its residual degrees.

Proof. For $j = 6$ and $M = 4$, the displayed segment is

$$\rho_4, \rho_5, \ldots, \rho_{12}.$$

If the two examples had the same terminal boundary segment, then lemma 4.183 would recover the same Step 3 entries on L_4 in both examples. This contradicts lemma 4.345, which gives different assignments on L_4 while the strict-left continuation-column subfilling and the residual degree vector are the same. Hence any deterministic terminal-boundary source must use target-window, right-boundary, or equivalent semistandard data beyond those strict-left quantities. □

Lemma 4.347 (The strict-left obstruction is a packet endpoint obstruction)**.** *In the two examples of lemma 4.344, the missing datum is already visible in the endpoint-defect data of the singleton deleted packets. More precisely, for the witness set* $S = \{1, 3\}$ *the deleted packets are* $J_1 = \{1\}$ *and* $J_2 = \{3\}$*. The packet* J_1 *has the same endpoint pair and the same local defect columns for* T_A *and* T_B*, namely*

$$\lambda_A^{(0)} = \lambda_B^{(0)} = \varnothing, \qquad \lambda_A^{(1)} = \lambda_B^{(1)} = (2), \qquad O_{J_1}(T_A) = O_{J_1}(T_B) = \{1, 2\}.$$

The packet J_2*, however, has different endpoint pairs and different local defect columns:*

$$\lambda_A^{(2)} = (2, 2), \quad \lambda_A^{(3)} = (3, 2, 1), \quad O_{J_2}(T_A) = \{1, 3\},$$

whereas

$$\lambda_B^{(2)} = (3, 1), \quad \lambda_B^{(3)} = (4, 1, 1), \quad O_{J_2}(T_B) = \{1, 4\}.$$

Consequently, a tight right-boundary decoder that uses the packet inverse of lemma 4.226 on these examples must obtain the J_2 endpoint-defect datum from the final-zone constraints or auxiliary right-boundary data before the local branch word can be interpreted.

Proof. The displayed tableaux in lemma 4.344 give the Pieri prefix shapes directly. For both tableaux, the two boxes carrying label 1 occupy the first row, so

$$\lambda_A^{(0)} = \lambda_B^{(0)} = \varnothing, \qquad \lambda_A^{(1)} = \lambda_B^{(1)} = (2).$$

Since $(2)' = (1,1)$, the local defect columns of this singleton packet are

$$\{c : (\lambda^{(1)})'_c - (\lambda^{(0)})'_c \equiv 1 \pmod 2\} = \{1,2\}$$

for both tableaux.

For T_A, after labels at most 2 the first two rows have length 2, and after label 3 the shape is $(3,2,1)$:

$$\lambda_A^{(2)} = (2,2), \qquad \lambda_A^{(3)} = (3,2,1).$$

Thus

$$(\lambda_A^{(2)})' = (2,2), \qquad (\lambda_A^{(3)})' = (3,2,1),$$

and the columns in which the conjugate endpoint difference is odd are 1 and 3.

For T_B, labels at most 2 give the shape $(3,1)$, and labels at most 3 give the shape $(4,1,1)$:

$$\lambda_B^{(2)} = (3,1), \qquad \lambda_B^{(3)} = (4,1,1).$$

Hence

$$(\lambda_B^{(2)})' = (2,1,1), \qquad (\lambda_B^{(3)})' = (3,1,1,1),$$

so the odd endpoint-difference columns are 1 and 4. These are exactly the local defect columns used by item (2) of lemma 4.226.

By the preceding lemma, the strict-left boundary segment and the witness set agree for the two examples, while the J_2 endpoint-defect datum differs. Therefore any decoder that proceeds through the local-branch recovery lemma must recover this datum from the right-boundary-zone side before applying the local packet inverse. ☐

Lemma 4.348 (Retained closure does not recover all tight-prefix packets)**.** *There is a non-full tight height example in which the retained boxes and their smallest column-even Young closure do not recover the deleted singleton packets. More precisely, there is a column-even semistandard tableau of content (2^8) with*

$$H_3(T) = \{1,3,5\}$$

such that the tight right-boundary zone has deleted labels $\{1,3,5\}$ and retained labels $\{2,4,6\}$, the prefix endpoint $\lambda^{(6)}$ is not column-even, and after deleting labels $\{1,3,5\}$ the smallest column-even Young diagram containing the retained boxes is strictly smaller than the original terminal shape.

Consequently, a tight-zone decoder cannot recover all deleted packets merely by taking the column-even closure of the retained labelled boxes, even when the endpoint-defect parity of the tight prefix is known. It must also recover or encode all-deleted even-column pairs inside the tight prefix.

Proof. Consider the tableau

$$T = \begin{array}{cccc} 1 & 1 & 4 & 6 \\ 2 & 2 & 7 & 8 \\ 3 & 3 & & \\ 4 & 5 & & \\ 5 & & & \\ 6 & & & \\ 7 & & & \\ 8 & & & \end{array}.$$

It has content (2^8), its rows are weakly increasing, and its columns are strictly increasing. Its terminal shape is

$$\lambda^{(8)} = (4,4,2,2,1,1,1,1), \qquad (\lambda^{(8)})' = (8,4,2,2),$$

so the terminal shape is column-even.

The successive prefix shapes have lengths

$$1,2,3,4,5,6,7,8.$$

Therefore

$$\min(3, \lceil \ell(\lambda^{(t)})/2 \rceil)$$

increases exactly at $t = 1,3,5$, and hence $H_3(T) = \{1,3,5\}$. Since $j = 8$ and $m = 3$, corollary 4.320 gives the tight right-boundary zone

$$Z_\gamma = \{3,4,5,6,7,8\},$$

whose deleted labels are $\{1,3,5\}$ and whose retained in-zone labels are $\{2,4,6\}$. The prefix endpoint through label 6 is

$$\lambda^{(6)} = (4,2,2,2,1,1), \qquad (\lambda^{(6)})' = (6,4,1,1),$$

so it is not column-even.

Delete the boxes carrying labels $1,3,5$. The remaining boxes include $(8,1)$ and $(2,4)$, so any Young diagram containing them contains the shape

$$\mu = (4,4,1,1,1,1,1,1).$$

Conversely, this shape contains every retained box. Its conjugate is

$$\mu' = (8,2,2,2),$$

so μ is already column-even. Hence μ is the smallest column-even Young diagram containing the retained boxes.

However the original terminal shape is strictly larger:

$$\lambda^{(8)} \setminus \mu = \{(3,2),(4,2)\}.$$

Both of these boxes are deleted boxes, carrying labels 3 and 5, respectively. They form an all-deleted pair in column 2, so this missing pair is invisible to the column parity of the retained closure. It is also not an endpoint defect of the tight prefix, since

$$(\lambda^{(6)})' = (6,4,1,1)$$

has odd columns only at 3 and 4. Thus endpoint-defect parity does not record the missing column-2 pair. The final assertion follows. □

Lemma 4.349 (Prefix endpoint and retained even labels do not recover the tight packets)**.** *There are two column-even semistandard tableaux of content* (2^9) *with the same tight height witness set*

$$H_4(T) = \{1,3,5,7\}$$

such that the following data agree:

1. *the tight-prefix endpoint* $\lambda^{(8)}$*;*
2. *the two boxes carrying each retained tight-prefix label* $2,4,6,8$*; and*
3. *the two boxes carrying the continuation label* 9*.*

Nevertheless the deleted singleton packets for the odd labels 5 *and* 7 *are different. Consequently, a final tight-zone decoder cannot recover the full deleted packet data from the prefix endpoint, retained even-label boxes, and continuation boxes alone; the repaired target window or the allowed branch word must also determine the residual odd-packet matching.*

Proof. Consider the two tableaux

$$T_A = \begin{array}{cccc} 1 & 1 & 4 & 7 \\ 2 & 2 & 6 & 9 \\ 3 & 3 & & \\ 4 & 5 & & \\ 5 & 8 & & \\ 6 & 9 & & \\ 7 & & & \\ 8 & & & \end{array} \qquad \text{and} \qquad T_B = \begin{array}{cccc} 1 & 1 & 4 & 5 \\ 2 & 2 & 6 & 9 \\ 3 & 3 & & \\ 4 & 7 & & \\ 5 & 8 & & \\ 6 & 9 & & \\ 7 & & & \\ 8 & & & \end{array}.$$

Both have weakly increasing rows, strictly increasing columns, and content (2^9). Their common terminal shape is

$$\lambda^{(9)} = (4,4,2,2,2,2,1,1), \qquad (\lambda^{(9)})' = (8,6,2,2),$$

so both terminal shapes are column-even.

For both tableaux the prefix endpoint through label 8 is

$$\lambda^{(8)} = (4,3,2,2,2,1,1,1).$$

The retained tight-prefix labels have the same positions in the two tableaux:

label	positions
2	$(2,1),(2,2)$
4	$(1,3),(4,1)$
6	$(2,3),(6,1)$
8	$(5,2),(8,1)$.

The continuation label also has the same positions,

$$9:\ (2,4),(6,2).$$

The deleted odd labels 1 and 3 agree:

$$1:\ (1,1),(1,2), \qquad 3:\ (3,1),(3,2).$$

But labels 5 and 7 differ. In T_A one has

$$5:\ (4,2),(5,1), \qquad 7:\ (1,4),(7,1),$$

whereas in T_B one has

$$5:\ (1,4),(5,1), \qquad 7:\ (4,2),(7,1).$$

Thus the two odd-packet assignments are distinct while the displayed prefix endpoint, retained even-label boxes, and continuation boxes are identical.

The successive prefix lengths are $1,2,\dots,8$ in both tableaux. Hence

$$\min(4, \lceil \ell(\lambda^{(t)})/2 \rceil)$$

increases exactly for $t = 1,3,5,7$, proving $H_4(T_A) = H_4(T_B) = \{1,3,5,7\}$. Since $j = 9$ and $m = 4$, corollary 4.320 identifies the final tight zone as the terminal block interval corresponding to labels $1,\dots,8$. The final sentence is exactly the displayed collision of packet data. □

Corollary 4.350 (Hall data alone do not recover the tight odd packets)**.** *In the two tableaux of lemma 4.349, the following data agree:*

(1) *the tight-prefix endpoint* $\lambda^{(8)}$*;*
(2) *the retained even-label boxes for labels* $2,4,6,8$*;*
(3) *the labelled continuation boxes;*
(4) *the endpoint-defect pairs of* $\lambda^{(8)}$*;*
(5) *the continuation incidence relation; and*
(6) *the canonical continuation Hall charge of lemma 4.335.*

Nevertheless the deleted odd-packet assignments differ. Consequently, a non-column-even tight-zone decoder cannot recover the residual odd-packet matching from the prefix endpoint, retained even boxes, continuation boxes, and Hall charge alone; any uniform large-zone carrier rule must use additional target-window or Step 3 boundary data, or else use the already counted branch word to select among the remaining packet assignments.

Proof. Lemma 4.349 states that the two examples have the same tight-prefix endpoint, the same retained even-label boxes, and the same continuation boxes, while their deleted odd-packet assignments differ.

The common prefix endpoint and common continuation boxes determine the same Pieri continuation segment by lemma 4.337. Applying lemma 4.513 to this common segment shows that the endpoint-defect pairs, the continuation incidence relation, and the canonical continuation Hall charge are also the same in the two examples. The different odd-packet assignments are exactly those displayed in lemma 4.349. □

Corollary 4.351 (Direct Hall-selected blocks do not recover the tight odd packets)**.** *In the two tableaux of lemma 4.349, the direct Hall-data package used in corollary 4.500 agrees up to the Hall-selected continuation pieces: the tight-prefix endpoint, endpoint-defect pairs, continuation-incidence family, canonical Hall charge, and Hall-selected continuation strips are the same. Nevertheless the deleted odd-packet assignments differ.*

Consequently the direct Hall data together with the Hall-selected continuation pieces are not, by themselves, a terminal source for the non-column-even tight zone. A uniform large-zone carrier or terminal-boundary rule must use actual target-window or Step 3 boundary data, or else use the already counted branch word to select among the remaining packet assignments.

Proof. The agreement of the tight-prefix endpoint, retained even-label boxes, and labelled continuation boxes is part of lemma 4.349. As in the proof of corollary 4.350, the common prefix endpoint and common continuation boxes determine the same Pieri continuation segment by lemma 4.337; hence lemma 4.513 gives the same endpoint-defect pairs, continuation incidences, and canonical Hall charge.

The Hall-selected continuation labels are therefore the same. Since all labelled continuation boxes agree, the two-box continuation strip attached to each selected Hall label also agrees. The deleted odd-packet assignments are different by lemma 4.349. Thus no deterministic rule depending only on the displayed direct Hall package and the selected continuation pieces can recover those packets. □

Lemma 4.352 (Height branch rank slack)**.** *In the height case of lemma 4.226, the local repair data recovered from a branch word depend only on its last h_α bits. Equivalently, if*

$$\beta, \widetilde{\beta} \in \{0,1\}^{2h_\alpha}$$

have

$$\beta_{h_\alpha+i} = \widetilde{\beta}_{h_\alpha+i} \qquad (1 \le i \le h_\alpha),$$

then the height-case event list, local defect pairs, and selected subset recovered by lemma 4.226 are the same for β and $\widetilde{\beta}$. Thus, in a height-zone decoder, the first h_α branch bits may be read as an auxiliary rank and then replaced by zeros before applying the local height branch recovery.

Proof. In the height case, item (1) of lemma 4.226 gives

$$e_i = 2(a_\alpha + i - 1) - 1 \qquad (1 \le i \le h_\alpha),$$

which contains no branch bit. Item (2) obtains the local defect columns from the endpoint partitions σ^- and σ^+, again with no branch bit. Item (3) reads the selected subset from the last h_α bits of the word. Therefore two words with the same last half recover the same three pieces of local repair data. Replacing the first half by zeros leaves the data used by the height local inverse unchanged, while making the word one of the height-branch words of lemma 4.225. □

Proposition 4.353 (Finite odd-packet rank fits the tight height branch budget)**.** *Work in the non-full tight height setting of corollary 4.320. Thus the final tight right-boundary zone has deleted labels*

$$1, 3, \ldots, 2m-1$$

and retained tight-prefix labels

$$2, 4, \ldots, 2m.$$

Suppose a parser-visible datum $D(P)$ determines the tight-prefix endpoint $\lambda^{(2m)}$, the two boxes carrying each retained label $2, 4, \ldots, 2m$, and the labelled continuation boxes. Suppose also that $D(P)$ determines a canonically ordered finite set

$$\Omega(D(P))$$

of possible assignments of the deleted odd-label singleton packets, that the true assignment $\omega(P)$ belongs to this set, and that

$$|\Omega(D(P))| \leq 2^m.$$

Then the first m bits of the tight-zone branch word

$$\beta_\gamma \in \{0,1\}^{2m}$$

can encode the lexicographic rank of $\omega(P)$ in $\Omega(D(P))$, while the last m bits remain available for the height local selected-defect data. From $D(P)$ and β_γ one recovers the full tight-prefix singleton packet data.

Proof. Order $\Omega(D(P))$ by the canonical order in the hypothesis. Since its cardinality is at most 2^m, the rank of the true assignment $\omega(P)$ is encoded by an m-bit binary word. Read this word from the first half of β_γ and choose the corresponding element of

$$\Omega(D(P)).$$

This recovers the two boxes of every deleted odd label $1, 3, \ldots, 2m-1$.

Together with the retained even-label boxes already contained in $D(P)$, these recovered odd packets reconstruct the whole Pieri prefix

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}$$

by adding the two boxes of labels $1, 2, \ldots, 2m$ in order. Hence every singleton deleted packet endpoint pair, its local defect columns, and the local paired defects are determined.

The last m bits of β_γ have not been used in the rank encoding. By lemma 4.352, the height local recovery depends on these last bits, together with the recovered endpoint data, exactly as in lemma 4.226. Thus the selected defect subset needed by the height local inverse is still recovered, and the first half of the branch word has carried only the auxiliary odd-packet rank. Therefore $D(P)$ and β_γ recover the full tight-prefix singleton packet data. □

Definition 4.354 (Tight odd-packet compatibility set)**.** Fix $m \geq 1$. A tight-prefix datum consists of a partition ν of size $4m$ and, for each $1 \leq r \leq m$, a two-box horizontal strip E_r whose boxes are intended to carry the retained even label $2r$. Put

$$f_r = (2r-1, 1) \qquad (1 \leq r \leq m).$$

Let $\Omega(\nu; E_1, \ldots, E_m)$ be the set of tuples

$$x = (x_1, \ldots, x_m)$$

of boxes of ν with the following properties:

(1) the sets
$$\{f_r : 1 \leq r \leq m\}, \qquad \{x_r : 1 \leq r \leq m\}, \qquad E_1, \ldots, E_m$$
are pairwise disjoint and have union the Young diagram of ν;

(2) for every r, the box x_r is not in column 1, and the two boxes f_r, x_r lie in distinct columns;

(3) if
$$\nu^{(0)} = \varnothing, \qquad \nu^{(2r-1)} = \nu^{(2r-2)} \cup \{f_r, x_r\}, \qquad \nu^{(2r)} = \nu^{(2r-1)} \cup E_r,$$
then each $\nu^{(t)}$ is a Young diagram, each displayed difference is a horizontal strip, and $\nu^{(2m)} = \nu$.

The set $\Omega(\nu; E_1, \ldots, E_m)$ is ordered lexicographically by the tuples $x_1, \ldots, x_m$ in row-column order.

Lemma 4.355 (The tight compatibility set contains the true odd packets)**.** *Work in the non-full tight height setting of corollary 4.320. Let*
$$\nu = \lambda^{(2m)}$$
be the tight-prefix endpoint, and let E_r be the two boxes carrying label $2r$ for $1 \le r \le m$. For each r, let x_r be the second box carrying the deleted odd label $2r-1$, the first one being the forced first-column box $f_r = (2r-1, 1)$. Then
$$(x_1, \ldots, x_m) \in \Omega(\nu; E_1, \ldots, E_m).$$
Consequently the compatibility set of definition 4.354 is a canonical parser-visible candidate set for the residual odd-packet assignment once $\lambda^{(2m)}$ and the retained even-label boxes are parser-visible.

Proof. By corollary 4.320, the deleted labels in the tight prefix are exactly $1, 3, \ldots, 2m-1$, and the retained labels in that prefix are exactly $2, 4, \ldots, 2m$. By corollary 4.328, the odd label $2r-1$ creates the row $2r-1$. Thus one of its two boxes is $f_r = (2r-1, 1)$; since the strip of a single label is horizontal, the other box x_r is not in column 1.

The boxes carrying the labels $1, \ldots, 2m$ are exactly the boxes of the prefix endpoint $\lambda^{(2m)} = \nu$. Each label occurs twice, so the forced odd first-column boxes, the odd second boxes x_r, and the even strips E_r are pairwise disjoint and have union ν. The actual Pieri prefix gives the recursive sequence
$$\lambda^{(0)}, \lambda^{(1)}, \ldots, \lambda^{(2m)},$$
and its successive differences are horizontal two-strips by lemma 4.97. This is exactly item (3) in definition 4.354 for the tuple $(x_1, \ldots, x_m)$. Therefore that tuple belongs to $\Omega(\nu; E_1, \ldots, E_m)$.

The final sentence follows because the definition of Ω uses only the displayed endpoint partition and retained even strips, together with the fixed row-column lexicographic order. ☐

Lemma 4.356 (Low-label boxes determine the tight packet datum)**.** *Work in the non-full tight height setting of corollary 4.320, and let*
$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(j)}$$
be the Pieri two-strip path of the underlying semistandard tableau. Suppose a datum determines, for each $1 \le a \le 2m$, the two boxes carrying semistandard label a. Then it determines the tight-prefix chain
$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)},$$
the endpoint $\nu = \lambda^{(2m)}$, the retained even strips
$$E_r = \lambda^{(2r)} \setminus \lambda^{(2r-1)} \qquad (1 \le r \le m),$$
and the true odd-packet assignment
$$(x_1, \ldots, x_m) \in \Omega(\nu; E_1, \ldots, E_m).$$

Proof. For $0 \le a \le 2m$, the shape $\lambda^{(a)}$ is the union of the boxes carrying labels at most a by lemma 4.97. Hence the displayed low-label box data determine the whole tight-prefix chain and its endpoint $\nu = \lambda^{(2m)}$. The retained even strip E_r is exactly the two-box set carrying label $2r$, so all labelled retained even strips are also determined.

It remains to identify the odd packets. By corollary 4.328, the step from $\lambda^{(2r-2)}$ to $\lambda^{(2r-1)}$ creates row $2r-1$ for $1 \le r \le m$. Since this step is a horizontal two-strip, one of the two boxes carrying label $2r-1$ is the forced first-column box

$$f_r = (2r-1, 1),$$

and the other box is determined by the same two-box label data; call it x_r. Applying lemma 4.355 to the endpoint ν, the retained strips E_r, and these odd second boxes gives

$$(x_1, \dots, x_m) \in \Omega(\nu; E_1, \dots, E_m).$$

Thus the true tight-prefix packet datum is determined. ☐

Lemma 4.357 (The singleton tight odd packet is forced)**.** *For $m = 1$, every element of*

$$\Omega(\nu; E_1)$$

has $x_1 = (1, 2)$. In particular

$$|\Omega(\nu; E_1)| \le 1 \le 2^1.$$

Proof. Let $(x_1) \in \Omega(\nu; E_1)$. In definition 4.354 one has $f_1 = (1, 1)$, and item (2) says that x_1 is not in column 1. Item (3) says that

$$\nu^{(1)} = \{(1, 1), x_1\}$$

is a Young diagram and that $\nu^{(1)} \setminus \nu^{(0)}$ is a horizontal strip. The only Young diagram with two boxes, containing $(1, 1)$ and another box outside column 1, is $\{(1, 1), (1, 2)\}$. Therefore $x_1 = (1, 2)$. The cardinality bound follows immediately; if the displayed set is empty the same bound is trivial. ☐

Corollary 4.358 (The singleton tight rank leaf is closed)**.** *In the setting of lemma 4.355, suppose $m = 1$. Let $D(P)$ be any parser-visible datum that determines $\lambda^{(2)}$, the retained strip E_1, and the labelled continuation boxes as in proposition 4.353. If one sets*

$$\Omega(D(P)) = \Omega(\lambda^{(2)}; E_1).$$

Then the membership, canonical-ordering, and cardinality hypotheses of proposition 4.353 hold. Thus the singleton tight right-boundary zone needs no refined outside-data ambiguity bound for its deleted odd packet.

Proof. Lemma 4.355 puts the true deleted odd-packet assignment in $\Omega(\lambda^{(2)}; E_1)$, while lemma 4.357 gives $|\Omega(\lambda^{(2)}; E_1)| \le 2^1$. The lexicographic order from definition 4.354 is canonical. These are precisely the membership, ordering, and cardinality hypotheses of proposition 4.353. ☐

Lemma 4.359 (The first two tight odd packets have binary prefix ambiguity)**.** *Fix $m \ge 2$. Let*

$$\Pi_2(\nu; E_1, \dots, E_m)$$

be the set of pairs (x_1, x_2) which occur as the first two coordinates of an element of $\Omega(\nu; E_1, \dots, E_m)$. Then

$$|\Pi_2(\nu; E_1, \dots, E_m)| \le 2.$$

In particular, for $m = 2$,

$$|\Omega(\nu; E_1, E_2)| \le 2 \le 2^2.$$

Proof. Let $(x_1, \dots, x_m) \in \Omega(\nu; E_1, \dots, E_m)$. Applying the proof of lemma 4.357 to the first odd step gives

$$x_1 = (1, 2), \qquad \nu^{(1)} = (2).$$

The diagram $\nu^{(3)}$ contains the forced cell $f_2 = (3, 1)$. Hence, by the Young property, it also contains $(2, 1)$. This cell is not in $\nu^{(1)}$, is not f_2, and cannot be x_2 because x_2 is not in column 1. Therefore $(2, 1) \in E_1$.

Write $E_1 = \{(2,1), y\}$. Since E_1 is a horizontal strip, y is not in column 1. Since

$$\nu^{(2)} = (2) \cup E_1$$

is a Young diagram with four cells, the only possibilities are

$$y = (1,3) \qquad \text{or} \qquad y = (2,2).$$

Indeed, a cell in the first row must be $(1,3)$, a cell in the second row must be $(2,2)$, and any lower-row or farther-right cell would force an additional missing predecessor cell.

If $y = (1,3)$, then $\nu^{(2)} = (3,1)$. The Young diagram

$$\nu^{(3)} = \nu^{(2)} \cup \{(3,1), x_2\}$$

can then have x_2 only at $(1,4)$ or $(2,2)$: these give the two Young diagrams $(4,1,1)$ and $(3,2,1)$, while every other cell outside $\nu^{(2)}$ and outside column 1 leaves a missing predecessor.

If $y = (2,2)$, then $\nu^{(2)} = (2,2)$. The same Young-diagram check gives

$$x_2 \in \{(1,3), (3,2)\},$$

corresponding to the two shapes $(3,2,1)$ and $(2,2,2)$.

Thus, for the fixed datum, x_1 is forced and x_2 has at most two possible values among all compatible tuples. Hence

$$|\Pi_2(\nu; E_1, \dots, E_m)| \le 2.$$

When $m = 2$, an element of $\Omega(\nu; E_1, E_2)$ has no coordinates beyond (x_1, x_2), so the same bound gives $|\Omega(\nu; E_1, E_2)| \le 2$. ☐

Lemma 4.360 (Binary prefix factorial bound)**.** *For every $m \ge 2$,*

$$|\Omega(\nu; E_1, \dots, E_m)| \le 2\,(m-2)!.$$

Proof. Fix a pair

$$(a_1, a_2) \in \Pi_2(\nu; E_1, \dots, E_m).$$

Consider an element $x = (x_1, \dots, x_m)$ of $\Omega(\nu; E_1, \dots, E_m)$ with

$$(x_1, x_2) = (a_1, a_2).$$

The union condition in definition 4.354 forces the remaining cells $x_3, \dots, x_m$ to be exactly the elements of

$$U(a_1, a_2) = \nu \setminus (\{f_1, \dots, f_m, a_1, a_2\} \cup E_1 \cup \cdots \cup E_m).$$

This set has cardinality $m-2$: the diagram ν has $4m$ cells, the forced first-column cells contribute m cells, the fixed even strips contribute $2m$ cells, and the fixed prefix cells a_1, a_2 contribute two cells. Therefore, after the first two coordinates are fixed, there are at most $(m-2)!$ possible ordered tuples $(x_3, \dots, x_m)$.

There are at most two possible first-two-coordinate pairs by lemma 4.359. Multiplying gives the displayed bound. ☐

Corollary 4.361 (The two-packet tight rank leaf is closed)**.** *In the setting of lemma 4.355, suppose $m = 2$. Let $D(P)$ be any parser-visible datum that determines $\lambda^{(4)}$, the retained strips E_1, E_2, and the labelled continuation boxes as in proposition 4.353. If one sets*

$$\Omega(D(P)) = \Omega(\lambda^{(4)}; E_1, E_2),$$

then the membership, canonical-ordering, and cardinality hypotheses of proposition 4.353 hold. Thus the two-packet tight right-boundary zone needs no refined outside-data ambiguity bound for its deleted odd packets.

Proof. Lemma 4.355 puts the true deleted odd-packet assignment in the displayed set. Meanwhile lemma 4.359 gives

$$|\Omega(\lambda^{(4)}; E_1, E_2)| \le 2 \le 2^2.$$

The lexicographic order from definition 4.354 is canonical. These are precisely the membership, ordering, and cardinality hypotheses of proposition 4.353. □

Corollary 4.362 (Small tight rank leaves are closed)**.** *In the setting of lemma 4.355, suppose $2 \le m \le 6$. Let $D(P)$ be any parser-visible datum that determines $\lambda^{(2m)}$, the retained strips $E_1, \ldots, E_m$, and the labelled continuation boxes as in proposition 4.353. If one sets*

$$\Omega(D(P)) = \Omega(\lambda^{(2m)}; E_1, \ldots, E_m),$$

then the membership, canonical-ordering, and cardinality hypotheses of proposition 4.353 hold. Thus tight right-boundary zones with at most six deleted odd packets need no refined outside-data ambiguity bound for their deleted odd packets.

Proof. The true assignment belongs to the displayed set by lemma 4.355, and its order is the canonical lexicographic order of definition 4.354. For the size bound, lemma 4.360 gives

$$|\Omega(\lambda^{(2m)}; E_1, \ldots, E_m)| \le 2\,(m-2)!.$$

For $2 \le m \le 6$ one checks directly that $2(m-2)! \le 2^m$. These are exactly the membership, ordering, and cardinality hypotheses of proposition 4.353. □

Proposition 4.363 (The seven-packet tight rank leaf is certified)**.** *In the setting of lemma 4.355, suppose $m = 7$. Let $D(P)$ be any parser-visible datum that determines $\lambda^{(14)}$, the retained strips $E_1, \ldots, E_7$, and the labelled continuation boxes as in proposition 4.353. If one sets*

$$\Omega(D(P)) = \Omega(\lambda^{(14)}; E_1, \ldots, E_7),$$

then the membership, canonical-ordering, and cardinality hypotheses of proposition 4.353 hold. Thus tight right-boundary zones with seven deleted odd packets need no refined outside-data ambiguity bound for their deleted odd packets.

Proof. Lemma 4.355 gives membership of the true assignment, and the ordering is the canonical lexicographic order of definition 4.354. It remains only to check the cardinality bound.

The exact C++ verifier

`character_ring_iter/post_m29_bc_tight_omega_small.cpp`

enumerates every tight-prefix datum by the recursive horizontal-two-strip definition of definition 4.354: at each odd step it adds the forced cell $f_r = (2r-1, 1)$ and one non-first-column cell, and at each even step it adds an arbitrary horizontal two-strip E_r. It groups the resulting tuples by the pair

$$(\lambda^{(2m)}; E_1, \ldots, E_m)$$

and records the maximum fiber size.

The archived 'optimus' run

`certificates/post_m29/bc_tight_omega_small_m7_certificate.log`

has SHA-256

`24063143ad7f4ac92259560dd11b1d8c`
`5291c544e92656e119b8b1a911b6df7d`.

The source file has SHA-256

`c6f03e177fbfcc9a3e7c055a10582445`
`6e27124bc915e2c20d8e437da1aff7b7`.

The log ends with `__EXIT_STATUS=0` and reports

m	`states`	`total_paths`	`max_omega`	`pow2`
7	3160108	7565247	20	128.

Thus every $m = 7$ tight compatibility set has size at most $20 \leq 2^7$. These are exactly the membership, ordering, and cardinality hypotheses of proposition 4.353. ☐

Proposition 4.364 (The eight-packet tight rank leaf is certified)**.** *In the setting of lemma 4.355, suppose $m = 8$. Let $D(P)$ be any parser-visible datum that determines $\lambda^{(16)}$, the retained strips $E_1, \ldots, E_8$, and the labelled continuation boxes as in proposition 4.353. If one sets*

$$\Omega(D(P)) = \Omega(\lambda^{(16)}; E_1, \ldots, E_8),$$

then the membership, canonical-ordering, and cardinality hypotheses of proposition 4.353 hold. Thus tight right-boundary zones with eight deleted odd packets need no refined outside-data ambiguity bound for their deleted odd packets.

Proof. Lemma 4.355 gives membership of the true assignment, and the ordering is the canonical lexicographic order of definition 4.354. It remains only to check the cardinality bound.

The exact C++ verifier

`character_ring_iter/post_m29_bc_tight_omega_packed.cpp`

uses the same recursive enumeration as proposition 4.363, but stores the fixed even strips in packed integer keys and expands each level with OpenMP on 'optimus'. It groups the resulting tuples by

$$(\lambda^{(2m)}; E_1, \ldots, E_m)$$

and records the maximum fiber size.

The archived run

`certificates/post_m29/bc_tight_omega_packed_m8_certificate.log`

has SHA-256

`1ab7eafd9308a50dc2ea4c2a946e2a9c`
`69514c2661a7166356c410fb818e96e7`.

The source file has SHA-256

`2fe5f8caa1f61536aaebe97c2f0b215f`
`fa7a55b7bc22c3f7b5441de42a1e9346`.

The log ends with `__EXIT_STATUS=0` and reports

m	`states`	`total_paths`	`max_omega`	`pow2`
8	44051883	152106391	45	256.

Thus every $m = 8$ tight compatibility set has size at most $45 \leq 2^8$. These are exactly the membership, ordering, and cardinality hypotheses of proposition 4.353. ☐

Corollary 4.365 (The tight odd-packet ambiguity bound closes the rank leaf)**.** *In the setting of lemma 4.355, suppose that the canonical compatibility set satisfies*

$$|\Omega(\lambda^{(2m)}; E_1, \ldots, E_m)| \leq 2^m.$$

Then the hypotheses of proposition 4.353 hold for the residual odd-packet assignment in the final tight right-boundary zone.

Proof. Take $D(P)$ to be the parser-visible datum containing $\lambda^{(2m)}$, the retained even-label boxes $E_1, \ldots, E_m$, and the labelled continuation boxes. Let

$$\Omega(D(P)) = \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

with the lexicographic order of definition 4.354. Lemma 4.355 proves that the true residual odd-packet assignment belongs to this set. The displayed cardinality bound is exactly the remaining size hypothesis of proposition 4.353. ☐

Corollary 4.366 (Refined tight-packet classes are the live rank criterion)**.** *Work in the setting of lemma 4.355. Suppose a parser-visible auxiliary datum $A(P)$, recovered without additional branch bits from the canonical halo/right-boundary parser data already counted in the replacement-zone package, determines a canonically ordered finite set*

$$\Omega_A(A(P)) \subseteq \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

such that the true residual odd-packet assignment belongs to $\Omega_A(A(P))$ and

$$|\Omega_A(A(P))| \leq 2^m.$$

Then the first m bits of the tight-zone branch word recover the true residual odd-packet assignment, and the last m bits remain available for the height selected-defect data. Consequently the rank hypothesis of proposition 4.353 holds for this refined parser-visible datum.

Proof. Use as $D(P)$ the parser-visible datum consisting of $\lambda^{(2m)}$, the retained even-label boxes $E_1, \ldots, E_m$, the labelled continuation boxes, and $A(P)$, and put

$$\Omega(D(P)) = \Omega_A(A(P)).$$

The hypotheses of the present corollary are exactly the membership, canonical-ordering, and cardinality hypotheses required in proposition 4.353. That proposition therefore gives the claimed recovery and leaves the last half of the branch word for the height selected-defect data. ☐

Proposition 4.367 (The refined $m \geq 9$ suffix is the remaining tight rank input)**.** *Work in the non-full tight height setting of corollary 4.320. Suppose that, whenever the final tight right-boundary zone has $m \geq 9$ deleted singleton packets, the branch-free canonical halo/right-boundary parser data determine an auxiliary datum $A(P)$ and a canonically ordered set*

$$\Omega_A(A(P)) \subseteq \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

such that the true residual odd-packet assignment belongs to $\Omega_A(A(P))$ and

$$|\Omega_A(A(P))| \leq 2^m.$$

Then the tight odd-packet rank hypothesis of proposition 4.353 holds for every non-full tight height right-boundary zone.

Proof. Let m be the number of deleted singleton packets in the final tight right-boundary zone. If $m = 1$, the assertion is corollary 4.358. If $2 \leq m \leq 6$, it is corollary 4.362. If $m = 7$ or $m = 8$, it is, respectively, proposition 4.363 and proposition 4.364. In the remaining case $m \geq 9$, the hypothesis of the present proposition is exactly the hypothesis of corollary 4.366, so that corollary applies. ☐

Definition 4.368 (Fixed-cell completions)**.** For a tight-prefix datum $(\nu; E_1, \ldots, E_m)$ as in definition 4.354, let

$$U(\nu; E) = \nu \setminus (\{f_r : 1 \leq r \leq m\} \cup E_1 \cup \cdots \cup E_m)$$

be the set of unfilled cells. Let

$$\mathcal{C}(\nu; E_1, \ldots, E_m)$$

be the set of bijections

$$\tau : U(\nu; E) \longrightarrow \{1, \ldots, m\}$$

such that the filling of ν defined by

$$f_r \mapsto 2r - 1, \qquad \tau^{-1}(r) \mapsto 2r - 1, \qquad E_r \mapsto 2r$$

is semistandard: rows are weakly increasing from left to right and columns are strictly increasing from top to bottom.

Lemma 4.369 (Omega equals completions)**.** *For every tight-prefix datum, the map*

$$(x_1, \dots, x_m) \longmapsto \big(\tau(x_r) = r\big)_{r=1}^{m}$$

is a bijection

$$\Omega(\nu; E_1, \dots, E_m) \longleftrightarrow \mathcal{C}(\nu; E_1, \dots, E_m).$$

Consequently

$$|\Omega(\nu; E_1, \dots, E_m)| = |\mathcal{C}(\nu; E_1, \dots, E_m)|.$$

Proof. Let $x = (x_1, \dots, x_m) \in \Omega(\nu; E_1, \dots, E_m)$. By item (1) of definition 4.354, the cells $x_1, \dots, x_m$ are exactly the cells of $U(\nu; E)$, so the displayed rule defines a bijection $\tau_x : U(\nu; E) \to \{1, \dots, m\}$. The recursive chain

$$\nu^{(0)} \subset \nu^{(1)} \subset \cdots \subset \nu^{(2m)} = \nu$$

in item (3) is the sequence of shapes formed by the cells whose assigned labels are at most $1, 2, \dots, 2m$. The successive differences are horizontal strips. By the Pieri-path equivalence lemma 4.97, the corresponding filling is semistandard. Hence $\tau_x \in \mathcal{C}(\nu; E_1, \dots, E_m)$.

Conversely, let $\tau \in \mathcal{C}(\nu; E_1, \dots, E_m)$ and write $x_r = \tau^{-1}(r)$. The definition of $U(\nu; E)$ gives the disjoint-union condition in item (1) of definition 4.354. Since f_r and x_r carry the same label $2r-1$ in a semistandard filling, they cannot lie in the same column; by definition x_r is not one of the forced first-column cells. This gives item (2). Finally, applying lemma 4.97 to the semistandard filling shows that the sublevel shapes for labels $1, \dots, 2m$ are Young diagrams and that each successive difference is the required horizontal two-strip. Thus $(x_1, \dots, x_m) \in \Omega(\nu; E_1, \dots, E_m)$.

The two constructions are inverse because x_r is, in both directions, the unique unfilled cell carrying the odd label $2r - 1$. The cardinality identity follows. ☐

Corollary 4.370 (Fixed-cell completion bound implies the unrefined Omega bound)**.** *If every tight-prefix datum satisfies*

$$|\mathcal{C}(\nu; E_1, \dots, E_m)| \le 2^m,$$

then every tight-prefix datum satisfies

$$|\Omega(\nu; E_1, \dots, E_m)| \le 2^m.$$

Proof. This is immediate from lemma 4.369. ☐

Lemma 4.371 (The first column is forced in any fixed-cell completion)**.** *Let $(\nu; E_1, \dots, E_m)$ be a tight-prefix datum with $\mathcal{C}(\nu; E_1, \dots, E_m) \ne \varnothing$. In every completion, the first-column cell in row i carries the label i. Consequently the height of ν is either $2m - 1$ or $2m$; for $1 \le r < m$ one has*

$$(2r, 1) \in E_r,$$

and, if $(2m, 1)$ belongs to ν, then $(2m, 1) \in E_m$.

Proof. Fix a completion and let T be its semistandard filling. Since $f_m = (2m - 1, 1)$ belongs to ν, the first column has height at least $2m - 1$. The entries in a column of a semistandard tableau are strictly increasing from top to bottom, and the forced cells satisfy

$$T(2r - 1, 1) = 2r - 1 \qquad (1 \le r \le m).$$

For $1 \le r < m$, the cell $(2r, 1)$ lies between the forced cells $(2r - 1, 1)$ and $(2r + 1, 1)$ in the same column. Its label is therefore strictly between $2r - 1$ and $2r + 1$, hence is exactly $2r$. The only cells with label $2r$ are the two cells of E_r, so $(2r, 1) \in E_r$.

If a row $2m$ is present, the first-column entry in that row is strictly larger than $2m - 1$. Since all labels are at most $2m$, it is $2m$, and so $(2m, 1) \in E_m$. A row below $2m$ would require a first-column label larger than $2m$, impossible. Thus the height is either $2m - 1$ or $2m$, and the first-column labels are exactly $1, 2, \dots, h$. ☐

Definition 4.372 (Reduced fixed-even growth chains)**.** Assume that $\mathcal{C}(\nu; E_1, \dots, E_m) \neq \varnothing$. For a non-first-column box $c = (i, j)$ with $j > 1$, write

$$\overline{c} = (i, j-1).$$

Let $\overline{\nu}$ be the diagram obtained from ν by deleting the first column, and put

$$\overline{E}_r = \{\overline{c} : \ c \in E_r, \ c \neq (2r, 1)\} \qquad (1 \leq r \leq m).$$

Let $\varepsilon_r = 1$ if $(2r, 1) \in E_r$, and $\varepsilon_r = 0$ otherwise. By lemma 4.371, $\varepsilon_r = 1$ for $r < m$, while $\varepsilon_m \in \{0, 1\}$.

The set

$$\mathcal{R}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)$$

consists of tuples

$$y = (y_1, \dots, y_m)$$

of boxes of $\overline{\nu}$ such that the singleton sets $\{y_r\}$ and the fixed sets $\overline{E}_r$ are pairwise disjoint and have union $\overline{\nu}$, and such that the recursively defined sets

$$\alpha_0 = \varnothing, \qquad \alpha_{r-\frac{1}{2}} = \alpha_{r-1} \cup \{y_r\}, \qquad \alpha_r = \alpha_{r-\frac{1}{2}} \cup \overline{E}_r$$

are Young diagrams for every r. In addition, $\alpha_{r-\frac{1}{2}}$ has at most $2r-1$ rows, α_r has at most $2r-1+\varepsilon_r$ rows, and each difference $\alpha_r \setminus \alpha_{r-\frac{1}{2}}$ is a horizontal strip. The set is ordered lexicographically by the tuple $y_1, \dots, y_m$ in row-column order.

Proposition 4.373 (Fixed-cell completions equal reduced growth chains)**.** *For every tight-prefix datum with $\mathcal{C}(\nu; E_1, \dots, E_m) \neq \varnothing$, the map*

$$\tau \longmapsto \left(\overline{\tau^{-1}(1)}, \dots, \overline{\tau^{-1}(m)}\right)$$

is a bijection

$$\mathcal{C}(\nu; E_1, \dots, E_m) \longleftrightarrow \mathcal{R}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m).$$

Consequently

$$|\mathcal{C}(\nu; E_1, \dots, E_m)| = |\mathcal{R}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)|.$$

Proof. Let τ be a fixed-cell completion and let T_τ be the corresponding semistandard filling. For each r, the cell $\tau^{-1}(r)$ is off the first column: otherwise the first column would contain two cells with the same label $2r-1$, namely f_r and $\tau^{-1}(r)$, contradicting strict column increase. By lemma 4.97, the sublevel sets of labels $1, \dots, 2m$ form a Pieri growth path. By lemma 4.371, the first-column cells in that path are exactly the cells with labels $1, 2, \dots, h$, where $h \in \{2m-1, 2m\}$. Deleting the first column from each sublevel diagram therefore leaves a sequence of Young diagrams. The odd step $2r-1$ loses the forced first-column cell f_r and retains exactly the single box $\overline{\tau^{-1}(r)}$; the even step $2r$ retains exactly the off-column fixed strip $\overline{E}_r$. Hence the tuple displayed in the statement belongs to $\mathcal{R}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)$.

Conversely, let $y \in \mathcal{R}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)$. Lift each $y_r = (i, j)$ to the box $x_r = (i, j+1)$ of ν. Start with the empty diagram. At odd step $2r-1$, adjoin $f_r = (2r-1, 1)$ and x_r; at even step $2r$, adjoin the two boxes of E_r. The row-support conditions in definition 4.372 say exactly that, after the appropriate first column of height $2r-1$ or $2r-1+\varepsilon_r$ is restored, the reduced diagrams $\alpha_{r-\frac{1}{2}}$ and α_r lift to Young diagrams. The odd differences are horizontal because x_r is off the first column, and the even differences are horizontal by the last condition in the definition of $\mathcal{R}$. Thus the lifted sequence is a Pieri path ending at ν.

Applying lemma 4.97 to this Pieri path gives a semistandard filling of ν in which f_r and x_r carry label $2r-1$ and the cells of E_r carry label $2r$. This filling defines an element of $\mathcal{C}(\nu; E_1, \dots, E_m)$ by setting $\tau(x_r) = r$. The two constructions are inverse, since deleting and restoring the first column does not change any off-column box. The cardinality identity follows. $\square$

Corollary 4.374 (Reduced growth-chain bound closes the fixed-cell leaf). *Suppose that every tight-prefix datum with $\mathcal{C}(\nu; E_1, \ldots, E_m) \neq \varnothing$ satisfies*

$$|\mathcal{R}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)| \leq 2^m.$$

Then every tight-prefix datum satisfies

$$|\mathcal{C}(\nu; E_1, \ldots, E_m)| \leq 2^m.$$

Proof. If $\mathcal{C}(\nu; E_1, \ldots, E_m) = \varnothing$, the conclusion is immediate. Otherwise it follows from proposition 4.373. ☐

Lemma 4.375 (Only the terminal reduced even strip may have size two). *For every reduced fixed-even datum arising from a nonempty fixed-cell completion set,*

$$|\overline{E}_r| = 1 \qquad (1 \leq r < m),$$

and

$$|\overline{E}_m| = 2 - \varepsilon_m \in \{1, 2\}.$$

Proof. Each original even strip E_r has exactly two boxes. By lemma 4.371, $(2r, 1) \in E_r$ for $1 \leq r < m$. Deleting the first column therefore removes exactly one box of E_r, leaving $|\overline{E}_r| = 1$.

For $r = m$, the first-column box $(2m, 1)$ belongs to E_m exactly when $\varepsilon_m = 1$. Hence deleting the first column removes ε_m boxes from the two-box strip E_m, leaving $|\overline{E}_m| = 2 - \varepsilon_m$. ☐

Definition 4.376 (Reverse reduced corner tree). Fix a reduced fixed-even datum $(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)$ arising from a nonempty fixed-cell completion set, with ε_r as in definition 4.372. A reverse reduced corner path is a sequence

$$\overline{\nu} = \beta_m \supset \beta_{m-1} \supset \cdots \supset \beta_0 = \varnothing$$

together with boxes $y_m, \ldots, y_1$ such that, for each r,

$$\beta_{r-\frac{1}{2}} := \beta_r \setminus \overline{E}_r$$

is a Young diagram, $\overline{E}_r$ has at most one box in each column of $\beta_r \setminus \beta_{r-\frac{1}{2}}$,

$$\ell(\beta_{r-\frac{1}{2}}) \leq 2r - 1, \qquad \ell(\beta_r) \leq 2r - 1 + \varepsilon_r,$$

the box y_r is a removable corner of $\beta_{r-\frac{1}{2}}$, and

$$\beta_{r-1} = \beta_{r-\frac{1}{2}} \setminus \{y_r\}.$$

Let

$$\mathcal{T}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)$$

be the set of such reverse reduced corner paths.

Lemma 4.377 (Reduced chains are reverse corner paths). *For every reduced fixed-even datum arising from a nonempty fixed-cell completion set, the assignment sending*

$$y = (y_1, \ldots, y_m) \in \mathcal{R}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)$$

to the reverse sequence

$$\beta_r = \alpha_r(y) \qquad (0 \leq r \leq m)$$

is a bijection

$$\mathcal{R}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m) \longleftrightarrow \mathcal{T}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m).$$

Consequently

$$|\mathcal{R}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)| = |\mathcal{T}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)|.$$

Proof. Let $y \in \mathcal{R}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)$, and let $\alpha_0, \alpha_{1/2}, \alpha_1, \dots, \alpha_m$ be the forward diagrams in definition 4.372. Put $\beta_r = \alpha_r$. Since

$$\alpha_r = \alpha_{r-\frac{1}{2}} \cup \overline{E}_r$$

and $\alpha_r \setminus \alpha_{r-\frac{1}{2}}$ is a horizontal strip, deleting $\overline{E}_r$ from β_r gives the Young diagram $\beta_{r-\frac{1}{2}} = \alpha_{r-\frac{1}{2}}$ and removes at most one box from each column. Since

$$\alpha_{r-\frac{1}{2}} = \alpha_{r-1} \cup \{y_r\}$$

is a Young diagram, deleting y_r from it gives the Young diagram α_{r-1}; hence y_r is a removable corner of $\beta_{r-\frac{1}{2}}$. Thus the reversed data define an element of $\mathcal{T}$.

Conversely, a reverse reduced corner path determines boxes $y_1, \dots, y_m$ by its removals. Reversing the displayed recurrence gives

$$\alpha_{r-\frac{1}{2}} = \alpha_{r-1} \cup \{y_r\}, \qquad \alpha_r = \alpha_{r-\frac{1}{2}} \cup \overline{E}_r,$$

and the hypotheses in definition 4.376 say exactly that these sets are Young diagrams, satisfy the same row-support bounds, and that the fixed even difference is a horizontal strip. The disjoint-union condition follows because the reverse path removes every box of $\overline{\nu}$ exactly once, either as one of the fixed sets $\overline{E}_r$ or as one of the corners y_r. Hence the tuple belongs to $\mathcal{R}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)$.

The two constructions are inverse term by term, so the cardinality identity follows. □

Corollary 4.378 (Reverse corner tree bound closes the reduced leaf)**.** *Suppose that every reduced fixed-even datum arising from a nonempty fixed-cell completion set satisfies*

$$|\mathcal{T}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)| \le 2^m.$$

Then every such datum satisfies

$$|\mathcal{R}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)| \le 2^m.$$

Proof. This is immediate from lemma 4.377. □

Definition 4.379 (Odd fixed-even standard tableaux)**.** Let $(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)$ be a reduced fixed-even datum arising from a nonempty fixed-cell completion set. Put

$$\gamma = \overline{\nu} \setminus \overline{E}_m.$$

By nonemptiness and definition 4.376, γ is a Young diagram. For $1 \le r < m$, let e_r be the unique cell of $\overline{E}_r$, whose existence is supplied by lemma 4.375.

Let

$$\mathcal{S}(\gamma; e_1, \dots, e_{m-1})$$

be the set of standard Young tableaux of shape γ with entries $1, \dots, 2m-1$ such that the cell e_r carries entry $2r$ for every $1 \le r < m$.

The same notation will also be used for any Young diagram δ of size $2r-1$ together with distinct cells $f_1, \dots, f_{r-1} \in \delta$: then $\mathcal{S}(\delta; f_1, \dots, f_{r-1})$ denotes the set of standard Young tableaux of shape δ with $2s$ in f_s for $1 \le s < r$. If the fixed entries are incompatible with standardness, this set is empty.

Lemma 4.380 (Reverse trees are odd fixed-even standard tableaux)**.** *For every reduced fixed-even datum arising from a nonempty fixed-cell completion set, the map sending a reverse reduced corner path to the tableau whose cell y_r carries $2r-1$ and whose cell e_r carries $2r$ for $1 \le r < m$ is a bijection*

$$\mathcal{T}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m) \longleftrightarrow \mathcal{S}(\gamma; e_1, \dots, e_{m-1}).$$

Consequently

$$|\mathcal{T}(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)| = |\mathcal{S}(\gamma; e_1, \dots, e_{m-1})|.$$

Proof. Let a reverse reduced corner path be given. Its first step removes the fixed terminal strip $\overline{E}_m$ from $\overline{\nu}$ and gives the Young diagram $\gamma = \beta_{m-\frac{1}{2}}$. Thereafter the reverse removals are, in order,

$$y_m,\ e_{m-1},\ y_{m-1},\ e_{m-2}, \ldots, e_1,\ y_1.$$

Equivalently, the forward construction of γ adds

$$y_1,\ e_1,\ y_2,\ e_2, \ldots, y_{m-1},\ e_{m-1},\ y_m.$$

Every intermediate set is a Young diagram by definition 4.376. Therefore placing $2r-1$ in y_r for $1 \le r \le m$ and $2r$ in e_r for $1 \le r < m$ gives a standard Young tableau of shape γ, with the required fixed even entries.

Conversely, let $S \in \mathcal{S}(\gamma; e_1, \ldots, e_{m-1})$. Let y_r be the cell carrying $2r-1$. The sublevel sets of a standard Young tableau are Young diagrams. Reading these sublevel diagrams in reverse shows that the cells

$$y_m,\ e_{m-1},\ y_{m-1}, \ldots, e_1,\ y_1$$

are removable in that order from γ. Restoring the fixed terminal strip $\overline{E}_m$ to γ gives the initial diagram $\overline{\nu}$, and the datum already supplies that this restoration is the valid terminal fixed horizontal strip of a reduced corner path. Thus the same cells define an element of $\mathcal{T}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)$.

The two constructions are inverse because the cells carrying the odd entries and the fixed even entries are read off uniquely. The cardinality identity follows. □

Corollary 4.381 (Odd fixed-even standard-tableau bound closes the reverse tree)**.** *Suppose that every datum of definition 4.379 satisfies*

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| \le 2^m.$$

Then every reduced fixed-even datum arising from a nonempty fixed-cell completion set satisfies

$$|\mathcal{T}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)| \le 2^m.$$

Proof. This is immediate from lemma 4.380. □

Definition 4.382 (Odd-cell interval extension poset)**.** Let $(\gamma; e_1, \ldots, e_{m-1})$ be a datum of definition 4.379. Put

$$U = \gamma \setminus \{e_1, \ldots, e_{m-1}\}.$$

Order boxes by the Young order:

$$(i,j) \preceq (i',j') \quad\iff\quad i \le i' \text{ and } j \le j'.$$

For $u \in U$, define

$$L(u) = 1 + \max\bigl(\{0\} \cup \{s:\ e_s \preceq u\}\bigr), \qquad R(u) = \min\bigl(\{m\} \cup \{s:\ u \preceq e_s\}\bigr).$$

Let

$$\mathcal{L}(\gamma; e_1, \ldots, e_{m-1})$$

be the set of bijections $\phi : U \to \{1, \ldots, m\}$ such that

$$L(u) \le \phi(u) \le R(u) \qquad (u \in U),$$

and, whenever $u \prec v$ in the Young order on U,

$$\phi(u) < \phi(v).$$

Lemma 4.383 (Fixed-even tableaux are interval-constrained extensions)**.** *For every datum of definition 4.379, the map*

$$S \longmapsto \phi_S, \qquad S(u) = 2\phi_S(u) - 1 \quad (u \in U),$$

is a bijection

$$\mathcal{S}(\gamma; e_1, \ldots, e_{m-1}) \longleftrightarrow \mathcal{L}(\gamma; e_1, \ldots, e_{m-1}).$$

Consequently

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| = |\mathcal{L}(\gamma; e_1, \ldots, e_{m-1})|.$$

Proof. Let $S \in \mathcal{S}(\gamma; e_1, \dots, e_{m-1})$. The non-fixed cells carry exactly the odd entries $1, 3, \dots, 2m-1$, so the displayed rule defines a bijection $\phi_S : U \to \{1, \dots, m\}$. If $u \prec v$ are both in U, then standardness gives $S(u) < S(v)$, hence $\phi_S(u) < \phi_S(v)$.

If $e_s \preceq u$, then standardness gives

$$2s = S(e_s) < S(u) = 2\phi_S(u) - 1,$$

so $\phi_S(u) \geq s+1$. Taking the largest such s gives $L(u) \leq \phi_S(u)$. If $u \preceq e_s$, then

$$2\phi_S(u) - 1 = S(u) < S(e_s) = 2s,$$

so $\phi_S(u) \leq s$. Taking the smallest such s gives $\phi_S(u) \leq R(u)$. Thus $\phi_S \in \mathcal{L}$.

Conversely, let $\phi \in \mathcal{L}(\gamma; e_1, \dots, e_{m-1})$. Fill e_s with $2s$ and fill $u \in U$ with $2\phi(u) - 1$. The entries are exactly $1, \dots, 2m-1$. If $u \prec v$ both lie in U, then $\phi(u) < \phi(v)$ gives increasing entries. If $e_s \preceq u$, the lower bound $L(u) \leq \phi(u)$ implies $\phi(u) \geq s+1$, so $2s < 2\phi(u) - 1$. If $u \preceq e_s$, the upper bound $\phi(u) \leq R(u)$ implies $\phi(u) \leq s$, so $2\phi(u) - 1 < 2s$. Finally, if $e_s \preceq e_t$, the datum is nonempty, so some tableau in $\mathcal{S}(\gamma; e_1, \dots, e_{m-1})$ exists; applying the previous paragraph to that tableau gives $s < t$. Hence the fixed even entries are also increasing along every comparable pair. The filling is therefore a standard Young tableau in $\mathcal{S}(\gamma; e_1, \dots, e_{m-1})$.

The two constructions are inverse because odd entries have the unique form $2r-1$. The cardinality identity follows. □

Corollary 4.384 (Interval-extension bound closes the fixed-even leaf)**.** *Suppose that every datum of definition 4.382 satisfies*

$$|\mathcal{L}(\gamma; e_1, \dots, e_{m-1})| \leq 2^m.$$

Then every datum of definition 4.379 satisfies

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})| \leq 2^m.$$

Proof. This is immediate from lemma 4.383. □

Definition 4.385 (Last-pair deletion candidates)**.** Let $(\gamma; e_1, \dots, e_{m-1})$ be a datum of definition 4.379, with $m \geq 2$, and put $e = e_{m-1}$. Let

$$Y(\gamma; e)$$

be the set of cells $y \in \gamma \setminus \{e_1, \dots, e_{m-1}\}$ such that y is a removable corner of γ and e is a removable corner of $\gamma \setminus \{y\}$. For $y \in Y(\gamma; e)$, write

$$\gamma_y = \gamma \setminus \{y, e\}.$$

Lemma 4.386 (Last-pair deletion recurrence)**.** *For every datum of definition 4.379 with $m \geq 2$,*

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})| = \sum_{y \in Y(\gamma; e_{m-1})} |\mathcal{S}(\gamma_y; e_1, \dots, e_{m-2})|.$$

Proof. Let $S \in \mathcal{S}(\gamma; e_1, \dots, e_{m-1})$, and let y be the cell carrying the largest entry $2m-1$. Since sublevel sets of a standard Young tableau are Young diagrams, y is a removable corner of γ. After removing y, the largest remaining entry is the fixed entry $2m-2$ in e_{m-1}, so e_{m-1} is a removable corner of $\gamma \setminus \{y\}$. Thus $y \in Y(\gamma; e_{m-1})$. Removing first y and then e_{m-1} leaves a standard tableau of shape γ_y whose fixed even entries are $2, 4, \dots, 2m-4$ at $e_1, \dots, e_{m-2}$.

Conversely, fix $y \in Y(\gamma; e_{m-1})$ and a tableau in $\mathcal{S}(\gamma_y; e_1, \dots, e_{m-2})$. Put $2m-2$ in e_{m-1} and $2m-1$ in y. The definition of $Y(\gamma; e_{m-1})$ says exactly that

$$\gamma_y \subset \gamma_y \cup \{e_{m-1}\} \subset \gamma_y \cup \{e_{m-1}, y\} = \gamma$$

are Young diagrams obtained by adjoining removable corners in the displayed order. Therefore adjoining the two largest entries preserves standardness and gives an element of $\mathcal{S}(\gamma; e_1, \dots, e_{m-1})$.

The two constructions are inverse, because the cell carrying $2m-1$ is uniquely recovered as y. Summing over y gives the displayed recurrence. □

Corollary 4.387 (Blocked last-pair choices are binary)**.** *In the setting of definition 4.385, if $e = e_{m-1}$ is not a removable corner of γ, then*

$$|Y(\gamma; e)| \leq 2.$$

More precisely, every $y \in Y(\gamma; e)$ is either immediately east or immediately south of e.

Proof. Write $e = (i, j)$. Since e is not removable in γ, at least one of $(i, j+1)$ and $(i+1, j)$ belongs to γ. Since e becomes removable after deleting y, every cell of γ immediately east or south of e must be deleted by that same one-cell deletion. Hence y is one of those two adjacent cells. This gives the displayed bound. □

Definition 4.388 (Fixed-even recursive envelope)**.** For a Young diagram γ of odd size, define $W(\gamma)$ recursively as follows. If $\gamma = (1)$, put $W(\gamma) = 1$. If $|\gamma| = 2m - 1$ with $m \geq 2$, and if $e \in \gamma$, let

$$\widehat{Y}(\gamma; e)$$

be the set of cells $y \in \gamma$ such that $y \neq e$, y is a removable corner of γ, and e is a removable corner of $\gamma \setminus \{y\}$. Put

$$W(\gamma) = \max_{e \in \gamma} \sum_{y \in \widehat{Y}(\gamma; e)} W(\gamma \setminus \{y, e\}).$$

Lemma 4.389 (The recursive envelope bounds fixed-even fibers)**.** *For every datum of definition 4.379,*

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| \leq W(\gamma).$$

Proof. We induct on m. For $m = 1$, the shape has one cell and the fixed-even fiber has cardinality at most one, which is $W((1))$.

Assume $m \geq 2$ and put $e = e_{m-1}$. By lemma 4.386,

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| = \sum_{y \in Y(\gamma; e)} |\mathcal{S}(\gamma_y; e_1, \ldots, e_{m-2})|.$$

For each $y \in Y(\gamma; e)$, the child shape is

$$\gamma_y = \gamma \setminus \{y, e\},$$

and the induction hypothesis bounds the corresponding summand by $W(\gamma_y)$. Moreover

$$Y(\gamma; e) \subseteq \widehat{Y}(\gamma; e),$$

because the latter set drops only the requirement that y avoid the earlier fixed even cells. Hence

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| \leq \sum_{y \in \widehat{Y}(\gamma; e)} W(\gamma \setminus \{y, e\}) \leq W(\gamma),$$

by the definition of the maximum in definition 4.388. □

Proposition 4.390 (The recursive fixed-even envelope is full-budget through twenty-eight)**.** *For every Young diagram γ of size $2m - 1$ with $1 \leq m \leq 28$,*

$$W(\gamma) \leq 2^{2m}.$$

Proof. The exact C++ verifier

`character_ring_iter/post_m29_bc_fixed_even_envelope_gmp.cpp`

implements definition 4.388 by dynamic programming over all partitions of $1, 3, \ldots, 55$. For each m, it enumerates every partition γ of $2m - 1$, computes $W(\gamma)$ from the previously computed values at size $2m - 3$, and records

$$\max_{|\gamma| = 2m-1} W(\gamma).$$

The archived 'optimus' run

`certificates/post_m29/bc_fixed_even_envelope_m28_gmp_certificate.log`

has SHA-256

$$\texttt{ff511546d3ffde8c831c4076ca125ff4}$$
$$\texttt{7bc6d0f9f1279f1e2d890a0d925292b0}.$$

The source file has SHA-256

$$\texttt{af7ce7efddaf8fbdc9ff0b54a54935e6}$$
$$\texttt{b1bc96aa58041b335d64e220a2150b17}.$$

The log ends with `__EXIT_STATUS=0` and `SUMMARY failures=0`. In the only range not already covered by the factorial estimate, it reports

m	$\max W(\gamma)$	2^{2m}
13	85787	67108864
14	370729	268435456
15	2037069	1073741824
16	10845675	4294967296
17	54725353	17179869184
18	286275138	68719476736
19	1788942066	274877906944
20	11349507850	1099511627776
21	67061584700	4398046511104
22	403479661259	17592186044416
23	2554342683425	70368744177664
24	18877135179266	281474976710656
25	133746685912501	1125899906842624
26	907074996503025	4503599627370496
27	6344159720142261	18014398509481984
28	45416126476990210	72057594037927936.

Every displayed maximum is at most the corresponding full budget, and the log also checks $1 \le m \le 12$. $\square$

Definition 4.391 (Recursive fixed-even envelope frontiers)**.** For $m \ge 1$, let

$$\mathcal{E}_m^{\mathrm{env}} = \{\gamma \vdash (2m-1) : W(\gamma) > 2^{2m}\}.$$

If $m \ge 2$, $\gamma \vdash (2m-1)$, and $e \in \gamma$, put

$$W^{\mathrm{top}}(\gamma; e) = \sum_{y \in \widehat{Y}(\gamma; e)} W(\gamma \setminus \{y, e\}),$$

and let

$$\mathcal{P}_m^{\mathrm{env}} = \{(\gamma, e) : \gamma \vdash (2m-1),\ e \in \gamma,\ W^{\mathrm{top}}(\gamma; e) > 2^{2m}\}.$$

If $m \ge 3$, $\gamma \vdash (2m-1)$, and $e, f \in \gamma$, put

$$W^{\mathrm{top},2}(\gamma; e, f) = \sum_{\substack{y \in \widehat{Y}(\gamma; e) \\ y \ne f,\ f \in \gamma \setminus \{y, e\}}} W^{\mathrm{top}}(\gamma \setminus \{y, e\}; f),$$

and let

$$\mathcal{P}_m^{\mathrm{env},2} = \{(\gamma, e, f) : \gamma \vdash (2m-1),\ e, f \in \gamma,\ W^{\mathrm{top},2}(\gamma; e, f) > 2^{2m}\}.$$

If $m \geq 4$, $\gamma \vdash (2m-1)$, and $e, f, g \in \gamma$, put

$$W^{\mathrm{top},3}(\gamma; e, f, g) = \sum_{\substack{y \in \widehat{Y}(\gamma;e) \\ y \neq f,\ y \neq g,\ f,g \in \gamma \setminus \{y,e\}}} W^{\mathrm{top},2}(\gamma \setminus \{y, e\}; f, g),$$

and let

$$\mathcal{P}_m^{\mathrm{env},3} = \{(\gamma, e, f, g) : \gamma \vdash (2m-1),\ e, f, g \in \gamma,\ W^{\mathrm{top},3}(\gamma; e, f, g) > 2^{2m}\}.$$

For $d \geq 4$, define recursively

$$W^{\mathrm{top},d}(\gamma; e_1, \dots, e_d) = \sum_{\substack{y \in \widehat{Y}(\gamma;e_1) \\ y \notin \{e_2,\dots,e_d\} \\ e_2,\dots,e_d \in \gamma \setminus \{y,e_1\}}} W^{\mathrm{top},d-1}(\gamma \setminus \{y, e_1\}; e_2, \dots, e_d),$$

and let

$$\mathcal{P}_m^{\mathrm{env},d} = \{(\gamma, e_1, \dots, e_d) : \gamma \vdash (2m-1),\ e_1, \dots, e_d \in \gamma,\ W^{\mathrm{top},d}(\gamma; e_1, \dots, e_d) > 2^{2m}\}.$$

Lemma 4.392 (The top-fixed recursive envelope bounds fixed-even fibers)**.** *For every datum of definition 4.379 with $m \geq 2$,*

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})| \leq W^{\mathrm{top}}(\gamma; e_{m-1}).$$

Proof. Put $e = e_{m-1}$. By lemma 4.386,

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})| = \sum_{y \in Y(\gamma;e)} |\mathcal{S}(\gamma_y; e_1, \dots, e_{m-2})|.$$

For each $y \in Y(\gamma; e)$,

$$\gamma_y = \gamma \setminus \{y, e\},$$

and lemma 4.389 bounds the corresponding lower fiber by $W(\gamma \setminus \{y, e\})$. Since

$$Y(\gamma; e) \subseteq \widehat{Y}(\gamma; e),$$

the displayed recurrence gives

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})| \leq \sum_{y \in \widehat{Y}(\gamma;e)} W(\gamma \setminus \{y, e\}) = W^{\mathrm{top}}(\gamma; e).$$

□

Lemma 4.393 (The top-two recursive envelope bounds fixed-even fibers)**.** *For every datum of definition 4.379 with $m \geq 3$,*

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})| \leq W^{\mathrm{top},2}(\gamma; e_{m-1}, e_{m-2}).$$

Proof. Put $e = e_{m-1}$ and $f = e_{m-2}$. By lemma 4.386,

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})| = \sum_{y \in Y(\gamma;e)} |\mathcal{S}(\gamma_y; e_1, \dots, e_{m-2})|.$$

Every $y \in Y(\gamma; e)$ avoids the earlier fixed even cells, so $y \neq f$, and $f \in \gamma_y = \gamma \setminus \{y, e\}$. Applying lemma 4.392 to the child datum gives

$$|\mathcal{S}(\gamma_y; e_1, \dots, e_{m-2})| \leq W^{\mathrm{top}}(\gamma_y; f).$$

Also $Y(\gamma; e) \subseteq \widehat{Y}(\gamma; e)$. Summing these estimates gives the displayed bound by the definition of $W^{\mathrm{top},2}(\gamma; e, f)$. □

Lemma 4.394 (The top-three recursive envelope bounds fixed-even fibers)**.** *For every datum of definition 4.379 with $m \geq 4$,*

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})| \leq W^{\mathrm{top},3}(\gamma; e_{m-1}, e_{m-2}, e_{m-3}).$$

Proof. Put $e = e_{m-1}$, $f = e_{m-2}$, and $g = e_{m-3}$. The last-pair recurrence gives

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| = \sum_{y \in Y(\gamma;e)} |\mathcal{S}(\gamma_y; e_1, \ldots, e_{m-2})|.$$

Every $y \in Y(\gamma; e)$ avoids f and g, and both cells remain in $\gamma_y = \gamma \setminus \{y, e\}$. By lemma 4.393,

$$|\mathcal{S}(\gamma_y; e_1, \ldots, e_{m-2})| \leq W^{\mathrm{top},2}(\gamma_y; f, g).$$

Using again $Y(\gamma; e) \subseteq \widehat{Y}(\gamma; e)$ and summing gives the asserted $W^{\mathrm{top},3}$ bound. □

Lemma 4.395 (The fixed-depth recursive envelope bounds fixed-even fibers)**.** *Let $d \geq 4$. For every datum of definition 4.379 with $m \geq d + 1$,*

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| \leq W^{\mathrm{top},d}(\gamma; e_{m-1}, e_{m-2}, \ldots, e_{m-d}).$$

Proof. We induct on d. For $d = 4$, put $e = e_{m-1}$. By lemma 4.386,

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| = \sum_{y \in Y(\gamma;e)} |\mathcal{S}(\gamma_y; e_1, \ldots, e_{m-2})|.$$

For $y \in Y(\gamma; e)$, the cells $e_{m-2}, e_{m-3}, e_{m-4}$ remain in $\gamma_y = \gamma \setminus \{y, e\}$. Hence lemma 4.394 applied to the child datum bounds the corresponding summand by

$$W^{\mathrm{top},3}(\gamma_y; e_{m-2}, e_{m-3}, e_{m-4}).$$

Since $Y(\gamma; e) \subseteq \widehat{Y}(\gamma; e)$, summing gives the definition of

$$W^{\mathrm{top},4}(\gamma; e_{m-1}, e_{m-2}, e_{m-3}, e_{m-4}).$$

Assume the assertion for $d-1 \geq 4$. The same last-pair recurrence and survival of the earlier fixed even cells give, for every $y \in Y(\gamma; e)$,

$$|\mathcal{S}(\gamma_y; e_1, \ldots, e_{m-2})| \leq W^{\mathrm{top},d-1}(\gamma_y; e_{m-2}, \ldots, e_{m-d}).$$

Using again $Y(\gamma; e) \subseteq \widehat{Y}(\gamma; e)$, the sum of these bounds is exactly

$$W^{\mathrm{top},d}(\gamma; e_{m-1}, e_{m-2}, \ldots, e_{m-d}),$$

by definition 4.391. □

Lemma 4.396 (The full-depth recursive envelope is the exact fixed-even fiber)**.** *For every datum of definition 4.379 with $m \geq 2$, define*

$$\mathfrak{W}_m(\gamma; e_1, \ldots, e_{m-1}) = \begin{cases} W^{\mathrm{top}}(\gamma; e_1), & m = 2, \\ W^{\mathrm{top},2}(\gamma; e_2, e_1), & m = 3, \\ W^{\mathrm{top},3}(\gamma; e_3, e_2, e_1), & m = 4, \\ W^{\mathrm{top},m-1}(\gamma; e_{m-1}, e_{m-2}, \ldots, e_1), & m \geq 5. \end{cases}$$

Then

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| = \mathfrak{W}_m(\gamma; e_1, \ldots, e_{m-1}).$$

Proof. We induct on m. For $m = 2$, the set $Y(\gamma; e_1)$ of definition 4.385 is exactly $\widehat{Y}(\gamma; e_1)$. For every $y \in Y(\gamma; e_1)$, the child $\gamma \setminus \{y, e_1\}$ has one cell, so its fixed-even fiber and its recursive envelope both have cardinality 1. The last-pair recurrence lemma 4.386 therefore gives the displayed identity with $W^{\mathrm{top}}(\gamma; e_1)$.

Assume $m \geq 3$ and the assertion has been proved for $m - 1$. Put $e = e_{m-1}$. In the last-pair recurrence, the summation set $Y(\gamma; e)$ is exactly the set of cells y appearing in the first summation of the corresponding full-depth envelope: y is a removable corner, e is removable after deleting y, and y avoids all earlier fixed even cells $e_1, \dots, e_{m-2}$, which then survive in the child $\gamma_y = \gamma \setminus \{y, e\}$. For every such child, the induction hypothesis identifies

$$|\mathcal{S}(\gamma_y; e_1, \dots, e_{m-2})|$$

with the lower full-depth envelope: $W^{\mathrm{top}}(\gamma_y; e_1)$ when $m = 3$, $W^{\mathrm{top},2}(\gamma_y; e_2, e_1)$ when $m = 4$, $W^{\mathrm{top},3}(\gamma_y; e_3, e_2, e_1)$ when $m = 5$, and

$$W^{\mathrm{top},m-2}(\gamma_y; e_{m-2}, \dots, e_1)$$

when $m \geq 6$. Substituting these identities into lemma 4.386 gives exactly the defining recursion for $\mathfrak{W}_m(\gamma; e_1, \dots, e_{m-1})$. □

Corollary 4.397 (The full-depth envelope is exact for broad odd-packet classes)**.** *Suppose $m \geq 5$, and suppose target-window data determine*

$$\lambda^{(2m)}, \qquad E_1, \dots, E_m.$$

Let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \dots, E_m).$$

If $\Omega_{\mathrm{tw}} \neq \varnothing$, let $(\gamma; e_1, \dots, e_{m-1})$ be the associated fixed-even datum. Then

$$|\Omega_{\mathrm{tw}}| = W^{\mathrm{top},m-1}(\gamma; e_{m-1}, e_{m-2}, \dots, e_1).$$

Consequently,

$$(\gamma, e_{m-1}, e_{m-2}, \dots, e_1) \notin \mathcal{P}_m^{\mathrm{env},m-1}$$

is equivalent to $|\Omega_{\mathrm{tw}}| \leq 2^{2m}$.

Proof. The cardinality identifications in lemmas 4.369, 4.377 and 4.380 and proposition 4.373 give

$$|\Omega_{\mathrm{tw}}| = |\mathcal{S}(\gamma; e_1, \dots, e_{m-1})|.$$

Since $m \geq 5$, lemma 4.396 identifies the right side with

$$W^{\mathrm{top},m-1}(\gamma; e_{m-1}, e_{m-2}, \dots, e_1).$$

The final equivalence is then the definition of $\mathcal{P}_m^{\mathrm{env},m-1}$. □

Proposition 4.398 (The recursive fixed-even envelope has a certified twenty-nine frontier)**.** *Let*

$$\gamma = (11, 9, 8, 7, 6, 5, 4, 3, 2, 1, 1).$$

Then

$$|\mathcal{E}_{29}^{\mathrm{env}}| = 144, \qquad |\mathcal{P}_{29}^{\mathrm{env}}| = 346, \qquad |\mathcal{P}_{29}^{\mathrm{env},2}| = 62, \qquad \mathcal{P}_{29}^{\mathrm{env},3} = \varnothing.$$

For every partition $\eta \vdash 57$ outside $\mathcal{E}_{29}^{\mathrm{env}}$,

$$W(\eta) \leq 2^{58}.$$

For every partition $\eta \vdash 57$ and cell $e \in \eta$ with $(\eta, e) \notin \mathcal{P}_{29}^{\mathrm{env}}$,

$$W^{\mathrm{top}}(\eta; e) \leq 2^{58}.$$

For every partition $\eta \vdash 57$ and cells $e, f \in \eta$ with $(\eta, e, f) \notin \mathcal{P}_{29}^{\mathrm{env},2}$,

$$W^{\mathrm{top},2}(\eta; e, f) \leq 2^{58}.$$

For every partition $\eta \vdash 57$ and cells $e, f, g \in \eta$,

$$W^{\mathrm{top},3}(\eta; e, f, g) \leq 2^{58}.$$

Moreover, $|\gamma| = 57 = 2 \cdot 29 - 1$, γ is not column-even, and

$$W(\gamma) = 367676108829086275 = \max_{\eta \vdash 57} W(\eta) > 288230376151711744 = 2^{58}.$$

Consequently the uniform full-budget assertion $W(\gamma) \le 2^{2m}$ cannot be extended from proposition 4.390 to all non-column-even shapes at $m = 29$ by the same recursive envelope. This proposition is an obstruction to the uniform envelope route and a positive certificate for the complement of the frontier; it gives no lower bound on the actual broad-Omega compatibility classes or on any fixed-even tableau fiber.

Proof. The size identity is immediate by summing the displayed row lengths. The conjugate diagram is again

$$\gamma' = (11, 9, 8, 7, 6, 5, 4, 3, 2, 1, 1),$$

so its first column has odd length. Hence γ is not column-even.

The exact C++ verifier

`character_ring_iter/post_m29_bc_fixed_even_envelope_m29_obstruction_gmp.cpp`

implements the recursion of definition 4.388 by dynamic programming over every partition of $1, 3, \ldots, 57$. It then checks the displayed shape, its non-column-even status, the exact value of $W(\gamma)$, the exact budget 2^{58}, that the displayed shape is a maximizer among partitions of 57, and that exactly 144 partitions of 57 have recursive-envelope value strictly above 2^{58}. It also computes $W^{\mathrm{top}}(\eta; e)$ for every cell e of every partition $\eta \vdash 57$, checks that exactly 346 pairs have top-fixed recursive-envelope value strictly above 2^{58}, and prints one `M29_TOP_PAIR_OVER_BUDGET` line for each such pair. Since $W^{\mathrm{top},2}(\eta; e, f) \le W^{\mathrm{top}}(\eta; e)$, it then computes $W^{\mathrm{top},2}$ for every possible next fixed even cell over the 346 top-fixed pairs and checks that exactly 62 triples remain above budget. Since $W^{\mathrm{top},3}(\eta; e, f, g) \le W^{\mathrm{top},2}(\eta; e, f)$, it then computes $W^{\mathrm{top},3}$ over those 62 triples and checks that no quadruple remains above budget.

The archived 'optimus' run

`certificates/post_m29/bc_fixed_even_envelope_m29_obstruction_gmp_certificate.log`

has SHA-256

`cf30ce672a1511201db50c17ecfe15a9`
`f6431fdb169cb617ef0c668940fd16a1`.

The source file has SHA-256

`2d7b99007d80e10f3a04c0c546ddb3ed`
`8c54ffa8adb5382ea9898a39aadd81b8`.

The log ends with `__EXIT_STATUS=0` and `SUMMARY failures=0`. Its frontier and target lines report

```
M29_OVER_BUDGET count=144, expected_count=144,
target_in_over_budget=yes,
M29_TOP_PAIR_OVER_BUDGET count=346, expected_count=346,
target_has_top_pair_over_budget=yes,
M29_TOP_TWO_OVER_BUDGET count=62, expected_count=62,
M29_TOP_THREE_OVER_BUDGET count=0, expected_count=0,
size=57, column_even=no, W=367676108829086275,
budget=2^58, budget_value=288230376151711744,
exceeds_budget=yes, target_is_m29_maximizer=yes.
```

These exact checks give the asserted cardinalities, the displayed maximum, and the strict inequality. Since the verifier enumerates every partition $\eta \vdash 57$ and records precisely those with $W(\eta) > 2^{58}$, every partition outside $\mathcal{E}_{29}^{\mathrm{env}}$ satisfies $W(\eta) \le 2^{58}$. Since it also enumerates every pair (η, e) with $\eta \vdash 57$ and $e \in \eta$, and records precisely those with $W^{\mathrm{top}}(\eta; e) > 2^{58}$, every pair outside $\mathcal{P}_{29}^{\mathrm{env}}$ satisfies the displayed top-fixed bound. The monotonicity inequalities in the preceding paragraph show that an over-budget top-two triple must lie over an over-budget top pair, and an over-budget top-three quadruple must lie over an over-budget top-two triple. They follow directly from the definitions:

W is the maximum over the top-fixed values, and each deeper fixed-envelope sum is obtained by restricting the preceding summation and replacing each child contribution by a value no larger than the preceding child contribution. The verifier records precisely these remaining over-budget triples and quadruples, giving $|\mathcal{P}_{29}^{\mathrm{env},2}| = 62$ and $\mathcal{P}_{29}^{\mathrm{env},3} = \varnothing$. □

Proposition 4.399 (The fixed-depth recursive envelope closes the thirty frontier)**.** *For $m = 30$,*

$$|\mathcal{E}_{30}^{\mathrm{env}}| = 11090.$$

The maximum recursive-envelope value among partitions of 59 is attained at

$$\gamma = (11, 10, 8, 7, 6, 5, 4, 3, 2, 2, 1),$$

where

$$W(\gamma) = 2963381126459933824 > 1152921504606846976 = 2^{60}.$$

Furthermore the successive over-budget fixed-depth frontiers have cardinalities

d	1	2	3	4	5	6	7
$\lvert\mathcal{P}_{30}^{\mathrm{env},d}\rvert$	44333	92312	60440	5438	408	18	0.

Consequently, for every partition $\eta \vdash 59$ and every seven distinct cells $e_1, \ldots, e_7 \in \eta$,

$$W^{\mathrm{top},7}(\eta; e_1, \ldots, e_7) \leq 2^{60}.$$

Proof. The exact C++ verifier

`character_ring_iter/post_m29_bc_fixed_even_envelope_m30_frontier_gmp.cpp`

implements definitions 4.388 and 4.391 by dynamic programming over every partition of $1, 3, \ldots, 59$. It first computes $W(\eta)$ for every $\eta \vdash 59$, checks the displayed maximum, and checks that exactly 11090 partitions have $W(\eta) > 2^{60}$. It then extends only the over-budget prefixes because

$$W^{\mathrm{top},d+1}(\eta; e_1, \ldots, e_{d+1}) \leq W^{\mathrm{top},d}(\eta; e_1, \ldots, e_d),$$

which follows directly from the fixed-depth recursive definition. Thus every over-budget depth-$(d+1)$ prefix lies over an over-budget depth-d prefix.

The archived 'optimus' run

`certificates/post_m29/bc_fixed_even_envelope_m30_frontier_gmp_certificate.log`

has SHA-256

`b540fe210ff63596bf4ca3b761e32da3`
`8eca42ca0c5340f1c3e25d41f09fb593`.

The source file has SHA-256

`eac9e5697b79ce291b8eabff388e41be`
`7ac0858ea536223f4fe1f3b264bd5d29`.

The log ends with `__EXIT_STATUS=0` and `SUMMARY remaining_count=0 failures=0`. Its exact count lines are

```
M30_OVER_BUDGET_SHAPES count=11090, expected_count=11090,
M30_MAX W=2963381126459933824, expected_W=2963381126459933824,
M30_TOP_DEPTH depth=1 count=44333, expected_count=44333,
M30_TOP_DEPTH depth=2 count=92312, expected_count=92312,
M30_TOP_DEPTH depth=3 count=60440, expected_count=60440,
M30_TOP_DEPTH depth=4 count=5438, expected_count=5438,
M30_TOP_DEPTH depth=5 count=408, expected_count=408,
M30_TOP_DEPTH depth=6 count=18, expected_count=18,
M30_TOP_DEPTH depth=7 count=0, expected_count=0.
```

These exact checks prove the displayed cardinalities and the empty depth-seven frontier. The final inequality is precisely $\mathcal{P}_{30}^{\mathrm{env},7} = \varnothing$. ☐

Proposition 4.400 (The fixed-depth recursive envelope closes the thirty-one frontier)**.** *For $m = 31$,*

$$|\mathcal{E}_{31}^{\mathrm{env}}| = 67038.$$

The maximum recursive-envelope value among partitions of 61 is attained at

$$\gamma = (12, 10, 9, 7, 6, 5, 4, 3, 2, 2, 1),$$

where

$$W(\gamma) = 22734772942354066748 > 4611686018427387904 = 2^{62}.$$

Furthermore the successive over-budget fixed-depth frontiers have cardinalities

d	1	2	3	4	5	6	7	8	9	10
$\lvert\mathcal{P}_{31}^{\mathrm{env},d}\rvert$	331969	1229238	2687932	2263824	499606	49486	6142	610	24	0.

Consequently, for every partition $\eta \vdash 61$ and every ten distinct cells $e_1, \ldots, e_{10} \in \eta$,

$$W^{\mathrm{top},10}(\eta; e_1, \ldots, e_{10}) \leq 2^{62}.$$

Proof. The exact C++ verifier

`character_ring_iter/post_m29_bc_fixed_even_envelope_m31_frontier_gmp.cpp`

implements definitions 4.388 and 4.391 by dynamic programming over every partition of $1, 3, \ldots, 61$. It first computes $W(\eta)$ exactly with GMP integers for every $\eta \vdash 61$, checks the displayed maximum, and checks that exactly 67038 partitions have $W(\eta) > 2^{62}$. For the fixed-depth frontier comparisons it computes the exact capped integer

$$\widetilde{W}^{\mathrm{top},d} = \min\{W^{\mathrm{top},d}, 2^{62} + 1\}$$

by the same recursive summation, stopping a summand only after an exact partial integer sum has exceeded 2^{62}. Therefore

$$\widetilde{W}^{\mathrm{top},d} > 2^{62} \iff W^{\mathrm{top},d} > 2^{62},$$

so the capped computation is exact for every over-budget membership test. As in the $m = 30$ verifier, monotonicity gives

$$W^{\mathrm{top},d+1}(\eta; e_1, \ldots, e_{d+1}) \leq W^{\mathrm{top},d}(\eta; e_1, \ldots, e_d),$$

so every over-budget depth-$(d + 1)$ prefix lies over an over-budget depth-d prefix.

The archived 'optimus' run

`certificates/post_m29/bc_fixed_even_envelope_m31_frontier_gmp_certificate.log`

has SHA-256

`7331b1cdd3ee34cd969d5456fd44ca3f`
`f5f814ba2f983a0043fda5eaa4e6d69f`.

The source file has SHA-256

`654537912ac6e9333a41aa5fcc698e19`
`33e59a9775c1fa1bf1f8e49a826fab0a`.

The log ends with `__EXIT_STATUS=0`, `__SHELL_STATUS=0`, and `SUMMARY remaining_count=0 failures=0`. Its exact count lines are

```
M31_OVER_BUDGET_SHAPES count=67038, expected_count=67038,
M31_MAX W=22734772942354066748, expected_W=22734772942354066748,
M31_TOP_DEPTH depth=1 count=331969, expected_count=331969,
M31_TOP_DEPTH depth=2 count=1229238, expected_count=1229238,
M31_TOP_DEPTH depth=3 count=2687932, expected_count=2687932,
M31_TOP_DEPTH depth=4 count=2263824, expected_count=2263824,
M31_TOP_DEPTH depth=5 count=499606, expected_count=499606,
M31_TOP_DEPTH depth=6 count=49486, expected_count=49486,
M31_TOP_DEPTH depth=7 count=6142, expected_count=6142,
M31_TOP_DEPTH depth=8 count=610, expected_count=610,
M31_TOP_DEPTH depth=9 count=24, expected_count=24,
M31_TOP_DEPTH depth=10 count=0, expected_count=0.
```

These exact checks prove the displayed cardinalities and the empty depth-ten frontier. The final inequality is precisely $\mathcal{P}_{31}^{\mathrm{env},10} = \varnothing$. □

Definition 4.401 (Commuting normal-form signature)**.** Let $(\gamma; e_1, \dots, e_{m-1})$ be a datum of definition 4.379. For $S \in \mathcal{S}(\gamma; e_1, \dots, e_{m-1})$ define its commuting normal-form signature $\kappa(S)$ recursively as follows.

If $m = 1$, the signature is the empty word. If $m \geq 2$, let y_m be the cell carrying $2m-1$ in S, put $e = e_{m-1}$, and let

$$\gamma_{y_m} = \gamma \setminus \{y_m, e\}.$$

Let S^- be the restriction of S to γ_{y_m}, with entries $1, \dots, 2m-3$. The first symbol of $\kappa(S)$ is y_m if e is a removable corner of γ, and is the symbol $*$ otherwise. The remaining symbols are $\kappa(S^-)$, computed for the smaller datum $(\gamma_{y_m}; e_1, \dots, e_{m-2})$.

For a word σ in these symbols, let

$$\mathcal{S}_\sigma(\gamma; e_1, \dots, e_{m-1})$$

denote the set of tableaux in $\mathcal{S}(\gamma; e_1, \dots, e_{m-1})$ with commuting normal-form signature σ.

Lemma 4.402 (One-step recovery of a commuting signature symbol)**.** *Let $m \geq 2$, let $S \in \mathcal{S}(\gamma; e_1, \dots, e_{m-1})$, let y_m be the cell carrying $2m-1$, put $e = e_{m-1}$, and put*

$$\gamma^- = \gamma \setminus \{y_m, e\}.$$

Then the first symbol of $\kappa(S)$ is $*$ if and only if e is not a removable corner of γ. If e is a removable corner of γ, then the first symbol of $\kappa(S)$ is the unique cell in

$$\gamma \setminus (\gamma^- \cup \{e\}).$$

Consequently, a parser that determines γ, e, and, in the commuting case, the lower shape γ^- determines the first symbol of the commuting normal-form signature.

Proof. The first assertion is exactly the recursive definition of definition 4.401: the first symbol is $*$ precisely in the non-removable case, and otherwise it is the top odd cell y_m.

In the removable case,

$$\gamma^- = \gamma \setminus \{y_m, e\}.$$

The cells y_m and e are distinct, because e is a fixed even cell while y_m carries the odd entry $2m-1$. Hence

$$\gamma \setminus (\gamma^- \cup \{e\}) = \{y_m\},$$

which proves the displayed recovery rule. The final sentence follows immediately from the two cases. ☐

Definition 4.403 (Odd sublevel shape chain)**.** Let $Q \in \mathcal{S}(\gamma; e_1, \ldots, e_{m-1})$. For $1 \le r \le m$, define

$$\Gamma_r(Q) = \{c \in \gamma : Q(c) \le 2r - 1\}.$$

Thus

$$\Gamma_1(Q) \subset \Gamma_2(Q) \subset \cdots \subset \Gamma_m(Q) = \gamma$$

is the chain of odd sublevel shapes of the fixed-even standard tableau.

Proposition 4.404 (Odd sublevel shapes determine the commuting signature)**.** *For every datum of definition 4.379, the commuting normal-form signature $\kappa(Q)$ of $Q \in \mathcal{S}(\gamma; e_1, \ldots, e_{m-1})$ is determined by*

$$\gamma, \quad e_1, \ldots, e_{m-1}, \quad \Gamma_1(Q), \ldots, \Gamma_m(Q).$$

Proof. We argue by induction on m. For $m = 1$ the signature is empty by definition 4.401.

Assume $m \ge 2$. Since $\Gamma_m(Q) = \gamma$, the data determine the current shape. They also determine the current fixed even cell $e = e_{m-1}$ and the lower shape

$$\gamma^- = \Gamma_{m-1}(Q).$$

Indeed $\Gamma_{m-1}(Q)$ is exactly the shape obtained from γ by removing the cells carrying the two largest entries $2m - 2$ and $2m - 1$, namely e_{m-1} and the top odd cell. Therefore lemma 4.402 determines the first symbol of $\kappa(Q)$ from the displayed data.

The restriction of Q to $\Gamma_{m-1}(Q)$ is an element of

$$\mathcal{S}(\Gamma_{m-1}(Q); e_1, \ldots, e_{m-2}),$$

and its odd sublevel shape chain is

$$\Gamma_1(Q), \ldots, \Gamma_{m-1}(Q).$$

By the induction hypothesis, these data determine the tail of the commuting normal-form signature. Together with the first symbol just determined, this determines all of $\kappa(Q)$. ☐

Corollary 4.405 (Target-window odd sublevel recovery closes the signature input)**.** *Work in the setting of proposition 4.506. Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the associated fixed-even datum

$$(\gamma; e_1, \ldots, e_{m-1})$$

and the odd sublevel shape chain

$$\Gamma_1, \ldots, \Gamma_m$$

of the fixed-even standard tableau associated to the true odd-packet assignment. Then the hypothesis of proposition 4.506 holds.

Proof. By proposition 4.404, the recovered fixed-even datum together with the recovered odd sublevel shape chain determines the commuting normal-form signature. Thus the deterministic-function hypothesis of proposition 4.506 is satisfied. ☐

Proposition 4.406 (Odd sublevels are reduced half-step diagrams)**.** *Let $(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)$ be a reduced fixed-even datum arising from a nonempty fixed-cell completion set, and let*

$$y = (y_1, \ldots, y_m) \in \mathcal{R}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m).$$

Let

$$\alpha_0, \alpha_{\frac{1}{2}}, \alpha_1, \ldots, \alpha_m$$

be the forward diagrams defined by definition 4.372. Let $Q \in \mathcal{S}(\gamma; e_1, \dots, e_{m-1})$ be the fixed-even standard tableau corresponding to y under

$$\mathcal{R} \longleftrightarrow \mathcal{T} \longleftrightarrow \mathcal{S}$$

from lemmas 4.377 and 4.380. Then, for every $1 \le r \le m$,

$$\Gamma_r(Q) = \alpha_{r-\frac{1}{2}}.$$

Proof. By lemma 4.380, the tableau Q places the odd entry $2s-1$ in y_s for $1 \le s \le m$ and the fixed even entry $2s$ in e_s, the unique cell of $\overline{E}_s$, for $1 \le s < m$. Therefore the sublevel set $\Gamma_r(Q)$ consists exactly of the cells

$$y_1, \dots, y_r, \qquad e_1, \dots, e_{r-1}.$$

On the other hand, the recursion in definition 4.372 gives

$$\alpha_{r-\frac{1}{2}} = \{y_1, \dots, y_r\} \cup \overline{E}_1 \cup \cdots \cup \overline{E}_{r-1}.$$

For $s < m$, lemma 4.375 identifies $\overline{E}_s = \{e_s\}$. Hence the two displayed sets agree for $r < m$.

For $r = m$, the same recursion gives

$$\alpha_{m-\frac{1}{2}} = \overline{\nu} \setminus \overline{E}_m = \gamma,$$

because $\alpha_m = \overline{\nu}$ and $\alpha_m = \alpha_{m-\frac{1}{2}} \cup \overline{E}_m$ is a disjoint union. Also $\Gamma_m(Q) = \gamma$ by definition 4.403. Thus the identity holds for every r. □

Lemma 4.407 (Retained even strips determine the reduced fixed-even datum)**.** *Let $(\nu; E_1, \dots, E_m)$ be a tight-prefix datum with*

$$\mathcal{C}(\nu; E_1, \dots, E_m) \neq \varnothing.$$

If a parser-visible datum determines ν and the labelled retained even strips $E_1, \dots, E_m$, then it determines the reduced fixed-even datum

$$(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m),$$

the indicators $\varepsilon_1, \dots, \varepsilon_m$, and the associated fixed-even standard-tableau datum

$$(\gamma; e_1, \dots, e_{m-1}).$$

Proof. The reduced shape $\overline{\nu}$ is obtained from ν by deleting the first column and shifting every remaining cell one column left. For each retained even strip E_r, the reduced strip is the deterministic set

$$\overline{E}_r = \{\overline{c} : \ c \in E_r, \ c \neq (2r, 1)\},$$

and

$$\varepsilon_r = \begin{cases} 1, & (2r, 1) \in E_r, \\ 0, & (2r, 1) \notin E_r. \end{cases}$$

Thus ν and the labelled strips determine $(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)$ and all ε_r.

Since the completion set is nonempty, lemma 4.375 says that $|\overline{E}_r| = 1$ for $1 \le r < m$. Hence each fixed even cell e_r is the unique cell of $\overline{E}_r$. Finally,

$$\gamma = \overline{\nu} \setminus \overline{E}_m$$

by definition 4.379. Therefore the same data determine $(\gamma; e_1, \dots, e_{m-1})$. □

Lemma 4.408 (Pieri prefix shapes are reduced half-step diagrams)**.** *Work in the non-full tight height setting of corollary 4.320. Let*

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}$$

be the true tight-prefix Pieri shape chain, let

$$\nu = \lambda^{(2m)}$$

be its endpoint, and let E_r be the two boxes carrying the retained even label $2r$. For each r, let x_r be the second box carrying the deleted odd label $2r-1$, so that the first one is $f_r = (2r-1, 1)$, and put $y_r = \overline{x_r}$. Let

$$\alpha_0, \alpha_{\frac{1}{2}}, \alpha_1, \ldots, \alpha_m$$

be the reduced forward diagrams attached to $y = (y_1, \ldots, y_m)$ by definition 4.372. For a Young diagram μ, write

$$\mathrm{del}_1(\mu) = \{\overline{c} : \ c \in \mu, \ c \text{ is not in the first column}\}.$$

Then, for every $1 \le r \le m$,

$$\mathrm{del}_1(\lambda^{(2r-1)}) = \alpha_{r-\frac{1}{2}}, \qquad \mathrm{del}_1(\lambda^{(2r)}) = \alpha_r.$$

Proof. By lemma 4.355, the tuple $(x_1, \ldots, x_m)$ belongs to $\Omega(\nu; E_1, \ldots, E_m)$. Under lemma 4.369 and proposition 4.373, its first-column deletion is exactly the element $y = (y_1, \ldots, y_m)$ of $\mathcal{R}(\overline{\nu}; \overline{E}_1, \ldots, \overline{E}_m)$ used to define the displayed diagrams α.

For a fixed r, the shape $\lambda^{(2r-1)}$ consists of the forced first-column odd boxes $f_1, \ldots, f_r$, the off-first-column odd boxes $x_1, \ldots, x_r$, and the even strips $E_1, \ldots, E_{r-1}$. Deleting the first column removes the forced odd boxes and, from each earlier even strip, removes exactly its first-column member when one is present. Therefore

$$\mathrm{del}_1(\lambda^{(2r-1)}) = \{y_1, \ldots, y_r\} \cup \overline{E}_1 \cup \cdots \cup \overline{E}_{r-1}.$$

This is $\alpha_{r-\frac{1}{2}}$ by the recursion in definition 4.372.

Similarly, $\lambda^{(2r)}$ is obtained from $\lambda^{(2r-1)}$ by adjoining the retained even strip E_r. Hence

$$\mathrm{del}_1(\lambda^{(2r)}) = \{y_1, \ldots, y_r\} \cup \overline{E}_1 \cup \cdots \cup \overline{E}_r = \alpha_r,$$

again by the same recursion. □

Corollary 4.409 (Target-window tight-prefix recovery closes the signature input)**.** *Work in the setting of proposition 4.506. Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the true tight-prefix Pieri shape chain

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}.$$

Then the hypothesis of proposition 4.506 holds.

Proof. The recovered shape chain determines

$$\nu = \lambda^{(2m)}$$

and, for $1 \le r \le m$, the retained even strip

$$E_r = \lambda^{(2r)} \setminus \lambda^{(2r-1)}.$$

By lemma 4.408, the same recovered chain determines the reduced half-step diagrams

$$\alpha_{\frac{1}{2}}, \alpha_{\frac{3}{2}}, \ldots, \alpha_{m-\frac{1}{2}}.$$

Thus the hypotheses of corollary 4.425 are satisfied, and that corollary gives the deterministic commuting-signature input. □

Corollary 4.410 (Target-window standard even chain recovery closes the signature input)**.** *Work in the setting of proposition 4.506, and let $U = \mathrm{std}(T)$ be the doubled standardization of the underlying semistandard tableau. Let*

$$\varnothing = \sigma^{(0)} \subset \sigma^{(1)} \subset \cdots$$

be the standard shape chain of U. Suppose that the data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the even standard-time endpoint chain

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \ldots, \sigma^{(4m)}.$$

Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.169, the even standard-time endpoints are exactly the semistandard Pieri shapes:

$$\sigma^{(2a)} = \lambda^{(a)} \qquad (0 \le a \le 2m).$$

Thus the recovered standard even chain determines the true tight-prefix Pieri shape chain

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}.$$

Applying corollary 4.409 gives the deterministic commuting-signature input. □

Corollary 4.411 (Target-window dual-RSK' carrier recovery closes the signature input)**.** *Work in the setting of proposition 4.506. Use the Step 3 dual-RSK' half-square growth diagram of lemma 4.125 for the underlying semistandard tableau, and let*

$$\varnothing = \sigma^{(0)} \subset \sigma^{(1)} \subset \cdots$$

be the standard shape chain of its doubled standardization. Suppose that the data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a Ferrers subarrangement E of this Step 3 diagram and determine the partition labels on the top/right boundary of E. Suppose moreover that the recovered cell entries and internal corner labels of E determine the even standard-time endpoints

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \ldots, \sigma^{(4m)}$$

by a deterministic readout fixed by the same data. Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.125, the Step 3 diagram is the fourth arbitrary-filling dual-RSK' growth diagram. Since the displayed data determine the shape of E and its top/right boundary labels, corollary 4.124 recovers, by a deterministic backward sweep, every cell entry and every corner label on or inside E. The deterministic readout in the hypothesis therefore recovers the listed even standard-time endpoints. Applying corollary 4.410 gives the deterministic commuting-signature input. □

Corollary 4.412 (Low-label carrier readout closes the signature input)**.** *Work in the setting of proposition 4.506. Use the Step 3 dual-RSK' half-square growth diagram of lemma 4.125 for the underlying semistandard tableau T. Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a Ferrers subarrangement E of this Step 3 diagram and determine the partition labels on the top/right boundary of E. Suppose moreover that the recovered cell entries and internal corner labels of E determine, for each semistandard label $1 \le a \le 2m$, the two boxes of T carrying label a. Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.125, the Step 3 diagram is the fourth arbitrary-filling dual-RSK' growth diagram. The displayed data determine E and its top/right boundary labels, so corollary 4.124 recovers the cell entries and internal corner labels of E. The low-label readout in the hypothesis therefore recovers the two boxes carrying each semistandard label $1, \ldots, 2m$. Applying lemma 4.170 with $M = 2m$ gives

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \ldots, \sigma^{(4m)}.$$

Thus the deterministic readout hypothesis of corollary 4.411 is satisfied, and that corollary gives the deterministic commuting-signature input. □

Corollary 4.413 (Step-2 prefix carrier criterion for target-window signatures)**.** *Work in the setting of proposition 4.506. Use the Step 2 symmetric matrix and the Step 3 dual-RSK' half-square growth diagram from* [8, Theorem 4, Steps 2–3]*. Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a Ferrers subarrangement E of the Step 3 diagram and determine the partition labels on the top/right boundary of E. Suppose moreover that the recovered cell entries and internal corner labels of E determine a Ferrers subarrangement C of the Step 2 first-variant Knuth growth diagram, the cell entries of C, and the left/bottom boundary labels of C, such that the forward boundary sweep on C contains

$$\lambda^{(0)}, \lambda^{(1)}, \dots, \lambda^{(2m)}$$

for the underlying semistandard tableau. Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.125, the Step 3 diagram is the fourth arbitrary-filling dual-RSK' growth diagram. Since the displayed data determine E and its top/right boundary labels, corollary 4.124 recovers the cell entries and internal corner labels of E. The additional hypothesis therefore determines the Step 2 subarrangement C, its cell entries, and its left/bottom boundary labels. Applying lemma 4.171 with $M = 2m$ recovers the two boxes carrying each semistandard label $1, \dots, 2m$. Hence the hypothesis of corollary 4.412 is satisfied, and that corollary gives the deterministic commuting-signature input. □

Corollary 4.414 (Step-3 half-matrix prefix criterion for target-window signatures)**.** *Work in the setting of proposition 4.506. Use the Step 2 symmetric matrix and the Step 3 dual-RSK' half-square growth diagram from* [8, Theorem 4, Steps 2–3]*. Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a Ferrers subarrangement E of the Step 3 diagram and determine the partition labels on the top/right boundary of E. Suppose moreover that the recovered cell entries and internal corner labels of E determine a Ferrers subarrangement C of the Step 2 first-variant Knuth growth diagram and the left/bottom boundary labels of C, such that every non-diagonal cell of C, or its transpose in the Step 2 symmetric matrix, lies in the recovered Step 3 subarrangement E, and such that the forward boundary sweep on C contains

$$\lambda^{(0)}, \lambda^{(1)}, \dots, \lambda^{(2m)}$$

for the underlying semistandard tableau. Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.125, the Step 3 diagram is the fourth arbitrary-filling dual-RSK' growth diagram. Since the displayed data determine E and its top/right boundary labels, corollary 4.124 recovers the cell entries and internal corner labels of E. The additional hypothesis therefore determines C and its left/bottom boundary labels. By lemma 4.127, the recovered Step 3 subfilling determines every Step 2 cell entry of C, using the symmetry of the Step 2 matrix for non-diagonal cells whose transposes lie in E; diagonal entries of C are zero in the Step 2 matrix. Thus the hypotheses of corollary 4.413 are satisfied, and that corollary gives the deterministic commuting-signature input. □

Corollary 4.415 (Step-2 Q-prefix strip criterion for target-window signatures)**.** *Work in the setting of proposition 4.506. Use the Step 2 symmetric matrix and the Step 3 dual-RSK' half-square growth diagram from* [8, Theorem 4, Steps 2–3]*. Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a Ferrers subarrangement E of the Step 3 diagram and determine the partition labels on the top/right boundary of E. Suppose moreover that, for the Step 2 Q-prefix strip C_{2m} of lemma 4.175, every non-diagonal cell of C_{2m}, or its transpose in the Step 2 symmetric matrix, lies in the recovered Step 3 subarrangement E. Then the hypothesis of proposition 4.506 holds.

Proof. The displayed data recover the Step 3 subfilling of E by corollary 4.124. The containment hypothesis and lemma 4.127 therefore determine every non-diagonal Step 2 entry in the Q-prefix strip C_{2m}, and the diagonal entries of that strip are known to be zero. Applying lemma 4.175 with $M = 2m$ recovers

$$\lambda^{(0)}, \lambda^{(1)}, \dots, \lambda^{(2m)}.$$

Then lemma 4.171 recovers the two boxes carrying each semistandard label $1, \dots, 2m$, so the hypothesis of corollary 4.412 is satisfied. That corollary gives the deterministic commuting-signature input. ☐

Corollary 4.416 (Low-label incidence carrier criterion for target-window signatures)**.** *Work in the setting of proposition 4.506. Use the Step 2 symmetric matrix and the Step 3 dual-RSK' half-square growth diagram from* [8, Theorem 4, Steps 2–3]. *Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a Ferrers subarrangement E of the Step 3 diagram and determine the partition labels on the top/right boundary of E. Suppose moreover that E contains the low-label incidence cell set H_{2m} of lemma 4.179. Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.179, the containment $H_{2m} \subseteq E$ implies that every non-diagonal cell of the Step 2 Q-prefix strip C_{2m} has its cell, or its transpose, in E. Hence corollary 4.415 applies and gives the deterministic commuting-signature input. ☐

Corollary 4.417 (Low-label incidence entries close target-window signatures)**.** *Work in the setting of proposition 4.506. Use the Step 2 symmetric matrix and the Step 3 dual-RSK' half-square growth diagram from* [8, Theorem 4, Steps 2–3]. *Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the Step 3 cell entries on every cell of H_{2m}. Then the hypothesis of proposition 4.506 holds.

Proof. Lemma 4.179 shows that the recovered entries on H_{2m} determine every non-diagonal Step 2 entry in the Q-prefix strip C_{2m}, while the diagonal entries of that strip are known zero by lemma 4.127. Therefore lemma 4.175 with $M = 2m$ recovers

$$\lambda^{(0)}, \lambda^{(1)}, \dots, \lambda^{(2m)}.$$

Applying lemma 4.171 then recovers the two boxes carrying each label $1, \dots, 2m$. Hence corollary 4.412 gives the deterministic commuting-signature input. ☐

Corollary 4.418 (Strict-left data reduce the incidence target to low-low entries)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the setting of proposition 4.506, with $Z_L = \{j-2m+1, \dots, j\}$. Let L_{2m} be the low-low cell set of lemma 4.182. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the Step 3 cell entries on every cell of L_{2m}. Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.182 with $M = 2m$,

$$H_{2m} = X_{2m} \sqcup L_{2m}.$$

The same lemma places X_{2m} inside the continuation-column subarrangement J_{2m}, and lemma 4.341 says that $\mathcal{D}_{\mathrm{out}}(P)$ determines all entries of J_{2m}. Hence the full target-window tuple determines all entries on X_{2m}. By hypothesis it also determines all entries on L_{2m}, and therefore all entries on H_{2m}. Applying corollary 4.417 gives the deterministic commuting-signature input. ☐

Corollary 4.419 (Terminal low-label boundary closes target-window signatures)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the setting of proposition 4.506, with* $Z_L = \{j-2m+1, \dots, j\}$*. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the terminal low-label Step 3 boundary segment

$$\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j}.$$

Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.183 with $M = 2m$, the displayed boundary segment determines every Step 3 cell entry on L_{2m}. Therefore the hypothesis of corollary 4.418 is satisfied, and that corollary gives the deterministic commuting-signature input. □

Corollary 4.420 (Branch-free terminal-zone decoding closes target-window signatures)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the setting of proposition 4.506, with* $Z_L = \{a_L, \dots, j\} = \{j-2m+1, \dots, j\}$*. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the original right-boundary zone subpath

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \dots, \rho_{2j}.$$

Then the hypothesis of proposition 4.506 holds.

Proof. The identity $a_L = j - 2m + 1$ gives

$$2a_L - 2 = 2(j-2m).$$

Thus the displayed right-boundary zone subpath is exactly the terminal low-label boundary segment

$$\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j}.$$

Applying corollary 4.419 gives the deterministic commuting-signature input. □

Corollary 4.421 (Strict-left carrier criterion for target-window signatures)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the setting of proposition 4.506, so the final right-boundary zone is*

$$Z_L = \{a_L, \dots, j\} = \{j-2m+1, \dots, j\}.$$

Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on its top/right boundary, and identifies a deterministic readout which, from the recovered cell entries and internal corner labels of E, returns a Step 2 first-variant Knuth subarrangement C and its left/bottom boundary labels, such that every non-diagonal cell of C, or its transpose in the Step 2 symmetric matrix, lies in E, and such that the forward boundary sweep on C contains

$$\lambda^{(0)}, \lambda^{(1)}, \dots, \lambda^{(2m)}.$$

Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.339, the outside datum $\mathcal{D}_{\mathrm{out}}(P)$ determines

$$\rho_0, \rho_1, \dots, \rho_{2(j-2m)}.$$

Therefore the displayed target-window tuple determines every input to the deterministic rule in the hypothesis once it is enlarged by $\mathcal{D}_{\mathrm{out}}(P)$, as in proposition 4.506. It follows that the full tuple

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determines the carrier E, its top/right boundary labels, and the deterministic Step 2 half-matrix prefix readout required in corollary 4.414. Applying corollary 4.414 gives the deterministic commuting-signature input. □

Corollary 4.422 (Strict-left Q-prefix strip criterion for target-window signatures)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the setting of proposition 4.506, with $Z_L = \{j-2m+1, \dots, j\}$. Suppose that, from*

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on its top/right boundary, and verifies that every cell of the non-diagonal part of the Step 2 Q-prefix strip C_{2m} has its cell, or its transpose in the Step 2 symmetric matrix, in E. Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.339, $\mathcal{D}_{\mathrm{out}}(P)$ determines

$$\rho_0, \rho_1, \dots, \rho_{2(j-2m)}.$$

Thus the full target-window tuple determines every input to the deterministic rule. It follows that the tuple determines E, its top/right boundary labels, and the Q-prefix strip containment hypothesis required in corollary 4.415. Applying that corollary gives the deterministic commuting-signature input. □

Corollary 4.423 (Strict-left low-label incidence carrier criterion)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the setting of proposition 4.506, with $Z_L = \{j-2m+1, \dots, j\}$. Suppose that, from*

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on its top/right boundary, and verifies

$$H_{2m} \subseteq E$$

for the low-label incidence cell set of lemma 4.179. Then the hypothesis of proposition 4.506 holds.

Proof. By lemma 4.339, $\mathcal{D}_{\mathrm{out}}(P)$ determines

$$\rho_0, \rho_1, \dots, \rho_{2(j-2m)}.$$

Therefore the full target-window tuple determines the carrier E, its top/right boundary labels, and the inclusion $H_{2m} \subseteq E$. Applying corollary 4.416 gives the deterministic commuting-signature input. □

Corollary 4.424 (Target-window half-step recovery closes the signature input)**.** *Work in the setting of proposition 4.506. Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the associated reduced fixed-even datum

$$(\overline{\nu}; \overline{E}_1, \dots, \overline{E}_m)$$

and the reduced half-step diagrams

$$\alpha_{\frac{1}{2}}, \alpha_{\frac{3}{2}}, \dots, \alpha_{m-\frac{1}{2}}$$

of the true reduced growth chain. Then the hypothesis of proposition 4.506 holds.

Proof. The reduced fixed-even datum determines

$$\gamma = \overline{\nu} \setminus \overline{E}_m \quad \text{and} \quad e_1, \dots, e_{m-1}$$

by definition 4.379. By proposition 4.406, the recovered half-step diagrams are exactly the odd sublevel shapes

$$\Gamma_1, \dots, \Gamma_m$$

of the associated fixed-even standard tableau. Therefore corollary 4.405 applies and gives the required deterministic signature input. □

Corollary 4.425 (Target-window retained-even half-step recovery closes the signature input)**.** *Work in the setting of proposition 4.506. Suppose that the data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the tight-prefix endpoint

$$\nu = \lambda^{(2m)}$$

and the retained even strips

$$E_1, \dots, E_m$$

of the true tight prefix, as well as the reduced half-step diagrams

$$\alpha_{\frac{1}{2}}, \alpha_{\frac{3}{2}}, \dots, \alpha_{m-\frac{1}{2}}$$

of the true reduced growth chain. Then the hypothesis of proposition 4.506 holds.

Proof. The true deleted odd-packet assignment belongs to $\Omega(\nu; E_1, \dots, E_m)$ by lemma 4.355. Hence lemma 4.369 gives

$$\mathcal{C}(\nu; E_1, \dots, E_m) \neq \varnothing.$$

Therefore lemma 4.407 applies to the recovered endpoint and retained strips, so the displayed target-window data determine the associated reduced fixed-even datum. Together with the recovered half-step diagrams, this is exactly the hypothesis of corollary 4.424; applying that corollary gives the deterministic commuting-signature input. □

Proposition 4.426 (Commuting normal-form fibers are binary)**.** *For every datum of definition 4.379 and every signature word σ,*

$$|\mathcal{S}_\sigma(\gamma; e_1, \dots, e_{m-1})| \leq 2^m.$$

Proof. We argue by induction on m. For $m = 1$ the shape has one cell and there is at most one standard tableau, so the bound is immediate.

Assume $m \geq 2$ and put $e = e_{m-1}$. If σ begins with a cell c, then the fiber is empty unless e is removable in γ and $c \in Y(\gamma; e)$. In the nonempty case, deleting the cell c carrying $2m-1$ and then deleting e gives an injection

$$\mathcal{S}_\sigma(\gamma; e_1, \dots, e_{m-1}) \hookrightarrow \mathcal{S}_{\sigma'}(\gamma_c; e_1, \dots, e_{m-2}),$$

where σ' is the tail of σ and $\gamma_c = \gamma \setminus \{c, e\}$. By induction the right-hand side has size at most 2^{m-1}, hence this case is bounded by 2^m.

It remains to treat the case where σ begins with $*$. If e is removable in γ, the fiber is empty by definition of the signature. If e is not removable, then the possible cells carrying $2m-1$ lie in $Y(\gamma; e)$, and

$$|Y(\gamma; e)| \leq 2$$

by corollary 4.387. Partitioning the fiber according to this top cell y and then deleting y and e gives injections into the tail-signature fibers

$$\mathcal{S}_{\sigma'}(\gamma_y; e_1, \dots, e_{m-2}) \qquad (y \in Y(\gamma; e)).$$

Each has size at most 2^{m-1} by induction, and there are at most two choices of y. Thus the $*$-fiber has size at most 2^m. $\square$

Corollary 4.427 (Parser-visible commuting signatures close the refined rank suffix)**.** *Work in the setting of proposition 4.367. Suppose that for every final tight right-boundary zone with $m \geq 9$ deleted singleton packets, the branch-free canonical halo/right-boundary parser data determine the commuting normal-form signature of the fixed-even standard tableau associated to the true odd-packet assignment by*

$$\Omega \longleftrightarrow \mathcal{C} \longleftrightarrow \mathcal{R} \longleftrightarrow \mathcal{T} \longleftrightarrow \mathcal{S}.$$

Then the hypothesis of proposition 4.367 holds. Consequently the tight odd-packet rank hypothesis of proposition 4.353 holds for every non-full tight height right-boundary zone.

Proof. Fix a tight zone with $m \geq 9$ and let κ be the parser-visible commuting normal-form signature supplied by the hypothesis. Define

$$\Omega_A(A(P))$$

to be the set of elements of $\Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$ whose associated fixed-even standard tableau has signature κ under the displayed chain of bijections. The true odd-packet assignment belongs to this set by the choice of κ. The bijections lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify this set with a commuting-normal-form fiber

$$\mathcal{S}_\kappa(\gamma; e_1, \ldots, e_{m-1}),$$

whose cardinality is at most 2^m by proposition 4.426. Thus the $m \geq 9$ hypothesis of proposition 4.367 is satisfied. That proposition combines this case with the already closed $m \leq 8$ cases and gives the claimed rank hypothesis. $\square$

Corollary 4.428 (Target-window refined ambiguity classes close the rank suffix)**.** *Work in the right-boundary auxiliary setting of proposition 4.253, and suppose the last zone $Z_L = \{a_L, \ldots, j\}$ is a non-full tight height zone with $m \geq 9$ deleted singleton packets. Let V_L^* be the replacement window parsed from the repaired path, and let $\mathcal{D}_{\mathrm{out}}(P)$ be the outside datum of proposition 4.253. Suppose the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the tight-prefix endpoint $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, the labelled continuation boxes, and a canonically ordered finite set

$$\Omega_{\mathrm{tw}} \subseteq \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

such that the true residual odd-packet assignment belongs to Ω_{tw} and

$$|\Omega_{\mathrm{tw}}| \leq 2^m.$$

Then the tight odd-packet rank hypothesis of proposition 4.353 holds for the final right-boundary zone.

Proof. The displayed target-window tuple is parser-visible without adding branch words: V_L^* is a subwindow of the already counted repaired path, the endpoint pair is returned by the parser, and $\mathcal{D}_{\mathrm{out}}(P)$ is recovered in proposition 4.253. Use as $D(P)$ the determined tight-prefix endpoint, retained even-label strips, labelled continuation boxes, and the determined set Ω_{tw}. The membership and size hypotheses are exactly those required by corollary 4.366. Applying that corollary gives the final-zone tight odd-packet rank hypothesis. $\square$

Corollary 4.429 (Target-window signature fibers are refined ambiguity classes)**.** *Work in the right-boundary auxiliary setting of proposition 4.253, and suppose the last zone $Z_L = \{a_L, \ldots, j\}$ is a non-full tight height zone with $m \geq 9$ deleted singleton packets. Let V_L^* be the replacement window* *parsed from the repaired path, and let $\mathcal{D}_{\mathrm{out}}(P)$ be the outside datum of proposition 4.253. Suppose the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, the labelled continuation boxes, and the commuting normal-form signature κ of the fixed-even standard tableau associated to the true odd-packet assignment. Define

$$\Omega_{\mathrm{tw}}(\kappa)$$

to be the set of elements of $\Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$ whose image under

$$\Omega \longleftrightarrow \mathcal{C} \longleftrightarrow \mathcal{R} \longleftrightarrow \mathcal{T} \longleftrightarrow \mathcal{S}$$

has commuting normal-form signature κ. Then this $\Omega_{\mathrm{tw}}(\kappa)$ is determined by the target-window data, contains the true residual odd-packet assignment, and satisfies

$$|\Omega_{\mathrm{tw}}(\kappa)| \leq 2^m.$$

Consequently the tight odd-packet rank hypothesis of proposition 4.353 holds for the final right-boundary zone.

Proof. The target-window data determine the ambient set $\Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$ and the signature κ, so they determine the displayed subset. The true odd-packet assignment belongs to it because κ is, by hypothesis, the signature of the associated true fixed-even standard tableau.

The bijections lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify $\Omega_{\mathrm{tw}}(\kappa)$ with a commuting-normal-form fiber

$$\mathcal{S}_\kappa(\gamma; e_1, \ldots, e_{m-1}).$$

By proposition 4.426, this fiber has cardinality at most 2^m. Thus corollary 4.428 applies. □

Corollary 4.430 (Bounded target-window signature lists give full-budget odd-packet classes)**.** *Work in the setting of corollary 4.429. Suppose the target-window data determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, the labelled continuation boxes, and a canonically ordered finite set*

$$\Sigma_{\mathrm{tw}}$$

of commuting normal-form signature words such that the true fixed-even standard tableau has signature in Σ_{tw} and

$$|\Sigma_{\mathrm{tw}}| \leq 2^m.$$

Define $\Omega_{\mathrm{tw}}(\Sigma_{\mathrm{tw}})$ to be the set of elements of $\Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$ whose image under

$$\Omega \longleftrightarrow \mathcal{C} \longleftrightarrow \mathcal{R} \longleftrightarrow \mathcal{T} \longleftrightarrow \mathcal{S}$$

has commuting normal-form signature in Σ_{tw}, ordered by the ambient lexicographic order on $\Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$. Then $\Omega_{\mathrm{tw}}(\Sigma_{\mathrm{tw}})$ is determined by the target-window data, contains the true residual odd-packet assignment, and satisfies

$$|\Omega_{\mathrm{tw}}(\Sigma_{\mathrm{tw}})| \leq 2^{2m}.$$

Proof. The target-window data determine the ambient odd-packet class, the signature list, and the displayed filtering rule, hence determine $\Omega_{\mathrm{tw}}(\Sigma_{\mathrm{tw}})$ with its inherited order. The true residual odd-packet assignment belongs to the set because the true fixed-even standard tableau has signature in Σ_{tw}.

For each $\sigma \in \Sigma_{\mathrm{tw}}$, the bijections lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify the σ-part of $\Omega_{\mathrm{tw}}(\Sigma_{\mathrm{tw}})$ with a subset of

$$\mathcal{S}_\sigma(\gamma; e_1, \ldots, e_{m-1}).$$

By proposition 4.426, each such fiber has cardinality at most 2^m. Summing over the signature list gives

$$|\Omega_{\mathrm{tw}}(\Sigma_{\mathrm{tw}})| \le |\Sigma_{\mathrm{tw}}|\, 2^m \le 2^m \cdot 2^m = 2^{2m}.$$

□

Corollary 4.431 (All-depth overflow realizes more than 2^m signatures)**.** *Work in the setting of corollary 4.397, and assume*

$$(\gamma, e_{m-1}, e_{m-2}, \ldots, e_1) \in \mathcal{P}_m^{\mathrm{env},m-1}.$$

Let $\Sigma(\Omega_{\mathrm{tw}})$ be the set of commuting normal-form signatures realized by the fixed-even standard tableaux corresponding to elements of Ω_{tw}. Then

$$|\Omega_{\mathrm{tw}}| > 2^{2m} \qquad \text{and} \qquad |\Sigma(\Omega_{\mathrm{tw}})| > 2^m.$$

Proof. The first inequality is the contrapositive of the final equivalence in corollary 4.397. If $|\Sigma(\Omega_{\mathrm{tw}})| \le 2^m$, then the elements of Ω_{tw} split into at most 2^m commuting-signature fibers. By proposition 4.426, each fiber has cardinality at most 2^m. Hence

$$|\Omega_{\mathrm{tw}}| \le 2^m \cdot 2^m = 2^{2m},$$

contradicting the first inequality. □

Corollary 4.432 (Target-window retained-box readout supplies the signature fiber)**.** *Work in the setting of corollary 4.429. Suppose the target-window data determine the tight-prefix endpoint $\lambda^{(2m)}$, the two boxes carrying each retained even label*

$$2, 4, \ldots, 2m,$$

the labelled continuation boxes, and the commuting normal-form signature κ of the true fixed-even standard tableau. Then the tight odd-packet rank hypothesis of proposition 4.353 holds for the final right-boundary zone.

Proof. For each $1 \le r \le m$, the two boxes carrying the retained even label $2r$ are exactly the retained strip E_r. Thus the target-window data determine

$$\lambda^{(2m)}, \quad E_1, \ldots, E_m,$$

the labelled continuation boxes, and κ. Applying corollary 4.429 gives the claimed rank hypothesis. □

Corollary 4.433 (Target-window even-chain readout supplies the signature fiber)**.** *Work in the setting of corollary 4.432, and let $U = \mathrm{std}(T)$ be the doubled standardization of the underlying semistandard tableau. Suppose the target-window data determine the even standard-time endpoint chain*

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \ldots, \sigma^{(4m)}$$

and the labelled continuation boxes. Then the tight odd-packet rank hypothesis of proposition 4.353 holds for the final right-boundary zone.

Proof. By lemma 4.169, the recovered even standard-time endpoints determine the true tight-prefix Pieri chain

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}.$$

In particular they determine the endpoint $\lambda^{(2m)}$ and, for $1 \le r \le m$, the retained even-label boxes

$$E_r = \lambda^{(2r)} \setminus \lambda^{(2r-1)}.$$

The proof of corollary 4.409 then gives a deterministic commuting normal-form signature κ from the same recovered prefix chain: the chain determines the retained strips and the reduced half-step diagrams, hence the fixed-even datum and odd sublevel chain. Thus the hypotheses of corollary 4.432 are satisfied. □

Corollary 4.434 (Target-window low-label boxes supply the signature fiber)**.** *Work in the setting of corollary 4.433. Suppose the target-window data determine, for each semistandard label $1 \leq a \leq 2m$, the two boxes carrying label a, and also determine the labelled continuation boxes for labels $2m+1, \ldots, j$. Then the tight odd-packet rank hypothesis of proposition 4.353 holds for the final right-boundary zone.*

Proof. By lemma 4.170 with $M = 2m$, the recovered boxes carrying labels $1, \ldots, 2m$ determine

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \ldots, \sigma^{(4m)}.$$

Together with the recovered labelled continuation boxes, this is exactly the hypothesis of corollary 4.433. Applying that corollary gives the claimed rank hypothesis. □

Corollary 4.435 (Target-window low-label boxes close the rank suffix as a singleton)**.** *Work in the right-boundary auxiliary setting of proposition 4.253, and suppose the last zone $Z_L = \{a_L, \ldots, j\}$ is a non-full tight height zone with $m \geq 9$ deleted singleton packets. Let V_L^* be the replacement window parsed from the repaired path, and let $\mathcal{D}_{\mathrm{out}}(P)$ be the outside datum of proposition 4.253. Suppose the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine, for each semistandard label $1 \leq a \leq 2m$, the two boxes carrying label a, and also determine the labelled continuation boxes for labels $2m+1, \ldots, j$. Then the tight odd-packet rank hypothesis of proposition 4.353 holds for the final right-boundary zone.

Proof. Lemma 4.356 shows that the recovered low-label boxes determine

$$\lambda^{(2m)}, \qquad E_1, \ldots, E_m, \qquad \omega(P) = (x_1, \ldots, x_m) \in \Omega(\lambda^{(2m)}; E_1, \ldots, E_m),$$

where $\omega(P)$ is the true deleted odd-packet assignment. Use the target-window data as the parser-visible datum $D(P)$ and put

$$\Omega(D(P)) = \{\omega(P)\}$$

with its unique order. This set is determined by $D(P)$, contains the true assignment, and has size $1 \leq 2^m$. The same data determine the labelled continuation boxes by hypothesis. Therefore all hypotheses of proposition 4.353 are satisfied. □

Corollary 4.436 (Target-window continuation sweep plus low-label boxes closes the rank suffix)**.** *Work in the setting of corollary 4.435. Suppose the target-window data determine, for each semistandard label $1 \leq a \leq 2m$, the two boxes carrying label a. Suppose moreover that the same data determine a Ferrers subarrangement C of the Step 2 first-variant Knuth growth diagram, the cell entries of C, and the left/bottom boundary labels of C, such that the forward boundary sweep on C contains*

$$\lambda^{(2m)}, \lambda^{(2m+1)}, \ldots, \lambda^{(j)}.$$

Then the tight odd-packet rank hypothesis of proposition 4.353 holds for the final right-boundary zone.

Proof. Lemma 4.173 with $M = 2m$ recovers the two boxes carrying every continuation label $2m + 1, \ldots, j$ from the displayed Step 2 continuation sweep. Together with the recovered low-label boxes, this is exactly the hypothesis of corollary 4.435. Applying that corollary gives the rank hypothesis. □

Corollary 4.437 (Strict-left plus low-low entries close the singleton rank route)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the right-boundary auxiliary setting of proposition 4.253, with $Z_L = \{j - 2m + 1, \ldots, j\}$. Let L_{2m} be the low-low cell set of lemma 4.182. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the Step 3 cell entries on every cell of L_{2m}. Then the tight odd-packet rank hypothesis of proposition 4.353 holds for the final right-boundary zone.

Proof. Put $M = 2m$. By lemma 4.341, the outside datum $\mathcal{D}_{\text{out}}(P)$ determines every Step 3 entry in the continuation-column subarrangement J_M, whose cells are exactly the off-diagonal unordered label pairs incident to some continuation label $M+1, \dots, j$. The hypothesis determines every entry on L_M, the cells whose two incident labels are both at most M.

Every off-diagonal unordered label pair in the Step 3 half-square is in exactly one of these two classes: either both labels are at most M, or at least one label is a continuation label. Hence the target-window data determine every Step 3 cell entry. By lemma 4.127, these entries determine every off-diagonal entry of the Step 2 symmetric matrix, and the diagonal entries are known to be zero. Thus the entries of the whole Step 2 first-variant Knuth diagram are determined.

Applying lemma 4.175 with $M = j$ recovers the complete Pieri boundary path

$$\lambda^{(0)}, \lambda^{(1)}, \dots, \lambda^{(j)}.$$

Then lemma 4.171 with $M = 2m$ recovers the two boxes carrying every label $1, \dots, 2m$, and lemma 4.173 with $M = 2m$ recovers the two boxes carrying every continuation label $2m+1, \dots, j$. The hypotheses of corollary 4.435 are therefore satisfied, and that corollary gives the rank hypothesis. ☐

Corollary 4.438 (Bounded low-low candidates fit the tight branch budget)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the right-boundary auxiliary setting of proposition 4.253, with $Z_L = \{j - 2m + 1, \dots, j\}$. Let L_{2m} be the low-low cell set of lemma 4.182. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a canonically ordered finite set

$$\mathcal{L}_{\text{tw}}$$

of possible Step 3 entry assignments on L_{2m}, that the true assignment belongs to this set, and that

$$|\mathcal{L}_{\text{tw}}| \le 2^m.$$

Then the first m bits of the tight-zone branch word

$$\beta_\gamma \in \{0,1\}^{2m}$$

can encode the rank of the true low-low assignment in $\mathcal{L}_{\text{tw}}$. From the target-window data and β_γ one recovers the full tight-prefix singleton packet data, and the last m bits remain available for the height selected-defect data.

Proof. Use the first m bits of β_γ to encode the rank of the true low-low assignment in the canonical order on $\mathcal{L}_{\text{tw}}$. Since $|\mathcal{L}_{\text{tw}}| \le 2^m$, this selects the true Step 3 entries on L_{2m}.

Put $M = 2m$. By lemma 4.341, the outside datum determines every Step 3 entry in the continuation-column subarrangement J_M. Combining those entries with the selected entries on L_M determines the whole Step 3 half-square, by the same dichotomy used in corollary 4.437: each off-diagonal unordered label pair either has both labels at most M, or is incident to a continuation label.

Lemma 4.127 then determines the whole Step 2 symmetric matrix, with zero diagonal. Applying lemma 4.175 with $M = j$ recovers

$$\lambda^{(0)}, \lambda^{(1)}, \dots, \lambda^{(j)}.$$

Thus lemma 4.171 with $M = 2m$ recovers the boxes carrying labels $1, \dots, 2m$, and lemma 4.173 with $M = 2m$ recovers the boxes carrying continuation labels $2m+1, \dots, j$. Finally, lemma 4.356 recovers the true deleted odd-packet assignment and the tight-prefix endpoint and retained even-label strips.

The last m bits of β_γ have not been used in the low-low rank encoding. By lemma 4.352, these last bits are still exactly the bits used by the height local recovery for the selected defect subset.

Therefore the target-window data and β_γ recover the full tight-prefix singleton packet data while preserving the height selected-defect recovery. □

Corollary 4.439 (Bounded low-low candidates recover the tight right-boundary subpath)**.** *Work in the setting of corollary 4.438, with*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}.$$

Suppose the target-window data determine the canonically ordered set $\mathcal{L}_{\mathrm{tw}}$ from corollary 4.438. Then from

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}, \quad \beta_L \in \{0,1\}^{2m}$$

one recovers the original right-boundary zone subpath

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \dots, \rho_{2j}.$$

Proof. Read the first m bits of β_L as the rank of the true low-low assignment in the canonical order on $\mathcal{L}_{\mathrm{tw}}$. Since $|\mathcal{L}_{\mathrm{tw}}| \le 2^m$, this selects the true Step 3 entries on L_{2m}.

Put $M = 2m$. By lemma 4.341, $\mathcal{D}_{\mathrm{out}}(P)$ determines every Step 3 entry in the continuation-column subarrangement J_M. The cell sets L_M and J_M cover the full Step 3 half-square: an unordered off-diagonal label pair either has both labels at most M, or is incident to a continuation label $M+1, \dots, j$. Hence the displayed data and the selected low-low assignment determine every cell entry of the full Step 3 half-square.

By lemma 4.126, those full Step 3 entries determine the whole boundary sequence

$$\rho_0, \rho_1, \dots, \rho_{2j}.$$

Since $a_L = j - 2m + 1$, the desired right-boundary subpath is the terminal segment

$$\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j} = \rho_{2a_L-2}, \rho_{2a_L-1}, \dots, \rho_{2j}.$$

□

Definition 4.440 (Degree-constrained target-window low-low candidates)**.** Work in the setting of corollary 4.438, and put $M = 2m$. By lemma 4.341, the outside datum determines the Step 3 entries in the continuation-column subarrangement J_M. For $1 \le a \le M$ define

$$c_a = \sum_{t=M+1}^{j} u_{\{a,t\}},$$

where $u_{\{a,t\}}$ is the recovered Step 3 entry on the unordered label pair $\{a, t\}$. Put

$$d_a = 2 - c_a.$$

The degree-constrained target-window low-low candidate set

$$\mathcal{L}_{\deg}(S, Z_L, V_L^*, \mathcal{D}_{\mathrm{out}}(P), \rho_{2a_L-2}, \rho_{2j})$$

is empty if some $d_a < 0$. Otherwise it is the set of all nonnegative integer assignments

$$\ell = (\ell_{ab})_{1 \le a < b \le M}$$

to the low-low cell set L_M satisfying

$$\sum_{b<a} \ell_{ba} + \sum_{a<b\le M} \ell_{ab} = d_a \qquad (1 \le a \le M).$$

It is ordered lexicographically by the list

$$(\ell_{12}, \ell_{13}, \dots, \ell_{M-1,M}).$$

Proposition 4.441 (Strict-left degrees prune the low-low supplier)**.** *In the setting of definition 4.440, the displayed target-window data determine the set $\mathcal{L}_{\text{deg}}$ as a fixed finite combinatorial object. The true Step 3 assignment on L_{2m} belongs to $\mathcal{L}_{\text{deg}}$, and $\mathcal{L}_{\text{deg}}$ is finite. Consequently, to apply corollary 4.438 it is enough to find a canonical target-window subset*

$$\mathcal{L}_{\text{tw}} \subseteq \mathcal{L}_{\text{deg}}$$

which contains the true low-low assignment and has cardinality at most 2^m.

Proof. The target-window data include $\mathcal{D}_{\text{out}}(P)$, and lemma 4.341 recovers from it every Step 3 entry in J_M. Hence the integers c_a and d_a, and therefore the lexicographically ordered set $\mathcal{L}_{\text{deg}}$, are determined by the displayed data.

Let $N = (N_{ab})$ be the true Step 2 symmetric matrix. By lemma 4.127, each recovered Step 3 entry on the unordered pair $\{a, t\}$ is the corresponding off-diagonal entry $N_{at} = N_{ta}$. By lemma 4.128, the a-th row of N has sum 2 and $N_{aa} = 0$. Therefore, for each $1 \le a \le M$,

$$\sum_{\substack{1 \le b \le M \\ b \ne a}} N_{ab} = 2 - \sum_{t=M+1}^{j} N_{at} = d_a.$$

The low-low Step 3 entries of the true tableau are exactly these N_{ab} with $1 \le a < b \le M$, again by lemma 4.127. They therefore satisfy the degree equations in definition 4.440, proving membership.

Finally each d_a is at most 2 for a true target-window datum, and in any case a nonempty candidate set has $0 \le \ell_{ab} \le \min(d_a, d_b)$. Thus only finitely many assignments can satisfy the displayed equations. If a canonical subset $\mathcal{L}_{\text{tw}}$ of this finite ambient set contains the true assignment and has size at most 2^m, then it satisfies the hypotheses of corollary 4.438, which gives the last assertion. $\square$

Proposition 4.442 (Sparse low-low residual degree fits the full final-zone budget)**.** *Work in the setting of definition 4.440. Let*

$$\mathcal{A}_{\text{tw}} = \{1 \le a \le 2m : d_a > 0\}, \qquad r = |\mathcal{A}_{\text{tw}}|, \qquad D = \sum_{a=1}^{2m} d_a.$$

Use the convention $(-1)!! = 1$. If

$$(D-1)!! \le 2^{2m},$$

then the degree-constrained set $\mathcal{L}_{\text{deg}}$ itself is a canonical full-budget low-low candidate set: it is determined by the target-window data, contains the true low-low Step 3 assignment on L_{2m}, and satisfies

$$|\mathcal{L}_{\text{deg}}| \le 2^{2m}.$$

Consequently the final zone satisfies the right-boundary inverse hypothesis of proposition 4.253 by proposition 4.478. In particular the same conclusion holds under the simpler sufficient condition

$$(2r-1)!! \le 2^{2m}.$$

Proof. The target-window data determine the integers d_a, hence determine $\mathcal{A}_{\text{tw}}$, r, and D. By proposition 4.441, they also determine the finite set $\mathcal{L}_{\text{deg}}$, and the true low-low assignment belongs to it.

It remains to bound the size of $\mathcal{L}_{\text{deg}}$. If $\mathcal{L}_{\text{deg}}$ is empty there is nothing to prove, so suppose it is nonempty. Then D is even, since every assignment in $\mathcal{L}_{\text{deg}}$ has total degree twice its number of low-low edges.

For each active label a, create d_a labelled ports $(a, 1), \dots, (a, d_a)$. Given an assignment $\ell \in \mathcal{L}_{\text{deg}}$, order the edge-ends incident to each vertex first by the other endpoint and then by multiplicity, and attach the ordered ports of that vertex to those ordered edge-ends. Pair the two ports belonging to each low-low edge. This gives a perfect matching of the D labelled ports. The construction is injective: from the matched port pairs and their vertex labels one recovers exactly the edge multiplicities ℓ_{ab} for $1 \le a < b \le 2m$.

The number of perfect matchings of D labelled ports is $(D-1)!!$. Hence

$$|\mathcal{L}_{\deg}| \le (D-1)!! \le 2^{2m},$$

with the displayed convention in the case $D = 0$. Taking $\mathcal{L}_{\mathrm{tw}} = \mathcal{L}_{\deg}$ in proposition 4.478 gives the stated right-boundary inverse.

For the final sentence, observe that $D \le 2r$ because $d_a = 2 - c_a$ in definition 4.440, with c_a a sum of nonnegative recovered Step 3 entries. Thus $(D-1)!! \le (2r-1)!!$, and the simpler condition implies the first displayed hypothesis. □

Proposition 4.443 (Degree-two residual labels improve the full final-zone budget)**.** *Work in the setting of definition 4.440. Put*

$$D = \sum_{a=1}^{2m} d_a, \qquad r_2 = \#\{1 \le a \le 2m : d_a = 2\},$$

and use the convention $(-1)!! = 1$*. If*

$$(D-1)!! \le 2^{2m + \lceil r_2/2 \rceil},$$

then the degree-constrained set $\mathcal{L}_{\deg}$ *itself is a canonical full-budget low-low candidate set: it is determined by the target-window data, contains the true low-low Step 3 assignment on* L_{2m}*, and satisfies*

$$|\mathcal{L}_{\deg}| \le 2^{2m}.$$

Consequently the final zone satisfies the right-boundary inverse hypothesis of proposition 4.253 by proposition 4.478.

Proof. The target-window data determine the integers d_a, hence determine D and r_2. By proposition 4.441, they also determine the finite set $\mathcal{L}_{\deg}$, and the true low-low assignment belongs to it.

It remains to bound the size of $\mathcal{L}_{\deg}$. If $\mathcal{L}_{\deg}$ is empty the cardinality bound is trivial, so suppose it is nonempty. Then D is even. For each label a, create d_a labelled ports. Nonemptiness gives $d_a \ge 0$, while $d_a = 2 - c_a$ and c_a is a sum of nonnegative recovered Step 3 entries give $d_a \le 2$. Hence every nonempty degree vector has $d_a \in \{0, 1, 2\}$.

For an assignment $\ell \in \mathcal{L}_{\deg}$, let $\mathcal{P}(\ell)$ be the set of perfect matchings of the D labelled ports which realize the multiplicities ℓ_{ab}: a matched port pair with vertex labels a and b contributes one unit to ℓ_{ab}. These sets $\mathcal{P}(\ell)$ are pairwise disjoint subsets of the set of all perfect matchings of the D ports.

For a fixed ℓ, the number of such labelled-port matchings is

$$|\mathcal{P}(\ell)| = \frac{\prod_{a=1}^{2m} d_a!}{\prod_{1 \le a < b \le 2m} \ell_{ab}!}.$$

The numerator is 2^{r_2}. The denominator is a power of two, with one factor 2 for each doubled edge $\ell_{ab} = 2$. A doubled edge consumes two labels of residual degree 2, so there are at most $\lfloor r_2/2 \rfloor$ doubled edges. Therefore

$$|\mathcal{P}(\ell)| \ge 2^{r_2 - \lfloor r_2/2 \rfloor} = 2^{\lceil r_2/2 \rceil}.$$

The total number of perfect matchings of D labelled ports is $(D-1)!!$. Since the subsets $\mathcal{P}(\ell)$ are disjoint,

$$|\mathcal{L}_{\deg}|\, 2^{\lceil r_2/2 \rceil} \le (D-1)!!.$$

The displayed hypothesis gives

$$|\mathcal{L}_{\deg}| \le 2^{2m}.$$

Taking $\mathcal{L}_{\mathrm{tw}} = \mathcal{L}_{\deg}$ in proposition 4.478 gives the stated right-boundary inverse. □

Definition 4.444 (Exact low-low degree-profile count)**.** For $c \geq 0$, let $\mathsf{C}(c)$ be the number of loopless labelled multigraphs on c vertices in which every vertex has degree 2. Thus $\mathsf{C}(0) = 1$, $\mathsf{C}(1) = 0$, and, for $c \geq 2$,

$$\mathsf{C}(c) = \sum_{s=2}^{c} \binom{c-1}{s-1} \operatorname{cyc}(s)\mathsf{C}(c-s),$$

where $\operatorname{cyc}(2) = 1$ and $\operatorname{cyc}(s) = (s-1)!/2$ for $s \geq 3$.

For $b, c \geq 0$, let $\mathsf{N}(b, c)$ be the number of loopless labelled multigraphs on a fixed vertex set $B \sqcup C$, with $|B| = b$ and $|C| = c$, in which every vertex of B has degree 1 and every vertex of C has degree 2. Equivalently, $\mathsf{N}(b, c) = 0$ for odd b, $\mathsf{N}(0, c) = \mathsf{C}(c)$, and, for even $b \geq 2$,

$$\mathsf{N}(b, c) = (b-1) \sum_{k=0}^{c} \binom{c}{k} k!\, \mathsf{N}(b-2, c-k).$$

Lemma 4.445 (The profile recurrence counts exactly)**.** *The recurrences in definition 4.444 compute the stated labelled multigraph counts.*

Proof. We first count $\mathsf{C}(c)$. In a loopless labelled multigraph whose vertices all have degree 2, every connected component is either a doubled edge on two vertices or a simple cycle on at least three vertices. Look at the component containing a distinguished vertex. If that component has size $s = 2$, the other vertex is chosen in $\binom{c-1}{1}$ ways and the component is the doubled edge. If $s \geq 3$, the other $s-1$ vertices are chosen in $\binom{c-1}{s-1}$ ways and the number of undirected cycles through the distinguished vertex on these s vertices is $(s-1)!/2$. The remaining $c-s$ vertices carry an independent object counted by $\mathsf{C}(c-s)$. Summing over s gives the displayed recurrence, with the initial values $\mathsf{C}(0) = 1$ and $\mathsf{C}(1) = 0$.

Now suppose $b > 0$. A graph with all vertices of B of degree 1 has no component consisting only of degree-2 vertices attached to a degree-1 vertex; every component containing a vertex of B is a path whose two endpoints are vertices of B and whose internal vertices lie in C. In particular b must be even, so $\mathsf{N}(b, c) = 0$ for odd b. For even $b \geq 2$, distinguish one vertex of B. Choose the other endpoint of its path in $b-1$ ways. If the path has k internal degree-2 vertices, choose those vertices in $\binom{c}{k}$ ways and order them along the path in $k!$ ways. Removing this path leaves an independent graph with $b-2$ degree-1 vertices and $c-k$ degree-2 vertices. This gives the displayed recurrence for $\mathsf{N}(b, c)$, with base $\mathsf{N}(0, c) = \mathsf{C}(c)$. The construction is reversible by taking the component of the distinguished B-vertex, so the count is exact. □

Proposition 4.446 (Exact degree profiles fit the full final-zone budget)**.** *Work in the setting of definition 4.440. Put*

$$r_1 = \#\{1 \leq a \leq 2m : d_a = 1\}, \qquad r_2 = \#\{1 \leq a \leq 2m : d_a = 2\}.$$

If

$$\mathsf{N}(r_1, r_2) \leq 2^{2m},$$

then the degree-constrained set $\mathcal{L}_{\deg}$ itself is a canonical full-budget low-low candidate set: it is determined by the target-window data, contains the true low-low Step 3 assignment on L_{2m}, and satisfies

$$|\mathcal{L}_{\deg}| \leq 2^{2m}.$$

Consequently the final zone satisfies the right-boundary inverse hypothesis of proposition 4.253 by proposition 4.478.

Proof. The target-window data determine the integers d_a, hence determine r_1, r_2, and the finite set $\mathcal{L}_{\deg}$. By proposition 4.441, the true low-low assignment belongs to $\mathcal{L}_{\deg}$.

If $\mathcal{L}_{\deg}$ is empty there is nothing to prove. Otherwise the degree equations give $d_a \geq 0$, while $d_a = 2 - c_a$ and the recovered entries c_a are nonnegative give $d_a \leq 2$. Thus every d_a lies in $\{0, 1, 2\}$.

Relabel the vertices with residual degree 1 as B and the vertices with residual degree 2 as C. An assignment

$$\ell = (\ell_{ab})_{1\le a<b\le 2m} \in \mathcal{L}_{\deg}$$

is exactly a loopless labelled multigraph on $B \sqcup C$: the edge multiplicity between a and b is ℓ_{ab}, the degree equations give degree 1 on B and degree 2 on C, and labels with $d_a = 0$ are isolated and irrelevant. Conversely every such multigraph gives one degree-compatible low-low assignment by reading its edge multiplicities. Thus

$$|\mathcal{L}_{\deg}| = \mathsf{N}(r_1, r_2).$$

The displayed hypothesis gives the full-budget bound. Taking $\mathcal{L}_{\mathrm{tw}} = \mathcal{L}_{\deg}$ in proposition 4.478 gives the stated right-boundary inverse. □

Proposition 4.447 (Partial low-low readouts with bounded residual profile fit the full budget)**.** *Work in the setting of definition 4.440, and suppose $m \ge 9$. Let*

$$\mathcal{E}_{\mathrm{tw}} \subseteq \{\{a,b\} : 1 \le a < b \le 2m\}$$

be a target-window-determined set of low-low cells, and suppose the same target-window data determine nonnegative integers

$$p_{\{a,b\}} \qquad (\{a,b\} \in \mathcal{E}_{\mathrm{tw}})$$

which agree with the true Step 3 entries on those cells. Put

$$d_a^{\mathcal{E}} = d_a - \sum_{\substack{1\le b\le 2m \\ b\ne a \\ \{a,b\}\in\mathcal{E}_{\mathrm{tw}}}} p_{\{a,b\}} \qquad (1 \le a \le 2m),$$

and

$$r_1^{\mathcal{E}} = \#\{a : d_a^{\mathcal{E}} = 1\}, \qquad r_2^{\mathcal{E}} = \#\{a : d_a^{\mathcal{E}} = 2\}.$$

If

$$\mathsf{N}(r_1^{\mathcal{E}}, r_2^{\mathcal{E}}) \le 2^{2m},$$

then the target-window data determine a canonical full-budget low-low candidate set containing the true low-low assignment. Consequently the final zone satisfies the right-boundary inverse hypothesis of proposition 4.253 by proposition 4.478.

Proof. Define

$$\mathcal{L}_{\mathrm{part}} = \{\ell \in \mathcal{L}_{\deg} : \ell_{ab} = p_{\{a,b\}} \text{ for every } \{a,b\} \in \mathcal{E}_{\mathrm{tw}}\}.$$

This set is a canonical function of the target-window data, with the inherited lexicographic order from $\mathcal{L}_{\deg}$. The true low-low assignment belongs to $\mathcal{L}_{\deg}$ by proposition 4.441, and it agrees with the displayed partial readout by hypothesis, so it belongs to $\mathcal{L}_{\mathrm{part}}$.

It remains to bound the size. If $\mathcal{L}_{\mathrm{part}}$ is empty there is nothing to prove, so suppose it is nonempty. The agreement equations show that, after removing the already read entries $p_{\{a,b\}}$, every remaining completion has residual degrees $d_a^{\mathcal{E}}$. Since a true completion exists, these residual degrees are nonnegative; because $d_a \le 2$ and the removed entries are nonnegative, they lie in $\{0,1,2\}$.

For $\ell \in \mathcal{L}_{\mathrm{part}}$, form a residual loopless multigraph by keeping edge multiplicity ℓ_{ab} on pairs not in $\mathcal{E}_{\mathrm{tw}}$ and edge multiplicity 0 on pairs in $\mathcal{E}_{\mathrm{tw}}$. This residual graph has degree $d_a^{\mathcal{E}}$ at label a. The map is injective, because the removed multiplicities $p_{\{a,b\}}$ are fixed and adding them back on $\mathcal{E}_{\mathrm{tw}}$ recovers the original ℓ. Therefore $\mathcal{L}_{\mathrm{part}}$ injects into the set of all loopless labelled multigraphs with $r_1^{\mathcal{E}}$ degree-one vertices and $r_2^{\mathcal{E}}$ degree-two vertices. By definition 4.444, that ambient set has cardinality $\mathsf{N}(r_1^{\mathcal{E}}, r_2^{\mathcal{E}})$. The displayed hypothesis gives

$$|\mathcal{L}_{\mathrm{part}}| \le 2^{2m}.$$

Taking $\mathcal{L}_{\mathrm{tw}} = \mathcal{L}_{\mathrm{part}}$ in proposition 4.478 gives the stated right-boundary inverse. □

Corollary 4.448 (Partial low-low readouts below the certified frontier fit the full budget)**.** *Work in the setting of proposition 4.447, with $m \geq 29$. Let $A(m)$ and $B(m)$ be the frontier functions of proposition 4.452, and put*

$$D^{\mathcal{E}} = \sum_{a=1}^{2m} d_a^{\mathcal{E}}.$$

If

$$r_1^{\mathcal{E}} + r_2^{\mathcal{E}} < A(m) \qquad \text{or} \qquad D^{\mathcal{E}} < B(m),$$

then the same full-budget low-low candidate conclusion and right-boundary inverse hold.

Proof. By the definitions of the frontier functions $A(m)$ and $B(m)$ in proposition 4.452,

$$\mathsf{N}(r_1^{\mathcal{E}}, r_2^{\mathcal{E}}) \leq 2^{2m}$$

under either displayed inequality. Applying proposition 4.447 gives the result. □

Corollary 4.449 (Partial low-low readouts with enough edge mass fit the full budget)**.** *Work in the setting of proposition 4.447, with $m \geq 29$, and put*

$$D = \sum_{a=1}^{2m} d_a, \qquad P_{\mathcal{E}} = \sum_{\{a,b\} \in \mathcal{E}_{\text{tw}}} p_{\{a,b\}}.$$

Let $B(m)$ be the total-degree frontier of proposition 4.452. If

$$2P_{\mathcal{E}} > D - B(m),$$

then the same full-budget low-low candidate conclusion and right-boundary inverse hold.

Proof. Every read low-low entry contributes once to each of its two endpoint residual degrees. Therefore

$$D^{\mathcal{E}} = \sum_{a=1}^{2m} d_a^{\mathcal{E}} = D - 2P_{\mathcal{E}}.$$

The displayed inequality gives $D^{\mathcal{E}} < B(m)$. Applying corollary 4.448 gives the result. □

Corollary 4.450 (Incidence-covered low labels below the certified frontier fit the full budget)**.** *Work in the setting of definition 4.440, with $m \geq 29$. Let*

$$U_{\text{tw}} \subseteq \{1, \ldots, 2m\}$$

be a target-window-determined set of low labels. Suppose the same target-window data determine the true Step 3 entries

$$p_{\{a,b\}} \qquad (1 \leq a < b \leq 2m,\ \{a,b\} \cap U_{\text{tw}} \neq \varnothing)$$

on every low-low cell incident to U_{tw}. Let $A(m)$ and $B(m)$ be the frontier functions of proposition 4.452. If

$$\#\{a \notin U_{\text{tw}} : d_a > 0\} < A(m) \qquad \text{or} \qquad \sum_{a \notin U_{\text{tw}}} d_a < B(m),$$

then the target-window data determine a canonical full-budget low-low candidate set containing the true low-low assignment. Consequently the final zone satisfies the right-boundary inverse hypothesis of proposition 4.253 by proposition 4.478.

Proof. Apply proposition 4.447 to the incidence readout

$$\mathcal{E}_{\text{tw}} = \{\{a,b\} : 1 \leq a < b \leq 2m,\ \{a,b\} \cap U_{\text{tw}} \neq \varnothing\}.$$

This is a target-window-determined low-low cell set, and the displayed entries agree with the true Step 3 entries by hypothesis.

Let $d_a^{\mathcal{E}}$ be the residual degrees after this readout. The true low-low assignment belongs to the corresponding partial candidate set, so these residual degrees are nonnegative. For $a \in U_{\mathrm{tw}}$, all low-low cells incident to a have been read. Since the true low-low assignment has low-low degree d_a at a, this gives

$$d_a^{\mathcal{E}} = 0.$$

For $a \notin U_{\mathrm{tw}}$, the removed incident entries are nonnegative, so

$$0 \le d_a^{\mathcal{E}} \le d_a.$$

Thus

$$r_1^{\mathcal{E}} + r_2^{\mathcal{E}} = \#\{a : d_a^{\mathcal{E}} > 0\} \le \#\{a \notin U_{\mathrm{tw}} : d_a > 0\}$$

and

$$D^{\mathcal{E}} = \sum_{a=1}^{2m} d_a^{\mathcal{E}} \le \sum_{a \notin U_{\mathrm{tw}}} d_a.$$

Under either displayed hypothesis, the residual active support or the residual total degree is below the certified frontier. Therefore corollary 4.448 applies to this incidence readout and gives the stated full-budget candidate set and right-boundary inverse. □

Corollary 4.451 (Large incidence-cover readouts fit the full budget)**.** *Work in the setting of corollary 4.450, and put*

$$D = \sum_{a=1}^{2m} d_a, \qquad D(U_{\mathrm{tw}}) = \sum_{a \in U_{\mathrm{tw}}} d_a.$$

If

$$|U_{\mathrm{tw}}| > 2m - A(m) \qquad \text{or} \qquad D(U_{\mathrm{tw}}) > D - B(m),$$

then the target-window data determine a canonical full-budget low-low candidate set containing the true low-low assignment, and the final zone satisfies the right-boundary inverse hypothesis of proposition 4.253.

Proof. If $|U_{\mathrm{tw}}| > 2m - A(m)$, then

$$\#\{a \notin U_{\mathrm{tw}} : d_a > 0\} \le 2m - |U_{\mathrm{tw}}| < A(m).$$

If $D(U_{\mathrm{tw}}) > D - B(m)$, then

$$\sum_{a \notin U_{\mathrm{tw}}} d_a = D - D(U_{\mathrm{tw}}) < B(m).$$

Thus either displayed hypothesis implies one of the two frontier hypotheses in corollary 4.450. Applying that corollary gives the asserted full-budget candidate set and right-boundary inverse. □

Proposition 4.452 (Exact low-low profile frontiers fit the full final-zone budget)**.** *Work in the setting of definition 4.440, with $m \ge 29$, and put*

$$r_1 = \#\{1 \le a \le 2m : d_a = 1\}, \qquad r_2 = \#\{1 \le a \le 2m : d_a = 2\}, \qquad D = \sum_{a=1}^{2m} d_a.$$

If

$$r_1 + r_2 \le 20,$$

then the degree-constrained set $\mathcal{L}_{\deg}$ itself is a canonical full-budget low-low candidate set. Consequently the final zone satisfies the right-boundary inverse hypothesis of proposition 4.253 by proposition 4.478.

More generally, define the exact active-support frontier

$$A(m) = \min\Big(\{\, u + v : 0 \le u, v,\ u + v \le 2m,\ \mathsf{N}(u, v) > 2^{2m} \,\} \cup \{2m + 1\}\Big).$$

If $r_1 + r_2 < A(m)$, then the same full-budget conclusion holds. Hence the remaining exact dense degree-profile branch has $r_1 + r_2 \geq A(m)$.

Define also the exact total-degree frontier

$$B(m) = \min\Big(\{\, u + 2v : 0 \leq u, v,\ u + v \leq 2m,\ \mathsf{N}(u, v) > 2^{2m} \,\} \cup \{4m + 1\}\Big).$$

If $D < B(m)$, then the same full-budget conclusion holds. Hence the remaining exact dense degree-profile branch also has $D \geq B(m)$.

The active-support threshold is sharp for the degree-profile count: at $(r_1, r_2) = (0, 21)$,

$$\mathsf{N}(0, 21) = 487197682625401876 0 > 288230376151711744 = 2^{58}.$$

Proof. The exact C++/GMP verifier

`character_ring_iter/post_m29_bc_low_low_profile_gmp.cpp`

implements the recurrences of definition 4.444 for $0 \leq r_1, r_2 \leq 436$, and checks all $29 \leq m \leq 218$ with $r_1 + r_2 \leq 2m$. In particular, every profile with $r_1 + r_2 \leq 20$ is included already in the $m = 29$ pass. The archived 'optimus' run

`certificates/post_m29/bc_low_low_profile_gmp_certificate.log`

has SHA-256

`cc1b41189d965a5e076d3524647caa5b`
`7f5c5102ca6a1d8b03c623c15c29dc40`.

The source file has SHA-256

`e99bc1a746f569763d13ef4421797019`
`a0162faea4f9b65c0ca5b51e31c3f50f`.

The log ends with `__EXIT_STATUS=0` and `SUMMARY checked_m=190 failures=0`. Its active-support line reports

`ACTIVE20 max_profile=(0,20),`

`max_count=237669544014377896,`

`budget_2^58=288230376151711744,` `active20_failures=0.`

Thus, whenever $r_1 + r_2 \leq 20$,

$$\mathsf{N}(r_1, r_2) \leq 237669544014377896 < 2^{58} \leq 2^{2m},$$

because $m \geq 29$. Applying proposition 4.446 gives the asserted full-budget low-low candidate set and right-boundary inverse.

By the definition of $A(m)$, if $r_1 + r_2 < A(m)$, then the exact profile count $\mathsf{N}(r_1, r_2)$ is at most 2^{2m}. Therefore proposition 4.446 again gives the full-budget conclusion. Likewise, since $D = r_1 + 2r_2$, the definition of $B(m)$ shows that $D < B(m)$ implies $\mathsf{N}(r_1, r_2) \leq 2^{2m}$, and the same proposition gives the full-budget conclusion.

For the residual range used in the post-$m = 29$ B/C tail, the same log reports one `ACTIVE_FRONTIER` and one `DEGREE_FRONTIER` line for every interval on which these exact frontiers are constant, over $29 \leq m \leq 218$. Its summary line records

`global_min_bad_active=21` and `global_min_bad_D=34`,

and the interval lines end at

`ACTIVE_FRONTIER m=218 min_dense_active=88`

and

`DEGREE_FRONTIER m=218 min_dense_D=152`.

These printed values are not additional hypotheses; they are the exact certificate readout of the two frontier functions just defined on the finite residual range.

The same log reports

```
FIRST_DENSE profile=(0,21),
count=4871976826254018760,     exceeds_2^58=yes,
```

which is the displayed sharpness statement. □

Corollary 4.453 (Continuation-heavy profiles fit the full final-zone budget)**.** *Work in the setting of definition 4.440, with $m \geq 29$, on a true final-zone datum. Let $A(m)$ and $B(m)$ be the active-support and total-degree frontiers defined in proposition 4.452. Put*

$$c_a = \sum_{t=2m+1}^{j} u_{\{a,t\}}, \qquad C = \sum_{a=1}^{2m} c_a, \qquad r_0 = \#\{1 \leq a \leq 2m : d_a = 0\}.$$

If

$$r_0 > 2m - A(m) \qquad or \qquad C > 4m - B(m),$$

then the degree-constrained set $\mathcal{L}_{\deg}$ itself is a canonical full-budget low-low candidate set. Consequently the final zone satisfies the right-boundary inverse hypothesis of proposition 4.253 by proposition 4.478.

Equivalently, the remaining exact dense degree-profile branch has

$$r_0 \leq 2m - A(m), \qquad C \leq 4m - B(m).$$

Thus at least $A(m)$ low labels have at most one continuation incidence, and the true low-low subfilling has at least $B(m)/2$ low-low edges.

Proof. Since we are on a true final-zone datum, the true low-low assignment belongs to $\mathcal{L}_{\deg}$ by proposition 4.441. Hence the residual degrees are $d_a \in \{0, 1, 2\}$, with $d_a = 2 - c_a$. Therefore

$$r_1 + r_2 = \#\{a : d_a > 0\} = 2m - r_0$$

and

$$D = \sum_{a=1}^{2m} d_a = \sum_{a=1}^{2m} (2 - c_a) = 4m - C.$$

If $r_0 > 2m - A(m)$, then $r_1 + r_2 < A(m)$, so proposition 4.452 gives the full-budget conclusion. If $C > 4m - B(m)$, then $D < B(m)$, and the same proposition again gives the full-budget conclusion.

Taking the contraposition of these two closed cases gives the displayed conditions in the remaining exact dense branch. For the final sentence, the first condition says that at least $A(m)$ labels have $d_a > 0$, equivalently $c_a \leq 1$. The true low-low assignment has total low-low degree D, so its number of low-low edges is $D/2$; in the remaining branch $D \geq B(m)$, giving at least $B(m)/2$ such edges. □

Corollary 4.454 (Zero-residual incidence covers are already exhausted)**.** *Work in the setting of definition 4.440, with $m \geq 29$, on a true final-zone datum. Let $A(m)$ and $B(m)$ be the frontier functions of proposition 4.452, and put*

$$U_0 = \{1 \leq a \leq 2m : d_a = 0\}, \qquad r_i = \#\{1 \leq a \leq 2m : d_a = i\} \quad (0 \leq i \leq 2), \qquad D = \sum_{a=1}^{2m} d_a.$$

Then the target-window data determine the true Step 3 entries on every low-low cell incident to U_0: all such entries are zero.

Moreover, if the incidence-cover full-budget tests of corollary 4.451 hold for U_0, or if the two complement tests of corollary 4.450 hold for U_0, then the full-budget low-low candidate conclusion already follows from proposition 4.452 and corollary 4.453. Equivalently, in the live exact dense branch where

$$\mathsf{N}(r_1, r_2) > 2^{2m}, \qquad r_0 \leq 2m - A(m), \qquad D \geq B(m),$$

the automatic cover U_0 satisfies none of those incidence-cover full-budget tests.

Proof. The set U_0 is determined by the residual degrees d_a, hence by the target-window data. Let ℓ^{true} be the true low-low Step 3 assignment. By proposition 4.441, it belongs to $\mathcal{L}_{\mathrm{deg}}$. If $a \in U_0$, then the low-low degree of label a in ℓ^{true} is $d_a = 0$. Since all Step 3 entries are nonnegative, every low-low entry incident to a is zero. Thus the target-window data determine those true incident entries.

For U_0, the active complement is exactly the set of labels with positive residual degree, so its size is $r_1 + r_2$. If this is below $A(m)$, then proposition 4.452 gives the full-budget conclusion. The residual degree on the complement is D, because $d_a = 0$ on U_0; if $D < B(m)$, the same proposition gives the full-budget conclusion. The covered-label test is

$$|U_0| = r_0 > 2m - A(m),$$

which is one of the two closed cases in corollary 4.453. Finally,

$$D(U_0) = \sum_{a \in U_0} d_a = 0,$$

so the covered-degree test $D(U_0) > D - B(m)$ is equivalent to $D < B(m)$, again closed by proposition 4.452.

The final sentence is the contraposition of the four closed alternatives just listed, together with the definition of the displayed live branch. ☐

Corollary 4.455 (Zero-only incidence covers are not live sources)**.** *Work in the setting of corollary 4.454, and suppose the live exact dense branch satisfies*

$$\mathsf{N}(r_1, r_2) > 2^{2m}, \qquad r_0 \le 2m - A(m), \qquad D \ge B(m).$$

Let $U_{\mathrm{tw}} \subseteq U_0$. Then its positive-residual projection is empty, and hence it satisfies none of the positive-residual two-threshold incidence-cover source tests

$$|U_+| > r_1 + r_2 - A(m) \qquad \text{or} \qquad \sum_{a \in U_+} d_a > D - B(m), \qquad U_+ = U_{\mathrm{tw}} \cap \{a : d_a > 0\}.$$

Hence every incidence-cover source in the live branch must include at least one positive-residual label.

Proof. Since $U_{\mathrm{tw}} \subseteq U_0$, its positive-residual projection is $U_+ = \varnothing$. By the definition of $A(m)$, the inequality $\mathsf{N}(r_1, r_2) > 2^{2m}$ gives

$$r_1 + r_2 \ge A(m).$$

Thus

$$|U_+| = 0 \le r_1 + r_2 - A(m),$$

so the positive-label-count threshold fails. Also $D \ge B(m)$, so

$$\sum_{a \in U_+} d_a = 0 \le D - B(m),$$

so the positive residual-degree threshold fails. Therefore a live incidence-cover source cannot be supported only on zero-residual labels. ☐

Corollary 4.456 (Incidence covers project to positive residual labels)**.** *Work in the setting of corollary 4.451. Suppose the target-window data determine a set*

$$U_{\mathrm{tw}} \subseteq \{1, \ldots, 2m\}$$

and determine the true Step 3 entries on every low-low cell incident to U_{tw}. Put

$$U_+ = \{a \in U_{\mathrm{tw}} : d_a > 0\}.$$

Then the same data determine U_+ and determine the true Step 3 entries on every low-low cell incident to U_+. Moreover, if any one of the four incidence-cover tests

$$\#\{a \notin U_{\rm tw} : d_a > 0\} < A(m), \qquad \sum_{a \notin U_{\rm tw}} d_a < B(m),$$
$$|U_{\rm tw}| > 2m - A(m), \qquad \sum_{a \in U_{\rm tw}} d_a > D - B(m)$$

holds for $U_{\rm tw}$, then U_+ satisfies at least one of the three positive-residual tests

$$\#\{a \notin U_+ : d_a > 0\} < A(m), \qquad \sum_{a \notin U_+} d_a < B(m), \qquad \sum_{a \in U_+} d_a > D - B(m).$$

Thus every incidence-cover source may be stated, without loss, with $U_{\rm tw} \subseteq \{a : d_a > 0\}$ and with only these three tests.

Proof. The residual degrees d_a are target-window determined in definition 4.440, so U_+ is determined. Since $U_+ \subseteq U_{\rm tw}$, the readout of all cells incident to $U_{\rm tw}$ includes all cells incident to U_+.

The positive labels outside U_+ are exactly the positive labels outside $U_{\rm tw}$. Hence

$$\#\{a \notin U_+ : d_a > 0\} = \#\{a \notin U_{\rm tw} : d_a > 0\}.$$

Also labels in $U_{\rm tw} \setminus U_+$ have residual degree zero, so

$$\sum_{a \notin U_+} d_a = \sum_{a \notin U_{\rm tw}} d_a \qquad \text{and} \qquad \sum_{a \in U_+} d_a = \sum_{a \in U_{\rm tw}} d_a.$$

Therefore the first, second, and fourth original tests imply respectively the first, second, and third positive-residual tests.

It remains only to handle the covered-label test. If $|U_{\rm tw}| > 2m - A(m)$, then

$$\#\{a \notin U_+ : d_a > 0\} = \#\{a \notin U_{\rm tw} : d_a > 0\} \le 2m - |U_{\rm tw}| < A(m),$$

so the first positive-residual test holds. This proves the asserted normalization. □

Corollary 4.457 (Positive incidence covers have two threshold tests)**.** *Work in the setting of corollary 4.456. Put*

$$r_i = \#\{1 \le a \le 2m : d_a = i\} \quad (i = 1, 2), \qquad r_+ = \#\{1 \le a \le 2m : d_a > 0\} = r_1 + r_2, \qquad D = \sum_{a=1}^{2m} d_a.$$

If

$$U_{\rm tw} \subseteq \{1 \le a \le 2m : d_a > 0\},$$

then the disjunction of the three positive-residual incidence-cover tests

$$\#\{a \notin U_{\rm tw} : d_a > 0\} < A(m), \qquad \sum_{a \notin U_{\rm tw}} d_a < B(m), \qquad \sum_{a \in U_{\rm tw}} d_a > D - B(m)$$

is equivalent to the disjunction of the two threshold alternatives

$$|U_{\rm tw}| > r_+ - A(m) \qquad \text{or} \qquad \sum_{a \in U_{\rm tw}} d_a > D - B(m).$$

Proof. Since $U_{\rm tw}$ contains only positive-residual labels,

$$\#\{a \notin U_{\rm tw} : d_a > 0\} = r_+ - |U_{\rm tw}|.$$

Thus the active-complement test is equivalent to

$$|U_{\rm tw}| > r_+ - A(m).$$

Also

$$\sum_{a \notin U_{\mathrm{tw}}} d_a = D - \sum_{a \in U_{\mathrm{tw}}} d_a,$$

so the complement residual-degree test is equivalent to the displayed covered-degree test. This proves the two-threshold form. ☐

Lemma 4.458 (Degree-two half-square fillings have empty endpoints)**.** *Use the Step 3 dual-RSK' half-square in the $m = 0$, content (2^j) specialization of lemma 4.118. Let u be a nonnegative integer filling of the Step 3 half-square, and let N be the symmetric zero-diagonal matrix obtained by reflecting u across the diagonal. If every row of N has sum 2, then the top/right boundary path produced from u by the forward dual-RSK' sweep satisfies*

$$\rho_0 = \rho_{2j} = \varnothing.$$

Proof. The row-sum hypothesis says exactly that the two equal semistandard boundary weights in Krattenthaler's Step 2 construction are (2^j). The matrix is symmetric and has zero diagonal, so it is one of the $m = 0$ inputs in [8, Theorem 4, Steps 2–3]: Step 3 uses one off-diagonal half of this matrix as the dual-RSK' half-square filling, as recorded in lemma 4.127. The boundary path is therefore one of the paths in lemma 4.118 with even-column endpoint parameter $m = 0$. That lemma gives initial and terminal partitions equal to $\varnothing$. ☐

Definition 4.459 (Endpoint-compatible low-low candidates)**.** Work in the setting of definition 4.440, and put $M = 2m$. For

$$\ell = (\ell_{ab})_{1 \le a < b \le M} \in \mathcal{L}_{\mathrm{deg}},$$

form a complete Step 3 half-square filling u^ℓ as follows. On the low-low cell set L_M, put $u^\ell_{\{a,b\}} = \ell_{ab}$. On each cell of the continuation-column cell set J_M, put u^ℓ equal to the entry recovered from $\mathcal{D}_{\mathrm{out}}(P)$ by lemma 4.341. These two subarrangements cover the full Step 3 half-square. Let

$$\rho_0^\ell, \rho_1^\ell, \dots, \rho_{2j}^\ell$$

be the top/right boundary sequence obtained from u^ℓ by the deterministic forward dual-RSK' sweep of lemma 4.126. The endpoint-compatible low-low candidate set

$$\mathcal{L}_{\mathrm{end}} \subseteq \mathcal{L}_{\mathrm{deg}}$$

is the set of those $\ell \in \mathcal{L}_{\mathrm{deg}}$ for which

$$\rho^\ell_{2(j-2m)} = \rho_{2a_L-2} \qquad \text{and} \qquad \rho^\ell_{2j} = \rho_{2j}.$$

It is ordered by the lexicographic order inherited from $\mathcal{L}_{\mathrm{deg}}$.

Proposition 4.460 (Endpoint compatibility gives no extra low-low pruning)**.** *Work in the setting of definition 4.459, on a nonempty true final-zone target-window datum. Then*

$$\mathcal{L}_{\mathrm{end}} = \mathcal{L}_{\mathrm{deg}}.$$

Consequently the endpoint-compatible filter does not give a source beyond the exact degree-profile closures of propositions 4.446 and 4.452.

Proof. The inclusion $\mathcal{L}_{\mathrm{end}} \subseteq \mathcal{L}_{\mathrm{deg}}$ is part of definition 4.459. Conversely take $\ell \in \mathcal{L}_{\mathrm{deg}}$, and form the complete Step 3 half-square filling u^ℓ.

First consider the endpoint at the cut between the strict-left continuation columns and the terminal low-low corner. The entries of u^ℓ on J_M are, by definition, the entries recovered from $\mathcal{D}_{\mathrm{out}}(P)$. The first $j - M$ strict-left columns of the Step 3 half-square are exactly J_M by lemma 4.181. The forward dual-RSK' sweep on this Ferrers subarrangement has fixed empty incoming boundary labels and is deterministic by lemma 4.121. Applying the same sweep to the true filling gives the true strict-left segment recovered in lemma 4.341; hence

$$\rho^{\ell}_{2(j-M)} = \rho_{2(j-M)}.$$

Since $M = 2m$ and $a_L = j - 2m + 1$, this is the first endpoint condition in definition 4.459.

It remains to check the terminal endpoint. From the entries of u^ℓ form the symmetric zero-diagonal matrix N^ℓ whose off-diagonal entries are the corresponding Step 3 half-square entries. For $1 \le a \le M$, the degree equations defining $\mathcal{L}_{\deg}$ give

$$\sum_{b\neq a} N^{\ell}_{ab} = d_a + c_a = 2.$$

For a continuation label $t > M$, every off-diagonal entry incident to t lies in J_M, so those entries are the true recovered entries. The true Step 2 matrix has row sum 2 by lemma 4.128. Thus every row of N^ℓ has sum 2.

Thus N^ℓ satisfies the hypotheses of lemma 4.458. Hence

$$\rho^{\ell}_{2j} = \varnothing = \rho_{2j},$$

where the last equality is the endpoint assertion of lemma 4.118 for the true path.

Both endpoint tests hold for the arbitrary $\ell \in \mathcal{L}_{\deg}$, so $\ell \in \mathcal{L}_{\mathrm{end}}$. This proves the reverse inclusion. The final sentence follows immediately: bounding $\mathcal{L}_{\mathrm{end}}$ is the same as bounding $\mathcal{L}_{\deg}$, which is exactly the degree-profile budget problem handled by the cited exact profile results. $\square$

Proposition 4.461 (Degree-only low-low pruning exceeds both branch budgets)**.** *The degree equations in definition 4.440 do not imply the branch-budget bound*

$$|\mathcal{L}_{\deg}| \le 2^m$$

by themselves. More precisely, for every $m \ge 3$ there is a residual degree vector on $M = 2m$ low labels for which the corresponding degree-constrained low-low ambient set has cardinality

$$(2m-1)!! > 2^m.$$

For every $m \ge 7$, the same examples also violate the full final-zone branch budget:

$$(2m-1)!! > 2^{2m}.$$

Proof. Take

$$d_1 = d_2 = \cdots = d_{2m} = 1.$$

An assignment

$$\ell = (\ell_{ab})_{1\le a<b\le 2m}$$

satisfying the degree equations of definition 4.440 for this vector is exactly a perfect matching on the vertex set $\{1,\ldots,2m\}$: every vertex has degree one, so each nonzero ℓ_{ab} equals 1, and the nonzero pairs partition the vertices. Conversely every perfect matching gives such an assignment.

The number of perfect matchings on $2m$ labelled vertices is

$$(2m-1)(2m-3)\cdots 3\cdot 1 = (2m-1)!!.$$

For $m = 3$ this is $15 > 8 = 2^3$. Passing from m to $m+1$ multiplies the left-hand side by $2m+1$ and the right-hand side by 2, so the strict inequality persists for every $m \ge 3$.

For the full final-zone budget, check the base case

$$13!! = 135135 > 16384 = 2^{14}.$$

Passing from m to $m+1$ again multiplies the left-hand side by $2m+1$, while 2^{2m} is multiplied by 4. Since $2m+1 > 4$ for $m \ge 7$, the strict inequality persists for every $m \ge 7$. $\square$

Corollary 4.462 (Degree-only terminal-boundary pruning exceeds the full branch budget)**.** *Work with the degree-constrained low-low ambient set of definition 4.440. The degree equations alone do not give a full-budget terminal-boundary candidate list. More precisely, for every $m \geq 7$ there is a residual degree vector on $M = 2m$ low labels such that any terminal-boundary list whose induced low-low assignments contain every degree-compatible assignment has cardinality greater than 2^{2m}.*

Proof. Use the vector

$$d_1 = d_2 = \cdots = d_{2m} = 1$$

from proposition 4.461. The corresponding degree-compatible low-low assignments are exactly the perfect matchings on $\{1, \ldots, 2m\}$, so there are

$$(2m-1)!!$$

of them, and this number is greater than 2^{2m} for $m \geq 7$.

By lemma 4.183, a terminal Step 3 boundary segment determines every Step 3 entry on L_{2m}. Hence one terminal boundary segment can induce at most one low-low assignment. Therefore any terminal-boundary list whose induced assignments contain all assignments in the displayed degree ambient set has at least $(2m-1)!!$ elements, which is larger than the full final-zone budget 2^{2m} for $m \geq 7$. ☐

Corollary 4.463 (Refined odd-packet classes induce bounded low-low candidates)**.** *Work in the setting of definition 4.440. Suppose the target-window data determine*

$$\lambda^{(2m)}, \qquad E_1, \ldots, E_m,$$

the labelled continuation boxes, and a canonically ordered set

$$\Omega_{\mathrm{tw}} \subseteq \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

which contains the true deleted odd-packet assignment and satisfies

$$|\Omega_{\mathrm{tw}}| \leq 2^m.$$

Let $\mathcal{L}_{\mathrm{tw}}$ be the set of candidates $\ell \in \mathcal{L}_{\mathrm{deg}}$ with the following property: after combining ℓ with the recovered continuation-column subfilling J_{2m}, the resulting full Step 3 half-square and Step 2 symmetric matrix reconstruct a semistandard tableau whose boxes for labels $2m+1, \ldots, j$ are the displayed continuation boxes, whose tight-prefix endpoint and retained even strips are the displayed $\lambda^{(2m)}$ and $E_1, \ldots, E_m$, and whose deleted odd-packet assignment belongs to Ω_{tw}. Order $\mathcal{L}_{\mathrm{tw}}$ by first ordering its image in Ω_{tw} and breaking ties lexicographically as in definition 4.440. Then $\mathcal{L}_{\mathrm{tw}}$ is determined by the target-window data, contains the true low-low assignment, and satisfies

$$|\mathcal{L}_{\mathrm{tw}}| \leq 2^m.$$

Consequently the hypotheses of corollary 4.438 hold for the final right-boundary zone.

Proof. Every object used in the definition of $\mathcal{L}_{\mathrm{tw}}$ is determined by the target-window data: the ambient set $\mathcal{L}_{\mathrm{deg}}$ by proposition 4.441, the continuation-column subfilling by lemma 4.341, and the displayed endpoint, retained strips, continuation boxes, and Ω_{tw} by hypothesis. Thus the set and its order are canonical functions of the target-window data.

The true low-low assignment lies in $\mathcal{L}_{\mathrm{deg}}$ by proposition 4.441. Combining it with the true continuation-column subfilling gives the true Step 3 half-square, hence the true Step 2 matrix by lemma 4.127. The reconstructed tableau is therefore the original one. Its displayed base data are the true base data, and its deleted odd-packet assignment belongs to Ω_{tw} by hypothesis. Hence the true low-low assignment belongs to $\mathcal{L}_{\mathrm{tw}}$.

It remains to prove the cardinality bound. Send $\ell \in \mathcal{L}_{\mathrm{tw}}$ to the deleted odd-packet assignment of the semistandard tableau reconstructed from the completed matrix. This map lands in Ω_{tw} by definition. It is injective. Indeed, if two candidates have the same image in Ω_{tw}, then they have the same deleted odd-label boxes; by the definition of $\mathcal{L}_{\mathrm{tw}}$ they also have the same retained even-label boxes $E_1, \dots, E_m$ and the same labelled continuation boxes. Therefore they determine the same labelled semistandard tableau. By lemma 4.174, that tableau determines a unique Step 2 symmetric matrix. The low-low entries are entries of this matrix, so the two candidates are equal.

Thus

$$|\mathcal{L}_{\text{tw}}| \le |\Omega_{\text{tw}}| \le 2^m.$$

The membership, canonical-ordering, and cardinality hypotheses of corollary 4.438 are now satisfied. □

Corollary 4.464 (Full-budget odd-packet classes induce full-budget low-low candidates)**.** *Work in the setting of definition 4.440. Suppose the target-window data determine*

$$\lambda^{(2m)}, \qquad E_1, \dots, E_m,$$

the labelled continuation boxes, and a canonically ordered set

$$\Omega_{\text{src}} \subseteq \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

which contains the true deleted odd-packet assignment and satisfies

$$|\Omega_{\text{src}}| \le 2^{2m}.$$

Then the same target-window data determine a canonically ordered set

$$\mathcal{L}_{\text{tw}} \subseteq \mathcal{L}_{\text{deg}}$$

of possible Step 3 entry assignments on L_{2m}, the true low-low assignment belongs to this set, and

$$|\mathcal{L}_{\text{tw}}| \le 2^{2m}.$$

Consequently, whenever the corresponding right-boundary or graph terminal setting is in force, this is a full-budget low-low candidate source in the sense of proposition 4.478 and corollary 4.203.

Proof. Define $\mathcal{L}_{\text{tw}}$ exactly as in corollary 4.463, with Ω_{src} in place of Ω_{tw}. The determinacy and true-membership parts of that proof use only that the target-window data determine the displayed base data and the displayed odd-packet class, and that the true odd-packet assignment belongs to the class. Hence the same argument shows that $\mathcal{L}_{\text{tw}}$ is determined and contains the true low-low assignment.

The same injection used there sends each candidate low-low assignment to the deleted odd-packet assignment of the reconstructed semistandard tableau. Its image lies in Ω_{src}, and injectivity follows again from lemma 4.174. Therefore

$$|\mathcal{L}_{\text{tw}}| \le |\Omega_{\text{src}}| \le 2^{2m}.$$

This is exactly the full-budget candidate-list hypothesis in the cited right-boundary and graph statements. □

Corollary 4.465 (Bounded signature lists induce full-budget low-low candidates)**.** *Work simultaneously in the settings of corollary 4.430 and definition 4.440. Under the hypotheses of corollary 4.430, the target-window data determine a canonically ordered set*

$$\mathcal{L}_{\text{tw}} \subseteq \mathcal{L}_{\text{deg}}$$

of possible Step 3 entry assignments on L_{2m}, the true low-low assignment belongs to this set, and

$$|\mathcal{L}_{\text{tw}}| \le 2^{2m}.$$

Proof. Corollary 4.430 constructs a target-window-determined odd-packet class

$$\Omega_{\text{tw}}(\Sigma_{\text{tw}}) \subseteq \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

which contains the true deleted odd-packet assignment and has cardinality at most 2^{2m}. Applying corollary 4.464 to this odd-packet class gives the asserted low-low candidate set. □

Corollary 4.466 (Target-window signature fibers induce bounded low-low candidates)**.** *Work in the right-boundary auxiliary setting of proposition 4.253, and suppose the last zone $Z_L = \{a_L, \dots, j\}$ is a non-full tight height zone with $m \geq 9$ deleted singleton packets. Let V_L^* be the replacement window parsed from the repaired path, and let $\mathcal{D}_{\mathrm{out}}(P)$ be the outside datum of proposition 4.253. Suppose the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \dots, E_m$, the labelled continuation boxes, and the commuting normal-form signature κ of the fixed-even standard tableau associated to the true odd-packet assignment. Then the same target-window data determine a canonical set

$$\mathcal{L}_{\mathrm{tw}} \subseteq \mathcal{L}_{\mathrm{deg}}$$

of possible low-low Step 3 assignments which contains the true low-low assignment and satisfies

$$|\mathcal{L}_{\mathrm{tw}}| \leq 2^m.$$

Consequently the hypotheses of corollary 4.438 hold for the final right-boundary zone.

Proof. Corollary 4.429 defines the target-window signature fiber

$$\Omega_{\mathrm{tw}}(\kappa) \subseteq \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

from the displayed data and proves that it is canonical, contains the true odd-packet assignment, and has cardinality at most 2^m. Apply corollary 4.463 with $\Omega_{\mathrm{tw}} = \Omega_{\mathrm{tw}}(\kappa)$. The resulting set $\mathcal{L}_{\mathrm{tw}}$ is the required bounded low-low candidate class, and corollary 4.438 then applies. □

Corollary 4.467 (Target-window signature fibers give the tight right-boundary inverse)**.** *Work in the setting of corollary 4.466, with*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}.$$

Under the hypotheses of corollary 4.466, from

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}, \quad \beta_L \in \{0,1\}^{2m}$$

one recovers the original right-boundary zone subpath

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \dots, \rho_{2j}.$$

Proof. Corollary 4.466 gives a canonical low-low candidate set $\mathcal{L}_{\mathrm{tw}}$ containing the true low-low assignment and satisfying

$$|\mathcal{L}_{\mathrm{tw}}| \leq 2^m.$$

Applying corollary 4.439 to this set recovers the displayed right-boundary subpath. □

Proposition 4.468 (Target-window signatures supply the right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \geq 9$ deleted singleton packets, and suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \dots, E_m$, the labelled continuation boxes, and the commuting normal-form signature κ of the true fixed-even standard tableau. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. By corollary 4.320, the final tight zone has

$$|Z_L \cap I(S)| = m.$$

The datum $A(P) = \mathcal{D}_{\text{out}}(P)$ is a fixed function of S and $\mathcal{D}_{\text{out}}(P)$, as required in proposition 4.253. The inverse is allowed to use V_L^*, the endpoint pair, the known interval Z_L, and this auxiliary datum. Together these are exactly the target-window data in the hypothesis, and hence determine the base endpoint, retained strips, continuation boxes, and κ.

Applying corollary 4.467 recovers

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \ldots, \rho_{2j}$$

from these data and the word $\beta_L \in \{0,1\}^{2m}$. Since $m = |Z_L \cap I(S)|$, this is precisely a word of the length allowed in proposition 4.253. Thus the displayed procedure is the required fixed right-boundary inverse. □

Proposition 4.469 (Even standard-time chains supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \ldots, j\} = \{j - 2m + 1, \ldots, j\}$$

with $m \geq 9$ deleted singleton packets. Let $U = \operatorname{std}(T)$ be the doubled standardization of the underlying semistandard tableau, and let

$$\varnothing = \sigma^{(0)} \subset \sigma^{(1)} \subset \cdots$$

be the standard shape chain of U. Suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the even standard-time endpoint chain

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \ldots, \sigma^{(4m)}$$

and the labelled continuation boxes for labels $2m+1, \ldots, j$. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\text{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. By lemma 4.169, the recovered even standard-time endpoints determine the true Pieri prefix

$$\lambda^{(0)}, \lambda^{(1)}, \ldots, \lambda^{(2m)}.$$

Therefore the same data determine

$$\lambda^{(2m)} \quad \text{and} \quad E_r = \lambda^{(2r)} \setminus \lambda^{(2r-1)} \qquad (1 \leq r \leq m).$$

The proof of corollary 4.409 applied to this recovered prefix chain gives the commuting normal-form signature κ of the true fixed-even standard tableau.

Thus the target-window data determine $\lambda^{(2m)}$, the retained strips $E_1, \ldots, E_m$, the labelled continuation boxes, and κ. Applying proposition 4.468 gives the required fixed right-boundary auxiliary inverse. □

Proposition 4.470 (Low-label box readouts supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.469. Suppose the target-window data determine, for each semistandard label $1 \leq a \leq 2m$, the two boxes carrying label a, and also determine the labelled continuation boxes for labels $2m+1, \ldots, j$. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\text{out}}(P)$.*

Proof. By lemma 4.170 with $M = 2m$, the recovered boxes carrying labels $1, \ldots, 2m$ determine the even standard-time endpoint chain

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \ldots, \sigma^{(4m)}.$$

Together with the recovered labelled continuation boxes, this is exactly the hypothesis of proposition 4.469. Applying that proposition gives the claimed right-boundary auxiliary inverse. $\square$

Corollary 4.471 (Signature-route readouts recover the large deleted-only graph)**.** *Work in the setting of corollary 4.203, and suppose $m \geq 9$. Suppose the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, and the labelled continuation boxes. Suppose in addition that the same data determine one of the following signature-route readouts: the commuting normal-form signature κ of the true fixed-even standard tableau; the even standard-time endpoint chain

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \ldots, \sigma^{(4m)}$$

of the doubled standardization; or, for every semistandard label $1 \leq a \leq 2m$, the two boxes carrying label a. Then Γ is recovered from

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, these target-window data, and the already budgeted final-zone word $\beta_L \in \{0,1\}^{2m}$.

Proof. First suppose the target-window data determine κ. Then corollary 4.466 constructs a canonical low-low candidate set $\mathcal{L}_{\text{tw}}$ containing the true low-low assignment and satisfying

$$|\mathcal{L}_{\text{tw}}| \leq 2^m \leq 2^{2m}.$$

Applying corollary 4.203 to this candidate set recovers Γ from the displayed graph data and the same final-zone word.

If the target-window data determine the even standard-time endpoint chain, corollary 4.410 shows that the hypothesis of proposition 4.506 holds. That hypothesis is precisely deterministic recovery of the true commuting normal-form signature from the displayed target-window tuple. This reduces the even-chain case to the preceding paragraph.

Finally suppose the target-window data determine the two boxes carrying each label $1 \leq a \leq 2m$. By lemma 4.170, with $M = 2m$, these boxes determine the even standard-time endpoint chain. The previous paragraph then reduces this case to the direct-signature case. $\square$

Proposition 4.472 (Low-low Step-3 entries supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.470. Let L_{2m} be the low-low cell set of lemma 4.182. Suppose the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the Step 3 cell entries on every cell of L_{2m}. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\text{out}}(P)$.

Proof. Put $M = 2m$. By lemma 4.341, the outside datum $\mathcal{D}_{\text{out}}(P)$ determines every Step 3 entry in the continuation-column subarrangement J_M. The hypothesis determines every entry on L_M. These two sets cover the full Step 3 half-square: an off-diagonal unordered label pair either has both labels at most M, or is incident to a continuation label $M+1, \ldots, j$.

Thus the displayed target-window data determine the whole Step 3 half-square. By lemma 4.127, this determines every off-diagonal entry of the Step 2 symmetric matrix, and the diagonal entries are zero. Applying lemma 4.175 with $M = j$ recovers the complete Pieri boundary path

$$\lambda^{(0)}, \lambda^{(1)}, \ldots, \lambda^{(j)}.$$

Then lemma 4.171 with $M = 2m$ recovers the two boxes carrying every label $1, \dots, 2m$, and lemma 4.173 with $M = 2m$ recovers the two boxes carrying every continuation label $2m+1, \dots, j$. Applying proposition 4.470 gives the claimed right-boundary auxiliary inverse. □

Proposition 4.473 (Terminal boundary supplies the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.472. Suppose the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the terminal low-label Step 3 boundary segment

$$\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j}.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$.

Proof. By lemma 4.183 with $M = 2m$, the displayed terminal boundary segment determines every Step 3 cell entry on L_{2m}. Thus the hypothesis of proposition 4.472 is satisfied, and that proposition gives the required fixed right-boundary auxiliary inverse. □

Proposition 4.474 (Full branch-budget terminal boundaries supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \geq 9$ deleted singleton packets. Suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a canonically ordered finite set

$$\mathcal{B}_{\mathrm{tw}}$$

of possible terminal low-label Step 3 boundary segments

$$\tau = (\tau_{2(j-2m)}, \tau_{2(j-2m)+1}, \dots, \tau_{2j}),$$

that the true segment

$$(\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j})$$

belongs to this set, and that

$$|\mathcal{B}_{\mathrm{tw}}| \leq 2^{2m}.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. By corollary 4.320,

$$|Z_L \cap I(S)| = m.$$

Use the $2m$ bits of β_L to encode the rank of the true terminal boundary segment in the canonical order on $\mathcal{B}_{\mathrm{tw}}$. Since $|\mathcal{B}_{\mathrm{tw}}| \leq 2^{2m}$, this selects the true segment.

The identity $a_L = j - 2m + 1$ gives

$$2a_L - 2 = 2(j - 2m).$$

Thus the selected segment is exactly the original right-boundary zone subpath

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \dots, \rho_{2j}.$$

With $A(P) = \mathcal{D}_{\mathrm{out}}(P)$, this is the required fixed right-boundary inverse. □

Corollary 4.475 (Full branch-budget terminal boundaries induce low-low candidates)**.** *Work in the non-full tight-height setting of corollary 4.320. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \geq 9$ deleted singleton packets, and let L_{2m} be the low-low cell set of lemma 4.182. Suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a canonically ordered finite set

$$\mathcal{B}_{\mathrm{tw}}$$

of possible terminal low-label Step 3 boundary segments

$$\tau = (\tau_{2(j-2m)}, \tau_{2(j-2m)+1}, \dots, \tau_{2j}),$$

that the true segment

$$(\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j})$$

belongs to this set, and that

$$|\mathcal{B}_{\mathrm{tw}}| \leq 2^{2m}.$$

Then the same target-window data determine a canonically ordered finite set

$$\mathcal{L}_{\mathrm{tw}}$$

of possible Step 3 entry assignments on L_{2m}, the true low-low assignment belongs to this set, and

$$|\mathcal{L}_{\mathrm{tw}}| \leq 2^{2m}.$$

Proof. For each boundary segment in $\mathcal{B}_{\mathrm{tw}}$, apply the deterministic backward sweep of corollary 4.124 to the terminal Ferrers subarrangement L_{2m} identified by lemma 4.183. This gives a canonically ordered image set

$$\mathcal{L}_{\mathrm{tw}},$$

ordered by the first boundary segment which produces the image, with duplicate images removed. Thus $\mathcal{L}_{\mathrm{tw}}$ is determined by the same target-window data and has cardinality at most $|\mathcal{B}_{\mathrm{tw}}| \leq 2^{2m}$.

Since the true terminal boundary segment belongs to $\mathcal{B}_{\mathrm{tw}}$, lemma 4.183, applied with $M = 2m$, shows that its image under the same backward sweep is exactly the true Step 3 assignment on L_{2m}. Therefore the true low-low assignment belongs to $\mathcal{L}_{\mathrm{tw}}$. ☐

Proposition 4.476 (Bounded terminal boundaries supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \geq 9$ deleted singleton packets. Suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a canonically ordered finite set

$$\mathcal{B}_{\mathrm{tw}}$$

of possible terminal low-label Step 3 boundary segments

$$\tau = (\tau_{2(j-2m)}, \tau_{2(j-2m)+1}, \dots, \tau_{2j}),$$

that the true segment

$$(\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j})$$

belongs to this set, and that

$$|\mathcal{B}_{\mathrm{tw}}| \leq 2^{m}.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. For each boundary segment in $\mathcal{B}_{\mathrm{tw}}$, apply the deterministic backward sweep of corollary 4.124 to the terminal Ferrers subarrangement L_{2m} identified in lemma 4.183. This gives a canonically ordered image set

$$\mathcal{L}_{\mathrm{tw}}$$

of Step 3 entry assignments on L_{2m}, ordered by first occurrence from $\mathcal{B}_{\mathrm{tw}}$ and with duplicate images removed.

The set $\mathcal{L}_{\mathrm{tw}}$ is determined by the target-window data. It has cardinality at most $|\mathcal{B}_{\mathrm{tw}}| \le 2^m$. Since the true terminal boundary segment belongs to $\mathcal{B}_{\mathrm{tw}}$, lemma 4.183 shows that the true low-low Step 3 assignment belongs to $\mathcal{L}_{\mathrm{tw}}$. Therefore the hypotheses of proposition 4.477 are satisfied, and that proposition gives the required fixed right-boundary auxiliary inverse with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$. □

Proposition 4.477 (Bounded low-low candidates supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \ge 9$ deleted singleton packets. Let L_{2m} be the low-low cell set of lemma 4.182. Suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a canonically ordered finite set

$$\mathcal{L}_{\mathrm{tw}}$$

of possible Step 3 entry assignments on L_{2m}, that the true assignment belongs to this set, and that

$$|\mathcal{L}_{\mathrm{tw}}| \le 2^m.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. The displayed hypotheses are exactly the bounded low-low candidate hypotheses of corollary 4.438. Therefore corollary 4.439 applies to the canonically ordered set $\mathcal{L}_{\mathrm{tw}}$ and recovers

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \dots, \rho_{2j}$$

from the target-window data and the word $\beta_L \in \{0,1\}^{2m}$. By corollary 4.320,

$$|Z_L \cap I(S)| = m,$$

so this word has exactly the length allowed by proposition 4.253. With $A(P) = \mathcal{D}_{\mathrm{out}}(P)$, the displayed recovery procedure is the required fixed right-boundary inverse. □

Proposition 4.478 (Full branch-budget low-low candidates supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \ge 9$ deleted singleton packets. Let L_{2m} be the low-low cell set of lemma 4.182. Suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine a canonically ordered finite set

$$\mathcal{L}_{\mathrm{tw}}$$

of possible Step 3 entry assignments on L_{2m}*, that the true assignment belongs to this set, and that*

$$|\mathcal{L}_{\mathrm{tw}}| \leq 2^{2m}.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L *from*

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. By corollary 4.320,

$$|Z_L \cap I(S)| = m.$$

Use the $2m$ bits of β_L to encode the rank of the true low-low assignment in the canonical order on $\mathcal{L}_{\mathrm{tw}}$. Since $|\mathcal{L}_{\mathrm{tw}}| \leq 2^{2m}$, this selects the true Step 3 entries on L_{2m}.

Put $M = 2m$. By lemma 4.341, $\mathcal{D}_{\mathrm{out}}(P)$ determines every Step 3 entry in the continuation-column subarrangement J_M. The selected entries determine the entries on L_M. The cell sets L_M and J_M cover the full Step 3 half-square: an unordered off-diagonal label pair either has both labels at most M, or is incident to a continuation label $M+1,\ldots,j$.

Therefore the displayed data and β_L determine every cell entry of the full Step 3 half-square. By lemma 4.126, these entries determine the whole boundary sequence

$$\rho_0, \rho_1, \ldots, \rho_{2j}.$$

Since $a_L = j - 2m + 1$, the desired right-boundary zone subpath is the terminal segment

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \ldots, \rho_{2j}.$$

With $A(P) = \mathcal{D}_{\mathrm{out}}(P)$, the displayed recovery procedure is the required fixed right-boundary inverse. □

Proposition 4.479 (Refined odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \ldots, j\} = \{j - 2m + 1, \ldots, j\}$$

with $m \geq 9$ *deleted singleton packets. Suppose the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$*, the retained even-label strips* $E_1, \ldots, E_m$*, the labelled continuation boxes, and a canonically ordered set*

$$\Omega_{\mathrm{tw}} \subseteq \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

which contains the true deleted odd-packet assignment and satisfies

$$|\Omega_{\mathrm{tw}}| \leq 2^m.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L *from*

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. The hypotheses are exactly the inputs needed for corollary 4.463. That corollary constructs from the displayed target-window data a canonically ordered low-low candidate set

$$\mathcal{L}_{\mathrm{tw}}$$

containing the true low-low assignment and satisfying $|\mathcal{L}_{\mathrm{tw}}| \le 2^m$. Applying proposition 4.477 to this set gives the required fixed right-boundary auxiliary inverse. □

Proposition 4.480 (Full-budget odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \ge 9$ deleted singleton packets. Suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \dots, E_m$, the labelled continuation boxes, and a canonically ordered set

$$\Omega_{\mathrm{tw}} \subseteq \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

which contains the true deleted odd-packet assignment and satisfies

$$|\Omega_{\mathrm{tw}}| \le 2^{2m}.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. Define $\mathcal{L}_{\mathrm{tw}} \subseteq \mathcal{L}_{\mathrm{deg}}$ exactly as in corollary 4.463, using the displayed set Ω_{tw}. The proof of that corollary shows that $\mathcal{L}_{\mathrm{tw}}$ is determined by the target-window data, contains the true low-low assignment, and injects into Ω_{tw}. Hence

$$|\mathcal{L}_{\mathrm{tw}}| \le |\Omega_{\mathrm{tw}}| \le 2^{2m}.$$

Applying proposition 4.478 to this low-low candidate set gives the required fixed right-boundary auxiliary inverse. □

Proposition 4.481 (Bounded signature lists supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.480. Suppose the target-window data determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \dots, E_m$, the labelled continuation boxes, and a canonically ordered finite set*

$$\Sigma_{\mathrm{tw}}$$

of commuting normal-form signature words such that the true fixed-even standard tableau has signature in Σ_{tw} and

$$|\Sigma_{\mathrm{tw}}| \le 2^m.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$.

Proof. Corollary 4.430 constructs from the displayed target-window data a canonically ordered set

$$\Omega_{\mathrm{tw}}(\Sigma_{\mathrm{tw}}) \subseteq \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

which contains the true deleted odd-packet assignment and has cardinality at most 2^{2m}. Therefore the hypotheses of proposition 4.480 hold with $\Omega_{\mathrm{tw}} = \Omega_{\mathrm{tw}}(\Sigma_{\mathrm{tw}})$, and that proposition gives the required fixed right-boundary auxiliary inverse. □

Proposition 4.482 (Nine-to-twelve broad odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.480, and suppose*

$$9 \leq m \leq 12.$$

Suppose the target-window data determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, and the labelled continuation boxes. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$.

Proof. Let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

with the lexicographic order of definition 4.354. The target-window data determine this set, and lemma 4.355 puts the true deleted odd-packet assignment in it.

By lemma 4.360,

$$|\Omega_{\mathrm{tw}}| \leq 2(m-2)!.$$

For $9 \leq m \leq 12$ one has

$$2(m-2)! \leq 2^{2m};$$

explicitly the left side is, for $m = 9, 10, 11, 12$,

$$10080, \quad 80640, \quad 725760, \quad 7257600,$$

while 2^{2m} is respectively

$$262144, \quad 1048576, \quad 4194304, \quad 16777216.$$

Thus the hypotheses of proposition 4.480 are satisfied by the unrefined compatibility set Ω_{tw}, and that proposition gives the stated inverse. $\square$

Proposition 4.483 (Thirteen-to-twenty-eight broad odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.480, and suppose*

$$13 \leq m \leq 28.$$

Suppose the target-window data determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, and the labelled continuation boxes. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$.

Proof. Let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

with the lexicographic order of definition 4.354. The target-window data determine this set, and lemma 4.355 puts the true deleted odd-packet assignment in it.

If Ω_{tw} is empty there is nothing to prove. Otherwise lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with the cardinality of the associated fixed-even standard-tableau fiber

$$\mathcal{S}(\gamma; e_1, \ldots, e_{m-1}).$$

By lemma 4.389,

$$|\Omega_{\mathrm{tw}}| = |\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})| \leq W(\gamma).$$

Since $13 \leq m \leq 28$, proposition 4.390 gives

$$W(\gamma) \leq 2^{2m}.$$

Thus the hypotheses of proposition 4.480 are satisfied by the unrefined compatibility set Ω_{tw}, and that proposition gives the stated inverse. $\square$

Proposition 4.484 (Top-fixed envelope non-frontier broad odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.480, and suppose $m \geq 13$. Suppose the target-window data determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \dots, E_m$, and the labelled continuation boxes. Let*

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

with the lexicographic order of definition 4.354. If $\Omega_{\mathrm{tw}} \neq \varnothing$, let $(\gamma; e_1, \dots, e_{m-1})$ be the associated fixed-even datum of definition 4.379, and assume

$$(\gamma, e_{m-1}) \notin \mathcal{P}_m^{\mathrm{env}}.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$.

Proof. The target-window data determine Ω_{tw}, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\mathrm{tw}} = \varnothing$, the assertion is vacuous. Otherwise, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})|.$$

By lemma 4.392,

$$|\Omega_{\mathrm{tw}}| \leq W^{\mathrm{top}}(\gamma; e_{m-1}).$$

Since $(\gamma, e_{m-1}) \notin \mathcal{P}_m^{\mathrm{env}}$, definition 4.391 gives

$$W^{\mathrm{top}}(\gamma; e_{m-1}) \leq 2^{2m}.$$

Thus the hypotheses of proposition 4.480 are satisfied by the unrefined compatibility set Ω_{tw}, and that proposition gives the stated inverse. □

Proposition 4.485 (Top-two envelope non-frontier broad odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.480, and suppose $m \geq 13$. Suppose the target-window data determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \dots, E_m$, and the labelled continuation boxes. Let*

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

with the lexicographic order of definition 4.354. If $\Omega_{\mathrm{tw}} \neq \varnothing$, let $(\gamma; e_1, \dots, e_{m-1})$ be the associated fixed-even datum of definition 4.379, and assume

$$(\gamma, e_{m-1}, e_{m-2}) \notin \mathcal{P}_m^{\mathrm{env},2}.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$.

Proof. The target-window data determine Ω_{tw}, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\mathrm{tw}} = \varnothing$, the assertion is vacuous. Otherwise, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})|.$$

By lemma 4.393,

$$|\Omega_{\mathrm{tw}}| \leq W^{\mathrm{top},2}(\gamma; e_{m-1}, e_{m-2}).$$

Since $(\gamma, e_{m-1}, e_{m-2}) \notin \mathcal{P}_m^{\mathrm{env},2}$, definition 4.391 gives

$$W^{\mathrm{top},2}(\gamma; e_{m-1}, e_{m-2}) \leq 2^{2m}.$$

Thus the hypotheses of proposition 4.480 are satisfied by the unrefined compatibility set Ω_{tw}, and that proposition gives the stated inverse. □

Proposition 4.486 (Fixed-depth envelope non-frontier broad odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.480. Let $d \geq 3$, and suppose $m \geq d+1$. Suppose the target-window data determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, and the labelled continuation boxes. Let*

$$\Omega_{\text{tw}} = \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

with the lexicographic order of definition 4.354. If $\Omega_{\text{tw}} \neq \varnothing$, let $(\gamma; e_1, \ldots, e_{m-1})$ be the associated fixed-even datum of definition 4.379, and assume

$$(\gamma, e_{m-1}, e_{m-2}, \ldots, e_{m-d}) \notin \mathcal{P}_m^{\text{env},d}.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\text{out}}(P)$.

Proof. The target-window data determine Ω_{tw}, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\text{tw}} = \varnothing$, the assertion is vacuous. Otherwise, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{m-1})|.$$

If $d = 3$, then lemma 4.394 gives

$$|\Omega_{\text{tw}}| \leq W^{\text{top},3}(\gamma; e_{m-1}, e_{m-2}, e_{m-3}).$$

If $d \geq 4$, then lemma 4.395 gives

$$|\Omega_{\text{tw}}| \leq W^{\text{top},d}(\gamma; e_{m-1}, e_{m-2}, \ldots, e_{m-d}).$$

In both cases, the displayed non-frontier hypothesis and definition 4.391 give

$$|\Omega_{\text{tw}}| \leq 2^{2m}.$$

Thus the hypotheses of proposition 4.480 are satisfied by the unrefined compatibility set Ω_{tw}, and that proposition gives the stated inverse. □

Proposition 4.487 (Twenty-nine top-three envelope broad odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.480. Suppose the target-window data determine $\lambda^{(58)}$, the retained even-label strips $E_1, \ldots, E_{29}$, and the labelled continuation boxes. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\text{out}}(P)$.*

Proof. Let

$$\Omega_{\text{tw}} = \Omega(\lambda^{(58)}; E_1, \ldots, E_{29})$$

with the lexicographic order of definition 4.354. The target-window data determine this set, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\text{tw}} = \varnothing$, the assertion is vacuous.

Assume $\Omega_{\text{tw}} \neq \varnothing$, and let $(\gamma; e_1, \ldots, e_{28})$ be the associated fixed-even datum. As above, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{28})|.$$

By lemma 4.394 and proposition 4.398,

$$|\Omega_{\text{tw}}| \leq W^{\text{top},3}(\gamma; e_{28}, e_{27}, e_{26}) \leq 2^{58}.$$

Thus the hypotheses of proposition 4.480 are satisfied by the unrefined compatibility set Ω_{tw}, and that proposition gives the stated inverse. □

Proposition 4.488 (Thirty fixed-depth envelope broad odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.480. Suppose the target-window data determine $\lambda^{(60)}$, the retained even-label strips $E_1, \ldots, E_{30}$, and the labelled continuation boxes. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\rm out}(P)$.*

Proof. Let

$$\Omega_{\rm tw} = \Omega(\lambda^{(60)}; E_1, \ldots, E_{30})$$

with the lexicographic order of definition 4.354. The target-window data determine this set, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\rm tw} = \varnothing$, the assertion is vacuous.

Assume $\Omega_{\rm tw} \neq \varnothing$, and let $(\gamma; e_1, \ldots, e_{29})$ be the associated fixed-even datum. As above, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{29})|.$$

By lemma 4.395 and proposition 4.399,

$$|\Omega_{\rm tw}| \leq W^{\rm top,7}(\gamma; e_{29}, e_{28}, e_{27}, e_{26}, e_{25}, e_{24}, e_{23}) \leq 2^{60}.$$

Thus the hypotheses of proposition 4.480 are satisfied by the unrefined compatibility set $\Omega_{\rm tw}$, and that proposition gives the stated inverse. □

Proposition 4.489 (Thirty-one fixed-depth envelope broad odd-packet classes supply the large right-boundary auxiliary inverse)**.** *Work in the setting of proposition 4.480. Suppose the target-window data determine $\lambda^{(62)}$, the retained even-label strips $E_1, \ldots, E_{31}$, and the labelled continuation boxes. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\rm out}(P)$.*

Proof. Let

$$\Omega_{\rm tw} = \Omega(\lambda^{(62)}; E_1, \ldots, E_{31})$$

with the lexicographic order of definition 4.354. The target-window data determine this set, and lemma 4.355 puts the true deleted odd-packet assignment in it. If $\Omega_{\rm tw} = \varnothing$, the assertion is vacuous.

Assume $\Omega_{\rm tw} \neq \varnothing$, and let $(\gamma; e_1, \ldots, e_{30})$ be the associated fixed-even datum. As above, lemmas 4.369, 4.377 and 4.380 and proposition 4.373 identify its cardinality with

$$|\mathcal{S}(\gamma; e_1, \ldots, e_{30})|.$$

By lemma 4.395 and proposition 4.400,

$$|\Omega_{\rm tw}| \leq W^{\rm top,10}(\gamma; e_{30}, e_{29}, e_{28}, e_{27}, e_{26}, e_{25}, e_{24}, e_{23}, e_{22}, e_{21}) \leq 2^{62}.$$

Thus the hypotheses of proposition 4.480 are satisfied by the unrefined compatibility set $\Omega_{\rm tw}$, and that proposition gives the stated inverse. □

Lemma 4.490 (The binary-prefix full-budget certificate stops at twelve)**.** *For every integer $m \geq 13$,*

$$2(m-2)! > 2^{2m}.$$

Consequently, the binary-prefix factorial estimate of lemma 4.360 by itself does not certify the full-branch budget inequality $|\Omega(\lambda^{(2m)}; E_1, \ldots, E_m)| \leq 2^{2m}$ in the range $m \geq 13$.

Proof. At $m = 13$,

$$2(m-2)! = 2 \cdot 11! = 79833600 > 67108864 = 2^{26} = 2^{2m}.$$

If

$$2(m-2)! > 2^{2m}$$

holds for some $m \geq 13$, then

$$\frac{2(m-1)!}{2^{2m+2}} = \frac{m-1}{4} \cdot \frac{2(m-2)!}{2^{2m}} > \frac{m-1}{4} \geq 3 > 1.$$

Thus the displayed strict inequality propagates from m to $m+1$, and induction proves it for every $m \geq 13$.

Lemma 4.360 gives only the upper estimate $|\Omega| \leq 2(m-2)!$. Since this upper estimate is already larger than the full budget 2^{2m} for $m \geq 13$, that estimate alone cannot supply the displayed budget inequality in this range. ☐

Proposition 4.491 (Strict-left low-label incidence carriers supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \geq 9$ deleted singleton packets. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$H_{2m} \subseteq E$$

for the low-label incidence cell set of lemma 4.179. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\text{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. By lemma 4.339, the outside datum $\mathcal{D}_{\text{out}}(P)$ determines

$$\rho_0, \rho_1, \dots, \rho_{2(j-2m)}.$$

Therefore the full target-window tuple determines every input to the deterministic rule in the hypothesis. It follows that the tuple determines the carrier E, its top/right boundary labels, and the inclusion $H_{2m} \subseteq E$.

Since the top/right boundary labels of E are determined, corollary 4.124 recovers every Step 3 cell entry on E. The inclusion $H_{2m} \subseteq E$ therefore gives every entry on H_{2m}. By lemma 4.182,

$$H_{2m} = X_{2m} \sqcup L_{2m},$$

so the entries on the low-low cell set L_{2m} are determined. Applying proposition 4.472 gives the required fixed right-boundary auxiliary inverse with $A(P) = \mathcal{D}_{\text{out}}(P)$. ☐

Proposition 4.492 (Strict-left low-low carriers supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \geq 9$ deleted singleton packets. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$L_{2m} \subseteq E$$

for the low-low cell set of lemma 4.182. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\text{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. By lemma 4.339, the outside datum $\mathcal{D}_{\rm out}(P)$ determines

$$\rho_0, \rho_1, \ldots, \rho_{2(j-2m)}.$$

Therefore the full target-window tuple determines every input to the deterministic rule in the hypothesis. It follows that the tuple determines the carrier E, its top/right boundary labels, and the inclusion $L_{2m} \subseteq E$.

Since the top/right boundary labels of E are determined, corollary 4.124 recovers every Step 3 cell entry on E. The inclusion $L_{2m} \subseteq E$ therefore gives every entry on L_{2m}. Applying proposition 4.472 gives the required fixed right-boundary auxiliary inverse with $A(P) = \mathcal{D}_{\rm out}(P)$. $\square$

Proposition 4.493 (Strict-left incidence-cover carriers supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of lemma 4.339 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \ldots, j\} = \{j-2m+1, \ldots, j\}$$

with $m \geq 29$ deleted singleton packets. Let $A(m)$ and $B(m)$ be the frontier functions of proposition 4.452. Let d_a be the residual low-low degrees of definition 4.440. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \ldots, \rho_{2(j-2m)},$$

a deterministic rule identifies a target-window-determined set

$$U_{\rm tw} \subseteq \{1 \leq a \leq 2m : d_a > 0\},$$

identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies that E contains every low-low cell incident to $U_{\rm tw}$:

$$\{\{a,b\} : 1 \leq a < b \leq 2m,\ \{a,b\} \cap U_{\rm tw} \neq \varnothing\} \subseteq E.$$

Assume moreover that one positive-residual incidence-cover threshold holds. Namely, put

$$r_i = \#\{1 \leq a \leq 2m : d_a = i\} \quad (i = 1,2), \qquad D = \sum_{a=1}^{2m} d_a, \qquad D(U_{\rm tw}) = \sum_{a \in U_{\rm tw}} d_a.$$

Assume one of

$$|U_{\rm tw}| > r_1 + r_2 - A(m) \qquad \text{or} \qquad D(U_{\rm tw}) > D - B(m).$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\rm out}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. By lemma 4.339, the outside datum $\mathcal{D}_{\rm out}(P)$ determines

$$\rho_0, \rho_1, \ldots, \rho_{2(j-2m)}.$$

Therefore the full target-window tuple determines every input to the deterministic rule in the hypothesis. It follows that the tuple determines $U_{\rm tw}$, the carrier E, its top/right boundary labels, and the displayed incidence inclusion.

Since the top/right boundary labels of E are determined, corollary 4.124 recovers every Step 3 cell entry on E. The displayed inclusion therefore gives a target-window partial low-low readout on every low-low cell incident to $U_{\rm tw}$. If

$$|U_{\rm tw}| > r_1 + r_2 - A(m),$$

then, because U_{tw} contains only positive-residual labels,

$$\#\{a \notin U_{\mathrm{tw}} : d_a > 0\} = r_1 + r_2 - |U_{\mathrm{tw}}| < A(m),$$

and corollary 4.450 supplies a canonical full-budget low-low candidate set containing the true low-low assignment. If instead $D(U_{\mathrm{tw}}) > D - B(m)$, then corollary 4.451 supplies the same candidate set. In all cases proposition 4.478 then gives the required fixed right-boundary inverse with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$. ☐

Proposition 4.494 (Column-even tight prefixes supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \ldots, j\} = \{j - 2m + 1, \ldots, j\}$$

with $m \geq 2$ deleted singleton packets, and let

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}$$

be the true tight-prefix Pieri path. Assume that $\lambda^{(2m)}$ is column-even. Choose the final-zone replacement window V_L^* and branch word

$$\beta_L \in \{0,1\}^{2m}$$

by the fixed odd-alternating height full-zone decoder of proposition 4.108 applied to this tight prefix. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P).$$

Proof. By corollary 4.328, the displayed prefix is an odd alternating height full-zone path with crossing set

$$\{1, 3, \ldots, 2m - 1\}.$$

Since $m \geq 2$ and $\lambda^{(2m)}$ is column-even, proposition 4.108 supplies a target path V_L^* with m two-step blocks, a word $\beta_L \in \{0,1\}^{2m}$, and a fixed decoder recovering the full prefix Pieri path from (V_L^*, β_L).

The recovered prefix path determines the boxes carrying the semistandard labels $1, \ldots, 2m$ by lemma 4.97. Restricting to these labels gives the same labelled prefix tableau as in the original tableau. Therefore lemma 4.174 determines the Step 2 symmetric matrix entries whose two labels are at most $2m$, and lemma 4.127 identifies these entries with the Step 3 cell entries on the low-low cell set L_{2m}.

The outside datum $A(P) = \mathcal{D}_{\mathrm{out}}(P)$ determines every Step 3 entry in the continuation-column subarrangement by lemma 4.341. Combining those entries with the recovered entries on L_{2m} determines every entry of the full Step 3 half-square: an unordered off-diagonal label pair either has both labels at most $2m$, or is incident to a continuation label. Applying lemma 4.126 recovers

$$\rho_0, \rho_1, \ldots, \rho_{2j}.$$

The required right-boundary subpath is its terminal segment

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \ldots, \rho_{2j},$$

because $a_L = j - 2m + 1$. ☐

Proposition 4.495 (Target-window Hall-augmented non-column-even carriers supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.327 and lemma 4.339 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \ldots, j\} = \{j - 2m + 1, \ldots, j\}$$

with $m \geq 9$ deleted singleton packets, and let

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}$$

be the true tight-prefix Pieri path. Assume that $\lambda^{(2m)}$ is not column-even. Suppose that the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the paired endpoint-defect set of lemma 4.331, the continuation-incidence sets

$$N(a) = \{t \in \{2m+1, \ldots, j\} : C_t \cap P_a \neq \varnothing\},$$

where P_a are the adjacent endpoint-defect pairs and C_t is the two-column set hit by continuation label t, as in lemmas 4.334 and 4.336, and the lexicographically least canonical Hall charge of lemma 4.335. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \ldots, \rho_{2(j-2m)},$$

together with $\lambda^{(2m)}$, the endpoint-defect pairs, the continuation incidences, the canonical continuation Hall charge, and the Hall-charged continuation blocks recovered by corollary 4.516, a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$L_{2m} \subseteq E$$

for the low-low cell set of lemma 4.182. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\text{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. The first recovery hypothesis makes $\lambda^{(2m)}$, the endpoint-defect pairs, the continuation-incidence sets, and the canonical Hall charge fixed functions of the displayed target-window data. The outside component $\mathcal{D}_{\text{out}}(P)$ also determines

$$\rho_0, \rho_1, \ldots, \rho_{2(j-2m)}$$

by lemma 4.339. Therefore the displayed target-window data determine the Hall-charged continuation blocks by corollary 4.516, and hence determine every input to the deterministic Hall-augmented carrier rule. They determine the carrier E, its top/right boundary labels, and the inclusion $L_{2m} \subseteq E$.

Since the top/right boundary labels of E are determined, corollary 4.124 recovers every Step 3 cell entry on E. The inclusion $L_{2m} \subseteq E$ therefore gives every entry on the low-low cell set L_{2m}. Applying proposition 4.472 gives the required fixed right-boundary auxiliary inverse with $A(P) = \mathcal{D}_{\text{out}}(P)$. ∎

Proposition 4.496 (Hall-augmented incidence-cover carriers supply the large right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.327 and lemma 4.339 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \ldots, j\} = \{j - 2m + 1, \ldots, j\}$$

with $m \geq 29$ deleted singleton packets, and let

$$\lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2m)}$$

be the true tight-prefix Pieri path. Assume that $\lambda^{(2m)}$ is not column-even. Suppose that the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$*, the paired endpoint-defect set of lemma 4.331, the continuation-incidence sets*

$$N(a) = \{t \in \{2m+1, \dots, j\} : C_t \cap P_a \neq \varnothing\},$$

where P_a *are the adjacent endpoint-defect pairs and* C_t *is the two-column set hit by continuation label* t*, as in lemmas 4.334 and 4.336, and the lexicographically least canonical Hall charge of lemma 4.335. Let* $A(m)$ *and* $B(m)$ *be the frontier functions of proposition 4.452. Let* d_a *be the residual low-low degrees of definition 4.440. Suppose that, from*

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

together with $\lambda^{(2m)}$*, the endpoint-defect pairs, the continuation incidences, the canonical continuation Hall charge, and the Hall-charged continuation blocks recovered by corollary 4.516, a deterministic rule identifies a target-window-determined set*

$$U_{\mathrm{tw}} \subseteq \{1 \leq a \leq 2m : d_a > 0\},$$

identifies a Ferrers subarrangement E *of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of* E*, and verifies*

$$\{\{a,b\} : 1 \leq a < b \leq 2m,\ \{a,b\} \cap U_{\mathrm{tw}} \neq \varnothing\} \subseteq E.$$

Assume moreover that one positive-residual incidence-cover threshold holds. Namely, put

$$r_i = \#\{1 \leq a \leq 2m : d_a = i\} \quad (i = 1, 2), \qquad D = \sum_{a=1}^{2m} d_a, \qquad D(U_{\mathrm{tw}}) = \sum_{a \in U_{\mathrm{tw}}} d_a.$$

Assume one of

$$|U_{\mathrm{tw}}| > r_1 + r_2 - A(m) \qquad \text{or} \qquad D(U_{\mathrm{tw}}) > D - B(m).$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L *from*

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. The first recovery hypothesis makes $\lambda^{(2m)}$, the endpoint-defect pairs, the continuation-incidence sets, and the canonical Hall charge fixed functions of the displayed target-window data. The outside component $\mathcal{D}_{\mathrm{out}}(P)$ also determines

$$\rho_0, \rho_1, \dots, \rho_{2(j-2m)}$$

by lemma 4.339. Therefore the displayed target-window data determine the Hall-charged continuation blocks by corollary 4.516, and hence every input to the deterministic Hall-augmented carrier rule. They determine U_{tw}, the carrier E, its top/right boundary labels, and the displayed incidence inclusion.

Since the top/right boundary labels of E are determined, corollary 4.124 recovers every Step 3 cell entry on E. Hence the target-window data determine the true Step 3 entries on every low-low cell incident to U_{tw}. In the positive-label-count threshold subcase, the positivity of U_{tw} gives

$$\#\{a \notin U_{\mathrm{tw}} : d_a > 0\} = r_1 + r_2 - |U_{\mathrm{tw}}| < A(m),$$

so corollary 4.450 gives a canonical full-budget low-low candidate set containing the true low-low assignment. In the positive residual-degree threshold subcase, corollary 4.451 gives the same candidate set. Finally proposition 4.478 gives the required fixed right-boundary inverse with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$. ∎

Proposition 4.497 (Target-window continuation segments supply the Hall-augmented carrier inverse)**.** *Work in the setting of proposition 4.495. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the Pieri continuation segment

$$\lambda^{(2m)} \subset \lambda^{(2m+1)} \subset \cdots \subset \lambda^{(j)}.$$

Using this recovered segment, form the endpoint-defect pairs, the continuation incidences, and the canonical continuation Hall charge of lemma 4.513, and recover the Hall-charged continuation blocks of corollary 4.515. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \ldots, \rho_{2(j-2m)},$$

together with the recovered continuation segment, its canonical Hall data, and the Hall-charged continuation blocks, a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$L_{2m} \subseteq E.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$.

Proof. The first partition of the recovered continuation segment is $\lambda^{(2m)}$. For each continuation label $t = 2m+1, \ldots, j$, the difference

$$\lambda^{(t)} \setminus \lambda^{(t-1)}$$

is exactly the pair of boxes carrying label t by lemma 4.97. Hence the target-window data determine the prefix endpoint and all labelled continuation boxes.

Lemma 4.513 shows that the same segment determines the endpoint-defect set, its canonical adjacent pairing, the continuation incidences, and the canonical Hall charge. Moreover, corollary 4.515 shows that the Hall-charged continuation blocks are fixed functions of this segment and the recovered outside datum. By lemma 4.339, $\mathcal{D}_{\mathrm{out}}(P)$ determines

$$\rho_0, \rho_1, \ldots, \rho_{2(j-2m)}.$$

Therefore the displayed target-window data determine every input to the deterministic carrier rule in the present proposition. They determine the carrier E, its top/right boundary labels, and the inclusion $L_{2m} \subseteq E$.

The top/right boundary labels of E determine every Step 3 entry on E by corollary 4.124. Hence every entry on the low-low cell set L_{2m} is determined. Applying proposition 4.472 gives the claimed right-boundary inverse with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$. □

Corollary 4.498 (Internal Q-cuts supply the Hall-augmented carrier inverse)**.** *Work in the setting of proposition 4.497. Let C_{2m}^+ be the Step 2 Q-suffix strip of lemma 4.176, and let C_{2m} be the corresponding Step 2 Q-prefix strip. Suppose that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine the ordered partition labels on the internal separating boundary between the Step 2 Q-prefix strip C_{2m} and the suffix strip C_{2m}^+. Use corollary 4.342 to recover the Pieri continuation segment and then form its canonical Hall data and Hall-charged continuation blocks. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \ldots, \rho_{2(j-2m)},$$

together with the recovered continuation segment, its canonical Hall data, and the Hall-charged continuation blocks, a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square of lemma 4.125, identifies the partition labels on the top/right boundary of E, and verifies

$$L_{2m} \subseteq E.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$.

Proof. Corollary 4.342 recovers the Pieri continuation segment from the displayed target-window data. The remaining hypotheses are then exactly the Hall-augmented carrier hypotheses of proposition 4.497. Applying that proposition gives the asserted right-boundary inverse. ☐

Corollary 4.499 (Internal Q-cut Hall carriers are the shared terminal source)**.** *Work simultaneously in the non-full tight-height right-boundary setting of corollary 4.498 and in the large deleted-only graph setting of corollary 4.220. Assume the target-window data determine the ordered partition labels on the internal separating boundary between the Step 2 Q-prefix strip* C_{2m} *and the suffix strip* C_{2m}^{+}. *Use corollary 4.342 to recover the Pieri continuation segment, form its canonical Hall data, and recover the Hall-charged continuation blocks. Suppose that the same recovered data and strict-left segment determine a Ferrers subarrangement* E *of the Step 3 dual-RSK' half-square, determine the partition labels on the top/right boundary of* E*, and verify*

$$L_{2m} \subseteq E.$$

Then the same source artifact has both consequences:

(1) *the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with* $A(P) = \mathcal{D}_{\mathrm{out}}(P)$;
(2) *in the large-complement graph branch, the deleted-only graph is recovered from*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, the recovered internal Q-cut, and this Hall-augmented carrier rule.

Proof. The first assertion is exactly corollary 4.498 with the displayed internal cut and Hall-augmented carrier rule.

The second assertion is exactly corollary 4.220 with the same internal cut and the same Hall-augmented carrier rule. Thus no separate continuation-segment or deleted-only graph source is used. ☐

Corollary 4.500 (Direct Hall-data carriers are a shared terminal source)**.** *Work simultaneously in the non-full tight-height right-boundary setting of proposition 4.495 and in the large deleted-only graph setting of corollary 4.217. Assume that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, *the paired endpoint-defect set, the continuation-incidence sets* $N(a)$, *and the canonical Hall charge. Let the Hall-charged continuation blocks be the blocks determined from this Hall charge and the outside datum by corollary 4.516. Suppose that, from*

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \ldots, \rho_{2(j-2m)},$$

together with those direct Hall data and the recovered Hall-charged continuation blocks, the same deterministic rule determines a Ferrers subarrangement E *of the Step 3 dual-RSK' half-square, determines the partition labels on the top/right boundary of* E*, and verifies*

$$L_{2m} \subseteq E.$$

Then the same source artifact has both consequences:

(1) *the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with* $A(P) = \mathcal{D}_{\mathrm{out}}(P)$;
(2) *in the large-complement graph branch, the deleted-only graph is recovered from*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum and this direct Hall-data carrier rule.

Proof. The first assertion is exactly proposition 4.495 with the displayed direct Hall-data carrier rule. The second assertion is exactly corollary 4.217 with the same direct Hall data, the same recovered Hall-charged continuation blocks, and the same carrier rule. ☐

Corollary 4.501 (Direct Hall-data incidence-cover carriers are a shared terminal source)**.** *Work simultaneously in the non-full tight-height right-boundary setting of proposition 4.496 and in the large deleted-only graph setting of corollary 4.218. Assume that the target-window data*

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the paired endpoint-defect set, the continuation-incidence sets $N(a)$, and the canonical Hall charge. Let the Hall-charged continuation blocks be the blocks determined from this Hall charge and the outside datum by corollary 4.516. Let d_a be the residual low-low degrees from the incidence-cover carrier setting. Suppose that, from

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

together with those direct Hall data and the recovered Hall-charged continuation blocks, the same deterministic rule determines a target-window-determined set

$$U_{\text{tw}} \subseteq \{1 \le a \le 2m : d_a > 0\},$$

determines a Ferrers subarrangement E of the Step 3 dual-RSK' half-square, determines the partition labels on the top/right boundary of E, verifies

$$\{\{a,b\} : 1 \le a < b \le 2m,\ \{a,b\} \cap U_{\text{tw}} \neq \varnothing\} \subseteq E,$$

and verifies one of the positive-residual incidence-cover thresholds in proposition 4.496 for this U_{tw}. Then the same source artifact has both consequences:

1. *the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\text{out}}(P)$;*
2. *in the large-complement graph branch, the deleted-only graph is recovered from*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum and this direct Hall-data incidence-cover carrier rule.

Proof. The first assertion is exactly proposition 4.496 with the displayed direct Hall-data incidence-cover carrier rule. The second assertion is exactly corollary 4.218 with the same direct Hall data, recovered Hall-charged continuation blocks, carrier, incidence cover, and incidence-cover frontier verification. ☐

Corollary 4.502 (Strict-left low-low carriers are a shared terminal source)**.** *Work simultaneously in the non-full tight-height right-boundary setting of proposition 4.492 and in the large deleted-only graph setting of corollary 4.215, and in the target-window signature setting of proposition 4.506. Suppose that, from*

$$S, \quad Z_L, \quad V_L^*, \quad \rho_{2a_L-2}, \rho_{2j}, \quad \rho_0, \rho_1, \dots, \rho_{2(j-2m)},$$

a deterministic rule identifies a Ferrers subarrangement E of the Step 3 dual-RSK' half-square, identifies the partition labels on the top/right boundary of E, and verifies

$$L_{2m} \subseteq E.$$

Then the same low-low carrier source has all five consequences:

1. *the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253 with $A(P) = \mathcal{D}_{\text{out}}(P)$;*

(2) *in the large-complement graph branch, the deleted-only graph is recovered from*

$$S, \quad \Gamma^*, \quad \rho(v) \in \{0,1\}^q, \quad \iota(A) \in \{0,1\}^q,$$

the recovered outside datum, the carrier E, its recovered top/right boundary labels, and the verified inclusion $L_{2m} \subseteq E$;

(3) *the target-window data determine the terminal low-low boundary segment*

$$\rho_{2(j-2m)}, \rho_{2(j-2m)+1}, \dots, \rho_{2j};$$

(4) *the target-window data determine the internal separating boundary between the Step 2 Q-prefix strip C_{2m} and the suffix strip C^+_{2m}, and hence determine the Pieri continuation segment.*

(5) *the target-window data satisfy the hypothesis of proposition 4.506.*

Proof. The first assertion is proposition 4.492, and the second is corollary 4.215.

The third assertion is corollary 4.185 with $M = 2m$.

For the fourth assertion, the recovered top/right boundary labels of E determine every Step 3 cell entry on E by corollary 4.124; the inclusion $L_{2m} \subseteq E$ therefore determines every entry on L_{2m}. Applying corollary 4.343 gives the internal Q-cut and the Pieri continuation segment.

By lemma 4.339, the outside datum $\mathcal{D}_{\text{out}}(P)$ determines the strict-left segment $\rho_0, \rho_1, \dots, \rho_{2(j-2m)}$. Hence the full target-window tuple determines every input to the same carrier rule, and therefore determines the same entries on L_{2m}. Thus corollary 4.418 gives the fifth assertion. ☐

Corollary 4.503 (Small tight compatibility sets recover the tight right-boundary subpath)**.** *Work in the right-boundary auxiliary setting of proposition 4.253 and the non-full tight-height setting of corollary 4.320. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $1 \le m \le 8$ deleted singleton packets. Suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \dots, E_m$, and the labelled continuation boxes. Then from these data and

$$\beta_L \in \{0,1\}^{2m}$$

one recovers the original right-boundary zone subpath

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \dots, \rho_{2j}.$$

Proof. Set

$$\Omega_{\text{tw}} = \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

with the lexicographic order of definition 4.354. This set is a canonical function of the displayed target-window data. The true deleted odd-packet assignment belongs to it by lemma 4.355.

It remains to record the small-zone cardinality bound. If $m = 1$, then lemma 4.357 gives $|\Omega_{\text{tw}}| \le 1 \le 2^m$. If $2 \le m \le 6$, then lemma 4.360 gives

$$|\Omega_{\text{tw}}| \le 2(m-2)! \le 2^m.$$

For $m = 7$ and $m = 8$, apply respectively propositions 4.363 and 4.364 to the displayed target-window datum as $D(P)$; the hypotheses on $D(P)$ are exactly the displayed recovery of $\lambda^{(2m)}$, $E_1, \dots, E_m$, and the labelled continuation boxes. These propositions give $|\Omega_{\text{tw}}| \le 2^m$ in the two certified cases.

Thus Ω_{tw} satisfies the membership, canonical-ordering, and cardinality hypotheses needed in corollary 4.463. That corollary gives a canonically ordered low-low candidate set $\mathcal{L}_{\text{tw}}$ containing the true low-low assignment and satisfying

$$|\mathcal{L}_{\text{tw}}| \le 2^m.$$

Applying corollary 4.439 to this candidate set recovers the displayed right-boundary subpath from the target-window data and β_L. □

Proposition 4.504 (Small tight compatibility sets supply the right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $1 \le m \le 8$ deleted singleton packets, and suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \dots, E_m$, and the labelled continuation boxes. Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. By corollary 4.320, the final tight zone has

$$|Z_L \cap I(S)| = m.$$

The datum $A(P) = \mathcal{D}_{\mathrm{out}}(P)$ is an allowed auxiliary input in proposition 4.253. Applying corollary 4.503 recovers

$$\rho_{2a_L-2}, \rho_{2a_L-1}, \dots, \rho_{2j}$$

from the allowed data and the word $\beta_L \in \{0,1\}^{2m}$. Since $m = |Z_L \cap I(S)|$, this is the word length allowed by the global right-boundary auxiliary criterion. □

Proposition 4.505 (Split target-window readouts supply the right-boundary auxiliary inverse)**.** *Work in the non-full tight-height setting of corollary 4.320 and in the right-boundary auxiliary setting of proposition 4.253. Suppose the final zone is*

$$Z_L = \{a_L, \dots, j\} = \{j - 2m + 1, \dots, j\}$$

with $m \ge 1$ deleted singleton packets. Suppose the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \dots, E_m$, and the labelled continuation boxes. If $m \ge 29$, let

$$\Omega_{\mathrm{tw}} = \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

with the lexicographic order of definition 4.354; when $\Omega_{\mathrm{tw}} \ne \varnothing$, let $(\gamma; e_1, \dots, e_{m-1})$ be the associated fixed-even datum of definition 4.379. In the $m \ge 29$ cases not already covered by the condition

$$\Omega_{\mathrm{tw}} = \varnothing \quad \textit{or} \quad (\Omega_{\mathrm{tw}} \ne \varnothing \quad \textit{and} \quad (m = 29 \quad \textit{or} \quad m = 30 \quad \textit{or} \quad m = 31 \quad \textit{or} \quad (\gamma, e_{m-1}) \notin \mathcal{P}_m^{\mathrm{env}} \quad \textit{or} \quad (\gamma, e_{m-1}, e_{m-2}) \notin \mathcal{P}_m^{\mathrm{env},2} \quad \textit{or} \quad \exists d,\ 3 \le d \le m-1,\ (\gamma, e_{m-1}, e_{m-2}, \dots, e_{m-d}) \notin \mathcal{P}_m^{\mathrm{env},d})),$$

suppose moreover that the same target-window data determine either the commuting normal-form signature κ of the true fixed-even standard tableau, or a canonically ordered finite set

$$\Sigma_{\mathrm{tw}}$$

of commuting normal-form signature words such that the true fixed-even standard tableau has signature in Σ_{tw} and

$$|\Sigma_{\mathrm{tw}}| \le 2^m.$$

Then the final zone satisfies the right-boundary inverse hypothesis in proposition 4.253: taking

$$A(P) = \mathcal{D}_{\mathrm{out}}(P),$$

there is a fixed inverse which recovers the original subpath on Z_L from

$$V_L^*, \quad \beta_L, \quad \rho_{2a_L-2}, \rho_{2j}, \quad Z_L, \quad A(P), \qquad \beta_L \in \{0,1\}^{2|Z_L \cap I(S)|}.$$

Proof. If $1 \le m \le 8$, the hypotheses are exactly those of proposition 4.504; applying that proposition gives the required fixed inverse.

If $9 \le m \le 12$, the hypotheses are exactly those of proposition 4.482; applying that proposition gives the required fixed inverse.

If $13 \le m \le 28$, the hypotheses are exactly those of proposition 4.483; applying that proposition gives the required fixed inverse.

Now assume $m \ge 29$. The target-window data determine the unrefined compatibility set. If this set is empty, there is no current fiber. If $m = 29$, then proposition 4.487 supplies the required fixed inverse. If $m = 30$, then proposition 4.488 supplies it. If $m = 31$, then proposition 4.489 supplies it. If $m \ge 32$ and $(\gamma, e_{m-1}) \notin \mathcal{P}_m^{\mathrm{env}}$, then proposition 4.484 supplies it. If $m \ge 32$ and $(\gamma, e_{m-1}, e_{m-2}) \notin \mathcal{P}_m^{\mathrm{env},2}$, then proposition 4.485 supplies it. If $m \ge 32$ and some $3 \le d \le m-1$ satisfies

$$(\gamma, e_{m-1}, e_{m-2}, \dots, e_{m-d}) \notin \mathcal{P}_m^{\mathrm{env},d},$$

then proposition 4.486 supplies it. In the remaining case, taking $d = m-1$ and applying corollary 4.397 shows that the unrefined compatibility set itself has cardinality $> 2^{2m}$. Thus the additional signature or signature-list hypothesis supplies the required refinement input. If the data determine κ, then proposition 4.468 applies. If they determine Σ_{tw}, then proposition 4.481 applies. In both cases we get the same fixed inverse conclusion with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$. □

Proposition 4.506 (Right-boundary target-window signatures close the refined suffix)**.** *Work in the right-boundary auxiliary setting of proposition 4.253, and suppose the last zone $Z_L = \{a_L, \dots, j\}$ is a non-full tight height zone with $m \ge 9$ deleted singleton packets. Let*

$$V_L^*$$

be the replacement window parsed from the repaired path, and let

$$\mathcal{D}_{\mathrm{out}}(P)$$

be the outside datum of proposition 4.253. Suppose the commuting normal-form signature of the fixed-even standard tableau associated to the true odd-packet assignment is a deterministic function of

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\mathrm{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}.$$

Then the hypothesis of proposition 4.367 holds for the final right-boundary zone.

Proof. The replacement window is a subwindow of the already counted repaired path P^*, and the endpoint pair is returned by the parser. The outside datum $\mathcal{D}_{\mathrm{out}}(P)$ is recovered in proposition 4.253 from the unchanged blocks and earlier recovered zones, using only branch words already present in that recovery. Thus the deterministic function in the hypothesis supplies the commuting normal-form signature without adding a new branch word. Applying corollary 4.427 gives the required refined suffix hypothesis. □

Lemma 4.507 (The broad fixed-even bound is false)**.** *The estimate*

$$|\mathcal{S}(\gamma; e_1, \dots, e_{m-1})| \le 2^m$$

does not hold for all compatible fixed-even data. More precisely, there is a reduced fixed-even datum with $m = 26$ for which

$$|\mathcal{S}(\gamma; e_1, \dots, e_{25})| = 254531860 > 2^{26}.$$

Proof. Take

$$\gamma = (12, 10, 8, 6, 4, 3, 3, 2, 2, 1)$$

and fix the even cells

r	1	2	3	4	5	6	7	8	9	10
e_r	$(1,2)$	$(2,1)$	$(2,2)$	$(3,2)$	$(4,1)$	$(2,4)$	$(1,5)$	$(6,1)$	$(1,7)$	$(1,8)$

r	11	12	13	14	15	16	17	18	19	20
e_r	$(5,2)$	$(7,1)$	$(1,9)$	$(5,3)$	$(2,6)$	$(3,5)$	$(4,5)$	$(2,7)$	$(2,8)$	$(4,6)$

r	21	22	23	24	25
e_r	$(5,4)$	$(1,11)$	$(9,1)$	$(3,8)$	$(10,1)$.

This datum is nonempty. One witness fills the odd labels $1, 3, \ldots, 51$ in the cells

r	1	2	3	4	5	6	7	8	9	10
y_r	$(1,1)$	$(1,3)$	$(1,4)$	$(3,1)$	$(2,3)$	$(4,2)$	$(5,1)$	$(1,6)$	$(3,3)$	$(4,3)$

r	11	12	13	14	15	16	17	18	19	20
y_r	$(3,4)$	$(4,4)$	$(8,1)$	$(6,2)$	$(2,5)$	$(7,2)$	$(1,10)$	$(8,2)$	$(3,6)$	$(6,3)$

r	21	22	23	24	25	26
y_r	$(2,9)$	$(3,7)$	$(7,3)$	$(9,2)$	$(1,12)$	$(2,10)$.

The sublevel diagrams are Young diagrams, so this is a standard tableau with the displayed fixed even entries. Adjoining the terminal singleton fixed strip $\overline{E}_{26} = \{(1,13)\}$ lifts it to a reduced fixed-even datum of definition 4.379.

The exact verifier

`character_ring_iter/post_m29_bc_fixed_even_counterexample.cpp`

evaluates the recurrence of lemma 4.386 for this datum. The archived ‘optimus’ log

`certificates/post_m29/bc_fixed_even_counterexample_certificate.log`

has SHA-256

`69921ed52cfc8ad47d922dad9baf4fda`
`d4bbc2cc95393e3f9c21c6a586166566`’

ends with `__EXIT_STATUS=0`, verifies the displayed witness path, and reports

`fixed_even_count=254531860`, `pow2_budget=67108864`.

Thus the displayed compatible datum has more than 2^{26} fixed-even standard tableaux. □

Definition 4.508 (Viable active odd cells)**.** In the setting of definition 4.382, let $I \subseteq U$ and $0 \le r < m$. Call I viable at depth r if $|I| = r$ and there exists $\phi \in \mathcal{L}(\gamma; e_1, \ldots, e_{m-1})$ such that

$$I = \{u \in U : \phi(u) \le r\}.$$

For a viable I of depth r, define the viable active set

$$A_{\text{viab}}(I, r+1)$$

to be the set of cells $u \in U \setminus I$ for which there exists $\phi \in \mathcal{L}(\gamma; e_1, \ldots, e_{m-1})$ satisfying

$$I = \{v \in U : \phi(v) \le r\}, \qquad \phi(u) = r + 1.$$

Proposition 4.509 (Binary viable branching closes the interval leaf)**.** *Suppose that, for every datum of definition 4.382, every viable set I of depth $0 \le r < m$ satisfies*

$$|A_{\text{viab}}(I, r+1)| \le 2.$$

Then every such datum satisfies

$$|\mathcal{L}(\gamma; e_1, \ldots, e_{m-1})| \le 2^m.$$

Proof. Build a rooted tree whose vertices at depth r are the viable sets I of depth r, with an edge

$$I \longrightarrow I \cup \{u\}$$

for each $u \in A_{\text{viab}}(I, r+1)$. By definition, every $\phi \in \mathcal{L}(\gamma; e_1, \ldots, e_{m-1})$ determines exactly one path from the root $\varnothing$ to a depth-m leaf, namely its sublevel sets

$$\{u : \phi(u) \leq r\} \qquad (0 \leq r \leq m).$$

Conversely, a depth-m path records, at each depth, the unique cell assigned the next label; since every edge is required to be viable, the resulting final bijection is an element of $\mathcal{L}(\gamma; e_1, \ldots, e_{m-1})$. Thus the leaves of this tree are in bijection with $\mathcal{L}(\gamma; e_1, \ldots, e_{m-1})$.

The hypothesis says that every vertex has at most two children. A rooted tree of depth m with branching at most two has at most 2^m leaves. Therefore $|\mathcal{L}(\gamma; e_1, \ldots, e_{m-1})| \leq 2^m$. □

Lemma 4.510 (Binary viable branching is too strong)**.** *The hypothesis of proposition 4.509 is not satisfied by all reduced fixed-even data arising from nonempty fixed-cell completion sets. More precisely, for $m = 6$ there is such a datum and a viable depth-3 set I for which*

$$|A_{\text{viab}}(I, 4)| = 3.$$

Proof. Consider the reduced fixed-even endpoint

$$\overline{\nu} = (5, 3, 3, 1, 1)$$

and the reduced fixed even strips

$$\overline{E}_1 = \{(1,2)\}, \quad \overline{E}_2 = \{(2,1)\}, \quad \overline{E}_3 = \{(3,1)\},$$

$$\overline{E}_4 = \{(2,3)\}, \quad \overline{E}_5 = \{(1,5)\}, \quad \overline{E}_6 = \{(3,3),(5,1)\}.$$

This reduced datum is obtained by deleting the first column from the tight-prefix datum with endpoint

$$\nu = (6, 4, 4, 2, 2, 1, 1, 1, 1, 1, 1)$$

and even strips

r	E_r
1	$\{(1,3),(2,1)\}$
2	$\{(2,2),(4,1)\}$
3	$\{(3,2),(6,1)\}$
4	$\{(2,4),(8,1)\}$
5	$\{(1,6),(10,1)\}$
6	$\{(3,4),(5,2)\}$.

It is nonempty, for instance by taking odd second boxes

$$(1,2), (1,4), (2,3), (1,5), (4,2), (3,3)$$

for the labels $1, 3, 5, 7, 9, 11$.

After removing the terminal fixed strip one has

$$\gamma = \overline{\nu} \setminus \overline{E}_6 = (5, 3, 2, 1),$$

with fixed even cells

$$e_1 = (1,2), \quad e_2 = (2,1), \quad e_3 = (3,1), \quad e_4 = (2,3), \quad e_5 = (1,5).$$

The odd-cell set is

$$U = \{(1,1),(1,3),(1,4),(2,2),(3,2),(4,1)\}.$$

The interval bounds of definition 4.382 are

u	$[L(u), R(u)]$
$(1,1)$	$[1,1]$
$(1,3)$	$[2,4]$
$(1,4)$	$[2,5]$
$(2,2)$	$[3,4]$
$(3,2)$	$[4,6]$
$(4,1)$	$[4,6]$.

Let

$$I = \{(1,1), (1,3), (2,2)\}.$$

It is viable at depth 3: the assignment

$$(1,1) \mapsto 1, \quad (1,3) \mapsto 2, \quad (2,2) \mapsto 3, \quad (1,4) \mapsto 4, \quad (3,2) \mapsto 5, \quad (4,1) \mapsto 6$$

belongs to $\mathcal{L}(\gamma; e_1, \ldots, e_5)$ and has depth-3 sublevel set I.

Moreover each of $(1,4)$, $(3,2)$, and $(4,1)$ is viable as the next cell. Indeed, the first displayed assignment has $(1,4)$ carrying 4; the assignment

$$(1,1) \mapsto 1, \quad (1,3) \mapsto 2, \quad (2,2) \mapsto 3, \quad (3,2) \mapsto 4, \quad (1,4) \mapsto 5, \quad (4,1) \mapsto 6$$

has $(3,2)$ carrying 4; and the assignment

$$(1,1) \mapsto 1, \quad (1,3) \mapsto 2, \quad (2,2) \mapsto 3, \quad (4,1) \mapsto 4, \quad (1,4) \mapsto 5, \quad (3,2) \mapsto 6$$

has $(4,1)$ carrying 4. Each assignment respects the displayed intervals and the Young-order inequalities on U. Thus

$$\{(1,4), (3,2), (4,1)\} \subseteq A_{\text{viab}}(I, 4).$$

No other cell remains outside I, so this viable active set has cardinality 3. □

Lemma 4.511 (Reverse corner steps are commuting or fixed-blocked)**.** *Let*

$$\beta_r \supset \beta_{r-\frac{1}{2}} \supset \beta_{r-1}$$

be the rth step of a reverse reduced corner path, and write

$$\overline{E}_r = \beta_r \setminus \beta_{r-\frac{1}{2}}, \qquad \{y_r\} = \beta_{r-\frac{1}{2}} \setminus \beta_{r-1}.$$

If y_r is a removable corner of β_r, then the two removals commute:

$$\beta_r \setminus \{y_r\}, \qquad (\beta_r \setminus \{y_r\}) \setminus \overline{E}_r = \beta_{r-1}$$

are Young diagrams, and $\overline{E}_r$ is still a horizontal strip in the intermediate diagram.

If $y_r = (i,j)$ is not a removable corner of β_r, then at least one of

$$(i, j+1), \qquad (i+1, j)$$

belongs to $\overline{E}_r$. In particular, when $\overline{E}_r$ is a singleton, there are at most two non-commuting possible positions for y_r.

Proof. Assume first that y_r is a removable corner of β_r. Removing a removable corner from a Young diagram leaves a Young diagram, so $\beta_r \setminus \{y_r\}$ is a Young diagram. Since $y_r \in \beta_{r-\frac{1}{2}}$, it is disjoint from $\overline{E}_r$, and hence

$$(\beta_r \setminus \{y_r\}) \setminus \overline{E}_r = \beta_{r-\frac{1}{2}} \setminus \{y_r\} = \beta_{r-1}.$$

The last diagram is a Young diagram by the definition of the reverse corner path. The set $\overline{E}_r$ has at most one box in each column by definition 4.376; this property is unchanged by removing the disjoint box y_r first. Thus $\overline{E}_r$ remains a horizontal strip for the swapped order.

Now suppose that $y_r = (i,j)$ is not a removable corner of β_r. Because y_r is a removable corner of $\beta_{r-\frac{1}{2}}$, neither $(i, j+1)$ nor $(i+1, j)$ belongs to $\beta_{r-\frac{1}{2}}$. Since y_r is not removable in β_r, at least one of those two adjacent boxes belongs to β_r. The difference between β_r and $\beta_{r-\frac{1}{2}}$ is exactly $\overline{E}_r$, so that adjacent box belongs to $\overline{E}_r$. If $\overline{E}_r$ is a singleton, the only possible non-commuting positions are therefore the box immediately west of it and the box immediately north of it. ☐

Corollary 4.512 (Branch overflow is commuting)**.** *In the setting of lemma 4.511, assume that $\overline{E}_r$ is a singleton. Put*

$$\beta_{r-\frac{1}{2}} = \beta_r \setminus \overline{E}_r$$

and let $B_r(\beta_r)$ be the set of removable corners y of $\beta_{r-\frac{1}{2}}$. Then the subset

$$N_r(\beta_r) = \{y \in B_r(\beta_r) : y \text{ is not a removable corner of } \beta_r\}$$

has cardinality at most two. Equivalently, if a reverse state has more than two possible odd removals after the fixed singleton removal, all but at most two of those removals commute with the fixed removal.

Proof. Write the singleton as $\overline{E}_r = \{e\}$, with $e = (i, j)$. By lemma 4.511, every $y \in N_r(\beta_r)$ must be one of the two adjacent positions

$$(i, j-1), \qquad (i-1, j),$$

with the nonexistent row or column omitted. Hence $|N_r(\beta_r)| \leq 2$. The final sentence is the same statement together with the commuting alternative in lemma 4.511. ☐

Lemma 4.513 (The Pieri continuation segment determines the tight Hall charge)**.** *In the setting of lemma 4.331, suppose one is given the Pieri continuation segment*

$$\lambda^{(2m)} \subset \lambda^{(2m+1)} \subset \cdots \subset \lambda^{(j)}.$$

Then this segment determines the endpoint-defect set O_{pre}, its canonical adjacent pairing $P_1, \ldots, P_r$, every continuation column set C_t for $2m < t \leq j$, the incidence relation

$$C_t \cap P_a \neq \varnothing,$$

and the canonical Hall charge of lemma 4.335.

Proof. The first partition in the displayed segment is $\lambda^{(2m)}$, so it determines

$$O_{\mathrm{pre}} = \{c \geq 1 : (\lambda^{(2m)})'_c \equiv 1 \pmod 2\}.$$

Listing this finite set increasingly determines its adjacent pairing, as in lemma 4.331. For each continuation label $t > 2m$, the two consecutive partitions $\lambda^{(t-1)} \subset \lambda^{(t)}$ determine

$$C_t = \{c \geq 1 : (\lambda^{(t)})'_c - (\lambda^{(t-1)})'_c = 1\}.$$

Therefore the incidence relation $C_t \cap P_a \neq \varnothing$ is determined. Finally, lemma 4.335 defines the Hall charge as the lexicographically least admissible injection for this incidence family, so that charge is determined by the same segment. ☐

Lemma 4.514 (Tight continuation labels are external to the tight zone)**.** *In the setting of corollary 4.320, write the tight height zone as*

$$Z_\gamma = \{a_\gamma, \ldots, j\}, \qquad a_\gamma = j - 2m + 1.$$

Let $H(S)$ and R be the one-block halo and unchanged complement of lemma 4.254. For a continuation label $t \in \{2m+1, \ldots, j\}$, put

$$r(t) = j - t + 1.$$

Then $r(t) \notin Z_\gamma$. Moreover, either $r(t) \in R$, or $r(t)$ belongs to a canonical halo zone Z_η with $b_\eta < a_\gamma$. Consequently, every label selected by the injection of lemma 4.334 corresponds to a block outside the tight zone: an unchanged complement block or a block carried by a strictly earlier halo-zone window.

Proof. For $t \geq 2m+1$ one has

$$r(t) = j - t + 1 \leq j - 2m = a_\gamma - 1,$$

so $r(t) \notin Z_\gamma$.

If $r(t) \notin H(S)$, then $r(t) \in R$ by the definition of the complement in lemma 4.254. If $r(t) \in H(S)$, it lies in a unique maximal consecutive component $Z_\eta = \{a_\eta, \ldots, b_\eta\}$ of $H(S)$. This component cannot be Z_γ, because $r(t) < a_\gamma$. The canonical halo zones are disjoint intervals, and Z_γ begins at a_γ, so any different component containing $r(t)$ must satisfy $b_\eta < a_\gamma$. Applying this to the labels in the image of the injection from lemma 4.334 proves the final sentence. □

Corollary 4.515 (Hall-charged continuation blocks are outside data)**.** *Work in the non-full tight-height right-boundary setting of lemma 4.339. Suppose that the Pieri continuation segment*

$$\lambda^{(2m)} \subset \lambda^{(2m+1)} \subset \cdots \subset \lambda^{(j)}$$

has been recovered, and let

$$\phi : \{1, \ldots, r\} \hookrightarrow \{2m+1, \ldots, j\}$$

be the canonical continuation Hall charge determined by lemma 4.513. For each endpoint-defect pair P_a put

$$b_a = j - \phi(a) + 1.$$

Then the outside datum $\mathcal{D}_{\text{out}}(P)$ determines the original two-step growth block

$$B_{b_a}(P) = (\rho_{2b_a-2}, \rho_{2b_a-1}, \rho_{2b_a}).$$

Consequently the canonically ordered list of Hall-charged continuation blocks is a fixed function of the recovered continuation segment and the outside datum.

Proof. The recovered continuation segment determines the canonical Hall charge by lemma 4.513. Since $\phi(a) \geq 2m+1$, the corresponding block index satisfies

$$b_a = j - \phi(a) + 1 \leq j - 2m.$$

Also $b_a \geq 1$ because $\phi(a) \leq j$. Thus the three boundary labels

$$\rho_{2b_a-2}, \qquad \rho_{2b_a-1}, \qquad \rho_{2b_a}$$

all lie in the strict-left segment

$$\rho_0, \rho_1, \ldots, \rho_{2(j-2m)}.$$

By lemma 4.339, this whole segment is determined by $\mathcal{D}_{\text{out}}(P)$. Therefore every block $B_{b_a}(P)$ selected by the Hall charge is recovered, and the final assertion follows by listing the defect pairs in their canonical order. □

Corollary 4.516 (Direct Hall charges select outside blocks)**.** *Work in the non-full tight-height right-boundary setting of lemma 4.339. Suppose that the paired endpoint-defect set and the canonical continuation Hall charge*

$$\phi : \{1, \ldots, r\} \hookrightarrow \{2m+1, \ldots, j\}$$

are recovered. For each endpoint-defect pair P_a put

$$b_a = j - \phi(a) + 1.$$

Then the outside datum $\mathcal{D}_{\text{out}}(P)$ determines the original two-step growth block

$$B_{b_a}(P) = (\rho_{2b_a-2}, \rho_{2b_a-1}, \rho_{2b_a}).$$

Consequently the canonically ordered list of Hall-charged continuation blocks is a fixed function of the recovered Hall charge and the outside datum, and does not require the full Pieri continuation segment.

Proof. The proof is the same block-location argument as in corollary 4.515, but the Hall charge is now an input rather than being computed from the continuation segment. Since

$$2m + 1 \le \phi(a) \le j,$$

the corresponding block index satisfies

$$1 \le b_a = j - \phi(a) + 1 \le j - 2m.$$

Thus the whole block $B_{b_a}(P)$ lies in the strict-left segment

$$\rho_0, \rho_1, \dots, \rho_{2(j-2m)}.$$

By lemma 4.339, this segment is determined by $\mathcal{D}_{\text{out}}(P)$. Listing the defect pairs in their canonical order gives the stated ordered list of blocks. □

Lemma 4.517 (Canonical halo zones carry frontier-event packets)**.** *Work in either the width or height setting of lemma 4.188. Let*

$$J_\beta = \{a_\beta, a_\beta + 1, \dots, b_\beta\}$$

be the maximal consecutive witness-label intervals in S, and let E_β denote the corresponding ordered frontier-event packet E^w_β or E^h_β. Put

$$K_\beta = \{j - b_\beta + 1, \dots, j - a_\beta + 1\} \subset I(S)$$

for the corresponding deleted block interval, and let

$$\widehat{Z}_\beta = \{\max(1, j - b_\beta), \dots, \min(j, j - a_\beta + 2)\}$$

be its one-block halo. Let Z_γ be the canonical halo replacement zones of lemma 4.254.

Then each interval K_β is contained in a unique Z_γ, and for each γ the set $Z_\gamma \cap I(S)$ is the disjoint union of the intervals K_β contained in Z_γ. Moreover

$$\sum_{K_\beta \subseteq Z_\gamma} |E_\beta| = |Z_\gamma \cap I(S)|.$$

The assignment of witness-label packets and deleted block intervals to a halo zone is determined by S; within each packet the frontier events occur in the same order as the labels.

Proof. By lemma 4.188, the maximal witness-label interval $J_\beta = \{a_\beta, \dots, b_\beta\}$ maps under the label-to-block conversion $s \mapsto j - s + 1$ to the deleted block interval $K_\beta = \{j - b_\beta + 1, \dots, j - a_\beta + 1\}$. The sets K_β are exactly the maximal consecutive components of $I(S)$, in the reversed order.

The one-block halo of $K_\beta = \{u, \dots, v\}$ inside $\{1, \dots, j\}$ is $\{\max(1, u-1), \dots, \min(j, v+1)\}$, which is precisely the displayed $\widehat{Z}_\beta$. The full halo $H(S)$ of lemma 4.254 is the union of these $\widehat{Z}_\beta$ over all maximal deleted block intervals. Therefore each $K_\beta \subset H(S)$ lies in a unique maximal consecutive component Z_γ of $H(S)$, and $Z_\gamma \cap I(S)$ is the disjoint union of the K_β contained in that component.

Again by lemma 4.188, $|E_\beta| = |J_\beta|$, and the label-to-block map is a bijection from J_β to K_β, so $|K_\beta| = |J_\beta| = |E_\beta|$. Summing over the intervals contained in a fixed Z_γ gives the displayed equality. The final order statement is also part of lemma 4.188: the elements of each E_β occur in the same order as the labels in J_β. □

Lemma 4.518 (Zone branch words split into packets)**.** *Work in the setting of lemma 4.517. Fix one canonical halo zone Z_γ, and let*

$$J_{\beta_1}, \dots, J_{\beta_m}$$

be the witness-label packets whose deleted block intervals K_{β_r} are contained in Z_γ, listed in increasing witness-label order. Put

$$h_r = |E_{\beta_r}| = |J_{\beta_r}| \qquad \textit{and} \qquad e_\gamma = \sum_{r=1}^{m} h_r.$$

Then the packet lengths $h_1, \dots, h_m$ and their order are determined by S. Moreover concatenation gives a bijection

$$\prod_{r=1}^{m} \{0,1\}^{2h_r} \longleftrightarrow \{0,1\}^{2e_\gamma}.$$

Consequently a single zone branch word in $\{0,1\}^{2e_\gamma}$ carries exactly the same branch data as the packet-local branch words used in lemmas 4.225 and 4.226 for the packets inside Z_γ.

Proof. The set S determines its maximal consecutive witness-label intervals J_β and their increasing order. By lemma 4.517, S also determines which of these packets are assigned to the fixed halo zone Z_γ, and $|E_{\beta_r}| = |J_{\beta_r}|$ for every r. Hence the ordered list $h_1, \dots, h_m$ is determined by S.

Concatenation sends

$$(\beta_{\beta_1}, \dots, \beta_{\beta_m}) \in \prod_{r=1}^{m} \{0,1\}^{2h_r}$$

to the word obtained by writing the packet words in the displayed order. The length of the concatenated word is

$$\sum_{r=1}^{m} 2h_r = 2e_\gamma.$$

Conversely, because the cut lengths $2h_1, \dots, 2h_m$ are determined by S, any word in $\{0,1\}^{2e_\gamma}$ is split uniquely into consecutive subwords of those lengths. These two operations are inverse to one another. The final sentence is exactly this bijection applied to the packet-local branch words of lemmas 4.225 and 4.226. □

Lemma 4.519 (Retained halo blocks expose packet endpoints)**.** *Let*

$$P = (\rho_0, \rho_1, \dots, \rho_{2j})$$

be a two-step growth path, let $S \subset \{1, \dots, j\}$, put $I = I(S)$, and let

$$Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$$

be one canonical halo zone. Write the maximal consecutive components of $Z_\gamma \cap I$ as

$$K_t = \{u_t, \dots, v_t\} \qquad (1 \le t \le m).$$

Suppose one is given S, the zone endpoint pair

$$\rho_{2a_\gamma - 2}, \qquad \rho_{2b_\gamma},$$

and, for every retained halo index $r \in Z_\gamma \setminus I$, the original two-step block

$$B_r(P) = (\rho_{2r-2}, \rho_{2r-1}, \rho_{2r}).$$

Then, for every packet K_t, the packet endpoint pair

$$\rho_{2u_t - 2}, \qquad \rho_{2v_t}$$

is determined by these data.

Proof. The set S determines I, the canonical halo zones, and the components $K_t = \{u_t, \dots, v_t\}$ of $Z_\gamma \cap I$.

Fix one component K_t. For its left endpoint, if $u_t = a_\gamma$, then $\rho_{2u_t-2} = \rho_{2a_\gamma-2}$ is the given left zone endpoint. If $u_t > a_\gamma$, then the preceding index $u_t - 1$ lies in Z_γ. By maximality of K_t as a component of $Z_\gamma \cap I$, this index is not in I, so the retained block $B_{u_t-1}(P)$ is among the given data. Its right endpoint is

$$\rho_{2(u_t-1)} = \rho_{2u_t-2},$$

which determines the desired left packet endpoint.

For the right endpoint, if $v_t = b_\gamma$, then $\rho_{2v_t} = \rho_{2b_\gamma}$ is the given right zone endpoint. If $v_t < b_\gamma$, then the following index $v_t + 1$ lies in $Z_\gamma \setminus I$ by the same maximality argument. The given retained block $B_{v_t+1}(P)$ has left endpoint

$$\rho_{2(v_t+1)-2} = \rho_{2v_t},$$

which determines the desired right packet endpoint. □

Lemma 4.520 (Packet endpoints need only adjacent halo blocks)**.** *In the setting of lemma 4.519, fix one packet $K_t = \{u_t, \dots, v_t\}$ in a canonical halo zone $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$. Its endpoint pair*

$$\rho_{2u_t-2}, \qquad \rho_{2v_t}$$

is determined by S, the zone endpoint pair

$$\rho_{2a_\gamma-2}, \qquad \rho_{2b_\gamma},$$

and only the following adjacent retained halo blocks, when they exist:

$$B_{u_t-1}(P) \quad \text{if } u_t > a_\gamma, \qquad B_{v_t+1}(P) \quad \text{if } v_t < b_\gamma.$$

Consequently, a zone-local inverse that calls packet inverses does not need all retained halo blocks merely to determine packet endpoints; it needs, for each packet, only the retained block immediately to its left or right, unless the corresponding side is the zone boundary.

Proof. The proof is the packetwise part of lemma 4.519. If $u_t = a_\gamma$, the left packet endpoint is the left zone endpoint. If $u_t > a_\gamma$, then $u_t - 1 \in Z_\gamma \setminus I$ by maximality of K_t as a component of $Z_\gamma \cap I$, and the right endpoint of $B_{u_t-1}(P)$ is

$$\rho_{2(u_t-1)} = \rho_{2u_t-2}.$$

This determines the left endpoint. The right endpoint is identical: if $v_t = b_\gamma$, use the right zone endpoint, and if $v_t < b_\gamma$, then $v_t + 1 \in Z_\gamma \setminus I$ and the left endpoint of $B_{v_t+1}(P)$ is

$$\rho_{2(v_t+1)-2} = \rho_{2v_t}.$$

No other retained halo block enters these formulas. □

Lemma 4.521 (All retained halo blocks are packet-adjacent)**.** *Let $S \subset \{1, \dots, j\}$, put $I = I(S)$, and let $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$ be one canonical halo zone. Write the maximal consecutive components of $Z_\gamma \cap I$ as*

$$K_t = \{u_t, \dots, v_t\} \qquad (1 \le t \le m).$$

Then

$$Z_\gamma \setminus I = \{u_t - 1 :\ 1 \le t \le m,\ a_\gamma \le u_t - 1\} \cup \{v_t + 1 :\ 1 \le t \le m,\ v_t + 1 \le b_\gamma\}.$$

Consequently, if a zone-local inverse uses lemma 4.520 to determine the endpoints of every packet K_t, then the adjacent retained block indices it may need are exactly all retained halo indices in $Z_\gamma \setminus I$. Any saving in visible retained blocks must therefore come from a merged rule that recovers some endpoints without exposing the corresponding retained block, or from a packet mechanism that is not called for every deleted component as a separate packet inverse.

Proof. The inclusion from right to left is immediate: if $a_\gamma \le u_t - 1$, then $u_t - 1 \in Z_\gamma$ and it is not in I because K_t is a component of $Z_\gamma \cap I$; similarly $v_t + 1 \in Z_\gamma \setminus I$ whenever $v_t + 1 \le b_\gamma$.

For the reverse inclusion, let $r \in Z_\gamma \setminus I$. Since Z_γ is a component of the one-block halo of I, there is some $i \in I$ with $|r - i| = 1$. The index i lies in a unique component $K_t = \{u_t, \dots, v_t\}$ of $Z_\gamma \cap I$. If $r = i - 1$, then $r \notin I$ forces $i = u_t$, so $r = u_t - 1$. If $r = i + 1$, then $r \notin I$ forces $i = v_t$, so $r = v_t + 1$. This proves the displayed equality.

The final statement is just the displayed equality read together with lemma 4.520: the left and right endpoint lookups over all packets use precisely the retained indices appearing in the two displayed sets, whose union is $Z_\gamma \setminus I$. □

Proposition 4.522 (Packet inverses assemble inside a halo zone)**.** *Let*

$$P = (\rho_0, \rho_1, \dots, \rho_{2j})$$

be a two-step growth path, let $S \subset \{1, \dots, j\}$, put $I = I(S)$, and let

$$Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$$

be one canonical halo zone. Write the maximal consecutive components of $Z_\gamma \cap I$ in increasing block order as

$$K_t = \{u_t, \dots, v_t\} \qquad (1 \le t \le m).$$

Suppose a zone-local datum consists of a replacement output V_γ^*, a zone branch word $\beta_\gamma \in \{0,1\}^{2e_\gamma}$, the endpoint pair $\rho_{2a_\gamma-2}, \rho_{2b_\gamma}$, and the interval Z_γ, with the following properties.

(1) *It recovers the original retained halo blocks*

$$B_r(P) = (\rho_{2r-2}, \rho_{2r-1}, \rho_{2r}) \qquad (r \in Z_\gamma \setminus I).$$

(2) *Using the deterministic splitting of lemma 4.518, it recovers a packet branch word β_{K_t} for each component K_t.*

(3) *For each K_t, it recovers a packet output U_t^* such that a fixed packet inverse recovers the deleted packet subpath*

$$(\rho_{2u_t-2}, \rho_{2u_t-1}, \dots, \rho_{2v_t})$$

from U_t^*, β_{K_t}, the packet endpoint pair $\rho_{2u_t-2}, \rho_{2v_t}$, and K_t.

Then the whole original zone subpath

$$(\rho_{2a_\gamma-2}, \rho_{2a_\gamma-1}, \dots, \rho_{2b_\gamma})$$

is recoverable from the zone-local datum.

Proof. The set S determines I, the canonical halo zones, and the ordered list of components K_t of $Z_\gamma \cap I$. By the first hypothesis, every retained halo block with index in $Z_\gamma \setminus I$ is recovered. Together with the given zone endpoint pair, these retained blocks determine the endpoint pair of every deleted packet by lemma 4.519.

By the second hypothesis, the zone branch word gives the branch word attached to each packet. Therefore the third hypothesis applies to every K_t and recovers each deleted packet subpath

$$(\rho_{2u_t-2}, \rho_{2u_t-1}, \dots, \rho_{2v_t}).$$

The packet intervals K_t and the retained singleton indices $Z_\gamma \setminus I$ partition Z_γ. Interleaving the recovered packet subpaths and the recovered retained blocks in increasing block order reconstructs

$$\rho_{2a_\gamma-2}, \rho_{2a_\gamma-1}, \dots, \rho_{2b_\gamma}.$$

The adjacent endpoints agree because each recovered piece carries the endpoint pair it had in the original path. Hence the zone subpath is recovered. □

Lemma 4.523 (Halo-zone outputs have no spare packet windows)**.** *Let $S \subset \{1, \dots, j\}$, put $I = I(S)$, and let*

$$Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$$

be one canonical halo zone. Write the maximal consecutive components of $Z_\gamma \cap I$ as $K_1, \dots, K_m$. Suppose a target replacement window V_γ^* for this zone has exactly

$$|Z_\gamma \setminus I|$$

two-step blocks. If a parser of V_γ^* returns, as pairwise disjoint subwindows,

(1) *one unchanged retained halo block for every $r \in Z_\gamma \setminus I$; and*

(2) *packet output windows U_t^* attached to the deleted packets K_t,*

then every packet output window U_t^* has half-length zero.

Consequently, a canonical halo-zone repair cannot expose all retained halo blocks as unchanged full windows and also expose any positive-length packet HKKO output as a separate subwindow. Any positive packet information must be encoded by modifying or merging retained halo windows, or by a zero-window inverse whose branch and endpoint data are sufficient for that restricted packet family.

Proof. There are exactly $|Z_\gamma \setminus I|$ retained halo indices. The parser in item (1) therefore already returns $|Z_\gamma \setminus I|$ pairwise disjoint two-step windows. Since V_γ^* itself has exactly this many two-step blocks, no additional disjoint packet output window can have positive half-length. Hence every U_t^* has half-length zero.

The consequence is the contrapositive applied to a proposed positive packet output: a positive separate packet subwindow would add at least one two-step block beyond the zone target count unless some retained halo window is not kept as an unchanged full window, or unless the packet is represented by a zero-window inverse rather than by a positive output window. ☐

Corollary 4.524 (Unchanged-retained zero-window packet parsers are not a uniform halo supplier)**.** *In the ambient two-step growth-path category, the fixed-witness canonical-halo supplier cannot be obtained by the following universal parser rule:*

1. *expose every retained halo block in $Z_\gamma \setminus I(S)$ unchanged as a disjoint subwindow of V_γ^*;*
2. *attach no positive-length packet output to any deleted packet; and*
3. *recover each deleted packet only from its endpoint pair, its known packet interval, and its packet branch word.*

Consequently, any canonical-halo supplier using the below-wall packet-splice interface must either merge or modify retained halo windows, use additional fixed-witness restrictions in the packet inverse, or recover packet-sensitive data from the replacement window.

Proof. If every retained halo block is exposed unchanged as a disjoint subwindow, lemma 4.523 forces every disjoint packet output to have half-length zero. Thus the proposed rule would need, already for a singleton packet, a fixed inverse from only the common endpoint pair, the known singleton interval, and a two-bit packet branch word.

Corollary 4.247 gives an explicit ambient family with common endpoint pair

$$(5,4,3,2,1), \qquad (5,4,3,2,1),$$

known singleton interval, and five possible deleted midpoints. A two-bit branch word has only four values, so such an endpoint-only zero-window inverse cannot be injective on that family. This contradicts the proposed universal parser rule. The final sentence lists the remaining alternatives after this endpoint-only zero-window route is removed. ☐

Lemma 4.525 (Halo-zone parser budget)**.** *Let $S \subset \{1,\dots,j\}$, put $I = I(S)$, and let*

$$Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$$

be one canonical halo zone. Write the maximal consecutive components of $Z_\gamma \cap I$ as $K_1, \dots, K_m$. Suppose a target replacement window V_γ^* for this zone has exactly

$$R_\gamma = |Z_\gamma \setminus I|$$

two-step blocks. If a parser of V_γ^* returns, as pairwise disjoint subwindows,

1. *unchanged retained halo blocks for the indices in a set $A \subseteq Z_\gamma \setminus I$; and*
2. *packet output windows U_t^* of half-length $N_t \geq 0$ for $1 \leq t \leq m$,*

then

$$|A| + \sum_{t=1}^{m} N_t \leq R_\gamma.$$

Consequently, if the packet output attached to K_t has the raw HKKO half-length $|K_t|$ for every t, then such disjoint outputs can fit only if

$$|A| + |Z_\gamma \cap I| \leq |Z_\gamma \setminus I|.$$

In particular, a zone with $|Z_\gamma \cap I| > |Z_\gamma \setminus I|$ cannot expose raw packet outputs for every deleted packet as disjoint target subwindows, even if no retained halo block is exposed unchanged.

Proof. The window V_γ^* contains exactly R_γ two-step blocks. The subwindows returned by the parser are pairwise disjoint. The unchanged retained blocks in item (1) therefore occupy $|A|$ two-step blocks, and the packet output windows in item (2) occupy $\sum_t N_t$ two-step blocks. Since all these subwindows lie inside V_γ^*, their total number of two-step blocks is at most R_γ, proving the first inequality.

If $N_t = |K_t|$ for every packet, then the packet intervals K_t are exactly the components of $Z_\gamma \cap I$, so

$$\sum_{t=1}^{m} N_t = \sum_{t=1}^{m} |K_t| = |Z_\gamma \cap I|.$$

Substitution gives the second displayed inequality. If $|Z_\gamma \cap I| > |Z_\gamma \setminus I|$, this inequality fails even for $A = \varnothing$, giving the final assertion. □

Corollary 4.526 (Packet compression budget)**.** *In the setting of lemma 4.525, suppose that every packet output half-length satisfies*

$$0 \leq N_t \leq |K_t|.$$

Define the total packet compression in the zone by

$$C_\gamma = \sum_{t=1}^{m} (|K_t| - N_t),$$

and put

$$\Delta_\gamma = |Z_\gamma \setminus I| - |Z_\gamma \cap I|.$$

Then any parser exposing unchanged retained halo blocks for $A \subseteq Z_\gamma \setminus I$ satisfies

$$|A| \leq \Delta_\gamma + C_\gamma.$$

Equivalently, exposing a retained halo blocks requires packet compression at least $a - \Delta_\gamma$ whenever $a > \Delta_\gamma$.

In particular, a negative-balance zone needs at least $-\Delta_\gamma$ packet compression even if it exposes no retained halo block, and exposing every retained halo block requires

$$C_\gamma \geq |Z_\gamma \cap I|.$$

Proof. Since the packet intervals K_t are the components of $Z_\gamma \cap I$,

$$\sum_{t=1}^{m} |K_t| = |Z_\gamma \cap I|.$$

The definition of C_γ gives

$$\sum_{t=1}^{m} N_t = |Z_\gamma \cap I| - C_\gamma.$$

Substituting this into the parser-budget inequality of lemma 4.525 yields

$$|A| + |Z_\gamma \cap I| - C_\gamma \leq |Z_\gamma \setminus I|.$$

Rearranging gives $|A| \leq \Delta_\gamma + C_\gamma$. The equivalent lower bound on C_γ is the same inequality rearranged with $|A| = a$.

For $A = \varnothing$ in a zone with $\Delta_\gamma < 0$, the inequality gives $0 \le \Delta_\gamma + C_\gamma$, hence $C_\gamma \ge -\Delta_\gamma$. If every retained halo block is exposed, then $|A| = |Z_\gamma \setminus I|$, so

$$C_\gamma \ge |Z_\gamma \setminus I| - \Delta_\gamma = |Z_\gamma \cap I|.$$

□

Corollary 4.527 (Raw packet outputs expose only the zone surplus)**.** *Work in the setting of lemma 4.525. Suppose every component K_t is assigned its raw HKKO packet output, and that this output has half-length $|K_t|$. Put*

$$\Delta_\gamma = |Z_\gamma \setminus I| - |Z_\gamma \cap I|.$$

Then any parser returning these raw packet outputs as pairwise disjoint subwindows can expose at most Δ_γ unchanged retained halo blocks:

$$|A| \le \Delta_\gamma.$$

In particular, if $\Delta_\gamma < 0$, raw packet outputs for all deleted components cannot fit even with no visible retained block; if $\Delta_\gamma = 0$, no unchanged retained halo block can be exposed.

Proof. The second displayed inequality in lemma 4.525 gives

$$|A| + |Z_\gamma \cap I| \le |Z_\gamma \setminus I|.$$

Rearranging gives $|A| \le \Delta_\gamma$. If $\Delta_\gamma < 0$, this is impossible because $|A| \ge 0$. If $\Delta_\gamma = 0$, it forces $|A| = 0$. □

Corollary 4.528 (Tight raw packet zones hide retained halo blocks)**.** *In the setting of lemma 4.525, suppose*

$$|Z_\gamma \cap I| = |Z_\gamma \setminus I|$$

and suppose the packet output attached to every component K_t has raw HKKO half-length $|K_t|$. Then any parser that returns these packet outputs as pairwise disjoint subwindows returns no unchanged retained halo block as a disjoint subwindow.

Consequently, in such a tight zone the endpoint-recovery mechanism of lemma 4.519 is unavailable unless the same target-size replacement blocks also decode the retained halo blocks by a merged rule.

Proof. The raw packet hypothesis gives

$$\sum_t N_t = |Z_\gamma \cap I|.$$

Since $|Z_\gamma \cap I| = |Z_\gamma \setminus I| = R_\gamma$, the budget inequality of lemma 4.525 gives

$$|A| + R_\gamma \le R_\gamma,$$

and hence $|A| = 0$. Thus no unchanged retained halo block is returned as a disjoint subwindow.

The consequence follows because lemma 4.519 requires the retained halo blocks inside $Z_\gamma \setminus I$ as input. If none of those blocks is exposed unchanged as a disjoint subwindow, that lemma can be used only after an additional merged decoder recovers the retained blocks from the same replacement window. □

Corollary 4.529 (Separate Step–3 carrier windows cannot be appended to unchanged retained blocks)**.** *Work in the setting of lemma 4.523. Suppose a replacement-window parser tries to satisfy the retained-block input of proposition 4.559 by returning every retained halo block $B_r(P)$, $r \in Z_\gamma \setminus I$, as an unchanged pairwise disjoint subwindow of V_γ^*. Suppose also that, for some deleted packet K_t, the same parser tries to satisfy the Step–3 carrier/readout input of that proposition by returning a positive-half-length pairwise disjoint packet subwindow from which the carrier E_t, its top/right boundary, and the deterministic Step–2 readout are decoded. Then no such parser exists.*

Consequently the carrier/readout hypotheses of proposition 4.560 cannot be supplied by appending separate positive Step–3 carrier windows to unchanged retained halo blocks. Any target-size source *theorem for that criterion must recover at least one of the retained blocks and the carrier/readout data by a merged or modified replacement-window decoder, or else use a zero-window packet inverse whose endpoint and branch data are already sufficient for the restricted packet family.*

Proof. The proposed positive Step–3 carrier subwindow is a packet output window in the sense of lemma 4.523. Since all retained halo blocks are also returned as unchanged pairwise disjoint subwindows, that lemma forces every packet output window to have half-length zero. This contradicts the assumed positive half-length of the carrier subwindow.

The consequence is the same budget obstruction applied to the packet-carrier input in proposition 4.559. The only remaining target-size possibilities are to decode retained blocks and carrier data from overlapping or modified replacement-window structure, or to replace the positive carrier output by a zero-window inverse whose data already separate the relevant packet family. ☐

Proposition 4.530 (Alternating full-zone decoder criterion)**.** *Fix $q \geq 1$, put $j = 2q$, and let*

$$S \in \big\{\{1, 3, \ldots, 2q-1\}, \{2, 4, \ldots, 2q\}\big\}.$$

Consider either fixed-witness fiber $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$ of proposition 4.543. Suppose that, for every tableau T in the chosen fiber, its growth path $P(T)$ is assigned data

$$P^*(T) \in \mathcal{P}_q^{\mathrm{even}}, \qquad \beta(T) \in \{0,1\}^{2q},$$

and that a fixed decoder depending only on S recovers $P(T)$ from $P^*(T)$ and $\beta(T)$. Then the chosen fixed-witness fiber has cardinality at most

$$2^{2q} s_q.$$

Moreover, for the canonical halo zones of S, this decoder is a single full-zone decoder: the only zone is $Z = \{1, \ldots, 2q\}$, its endpoint pair is $\varnothing, \varnothing$, its replacement output has q two-step blocks, and the q deleted components are singleton packets.

Proof. The recovery hypothesis makes the assignment

$$T \longmapsto \big(P^*(T), \beta(T)\big)$$

injective on the chosen fixed-witness fiber. By lemmas 4.65 and 4.118, $|\mathcal{P}_q^{\mathrm{even}}| = s_q$, and there are 2^{2q} possible branch words. Thus the chosen fiber has size at most $2^{2q} s_q$.

By lemma 4.256, the canonical halo construction for this S has the single zone $Z = \{1, \ldots, 2q\}$, with $|Z \setminus I(S)| = q$, and the deleted components are q singletons. Since a growth path starts and ends at $\varnothing$ by lemma 4.118, the endpoint pair of this full zone is $\varnothing, \varnothing$. The target path $P^*(T) \in \mathcal{P}_q^{\mathrm{even}}$ has exactly q two-step blocks, so it is precisely the target-size replacement output for this single zone. ☐

Proposition 4.531 (Count-compatible halo-zone packages give local inverses)**.** *Let*

$$P = (\rho_0, \rho_1, \ldots, \rho_{2j})$$

be a two-step growth path, let $S \subset \{1, \ldots, j\}$, put $I = I(S)$, and let

$$Z_\gamma = \{a_\gamma, \ldots, b_\gamma\}$$

be one canonical halo zone. Write the maximal consecutive components of $Z_\gamma \cap I$ in increasing block order as

$$K_t = \{u_t, \ldots, v_t\} \qquad (1 \leq t \leq m).$$

Put $e_\gamma = |Z_\gamma \cap I|$. Suppose a replacement output V_γ^* has exactly $|Z_\gamma \setminus I|$ two-step blocks, and suppose that from

$$S,\ Z_\gamma,\ V_\gamma^*,\ \beta_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}, \qquad \beta_\gamma \in \{0,1\}^{2e_\gamma},$$

a deterministic decoder recovers the following data.

(1) *The original retained halo blocks*

$$B_r(P) = (\rho_{2r-2}, \rho_{2r-1}, \rho_{2r}) \qquad (r \in Z_\gamma \setminus I).$$

(2) *For every packet K_t, a packet output U_t^*, possibly of half-length zero.*

(3) *For every packet K_t, after splitting β_γ by the deterministic cuts of lemma 4.518, a fixed packet inverse recovers*

$$(\rho_{2u_t-2}, \rho_{2u_t-1}, \dots, \rho_{2v_t})$$

from U_t^*, the packet branch word β_{K_t}, the packet endpoint pair $\rho_{2u_t-2}, \rho_{2v_t}$ determined from the retained blocks and the zone endpoint pair, and the known packet interval K_t.

Then a fixed zone-local inverse recovers the whole original subpath

$$(\rho_{2a_\gamma-2}, \rho_{2a_\gamma-1}, \dots, \rho_{2b_\gamma})$$

from V_γ^*, β_γ, the zone endpoint pair, and Z_γ.

Proof. The decoder in item (1) recovers every retained halo block. Hence lemma 4.519 determines the endpoint pair for every packet K_t from the recovered retained blocks and the zone endpoint pair. By lemma 4.518, the single word β_γ splits into the packet branch words β_{K_t} with no extra branch data. Item (3) then recovers every deleted packet subpath.

The packet components K_t and the retained singleton indices $Z_\gamma \setminus I$ partition Z_γ. Interleaving the recovered packet subpaths and retained blocks in increasing block order reconstructs the displayed zone subpath. This procedure depends only on the fixed decoder, the deterministic packet ordering, and the packet inverses, so it is the required zone-local inverse. ☐

Proposition 4.532 (Anchor-pair packages recover positive height halo zones)**.** *Work in the height setting of corollary 4.322, and let*

$$P = (\rho_0, \rho_1, \dots, \rho_{2j})$$

be the corresponding growth path. Write

$$Z_\gamma \cap I = \{u_1, \dots, u_m\}, \qquad u_1 < \cdots < u_m,$$

and let A_γ be the canonical right-anchor set of corollary 4.322. Suppose a replacement window V_γ^* has exactly $|Z_\gamma \setminus I|$ two-step blocks and that, from

$$S,\ Z_\gamma,\ V_\gamma^*,\ \beta_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma}, \qquad \beta_\gamma \in \{0,1\}^{2m},$$

a deterministic parser decomposes V_γ^* into

(1) *unchanged anchor blocks*

$$B_r(P) = (\rho_{2r-2}, \rho_{2r-1}, \rho_{2r}) \qquad (r \in A_\gamma),$$

in their block order; and

(2) *one merged two-step window M_t for each deleted singleton u_t, listed in increasing t.*

Assume that the deterministic split of β_γ into consecutive two-bit words β_t has the following local inverse property: for each t, a fixed pair inverse recovers the consecutive original two-block segment

$$\rho_{2u_t-4}, \rho_{2u_t-3}, \rho_{2u_t-2}, \rho_{2u_t-1}, \rho_{2u_t}$$

from M_t, β_t, the zone endpoint pair, S, Z_γ, and t. Then the whole original zone subpath

$$\rho_{2a_\gamma-2}, \rho_{2a_\gamma-1}, \dots, \rho_{2b_\gamma}$$

is recoverable from V_γ^*, β_γ, the zone endpoint pair, and Z_γ.

Proof. The branch word split into two-bit subwords is determined by the ordered singleton list $u_1 < \cdots < u_m$, hence by S and Z_γ. Also

$$|A_\gamma| + m = |Z_\gamma \setminus I|$$

by corollary 4.322, so the displayed parser uses exactly the target-size number of two-step windows. By corollary 4.323,

$$(Z_\gamma \setminus I) \setminus A_\gamma = \{u_t - 1 : \ 1 \le t \le m\}.$$

Together with $Z_\gamma \cap I = \{u_1, \ldots, u_m\}$, this says that the block indices in Z_γ are partitioned into the anchor singleton blocks A_γ and the consecutive pairs

$$\{u_t - 1, u_t\} \qquad (1 \le t \le m).$$

The parser recovers the original anchor blocks. For each t, the assumed pair inverse recovers the original two consecutive blocks with indices $u_t - 1$ and u_t. Interleaving the recovered anchor blocks and recovered two-block pair segments in increasing block order therefore reconstructs every block of the original zone. The endpoint overlaps agree because each recovered piece is recovered as a subpath of the original growth path with its original endpoints. Thus the displayed zone subpath is uniquely recovered. □

Corollary 4.533 (Retained-only anchor-pair windows are not universal)**.** *The pair inverse in proposition 4.532 cannot, in the ambient growth-path category, be replaced by the rule which takes M_t to be only the unchanged left-neighbour retained block*

$$B_{u_t - 1}(P)$$

and then tries to recover the deleted singleton block from the retained block, the outer endpoint pair, the known pair position, and a two-bit word. Equivalently, a valid merged pair-window construction must either modify the left-neighbour block, carry deleted-sensitive target data, or use additional fixed-witness restrictions beyond the ambient growth-path axioms.

Proof. Let $\mu = (5, 4, 3, 2, 1)$, and fix one removable corner c_0 of μ. Let

$$U^0 = \mu, \qquad U^1 = \mu \setminus \{c_0\}, \qquad U^2 = \mu$$

be the retained left-neighbour block. This block has vertical-strip adjacent moves and weight

$$|U^0| - 2|U^1| + |U^2| = 2.$$

Now append a deleted singleton block from μ back to μ by choosing any one of the five removable corners c of μ:

$$U^2 = \mu, \qquad U^3 = \mu \setminus \{c\}, \qquad U^4 = \mu.$$

Each choice of c gives a valid two-step deleted singleton block of weight 2, and all five choices have the same retained block $B_{u_t - 1}$ and the same outer endpoint pair $(U^0, U^4) = (\mu, \mu)$. A two-bit word has only four values, so those data cannot encode all five possibilities. Therefore an inverse using only the unchanged retained block, the endpoint data, the pair position, and two bits cannot recover all such two-block segments. □

Corollary 4.534 (One-block ambient pair windows are not universal)**.** *Even if the merged pair window in proposition 4.532 is allowed to be an arbitrary single two-step growth block with the same outer endpoint pair as the original two-block segment, no ambient inverse can recover all such two-block segments from that one-block window and a two-bit word. Equivalently, a target-size one-block anchor-pair construction must use additional fixed-witness restrictions, neighbouring parser data, or nonlocal right-boundary data; a generic same-endpoint one-block replacement is not enough in the ambient growth-path category.*

Proof. Again let
$$\mu = (5, 4, 3, 2, 1).$$
For every ordered pair (c, d) of removable corners of μ, define a two-block segment
$$U^0 = \mu, \qquad U^1 = \mu \setminus \{c\}, \qquad U^2 = \mu, \qquad U^3 = \mu \setminus \{d\}, \qquad U^4 = \mu.$$
Both blocks have vertical-strip adjacent moves and weight 2, so this gives 25 distinct ambient two-block segments with common outer endpoint pair (μ, μ).

Now consider any single two-step growth block
$$M = (\mu, X, \mu)$$
with the same outer endpoint pair and weight 2. The weight condition gives
$$|\mu| - 2|X| + |\mu| = 2,$$
so $|X| = |\mu| - 1$. Since both adjacent moves are vertical-strip moves and $X \subseteq \mu$, the partition X is obtained from μ by deleting one removable corner. Thus there are only five possible same-endpoint one-block windows M.

Together with a two-bit word, such data have at most $5 \cdot 4 = 20$ possible values, while the ambient two-block family above has 25 members. Therefore no inverse using only a same-endpoint one-block replacement window and two branch bits can recover all ambient two-block segments. The final statement is the same counting obstruction applied to a proposed target-size anchor-pair merger before any fixed-witness or extra parser data have been used. □

Proposition 4.535 (Below-wall packet splices give count-compatible halo-zone inverses)**.** *Work in the setting of proposition 4.531. For each deleted packet $K_t = \{u_t, \dots, v_t\}$ in $Z_\gamma \cap I$, let J_t be the corresponding maximal witness-label interval, and put $h_t = |J_t|$. Interpret each packet in the local branch-recovery coordinates of lemma 4.226; the packet branch word obtained from lemma 4.518 is the local branch word for this datum. Suppose that the replacement-window decoder, from*
$$S,\ Z_\gamma,\ V_\gamma^*,\ \beta_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma},$$
recovers the retained halo blocks
$$B_r(P) \qquad (r \in Z_\gamma \setminus I),$$
so that the packet endpoint pair
$$\mu_t = \rho_{2u_t-2}, \qquad \nu_t = \rho_{2v_t}$$
is determined for every K_t by lemma 4.519. Suppose further that, for every packet K_t, the canonical marked packet preimage
$$W_t = W(\mu_t, \nu_t, \beta_{K_t})$$
of lemma 4.241 admits a fixed packet readout which recovers the deleted packet subpath
$$(\rho_{2u_t-2}, \rho_{2u_t-1}, \dots, \rho_{2v_t})$$
from W_t, the packet endpoint pair $\rho_{2u_t-2}, \rho_{2v_t}$, and the known packet interval K_t. No packet-output datum is required from the replacement-window decoder. Then a fixed zone-local inverse recovers the whole original subpath
$$(\rho_{2a_\gamma-2}, \rho_{2a_\gamma-1}, \dots, \rho_{2b_\gamma})$$
from V_γ^*, β_γ, the zone endpoint pair, and Z_γ.

Proof. By lemma 4.518, the zone branch word β_γ splits, with cuts determined by S, into packet branch words
$$\beta_{K_t} \in \{0, 1\}^{2h_t}.$$
The recovered retained halo blocks and the zone endpoint pair determine
$$\mu_t = \rho_{2u_t-2}, \qquad \nu_t = \rho_{2v_t}$$

for every packet by lemma 4.519. For a fixed packet K_t, let $w_t = w(\mu_t, \nu_t)$ be the canonical wall of lemma 4.239. Applying lemma 4.241 to the packet endpoint pair μ_t, ν_t and the branch word β_{K_t} gives the canonical output W_t^* and its marked preimage W_t. The same lemma recovers W_t from

$$W_t^*, \quad \beta_{K_t}, \quad w_t, \quad h_t, \quad \mu_t.$$

The fixed packet readout in the hypothesis then recovers the deleted packet subpath from W_t, the packet endpoint pair, and K_t.

Thus every packet has the fixed packet inverse required in item (3) of proposition 4.531. The retained halo blocks are recovered by hypothesis, and the packet outputs for item (2) are the canonically constructed W_t^*. Applying proposition 4.531 gives the desired zone-local inverse. □

Proposition 4.536 (Label-box packet readouts give halo-zone inverses)**.** *Work in the setting of proposition 4.531. For each deleted packet $K_t = \{u_t, \dots, v_t\}$ in $Z_\gamma \cap I$, let J_t be the corresponding maximal witness-label interval, and put $h_t = |J_t|$. Interpret each packet in the local branch-recovery coordinates of lemma 4.226; the packet branch word obtained from lemma 4.518 is the local branch word for this datum.*

Suppose that the replacement-window decoder, from

$$S,\ Z_\gamma,\ V_\gamma^*,\ \beta_\gamma, \quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma},$$

recovers the retained halo blocks

$$B_r(P) \qquad (r \in Z_\gamma \setminus I),$$

and, for every packet K_t, recovers a datum A_t which determines, in the semistandard tableau encoded by the local packet, the two boxes carrying each deleted label in J_t. Then a fixed zone-local inverse recovers the whole original subpath

$$(\rho_{2a_\gamma-2}, \rho_{2a_\gamma-1}, \dots, \rho_{2b_\gamma})$$

from V_γ^*, β_γ, the zone endpoint pair, and Z_γ.

Proof. By lemma 4.518, the zone branch word β_γ splits, with cuts determined by S, into packet branch words

$$\beta_{K_t} \in \{0,1\}^{2h_t}.$$

The recovered retained halo blocks and the zone endpoint pair determine

$$\rho_{2u_t-2}, \qquad \rho_{2v_t}$$

for every packet by lemma 4.519. For a fixed packet K_t, the known interval J_t, the left endpoint ρ_{2u_t-2} in local coordinates, the local branch word β_{K_t}, and the recovered label-box datum A_t satisfy the hypotheses of corollary 4.555. That corollary recovers the whole standardized local packet subpath. Under the local branch-recovery interpretation of the packet, this is the deleted subpath

$$(\rho_{2u_t-2}, \rho_{2u_t-1}, \dots, \rho_{2v_t}).$$

Thus every deleted packet subpath is recovered from the replacement-window decoder output and the already budgeted branch word. The packet components K_t and the retained singleton indices $Z_\gamma \setminus I$ partition Z_γ. Interleaving the recovered packet subpaths and retained blocks in increasing block order reconstructs the displayed zone subpath. □

Proposition 4.537 (Label-box halo repairs give the fixed-witness bound)**.** *Fix $q \geq 1$, $j \geq q$, and a q-element set $S \subset \{1, \dots, j\}$. Let Z_γ be the canonical halo replacement zones of lemma 4.254, with each zone written as $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$, and consider either fixed-witness fiber $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$ of proposition 4.543. For each zone put*

$$e_\gamma = |Z_\gamma \cap I(S)|.$$

Suppose that, for every T in the chosen fiber, there is a repaired path $P^*(T) \in \mathcal{P}_{j-q}^{\mathrm{even}}$ and a parser depending only on S with the following properties.

(1) *The parser decomposes $P^*(T)$ into the original retained blocks with indices outside the halo zones, unchanged and in their original order, and local replacement windows $V^*_\gamma(T)$ for the zones Z_γ.*
(2) *Each $V^*_\gamma(T)$ has exactly $|Z_\gamma \setminus I(S)|$ two-step blocks.*
(3) *For each zone there is a branch word*

$$\beta_\gamma(T) \in \{0,1\}^{2e_\gamma}$$

and, for every zone, the endpoint pair $\rho_{2a_\gamma-2}, \rho_{2b_\gamma}$ is the one determined from the parsed complement blocks and the fixed endpoints of the growth-path model by lemma 4.255.
(4) *For each zone, the replacement-window decoder satisfies the hypotheses of proposition 4.536 for $V^*_\gamma(T)$, $\beta_\gamma(T)$, the zone endpoint pair, and the known interval Z_γ.*

Then the chosen fixed-witness fiber has cardinality at most

$$2^{2q} s_{j-q}.$$

Proof. By lemma 4.517, for each canonical halo zone the number $e_\gamma = |Z_\gamma \cap I(S)|$ is also the sum of the frontier-event packet lengths used in proposition 4.540. The endpoint pair of each zone is determined from the parsed complement blocks by lemma 4.255. For each zone, item (4) and proposition 4.536 give a fixed local inverse recovering the original subpath on Z_γ from $V^*_\gamma(T)$, $\beta_\gamma(T)$, the determined endpoint pair, and the known zone interval. Therefore the hypotheses of proposition 4.540 hold, and that proposition gives the displayed fixed-witness bound. □

Proposition 4.538 (Below-wall packet-splice halo repairs give the fixed-witness bound)**.** *Fix $q \geq 1$, $j \geq q$, and a q-element set $S \subset \{1, \dots, j\}$. Let Z_γ be the canonical halo replacement zones of lemma 4.254, with each zone written as $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$, and consider either fixed-witness fiber $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$ of proposition 4.543. For each zone put*

$$e_\gamma = |Z_\gamma \cap I(S)|.$$

Suppose that, for every T in the chosen fiber, there is a repaired path $P^*(T) \in \mathcal{P}^{\mathrm{even}}_{j-q}$ and a parser depending only on S with the following properties.

(1) *The parser decomposes $P^*(T)$ into the original retained blocks with indices outside the halo zones, unchanged and in their original order, and local replacement windows $V^*_\gamma(T)$ for the zones Z_γ.*
(2) *Each $V^*_\gamma(T)$ has exactly $|Z_\gamma \setminus I(S)|$ two-step blocks.*
(3) *For each zone there is a branch word*

$$\beta_\gamma(T) \in \{0,1\}^{2e_\gamma}$$

and, for every zone, the endpoint pair $\rho_{2a_\gamma-2}, \rho_{2b_\gamma}$ is the one determined from the parsed complement blocks and the fixed endpoints of the growth-path model by lemma 4.255.
(4) *For each zone, the replacement-window decoder satisfies the hypotheses of proposition 4.535 for $V^*_\gamma(T)$, $\beta_\gamma(T)$, the zone endpoint pair, and the known interval Z_γ.*

Then the chosen fixed-witness fiber has cardinality at most

$$2^{2q} s_{j-q}.$$

Proof. By lemma 4.517, for each canonical halo zone the number $e_\gamma = |Z_\gamma \cap I(S)|$ is also the sum of the frontier-event packet lengths used in proposition 4.540. Lemma 4.255 applied to the parsed complement blocks and the fixed endpoints of lemma 4.118 determines the endpoint pair of every canonical halo zone. For each zone, item (4) and proposition 4.535 give a fixed local inverse recovering the original subpath on Z_γ from $V^*_\gamma(T)$, $\beta_\gamma(T)$, the zone endpoint pair, and the known zone interval. Augmenting the parser by the endpoint pairs determined above, items (1)–(3) are exactly the parser, target-window, branch-word, and local inverse hypotheses of proposition 4.540. That proposition applies and gives the displayed fixed-witness bound. □

Corollary 4.539 (Below-wall packet splices supply canonical halo repair data). *Work in the setting of proposition 4.538. If the parser, replacement windows, branch words, and packet readouts satisfy the hypotheses of that proposition, then the same data satisfy the hypotheses of proposition 4.540.*

Proof. By lemma 4.517, for each canonical halo zone one has

$$e_\gamma = |Z_\gamma \cap I(S)|.$$

Thus the branch words have the lengths required in proposition 4.540. The parser and replacement-window length hypotheses are identical in the two criteria. The zone endpoint pair is determined from the parsed complement blocks by lemma 4.255. Finally, item (4) of proposition 4.538 and proposition 4.535 give, for each zone, a fixed local inverse recovering the original zone subpath from the replacement window, the branch word, the determined endpoint pair, and the known zone interval. These are exactly the remaining local-inverse hypotheses of proposition 4.540. ☐

Proposition 4.540 (Canonical halo local repair criterion). *Fix $q \geq 1$, $j \geq q$, and a q-element set $S \subset \{1, \dots, j\}$. Let Z_γ be the canonical halo replacement zones of lemma 4.254, with each zone written as $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$, and consider either fixed-witness fiber $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$ of proposition 4.543. For a zone Z_γ, let*

$$e_\gamma = \sum_{K_\beta \subseteq Z_\gamma} |E_\beta|$$

using the frontier-event packets assigned by lemma 4.517. Suppose that, for every T in the chosen fiber, there is a repaired path $P^*(T) \in \mathcal{P}^{\text{even}}_{j-q}$ and a parser depending only on S with the following properties.

1. *The parser decomposes $P^*(T)$ into the original retained blocks with indices outside the halo zones, unchanged and in their original order, and local replacement windows $V^*_\gamma(T)$ for the zones Z_γ.*
2. *Each $V^*_\gamma(T)$ has exactly $|Z_\gamma \setminus I(S)|$ two-step blocks.*
3. *For each γ there is a branch word*

$$\beta_\gamma(T) \in \{0,1\}^{2e_\gamma}$$

and a fixed local inverse recovers the original subpath on Z_γ from $V^*_\gamma(T)$, $\beta_\gamma(T)$, the endpoint pair $\rho_{2a_\gamma - 2}, \rho_{2b_\gamma}$ determined from the parsed complement blocks by lemma 4.255, and the known interval Z_γ.

Then the chosen fixed-witness fiber has cardinality at most

$$2^{2q} s_{j-q}.$$

Proof. By lemma 4.517,

$$e_\gamma = |Z_\gamma \cap I(S)|$$

for each canonical halo zone. Hence the branch words in the statement have the form required in proposition 4.541. The canonical halo zones satisfy the deterministic-zone and block-count hypotheses by lemma 4.254. The three displayed properties, together with lemma 4.255, canonically augment the parser by the endpoint pair of every zone. After this deterministic augmentation they are exactly the remaining parser, target, and local-inverse hypotheses of proposition 4.541. Applying that proposition gives the claimed bound. ☐

Proposition 4.541 (Replacement-zone criterion for fixed-witness fibers). *Fix $q \geq 1$, $j \geq q$, and a q-element set $S \subset \{1, \dots, j\}$. Consider either fixed-witness fiber $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$ of proposition 4.543. Suppose that, for every tableau T in the chosen fiber, its growth path $P(T)$ is assigned deterministic replacement zones Z_α satisfying the hypotheses of proposition 4.252, with*

$$I(S) = \{j - s + 1 : s \in S\}$$

and with branch words

$$\beta_\alpha(T) \in \{0,1\}^{2|Z_\alpha \cap I(S)|}.$$

Assume also that the repaired path $P^*(T)$ lies in $\mathcal{P}^{\text{even}}_{j-q}$. Then the chosen fixed-witness fiber has cardinality at most

$$2^{2q} s_{j-q}.$$

Proof. By proposition 4.252, the data

$$\Big(P^*(T), (\beta_\alpha(T))_\alpha\Big)$$

recover the original growth path $P(T)$ from S and hence recover T by the growth-path encoding. Therefore the assignment from the chosen fixed-witness fiber to these data is injective.

The zones are disjoint and cover every element of $I(S)$ exactly once, so

$$\sum_\alpha |Z_\alpha \cap I(S)| = |I(S)| = q.$$

Thus the total number of possible branch-word tuples is

$$\prod_\alpha 2^{2|Z_\alpha \cap I(S)|} = 2^{2q}.$$

The possible repaired paths are contained in $\mathcal{P}^{\text{even}}_{j-q}$, whose cardinality is s_{j-q} by lemmas 4.65 and 4.118. Multiplying these two counts gives the claimed bound. □

Proposition 4.542 (Right-boundary auxiliary fixed-witness criterion)**.** *Fix $q \geq 1$, $j \geq q$, and a q-element set $S \subset \{1, \dots, j\}$. Consider either fixed-witness fiber $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$ of proposition 4.543. Suppose that, for every tableau T in the chosen fiber, its growth path $P(T)$ is assigned deterministic replacement zones*

$$Z_1, \dots, Z_L$$

which satisfy the hypotheses of proposition 4.253, with

$$I(S) = \{j - s + 1 : s \in S\}, \qquad Z_L = \{a_L, \dots, j\},$$

and with branch words

$$\beta_\alpha(T) \in \{0,1\}^{2|Z_\alpha \cap I(S)|}.$$

Assume also that the repaired path $P^*(T)$ lies in $\mathcal{P}^{\text{even}}_{j-q}$. Then the chosen fixed-witness fiber has cardinality at most

$$2^{2q} s_{j-q}.$$

Proof. By proposition 4.253, the data

$$\Big(P^*(T), (\beta_\alpha(T))_{\alpha=1}^L\Big)$$

recover the original growth path $P(T)$ from S and hence recover T. Thus the assignment from the chosen fixed-witness fiber to these data is injective.

The replacement zones are disjoint and cover every element of $I(S)$ exactly once, so

$$\sum_{\alpha=1}^L |Z_\alpha \cap I(S)| = |I(S)| = q.$$

The total number of branch-word tuples is therefore

$$\prod_{\alpha=1}^L 2^{2|Z_\alpha \cap I(S)|} = 2^{2q}.$$

The repaired paths lie in $\mathcal{P}^{\text{even}}_{j-q}$, whose cardinality is s_{j-q} by lemmas 4.65 and 4.118. Multiplying gives the displayed bound. □

Proposition 4.543 (Jointwise repair criterion for fixed witnesses)**.** *Fix $q \geq 1$, $j \geq q$, and a q-element set $S \subset \{1, \dots, j\}$. Define the two fixed-witness fibers*

$$\mathcal{F}^w(S) = \{T : \lambda(T)' \text{ even},\ \lambda_1(T) \geq 2q,\ W_q(T) = S\}$$

and

$$\mathcal{F}^h(S) = \{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\}.$$

Let $\mathcal{P}^{\mathrm{even}}_{j-q}$ be the set of column-even two-step growth paths corresponding to semistandard tableaux of content (2^{j-q}).

Consider either $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$. Suppose that for every T in the chosen fiber, after decomposing S into maximal consecutive label intervals J_α and taking the corresponding per-joint event sets $E_\alpha(T)$ of lemma 4.188, there is an assignment

$$T \longmapsto \Big(P^*(T), (\beta_\alpha(T))_{\alpha=1}^L\Big)$$

such that $P^*(T) \in \mathcal{P}^{\mathrm{even}}_{j-q}$,

$$\beta_\alpha(T) \in \{0,1\}^{2|J_\alpha|},$$

and T is uniquely recoverable from S, $P^*(T)$, and the words $\beta_\alpha(T)$. Then the chosen fixed-witness fiber has cardinality at most

$$2^{2q} s_{j-q}.$$

Proof. By lemma 4.188, the sets $E_\alpha(T)$ contain one frontier event for each label in the corresponding interval J_α. Since the intervals J_α partition S, one has

$$\sum_{\alpha=1}^{L} |E_\alpha(T)| = \sum_{\alpha=1}^{L} |J_\alpha| = |S| = q.$$

Therefore the total number of possible branch-word tuples is

$$\prod_{\alpha=1}^{L} 2^{2|J_\alpha|} = 2^{2q}.$$

The recovery hypothesis makes the displayed assignment injective from the chosen fiber into

$$\mathcal{P}^{\mathrm{even}}_{j-q} \times \prod_{\alpha=1}^{L} \{0,1\}^{2|J_\alpha|}.$$

Finally, lemmas 4.65 and 4.118 identify $|\mathcal{P}^{\mathrm{even}}_{j-q}|$ with the stable moment s_{j-q}. Multiplying the two counts gives the claimed bound. $\square$

Corollary 4.544 (Defect bound for fixed residual witnesses)**.** *In either fixed-witness fiber appearing in proposition 4.563, deleting the two copies of every label in the fixed witness set S leaves a labelled retained subdiagram of content (2^{j-q}) whose remaining row and column inequalities are inherited from the original tableau and whose column occupancies are odd in at most $2q$ columns.*

Proof. Apply lemma 4.115 with $|S| = q$. $\square$

Lemma 4.545 (Alternating frontiers after witness deletion)**.** *Let T be as in lemma 4.95. If $\lambda_1(T) \geq 2q$ and $S = W_q(T)$, then in the retained subdiagram $T \setminus S$ no box remains at any of the top-row positions*

$$(1,1), (1,3), \dots, (1, 2q-1).$$

Consequently, among the first $2q$ positions of the top row, at most q boxes remain.

If $\ell(\lambda(T)) \geq 2q$ and $S = H_q(T)$, then in the retained subdiagram $T \setminus S$ no box remains at any of the first-column positions

$$(1,1), (3,1), \dots, (2q-1, 1).$$

Consequently, among the first $2q$ positions of the first column, at most q boxes remain.

Proof. In the width case, the labels occupying $(1,1),(1,3),\dots,(1,2q-1)$ are exactly the labels in $W_q(T)=S$ by definition, so those boxes are deleted when forming $T\setminus S$. The only top-row positions among the first $2q$ that can remain are therefore the even positions $2,4,\dots,2q$, of which there are q.

The height case is identical with the first column in place of the first row: the boxes at $(1,1),(3,1),\dots,(2q-1,1)$ carry exactly the labels in $H_q(T)=S$, and only the even row positions among $1,\dots,2q$ can remain. □

Lemma 4.546 (Alternating frontier retention is local)**.** *Let T be as in lemma 4.95.*

In the width case, assume $\lambda_1(T)\ge 2q$, put $s_r=T(1,2r-1)$ for $1\le r\le q$, and set $S=W_q(T)$. For $1\le r\le q$, put $a_r=T(1,2r)$. After deleting the labels in S, the box $(1,2r-1)$ is absent. The box $(1,2r)$ is absent if and only if either $a_r=s_r$ or $r<q$ and $a_r=s_{r+1}$; otherwise $(1,2r)$ remains.

In the height case, assume $\ell(\lambda(T))\ge 2q$, put $t_r=T(2r-1,1)$ for $1\le r\le q$, and set $S=H_q(T)$. After deleting the labels in S, the boxes

$$(1,1),(3,1),\dots,(2q-1,1)$$

are absent, while the boxes

$$(2,1),(4,1),\dots,(2q,1)$$

all remain. Thus exactly q boxes remain among the first $2q$ first-column positions.

Proof. The odd width boxes carry precisely the labels in S by definition of $W_q(T)$, so they are absent after deleting S. Since the first row is weakly increasing,

$$s_r\le a_r\le s_{r+1}\qquad(1\le r<q),$$

and $s_q\le a_q$. The labels $s_1,\dots,s_q$ are strictly increasing by lemma 4.95. Hence, for $r<q$, the only labels from S which can equal a_r are s_r and s_{r+1}; for $r=q$, the only possible label from S equal to a_q is s_q. A box is deleted exactly when its label lies in S, proving the width statement.

For height, the odd first-column boxes carry exactly the labels in S and are therefore deleted. Let $b_r=T(2r,1)$ for $1\le r\le q$. The first column is strictly increasing. Thus, for $r<q$,

$$t_r<b_r<t_{r+1},$$

and for $r=q$ one has $t_q<b_q$. Therefore no b_r belongs to $\{t_1,\dots,t_q\}=S$. The even first-column boxes are not deleted, and the last assertion follows. □

Corollary 4.547 (Width orientation forces retention)**.** *In the width setting of lemma 4.546, let $\epsilon_r^w(T)$ be the orientation bits of lemma 4.140. For $1\le r<q$, if the box $(1,2r)$ is deleted when passing from T to $T\setminus W_q(T)$, then*

$$\epsilon_r^w(T)=0\quad\textit{or}\quad\epsilon_{r+1}^w(T)=1.$$

Equivalently, if $\epsilon_r^w(T)=1$ and $\epsilon_{r+1}^w(T)=0$, then $(1,2r)$ remains in $T\setminus W_q(T)$. For the final slot, if $(1,2q)$ is deleted then $\epsilon_q^w(T)=0$; equivalently $\epsilon_q^w(T)=1$ forces $(1,2q)$ to remain.

Proof. Write $s_r=T(1,2r-1)$ and $a_r=T(1,2r)$. By lemma 4.546, deletion of $(1,2r)$ with $r<q$ means that $a_r=s_r$ or $a_r=s_{r+1}$. If $a_r=s_r$, then the two first-row copies of the label s_r occupy columns $2r-1$ and $2r$, so the frontier copy at $(1,2r-1)$ is the leftmost copy. Hence lemma 4.140 gives $\epsilon_r^w(T)=0$. If $a_r=s_{r+1}$, then the two first-row copies of the label s_{r+1} occupy columns $2r$ and $2r+1$, so the frontier copy at $(1,2r+1)$ is the rightmost copy. Thus $\epsilon_{r+1}^w(T)=1$ by the same lemma.

For $r=q$, lemma 4.546 says that deletion of $(1,2q)$ can only occur when $a_q=s_q$. Then the frontier copy at $(1,2q-1)$ is the leftmost copy, so $\epsilon_q^w(T)=0$. The equivalent retention statements are the contrapositives. □

Corollary 4.548 (Orientation bits charge width boundary deletions)**.** *In the width setting of lemma 4.546, decompose $S = W_q(T)$ into maximal consecutive label intervals J_α. Let $s_r = T(1, 2r-1)$ and let $\epsilon_r^w(T)$ be as in lemma 4.140. If the second copy of s_r lies in the first row, then the additional deleted first-row box carrying s_r is*

$$(1, 2r) \quad \textit{when } \epsilon_r^w(T) = 0,$$

and is

$$(1, 2r-2) \quad \textit{when } \epsilon_r^w(T) = 1,$$

with the second alternative omitted for $r = 1$. Hence every deleted even top-row box whose label lies in J_α is charged to the unique frontier-event slot of that label in J_α, and the possible adjacent position is determined by the first half of the local branch word.

In the height setting, no even first-column box among $(2, 1), (4, 1), \ldots, (2q, 1)$ is deleted by $H_q(T)$.

Proof. The first assertion is exactly the local second-copy statement in lemma 4.140: the frontier copy of s_r is the leftmost copy when $\epsilon_r^w(T) = 0$ and the rightmost copy when $\epsilon_r^w(T) = 1$, and the only possible first-row positions for the other copy are the adjacent columns $2r$ and $2r-2$. Since the witness labels in J_α are distinct, each deleted even top-row box carrying a label from J_α belongs to exactly one such label, and therefore to its unique frontier-event slot. The first half of the local branch word records these orientation bits by lemma 4.225 and lemma 4.226. The height assertion is the height half of lemma 4.546. □

Lemma 4.549 (Local position of width-witness second copies)**.** *Let T be a semistandard tableau of content (2^j) with $\lambda_1(T) \geq 2q$. Put $s_r = T(1, 2r-1)$ for $1 \leq r \leq q$. If the second copy of s_r occurs in the first row, then it occurs in column $2r-2$ or column $2r$, omitting column 0 when $r = 1$.*

Proof. The first row is weakly increasing and the label s_r occurs exactly twice in the whole tableau. Suppose first that another copy of s_r occurs in the first row at a column $c < 2r-1$. If $c \leq 2r-3$, then weak increase forces every entry from column c through column $2r-1$ to equal s_r, giving at least three copies. Hence $c = 2r-2$. Similarly, if another copy occurs at $c > 2r-1$ and $c \geq 2r+1$, then every entry from column $2r-1$ through column c equals s_r, again giving at least three copies. Hence $c = 2r$. □

Lemma 4.550 (Height-witness second copies avoid the first column)**.** *Let T be a semistandard tableau of content (2^j) with $\ell(\lambda(T)) \geq 2q$. Put $s_r = T(2r-1, 1)$ for $1 \leq r \leq q$. The second copy of s_r is not in the first column.*

Proof. Columns in a semistandard tableau are strictly increasing, so the first-column entries are pairwise distinct. Since s_r already occurs at $(2r-1, 1)$, its second copy cannot occur elsewhere in that column. □

Lemma 4.551 (Local branch words determine boundary witness readouts)**.** *Work in the setting of lemma 4.188 and fix a maximal witness-label interval*

$$J_\alpha = \{a_\alpha, \ldots, b_\alpha\} \subset S.$$

Let $h_\alpha = |J_\alpha|$ and let

$$\beta_\alpha = (\beta_1, \ldots, \beta_{2h_\alpha})$$

be the local branch word of lemma 4.225. In the width case, write $S = W_q(T) = \{s_1 < \cdots < s_q\}$ and let p_α be determined by $s_{p_\alpha} = a_\alpha$. For $1 \leq i \leq h_\alpha$, put $r = p_\alpha + i - 1$. The deleted witness label $s_r = a_\alpha + i - 1$ always deletes the bad-boundary box

$$(1, 2r-1).$$

If its second copy is also in the first row, then the only additional deleted bad-boundary box for this label is

$$(1, 2r) \quad \textit{when } \beta_i = 0, \qquad (1, 2r-2) \quad \textit{when } \beta_i = 1,$$

with the second alternative omitted for $r = 1$.

In the height case, write $S = H_q(T) = \{t_1 < \cdots < t_q\}$ and let p_α be determined by $t_{p_\alpha} = a_\alpha$. For $1 \le i \le h_\alpha$, put $r = p_\alpha + i - 1$. The deleted witness label $t_r = a_\alpha + i - 1$ deletes the bad-boundary box

$$(2r-1, 1),$$

and its second copy is not in the first column. Thus the bad-boundary part of each local deleted packet uses no branch data beyond the first half of β_α in the width case, and no branch data in the height case.

Proof. In the width case, lemma 4.189 gives $s_{p_\alpha+i-1} = a_\alpha + i - 1$ for $1 \le i \le h_\alpha$. The box $(1, 2r-1)$ carries s_r by the definition of $W_q(T)$, so it is deleted when the witness labels in S are deleted. By lemma 4.225, the first half of β_α is exactly the orientation word $\epsilon_r^w(T)$ along the ordered labels of J_α. Hence $\beta_i = \epsilon_r^w(T)$. If the second copy of s_r is also in the first row, lemma 4.140 places it at $(1, 2r)$ when $\epsilon_r^w(T) = 0$ and at $(1, 2r-2)$ when $\epsilon_r^w(T) = 1$, with the unavailable column 0 omitted. This proves the width assertions.

The height proof is identical up to the boundary location. Now lemma 4.189 gives $t_{p_\alpha+i-1} = a_\alpha + i - 1$, and the definition of $H_q(T)$ says that $(2r-1, 1)$ carries t_r, so it is deleted. The second copy of t_r is not in the first column by lemma 4.550. Therefore no additional first-column boundary position has to be selected. ☐

Corollary 4.552 (Bad-boundary packets need no interior branch source)**.** *In the setting of lemma 4.551, the data*

$$J_\alpha, \qquad \beta_\alpha, \qquad S$$

determine the entire bad-boundary part of the deleted local packet:

1. *in the width case, for each label $s_r \in J_\alpha$, the forced odd frontier box $(1, 2r-1)$ is deleted, and the first half of β_α determines whether a possible additional first-row copy of the same label is at $(1, 2r)$ or at $(1, 2r-2)$;*
2. *in the height case, for each label $t_r \in J_\alpha$, the forced odd frontier box $(2r-1, 1)$ is deleted, and there is no additional first-column copy to select.*

Consequently any remaining local packet theorem for the fixed-witness repair only has to reconstruct the off-boundary/interior part of the packet and the joint parity repair; the bad row or column frontier consumes no branch data beyond the already allocated local word.

Proof. In the width case, lemma 4.551 identifies the rank r of every label in J_α from S and the interval J_α, proves that $(1, 2r-1)$ is always deleted, and gives the two possible adjacent positions for a second first-row copy as the two values of the corresponding first-half bit of β_α. In the height case, the same lemma identifies r, proves that $(2r-1, 1)$ is deleted, and proves that the second copy is not in column 1. These are exactly the listed bad-boundary readouts, and the final sentence is the same statement expressed as a restriction on the remaining local packet theorem. ☐

Corollary 4.553 (Boundary-supported width packets have a fixed local inverse)**.** *Work in the width case of lemma 4.551. Write*

$$J_\alpha = \{a_\alpha, \dots, b_\alpha\}, \qquad S = W_q(T) = \{s_1 < \cdots < s_q\},$$

and let p_α be determined by $s_{p_\alpha} = a_\alpha$. Suppose that, for every $1 \le i \le h_\alpha = |J_\alpha|$, the second copy of $s_{p_\alpha+i-1}$ also lies in the first row. Then the data

$$S, \quad J_\alpha, \quad \sigma^{(2a_\alpha-2)}, \quad \beta_\alpha \in \{0,1\}^{2h_\alpha}$$

determine the whole standardized local packet subpath

$$\sigma^{(2a_\alpha-2)}, \sigma^{(2a_\alpha-1)}, \dots, \sigma^{(2b_\alpha)}.$$

Consequently a width local packet with no off-boundary deleted witness copy has no remaining packet-inverse ambiguity after the local branch word is known.

Proof. For $1 \leq i \leq h_\alpha$, put $r = p_\alpha + i - 1$ and $s = s_r = a_\alpha + i - 1$. By lemma 4.189, these are exactly the witness labels in the interval J_α, in increasing order. By lemma 4.551, the first half of β_α determines the orientation bit $\epsilon_r^w(T)$. The frontier box $(1, 2r-1)$ is deleted. Under the boundary-supported hypothesis the other deleted box carrying s is also in the first row, and lemma 4.140 gives the complete standardized placement:

$\epsilon_r^w(T)$	box carrying $2s-1$	box carrying $2s$
0	$(1, 2r-1)$	$(1, 2r)$
1	$(1, 2r-2)$	$(1, 2r-1)$.

The second row of the table is unavailable when $r = 1$ and the displayed column $2r-2$ is not a box; this is simply an inconsistent branch value for an actual boundary-supported packet.

Thus the displayed data determine, for each standard time $2s-1, 2s$ with $s \in J_\alpha$, the unique box added at that time. Starting from the known shape $\sigma^{(2a_\alpha - 2)}$ and adjoining these determined boxes in the standard-time order

$$2a_\alpha - 1, 2a_\alpha, \dots, 2b_\alpha$$

recovers every intermediate shape through $\sigma^{(2b_\alpha)}$. This is the claimed local inverse. The final sentence restates that, in this subcase, corollary 4.552 has already recovered all deleted witness boxes, so there is no off-boundary packet datum left to decode. ☐

Corollary 4.554 (Second-copy readout)**.** *Work in the setting of lemma 4.551. In the width case, write*

$$J_\alpha = \{a_\alpha, \dots, b_\alpha\}, \qquad S = W_q(T) = \{s_1 < \cdots < s_q\},$$

and let p_α be determined by $s_{p_\alpha} = a_\alpha$. Suppose that an auxiliary datum A_α determines the subset

$$R_\alpha^w(T) = \{\, s_r \in J_\alpha : \text{ the second copy of } s_r \text{ is not in the first row}\,\}$$

and, for every $s_r \in R_\alpha^w(T)$, the box occupied by this second copy. Then

$$S, \quad J_\alpha, \quad \sigma^{(2a_\alpha-2)}, \quad \beta_\alpha, \quad A_\alpha$$

determine the whole standardized local packet subpath

$$\sigma^{(2a_\alpha-2)}, \sigma^{(2a_\alpha-1)}, \dots, \sigma^{(2b_\alpha)}.$$

In the height case, write

$$J_\alpha = \{a_\alpha, \dots, b_\alpha\}, \qquad S = H_q(T) = \{t_1 < \cdots < t_q\},$$

and let p_α be determined by $t_{p_\alpha} = a_\alpha$. Suppose that an auxiliary datum A_α determines, for every $t_r \in J_\alpha$, the box occupied by the second copy of t_r. Then the same data determine the whole standardized local packet subpath.

Consequently the packet-sensitive part not supplied by the bad-boundary readout is exactly the second-copy readout: in width, only the off-first-row second copies have to be exposed, and in height all second copies have to be exposed.

Proof. Consider first the width case. For $1 \leq i \leq h_\alpha$, put $r = p_\alpha + i - 1$ and $s = s_r = a_\alpha + i - 1$. By lemma 4.189, these are exactly the labels of J_α in increasing order. The first half of β_α is the orientation word, so

$$\epsilon_r^w(T) = \beta_i$$

by lemma 4.551. The frontier box $(1, 2r-1)$ is always deleted. If $s_r \notin R^w_\alpha(T)$, then the second copy is also in the first row, and lemma 4.140 places the adjacent standard pair as

$\epsilon^w_r(T)$	box carrying $2s-1$	box carrying $2s$
0	$(1, 2r-1)$	$(1, 2r)$
1	$(1, 2r-2)$	$(1, 2r-1)$.

If $s_r \in R^w_\alpha(T)$, then A_α supplies the off-first-row second box, call it c_r. The definition of $\epsilon^w_r(T)$ gives the remaining standardized placement:

$\epsilon^w_r(T)$	box carrying $2s-1$	box carrying $2s$
0	$(1, 2r-1)$	c_r
1	c_r	$(1, 2r-1)$.

Thus the displayed data determine, for every $s \in J_\alpha$, the unique box added at the two standard times $2s-1$ and $2s$.

In the height case, the same rank argument gives $t_{p_\alpha+i-1} = a_\alpha + i - 1$. By lemma 4.140, the frontier box $(2r-1, 1)$ carries $2t_r - 1$, and lemma 4.550 says the second copy is not in the first column. The datum A_α supplies its box; that box carries $2t_r$. Hence the data determine the unique box added at the two standard times $2t_r - 1$ and $2t_r$ for every label in J_α.

Starting from the known left endpoint $\sigma^{(2a_\alpha-2)}$ and adjoining the determined boxes in the order

$$2a_\alpha - 1, 2a_\alpha, \ldots, 2b_\alpha$$

recovers every intermediate shape through $\sigma^{(2b_\alpha)}$ in both cases. The final sentence is just the same reconstruction statement with the already recovered bad-boundary boxes removed from the list of missing data. □

Corollary 4.555 (Label-box readout supplies second-copy readout)**.** *Work in the setting of lemma 4.551. In the width case, suppose that an auxiliary datum A_α determines, for every label $s \in J_\alpha$, the two boxes of T carrying the semistandard label s. Then*

$$S, \quad J_\alpha, \quad \sigma^{(2a_\alpha-2)}, \quad \beta_\alpha, \quad A_\alpha$$

determine the whole standardized local packet subpath.

In the height case, suppose instead that A_α determines, for every label $t \in J_\alpha$, the two boxes of T carrying the semistandard label t. Then the same data determine the whole standardized local packet subpath.

Consequently a target-size parser which exposes the two deleted semistandard boxes for every label in the local joint supplies the second-copy readout of corollary 4.554; no further packet-local branch source is needed for the original standardized subpath.

Proof. Consider first the width case. Write

$$S = W_q(T) = \{s_1 < \cdots < s_q\},$$

and let p_α be determined by $s_{p_\alpha} = a_\alpha$. For $s_r \in J_\alpha$, the rank r is determined by S and J_α, and the frontier box is

$$f_r = (1, 2r-1)$$

by lemma 4.551. The datum A_α supplies the two boxes carrying s_r; removing the known frontier box f_r from this two-element set determines the other box c_r. The label s_r belongs to $R^w_\alpha(T)$ exactly when c_r is not in the first row, and in that case A_α determines the required off-first-row box c_r. Thus A_α supplies the width second-copy readout required by corollary 4.554, and that corollary recovers the local packet subpath.

The height case is the same. Write

$$S = H_q(T) = \{t_1 < \cdots < t_q\},$$

and let p_α be determined by $t_{p_\alpha} = a_\alpha$. For $t_r \in J_\alpha$, the rank r is determined by S and J_α, and the frontier box is

$$f_r = (2r - 1, 1).$$

By lemma 4.140, this frontier box carries the standard entry $2t_r - 1$. Hence the other box among the two boxes supplied by A_α is the box carrying $2t_r$ in the doubled standardization. This is exactly the height second-copy readout required by corollary 4.554. Applying that corollary recovers the local packet subpath. The final sentence is the same implication stated in parser-facing language. □

Corollary 4.556 (Step-2 interval readout supplies local packet inverses)**.** *Work in the setting of lemma 4.551, and let*

$$J_\alpha = \{a_\alpha, \dots, b_\alpha\}.$$

Suppose that an auxiliary datum A_α determines the Step 2 Knuth boundary interval

$$\lambda^{(a_\alpha - 1)}, \lambda^{(a_\alpha)}, \dots, \lambda^{(b_\alpha)}$$

of the underlying semistandard tableau. Then

$$S, \quad J_\alpha, \quad \sigma^{(2a_\alpha - 2)}, \quad \beta_\alpha, \quad A_\alpha$$

determine the whole standardized local packet subpath in both the width and height cases.

The same conclusion holds if A_α determines a Step 2 first-variant Ferrers subarrangement whose forward boundary sweep contains the displayed interval.

Proof. By lemma 4.172, the displayed Step 2 boundary interval determines the two semistandard boxes carrying every label in J_α. The same is true in the subarrangement form by the final assertion of that lemma. Applying corollary 4.555 gives the local packet inverse in either family. □

Proposition 4.557 (Step-2 interval packet readouts give halo-zone inverses)**.** *Work in the setting of proposition 4.531. For each deleted packet $K_t = \{u_t, \dots, v_t\}$ in $Z_\gamma \cap I$, let*

$$J_t = \{c_t, \dots, d_t\}$$

be the corresponding maximal witness-label interval, and put $h_t = |J_t|$. Interpret each packet in the local branch-recovery coordinates of lemma 4.226; the packet branch word obtained from lemma 4.518 is the local branch word for this datum.

Suppose that the replacement-window decoder, from

$$S,\ Z_\gamma,\ V_\gamma^*,\ \beta_\gamma, \quad \rho_{2a_\gamma - 2}, \rho_{2b_\gamma},$$

recovers the retained halo blocks

$$B_r(P) \qquad (r \in Z_\gamma \setminus I),$$

and, for every packet K_t, recovers a datum A_t which determines either the Step 2 Knuth boundary interval

$$\lambda^{(c_t - 1)}, \lambda^{(c_t)}, \dots, \lambda^{(d_t)}$$

of the semistandard tableau encoded by the local packet, or a Step 2 first-variant Ferrers subarrangement whose forward boundary sweep contains this interval. Then a fixed zone-local inverse recovers the whole original subpath

$$(\rho_{2a_\gamma - 2}, \rho_{2a_\gamma - 1}, \dots, \rho_{2b_\gamma})$$

from V_γ^*, β_γ, the zone endpoint pair, and Z_γ.

Proof. By lemma 4.518, the zone branch word β_γ splits, with cuts determined by S, into packet branch words

$$\beta_{K_t} \in \{0, 1\}^{2h_t}.$$

The recovered retained halo blocks and the zone endpoint pair determine

$$\rho_{2u_t - 2}, \qquad \rho_{2v_t}$$

for every packet by lemma 4.519. For a fixed packet K_t, the known interval J_t, the left endpoint ρ_{2u_t-2} in local coordinates, the local branch word β_{K_t}, and the recovered Step 2 interval datum A_t satisfy the hypotheses of corollary 4.556. That corollary recovers the whole standardized local packet subpath. Under the local branch-recovery interpretation of the packet, this is the deleted subpath

$$(\rho_{2u_t-2}, \rho_{2u_t-1}, \dots, \rho_{2v_t}).$$

Thus every deleted packet subpath is recovered from the replacement-window decoder output and the already budgeted branch word. The packet components K_t and the retained singleton indices $Z_\gamma \setminus I$ partition Z_γ. Interleaving the recovered packet subpaths and retained blocks in increasing block order reconstructs the displayed zone subpath. □

Proposition 4.558 (Step-2 interval halo repairs give the fixed-witness bound). *Fix $q \geq 1$, $j \geq q$, and a q-element set $S \subset \{1, \dots, j\}$. Let Z_γ be the canonical halo replacement zones of lemma 4.254, with each zone written as $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$, and consider either fixed-witness fiber $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$ of proposition 4.543. For each zone put*

$$e_\gamma = |Z_\gamma \cap I(S)|.$$

Suppose that, for every T in the chosen fiber, there is a repaired path $P^*(T) \in \mathcal{P}^{\mathrm{even}}_{j-q}$ and a parser depending only on S with the following properties.

1. *The parser decomposes $P^*(T)$ into the original retained blocks with indices outside the halo zones, unchanged and in their original order, and local replacement windows $V^*_\gamma(T)$ for the zones Z_γ.*
2. *Each $V^*_\gamma(T)$ has exactly $|Z_\gamma \setminus I(S)|$ two-step blocks.*
3. *For each zone there is a branch word*
$$\beta_\gamma(T) \in \{0,1\}^{2e_\gamma}$$
and, for every zone, the endpoint pair $\rho_{2a_\gamma-2}, \rho_{2b_\gamma}$ is the one determined from the parsed complement blocks and the fixed endpoints of the growth-path model by lemma 4.255.
4. *For each zone, the replacement-window decoder satisfies the hypotheses of proposition 4.557 for $V^*_\gamma(T)$, $\beta_\gamma(T)$, the zone endpoint pair, and the known interval Z_γ.*

Then the chosen fixed-witness fiber has cardinality at most

$$2^{2q} s_{j-q}.$$

Proof. By lemma 4.517, for each canonical halo zone the number $e_\gamma = |Z_\gamma \cap I(S)|$ is also the sum of the frontier-event packet lengths used in proposition 4.540. The endpoint pair of each zone is determined from the parsed complement blocks by lemma 4.255. For each zone, item (4) and proposition 4.557 give a fixed local inverse recovering the original subpath on Z_γ from $V^*_\gamma(T)$, $\beta_\gamma(T)$, the determined endpoint pair, and the known zone interval. Therefore the hypotheses of proposition 4.540 hold, and that proposition gives the displayed fixed-witness bound. □

Proposition 4.559 (Step-3 carrier packet readouts give halo-zone inverses). *Work in the setting of proposition 4.531. For each deleted packet $K_t = \{u_t, \dots, v_t\}$ in $Z_\gamma \cap I$, let*

$$J_t = \{c_t, \dots, d_t\}$$

be the corresponding maximal witness-label interval, and put $h_t = |J_t|$. Interpret each packet in the local branch-recovery coordinates of lemma 4.226; the packet branch word obtained from lemma 4.518 is the local branch word for this datum.

Suppose that the replacement-window decoder, from

$$S,\ Z_\gamma,\ V^*_\gamma,\ \beta_\gamma,\quad \rho_{2a_\gamma-2}, \rho_{2b_\gamma},$$

recovers the retained halo blocks

$$B_r(P) \qquad (r \in Z_\gamma \setminus I).$$

Suppose further that, for every packet K_t, the same decoder recovers a Ferrers subarrangement E_t of the Step 3 dual-RSK' half-square and the partition labels on the top/right boundary of E_t, and identifies a deterministic readout which, from the recovered cell entries and internal corner labels of E_t, returns a Step 2 first-variant Knuth subarrangement C_t and its left/bottom boundary labels, such that every non-diagonal cell of C_t, or its transpose in the Step 2 symmetric matrix, lies in E_t, and such that the forward boundary sweep on C_t contains

$$\lambda^{(c_t-1)}, \lambda^{(c_t)}, \dots, \lambda^{(d_t)}.$$

Then a fixed zone-local inverse recovers the whole original subpath

$$(\rho_{2a_\gamma-2}, \rho_{2a_\gamma-1}, \dots, \rho_{2b_\gamma})$$

from V_γ^*, β_γ, the zone endpoint pair, and Z_γ.

Proof. The recovered top/right boundary of E_t determines every cell entry and internal corner label of E_t by corollary 4.124. The deterministic readout in the hypothesis therefore determines C_t and its left/bottom boundary labels. By lemma 4.127, the recovered Step 3 subfilling determines every non-diagonal Step 2 entry of C_t, using symmetry when the transpose cell lies in E_t; diagonal entries are zero by lemma 4.128. Hence the decoder determines a Step 2 first-variant subarrangement with known entries and known left/bottom boundary labels whose forward boundary sweep contains

$$\lambda^{(c_t-1)}, \lambda^{(c_t)}, \dots, \lambda^{(d_t)}$$

for every packet K_t.

These are exactly the packet interval data required in proposition 4.557. The retained halo blocks are recovered by hypothesis, so applying that proposition gives the asserted fixed zone-local inverse. □

Proposition 4.560 (Step-3 carrier halo repairs give the fixed-witness bound)**.** *Fix $q \geq 1$, $j \geq q$, and a q-element set $S \subset \{1, \dots, j\}$. Let Z_γ be the canonical halo replacement zones of lemma 4.254, with each zone written as $Z_\gamma = \{a_\gamma, \dots, b_\gamma\}$, and consider either fixed-witness fiber $\mathcal{F}^w(S)$ or $\mathcal{F}^h(S)$ of proposition 4.543. For each zone put*

$$e_\gamma = |Z_\gamma \cap I(S)|.$$

Suppose that, for every T in the chosen fiber, there is a repaired path $P^*(T) \in \mathcal{P}_{j-q}^{\mathrm{even}}$ and a parser depending only on S with the following properties.

1. *The parser decomposes $P^*(T)$ into the original retained blocks with indices outside the halo zones, unchanged and in their original order, and local replacement windows $V_\gamma^*(T)$ for the zones Z_γ.*
2. *Each $V_\gamma^*(T)$ has exactly $|Z_\gamma \setminus I(S)|$ two-step blocks.*
3. *For each zone there is a branch word*

   $$\beta_\gamma(T) \in \{0,1\}^{2e_\gamma}$$

   and, for every zone, the endpoint pair $\rho_{2a_\gamma-2}, \rho_{2b_\gamma}$ is the one determined from the parsed complement blocks and the fixed endpoints of the growth-path model by lemma 4.255.
4. *For each zone, the replacement-window decoder satisfies the hypotheses of proposition 4.559 for $V_\gamma^*(T)$, $\beta_\gamma(T)$, the zone endpoint pair, and the known interval Z_γ.*

Then the chosen fixed-witness fiber has cardinality at most

$$2^{2q} s_{j-q}.$$

Proof. For each zone, item (4) and proposition 4.559 give a fixed local inverse recovering the original subpath on Z_γ from $V_\gamma^*(T)$, $\beta_\gamma(T)$, the endpoint pair, and the known zone interval. The other items are exactly the parser, target-size, and endpoint hypotheses of proposition 4.540. Applying that proposition gives the displayed fixed-witness bound. □

Corollary 4.561 (Local branch words leave only segment realization)**.** *Work in the setting of lemma 4.551, and put*

$$\sigma^- = \sigma^{(2a_\alpha - 2)}, \qquad \sigma^+ = \sigma^{(2b_\alpha)}.$$

The data

$$S, \quad J_\alpha, \quad \sigma^-, \quad \sigma^+, \quad \beta_\alpha \in \{0,1\}^{2|J_\alpha|}$$

determine all discrete local packet data used by the HKKO repair:

(1) *the bad-boundary packet readout of corollary 4.552;*
(2) *the ordered frontier-event list;*
(3) *the paired local parity-defect slots; and*
(4) *the selected subset of those paired defect slots.*

Consequently the remaining local packet theorem has no further event, defect, selection, or bad-boundary data to store. It must construct the zero-free local segment, place the recovered selected defect slots on eligible boundary-up steps as in lemma 4.228, and prove the off-boundary/interior packet inverse.

Proof. Corollary 4.552 gives item (1) from S, J_α, and β_α. The three conclusions of lemma 4.226, applied with the displayed endpoints σ^- and σ^+, give items (2), (3), and (4). Finally, lemma 4.228 states exactly what remains once those recovered event and selection data are fixed: a zero-free segment with the selected slots realized at eligible boundary-up steps. The off-boundary packet inverse is the part not covered by the bad-boundary readout. □

Corollary 4.562 (Below-wall packets discharge the HKKO segment leaf)**.** *Work in the setting of corollary 4.561. Let μ, ν be an endpoint pair for the local deleted packet, put $h = |J_\alpha|$, and set*

$$w = w(\mu, \nu)$$

as in lemma 4.239. Then

$$\mu, \quad \nu, \quad J_\alpha, \quad \beta_\alpha$$

canonically determine a below-wall w-up-down packet segment $W(\mu, \nu, \beta_\alpha)$, the selected HKKO event indices corresponding to the selected paired-defect subset recovered from β_α, and the unmarked HKKO boundary-event output $W^*(\mu, \nu, \beta_\alpha)$. The marked preimage $W(\mu, \nu, \beta_\alpha)$ is recoverable from

$$W^*(\mu, \nu, \beta_\alpha), \quad \beta_\alpha, \quad w, \quad h, \quad \mu.$$

After the coefficient extraction of lemma 4.129, the selected event indices are eligible boundary-up marks in the vacillating model of lemma 4.134.

Consequently, once a target-size parser has recovered the packet endpoint pair, the local theorem no longer has to construct a separate HKKO segment, marking injection, or wall parameter. The remaining packet-specific input is a target-size parser/merger and fixed off-boundary/interior readout that recovers the original deleted packet subpath from the recovered target-window data and the canonical marked packet preimage.

Proof. Corollary 4.561 and lemma 4.226 recover the selected subset M_α of paired defect slots from the second half of β_α. By lemma 4.239, the displayed integer $w = w(\mu, \nu)$ satisfies

$$\ell(\mu) < w, \qquad \ell(\nu) < w,$$

and is determined by the endpoint pair. Therefore proposition 4.240 applies with this endpoint pair, $h = h_\alpha$, and the branch word β_α. It supplies the canonical assigned event-index list, the below-wall segment $W(\mu, \nu, \beta_\alpha)$ realizing the subset M_α, the unmarked boundary-event splice output $W^*(\mu, \nu, \beta_\alpha)$, and the displayed recovery of the marked preimage.

Lemma 4.129 identifies the corresponding coefficient-extracted marked-vacillating events with eligible boundary-up steps, and lemma 4.134 records the same admissibility condition in the marked vacillating class. Thus the segment and marking part of the local packet theorem is supplied by the cited below-wall package after the packet endpoints are known. What is not supplied by that package is a target-size replacement-window decoder or a readout of the original off-boundary packet subpath; those are exactly the remaining inputs stated in the consequence. □

Proposition 4.563 (Bad-shape fiber reduction)**.** *Fix residual $q = b+1$. Suppose that, for every j in the residual window, every q-element subset $S \subset \{1,\dots,j\}$ satisfies the two fiber estimates*

$$\#\{T : \lambda(T)' \text{ even},\ \lambda_1(T) \geq 2q,\ W_q(T) = S\} \leq 2^{2q} s_{j-q},$$

and

$$\#\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\} \leq 2^{2q} s_{j-q}.$$

Then the two bad-shape estimates in proposition 4.811 hold for this q and j.

Proof. By lemma 4.95, each width-bad tableau has exactly one canonical witness $W_q(T)$, and each height-bad tableau has exactly one canonical witness $H_q(T)$. Partitioning the two bad sets by that witness and applying the displayed fiber estimates gives

$$\sum_{\substack{\lambda' : \text{even} \\ \lambda_1 \geq 2q}} a_j(\lambda) \leq \sum_{\substack{S \subset \{1,\dots,j\} \\ |S| = q}} 2^{2q} s_{j-q} = 2^{2q} \binom{j}{q} s_{j-q},$$

and the height estimate is identical with H_q in place of W_q. □

Proposition 4.564 (The 2^q fixed-height-fiber target is false)**.** *Let $q = 15$, $j = 30$, and*

$$S = \{1, 3, 5, \dots, 29\}.$$

Then the height fixed-witness fiber satisfies

$$\#\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 30,\ H_{15}(T) = S\} = 6190283353629375,$$

whereas

$$2^{15} s_{15} = 4825650970886144.$$

In particular, the estimate

$$\#\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\} \leq 2^{q} s_{j-q}$$

is false even at the first residual boundary $q = 15$, $j = 30$.

Proof. Write the displayed height fiber as $\mathcal{F}^h(S)$. By corollary 4.114, the displayed fiber is equinumerous with the column-even Pieri two-strip paths of length 30 with crossing set $C^h_{15} = S$. The exact GMP/OpenMP obstruction replay

`certificates/post_m29/bc_fixed_witness_2q_obstruction_gmp_certificate.log`

enumerates this path fiber and independently recomputes s_{15}. It runs on `optimus` with 64 OpenMP threads, uses exact `mpz_class` arithmetic, reports

$$s_{15} = 147267180508,$$
$$2^{15} s_{15} = 4825650970886144,$$
$$\#\mathcal{F}^h(S) = 6190283353629375,$$

and verifies `height_exceeds_2powq_rhs=yes`, `SUMMARY failures=0`, and `__EXIT_STATUS=0`. Its SHA-256 is

`cc303f3bffd77ba4d8954054c31bc99c`
`cd5c406f08e3a397ca0f37b759398436`.

The final displayed inequality is therefore contradicted by this exact integer count. □

Lemma 4.565 (B-boundary width-bad paths have a three-step encoding)**.** *Let $q \geq 1$, and let $\lambda^\bullet$ be a column-even Pieri two-strip path of length $2q$ such that*

$$\lambda_1^{(2q)} \geq 2q.$$

Then

$$\lambda^{(2q)} = (2q, 2q),$$

and the number of such paths is at most 3^{2q}.

Proof. The terminal partition has size $4q$. Since $(\lambda^{(2q)})'$ has only even parts, every nonzero column has height at least 2. Hence

$$4q = |\lambda^{(2q)}| \geq 2\lambda_1^{(2q)} \geq 4q.$$

Thus $\lambda_1^{(2q)} = 2q$, every nonzero column has height exactly 2, and the terminal partition is $(2q, 2q)$.

Every intermediate partition is contained in this two-row rectangle. At each Pieri step, the horizontal two-strip can therefore add its two boxes in only one of three row patterns: two boxes in the first row, one box in each row, or two boxes in the second row. Sending a path to this word in a three-letter alphabet is injective, because the current partition and the row pattern determine the next partition. Hence there are at most 3^{2q} such paths. □

Lemma 4.566 (Stable moments dominate the boundary width count)**.** *For every $q \geq 15$,*

$$2 \cdot 3^{2q} \leq s_{2q}.$$

Proof. By lemma 4.65, s_{2q} counts loopless labelled degree-two multigraphs on $2q$ vertices. It therefore contains all labelled simple cycles, whose number is

$$\frac{(2q-1)!}{2}.$$

It is enough to prove

$$4 \cdot 3^{2q} \leq (2q-1)!.$$

At $q = 15$,

$$4 \cdot 3^{30} = 823564528378596 < 8841761993739701954543616000000 = 29!.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by 9, while the right side is multiplied by

$$(2q)(2q+1) > 9.$$

Thus induction proves the inequality for every $q \geq 15$. □

Proposition 4.567 (Residual B-boundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q} \leq \delta_{2q}^{B_b}.$$

Proof. By lemma 4.94 and the coefficient interpretation in lemma 4.95, $|\delta_{2q}^{B_b}|$ is the number of semistandard tableaux T of content (2^{2q}) whose shape $\lambda(T)$ has $\lambda(T)'$ even and $\lambda_1(T) \geq 2q$. Under lemma 4.97, these tableaux are exactly the column-even Pieri two-strip paths of length $2q$ with terminal first row at least $2q$. By lemma 4.565, their number is at most 3^{2q}. Since $q \geq 15$, lemma 4.566 gives

$$|\delta_{2q}^{B_b}| \leq 3^{2q} \leq \frac{1}{2}s_{2q}.$$

This proves the stated lower bound. □

Lemma 4.568 (Stable moments dominate the first B nonboundary width count)**.** *For every $q \geq 15$,*

$$2 \cdot 10^{2q+1} \leq s_{2q+1}.$$

Proof. By lemma 4.65, s_{2q+1} counts loopless labelled degree-two multigraphs on $2q+1$ vertices. It therefore contains all labelled simple cycles, whose number is

$$\frac{(2q)!}{2}.$$

It is enough to prove

$$4 \cdot 10^{2q+1} \leq (2q)!.$$

At $q = 15$,

$$4 \cdot 10^{31} < 265252859812191058636308480000000 = 30!.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by 100, while the right side is multiplied by

$$(2q+1)(2q+2) \geq 31 \cdot 32 > 100.$$

Thus induction proves the inequality for every $q \geq 15$. □

Lemma 4.569 (First B nonboundary width-bad paths have a ten-step encoding)**.** *Let $q \geq 1$, and let $\lambda^\bullet$ be a column-even Pieri two-strip path of length $2q+1$ such that*

$$\lambda_1^{(2q+1)} \geq 2q.$$

Then the terminal partition is either

$$(2q+1, 2q+1) \qquad \text{or} \qquad (2q, 2q, 1, 1),$$

and the number of such paths is at most 10^{2q+1}.

Proof. The terminal partition has size $4q+2$. Since $(\lambda^{(2q+1)})'$ has only even parts, every nonzero column has height at least 2. Hence the terminal first row is either $2q$ or $2q+1$. If it is $2q+1$, all columns have height exactly 2, so the terminal partition is $(2q+1, 2q+1)$. If it is $2q$, the columns of height at least 2 account for $4q$ boxes. The remaining two boxes must form one more even column-height contribution, hence the terminal partition is $(2q, 2q, 1, 1)$.

Thus every intermediate partition has at most four rows. At each Pieri step, record the unordered pair of rows, with repetition allowed, receiving the two new boxes. There are $\binom{4+1}{2} = 10$ possible row patterns, and the current partition together with this row pattern determines the next partition: the boxes are added at the right ends of the selected rows. Therefore the map from paths to row-pattern words is injective, giving at most 10^{2q+1} paths. □

Proposition 4.570 (Residual B first nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2} s_{2q+1} \leq \delta_{2q+1}^{B_b}.$$

Proof. By lemma 4.94 and the coefficient interpretation in lemma 4.95, $|\delta_{2q+1}^{B_b}|$ is the number of semistandard tableaux T of content (2^{2q+1}) whose shape $\lambda(T)$ has $\lambda(T)'$ even and $\lambda_1(T) \geq 2q$. Under lemma 4.97, these tableaux are exactly the column-even Pieri two-strip paths of length $2q+1$ with terminal first row at least $2q$. By lemma 4.569, their number is at most 10^{2q+1}. Since $q \geq 15$, lemma 4.568 gives

$$|\delta_{2q+1}^{B_b}| \leq 10^{2q+1} \leq \frac{1}{2} s_{2q+1}.$$

This proves the stated lower bound. □

Lemma 4.571 (Second B nonboundary width-bad paths have sparse exceptional rows)**.** *Let $q \geq 1$, and let $\lambda^\bullet$ be a column-even Pieri two-strip path of length $2q+2$ such that*

$$\lambda_1^{(2q+2)} \geq 2q.$$

Then the number of such paths is at most

$$5(2q+2)^4\,21^4\,3^{2q+2}.$$

Proof. The terminal partition has size $4q+4$. Since $(\lambda^{(2q+2)})'$ has only even parts, every nonzero column has height at least 2. Hence the terminal first row $W=\lambda_1^{(2q+2)}$ satisfies

$$2q \le W \le 2q+2.$$

The number of terminal boxes below the second row is

$$|\lambda^{(2q+2)}| - 2W \le 4.$$

Thus the terminal partition has at most six nonzero rows, and at most four Pieri steps add a box below the second row.

Encode a path as follows. First record the exceptional set E of steps which add at least one box below the second row. It has size at most 4. For a nonexceptional step, record which unordered pair of rows among $\{1,2\}$, with repetition allowed, receives the two boxes; there are three choices. For an exceptional step, record the unordered pair of rows among the at most six terminal rows; there are $\binom{6+1}{2}=21$ choices. The current partition together with this row pair determines the next partition, because the new boxes are added at the right ends of the selected rows. Therefore the encoding is injective.

For $n=2q+2$, the number of codes is at most

$$\sum_{e=0}^{4}\binom{n}{e}3^{n-e}21^e \le 5n^4\,21^4\,3^n,$$

which is the displayed bound. □

Lemma 4.572 (Stable moments dominate the second B nonboundary width count)**.** *For every $q \ge 15$,*

$$10(2q+2)^4\,21^4\,3^{2q+2} \le s_{2q+2}.$$

Proof. By lemma 4.65, s_{2q+2} counts loopless labelled degree-two multigraphs on $2q+2$ vertices. It therefore contains all labelled simple cycles, whose number is

$$\frac{(2q+1)!}{2}.$$

It is enough to prove

$$20(2q+2)^4\,21^4\,3^{2q+2} \le (2q+1)!.$$

At $q=15$,

$$\begin{aligned} 20\cdot 32^4\cdot 21^4\cdot 3^{32} &= 7557658063102958937487441920,\\ 31! &= 8222838654177922817725562880000000. \end{aligned}$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\left(\frac{2q+4}{2q+2}\right)^4 < 12,$$

while the right side is multiplied by

$$(2q+2)(2q+3) \ge 32\cdot 33 > 12.$$

Thus induction proves the inequality for every $q \ge 15$. □

Proposition 4.573 (Residual B second nonboundary lower correction)**.** *For every residual row B_b with $14 \le b \le 123$, put $q=b+1$ and write $s_j=m_j^{\mathrm{st}}$ and $m_j^{B_b}=s_j+\delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+2} \le \delta_{2q+2}^{B_b}.$$

Proof. By lemma 4.94 and the coefficient interpretation in lemma 4.95, $|\delta_{2q+2}^{B_b}|$ is the number of semistandard tableaux T of content (2^{2q+2}) whose shape $\lambda(T)$ has $\lambda(T)'$ even and $\lambda_1(T) \geq 2q$. Under lemma 4.97, these tableaux are exactly the column-even Pieri two-strip paths of length $2q+2$ with terminal first row at least $2q$. By lemma 4.571, their number is at most

$$5(2q+2)^4\, 21^4\, 3^{2q+2}.$$

Since $q \geq 15$, lemma 4.572 gives

$$|\delta_{2q+2}^{B_b}| \leq 5(2q+2)^4\, 21^4\, 3^{2q+2} \leq \frac{1}{2} s_{2q+2}.$$

This proves the stated lower bound. □

Lemma 4.574 (Third B nonboundary width-bad paths have sparse exceptional rows)**.** *Let $q \geq 1$, and let $\lambda^\bullet$ be a column-even Pieri two-strip path of length $2q+3$ such that*

$$\lambda_1^{(2q+3)} \geq 2q.$$

Then the number of such paths is at most

$$\frac{7}{720}(2q+3)^6\, 12^6\, 3^{2q+3}.$$

Proof. Put $n = 2q+3$. The terminal partition has size $2n = 4q+6$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+6-2(2q) = 6.$$

Hence the terminal partition has at most eight nonzero rows, and at most six Pieri steps add a box below the second row.

As in lemma 4.571, record the exceptional set E of below-second-row steps. At nonexceptional steps there are three unordered row-pair choices in the first two rows. At exceptional steps there are at most $\binom{8+1}{2} = 36$ unordered row-pair choices. This row-pair word reconstructs the path recursively, so the encoding is injective. Therefore the number of paths is at most

$$\sum_{e=0}^{6} \binom{n}{e} 3^{n-e} 36^e = 3^n \sum_{e=0}^{6} \binom{n}{e} 12^e.$$

Since $\binom{n}{e} \leq n^e/e!$, it remains to bound the elementary sum with these larger summands. For $0 \leq e \leq 6$, the terms

$$\frac{n^e 12^e}{e!}$$

increase with e, since $12n/(e+1) > 1$. Thus the last display is at most

$$3^n \cdot 7\, \frac{n^6 12^6}{6!} = \frac{7}{720} n^6\, 12^6\, 3^n,$$

as claimed. □

Lemma 4.575 (Stable moments dominate the third B nonboundary width count)**.** *For every $q \geq 15$,*

$$\frac{7}{360}(2q+3)^6\, 12^6\, 3^{2q+3} \leq s_{2q+3}.$$

Proof. By lemma 4.65, s_{2q+3} contains all labelled simple cycles on $2q+3$ vertices, whose number is $(2q+2)!/2$. It is enough to prove

$$7(2q+3)^6\, 12^6\, 3^{2q+3} \leq 180(2q+2)!.$$

At $q = 15$,

$$\begin{aligned} 7 \cdot 33^6 \cdot 12^6 \cdot 3^{33} &= 150061941592511735750704505081856, \\ 180 \cdot 32! &= 47363550648064835430099242188800000000. \end{aligned}$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\left(\frac{2q+5}{2q+3}\right)^6 < 13,$$

while the right side is multiplied by

$$(2q+3)(2q+4) \geq 33 \cdot 34 > 13.$$

Induction proves the result for every $q \geq 15$. □

Proposition 4.576 (Residual B third nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+3} \leq \delta_{2q+3}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+3}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+3$ with terminal first row at least $2q$. By lemma 4.574, this number is at most

$$\frac{7}{720}(2q+3)^6\, 12^6\, 3^{2q+3}.$$

Since $q \geq 15$, lemma 4.575 gives

$$|\delta_{2q+3}^{B_b}| \leq \frac{1}{2}s_{2q+3}.$$

This proves the stated lower bound. □

Lemma 4.577 (Fourth B nonboundary width-bad paths have sparse exceptional rows)**.** *Let $q \geq 1$, and let $\lambda^\bullet$ be a column-even Pieri two-strip path of length $2q+4$ such that*

$$\lambda_1^{(2q+4)} \geq 2q.$$

Then the number of such paths is at most

$$\frac{9}{8!}(2q+4)^8\, 55^8\, 3^{2q-4}.$$

Proof. Put $n = 2q+4$. The terminal partition has size $2n = 4q+8$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q + 8 - 2(2q) = 8.$$

Hence the terminal partition has at most ten nonzero rows, and at most eight Pieri steps add a box below the second row.

Record the exceptional set E of below-second-row steps. At a nonexceptional step there are three unordered row-pair choices in the first two rows. At an exceptional step there are at most $\binom{10+1}{2} = 55$ unordered row-pair choices among terminal rows. The current partition and the recorded row pair determine the next partition, so the encoding is injective. Therefore the number of paths is at most

$$\sum_{e=0}^{8}\binom{n}{e}3^{n-e}55^e = 3^n\sum_{e=0}^{8}\binom{n}{e}\left(\frac{55}{3}\right)^e.$$

Using $\binom{n}{e} \leq n^e/e!$, the e-th summand is at most

$$\frac{n^e 55^e 3^{n-e}}{e!}.$$

These upper summands increase for $0 \le e < 8$, because

$$\frac{55n}{3(e+1)} > 1.$$

Thus the sum is bounded by nine times its final upper summand, giving

$$\frac{9}{8!} n^8 \, 55^8 \, 3^{n-8} = \frac{9}{8!} (2q+4)^8 \, 55^8 \, 3^{2q-4}.$$

□

Lemma 4.578 (Stable moments dominate the fourth B nonboundary width count)**.** *For every $q \ge 15$,*

$$\frac{18}{8!} (2q+4)^8 \, 55^8 \, 3^{2q-4} \le s_{2q+4}.$$

Proof. By lemma 4.65, s_{2q+4} contains all labelled simple cycles on $2q+4$ vertices, whose number is $(2q+3)!/2$. It is enough to prove

$$18(2q+4)^8 \, 55^8 \, 3^{2q-4} \le \frac{8!}{2} (2q+3)!.$$

At $q = 15$,

$$18 \cdot 34^8 \cdot 55^8 \cdot 3^{26} = 684160474040780221038402137953620000000,$$
$$\frac{8!}{2} 33! = 1750556831952476317496467991298048000000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9 \left(\frac{2q+6}{2q+4} \right)^8 < 16,$$

while the right side is multiplied by

$$(2q+4)(2q+5) \ge 34 \cdot 35 > 16.$$

Induction proves the result for every $q \ge 15$. □

Proposition 4.579 (Residual B fourth nonboundary lower correction)**.** *For every residual row B_b with $14 \le b \le 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2} s_{2q+4} \le \delta_{2q+4}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+4}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+4$ with terminal first row at least $2q$. By lemma 4.577, this number is at most

$$\frac{9}{8!} (2q+4)^8 \, 55^8 \, 3^{2q-4}.$$

Since $q \ge 15$, lemma 4.578 gives

$$|\delta_{2q+4}^{B_b}| \le \frac{1}{2} s_{2q+4}.$$

This proves the stated lower bound. □

Lemma 4.580 (Fifth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+5)}$$

with

$$\lambda_1^{(2q+5)} \ge 2q$$

is at most

$$36864\binom{2q+5}{10}20^{10}3^{2q-5}.$$

Proof. Put $n=2q+5$. The terminal partition has size $2n=4q+10$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+10-2(2q)=10.$$

Hence the terminal partition has at most twelve nonzero rows.

Encode a path by its row-pair word, but keep track of the number of boxes put below row 2. A step with no such box has one of the three top-row pair types. A step with exactly one such box has at most $2\cdot 10=20$ choices: one top row and one of the ten possible lower terminal rows. A step with two such boxes has at most $\binom{10+1}{2}=55$ choices. The current partition and this row-pair type reconstruct the next partition, so the encoding is injective. Thus the number of paths is bounded by the sum of the coefficients of $x^0,\dots,x^{10}$ in

$$(3+20x+55x^2)^n.$$

For a monomial with a one-lower-box steps and c two-lower-box steps, $a+2c\le 10$, its contribution is

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}20^a55^c.$$

Since $n\ge 35$ and $a+c\le 10$,

$$\binom{n}{a+c}\le\binom{n}{10},\qquad \binom{a+c}{c}\le 2^{10}.$$

Also $55\le 20^2/3$, so

$$3^{n-a-c}20^a55^c\le 20^{a+2c}3^{n-a-2c}\le 20^{10}3^{n-10}.$$

There are 36 pairs (a,c) with $a,c\ge 0$ and $a+2c\le 10$. Multiplying these bounds gives

$$36\cdot 2^{10}\binom{n}{10}20^{10}3^{n-10}=36864\binom{2q+5}{10}20^{10}3^{2q-5}.$$

$\square$

Lemma 4.581 (Stable moments dominate the fifth B nonboundary width count)**.** *For every $q\ge 15$,*

$$73728\binom{2q+5}{10}20^{10}3^{2q-5}\le s_{2q+5}.$$

Proof. By lemma 4.65, s_{2q+5} contains all labelled simple cycles on $2q+5$ vertices, whose number is $(2q+4)!/2$. It is enough to prove

$$147456\binom{2q+5}{10}20^{10}3^{2q-5}\le (2q+4)!.$$

At $q=15$,

$$147456\binom{35}{10}20^{10}3^{25}=2348646797088236025479902003200000000000,$$

$$34!=295232799039604140847618609643520000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+7)(2q+6)}{(2q-3)(2q-4)}<18,$$

while the right side is multiplied by

$$(2q+5)(2q+6)\ge 35\cdot 36>18.$$

Induction proves the result for every $q \geq 15$. □

Proposition 4.582 (Residual B fifth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+5} \leq \delta_{2q+5}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+5}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+5$ with terminal first row at least $2q$. By lemma 4.580, this number is at most

$$36864\binom{2q+5}{10}20^{10}3^{2q-5}.$$

Since $q \geq 15$, lemma 4.581 gives

$$|\delta_{2q+5}^{B_b}| \leq \frac{1}{2}s_{2q+5}.$$

This proves the stated lower bound. □

Lemma 4.583 (Sixth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \geq 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+6)}$$

with

$$\lambda_1^{(2q+6)} \geq 2q$$

is at most

$$6\binom{2q+6}{12}24^{12}3^{2q-6}.$$

Proof. Put $n = 2q+6$. The terminal partition has size $2n = 4q+12$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q + 12 - 2(2q) = 12.$$

Hence the terminal partition has at most fourteen nonzero rows.

Encode a path by its row-pair word, keeping the number of boxes placed below row 2. A step with no lower box has one of three top-row choices. A step with exactly one lower box has at most $2 \cdot 12 = 24$ choices. A step with two lower boxes has at most $\binom{12+1}{2} = 78$ choices. The current partition and the row-pair choice recover the next partition, so the encoding is injective. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{12}$ in

$$(3 + 24x + 78x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a + 2c \leq 12$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}24^a78^c.$$

Since $n \geq 36$ and $a + c \leq 12$,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{12}} \leq \frac{12!}{a!\,c!\,25^{12-a-c}}.$$

Also

$$3^{n-a-c}24^a78^c = 24^{12}3^{n-12}\left(\frac{13}{32}\right)^c 8^{a+2c-12}.$$

Therefore the coefficient sum is at most

$$\binom{n}{12}24^{12}3^{n-12}\sum_{\substack{a,c\geq 0\\ a+2c\leq 12}}\frac{12!}{a!\,c!\,25^{12-a-c}}\left(\frac{13}{32}\right)^{c}8^{a+2c-12}.$$

The finite rational sum is

$$\frac{880987349156210452990561}{16000000000000000000000000}<6.$$

This gives the displayed bound. □

Lemma 4.584 (Stable moments dominate the sixth B nonboundary width count)**.** *For every $q\geq 15$,*

$$12\binom{2q+6}{12}24^{12}3^{2q-6}\leq s_{2q+6}.$$

Proof. By lemma 4.65, s_{2q+6} contains all labelled simple cycles on $2q+6$ vertices, whose number is $(2q+5)!/2$. It is enough to prove

$$24\binom{2q+6}{12}24^{12}3^{2q-6}\leq(2q+5)!.$$

At $q=15$,

$$24\binom{36}{12}24^{12}3^{24}=3098480522063423281008179759743500288 00,$$
$$35!=10333147966386144929666651337523200000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+8)(2q+7)}{(2q-4)(2q-5)}<20,$$

because $44q^2-630q-104>0$ for $q\geq 15$. The right side is multiplied by

$$(2q+6)(2q+7)\geq 36\cdot 37>20.$$

Induction proves the result. □

Proposition 4.585 (Residual B sixth nonboundary lower correction)**.** *For every residual row B_b with $14\leq b\leq 123$, put $q=b+1$ and write $s_j=m_j^{\mathrm{st}}$ and $m_j^{B_b}=s_j+\delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+6}\leq\delta_{2q+6}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+6}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+6$ with terminal first row at least $2q$. By lemma 4.583, this number is at most

$$6\binom{2q+6}{12}24^{12}3^{2q-6}.$$

Since $q\geq 15$, lemma 4.584 gives

$$|\delta_{2q+6}^{B_b}|\leq\frac{1}{2}s_{2q+6}.$$

This proves the stated lower bound. □

Lemma 4.586 (Seventh B nonboundary width-bad paths have a lower-box code)**.** *For every $q \geq 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+7)}$$

with

$$\lambda_1^{(2q+7)} \geq 2q$$

is at most

$$11\binom{2q+7}{14}28^{14}3^{2q-7}.$$

Proof. Put $n = 2q+7$. The terminal partition has size $2n = 4q+14$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q + 14 - 2(2q) = 14.$$

Hence the terminal partition has at most sixteen nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 14 = 28$ choices, and a step with two lower boxes has at most $\binom{14+1}{2} = 105$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{14}$ in

$$(3 + 28x + 105x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a + 2c \leq 14$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}28^a105^c.$$

Since $n \geq 37$, $a+c \leq 14$, and the fourteen factors in the quotient $\binom{n}{14}/\binom{n}{a+c}$ which are not cancelled are all at least 24,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{14}} \leq \frac{14!}{a!\,c!\,24^{14-a-c}}.$$

Also

$$3^{n-a-c}28^a105^c = 28^{14}3^{n-14}\left(\frac{45}{112}\right)^c\left(\frac{28}{3}\right)^{a+2c-14}.$$

Therefore the coefficient sum is at most

$$\binom{n}{14}28^{14}3^{n-14}\sum_{\substack{a,c\geq 0\\ a+2c\leq 14}}\frac{14!}{a!\,c!\,24^{14-a-c}}\left(\frac{45}{112}\right)^c\left(\frac{28}{3}\right)^{a+2c-14}.$$

The finite rational sum is

$$\frac{79869422716689772849072285277}{7978958832736206792937177088} < 11.$$

This gives the displayed bound. ∎

Lemma 4.587 (Stable moments dominate the seventh B nonboundary width count above the first row)**.** *For every $q \geq 16$,*

$$22\binom{2q+7}{14}28^{14}3^{2q-7} \leq s_{2q+7}.$$

Proof. By lemma 4.65, s_{2q+7} contains all labelled simple cycles on $2q+7$ vertices, whose number is $(2q+6)!/2$. It is enough to prove

$$44\binom{2q+7}{14}28^{14}3^{2q-7} \le (2q+6)!.$$

At $q=16$,

$$\begin{aligned} 44\binom{39}{14}28^{14}3^{25} &= 1023829241393349569610919444974346354902958 08, \\ 38! &= 523022617466601111760007224100074291200000000. \end{aligned}$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+9)(2q+8)}{(2q-5)(2q-6)} < 23,$$

because $56q^2 - 812q + 42 > 0$ for $q \ge 16$. The right side is multiplied by

$$(2q+7)(2q+8) \ge 39 \cdot 40 > 23.$$

Induction proves the result. ☐

Lemma 4.588 (Exact B_{14} seventh nonboundary bad-shape count)**.** *For $q = 15$, the number of column-even Pieri two-strip paths of length 37 with terminal first row at least 30 is*

$$617001871685935689564870 5996.$$

Moreover

$$2 \cdot 6170018716859356895648705996 \le s_{37}.$$

Proof. The exact C++/GMP replay

`certificates/post_m29/bc_b14_j37_badshape_gmp_certificate.log`

implements the following finite reverse Pieri recursion. It initializes the terminal set to the 45 partitions

$$(\mu_1, \mu_1, \mu_2, \mu_2, \ldots), \qquad |\mu| = 37, \qquad \mu_1 \ge 30.$$

At each reverse step it removes either two boxes from one row, or one box from two rows whose removed boxes have different columns, and keeps exactly those predecessors which remain partitions. This is precisely the inverse of adding a horizontal two-strip, so after 37 reverse steps the coefficient of the empty partition is the displayed number of Pieri paths.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`bf212c9d15e9b1953f99b01aff69af74`
`b7e4a0a78ae24b769ba2fcd502fa046b`.

It reports

$$s_{37} = 991021752128621887393916677761923440691808$$

from the recurrence of lemma 4.67, and verifies the positive margin

$$s_{37} - 2 \cdot 6170018716859356895648705996 = 991021752128609547356482959048132143279816.$$

This proves both assertions. ☐

Proposition 4.589 (Residual B seventh nonboundary lower correction)**.** *For every residual row B_b with $14 \le b \le 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+7} \le \delta_{2q+7}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+7}|$ is the number of column-even Pieri two-strip paths of length $2q+7$ with terminal first row at least $2q$. If $q = 15$, this is bounded by $s_{37}/2$ by lemma 4.588. If $q \geq 16$, lemma 4.586 bounds this number by

$$11\binom{2q+7}{14}28^{14}3^{2q-7},$$

and lemma 4.587 absorbs that count into $s_{2q+7}/2$. In all cases $|\delta^{B_b}_{2q+7}| \leq s_{2q+7}/2$, which proves the stated lower bound. ☐

Lemma 4.590 (Eighth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \geq 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+8)}$$

with

$$\lambda^{(2q+8)}_1 \geq 2q$$

is at most

$$21\binom{2q+8}{16}32^{16}3^{2q-8}.$$

Proof. Put $n = 2q+8$. The terminal partition has size $2n = 4q+16$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q + 16 - 2(2q) = 16.$$

Hence the terminal partition has at most eighteen nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 16 = 32$ choices, and a step with two lower boxes has at most $\binom{16+1}{2} = 136$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{16}$ in

$$(3 + 32x + 136x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a + 2c \leq 16$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}32^a136^c.$$

Since $n \geq 38$, $a + c \leq 16$, and the sixteen factors in the quotient $\binom{n}{16}/\binom{n}{a+c}$ which are not cancelled are all at least 23,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{16}} \leq \frac{16!}{a!\,c!\,23^{16-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{16}32^{16}3^{n-16}\sum_{\substack{a,c\geq 0\\ a+2c\leq 16}}\frac{16!}{a!\,c!\,23^{16-a-c}}\frac{3^{16-a-c}136^c}{32^{16-a}}.$$

The finite rational sum is

$$\frac{4526978891280486763545625577579279602498403}{226253389683665863434374437380034188541952} < 21.$$

This gives the displayed bound. ☐

Lemma 4.591 (Stable moments dominate the eighth B nonboundary width count above the first row)**.** *For every $q \geq 18$,*

$$42\binom{2q+8}{16}32^{16}3^{2q-8} \leq s_{2q+8}.$$

Proof. By lemma 4.65, s_{2q+8} contains all labelled simple cycles on $2q+8$ vertices, whose number is $(2q+7)!/2$. It is enough to prove

$$84\binom{2q+8}{16}32^{16}3^{2q-8} \leq (2q+7)!.$$

At $q = 18$,

$$84\binom{44}{16}32^{16}3^{28} = 9680839112802527247642604013895333981167595930255536,$$
$$43! = 60415263063373835637355132068513997507264512000000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+10)(2q+9)}{(2q-6)(2q-7)} < 23,$$

because $56q^2 - 940q + 156 > 0$ for $q \geq 18$. The right side is multiplied by

$$(2q+8)(2q+9) \geq 44 \cdot 45 > 23.$$

Induction proves the result. ☐

Lemma 4.592 (Exact first three eighth B nonboundary bad-shape counts)**.** *For $q = 15, 16, 17$, the number of column-even Pieri two-strip paths of length $2q+8$ with terminal first row at least $2q$ is at most $s_{2q+8}/2$.*

Proof. The exact C++/GMP/OpenMP replay

```
certificates/post_m29/bc_b_eighth_frontier_gmp_certificate.log
```

implements the same reverse Pieri recursion as lemma 4.588, now for the three cases $q = 15, 16, 17$, $j = 2q+8$. It initializes the terminal set to the column-even width-bad partitions

$$(\mu_1, \mu_1, \mu_2, \mu_2, \ldots), \qquad |\mu| = j, \qquad \mu_1 \geq 2q,$$

and reverses horizontal two-strip additions for j steps.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`5d05cb22e6586256710a9d062f898f67`
`f3537c88425ad5b3d3d3ebcecd8972ed`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	38	7273729317904420362730618695531	371630283926365238456764504124209593715917786
16	40	155002365324834308572576825681005	565149003794044140741950854381686635870121608846
17	42	3145163595517583154719821722291500	9498630233337573143836003575168282228037177187235 6.

Each margin is positive, proving the claimed half-stable bound. ☐

Proposition 4.593 (Residual B eighth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+8} \leq \delta_{2q+8}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+8}|$ is the number of column-even Pieri two-strip paths of length $2q+8$ with terminal first row at least $2q$. If $q = 15, 16, 17$, this is bounded by $s_{2q+8}/2$ by lemma 4.592. If $q \geq 18$, lemma 4.590 bounds this number by

$$21\binom{2q+8}{16}32^{16}3^{2q-8},$$

and lemma 4.591 absorbs that count into $s_{2q+8}/2$. In all cases $|\delta^{B_b}_{2q+8}| \leq s_{2q+8}/2$, which proves the stated lower bound. $\square$

Lemma 4.594 (Ninth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \geq 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+9)}$$

with

$$\lambda_1^{(2q+9)} \geq 2q$$

is at most

$$45\binom{2q+9}{18}36^{18}3^{2q-9}.$$

Proof. Put $n = 2q+9$. The terminal partition has size $2n = 4q+18$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q + 18 - 2(2q) = 18.$$

Hence the terminal partition has at most twenty nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 18 = 36$ choices, and a step with two lower boxes has at most $\binom{18+1}{2} = 171$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{18}$ in

$$(3 + 36x + 171x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a + 2c \leq 18$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}36^a171^c.$$

Since $n \geq 39$, $a+c \leq 18$, and the eighteen factors in the quotient $\binom{n}{18}/\binom{n}{a+c}$ which are not cancelled are all at least 22,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{18}} \leq \frac{18!}{a!\,c!\,22^{18-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{18}36^{18}3^{n-18}\sum_{\substack{a,c\geq 0\\ a+2c\leq 18}}\frac{18!}{a!\,c!\,22^{18-a-c}}\frac{3^{18-a-c}171^c}{36^{18-a}}.$$

The finite rational sum is

$$\frac{36191780928311296355762228472388 6093}{8204044929474043719393332089061376} < 45.$$

This gives the displayed bound. $\square$

Lemma 4.595 (Stable moments dominate the ninth B nonboundary width count above the first row)**.** *For every $q \geq 19$,*

$$90\binom{2q+9}{18}36^{18}3^{2q-9} \leq s_{2q+9}.$$

Proof. By lemma 4.65, s_{2q+9} contains all labelled simple cycles on $2q+9$ vertices, whose number is $(2q+8)!/2$. It is enough to prove

$$180\binom{2q+9}{18}36^{18}3^{2q-9} \leq (2q+8)!.$$

At $q = 19$,

$$\begin{aligned} 180\binom{47}{18}36^{18}3^{29} &= 582132191395269676126538459972138264700438046490335641600, \\ 46! &= 5502622159812088949850305428800254892961651752960000000000. \end{aligned}$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+11)(2q+10)}{(2q-7)(2q-8)} < 23,$$

because $56q^2 - 1068q + 298 > 0$ for $q \geq 19$. The right side is multiplied by

$$(2q+9)(2q+10) \geq 47 \cdot 48 > 23.$$

Induction proves the result. $\square$

Lemma 4.596 (Exact first four ninth B nonboundary bad-shape counts)**.** *For $q = 15, 16, 17, 18$, the number of column-even Pieri two-strip paths of length $2q+9$ with terminal first row at least $2q$ is at most $s_{2q+9}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_ninth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $q = 15, 16, 17, 18$, $j = 2q+9$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`3459b8f1c779c878c9a1666f18b27a03`
`548202729f4ddaa23399b6f2c11c242c`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	39	81596655286601105266416444390197	1430766394625961748437041706576357337819220214
16	41	1926342911785452502271099095125800	2288839508814961324669405817236421254438863593520
17	43	43077718528836304303270590063965610	4036896628145341970914689003048977156793546166686156
18	45	917849214459918430646886583313786946	7814348742732519466761141196680536729520495482426158780.

Each margin is positive, proving the claimed half-stable bound. $\square$

Proposition 4.597 (Residual B ninth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+9} \leq \delta_{2q+9}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+9}|$ is the number of column-even Pieri two-strip paths of length $2q+9$ with terminal first row at least $2q$. If $q = 15, 16, 17, 18$, this is bounded by $s_{2q+9}/2$ by lemma 4.596. If $q \geq 19$, lemma 4.594 bounds this number by

$$45\binom{2q+9}{18}36^{18}3^{2q-9},$$

and lemma 4.595 absorbs that count into $s_{2q+9}/2$. In all cases $|\delta^{B_b}_{2q+9}| \leq s_{2q+9}/2$, which proves the stated lower bound. $\square$

Lemma 4.598 (Tenth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \geq 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+10)}$$

with

$$\lambda_1^{(2q+10)} \geq 2q$$

is at most

$$108\binom{2q+10}{20}40^{20}3^{2q-10}.$$

Proof. Put $n = 2q+10$. The terminal partition has size $2n = 4q+20$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q + 20 - 2(2q) = 20.$$

Hence the terminal partition has at most twenty-two nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 20 = 40$ choices, and a step with two lower boxes has at most $\binom{20+1}{2} = 210$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{20}$ in

$$(3 + 40x + 210x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a + 2c \leq 20$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}40^a 210^c.$$

Since $n \geq 40$, $a + c \leq 20$, and the twenty factors in the quotient $\binom{n}{20}/\binom{n}{a+c}$ which are not cancelled are all at least 21,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{20}} \leq \frac{20!}{a!\,c!\,21^{20-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{20}40^{20}3^{n-20}\sum_{\substack{a,c\geq 0\\ a+2c\leq 20}}\frac{20!}{a!\,c!\,21^{20-a-c}}\frac{3^{20-a-c}210^c}{40^{20-a}}.$$

The finite rational sum is

$$\frac{117722605237864079789280365964598334253029}{1092809866779230061199360000000000000000} < 108.$$

This gives the displayed bound. $\square$

Lemma 4.599 (Stable moments dominate the tenth B nonboundary width count above the first row)**.** *For every $q \ge 20$,*

$$216\binom{2q+10}{20}40^{20}3^{2q-10} \le s_{2q+10}.$$

Proof. By lemma 4.65, s_{2q+10} contains all labelled simple cycles on $2q+10$ vertices, whose number is $(2q+9)!/2$. It is enough to prove

$$432\binom{2q+10}{20}40^{20}3^{2q-10} \le (2q+9)!.$$

At $q = 20$,

$$\begin{aligned} 432\binom{50}{20}40^{20}3^{30} &= 46090497469859536932866335038343828262092800000000000000000000, \\ 49! &= 608281864034267560872252163321295376887552831379210240000000000. \end{aligned}$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+12)(2q+11)}{(2q-8)(2q-9)} < 25,$$

because $64q^2 - 1264q + 612 > 0$ for $q \ge 20$. The right side is multiplied by

$$(2q+10)(2q+11) \ge 50 \cdot 51 > 25.$$

Induction proves the result. □

Lemma 4.600 (Exact first five tenth B nonboundary bad-shape counts)**.** *For $q = 15, 16, 17, 18, 19$, the number of column-even Pieri two-strip paths of length $2q+10$ with terminal first row at least $2q$ is at most $s_{2q+10}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_tenth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $q = 15, 16, 17, 18, 19$, $j = 2q+10$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`abf7ac5edc1f162794cea23d37343d9c`
`7dda9415a30f8eab1a622884ebe7b49c`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	40	8794106710689188840998694148435610	56514900379386856861246772027348380714080425836
16	42	229375444845227744316056074438167665	94986302333375273316503064399710821112187239995326
17	44	5638833915013514621203873625181397292	175604143934124940100093706905280814409929173685827432
18	46	131485817886217717873410741865804465230	355551352238135109999385991741420958490914665526325828260
19	48	29238044981357953850200647177727104507466	785317984503258803849482363853038965238893379037105540047660.

Each margin is positive, proving the claimed half-stable bound. □

Proposition 4.601 (Residual B tenth nonboundary lower correction)**.** *For every residual row B_b with $14 \le b \le 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+10} \le \delta_{2q+10}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+10}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+10$ with terminal first row at least $2q$. If $q = 15, 16, 17, 18, 19$, this is bounded by $s_{2q+10}/2$ by lemma 4.600. If $q \geq 20$, lemma 4.598 bounds this number by

$$108\binom{2q+10}{20}40^{20}3^{2q-10},$$

and lemma 4.599 absorbs that count into $s_{2q+10}/2$. In all cases $|\delta_{2q+10}^{B_b}| \leq s_{2q+10}/2$, which proves the stated lower bound. □

Lemma 4.602 (Eleventh B nonboundary width-bad paths have a lower-box code). *For every $q \geq 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+11)}$$

with

$$\lambda_1^{(2q+11)} \geq 2q$$

is at most

$$293\binom{2q+11}{22}44^{22}3^{2q-11}.$$

Proof. Put $n = 2q+11$. The terminal partition has size $2n = 4q+22$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+22-2(2q) = 22.$$

Hence the terminal partition has at most twenty-four nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 22 = 44$ choices, and a step with two lower boxes has at most $\binom{22+1}{2} = 253$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{22}$ in

$$(3+44x+253x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \leq 22$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}44^a253^c.$$

Since $n \geq 41$, $a+c \leq 22$, and the twenty-two factors in the quotient $\binom{n}{22}/\binom{n}{a+c}$ which are not cancelled are all at least 20,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{22}} \leq \frac{22!}{a!\,c!\,20^{22-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{22}44^{22}3^{n-22}\sum_{\substack{a,c\geq 0\\ a+2c\leq 22}}\frac{22!}{a!\,c!\,20^{22-a-c}}\frac{3^{22-a-c}253^c}{44^{22-a}}.$$

The finite rational sum is

$$\frac{443599768741143824547304978674911318281273800637175688599}{151489831324589016485763696059796684800000000000000000000} < 293.$$

This gives the displayed bound. □

Lemma 4.603 (Stable moments dominate the eleventh B nonboundary width count above the first row)**.** *For every $q \geq 22$,*

$$586\binom{2q+11}{22}44^{22}3^{2q-11} \leq s_{2q+11}.$$

Proof. By lemma 4.65, s_{2q+11} contains all labelled simple cycles on $2q+11$ vertices, whose number is $(2q+10)!/2$. It is enough to prove

$$1172\binom{2q+11}{22}44^{22}3^{2q-11} \leq (2q+10)!.$$

At $q = 22$,

$$1172\binom{55}{22}44^{22}3^{33} = 1213713917193990010668961581532488681057801025353038020317275341455360\text{0},$$

$$54! = 230843697339241380472092742683027581083278564571807941132288000000000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+13)(2q+12)}{(2q-9)(2q-10)} < 27,$$

because $72q^2 - 1476q + 1026 > 0$ for $q \geq 22$. The right side is multiplied by

$$(2q+11)(2q+12) \geq 55 \cdot 56 > 27.$$

Induction proves the result. ☐

Lemma 4.604 (Exact first seven eleventh B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 21$, the number of column-even Pieri two-strip paths of length $2q+11$ with terminal first row at least $2q$ is at most $s_{2q+11}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_eleventh_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 21$, $j = 2q+11$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`de69669b8df7784d87b33232131fee8f`
`a6eb22c830fdcc8eeb066d851a409c0c`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	41	917406516916434909318039084336626480	22888395088131303643213965183227897184683805921 60
16	43	26369959387638196716572586608961318825	4036896628145289317151350784328152618161508371979726
17	45	710983441612846257292453314350422789848	781434874273251804662995639990785900590763994820815 2976
18	47	18104254770555721644360125274088694749197	16533071956148091606739412756578627122586773576653757319078
19	49	437907984655019292950173937746451968524840	3808778878472691457866558359391036146957103131281890865735 7232
20	51	10111325553336459132478426731966206572544775	952093418618122706949507176425805565113395816466544133326065120 50
21	53	223820420460688222608586493645429584684736700	2574207999615222564214363856678435451035603220262568279765787283228 24.

Each margin is positive, proving the claimed half-stable bound. ☐

Proposition 4.605 (Residual B eleventh nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+11} \leq \delta_{2q+11}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+11}|$ is the number of column-even Pieri two-strip paths of length $2q+11$ with terminal first row at least $2q$. If $15 \le q \le 21$, this is bounded by $s_{2q+11}/2$ by lemma 4.604. If $q \ge 22$, lemma 4.602 bounds this number by

$$293\binom{2q+11}{22}44^{22}3^{2q-11},$$

and lemma 4.603 absorbs that count into $s_{2q+11}/2$. In all cases $|\delta^{B_b}_{2q+11}| \le s_{2q+11}/2$, which proves the stated lower bound. ☐

Lemma 4.606 (Twelfth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+12)}$$

with

$$\lambda_1^{(2q+12)} \ge 2q$$

is at most

$$892\binom{2q+12}{24}48^{24}3^{2q-12}.$$

Proof. Put $n = 2q+12$. The terminal partition has size $2n = 4q+24$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+24-2(2q) = 24.$$

Hence the terminal partition has at most twenty-six nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 24 = 48$ choices, and a step with two lower boxes has at most $\binom{24+1}{2} = 300$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{24}$ in

$$(3+48x+300x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 24$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}48^a300^c.$$

Since $n \ge 42$, $a+c \le 24$, and the twenty-four factors in the quotient $\binom{n}{24}/\binom{n}{a+c}$ which are not cancelled are all at least 19,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{24}} \le \frac{24!}{a!\,c!\,19^{24-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{24}48^{24}3^{n-24}\sum_{\substack{a,c\ge 0\\ a+2c\le 24}}\frac{24!}{a!\,c!\,19^{24-a-c}}\frac{3^{24-a-c}300^c}{48^{24-a}}.$$

The finite rational sum is

$$\frac{4341923919103137819106764321052880725980324389105539729}{4870263973198688168897112743383371412545482895917056} < 892.$$

This gives the displayed bound. ☐

Lemma 4.607 (Stable moments dominate the twelfth B nonboundary width count above the first row)**.** *For every $q \geq 23$,*

$$1784\binom{2q+12}{24}48^{24}3^{2q-12} \leq s_{2q+12}.$$

Proof. By lemma 4.65, s_{2q+12} contains all labelled simple cycles on $2q+12$ vertices, whose number is $(2q+11)!/2$. It is enough to prove

$$3568\binom{2q+12}{24}48^{24}3^{2q-12} \leq (2q+11)!.$$

At $q = 23$,

$$3568\binom{58}{24}48^{24}3^{34} = 17085925815413297276694044456077086105282622055811413510717226451649062502400,$$
$$57! = 40526919504877216755680601905432322134980384796226602145184481280000000000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+14)(2q+13)}{(2q-10)(2q-11)} < 27,$$

because $8q^2 - 180q + 148 > 0$ for $q \geq 23$. The right side is multiplied by

$$(2q+12)(2q+13) \geq 58 \cdot 59 > 27.$$

Induction proves the result. $\square$

Lemma 4.608 (Exact first eight twelfth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 22$, the number of column-even Pieri two-strip paths of length $2q+12$ with terminal first row at least $2q$ is at most $s_{2q+12}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twelfth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 22$, $j = 2q+12$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`027740d3f1fc0d56fcb27fa4e8d1f32a`
`35bf7f4d0938ae970eeaadef8c4ba875`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	42	93189699623293897062273042455009001300	94986302333189352668146167061074907139426098328056
16	44	2944970066850265258433435574323853045000	175604143934119061437627836401793189946527776342532016
17	46	86893906123913310142005123703778431425750	355551352238134936474545379687236110227488741701071907220
18	48	2411360700038598865341416512503696751025757	785317984503258799032608572772112825326100483465166247011078
19	50	63325607009226634814812586877483266881504446	1885339339071848339887681776626879027529213088016666282409649732
20	52	1582080760407312019816362520545354254202411288	4903266810544563254692169491537040013954573481350200637091863636048
21	54	37772835182389352001906800037722047354320693120	13771977065833964133817259347577277238395987638845438215079109063115904
22	56	865247986062810751415300150963091108347321056935	41656588897766692204622473783877622959545445539284227823934198657668262450.

Each margin is positive, proving the claimed half-stable bound. $\square$

Proposition 4.609 (Residual B twelfth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+12} \leq \delta_{2q+12}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+12}|$ is the number of column-even Pieri two-strip paths of length $2q+12$ with terminal first row at least $2q$. If $15 \le q \le 22$, this is bounded by $s_{2q+12}/2$ by lemma 4.608. If $q \ge 23$, lemma 4.606 bounds this number by

$$892\binom{2q+12}{24}48^{24}3^{2q-12},$$

and lemma 4.607 absorbs that count into $s_{2q+12}/2$. In all cases $|\delta^{B_b}_{2q+12}| \le s_{2q+12}/2$, which proves the stated lower bound. □

Lemma 4.610 (Thirteenth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+13)}$$

with

$$\lambda_1^{(2q+13)} \ge 2q$$

is at most

$$3063\binom{2q+13}{26}52^{26}3^{2q-13}.$$

Proof. Put $n = 2q+13$. The terminal partition has size $2n = 4q+26$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+26-2(2q) = 26.$$

Hence the terminal partition has at most twenty-eight nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2\cdot 26 = 52$ choices, and a step with two lower boxes has at most $\binom{26+1}{2} = 351$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0,\dots,x^{26}$ in

$$(3+52x+351x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 26$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}52^a351^c.$$

Since $n \ge 43$, $a+c \le 26$, and the twenty-six factors in the quotient $\binom{n}{26}/\binom{n}{a+c}$ which are not cancelled are all at least 18,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{26}} \le \frac{26!}{a!\,c!\,18^{26-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{26}52^{26}3^{n-26}\sum_{\substack{a,c\ge 0\\ a+2c\le 26}}\frac{26!}{a!\,c!\,18^{26-a-c}}\frac{3^{26-a-c}351^c}{52^{26-a}}.$$

The exact replay

`certificates/post_m29/bc_thirteenth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{257830552209451876888921936028191299953704615582650 6667}{841841424745431191739613263078552619810374977323008} < 3063.$$

This gives the displayed bound. □

Lemma 4.611 (Stable moments dominate the thirteenth B nonboundary width count above the first row)**.** *For every $q \geq 25$,*

$$6126\binom{2q+13}{26}52^{26}3^{2q-13} \leq s_{2q+13}.$$

Proof. By lemma 4.65, s_{2q+13} contains all labelled simple cycles on $2q+13$ vertices, whose number is $(2q+12)!/2$. It is enough to prove

$$12252\binom{2q+13}{26}52^{26}3^{2q-13} \leq (2q+12)!.$$

At $q = 25$,

$$12252\binom{63}{26}52^{26}3^{37} = 81406928799276083967394213974118443693122573324810575066474741340483607910274197094\,4,$$

$$62! = 31469973260387937525653122354950764088012280797258232192163168247821107200000000000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+15)(2q+14)}{(2q-11)(2q-12)} < 27,$$

because $8q^2 - 196q + 186 > 0$ for $q \geq 25$. The right side is multiplied by

$$(2q+13)(2q+14) \geq 63 \cdot 64 > 27.$$

Induction proves the result. ∎

Lemma 4.612 (Exact first ten thirteenth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 24$, the number of column-even Pieri two-strip paths of length $2q+13$ with terminal first row at least $2q$ is at most $s_{2q+13}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_thirteenth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 24$, $j = 2q+13$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`6f6ffcfb724ca85fa51310916cd2451e`
`4b38d08bcf0dd224719ce3d8b809e3e9`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	43	9261959617731322839939927285357323722740	4036896628126818137834663415041705908764011647171896
16	45	321085018576437107884314070871667147756360	7814348742731877298559686751384604962672525314758219952
17	47	10346577483470970761767098836420684066750148	16533071956148070949792955355748546877109351283463013317176
18	49	312316199642079406316926021804798245675155352	38087788784726913954909000279061587421619335578715321244096208
19	51	8889079614417722669504600448536456057226101910	95209341861812270677192781064851784090595337603045433631299397780
20	53	239889630231558154001173136014867176902705974938	257420799961522256420957054048221350172003192927214384481942685846348
21	55	6168070230594284508518829816148238282986612855740	750570909947066995055412314636054651557333090807838685620396595934109448
22	57	151734536869315297557940713054485939491583508893664	2353592104550586210415883410043731104479495613189519840996942501170379956800
23	59	3584278928823864849821504396330622763576445058140532	7916863333164200256454983208040696668961580499915095656840060749633498643359384
24	61	81563273739758344575205491568003548506654429986144278	28498624571845034398726355851767445640622747933084436782816994579034437113257216724.

Each margin is positive, proving the claimed half-stable bound. ∎

Proposition 4.613 (Residual B thirteenth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+13} \leq \delta_{2q+13}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+13}|$ is the number of column-even Pieri two-strip paths of length $2q+13$ with terminal first row at least $2q$. If $15 \le q \le 24$, this is bounded by $s_{2q+13}/2$ by lemma 4.612. If $q \ge 25$, lemma 4.610 bounds this number by

$$3063\binom{2q+13}{26}52^{26}3^{2q-13},$$

and lemma 4.611 absorbs that count into $s_{2q+13}/2$. In all cases $|\delta^{B_b}_{2q+13}| \le s_{2q+13}/2$, which proves the stated lower bound. □

Lemma 4.614 (Fourteenth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+14)}$$

with

$$\lambda_1^{(2q+14)} \ge 2q$$

is at most

$$11979\binom{2q+14}{28}56^{28}3^{2q-14}.$$

Proof. Put $n = 2q+14$. The terminal partition has size $2n = 4q+28$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+28-2(2q) = 28.$$

Hence the terminal partition has at most thirty nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 28 = 56$ choices, and a step with two lower boxes has at most $\binom{28+1}{2} = 406$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{28}$ in

$$(3+56x+406x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 28$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}56^a406^c.$$

Since $n \ge 44$, $a+c \le 28$, and the twenty-eight factors in the quotient $\binom{n}{28}/\binom{n}{a+c}$ which are not cancelled are all at least 17,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{28}} \le \frac{28!}{a!\,c!\,17^{28-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{28}56^{28}3^{n-28}\sum_{\substack{a,c\ge 0\\ a+2c\le 28}}\frac{28!}{a!\,c!\,17^{28-a-c}}\frac{3^{28-a-c}406^c}{56^{28-a}}.$$

The exact replay

`certificates/post_m29/bc_fourteenth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{2206251862522460035843841746796935620083943371000444640299878209563705578207}{184180472053899552005790127718892583988774500112337828541287665105895424} < 11979.$$

This gives the displayed bound. □

Lemma 4.615 (Stable moments dominate the fourteenth B nonboundary width count above the first row)**.** *For every $q \geq 26$,*

$$23958\binom{2q+14}{28}56^{28}3^{2q-14} \leq s_{2q+14}.$$

Proof. By lemma 4.65, s_{2q+14} contains all labelled simple cycles on $2q+14$ vertices, whose number is $(2q+13)!/2$. It is enough to prove

$$47916\binom{2q+14}{28}56^{28}3^{2q-14} \leq (2q+13)!.$$

At $q = 26$,

$$47916\binom{66}{28}56^{28}3^{38} = 1965920218394549173531718720316381901912412245557438940932955044811503832120275636615905280,$$
$$65! = 8247650592082470666723170306785496252186258551345437492922123134388955774976000000000000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+16)(2q+15)}{(2q-12)(2q-13)} < 27,$$

because $8q^2 - 212q + 228 > 0$ for $q \geq 26$. The right side is multiplied by

$$(2q+14)(2q+15) \geq 66 \cdot 67 > 27.$$

Induction proves the result. ☐

Lemma 4.616 (Exact first eleven fourteenth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 25$, the number of column-even Pieri two-strip paths of length $2q+14$ with terminal first row at least $2q$ is at most $s_{2q+14}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_fourteenth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 25$, $j = 2q+14$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`315dcc3f51bd178dba5b186d4cb4f3e0`
`0c636284f3b43d8428915d46a90af82b`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	44	9042414995108941187958696634815204057775720	1756041439323164683787397486947183174907133832370705 76
16	46	343165455790745769647624257433029445348921 40	3555513522380664771711994783599268693862495433688649744 40
17	48	12053445324275553213495248165138862344157370 08	785317984503256393166265117738667856959300480700090917588576
18	50	3950525599616764446178244672137901114796234 3549	1885339339071848261003820998310043373593944819013610520247971526
19	52	1216565694103918841752398350801977668931714967796	49032668105445632522622022648500169544894095047873360077368385230 32
20	54	354092053904441393447764018464634103418567831541 08	1377197706583396413374651648246675373841043864875258683849010413819 3928
21	56	979048556886031667100003765498860377735567368606600	4165658889776669220462051741725982302183274836235353202936094421757316 3120
22	58	2582947847161549773036814241928641197338296105838 5799	1353312643542805965252603433729529084652925720728084150837287612560121718928 50
23	60	652716666090191470295272747661134251183413102876318760	471052485162704317319812794664974957053085989881678780643648524177163566557148336
24	62	1585271687961012294605026680622073418053540056839279 2238	1752662440577781575209503668206421439574564463157839529802025182010505987910969154980
25	64	37115439532714123939107531016989430265628603460564643186 4	6955857144727139970751971374392861008233250096453707124089936965869531818002462773466128.

Each margin is positive, proving the claimed half-stable bound. ☐

Proposition 4.617 (Residual B fourteenth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+14} \leq \delta_{2q+14}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+14}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+14$ with terminal first row at least $2q$. If $15 \le q \le 25$, this is bounded by $s_{2q+14}/2$ by lemma 4.616. If $q \ge 26$, lemma 4.614 bounds this number by

$$11979\binom{2q+14}{28}56^{28}3^{2q-14},$$

and lemma 4.615 absorbs that count into $s_{2q+14}/2$. In all cases $|\delta_{2q+14}^{B_b}| \le s_{2q+14}/2$, which proves the stated lower bound. □

Lemma 4.618 (Fifteenth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+15)}$$

with

$$\lambda_1^{(2q+15)} \ge 2q$$

is at most

$$53918\binom{2q+15}{30}60^{30}3^{2q-15}.$$

Proof. Put $n = 2q+15$. The terminal partition has size $2n = 4q+30$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+30-2(2q) = 30.$$

Hence the terminal partition has at most thirty-two nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 30 = 60$ choices, and a step with two lower boxes has at most $\binom{30+1}{2} = 465$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{30}$ in

$$(3+60x+465x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 30$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}60^a465^c.$$

Since $n \ge 45$, $a+c \le 30$, and the thirty factors in the quotient $\binom{n}{30}/\binom{n}{a+c}$ which are not cancelled are all at least 16,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{30}} \le \frac{30!}{a!\,c!\,16^{30-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{30}60^{30}3^{n-30}\sum_{\substack{a,c\ge 0\\ a+2c\le 30}}\frac{30!}{a!\,c!\,16^{30-a-c}}\,\frac{3^{30-a-c}465^c}{60^{30-a}}.$$

The exact replay

`certificates/post_m29/bc_fifteenth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{14677725404098265068620724855748750166905094501498612189320160692921}{272225893536750770770699685945414569164800000000000000000000000} < 53918.$$

This gives the displayed bound. □

Lemma 4.619 (Stable moments dominate the fifteenth B nonboundary width count above the first row)**.** *For every $q \geq 28$,*

$$107836\binom{2q+15}{30}60^{30}3^{2q-15} \leq s_{2q+15}.$$

Proof. By lemma 4.65, s_{2q+15} contains all labelled simple cycles on $2q+15$ vertices, whose number is $(2q+14)!/2$. It is enough to prove

$$215672\binom{2q+15}{30}60^{30}3^{2q-15} \leq (2q+14)!.$$

At $q = 28$,

$$\begin{aligned} 215672\binom{71}{30}60^{30}3^{41} &= 1666775489268343282157788991449822823533337788585868663641529521274880000000000000000000000000000000, \\ 70! &= 11978571669969891796072783721689098736458938142546425857555362864628009582789845319680000000000000000. \end{aligned}$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+17)(2q+16)}{(2q-13)(2q-14)} < 27,$$

because $8q^2 - 228q + 274 > 0$ for $q \geq 28$. The right side is multiplied by

$$(2q+15)(2q+16) \geq 71 \cdot 72 > 27.$$

Induction proves the result. ∎

Lemma 4.620 (Exact first thirteen fifteenth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 27$, the number of column-even Pieri two-strip paths of length $2q+15$ with terminal first row at least $2q$ is at most $s_{2q+15}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_fifteenth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 27$, $j = 2q+15$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`5f829fb541bf8d84f097ff14aa6be272`
`508554407503bd74b37532e62c1484b9`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	45	870039209965529065972576824962654844710807770	7814348742558511626603733812405858225821737680111571132
16	47	360753612523022666673655079245242300045090413	165330719561408765706974618443565973097221192788302450009246
17	49	137866801862434269213014351322071153961572141304	380877887847266388459376746946819740267687350460038894501243304
18	51	4897938044505836438911625788422682007179764173625	95209341861812260899094851282014351606352961654754331386223254350
19	53	162938890183785316560360745867975853262168456585696	257420799961522256095559052941113833359284047463292412311411184624832
20	55	5107253627350791334756002476104885485190100308831416	750570909947066995045210143521814257456838123515261211126582368542158096
21	57	151642695284223916189224144734975201369045203750659540	2353592104550586210415580428122236395277712280781475999566083393929896425048
22	59	4284903864622671457145546904202741371329180364475710597	7916863333164200256454974645401525281266395908464296044018563618425659808219254
23	61	115692900449087199221605049442686667091810004951400197760	28498624571845034398726355620544771289927866179024748880579667492427736070429109760
24	63	2995482956282684852310456475723829592541709520358042539396	109541225789834175931658495490412553634851132915530955435886810144674617471147133041144
25	65	74609001886847997188501399507450228099031611764174645505536	448652128232881537732129952277643155760149930451547351191388653100147393271097763359032320
26	67	1792659116038401166877320912707051090756263525315000428518861	1954211164681443257926522972458387998748966177409228472342412732878018653927895874991504704870
27	69	41655688006298221943407882203261078530441457103239477150150316	9035761317088612444238991255468019372218605875608789660379001392822365211344779616325486454823080.

Each margin is positive, proving the claimed half-stable bound. ∎

Proposition 4.621 (Residual B fifteenth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+15} \leq \delta_{2q+15}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+15}|$ is the number of column-even Pieri two-strip paths of length $2q+15$ with terminal first row at least $2q$. If $15 \le q \le 27$, this is bounded by $s_{2q+15}/2$ by lemma 4.620. If $q \ge 28$, lemma 4.618 bounds this number by

$$53918\binom{2q+15}{30}60^{30}3^{2q-15},$$

and lemma 4.619 absorbs that count into $s_{2q+15}/2$. In all cases $|\delta^{B_b}_{2q+15}| \le s_{2q+15}/2$, which proves the stated lower bound. ☐

Lemma 4.622 (Sixteenth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+16)}$$

with

$$\lambda_1^{(2q+16)} \ge 2q$$

is at most

$$282957\binom{2q+16}{32}64^{32}3^{2q-16}.$$

Proof. Put $n = 2q+16$. The terminal partition has size $2n = 4q+32$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+32-2(2q) = 32.$$

Hence the terminal partition has at most thirty-four nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 32 = 64$ choices, and a step with two lower boxes has at most $\binom{32+1}{2} = 528$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{32}$ in

$$(3+64x+528x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 32$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}64^a528^c.$$

Since $n \ge 46$, $a+c \le 32$, and the thirty-two factors in the quotient $\binom{n}{32}/\binom{n}{a+c}$ which are not cancelled are all at least 15,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{32}} \le \frac{32!}{a!\,c!\,15^{32-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{32}64^{32}3^{n-32}\sum_{\substack{a,c\ge 0\\ a+2c\le 32}}\frac{32!}{a!\,c!\,15^{32-a-c}}\frac{3^{32-a-c}528^c}{64^{32-a}}.$$

The exact replay

`certificates/post_m29/bc_sixteenth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{246489541168394136468483634206198590453034992597677288160669567006157511}{8711228593176024664662389950253266213273600000000000000000000000000} < 282957.$$

This gives the displayed bound. ☐

Lemma 4.623 (Stable moments dominate the sixteenth B nonboundary width count above the first row)**.** *For every $q \geq 29$,*

$$565914\binom{2q+16}{32}64^{32}3^{2q-16} \leq s_{2q+16}.$$

Proof. By lemma 4.65, s_{2q+16} contains all labelled simple cycles on $2q+16$ vertices, whose number is $(2q+15)!/2$. It is enough to prove

$$1131828\binom{2q+16}{32}64^{32}3^{2q-16} \leq (2q+15)!.$$

At $q = 29$,

$$1131828\binom{74}{32}64^{32}3^{42} = 695557115329569709224150063682778061838037342308385804626953595512549529256644715070119059644382734974976,$$
$$73! = 4470115461512684340891257138125051110076800700282905015819080092370422104067183317016903680000000000000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+18)(2q+17)}{(2q-14)(2q-15)} < 30,$$

because $84q^2 - 2370q + 3546 > 0$ for $q \geq 29$. The right side is multiplied by

$$(2q+16)(2q+17) \geq 74 \cdot 75 > 30.$$

Induction proves the result. $\square$

Lemma 4.624 (Exact first fourteen sixteenth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 28$, the number of column-even Pieri two-strip paths of length $2q+16$ with terminal first row at least $2q$ is at most $s_{2q+16}/2$.*

Proof. The exact C++/GMP/OpenMP replay

```
certificates/post_m29/bc_sixteenth_frontier_gmp_certificate.log
```

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 28$, $j = 2q+16$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`9276304d5e5aff2105d4267604664e9f`
`14dbcc42df60e0a643c335b028250009`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	46	8272983354280527593468052465966387975898819215	355551352221589143553796572326920289305803373306137120290
16	48	374098023898022085704780517499191373641504735747	785317984502510607807533928677900994973935125725276739591098
17	50	15528496562219269103802590570633462837432040098832	1885339339071817283021208552107124691977696994845957952092460960
18	52	597029860046538795531886347547448881528802399385354	4903266810544562060635613559980263574221511111493528287995469687916
19	54	21423293633682332357729677671046428807332880396861126	13771977065833964090970747625882977301640636110352656044508056910779892
20	56	722136871516149082858615658006670079745042801148811140	41656588897766692203178201771341296919449717053871189590626329750012754040
21	58	22994558956786999982315266809317628071410159794070017603	135331264354280596525214405913996277696323402275474618400409887702346148629242
22	60	695043507649755527397768535849816623690533710220113245909	471052485162704317319811405883392989722414134935152576332669645476569332083294038
23	62	20027377511752658955477085619459584786840264485171632719887	1752662440577781575209503628183371849828466798095768824495297076691047818704489299682
24	64	552183555309122185320016983329451126500219486553090833065783	6955857144727139970751971373289236206405660008292455308051374501474405416965492400198290
25	66	14615891637018864121415359376808915816497044412622226036971340	29386673299901708964749475366730337406328925449922948207233125442364428752735436190800851560
26	68	372499764620854488855056835541977547662825473236779430400948538	131909085289227211701789514679798183330284404718062127535641201435022833027609889543630800130956
27	70	9164865545096087327418774390016633212464178987982649947882888584	627984678371728270376818250246812063500373154593071883732187585120608215213669982361566356229572336
28	72	218197243528751072881974686848428246042581074443937961025749863594	3165506839235730069039027353629446009170081519012503274768331305040780096639467941388157166991803039916.

Each margin is positive, proving the claimed half-stable bound. $\square$

Proposition 4.625 (Residual B sixteenth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+16} \leq \delta_{2q+16}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+16}|$ is the number of column-even Pieri two-strip paths of length $2q+16$ with terminal first row at least $2q$. If $15 \le q \le 28$, this is bounded by $s_{2q+16}/2$ by lemma 4.624. If $q \ge 29$, lemma 4.622 bounds this number by

$$282957\binom{2q+16}{32}64^{32}3^{2q-16},$$

and lemma 4.623 absorbs that count into $s_{2q+16}/2$. In all cases $|\delta^{B_b}_{2q+16}| \le s_{2q+16}/2$, which proves the stated lower bound. □

Lemma 4.626 (Seventeenth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+17)}$$

with

$$\lambda_1^{(2q+17)} \ge 2q$$

is at most

$$1759061\binom{2q+17}{34}68^{34}3^{2q-17}.$$

Proof. Put $n = 2q+17$. The terminal partition has size $2n = 4q+34$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+34-2(2q) = 34.$$

Hence the terminal partition has at most thirty-six nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 34 = 68$ choices, and a step with two lower boxes has at most $\binom{34+1}{2} = 595$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{34}$ in

$$(3+68x+595x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 34$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}68^a595^c.$$

Since $n \ge 47$, $a+c \le 34$, and the thirty-four factors in the quotient $\binom{n}{34}/\binom{n}{a+c}$ which are not cancelled are all at least 14,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{34}} \le \frac{34!}{a!\,c!\,14^{34-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{34}68^{34}3^{n-34}\sum_{\substack{a,c\ge 0\\ a+2c\le 34}}\frac{34!}{a!\,c!\,14^{34-a-c}}\frac{3^{34-a-c}595^c}{68^{34-a}}.$$

The exact replay

`certificates/post_m29/bc_seventeenth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{110837640989552376651072260993779108203216051363825886336192281558360154094119521897428034229}{63009550437898855669184790546695665525474533007607651696497530576675910920262370983936} < 1759061.$$

This gives the displayed bound. □

Lemma 4.627 (Stable moments dominate the seventeenth B nonboundary width count above the first row)**.** *For every $q \geq 31$,*

$$3518122\binom{2q+17}{34}68^{34}3^{2q-17} \leq s_{2q+17}.$$

Proof. By lemma 4.65, s_{2q+17} contains all labelled simple cycles on $2q+17$ vertices, whose number is $(2q+16)!/2$. It is enough to prove

$$7036244\binom{2q+17}{34}68^{34}3^{2q-17} \leq (2q+16)!.$$

At $q = 31$,

$$7036244\binom{79}{34}68^{34}3^{45} = 106355916925065248010489952553960701708665860237767524896895162233289562428870769777751176444776643393303389143040,$$
$$78! = 11324281178206297831457521158732046228731749579488251990048962825668835325234200766245086213177344000000000000000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+19)(2q+18)}{(2q-15)(2q-16)} < 30,$$

because $84q^2 - 2526q + 4122 > 0$ for $q \geq 31$. The right side is multiplied by

$$(2q+17)(2q+18) \geq 79 \cdot 80 > 30.$$

Induction proves the result. □

Lemma 4.628 (Exact first sixteen seventeenth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 30$, the number of column-even Pieri two-strip paths of length $2q+17$ with terminal first row at least $2q$ is at most $s_{2q+17}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_seventeenth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 30$, $j = 2q+17$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`a1241f73bdcbce52caf77e4ec67c6ef4`
`44661548ce303dfae09cf932a1a100a1`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	47	779230830328282566194944718451483846088992348189	1653307195458962998229135716530018097440405643265316212109 4
16	49	38360387184944232391876469096785250109518667246360	380877887846501938051715110984366471172778086880927752599 14192
17	51	1726692115725507500481070313173742205607315188935400	952093418618088173107394892786862672889781910157071311153737 30800
18	53	71735386427900995474187612996166772277800826683386414	2574207999615221129506639775066935181047795468656995632340947 31023396
19	55	2772571726861316706277840896757520378655743571097478742	7505709099470669895102811970538824275706683349524302247854754 26964863444
20	57	100370231018083149740520465136960860287328877535131642104	2353592104550586210215143251476638543629049798797024229394163 7292671344599 20
21	59	3423218908643343883538262352580350165137385093654041899679	7916863333164200256448136777391967838842233674852943749171031 506599080675841090
22	61	110553640659569321617408193271531514298313876556448290129830	2849862457184503439872613474464925304945902980584830470292440 50482946330766492456 20
23	63	3395819242224205352880062612877433058696841923927433331308 2655	1095412257898341759316584887047650350633094450188996108386732 95392919557963196591954626
24	65	99596157540943834973370831784468736492783160020751304567686720	4486521282328815377321299520786000586820359564991826904214660 8057077913645312350351466 9952
25	67	2798759702591853244197525420893125257124671570527511663162590028	1954211164681443257926522972452794064662014547723167176142451 896465950823313891481666036562536
26	69	75586918786910147053521816336294231373415979198115500561207963742	9035761317088612444238991255467868281692408067911126503562093 2108817754402692975918033183391962 28
27	71	1967329436741387607092007799580668990476115949062045477478124752560	4427287033724369808121418522720788874341902242824510865624649 08361702250635572839094934763173859227 20
28	73	49468245069462752490490464045768006352591715268463435477641465370248	2294990102513359737735742433500482291843936443540372583892579 05108061108935923733607882120367436505963248
29	75	1204367164349240206919084144787606267144533904932810904500278351782883	1256676780468867042847317084437511663920163009273552752219055 92325869356615825643511250874852907061053718380 2
30	77	28447654093505078593061960117489559777962687009967279148071918716382106	7258238000562610837751823684753927597930978364363379670268537 7141814946631427768455624885108999553522637910253 56.

Each margin is positive, proving the claimed half-stable bound. □

Proposition 4.629 (Residual B seventeenth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+17} \leq \delta_{2q+17}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+17}|$ is the number of column-even Pieri two-strip paths of length $2q+17$ with terminal first row at least $2q$. If $15 \le q \le 30$, this is bounded by $s_{2q+17}/2$ by lemma 4.628. If $q \ge 31$, lemma 4.626 bounds this number by

$$1759061\binom{2q+17}{34}68^{34}3^{2q-17},$$

and lemma 4.627 absorbs that count into $s_{2q+17}/2$. In all cases $|\delta^{B_b}_{2q+17}| \le s_{2q+17}/2$, which proves the stated lower bound. ☐

Lemma 4.630 (Eighteenth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+18)}$$

with

$$\lambda_1^{(2q+18)} \ge 2q$$

is at most

$$13210035\binom{2q+18}{36}72^{36}3^{2q-18}.$$

Proof. Put $n = 2q+18$. The terminal partition has size $2n = 4q+36$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+36-2(2q) = 36.$$

Hence the terminal partition has at most thirty-eight nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 36 = 72$ choices, and a step with two lower boxes has at most $\binom{36+1}{2} = 666$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{36}$ in

$$(3+72x+666x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 36$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}72^a666^c.$$

Since $n \ge 48$, $a+c \le 36$, and the thirty-six factors in the quotient $\binom{n}{36}/\binom{n}{a+c}$ which are not cancelled are all at least 13,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{36}} \le \frac{36!}{a!\,c!\,13^{36-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{36}72^{36}3^{n-36}\sum_{\substack{a,c\ge 0\\ a+2c\le 36}}\frac{36!}{a!\,c!\,13^{36-a-c}}\frac{3^{36-a-c}666^c}{72^{36-a}}.$$

The exact replay

`certificates/post_m29/bc_eighteenth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{217020849742425673414891610438834185052553081173308557629058562834339849372 73}{1642848430676142439453686466493927838235149844032662468884099699310592} < 13210035.$$

This gives the displayed bound. ☐

Lemma 4.631 (Stable moments dominate the eighteenth B nonboundary width count above the first row)**.** *For every $q \geq 32$,*

$$26420070\binom{2q+18}{36}72^{36}3^{2q-18} \leq s_{2q+18}.$$

Proof. By lemma 4.65, s_{2q+18} contains all labelled simple cycles on $2q+18$ vertices, whose number is $(2q+17)!/2$. It is enough to prove

$$52840140\binom{2q+18}{36}72^{36}3^{2q-18} \leq (2q+17)!.$$

At $q = 32$,

$$52840140\binom{82}{36}72^{36}3^{46} = 795121139679439680748590041906633953130656619006560341265646903093524728495031336078488722393875949486369900038245580800,$$
$$81! = 5797126020747367985879734231578109105412357244731625958745865049716390179693892056256184534249745940480000000000000000000.$$

The first integer is smaller than the second. If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+20)(2q+19)}{(2q-16)(2q-17)} < 30,$$

because $84q^2 - 2682q + 4740 > 0$ for $q \geq 32$. The right side is multiplied by

$$(2q+18)(2q+19) \geq 82 \cdot 83 > 30.$$

Induction proves the result. $\square$

Lemma 4.632 (Exact first seventeen eighteenth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 31$, the number of column-even Pieri two-strip paths of length $2q+18$ with terminal first row at least $2q$ is at most $s_{2q+18}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_eighteenth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 31$, $j = 2q+18$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`2c1768d73f6e84f175648fe543b44e67`
`f791c9221e749ff42253c8a69a6ed7a0`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	48	7284720980501933258600250850309812097642820791334	785317984357564384245291307677305539002737266519703333235924
16	50	3897671958418006847914295217951096754733373051015315	1885339339064052996097496976949503706722936068262166070070627994
17	52	189956451904271257973367176544772686298983735393295632	4903266810544183341791525110541907903641116661018693378129481867360
18	54	8515222682240817384288986784949072823253006669037635654	13771977065833947103371970411612873439126421554299867153160479629230836
19	56	354009388401361339571139085869943006414905915046298093568	41656588897766691496603698711650915942888776629998516920304585259714189184
20	58	13745840672301965992126714036868470336812979278255456156804	135331264354280596497768713687305919712034604735356312982926749465423376350840
21	60	501537005858674333330084220305394292725502806422776823786938	471052485162704317318809721958690940566808762031613487380466020931144218662211980
22	62	17286220248116620360011696433092366670786327505855441698376089	1752662440577781575209469095797630640093064685657073878681125077716565078164357987278
23	64	565412260020227317349094931173175026051291206417829480409450806	6955857144727139970751970243569083276569269680742626928360224651892431554412713247428244
24	66	17622233793457998802606163139381904487039919418495500636414785527	29386673299901708964749475331515101602686965572953452647222934299917584004569679370045223186
25	68	525227120153705052445522264916256422275333828462705468427530311688	131909085289227211701789514678748474089506236322148793119479772545567491021630952165636541404656
26	70	15017983315275271514943772179439551173264045729072445515932158299334	627984678371728270376818250246782045863473694242216651025377486051528112050569813435834387678750836
27	72	413137929816968194764470941137967440190873604534173610774411341339804	3165506839235730069039027353629445183330616372133615891590398402802756206977421021928811540220620087496
28	74	10962547901933364527319535364053039700674396423197629467092052579347946	16868160644365457941953337882591367932501288052108501658663011812498906384019434162381687962487460918312492
29	76	281232323362466735177620928843846456166533731604285601669187439420430570	94879010830044980231973613148711343553951153789424079612257058436848957410956389901125932638346783683138095660
30	78	6989805140556529916322470879509894906924056264106263867936760916148947281	562512973617582593017381142149439039365735839082232808233625389602883250292824675325085827160750052651887880339806
31	80	168630546042345400047058712680532671697137285643783300438293211852492794558	3510497287734778986037296352864815062600027653536998711627383460335903484235467656770090691961806799561483596096398980.

Each margin is positive, proving the claimed half-stable bound. $\square$

Proposition 4.633 (Residual B eighteenth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+18} \leq \delta_{2q+18}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+18}|$ is the number of column-even Pieri two-strip paths of length $2q+18$ with terminal first row at least $2q$. If $15 \le q \le 31$, this is bounded by $s_{2q+18}/2$ by lemma 4.632. If $q \ge 32$, lemma 4.630 bounds this number by

$$13210035\binom{2q+18}{36}72^{36}3^{2q-18},$$

and lemma 4.631 absorbs that count into $s_{2q+18}/2$. In all cases $|\delta^{B_b}_{2q+18}| \le s_{2q+18}/2$, which proves the stated lower bound. □

Lemma 4.634 (Nineteenth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+19)}$$

with

$$\lambda_1^{(2q+19)} \ge 2q$$

is at most

$$122788879\binom{2q+19}{38}76^{38}3^{2q-19}.$$

Proof. Put $n = 2q+19$. The terminal partition has size $2n = 4q+38$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+38-2(2q) = 38.$$

Hence the terminal partition has at most forty nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 38 = 76$ choices, and a step with two lower boxes has at most $\binom{38+1}{2} = 741$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{38}$ in

$$(3+76x+741x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 38$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}76^a741^c.$$

Since $n \ge 49$, $a+c \le 38$, and the thirty-eight factors in the quotient $\binom{n}{38}/\binom{n}{a+c}$ which are not cancelled are all at least 12,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{38}} \le \frac{38!}{a!\,c!\,12^{38-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{38}76^{38}3^{n-38}\sum_{\substack{a,c\ge 0\\ a+2c\le 38}}\frac{38!}{a!\,c!\,12^{38-a-c}}\frac{3^{38-a-c}741^c}{76^{38-a}}.$$

The exact replay

`certificates/post_m29/bc_nineteenth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{2212066407882542391457675604818888277620094404170788518714556367694290791259505245563634 91}{18015201621076410377909613578870418405491244462768730345873948186674870834503352 32} < 122788879.$$

This gives the displayed bound. □

Lemma 4.635 (Stable moments dominate the nineteenth B nonboundary width count above the first row)**.** *For every $q \geq 34$,*

$$245577758\binom{2q+19}{38}76^{38}3^{2q-19} \leq s_{2q+19}.$$

Proof. By lemma 4.65, s_{2q+19} contains all labelled simple cycles on $2q+19$ vertices, whose number is $(2q+18)!/2$. It is enough to prove

$$491155516\binom{2q+19}{38}76^{38}3^{2q-19} \leq (2q+18)!.$$

At $q = 34$,

$$491155516\binom{87}{38}76^{38}3^{49} = 2302885141104466213804101148094432815932621465512417825886455449771111171092717096776659883403545939985052017964291967108408934400,$$
$$86! = 24227095383672732381765523203441259715284870552429381750838764496720162249742450276789464634901319465571660595200000000000000000000.$$

The first integer is smaller than the second by

$$2399680686956228576038511308863181643369160840587813996825011895174305107864973318001280475149777352558660857723570803289159106 5600.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+21)(2q+20)}{(2q-17)(2q-18)} < 30,$$

because $84q^2 - 2838q + 5400 > 0$ for $q \geq 34$. The right side is multiplied by

$$(2q+19)(2q+20) \geq 87 \cdot 88 > 30.$$

Induction proves the result. □

Lemma 4.636 (Exact first nineteen nineteenth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 33$, the number of column-even Pieri two-strip paths of length $2q+19$ with terminal first row at least $2q$ is at most $s_{2q+19}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_nineteenth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 33$, $j = 2q+19$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`519d2680d84e75ce37567e5b89e5b1d7`
`3642f0ddf74f02f115aeb728df4168dd`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	49	6770832467644019079564113437221206469804889256556352	38087788771185249644253361404092173181028966248702034081294208
16	51	393122822418047174334190323048832027972488753247225875	95209341861026025050134845945018848783506874444173368239257149850
17	53	20713609885788249141195428390914016422967880019324583232	257420799961480829201665256810402075623223711166398183075709448629760
18	55	1000513078990890186889315546888119042022674618893259218256	750570909947064994029266668996142061495256352229386936747724782641384416
19	57	44683654959956359626290956640694641622888432782519135205912	2353592104550586121048573793600085590528177447681662704191955919299127332304
20	59	1858704376151413331518423923037635366002147196947105461916934	7916863333164200252737574462906427863572462353482833717497011882892177835806580
21	61	72467756181056332993253942006277920932854289443127449327382990	28498624571845034398581420339568459522115756738222291924087293097161491074574739300
22	63	2662742091452354373351765052132387893149373255693957805007608594	109541225789834175931653170012220614838669799090047603611548170583246648596253202902748
23	65	926492488348622948711710561773796924699426806473555334065212980416	448652128232881537732129766979294704039333884103811999231018613670984161783957982224082560
24	67	3065580383564886079043046133623641868554505565519696386236636779095	1954211164681443257926522966327230816937426096125469959736954409871189035415553732519088184402
25	69	96819085800680404234392380763172336679155449656053883968023472031752	9035761317088612444238991255274381283928621079736448785668421125986211372915586054868393811060208
26	71	2928382315052539397305817789315759122543700932218718904280851982211510	44272870337243698081214185227202035913447790832225711204683458479263118613924744595775869569671004820
27	73	85074927202479445497105355919110590965014947669202147642568533027500384	229499010251335973773574243350048059133475729534071769159526994978415191611211825740513706185653381702976
28	75	2380343018946518105665481853782505830247110834040589189899097933395525850	1256676780468867042847317084437511659161885705709215021301930383983257118198323834841195990539875300449697868
29	77	64296161734029197024019023043412600623223069241190599527445629315215210058	7258238000562610837751823684753927597802442936203508286377685792014904441015886632454127246403360237470793369452
30	79	1680283079261180654459758454356345266839516313276107630481313587080286524901	44157232836692700825063754574781471258469541645948190214366928899160825727087749511175294024150102699910285977546806
31	81	42568609323106085115368512864181140363369184496892098629888240297497728680920	282594820649086173955053884185975495344284216876528810508051194103365072761732733894645251911812624014809789051928811600
32	83	1047339729772837773239753504133535275882447037159770705867392734527969102456219	19000094241916970117931209240138935340233911256943486387578560442324438888619246289896495761665556156934238477666017200951498
33	85	25066345850201404826353089982459865488949766200157456123373154639121945830921160	13406572350450220333476929689717057966687138989769754346719747084213925159133552597982651773589939258393987499432673350091612272.

Each margin is positive, proving the claimed half-stable bound. □

Proposition 4.637 (Residual B nineteenth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+19} \leq \delta_{2q+19}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+19}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+19$ with terminal first row at least $2q$. If $15 \le q \le 33$, this is bounded by $s_{2q+19}/2$ by lemma 4.636. If $q \ge 34$, lemma 4.634 bounds this number by

$$122788879\binom{2q+19}{38}76^{38}3^{2q-19},$$

and lemma 4.635 absorbs that count into $s_{2q+19}/2$. In all cases $|\delta_{2q+19}^{B_b}| \le s_{2q+19}/2$, which proves the stated lower bound. □

Lemma 4.638 (Twentieth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+20)}$$

with

$$\lambda_1^{(2q+20)} \ge 2q$$

is at most

$$1456821633\binom{2q+20}{40}80^{40}3^{2q-20}.$$

Proof. Put $n = 2q+20$. The terminal partition has size $2n = 4q+40$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+40-2(2q) = 40.$$

Hence the terminal partition has at most forty-two nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 40 = 80$ choices, and a step with two lower boxes has at most $\binom{40+1}{2} = 820$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{40}$ in

$$(3+80x+820x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 40$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}80^a820^c.$$

Since $n \ge 50$, $a+c \le 40$, and the forty factors in the quotient $\binom{n}{40}/\binom{n}{a+c}$ which are not cancelled are all at least 11,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{40}} \le \frac{40!}{a!\,c!\,11^{40-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{40}80^{40}3^{n-40}\sum_{\substack{a,c\ge 0\\ a+2c\le 40}}\frac{40!}{a!\,c!\,11^{40-a-c}}\frac{3^{40-a-c}820^c}{80^{40-a}}.$$

The exact replay

`certificates/post_m29/bc_twentieth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{122649374612097167149545215376363490924816633787992253399774932728906545934504781958350531813805715044815 51}{8418969889721232717238251192735471087049751188699919746359969710080000000000000000000000000000000} < 1456821633.$$

This gives the displayed bound. □

Lemma 4.639 (Stable moments dominate the twentieth B nonboundary width count above the first row)**.** *For every $q \geq 35$,*

$$2913643266\binom{2q+20}{40}80^{40}3^{2q-20} \leq s_{2q+20}.$$

Proof. By lemma 4.65, s_{2q+20} contains all labelled simple cycles on $2q+20$ vertices, whose number is $(2q+19)!/2$. It is enough to prove

$$5827286532\binom{2q+20}{40}80^{40}3^{2q-20} \leq (2q+19)!.$$

At $q = 35$,

$$5827286532\binom{90}{40}80^{40}3^{50} = 33292348968955683792042241345295163797073207326342395650333540880316834786734306571664830654054400000000000000000000000000000000000000000,$$

$$89! = 16507955160908461081216919262453619309839666236496541854913520707833171034378509739399912570787600662729080382999756800000000000000000000.$$

The first integer is smaller than the second by

$$1317872026401289270201269512792410293013234550386230228988016661980148755570507908223342950538216066272908038299975680000000000000000000.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+22)(2q+21)}{(2q-18)(2q-19)} < 30,$$

because $84q^2 - 2994q + 6102 > 0$ for $q \geq 35$. The right side is multiplied by

$$(2q+20)(2q+21) \geq 90 \cdot 91 > 30.$$

Induction proves the result. $\square$

Lemma 4.640 (Exact first twenty twentieth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 34$, the number of column-even Pieri two-strip paths of length $2q+20$ with terminal first row at least $2q$ is at most $s_{2q+20}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentieth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 34$, $j = 2q+20$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`431f9748488a0ab0453777cc498a2260`
`996f1ddcf8b393545d30eaed4c291232`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	50	626601681181840133819216429897487931989295147764905498	1885339337818644977650652723006899437363862397793042520642847628
16	52	39420612603703867392627355783715290645400771435211691290	4903266810465722029487925918272599926426775625100489802729845076044
17	54	2242446146464775064275562338286456935754106993349472655980	13771977065829479241524405343119090892423418538574005445187118759190184
18	56	116558337698592557180247131336960516811482291550872041365600	41656588897766459087947078329259234590904274448850906785533313608227645120
19	58	5585210170121879526561125922429194385885745060805002368176460	135331264354280585354840054788150798574036187950704481885062586411929552311528
20	60	248599509132885821679577291704863685290633807256444573042201325	471052485162704316822613777704636645874314347062496905384649412031100626225383206
21	62	10345700647858013251402199244799072050890211369889928665516259466	1752662440577781575188812266942410846830980310560341919312686227631796931716722220524
22	64	404836600016733576711103055510191898670645733628688399470753645445	6955857144727139970751161701193569849870482172821468890912935463007587013272732559038966
23	66	14969678123642353715447551104423952372380690664212299390659112081830	29386673299901708964749445427403321904895532283063570078081998512616094416961899324650630580
24	68	525354764503926380135728268608709883616923564834003848150244395360974	131909085289227211701789513629089399321960886155583300431892518156271029010548666802002811306084
25	70	17566197089705483436441868884632067546570010121532303339011809074961848	627984678371728270376818250211679687650693277912362800800472230060734619898963351648842633845425808
26	72	561539692065488545749386866580206855352144207401671492062135109927629398	3165506839235730069039027353628322930222345028978506646799120265026932300309826747291908818823447508308
27	74	17214551458069968350433713444653484281812625806975750382245432515629949524	16868160644365457941953337882591333525323467716038529845875193233636422160116613057276182405806534817109336
28	76	507484204295461464937454166627315706838195136741042158389386084365845856475	94879010830044980231973613148711342539545209845226083207703967039905236646899183882250187062912989830287243850
29	78	14422556961003995248296886797113645544851487860035467470064996179205139251148	562512973617582593017381142149439039336904704770505930796864260950414978992935547717543104748355933815309899732072
30	80	396035212304441030625036839495664726356475522204673041377484731929117410978937	3510497287734778986037296352864815062599235920373481914256933480774337515848098100000252633445652706684049066260030222
31	82	10528981632129397811017258557478103262553686570594685961906226176015171593146138	23031461318491612837555179058823773176080157469237699341298896496597194266834837470553712350530985944750395962058801782220
32	84	271525326374832491253485074328530657045777479745138428600205017352812738426359874	158657763193874483403838398425529131802372689295122009945235536093107998553790399053774178262194660874425019778752931695953788
33	86	6803720890658077113591301726343365291891097789940838630961777707393423990642895000	1146261223216104476113850870504823904206973429562561380739229608352457616031721661870486480423783207958751382467015901495460849360
34	88	165910482327388070198645559912579287410075267024122936747501078593601271337767995276	8675754396856524876368104879696901763819780995644414859957047940170808748356386666889663394705181155539134163048350281405869496609000.

Each margin is positive, proving the claimed half-stable bound. $\square$

Proposition 4.641 (Residual B twentieth nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+20} \leq \delta_{2q+20}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+20}|$ is the number of column-even Pieri two-strip paths of length $2q+20$ with terminal first row at least $2q$. If $15 \le q \le 34$, this is bounded by $s_{2q+20}/2$ by lemma 4.640. If $q \ge 35$, lemma 4.638 bounds this number by

$$1456821633\binom{2q+20}{40}80^{40}3^{2q-20},$$

and lemma 4.639 absorbs that count into $s_{2q+20}/2$. In all cases $|\delta^{B_b}_{2q+20}| \le s_{2q+20}/2$, which proves the stated lower bound. ☐

Lemma 4.642 (Twenty-first B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+21)}$$

with

$$\lambda_1^{(2q+21)} \ge 2q$$

is at most

$$22953167364\binom{2q+21}{42}84^{42}3^{2q-21}.$$

Proof. Put $n = 2q+21$. The terminal partition has size $2n = 4q+42$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+42-2(2q) = 42.$$

Hence the terminal partition has at most forty-four nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 42 = 84$ choices, and a step with two lower boxes has at most $\binom{42+1}{2} = 903$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{42}$ in

$$(3+84x+903x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 42$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}84^a903^c.$$

Since $n \ge 51$, $a+c \le 42$, and the forty-two factors in the quotient $\binom{n}{42}/\binom{n}{a+c}$ which are not cancelled are all at least 10,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{42}} \le \frac{42!}{a!\,c!\,10^{42-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{42}84^{42}3^{n-42}\sum_{\substack{a,c\ge 0\\ a+2c\le 42}}\frac{42!}{a!\,c!\,10^{42-a-c}}\frac{3^{42-a-c}903^c}{84^{42-a}}.$$

The exact replay

`certificates/post_m29/bc_twentyfirst_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{1096457568590064650301360788749759942430223535847866944918020801891074989549823633723676719}{47769336197201542165283531601931133238466576384000000000000000000000000000000000} < 22953167364.$$

This gives the displayed bound. ☐

Lemma 4.643 (Stable moments dominate the twenty-first B nonboundary width count above the first row)**.** *For every $q \geq 37$,*

$$45906334728\binom{2q+21}{42}84^{42}3^{2q-21} \leq s_{2q+21}.$$

Proof. By lemma 4.65, s_{2q+21} contains all labelled simple cycles on $2q+21$ vertices, whose number is $(2q+20)!/2$. It is enough to prove

$$91812669456\binom{2q+21}{42}84^{42}3^{2q-21} \leq (2q+20)!.$$

At $q = 37$,

$$91812669456\binom{95}{42}84^{42}3^{53} = 2020952329702038580056883671765353532240490512901670259202664882974226156991067846672183897991598578931407451396781826335819778114297216850984960,$$
$$94! = 1087366156656743080273652852567866010041868035801828723074973744340451998694179276302291092145834154585608656512023853405306880000000000000000000000.$$

The first integer is smaller than the second by

$$1067156633359722694473084015850212474719463130672812020482947095510709737124268597835569253165918168796294581998056035141948682218857027831490150 40.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+23)(2q+22)}{(2q-19)(2q-20)} < 30,$$

because $84q^2 - 3150q + 6846 > 0$ for $q \geq 37$. The right side is multiplied by

$$(2q+21)(2q+22) \geq 95 \cdot 96 > 30.$$

Induction proves the result. $\square$

Lemma 4.644 (Exact first twenty-two twenty-first B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 36$, the number of column-even Pieri two-strip paths of length $2q+21$ with terminal first row at least $2q$ is at most $s_{2q+21}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyfirst_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 36$, $j = 2q+21$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`a5593e2a97e9cadcf10f028940d40366`
`500494e1c24910ccb6d39bf66334659c`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	51	578115439779792382089129815195113774832073880128079972 50	95209341746189182739012463875861266390581783533703569720135607100
16	53	3935271444298477943782795499450299610411496135409790836638	257420799953651713532839877421118875481104939978421126564928516122948
17	55	241359790403996572016624122823037347552325345755026539476832	7505709099465842754746166576324825918807040537723663314054525160808 67264
18	57	13483281043990464341359276425635930103373985735409424547427888	2353592104550559243853795732584122124557239457210739201997350665488302888352
19	59	692386586161110504840591158478703586672029913275563393651758692	791686333316419887168181089298824484542699324215093110544147972565960145 6123064
20	61	32940040994191698899954268857577120885406919305555183359844313880	2849862457184503433284627386354717438819372690708060599513916306493737925354087 7520
21	63	1461676232296968493641854994839517358525314295674909346240332096712	1095412257898341759287351430318095825601327926304728336702838407384082178193825539265 12
22	65	60851792748979218669089820745968269933777857355269431707076117488512	448652128232881537732008248692294415326585446804432417450535997841634917631211960415066368
23	67	2388995341463768143872970292900698372394645088033353666095569744142086	1954211164681443257926518194467708656530910508270976425587493357690024099747614313852873458420
24	69	88845432633654924642683731090191307380513016866522699304144400735686930	903576131708861244423899107777715418822013260283777136681215103831848853918229521451563928374985 2
25	71	3142436098248435450700340835367906594534784980700914457176590699368430280	442728703372436980812141852209230204815810250096196411695262768084386360543873531192312498748985672 80
26	73	106084616069733522991706312324435603742287840792452964287794263878255227336	22949901025133597377357424334983606005119066744697935074558996199211254595952225821623340279496292624907 2
27	75	3429043748809728020773262400360296509803773781663534433422854159153451438292	1256676780468867042847317084437504805835074124146209686108093370955249171144982175853507524629752860337872984
28	77	106430978497767131004211612045042701677382712222829596545853817344936869354405 0	725823800056261083775182368475392738506907826413730467200249974801632628749690831434457752838190459736383670146 8
29	79	3180209606591683066722938312879631206989185811412007356779413941353748937520548	441572328366927008250637545747814712521124829989233464422299717903102760036430569209778315258522374443769486755555 12
30	81	91695611909720121588087263800545921240894401059068291433938283249932971054409680	282594820649086173955053884185975495344100910789928016477045250312789709281531671830896109113142007224790518105277354080
31	83	2556620978845164037472442365575197853546037465279694388543124435075230258839889258	1900094241916970117931209240138935340233906145796208156796161043919214746490609749586459276596320482420153796261437726085420
32	85	69063341728972379136179074439017761543446116942228451918639682453152822122876000818	134065723504502203334769296897170579666871388516932035804753916215084824610629492420683174215333503333613689024052729960014529 56
33	87	1810751404550169304474456329896888502972674997950825632252450415579385587625360391760	99151536294250092657629161470485213770197136711695005130820868468289390882972150353866185554243407324584724810085001079294885522272
34	89	46153312083789859378846255075109804869964874276638518172691114855747721346017012231552	7678038341547969493397007744904375881226137813591756797430466763602705469153335073124830968691696462631811277025937141781829560637370 88
35	91	1145317127352153227002215852642636270280448741968539720685823104969892401118932253522220	62190124855617201691251757117470727466161015730336794593389263590901992474022027077197592337154617309502421909511104806741070512190623840 40
36	93	27709125359939135035710749040141509792026747440364037302700734332288341785668113871241340	5263611450868428102305746298324716844862795911849820294036162657203204489363201996698206282918389317653371474445647821936939175571695727422234 4.

Each margin is positive, proving the claimed half-stable bound. $\square$

Proposition 4.645 (Residual B twenty-first nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+21} \leq \delta_{2q+21}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+21}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+21$ with terminal first row at least $2q$. If $15 \le q \le 36$, this is bounded by $s_{2q+21}/2$ by lemma 4.644. If $q \ge 37$, lemma 4.642 bounds this number by

$$22953167364\binom{2q+21}{42}84^{42}3^{2q-21},$$

and lemma 4.643 absorbs that count into $s_{2q+21}/2$. In all cases $|\delta_{2q+21}^{B_b}| \le s_{2q+21}/2$, which proves the stated lower bound. ☐

Lemma 4.646 (Twenty-second B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+22)}$$

with

$$\lambda_1^{(2q+22)} \ge 2q$$

is at most

$$505942192849\binom{2q+22}{44}88^{44}3^{2q-22}.$$

Proof. Put $n = 2q+22$. The terminal partition has size $2n = 4q+44$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+44-2(2q) = 44.$$

Hence the terminal partition has at most forty-six nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 44 = 88$ choices, and a step with two lower boxes has at most $\binom{44+1}{2} = 990$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{44}$ in

$$(3+88x+990x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 44$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}88^a990^c.$$

Since $n \ge 52$, $a+c \le 44$, and the forty-four factors in the quotient $\binom{n}{44}/\binom{n}{a+c}$ which are not cancelled are all at least 9,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{44}} \le \frac{44!}{a!\,c!\,9^{44-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{44}88^{44}3^{n-44}\sum_{\substack{a,c\ge 0\\ a+2c\le 44}}\frac{44!}{a!\,c!\,9^{44-a-c}}\frac{3^{44-a-c}990^c}{88^{44-a}}.$$

The exact replay

```
certificates/post_m29/bc_twentysecond_frontier_gmp_certificate.log
```

prints this finite rational sum as

$$\frac{4803627013891462939106053200700607958220666324777824022294070839997596420617544275379521860 41}{949441869406064352643569097126964670279561525103351211617261691143903975323992064} < 505942192849.$$

This gives the displayed bound. ☐

Lemma 4.647 (Stable moments dominate the twenty-second B nonboundary width count above the first row)**.** *For every $q \geq 38$,*

$$1011884385698\binom{2q+22}{44}88^{44}3^{2q-22} \leq s_{2q+22}.$$

Proof. By lemma 4.65, s_{2q+22} contains all labelled simple cycles on $2q+22$ vertices, whose number is $(2q+21)!/2$. It is enough to prove

$$2023768771396\binom{2q+22}{44}88^{44}3^{2q-22} \leq (2q+21)!.$$

At $q = 38$,

$$2023768771396\binom{98}{44}88^{44}3^{54} = 65222480362836809952971880455666389153083387061070787188578773411432174644663894576003463013085469685050442643017163501545982315458846527106223293071360,$$
$$97! = 96192759682482119853328425949563698712343813919172976158104477319333745612481875498805879175589072651261284189679678167647067832320000000000000000000000.$$

The first integer is smaller than the second by

$$30970279319645309900356545493897309559260426858102188969525703907901570967817980922802416162503602966210841546662514666101085516861153472893776706928640.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+24)(2q+23)}{(2q-20)(2q-21)} < 30,$$

because $84q^2 - 3306q + 7632 > 0$ for $q \geq 38$. The right side is multiplied by

$$(2q+22)(2q+23) \geq 98 \cdot 99 > 30.$$

Induction proves the result. ☐

Lemma 4.648 (Exact first twenty-three twenty-second B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 37$, the number of column-even Pieri two-strip paths of length $2q+22$ with terminal first row at least $2q$ is at most $s_{2q+22}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentysecond_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 37$, $j = 2q+22$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`c2db04c00a2fcf719ada1acdb7dadc65`
`e468bbaa1c87e3dc9a5b9acd34d653a4`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	52	5323446653597160265034771100537264465709989137912479705596	4903266799897669947501013122988312436919677274971313069775309047432
16	54	391554862972481386596424143010396044394295401319384620857534	137719770650508544078723721200547937310791993636569228565350484627870 76
17	56	25859547932757013356832635997528874181577973416133615996397600	41656588897714973108756961487659929813171890621521373803284148120317581120
18	58	1550742762553659348667394009433698976116366730276697043739919936	1353312643542774950397352877132125169082691654111410209230921546278468088245 76
19	60	85242895433240203918806916229151814537253279743436583143929952152	471052485162704146834021929490000451619636472168595201459357539670823484449881552
20	62	4329962869966090601536843587787934103162815823735156489170666541261	1752662440577781566549577928305945670260097533474071857088835002901263810706421656934
21	64	204663057519903431651033174009112472574474979975894933913926071072724	695585714472713997034264525935379645372183803091426432956132679452305452224382192418440 8
22	66	9056239950446667193743574468809630675156045150849828616746594261440965	29386673299901708964731362886858675855215476029228159664636431183695723184327187454351912310
23	68	377141522624646355140328840518497494512116732940003787624669230532222948	1319090852892272117017887603967536790371033657693588006543232611566522786709811137640305375821 36
24	70	14850078186731138423985248674917656424793646087051943648719983451992745202	627984678371728270376818220546655708367827396814749188734423515566582466038140660886899348009859100
25	72	555149081642615737288848298400293225447491129358822235748371195596699510096	31655068392357300690390273525191478463212445314923088237316942278426543300069856187792906978499037469 12
26	74	19776026674693781692086274041223555144906782028524551930055135052232191346902	1686816064436545794195333788255181590107699629259105816521963543031517222131117790491683662656710169431458 0
27	76	673505255032373499948674138000900279194668169367082847349408761371139557634200	94879010830044980231973613148709996544003553689149115734336298493978260986950723198639805024162416282863688400
28	78	21993526020951550616792268240249754700707318613095492392253147020571791206369995	562512973617582593017381142149438995378697776789412687708921554045132868668001297246629255182191885030137765494378
29	80	690485903339611044427013997950184870025933019447977651797016217715895573752135316	3510497287734778986037296352864815061219056184118868707464155558552959105248945012148706675934375240716116153577717464
30	82	20891324507642358784525690594672942219409198560531818203698435293279516475388186937	230314613184916128375551790588237731760383958781856788825254671499249633386025441808057899034955128866161889594512117006 22
31	84	610489260918378676975516456004524172821749473928682746517197893039353161784257326908	158657763193874483403838398425529131802371468859650825937546567567166138162506069646381280386978483679048975778054840034019720
32	86	17264609322988471549439305743530068322150090948951837637129588102542524243091881345590	114626122321610447611385087050482390420697339504695017654360266370102873242427174815408677809998961096149873279675426329298394818 0
33	88	473362465274461427761811856572603603536037011412988634669670804610877073925734374493952	86757543968565248763681048796969017638197809946980217503727798607875823263310046183924095224132521320732875559837833360970762836116 48
34	90	12604196146271047822033601895687909401492047354315172216664616565100506719981511047858780	687184059649688758423828544676235495027684950621848454038988267902266782459165873582001823365806331965153791725371448667405310093619781 84
35	92	326428585335249770483542621184451051418483064466405760480428587999234699059605687441469656	5690393544280648095845017548539990210234550727620592988816684784595250960271193272475572291948968705904950147459230984340615679676814055788 32
36	94	8234325934363867839569348497508591288886234966633995120675510643422203724946419246833102165	49214743738562151235521852189485545974675578372577663471242034916839058738693843627263674142902283742059126676360646526158760924920604377045128 54
37	96	202583023176814053897477663683490322926799724426356872929411083825682875168306496912004395540	4441503572860413503297050674203655502578294871702635432273671647106030907704485025183193103131432255404126377340743528432083532821972510437058502140 0.

Each margin is positive, proving the claimed half-stable bound. ☐

Proposition 4.649 (Residual B twenty-second nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+22} \leq \delta_{2q+22}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+22}|$ is the number of column-even Pieri two-strip paths of length $2q+22$ with terminal first row at least $2q$. If $15 \le q \le 37$, this is bounded by $s_{2q+22}/2$ by lemma 4.648. If $q \ge 38$, lemma 4.646 bounds this number by

$$505942192849\binom{2q+22}{44}88^{44}3^{2q-22},$$

and lemma 4.647 absorbs that count into $s_{2q+22}/2$. In all cases $|\delta^{B_b}_{2q+22}| \le s_{2q+22}/2$, which proves the stated lower bound. $\square$

Lemma 4.650 (Twenty-third B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+23)}$$

with

$$\lambda_1^{(2q+23)} \ge 2q$$

is at most

$$16741606354773\binom{2q+23}{46}92^{46}3^{2q-23}.$$

Proof. Put $n = 2q+23$. The terminal partition has size $2n = 4q+46$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q + 46 - 2(2q) = 46.$$

Hence the terminal partition has at most forty-eight nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 46 = 92$ choices, and a step with two lower boxes has at most $\binom{46+1}{2} = 1081$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{46}$ in

$$(3 + 92x + 1081x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a + 2c \le 46$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}92^a1081^c.$$

Since $n \ge 53$, $a+c \le 46$, and the forty-six factors in the quotient $\binom{n}{46}/\binom{n}{a+c}$ which are not cancelled are all at least 8,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{46}} \le \frac{46!}{a!\,c!\,8^{46-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{46}92^{46}3^{n-46}\sum_{\substack{a,c\ge 0\\ a+2c\le 46}}\frac{46!}{a!\,c!\,8^{46-a-c}}\frac{3^{46-a-c}1081^c}{92^{46-a}}.$$

The exact replay

`certificates/post_m29/bc_twentythird_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{5413287532978551318097678891722127088655784535993669973030234234250361370376877380234841755619418027188786488040818356483827177779}{32334337686989351435388800602448850739648892465473535343460139726871239954850817547557293401079306480694122071562 6496} < 16741606354773.$$

This gives the displayed bound. $\square$

Lemma 4.651 (Stable moments dominate the twenty-third B nonboundary width count above the first row)**.** *For every $q \geq 40$,*

$$33483212709546\binom{2q+23}{46}92^{46}3^{2q-23} \leq s_{2q+23}.$$

Proof. By lemma 4.65, s_{2q+23} contains all labelled simple cycles on $2q+23$ vertices, whose number is $(2q+22)!/2$. It is enough to prove

$$66966425419092\binom{2q+23}{46}92^{46}3^{2q-23} \leq (2q+22)!.$$

At $q = 40$,

$$66966425419092\binom{103}{46}92^{46}3^{57} = 100799373482071927756100447624964335717029457547995464555007687269102068139512017998152522370498302975581376576947997835914189117769465601974511263680494378680320,$$
$$102! = 961446671503512660926865558697259548455355905059659464369444714048531715130254590603314961882364451384985595980362059157503710042865532928000000000000000000000000.$$

The first integer is smaller than the second by

$$860647298021440733170765111072295212738326447511663999814437026779429646990742572605162439511866148409404219403414061321589520925096067326025488736319505621319680.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+25)(2q+24)}{(2q-21)(2q-22)} < 30,$$

because $84q^2 - 3498q + 8100 > 0$ for $q \geq 40$. The right side is multiplied by

$$(2q+23)(2q+24) \geq 103 \cdot 104 > 30.$$

Induction proves the result. □

Lemma 4.652 (Exact first twenty-five twenty-third B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 39$, the number of column-even Pieri two-strip paths of length $2q+23$ with terminal first row at least $2q$ is at most $s_{2q+23}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentythird_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 39$, $j = 2q+23$. It exits with status 0, reports `SUMMARY failures=0` in each partition, and has SHA-256

`e095f8a87dbf4718ab5ac066d438ca1b`
`f6b32452d617172d90acd926094f717d`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	53	489720029079333857613134125714875124856232927111053146129896	257420798982082198262769118082416215050255289486778264613641805536432
16	55	38870934384506456534228679227956458714632949649039646028420588	750570909869325126286411737708058481670437211038205082798883277102979752
17	57	2760972531623856623586360234296180553041385386441903402894257432	2353592104545064265352636000265632122641498367964863179195937677531609229264
18	59	177529317063313594727013993839354605142600809019023903692310588212	7916863333163845197820856588019800498621631490347819247883268228979004138464024
19	61	10434766599703968803394804221878935599357976145166175185279618980446	28498624571845013529193156443992965398493820864363649050000711343697375413991544388
20	63	565345216504047943127435592612320105186245465113366840852839369823961	109541225789834174800968062488307633292545317395511658014843539103024354806184478472014
21	65	28436452122494546784503973932702955134312420993458055393779840223891968	448652128232881537675257048032803280194915678580518443721778711569429345707066432202259456
22	67	1336195933930798511146548004105498833028709271000610258638238376308898757	1954211164681443257923850580590529987044905158203351229318180728438198945937670028239743945078
23	69	58974092244345858142200133092969801790833911541634513030662754818190165252	9035761317088612444238873307283530763813697487939047607591184131521439003201632497294804374793208
24	71	2456569840020508957073775728150479642048185471841740547517048174713983263458	44272870337243698081214180314068212637059981763468866539302805901137262332308086999488081845668900924
25	73	96987495191309572433445415443111879196267468508045127977093262561694838645616	229499010251335973773574243156073238900710989626071932484015074807062184625016908190622466199329759412512
26	75	3643014674651632199039247569020925167825601643487772678803453906243107112389334	1256676780468867042847317084430225634573268479204172737494852241212617575405569957564767462525584953015970900
27	77	130626558654627780336785493356319144285233277019286563030236447231223641928726792	7258238000562610837751823684753666344813726004110893109439011199813159175708294197810447972563789048817366335984
28	79	4484824408626513520837247625031122901074027004243550353290153127575853286119227960	44157232836692700825063754574781462288824084959079670901180598353823736269567420056701139659104810175377874312140688
29	81	147839993704151891097673114935981535868645820113678338225334706966064166592768329640	282594820649086173955053884185975495048604314605443674524875194968427729386721820405657569245340469859162050861849514160
30	83	4690810571265634452525230604643372066452349462118325152714379364500745954517147932817	1900094241916970117931209240138935340224529637895634578219185527594659152753412142737153185679668809940022454812921109998302
31	85	143579375783345675129915376399568094232061561510997014964985636510055593581476759259076	13406572350450220333476929689717057966686851831068320347069890063113832202910008205837529311960324199367713697523754288234936440
32	87	4248316050658717351663544273621748138012165488984367502227485861088874554402804651204618	99151536294250092657629161470485213770197128218684406622486503750149756299269651335480557581410054134117903463494663448936303896556
33	89	121741397701642803189252770380994743350843307316009993882328516506636481279153495650343536	767803834154796949339700774490437588122613781115785190963928649700522698663494240220536411985702935534869476919032247062567998787513120
34	91	3384581136537593661899995441061156486986342192260755133345315750613170930637552174185364260	6219012485561720169125175711747072746616101573026912587700105876072853260951785680018327110228424157762992931096094078597634184735198699960
35	93	91433429026135885153051601448881562353933632458095125897796643809931783337877280108972533346	52636114508684281023057462983247168448627959118498020128921825020140008861850620284140374545972471867009993756570327020379399572492967071638332
36	95	2403659572979022768206316866500445519920780823246805116765721775040200192632674950996587156935	465079114888095472747378954931122536275227942736325900402962100465988261414976812349575413009986465099649036282868029083392176506913941910130566258
37	97	61573662580028538121138960264906234970216539335277488869959450210655901086720201850042978311904	4286049101830469787585519062310491636993132895599321887180676975724911052612077176435617242577630148038138756746834614683804694344830243614261847507520
38	99	1538912441716013944498604310705441375163609522812461895354428981430433150377490598407851632893185	41162110219623827986714077844387249387887643477052640993124155319052740791524607978035634866636003958801759691664111820942413257561151865783948334858360318
39	101	37569326911962077279113652639608882499403096094587221496093838457684817231354420086304461819573884	411610493361212659403148807521076157229850906012077840841555796551366675987937672908423250486030980537421626324382116463164316234745031427737901636673153889032.

Each margin is positive, proving the claimed half-stable bound. □

Proposition 4.653 (Residual B twenty-third nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+23} \leq \delta_{2q+23}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta^{B_b}_{2q+23}|$ is the number of column-even Pieri two-strip paths of length $2q+23$ with terminal first row at least $2q$. If $15 \le q \le 39$, this is bounded by $s_{2q+23}/2$ by lemma 4.652. If $q \ge 40$, lemma 4.650 bounds this number by

$$16741606354773\binom{2q+23}{46}92^{46}3^{2q-23},$$

and lemma 4.651 absorbs that count into $s_{2q+23}/2$. In all cases $|\delta^{B_b}_{2q+23}| \le s_{2q+23}/2$, which proves the stated lower bound. $\square$

Lemma 4.654 (Twenty-fourth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+24)}$$

with

$$\lambda_1^{(2q+24)} \ge 2q$$

is at most

$$917878401790345\binom{2q+24}{48}96^{48}3^{2q-24}.$$

Proof. Put $n = 2q+24$. The terminal partition has size $2n = 4q+48$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+48-2(2q) = 48.$$

Hence the terminal partition has at most fifty nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 48 = 96$ choices, and a step with two lower boxes has at most $\binom{48+1}{2} = 1176$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{48}$ in

$$(3+96x+1176x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 48$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}96^a1176^c.$$

Since $n \ge 54$, $a+c \le 48$, and the forty-eight factors in the quotient $\binom{n}{48}/\binom{n}{a+c}$ which are not cancelled are all at least 7,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{48}} \le \frac{48!}{a!\,c!\,7^{48-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{48}96^{48}3^{n-48}\sum_{\substack{a,c\ge 0\\ a+2c\le 48}}\frac{48!}{a!\,c!\,7^{48-a-c}}\frac{3^{48-a-c}1176^c}{96^{48-a}}.$$

The exact replay

`certificates/post_m29/bc_twentyfourth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{718988576413645698365593997653011467970623320513504394866582874345402208172045549034764439557490514252288862 3}{7833157148171707308011570939888586941509981933966218053272652442206028867904865324503574511616} < 917878401790345.$$

This gives the displayed bound. $\square$

Lemma 4.655 (Stable moments dominate the twenty-fourth B nonboundary width count above the first row)**.** *For every $q \geq 42$,*

$$1835756803580690\binom{2q+24}{48}96^{48}3^{2q-24} \leq s_{2q+24}.$$

Proof. By lemma 4.65, s_{2q+24} contains all labelled simple cycles on $2q+24$ vertices, whose number is $(2q+23)!/2$. It is enough to prove

$$3671513607161380\binom{2q+24}{48}96^{48}3^{2q-24} \leq (2q+23)!.$$

At $q = 42$, the left side is

2812900922340715779735034210441384326940779536622725930632011526173967860443865339898462554335587373172386120379474790454281740718041588004973774202190975508038721470464 00,

whereas the right side is

1226520203196137939351751701038733888713156815438294505265325141201353532492214424903465861328705906193374391671931856038096650652042000036817534976000000000000000000000000000.

The second integer exceeds the first by

119839119397273078155440135893432004544437490200720672459590050259396138538877757715044812357853500324616505304681371081335538332448615841567677972339780902449196127852953600.

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+26)(2q+25)}{(2q-22)(2q-23)} < 30,$$

because $84q^2 - 3618q + 9330 > 0$ for $q \geq 42$. The right side is multiplied by

$$(2q+24)(2q+25) \geq 108 \cdot 109 > 30.$$

Induction proves the result. □

Lemma 4.656 (Exact first twenty-seven twenty-fourth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 41$, the number of column-even Pieri two-strip paths of length $2q+24$ with terminal first row at least $2q$ is at most $s_{2q+24}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyfourth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 41$, $j = 2q+24$. It exits with status 0, reports `SUMMARY failures=0` in each partition, and has SHA-256

`ccd56620d91d6a3c2c9e23673211e25d`
`cb5e1ae5d1a4ddc71e699a7bf80fab4a`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	54	450452639396824907986258885869805983432755408207098856217487 26	13771976975743436254452353295995864843138794765894432017754046461004692
16	56	38535956185863621806677741835318538445600423118643074376095321 20	41656588890059500967449751153038046718103240680764445126387800477091312080
17	58	294039258448642901148097883499242498063650162962918317768929232232	135331264353692518008363109229613655929289547812965953330626871386396430199984
18	60	20249959191843444305833437598392302381278777523904378211166683503100	471052485162663817401429109081796622358272145867461718410869217787567438942779656
19	62	1271390358886551502663758823963550766753045144774576745906496942946037	1752662440577779032428785895135121545816136781948406557324177101218084976053868847382
20	64	733984620461132620129012983125472424253763121999511726439809259451257 39	6955857144727139823955047282167079291221307753838004423957652354572499102109822176078378
21	66	39250884667315008482656359629887541017326804296966645138647896397303855 00	29386673299901708956899298433296567493071691252188270722521382414604093813831101363414023240
22	68	1956795806859427839136219642103027078860389227054425089868562657434100772 22	131909085289227211701398155518427042761986402498619232233540207544707273660582650571004781873588
23	70	914545868485807523996287853600076004230365953401412843108016310879758771090 4	627984678371728270376799959329442365679624319028174537049651757834806611885165798000648656819927696
24	72	4026880183527004628072204140257488769818159094482091530411906928516102462485 62	3165506839235730069039026548253409304205550391629177369034526715105817692306324007894647385822810269980
25	74	167775848760165096815205812468270571472885383964387021522813111848682584186066200	16868160644365457941953337849036198202393364492934068219647968246027908992490822704214323199306397704875984
26	76	66400914749770764150910324276146267299132896045479589854953034020063414993976330 0	94879010830044980231973613147383325259518203153130806597089374948553992402379865567235443161283890262099430200
27	78	2505019892982289952797613534155274214479323064278979460823305874936869292560522816 7	562512973617582593017381142149388938967890172893457969022774929060352683621352943848397573570987188787868967778034
28	80	90364211179961887054997076493466665833811416160111285637022620073303985266084537884 5	3510497287734778986037296352864813255315804391560349696376653685120012169072487848818949239076006210068201979391230406
29	82	3125742193746311136136265984076212723156514807915026067988110980305291267834668043926 6	23031461318491612837555179058823773113565334652274740877369198849590054759911066419626332180140690151096922635708627195964
30	84	103941555082382070726341588362858285038586694026112528318432613168306253688420596783317 2	158657763193874483403838398425529131800293858736525021280373686832820981510079979264806887186102865811468929410609996613007192
31	86	333051716470090039185594351589587725518904890642873659798142922018287642901651271651529 30	1146261223216104476113850870504823904206906819232874804512708643709322301939304611476140101271705256636271280353222419222416333500
32	88	1030478358849312857260381531499271084474857808305628541123443156025862460289958735586143358	86757543968565248763681048796969017638197789346880289822959881955481430409340428757488669339821471545263171134806169040310738603122836
33	90	308468455399261958805061990035404293332652015025611229897572640511246210763969958224697797 94	6871840596496887584238285446762354950276849500051636271633897712485834744262710451047276342616701756143418030341810372738650238651813 6156
34	92	894947929676122083872901537759812156760573140029404763312322181492051760951134902103369589150	56903935442806480958450175485399902102345507276027005587948690479132026003684207183613891988096699387539133124011499338155741737484954933 9844
35	94	2520626261460852104853907721106493905429416181357001796487205799882517283828533719997854645379 9	4921474373856215123552185218948555459746755783725771595106762614336954447485365922786544140373907116743797316487108828911460697171049897427780 9586
36	96	6902335744627896925948507327677033534465015141458419536173651796470406402576184991390200326832 75	44415035728604135032970506742036555025782948717026352942674733591834551995138340043792204784164893711210070284535898157890920563319339820154528445930
37	98	18401969786379726509397550928547608668849022060215237021158376690029263548754575191596046896961224	417889612984311197972881691783935928096937375597305445986352710733921306024321448990581987404762569297015318862011729845392458401771406744610731284750960
38	100	47826900384456256960436906225832856246699847645480578678654838139433726346021440889899727524836930 6	4095628359235226170344431641109679151351419816735792386999621199938760398734934237313071615798391265669544725547987428986378191175226993278828223708741378764
39	102	1213217592880197477962411281678688475821059662579988584913335863145214524854499170861513332429434425 8	41778449637156136111589911062266613865092602309151061119068884600495926616345005830809526260844732371588482361350868475689895706862246077865474658190283617108860
40	104	30070881866087652254830210378261325313263810265132619448661501354096568773062028106769283715701637292 8	44321681420933857115734735790500905594590887486903642745184242901913948052315021813898844631576581046877708264778939459109258988530701086793214520328994585020573760 0
41	106	72902778566935127880269939046740257388434093778692106028524627888594363761471231844821972311036385395 60	4886351313102937424311841336468296380298521371436515901130417647337153440439863468297240623754779583629242523324475917211904198485224788159926080471121817552593787389360.

Each margin is positive, proving the claimed half-stable bound. □

Proposition 4.657 (Residual B twenty-fourth nonboundary lower correction)**.** *For every residual row B_b with $14 \le b \le 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+24} \le \delta_{2q+24}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+24}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+24$ with terminal first row at least $2q$. If $15 \le q \le 41$, this is bounded by $s_{2q+24}/2$ by lemma 4.656. If $q \ge 42$, lemma 4.654 bounds this number by

$$917878401790345\binom{2q+24}{48}96^{48}3^{2q-24},$$

and lemma 4.655 absorbs that count into $s_{2q+24}/2$. In all cases $|\delta_{2q+24}^{B_b}| \le s_{2q+24}/2$, which proves the stated lower bound. ☐

Lemma 4.658 (Twenty-fifth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+25)}$$

with

$$\lambda_1^{(2q+25)} \ge 2q$$

is at most

$$96319792639339148\binom{2q+25}{50}100^{50}3^{2q-25}.$$

Proof. Put $n = 2q+25$. The terminal partition has size $2n = 4q+50$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+50-2(2q) = 50.$$

Hence the terminal partition has at most fifty-two nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 50 = 100$ choices, and a step with two lower boxes has at most $\binom{50+1}{2} = 1275$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{50}$ in

$$(3+100x+1275x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 50$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}100^a1275^c.$$

Since $n \ge 55$, $a+c \le 50$, and the fifty factors in the quotient $\binom{n}{50}/\binom{n}{a+c}$ which are not cancelled are all at least 6,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{50}} \le \frac{50!}{a!\,c!\,6^{50-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{50}100^{50}3^{n-50}\sum_{\substack{a,c\ge 0\\ a+2c\le 50}}\frac{50!}{a!\,c!\,6^{50-a-c}}\frac{3^{50-a-c}1275^c}{100^{50-a}}.$$

The exact replay

`certificates/post_m29/bc_twentyfifth_frontier_gmp_certificate.log` prints this finite rational sum as

$$\frac{31562069652058651724050564331116427866898930990810385836335683810816344863875220220375406369645430784395169909}{32768000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000} < 96319792639339148.$$

This gives the displayed bound. $\square$

Lemma 4.659 (Stable moments dominate the twenty-fifth B nonboundary width count above the first row)**.** *For every $q \geq 44$,*

$$192639585278678296\binom{2q+25}{50}100^{50}3^{2q-25} \leq s_{2q+25}.$$

Proof. By lemma 4.65, s_{2q+25} contains all labelled simple cycles on $2q+25$ vertices, whose number is $(2q+24)!/2$. It is enough to prove

$$385279170557356592\binom{2q+25}{50}100^{50}3^{2q-25} \leq (2q+24)!.$$

At $q = 44$, the left side is

$$1631698339603928414165194585139136415150069360512059164383537511536124994126576640000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000,$$

whereas the right side is

$$197450685722107402353682037275992488341277868034975337796656295094902858969771811440894224355027779366597957338237853638272334919686385621811850780464277094400000000000000000000000000000000.$$

The second integer exceeds the first by

$$195818987382503473939516842690853351926127798674463278632272757583366733975645234800894224355027779366597957338237853638272334919686385621811850780464277094400000000000000000000000000000000.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+27)(2q+26)}{(2q-23)(2q-24)} < 30,$$

because $84q^2 - 3774q + 10242 > 0$ for $q \geq 44$. The right side is multiplied by

$$(2q+25)(2q+26) \geq 113 \cdot 114 > 30.$$

Induction proves the result. $\square$

Lemma 4.660 (Exact first twenty-nine twenty-fifth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 43$, the number of column-even Pieri two-strip paths of length $2q+25$ with terminal first row at least $2q$ is at most $s_{2q+25}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyfifth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 43$, $j = 2q+25$. It exits with status 0, reports `SUMMARY failures=0` in each B partition, and has SHA-256

`1bd9d79c07c867c4d2d40c446cc351f3`
`a2fbd100861b423ff9e32c0bfc41ed91`.

It verifies the following exact margins:

q	j	$\#\mathrm{bad}(q,j)$	$s_j - 2\#\mathrm{bad}(q,j)$
15	55	4145931517461905557858975498460147368518189723816827644791522010	750570901655203960131613535058564843206055391431091534463307279576776908
16	57	381825147256219851645212554161959212798369909875907096220856947944	2353592103786935915903444010222379734786172303474206130217007291895683848240
17	59	31262241721162037041241938421324025149111199451620716290903589710968	7916863333101675773012659141126770649766662149259882050598064844204581580218512
18	61	2303513684154537786512596537819715085137130432054160558264571499792840	28498624571840427371358046776357546995026625192064573505088893354931216830229919600
19	63	154331061421702794859302866377271197814492106835326405573603958567232701	109541225789833867269535652090813800941683748077756239403120798676946889303946083654534
20	65	9484773290242459222057202928554516041650527645222023360762614503785816096	448652128232881518762583371792873929649517769336892270689348263112298734969397105078411200
21	67	538765609706049811735637740933424335650982597262166834434347342128085157340	1954211164681443256848991753046291960595922772344713555682272952455866497586251820736191427912
22	69	28472554463555590153447734908385862845491061556375669678159717963940642743534	9035761317088612444182046146540908275223086418388461821503783676231770932871374386876559469636644
23	71	1407999823983756896957975237043370441410192679057679962625726147821044728467240	44272870337243698081211369227559925164283979960546236099379269612150090655863930581288789184178493360
24	73	65482513596139903435397956148551264351869178182153461043978953484141568164518232	229499010251335973773574112385021037003523263697050466267710129461240757334185076186902023039583107667280
25	75	2877000945746070073513592998559095898913681456515908712069980686077444195516288104	1256676780468867042847317078683509772430431596575065235514702293720905865660728078782412998183182776208173360
26	77	119888756626776632089808575502343737859968173294893746848818950024528282238973333612	7258238000562610837751823684514150084677481995491949529421036362381793295672545277238870545409194931623277122344
27	79	4755433453302025540806807094223694719079936824229588911800371943793040789295397920941	44157232836692700825063754574771960391566298161025098961487401026631378543972969365978245495523479245505855754754726
28	81	180122261004411799850782300882024397519889210366410200764974162892136015474289706437920	282594820649086173955053884185975135099762293190147755155619660791595761345592727812612715747684099519259435467973297600
29	83	6533814889183911337086297668211913451588726206575274598861572338785122326920449835991954	1900094241916970117931209240138935327166281480670343172951640652380118993708864428510840638261952891098779292881055733880028
30	85	227576088845539798983484324827616614867063917040530292163791999671686785188166588936377358	13406572350450220333476929689717057966231986812128807440453181244211398109364344494879490721662670171297360238334584063880699876
31	87	7629243383042814630196006231888299575515378431998099393346712401121861451922225008034154330	99151536294250092657629161470485213770181878228550422310660815064774526943614896602947539351358365164287381918339927804529537997132
32	89	246703306024066097803373166592171570021148679547491900777820203439034237196358256869385924964	767803834154796949339700774490437588122613287952655938235018649332695055082343684624863931633921367659119631863830816904361251316350264
33	91	7710283187893013835784344358602922029642325400191151143795062012174421286757753724395062592270	6219012485561720169125175711747072746616101557613115374187153391827964534628063933707648994230643380439559538248477846943401840293444243940
34	93	233324004198302830434854312274352246658842166040188021897341170251820615304989830564682131613676	52636114508684281023057462983247168448627959118031554987383271631040606340504813553947398081157012013467107009354305653336095665923820753477672
35	95	6848108068247151965795627395858831467260786613484327219811180368085085014244250337430578107544895	465079114888095472747378954931122536275227942736312208994144752120102206572818827687531931278321142938819647453575408993764073271588982747089790338
36	97	195243007857585300168028972108265141626487986830143959392983018240227775048035735368323967882264544	4286049101830469787585519062310491636993132895599321496817808585714367792796410880435146459542089566420774948520717034649566400446799910666412039602240
37	99	5414954288680224882813114357431128639023531061721428660661337850502110298527677652548871833485814200	41162110219623827986714077844387249387887643477052640982297324566575723053787376471794788471339269054404527294132144977901053526806551541883020371152518288
38	101	146286237117273352894547699227239310976202885033796329383086113103977967509724924541573727667937924550	411610493361212659403148807521076157229850906012077840841263299215785953206303135737274051081843573573546222840058936424633275669360044287494926790260917187700
39	103	3854208145235496907201420896374848214274728530488552179052146178754920439872389117617201368228523392174	4282289551634348924637847939334988424968279064470634107325419718782403448969788932329158846270183968871956233820510706247424515595115005373062980142668165291697828
40	105	99148506255084971187483653672387066123475937428263609916771914087900986624591806679913530709013275727664	46316141414884575423512785844162280630709788031116921318799723858789467181092314057828795308755402478761745913159139545969528577914828290659084552753349830883202224544
41	107	2492980620950423184030911529930656255902260845725546617956542306845227660381862220931430772388347754440216	520396248607814162823456898918379242662159520014922891460847538243510608836810618504477719528362625247830625823194172205046442989452872079233110065580070694962664054412240
42	109	61328864800734725452304195789836622444459461915696851008253808474360134995239026619488849442774457977304792	6069768027148561641469987059305686919203463935482378676309774609902944012654714526503114749223079939184754909539962195692260581280715316990819031184935102761658602563986149968
43	111	1477500817551458801892338405894396207044666417511868155552612189932909325352000260541969253705692496658370785	73442633136598658548124378676446602820000468916554313883241994137346871713887965773926070832575361847114241104749542170651189054330469960695541408770995495301695217983881475890750.

Each margin is positive, proving the claimed half-stable bound. $\square$

Proposition 4.661 (Residual B twenty-fifth nonboundary lower correction)**.** *For every residual row B_b with $14 \le b \le 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+25} \le \delta_{2q+25}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+25}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+25$ with terminal first row at least $2q$. If $15 \le q \le 43$, this is bounded by $s_{2q+25}/2$ by lemma 4.660. If $q \ge 44$, lemma 4.658 bounds this number by

$$96319792639339148\binom{2q+25}{50}100^{50}3^{2q-25},$$

and lemma 4.659 absorbs that count into $s_{2q+25}/2$. In all cases $|\delta_{2q+25}^{B_b}| \le s_{2q+25}/2$, which proves the stated lower bound. $\square$

Lemma 4.662 (Twenty-sixth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+26)}$$

with

$$\lambda_1^{(2q+26)} \ge 2q$$

is at most

$$24185591292956608915\binom{2q+26}{52}104^{52}3^{2q-26}.$$

Proof. Put $n = 2q+26$. The terminal partition has size $2n = 4q+52$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q + 52 - 2(2q) = 52.$$

Hence the terminal partition has at most fifty-four nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 52 = 104$ choices, and a step with two lower boxes has at most $\binom{52+1}{2} = 1378$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{52}$ in

$$(3 + 104x + 1378x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a + 2c \le 52$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}104^a 1378^c.$$

Since $n \ge 56$, $a + c \le 52$, and the factors in the quotient $\binom{n}{52}/\binom{n}{a+c}$ which are not cancelled are all at least 5,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{52}} \le \frac{52!}{a!\,c!\,5^{52-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{52}104^{52}3^{n-52}\sum_{\substack{a,c\ge 0\\ a+2c\le 52}}\frac{52!}{a!\,c!\,5^{52-a-c}}\frac{3^{52-a-c}1378^c}{104^{52-a}}.$$

The exact replay

`certificates/post_m29/bc_twentysixth_frontier_gmp_certificate.log`

prints this finite rational sum as

$$\frac{10515921361294745138317260438626617216441934753953658489934041385438143468232284823955926238983677843094936348125098252699625877660467}{43480108606471073675479211016723925095809380714889424442110924523084709888000000000000000000000000000000000000000000000} < 2418559129295660891 5.$$

This gives the displayed bound. ☐

Lemma 4.663 (Stable moments dominate the twenty-sixth B nonboundary width count above the first row)**.** *For every $q \geq 46$,*

$$48371182585913217830\binom{2q+26}{52}104^{52}3^{2q-26} \leq s_{2q+26}.$$

Proof. By lemma 4.65, s_{2q+26} contains all labelled simple cycles on $2q+26$ vertices, whose number is $(2q+25)!/2$. It is enough to prove

$$96742365171826435660\binom{2q+26}{52}104^{52}3^{2q-26} \leq (2q+25)!.$$

At $q = 46$, the left side is

$$2451863079196722366387922463618790851845178863000141764405243392241740881687916651889383264324547486389664519299314124497486834373763218892312942559330995686456687930973399372976314963197952 0,$$

whereas the right side is

$$3969937160808720895401959629498630647790406360168322301129748464310422041758630649341780708631240196854767624444057168110272995649603642560353748940315749184568295424000000000000000000000000000.$$

The second integer exceeds the first by

$$3945418530016753671738080404862442739271954571538320883485696030388004632941751482822886875987994721990870979251064026865298127305866010371430619514722439227703728544690266006270236850368020480.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+28)(2q+27)}{(2q-24)(2q-25)} < 30,$$

because $84q^2 - 3930q + 11196 > 0$ for $q \geq 46$. The right side is multiplied by

$$(2q+26)(2q+27) \geq 118 \cdot 119 > 30.$$

Induction proves the result. ☐

Lemma 4.664 (Exact first thirty-one twenty-sixth B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 45$, the number of column-even Pieri two-strip paths of length $2q+26$ with terminal first row at least $2q$ is at most $s_{2q+26}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentysixth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 45$, $j = 2q+26$. It exits with status 0 in each B partition, reports `SUMMARY failures=0` in each B partition, and has SHA-256

`6a107c7c20c107e6a71db9aa65cf2884`
`df0e1e94db82c20d7d3f1ae3acc4c07d`.

The transcript contains exactly thirty-one `B_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+26} - 2\#\mathrm{bad}(q, 2q+26).$$

Thus every displayed bad-shape count is at most $s_{2q+26}/2$. ☐

Proposition 4.665 (Residual B twenty-sixth nonboundary lower correction)**.** *For every residual row B_b with $14 \le b \le 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+26} \le \delta_{2q+26}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+26}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+26$ with terminal first row at least $2q$. If $15 \le q \le 45$, this is bounded by $s_{2q+26}/2$ by lemma 4.664. If $q \ge 46$, lemma 4.662 bounds this number by

$$24185591292956608915\binom{2q+26}{52}104^{52}3^{2q-26},$$

and lemma 4.663 absorbs that count into $s_{2q+26}/2$. In all cases $|\delta_{2q+26}^{B_b}| \le s_{2q+26}/2$, which proves the stated lower bound. □

Lemma 4.666 (Twenty-seventh B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+27)}$$

with

$$\lambda_1^{(2q+27)} \ge 2q$$

is at most

$$211135895942771652705 89\binom{2q+27}{54}108^{54}3^{2q-27}.$$

Proof. Put $n = 2q+27$. The terminal partition has size $2n = 4q+54$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q + 54 - 2(2q) = 54.$$

Hence the terminal partition has at most fifty-six nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 54 = 108$ choices, and a step with two lower boxes has at most $\binom{54+1}{2} = 1485$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{54}$ in

$$(3 + 108x + 1485x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a + 2c \le 54$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}108^a1485^c.$$

Since $n \ge 57$, $a + c \le 54$, and the factors in the quotient $\binom{n}{54}/\binom{n}{a+c}$ which are not cancelled are all at least 4,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{54}} \le \frac{54!}{a!\,c!\,4^{54-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{54}108^{54}3^{n-54}\sum_{\substack{a,c\ge 0\\ a+2c\le 54}}\frac{54!}{a!\,c!\,4^{54-a-c}}\frac{3^{54-a-c}1485^c}{108^{54-a}}.$$

The exact replay

`certificates/post_m29/bc_twentyseventh_frontier_gmp_certificate.log` prints this finite rational sum as

$$\frac{2627146205381501605105934114888073677477007311327886945300155746611238154833874468430700303859194480319027224499}{1244291594117936781313208413006796910681543749897004193933993038064379279628652013955317776} < 21113589594277165270589.$$

This gives the displayed bound. ☐

Lemma 4.667 (Stable moments dominate the twenty-seventh B nonboundary width count above the first row)**.** *For every $q \geq 48$,*

$$4222717918855433054117 8\binom{2q+27}{54}108^{54}3^{2q-27} \leq s_{2q+27}.$$

Proof. By lemma 4.65, s_{2q+27} contains all labelled simple cycles on $2q+27$ vertices, whose number is $(2q+26)!/2$. It is enough to prove

$$8445435837710866108235 6\binom{2q+27}{54}108^{54}3^{2q-27} \leq (2q+26)!.$$

At $q = 48$, the left side is

$$138258098978895487604667122962018484751331409313194022878328692091635120860287318602200868984061254416394370905843317937549708949878538918538977090725263119007224290021533620981815264487289287906164736 0,$$

whereas the right side is

$$987504420083360136241157987144820801256440413697835960595847005026767145720501436490337964277450422940710230505796264047365129395968426948958213782106200133880547472147952435200000000000000000000000000 00.$$

The second integer exceeds the first by

$$973678610185470587480691274848618952781307272766516558308014135817603633634472704630117877379044297499070793415211932253610158500980573057104316073033673821979825043145799073101818473551271071209383526 40.$$

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+29)(2q+28)}{(2q-25)(2q-26)} < 30,$$

because $84q^2 - 4086q + 12192 > 0$ for $q \geq 48$. The right side is multiplied by

$$(2q+27)(2q+28) \geq 123 \cdot 124 > 30.$$

Induction proves the result. ☐

Lemma 4.668 (Exact first thirty-three twenty-seventh B nonboundary bad-shape counts)**.** *For $15 \leq q \leq 47$, the number of column-even Pieri two-strip paths of length $2q+27$ with terminal first row at least $2q$ is at most $s_{2q+27}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyseventh_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \leq q \leq 47$, $j = 2q+27$. It exits with status 0 in each B partition, reports `SUMMARY failures=0` in each B partition, and has SHA-256

`67df773882805ff4321cd3659e16b1b9`
`985db56a76a94d7136448011ef30f80d`.

The transcript contains exactly thirty-three `B_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+27} - 2\#\mathrm{bad}(q, 2q+27).$$

Thus every displayed bad-shape count is at most $s_{2q+27}/2$. ☐

Proposition 4.669 (Residual B twenty-seventh nonboundary lower correction)**.** *For every residual row B_b with $14 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+27} \leq \delta_{2q+27}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+27}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+27$ with terminal first row at least $2q$. If $15 \le q \le 47$, this is bounded by $s_{2q+27}/2$ by lemma 4.668. If $q \ge 48$, lemma 4.666 bounds this number by

$$21113589594277165270589\binom{2q+27}{54}108^{54}3^{2q-27},$$

and lemma 4.667 absorbs that count into $s_{2q+27}/2$. In all cases $|\delta_{2q+27}^{B_b}| \le s_{2q+27}/2$, which proves the stated lower bound. $\square$

Lemma 4.670 (Twenty-eighth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+28)}$$

with

$$\lambda_1^{(2q+28)} \ge 2q$$

is at most

$$129062907305803474509175447\binom{2q+28}{56}112^{56}3^{2q-28}.$$

Proof. Put $n = 2q+28$. The terminal partition has size $2n = 4q+56$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+56-2(2q) = 56.$$

Hence the terminal partition has at most fifty-eight nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 56 = 112$ choices, and a step with two lower boxes has at most $\binom{56+1}{2} = 1596$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \dots, x^{56}$ in

$$(3+112x+1596x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a+2c \le 56$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}112^a1596^c.$$

Since $n \ge 58$, $a+c \le 56$, and the factors in the quotient $\binom{n}{56}/\binom{n}{a+c}$ which are not cancelled are all at least 3,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{56}} \le \frac{56!}{a!\,c!\,3^{56-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{56}112^{56}3^{n-56}\sum_{\substack{a,c\ge 0\\ a+2c\le 56}}\frac{56!}{a!\,c!\,3^{56-a-c}}\frac{3^{56-a-c}1596^c}{112^{56-a}}.$$

The exact replay

`certificates/post_m29/bc_twentyeighth_b_absorption_gmp_certificate.log`

prints this finite rational sum as

$$\frac{2025529548764936535663219997915257739427053097450973064895015969532428326966504020337406927050447841928760131584542621}{15694126151719317999571618173922000017876418395539549210232909869689551256045946062675902464} < 129062907305803474509175447.$$

This gives the displayed bound. $\square$

Lemma 4.671 (Exact twenty-eighth B variable lower-box absorption)**.** *For* $42 \le q \le 49$*, put*

$$R_q = \sum_{\substack{a,c\ge 0\\ a+2c\le 56}} \frac{56!}{a!\,c!\,(2q-27)^{56-a-c}} \frac{3^{56-a-c}1596^c}{112^{56-a}}.$$

Then

$$2R_q\binom{2q+28}{56}112^{56}3^{2q-28} \le s_{2q+28}.$$

Proof. For a fixed displayed q, the proof of lemma 4.670 may use the sharper lower bound $2q-27$ for every uncancelled quotient factor. Hence the corresponding bad-shape count is at most

$$R_q\binom{2q+28}{56}112^{56}3^{2q-28}.$$

The exact C++/GMP replay

`certificates/post_m29/bc_twentyeighth_b_absorption_gmp_certificate.log`

cross-multiplies the eight rational inequalities exactly, exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`c1127264a1dacd0dab0341a2c0afbfb1`
`1f76063c9f8531c5981953d7a104aa4a`.

Its transcript contains exactly eight `B_VARIABLE_ABSORPTION` records, one for each displayed q, and each record prints a positive exact margin numerator. This proves the displayed inequalities. ☐

Lemma 4.672 (Stable moments dominate the twenty-eighth B nonboundary width count above the first row)**.** *For every* $q \ge 50$*,*

$$258125814611606949018350894\binom{2q+28}{56}112^{56}3^{2q-28} \le s_{2q+28}.$$

Proof. By lemma 4.65, s_{2q+28} contains all labelled simple cycles on $2q+28$ vertices, whose number is $(2q+27)!/2$. It is enough to prove

$$516251629223213898036701788\binom{2q+28}{56}112^{56}3^{2q-28} \le (2q+27)!.$$

At $q = 50$, the left side is

5876199327924858412976987097324933219807464980384957794586635134215694626654944360574331680115622055293509864674650044145647350370114347233179219658083684215342605280347469529785533191127911803468226002321145856 00,

whereas the right side is

30126600184576595448099770775270596923241649186736217990533469005966672076184808090678606920977137619846097799457727839655638510333007723262977730878518699825002706617912441225976217600000000000000000000000000000000.

The second integer exceeds the first by

24250400856651737035122783677945663703434184206351260195946833871750977449529863730104275240861515564552587934783077795509991159962893376029798511220435015609660101337564971696190684408872088196531773997678854144 00.

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+30)(2q+29)}{(2q-26)(2q-27)} < 30,$$

because $84q^2 - 4242q + 13230 > 0$ for $q \ge 50$. The right side is multiplied by

$$(2q+28)(2q+29) \ge 128 \cdot 129 > 30.$$

Induction proves the result. ☐

Lemma 4.673 (Exact first twenty-seven twenty-eighth B nonboundary bad-shape counts)**.** *For $15 \le q \le 41$, the number of column-even Pieri two-strip paths of length $2q+28$ with terminal first row at least $2q$ is at most $s_{2q+28}/2$.*

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyeighth_frontier_gmp_certificate.log`

implements the reverse Pieri recursion of lemma 4.592 for $15 \le q \le 41$, $j = 2q+28$. It exits with status 0 in each B partition, reports `SUMMARY failures=0` in each B partition, and has SHA-256

`400e6b4c50b78e1c490ce93daca73864`
`a2ebfc3380e93c50ac7d502205299b0e`.

The transcript contains exactly twenty-seven `B_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+28} - 2\#\mathrm{bad}(q, 2q+28).$$

Thus every displayed bad-shape count is at most $s_{2q+28}/2$. ☐

Proposition 4.674 (Residual B twenty-eighth nonboundary lower correction)**.** *For every residual row B_b with $14 \le b \le 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+28} \le \delta_{2q+28}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+28}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+28$ with terminal first row at least $2q$. If $15 \le q \le 41$, this is bounded by $s_{2q+28}/2$ by lemma 4.673. If $42 \le q \le 49$, the same bound follows from lemma 4.671. If $q \ge 50$, lemma 4.670 bounds this number by

$$1290629073058034745091754447\binom{2q+28}{56}112^{56}3^{2q-28},$$

and lemma 4.672 absorbs that count into $s_{2q+28}/2$. In all cases $|\delta_{2q+28}^{B_b}| \le s_{2q+28}/2$, which proves the stated lower bound. ☐

Lemma 4.675 (Twenty-ninth B nonboundary width-bad paths have a lower-box code)**.** *For every $q \ge 15$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+29)}$$

with

$$\lambda_1^{(2q+29)} \ge 2q$$

is at most

$$26657057892939326657049273688667\binom{2q+29}{58}116^{58}3^{2q-29}.$$

Proof. Put $n = 2q+29$. The terminal partition has size $2n = 4q+58$. Since its columns have even lengths and its first row has length at least $2q$, the number of boxes below the second row is at most

$$4q+58-2(2q) = 58.$$

Hence the terminal partition has at most sixty nonzero rows.

As in lemma 4.583, encode a path by the row-pair word and track lower boxes. A step with no lower box has three choices, a step with one lower box has at most $2 \cdot 58 = 116$ choices, and a step with two lower boxes has at most $\binom{58+1}{2} = 1711$ choices. Thus the number of paths is bounded by the sum of the coefficients of $x^0, \ldots, x^{58}$ in

$$(3+116x+1711x^2)^n.$$

The monomial with a one-lower-box steps and c two-lower-box steps, where $a + 2c \le 58$, contributes

$$\binom{n}{a+c}\binom{a+c}{c}3^{n-a-c}116^{a}1711^{c}.$$

Since $n \ge 59$, $a + c \le 58$, and the factors in the quotient $\binom{n}{58}/\binom{n}{a+c}$ which are not cancelled are all at least 2,

$$\frac{\binom{n}{a+c}\binom{a+c}{c}}{\binom{n}{58}} \le \frac{58!}{a!\,c!\,2^{58-a-c}}.$$

Therefore the coefficient sum is at most

$$\binom{n}{58}116^{58}3^{n-58}\sum_{\substack{a,c\ge 0\\ a+2c\le 58}}\frac{58!}{a!\,c!\,2^{58-a-c}}\frac{3^{58-a-c}1711^{c}}{116^{58-a}}.$$

The exact replay

`certificates/post_m29/bc_twentyninth_b_absorption_gmp_certificate.log`

prints this finite rational sum as

$$\frac{277779106855808268919410844062670919179911198214031840717487498638936112277059652420979367222105109134880580581288858691889356285997892023429345943573}{10420471305251725205780667230686101341858409721635912006716989825031102329848667800732237308856347868892719711199952896} < 26657057892939326657049273688667.$$

This gives the displayed bound. □

Lemma 4.676 (Exact twenty-ninth B variable lower-box absorption)**.** *For* $43 \le q \le 52$, *put*

$$R_q = \sum_{\substack{a,c\ge 0\\ a+2c\le 58}}\frac{58!}{a!\,c!\,(2q-28)^{58-a-c}}\frac{3^{58-a-c}1711^{c}}{116^{58-a}}.$$

Then

$$2R_q\binom{2q+29}{58}116^{58}3^{2q-29} \le s_{2q+29}.$$

Proof. For a fixed displayed q, the proof of lemma 4.675 may use the sharper lower bound $2q - 28$ for every uncancelled quotient factor. Hence the corresponding bad-shape count is at most

$$R_q\binom{2q+29}{58}116^{58}3^{2q-29}.$$

The exact C++/GMP replay

`certificates/post_m29/bc_twentyninth_b_absorption_gmp_certificate.log`

cross-multiplies the ten rational inequalities exactly, exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`8fba7d987f90866756d4e1edaca8395d`
`780c116c775f14c9d8718e848732f355`.

Its transcript contains exactly ten `B_VARIABLE_ABSORPTION` records, one for each displayed q, and each record prints a positive exact margin numerator. This proves the displayed inequalities. □

Lemma 4.677 (Stable moments dominate the twenty-ninth B nonboundary width count above the first row)**.** *For every* $q \ge 53$*,*

$$5331411578587865331409854737733 4\binom{2q+29}{58}116^{58}3^{2q-29} \le s_{2q+29}.$$

Proof. By lemma 4.65, s_{2q+29} contains all labelled simple cycles on $2q+29$ vertices, whose number is $(2q+28)!/2$. It is enough to prove

$$1066282315717573066281970947546668\binom{2q+29}{58}116^{58}3^{2q-29} \leq (2q+28)!.$$

At $q = 53$, the left side is

25206535420665767692611480292597373756463682658724368466566545273591351275874249433979910866049991473286626550442039862589966993526530405025464115439317732081102736887908608167839767846871260571678323214994718937561146935214080 0,

whereas the right side is

1992942746161518876737324194182948445222558439641379420268181650403332765909000151088003004705990626788174304488982644980314383013435909872030129915047718433336536425741860482713199069412263880294400000000000000000000000000000000.

The second integer exceeds the first by

1740877391954861199811209391256974707657921613054135735602516197667419253150257656748203896045490712055308038984562246354414713078170605821775488760654541112525509056862774401034801390943551274577616767850052810624388530647859200.

If the displayed factorial inequality holds at q, then at $q+1$ the left side is multiplied by

$$9\frac{(2q+31)(2q+30)}{(2q-27)(2q-28)} < 30,$$

because $84q^2 - 4398q + 14310 > 0$ for $q \geq 53$. The right side is multiplied by

$$(2q+29)(2q+30) \geq 135 \cdot 136 > 30.$$

Induction proves the result. $\square$

Proposition 4.678 (Residual B twenty-ninth nonboundary lower correction above the exact frontier)**.** *For every residual row B_b with $42 \leq b \leq 123$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$. Then*

$$-\frac{1}{2}s_{2q+29} \leq \delta_{2q+29}^{B_b}.$$

Proof. By lemma 4.94, the coefficient interpretation in lemma 4.95, and the Pieri-path bijection lemma 4.97, $|\delta_{2q+29}^{B_b}|$ is the number of column-even Pieri two-strip paths of length $2q+29$ with terminal first row at least $2q$. If $43 \leq q \leq 52$, this number is bounded by $s_{2q+29}/2$ by lemma 4.676. If $q \geq 53$, lemma 4.675 bounds this number by

$$266570578929393266570492736886667\binom{2q+29}{58}116^{58}3^{2q-29},$$

and lemma 4.677 absorbs that count into $s_{2q+29}/2$. In all cases $|\delta_{2q+29}^{B_b}| \leq s_{2q+29}/2$, which proves the stated lower bound. $\square$

Lemma 4.679 (C-boundary height-bad paths are alternating)**.** *Let $q \geq 1$, and let $\lambda^\bullet$ be a Pieri two-strip path of length $2q$ such that*

$$\ell(\lambda^{(2q)}) \geq 2q.$$

Then

$$\ell(\lambda^{(t)}) = t \qquad (0 \leq t \leq 2q)$$

and

$$C_q^h(\lambda^\bullet) = \{1, 3, \ldots, 2q-1\}.$$

Proof. A horizontal two-strip can create at most one new row. Thus the height sequence starts at 0, increases by at most 1 at each of the $2q$ steps, and ends at height at least $2q$. Hence every step creates exactly one new row, giving $\ell(\lambda^{(t)}) = t$ for all t.

With this height formula,

$$\min(q, \lceil \ell(\lambda^{(t)})/2 \rceil) = \min(q, \lceil t/2 \rceil).$$

The displayed statistic increases exactly at the odd times $t = 1, 3, \ldots, 2q-1$ and is constant at the even times. This is precisely the claimed height-crossing set. ☐

Proposition 4.680 (Residual C-boundary lower correction). *For every residual row C_b with $20 \leq b \leq 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q} \leq \delta_{2q}^{C_b}.$$

Proof. By lemma 4.94 and the coefficient interpretation in lemma 4.95, $|\delta_{2q}^{C_b}|$ is the number of semistandard tableaux T of content (2^{2q}) whose shape $\lambda(T)$ has $\lambda(T)'$ even and $\ell(\lambda(T)) \geq 2q$. Under the Pieri-path bijection of lemma 4.97, every such tableau gives a path satisfying the hypothesis of lemma 4.679. By lemma 4.98, all of these tableaux therefore lie in the single height fixed-witness fiber with

$$S = \{1, 3, \ldots, 2q-1\}.$$

Conversely, the defining height fixed-witness fiber already has $\ell(\lambda(T)) \geq 2q$. Hence the bad-shape set is exactly that fiber.

Since $q \geq 21$, corollary 4.110 gives

$$|\delta_{2q}^{C_b}| \leq 2^{2q}s_q.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the stronger boundary inequality

$$2^{2q+1}\binom{2q}{q}s_q \leq s_{2q}$$

for every residual C_b row, reports `failures=0`, and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Since $\binom{2q}{q} \geq 1$, the preceding two inequalities give $|\delta_{2q}^{C_b}| \leq s_{2q}/2$. This implies the stated lower bound. ☐

Lemma 4.681 (Even-column standard tableaux have a stable graph code). *Let s_n be the stable Littlewood moment sequence of lemma 4.65. For every $n \geq 2$, the number of standard Young tableaux of size $2n$ whose columns all have even length is at most*

$$2^{2n}s_n.$$

Proof. Let U be such a tableau. The Robinson–Schensted inverse applied to (U, U) gives an involution π on $\{1, \ldots, 2n\}$. By Rains' fixed-point/odd-column form of Schensted's theorem [5, Lemma 3.3], the fixed points of π are counted by the odd columns of U. Hence π is fixed-point-free.

Contract the adjacent pairs

$$\{1, 2\}, \{3, 4\}, \ldots, \{2n-1, 2n\}$$

to vertices $1, \ldots, n$. The matching π becomes a labelled degree-two multigraph Γ° on n vertices, with loops allowed. Let L be its loop set. By lemma 4.107, Γ° is recoverable from the loopless repaired graph $\mathcal{R}(\Gamma^\circ)$ and L. There are s_n choices for the repaired graph by lemma 4.65, and 2^n choices for L.

After Γ° is known, one further bit at each vertex records which of the two canonically ordered incident edge-slots receives the odd port $2v-1$; the port $2v$ receives the other slot. These n bits recover the original matching π, hence recover U by Schensted insertion. Thus U injects into a loopless stable graph, a loop-set word, and a slot word, giving at most $s_n 2^n 2^n = 2^{2n}s_n$ possibilities. ☐

Lemma 4.682 (Stable moments are monotone after two)**.** *For the stable Littlewood moment sequence s_n of lemma 4.65,*

$$s_n \le s_{n+1} \qquad (n \ge 2).$$

Proof. By lemma 4.65, s_n counts loopless labelled degree-two multigraphs on $\{1, \ldots, n\}$, with parallel edges allowed. For such a graph, choose the lexicographically least edge ab, with parallel edges ordered by multiplicity. Delete this edge and insert the new vertex $n+1$ by adding the two edges $a(n+1)$ and $(n+1)b$. The result is again loopless and degree two at every vertex.

The construction is injective: from its output, delete the vertex $n+1$ and its two incident edges, then restore one edge between its two neighbours. This recovers the original graph. Hence the number of graphs on n vertices is at most the number on $n+1$ vertices. □

Lemma 4.683 (C first-nonboundary height-bad paths have a one-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+1)}$$

with

$$\ell(\lambda^{(2q+1)}) \ge 2q$$

is at most

$$(2q+1)2^{2q+2}s_{q+1}.$$

Proof. The terminal height is even, because $(\lambda^{(2q+1)})'$ has only even parts. A Pieri two-strip can create at most one new row, so the terminal height is at most $2q+1$. The displayed lower bound therefore forces

$$\ell(\lambda^{(2q+1)}) = 2q.$$

Consequently exactly one step, call it τ, does not create a new row, and every other step creates one new row.

Delete the first column from every $\lambda^{(t)}$, shift the remaining boxes one column to the left, and call the resulting partition $\nu^{(t)}$. If $t \ne \tau$, the step t creates a new first-column box and one off-column box, so $\nu^{(t)}/\nu^{(t-1)}$ has size 1. At $t = \tau$, no first-column box is created and the two boxes of the Pieri strip are both off the first column, so $\nu^{(\tau)}/\nu^{(\tau-1)}$ is a horizontal strip of size 2. Filling the boxes of each skew difference with the label t gives a semistandard tableau U of size $2q+2$, with the label τ appearing twice and every other label appearing once. Its shape has only even columns, since it is obtained from the column-even terminal shape by deleting the first column, whose height is $2q$.

The pair (τ, U) recovers the original path: the subtableau of labels at most t recovers $\nu^{(t)}$, and the first column has height t for $t < \tau$ and height $t-1$ for $t \ge \tau$. Adding that first column back and shifting $\nu^{(t)}$ one column to the right reconstructs $\lambda^{(t)}$.

Standardize U by replacing the two τ-entries by τ and $\tau+1$, in left-to-right order, and increasing every later label by one. This gives a standard Young tableau of the same even-column shape. Since τ is recorded, merging the entries $\tau, \tau+1$ and decreasing later entries recovers U. Thus the original paths inject into the choice of $\tau \in \{1, \ldots, 2q+1\}$ and an even-column standard tableau of size $2q+2$. Applying lemma 4.681 with $n = q+1$ gives the claimed bound. □

Lemma 4.684 (The central binomial factor absorbs the one-pause loss)**.** *For every $q \ge 3$,*

$$4(2q+1) \le \binom{2q+1}{q}.$$

Proof. At $q = 3$, this is $28 \le 35$. If the inequality holds at q, then

$$\frac{\binom{2q+3}{q+1}}{\binom{2q+1}{q}} = \frac{(2q+3)(2q+2)}{(q+1)(q+2)} \ge \frac{2q+3}{2q+1},$$

because $2(2q+1) \geq q+2$. Hence

$$\binom{2q+3}{q+1} \geq 4(2q+1)\frac{2q+3}{2q+1} = 4(2q+3).$$

Induction proves the claim. $\square$

Proposition 4.685 (Residual C first nonboundary lower correction). *For every residual row C_b with $20 \leq b \leq 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+1} \leq \delta_{2q+1}^{C_b}.$$

Proof. By lemma 4.94 and the coefficient interpretation in lemma 4.95, $|\delta_{2q+1}^{C_b}|$ is the number of semistandard tableaux T of content (2^{2q+1}) whose shape $\lambda(T)$ has $\lambda(T)'$ even and $\ell(\lambda(T)) \geq 2q$. Under lemma 4.97, these tableaux are exactly the column-even Pieri two-strip paths of length $2q+1$ with terminal height at least $2q$. Therefore lemma 4.683 gives

$$|\delta_{2q+1}^{C_b}| \leq (2q+1)2^{2q+2}s_{q+1}.$$

Since $q \geq 21$, lemma 4.684 gives

$$(2q+1)2^{2q+2}s_{q+1} \leq 2^{2q}\binom{2q+1}{q}s_{q+1}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+1}{q}s_{q+1} \leq s_{2q+1}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Combining the displayed inequalities gives $|\delta_{2q+1}^{C_b}| \leq s_{2q+1}/2$, and hence the stated lower bound. $\square$

Lemma 4.686 (C second-nonboundary height-bad paths have a two-pause code). *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+2)}$$

with

$$\ell(\lambda^{(2q+2)}) \geq 2q$$

is at most

$$\left(4 + 16\binom{2q+2}{2}\right)2^{2q}s_{q+2}.$$

Proof. The terminal height is even and at most $2q+2$. Thus the terminal height is either $2q+2$ or $2q$.

If the terminal height is $2q+2$, every step creates a new row. Deleting the first column and shifting the remaining boxes one column left gives an even-column standard tableau of size $2q+2$, and this deletion map is injective by the same reconstruction used in lemma 4.683. Therefore this case has at most

$$2^{2q+2}s_{q+1} \leq 2^{2q+2}s_{q+2}$$

paths, by lemmas 4.681 and 4.682.

If the terminal height is $2q$, exactly two steps fail to create a new row. Let $P \subset \{1, \dots, 2q+2\}$ be this two-element pause set. Deleting the first column and shifting left gives a semistandard tableau of size $2q+4$ whose shape has only even columns, whose labels in P occur twice, and whose other labels occur once. The pair (P, U) recovers the original path: the labels at most t recover the shifted off-column shape at time t, and the first-column height is $t - \#(P \cap \{1, \dots, t\})$.

For fixed P, standardize U by replacing the two equal entries for each label in P by consecutive labels, in left-to-right order, and shifting later labels accordingly. Since P is recorded, merging those consecutive pairs recovers U. Hence, for each P, the paths inject into even-column standard tableaux of size $2q+4$. By lemma 4.681, this gives at most

$$\binom{2q+2}{2} 2^{2q+4} s_{q+2}$$

paths in the two-pause case.

Adding the no-pause and two-pause bounds gives the displayed estimate. □

Lemma 4.687 (The central binomial factor absorbs the two-pause loss)**.** *For every $q \geq 6$,*

$$4 + 16\binom{2q+2}{2} \leq \binom{2q+2}{q}.$$

Proof. At $q = 6$, the inequality is

$$1460 \leq 3003 = \binom{14}{6}.$$

Put $R_q = 4 + 16\binom{2q+2}{2} = 32q^2 + 48q + 20$. For $q \geq 1$,

$$\frac{\binom{2q+4}{q+1}}{\binom{2q+2}{q}} = \frac{(2q+4)(2q+3)}{(q+2)^2} = 4 - \frac{2}{q+2} > 3,$$

whereas

$$R_{q+1} < 3R_q$$

is equivalent to $64q^2 + 32q - 40 > 0$. Thus the binomial side grows by a larger factor than R_q from q to $q+1$, and induction from $q = 6$ proves the claim. □

Proposition 4.688 (Residual C second nonboundary lower correction)**.** *For every residual row C_b with $20 \leq b \leq 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2} s_{2q+2} \leq \delta_{2q+2}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+2}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+2$ with terminal height at least $2q$. By lemma 4.686,

$$|\delta_{2q+2}^{C_b}| \leq \left(4 + 16\binom{2q+2}{2}\right) 2^{2q} s_{q+2}.$$

Since $q \geq 21$, lemma 4.687 gives

$$|\delta_{2q+2}^{C_b}| \leq 2^{2q}\binom{2q+2}{q} s_{q+2}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+2}{q} s_{q+2} \leq s_{2q+2}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta^{C_b}_{2q+2}| \le s_{2q+2}/2$, which implies the claimed lower bound. ☐

Lemma 4.689 (C third-nonboundary height-bad paths have a three-pause code). *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+3)}$$

with

$$\ell(\lambda^{(2q+3)}) \ge 2q$$

is at most

$$\left(16(2q+3) + 64\binom{2q+3}{3}\right)2^{2q}s_{q+3}.$$

Proof. The terminal height is even and at most $2q+3$. Thus it is either $2q+2$ or $2q$.

If the terminal height is $2q+2$, exactly one step fails to create a new row. Recording this pause, deleting the first column, shifting left, and standardizing the repeated pause label gives an injection into the choice of a pause and an even-column standard tableau of size $2q+4$, just as in lemma 4.683. This contributes at most

$$(2q+3)2^{2q+4}s_{q+2} \le 16(2q+3)2^{2q}s_{q+3}$$

paths, using lemmas 4.681 and 4.682.

If the terminal height is $2q$, exactly three steps fail to create a new row. Recording the three-element pause set, deleting the first column, shifting left, and standardizing the three repeated labels gives an injection into the choice of that pause set and an even-column standard tableau of size $2q+6$. This contributes at most

$$\binom{2q+3}{3}2^{2q+6}s_{q+3} = 64\binom{2q+3}{3}2^{2q}s_{q+3}$$

paths. Adding the two cases gives the claim. ☐

Lemma 4.690 (The central binomial factor absorbs the three-pause loss). *For every $q \ge 21$,*

$$16(2q+3) + 64\binom{2q+3}{3} \le \binom{2q+3}{q}.$$

Proof. At $q = 21$,

$$16 \cdot 45 + 64\binom{45}{3} = 908880 < 3773655750150 = \binom{45}{21}.$$

Let

$$P(q) = 16(2q+3) + 64\binom{2q+3}{3}.$$

For $q \ge 21$,

$$P(q+1) < 2P(q),$$

because

$$2P(q) - P(q+1) = \frac{16}{3}(16q^3 - 94q - 93) > 0.$$

On the other hand,

$$\frac{\binom{2q+5}{q+1}}{\binom{2q+3}{q}} = \frac{(2q+5)(2q+4)}{(q+1)(q+4)} > 3.$$

Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 21$ proves the inequality. ☐

Proposition 4.691 (Residual C third nonboundary lower correction). *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+3} \le \delta_{2q+3}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+3}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+3$ with terminal height at least $2q$. By lemma 4.689,

$$|\delta_{2q+3}^{C_b}| \le \left(16(2q+3) + 64\binom{2q+3}{3}\right)2^{2q}s_{q+3}.$$

Since $q \ge 21$, lemma 4.690 gives

$$|\delta_{2q+3}^{C_b}| \le 2^{2q}\binom{2q+3}{q}s_{q+3}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+3}{q}s_{q+3} \le s_{2q+3}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+3}^{C_b}| \le s_{2q+3}/2$, which implies the claimed lower bound. □

Lemma 4.692 (C fourth-nonboundary height-bad paths have a four-pause code). *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+4)}$$

with

$$\ell(\lambda^{(2q+4)}) \ge 2q$$

is at most

$$\left(16 + 64\binom{2q+4}{2} + 256\binom{2q+4}{4}\right)2^{2q}s_{q+4}.$$

Proof. The terminal height is even and at most $2q+4$. Thus it is

$$2q+4, \qquad 2q+2, \qquad \text{or} \qquad 2q.$$

If the terminal height is $2q+4$, every step creates a new row. Deleting the first column and shifting left gives an injection into even-column standard tableaux of size $2q+4$, as in lemma 4.686. This gives at most

$$2^{2q+4}s_{q+2} \le 16\,2^{2q}s_{q+4}$$

paths by lemmas 4.681 and 4.682.

If the terminal height is $2q+2$, exactly two steps fail to create a new row. Recording the two-element pause set, deleting the first column, shifting left, and standardizing the two repeated pause labels injects the paths into the choice of the pause set and an even-column standard tableau of size $2q+6$. This contributes at most

$$\binom{2q+4}{2}2^{2q+6}s_{q+3} \le 64\binom{2q+4}{2}2^{2q}s_{q+4}$$

paths.

If the terminal height is $2q$, exactly four steps fail to create a new row. The same deletion and standardization construction with the four-element pause set injects the paths into the choice of that pause set and an even-column standard tableau of size $2q+8$. This contributes at most

$$\binom{2q+4}{4}2^{2q+8}s_{q+4} = 256\binom{2q+4}{4}2^{2q}s_{q+4}$$

paths. Adding the three cases gives the claim. $\square$

Lemma 4.693 (The central binomial factor absorbs the four-pause loss)**.** *For every $q \geq 21$,*

$$16 + 64\binom{2q+4}{2} + 256\binom{2q+4}{4} \leq \binom{2q+4}{q}.$$

Proof. At $q = 21$,

$$16 + 64\binom{46}{2} + 256\binom{46}{4} = 41841616 < 6943526580276 = \binom{46}{21}.$$

Let

$$P(q) = 16 + 64\binom{2q+4}{2} + 256\binom{2q+4}{4}.$$

For $q \geq 21$,

$$P(q+1) < 2P(q),$$

because

$$2P(q) - P(q+1) = \frac{16}{3}\big(32q^4 + 32q^3 - 368q^2 - 932q - 657\big) > 0.$$

Indeed the quartic in parentheses is positive at $q = 21$, where it equals 6337227, and its derivative is increasing and positive for $q \geq 21$. On the other hand,

$$\frac{\binom{2q+6}{q+1}}{\binom{2q+4}{q}} = \frac{(2q+6)(2q+5)}{(q+1)(q+5)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 4q + 15$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 21$ proves the inequality. $\square$

Proposition 4.694 (Residual C fourth nonboundary lower correction)**.** *For every residual row C_b with $20 \leq b \leq 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+4} \leq \delta_{2q+4}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+4}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+4$ with terminal height at least $2q$. By lemma 4.692,

$$|\delta_{2q+4}^{C_b}| \leq \left(16 + 64\binom{2q+4}{2} + 256\binom{2q+4}{4}\right)2^{2q}s_{q+4}.$$

Since $q \geq 21$, lemma 4.693 gives

$$|\delta_{2q+4}^{C_b}| \leq 2^{2q}\binom{2q+4}{q}s_{q+4}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+4}{q}s_{q+4} \leq s_{2q+4}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta^{C_b}_{2q+4}| \le s_{2q+4}/2$, which implies the claimed lower bound. □

Lemma 4.695 (C fifth-nonboundary height-bad paths have a five-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+5)}$$

with

$$\ell(\lambda^{(2q+5)}) \ge 2q$$

is at most

$$\left(64(2q+5) + 256\binom{2q+5}{3} + 1024\binom{2q+5}{5}\right)2^{2q}s_{q+5}.$$

Proof. The terminal height is even and at most $2q+5$. Thus it is

$$2q+4, \qquad 2q+2, \qquad \text{or} \qquad 2q.$$

If the terminal height is $2q+4$, exactly one step fails to create a new row. Recording the pause, deleting the first column, shifting left, and standardizing the repeated pause label injects the paths into the choice of that pause and an even-column standard tableau of size $2q+6$. This contributes at most

$$(2q+5)2^{2q+6}s_{q+3} \le 64(2q+5)2^{2q}s_{q+5}$$

paths.

If the terminal height is $2q+2$, exactly three steps fail to create a new row. The same deletion and standardization construction gives at most

$$\binom{2q+5}{3}2^{2q+8}s_{q+4} \le 256\binom{2q+5}{3}2^{2q}s_{q+5}$$

paths.

If the terminal height is $2q$, exactly five steps fail to create a new row. Recording the five-element pause set and standardizing the five repeated labels injects the paths into the choice of that set and an even-column standard tableau of size $2q+10$. This contributes at most

$$\binom{2q+5}{5}2^{2q+10}s_{q+5} = 1024\binom{2q+5}{5}2^{2q}s_{q+5}$$

paths. Adding the three cases gives the claim. □

Lemma 4.696 (The central binomial factor absorbs the five-pause loss)**.** *For every $q \ge 21$,*

$$64(2q+5) + 256\binom{2q+5}{3} + 1024\binom{2q+5}{5} \le \binom{2q+5}{q}.$$

Proof. At $q = 21$,

$$64 \cdot 47 + 256\binom{47}{3} + 1024\binom{47}{5} = 1574907584 < 12551759587422 = \binom{47}{21}.$$

Let

$$P(q) = 64(2q+5) + 256\binom{2q+5}{3} + 1024\binom{2q+5}{5}.$$

For $q \ge 21$,

$$P(q+1) < 2P(q),$$

because

$$2P(q) - P(q+1) = \frac{64}{15}(64q^5 + 160q^4 - 1120q^3 - 5560q^2 - 9054q - 5415) > 0.$$

Indeed the polynomial in parentheses is positive at $q = 21$, where it equals 279479595. Its derivative equals 66436626 at $q = 21$, and its second derivative is positive for $q \geq 21$, so the polynomial remains positive thereafter. On the other hand,

$$\frac{\binom{2q+7}{q+1}}{\binom{2q+5}{q}} = \frac{(2q+7)(2q+6)}{(q+1)(q+6)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 5q + 24$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 21$ proves the inequality. ☐

Proposition 4.697 (Residual C fifth nonboundary lower correction)**.** *For every residual row C_b with $20 \leq b \leq 217$, put $q = b + 1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2} s_{2q+5} \leq \delta_{2q+5}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+5}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+5$ with terminal height at least $2q$. By lemma 4.695,

$$|\delta_{2q+5}^{C_b}| \leq \left(64(2q+5) + 256\binom{2q+5}{3} + 1024\binom{2q+5}{5}\right) 2^{2q} s_{q+5}.$$

Since $q \geq 21$, lemma 4.696 gives

$$|\delta_{2q+5}^{C_b}| \leq 2^{2q}\binom{2q+5}{q} s_{q+5}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+5}{q} s_{q+5} \leq s_{2q+5}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+5}^{C_b}| \leq s_{2q+5}/2$, which implies the claimed lower bound. ☐

Lemma 4.698 (C sixth-nonboundary height-bad paths have a six-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+6)}$$

with

$$\ell(\lambda^{(2q+6)}) \geq 2q$$

is at most

$$\left(64 + 256\binom{2q+6}{2} + 1024\binom{2q+6}{4} + 4096\binom{2q+6}{6}\right) 2^{2q} s_{q+6}.$$

Proof. The terminal height is even and at most $2q+6$. Thus it is

$$2q+6, \qquad 2q+4, \qquad 2q+2, \qquad \text{or} \qquad 2q.$$

If the terminal height is $2q+6$, every step creates a new row. Deleting the first column and shifting left gives an injection into even-column standard tableaux of size $2q+6$. This gives at most

$$2^{2q+6}s_{q+3} \le 64\,2^{2q}s_{q+6}$$

paths, by lemmas 4.681 and 4.682.

If the terminal height is $2q+4$, exactly two steps fail to create a new row. Recording the two-element pause set, deleting the first column, shifting left, and standardizing the two repeated pause labels injects the paths into the choice of the pause set and an even-column standard tableau of size $2q+8$. This contributes at most

$$\binom{2q+6}{2}2^{2q+8}s_{q+4} \le 256\binom{2q+6}{2}2^{2q}s_{q+6}$$

paths, by the same two lemmas.

If the terminal height is $2q+2$, exactly four steps fail to create a new row. The same deletion and standardization construction gives at most

$$\binom{2q+6}{4}2^{2q+10}s_{q+5} \le 1024\binom{2q+6}{4}2^{2q}s_{q+6}$$

paths, again by the same two lemmas.

If the terminal height is $2q$, exactly six steps fail to create a new row. Recording the six-element pause set and standardizing the six repeated labels injects the paths into the choice of that set and an even-column standard tableau of size $2q+12$. This contributes at most

$$\binom{2q+6}{6}2^{2q+12}s_{q+6} = 4096\binom{2q+6}{6}2^{2q}s_{q+6}$$

paths, by lemma 4.681. Adding the four cases gives the claim. $\square$

Lemma 4.699 (The central binomial factor absorbs the six-pause loss)**.** *For every $q \ge 21$,*

$$64+256\binom{2q+6}{2}+1024\binom{2q+6}{4}+4096\binom{2q+6}{6} \le \binom{2q+6}{q}.$$

Proof. At $q=21$,

$$64+256\binom{48}{2}+1024\binom{48}{4}+4096\binom{48}{6} = 50463651904 < 22314239266528 = \binom{48}{21}.$$

Let

$$P(q) = 64+256\binom{2q+6}{2}+1024\binom{2q+6}{4}+4096\binom{2q+6}{6}.$$

For $q \ge 21$,

$$P(q+1) < 2P(q),$$

because

$$2P(q)-P(q+1) = \frac{64}{45}(256q^6+1152q^5-5600q^4-50880q^3-143696q^2-189852q-103275) > 0.$$

Indeed the polynomial in parentheses equals 25033257945 at $q=21$, its derivative equals 7112403972 at $q=21$, and its second derivative is

$$32(240q^4+720q^3-2100q^2-9540q-8981),$$

which is positive for $q \geq 21$. On the other hand,

$$\frac{\binom{2q+8}{q+1}}{\binom{2q+6}{q}} = \frac{(2q+8)(2q+7)}{(q+1)(q+7)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 6q + 35$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 21$ proves the inequality. □

Proposition 4.700 (Residual C sixth nonboundary lower correction). *For every residual row C_b with $20 \leq b \leq 217$, put $q = b + 1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+6} \leq \delta_{2q+6}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+6}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q + 6$ with terminal height at least $2q$. By lemma 4.698,

$$|\delta_{2q+6}^{C_b}| \leq \left(64 + 256\binom{2q+6}{2} + 1024\binom{2q+6}{4} + 4096\binom{2q+6}{6}\right)2^{2q}s_{q+6}.$$

Since $q \geq 21$, lemma 4.699 gives

$$|\delta_{2q+6}^{C_b}| \leq 2^{2q}\binom{2q+6}{q}s_{q+6}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+6}{q}s_{q+6} \leq s_{2q+6}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+6}^{C_b}| \leq s_{2q+6}/2$, which implies the claimed lower bound. □

Lemma 4.701 (C seventh-nonboundary height-bad paths have a seven-pause code). *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+7)}$$

with

$$\ell(\lambda^{(2q+7)}) \geq 2q$$

is at most

$$\left(256(2q+7) + 1024\binom{2q+7}{3} + 4096\binom{2q+7}{5} + 16384\binom{2q+7}{7}\right)2^{2q}s_{q+7}.$$

Proof. The terminal height is even and at most $2q + 7$. Thus it is

$$2q+6, \qquad 2q+4, \qquad 2q+2, \qquad \text{or} \qquad 2q.$$

If the terminal height is $2q + 6$, exactly one step fails to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause label injects the paths into the choice of that pause and an even-column standard tableau of size $2q + 8$. This contributes at most

$$(2q+7)2^{2q+8}s_{q+4} \leq 256(2q+7)2^{2q}s_{q+7}$$

paths by lemmas 4.681 and 4.682.

If the terminal height is $2q+4$, exactly three steps fail to create a new row. The same deletion and standardization construction gives at most

$$\binom{2q+7}{3}2^{2q+10}s_{q+5} \le 1024\binom{2q+7}{3}2^{2q}s_{q+7}$$

paths, by the same two lemmas.

If the terminal height is $2q+2$, exactly five steps fail to create a new row. The same construction gives at most

$$\binom{2q+7}{5}2^{2q+12}s_{q+6} \le 4096\binom{2q+7}{5}2^{2q}s_{q+7}$$

paths, by the same two lemmas.

If the terminal height is $2q$, exactly seven steps fail to create a new row. Recording the seven-element pause set and standardizing the seven repeated labels injects the paths into the choice of that set and an even-column standard tableau of size $2q+14$. This contributes at most

$$\binom{2q+7}{7}2^{2q+14}s_{q+7} = 16384\binom{2q+7}{7}2^{2q}s_{q+7}$$

paths by lemma 4.681. Adding the four cases gives the claim. □

Lemma 4.702 (The central binomial factor absorbs the seven-pause loss)**.** *For every $q \ge 21$,*

$$256(2q+7) + 1024\binom{2q+7}{3} + 4096\binom{2q+7}{5} + 16384\binom{2q+7}{7} \le \binom{2q+7}{q}.$$

Proof. At $q = 21$,

$$256 \cdot 49 + 1024\binom{49}{3} + 4096\binom{49}{5} + 16384\binom{49}{7} = 1415224643840 < 39049918716424 = \binom{49}{21}.$$

Let

$$P(q) = 256(2q+7) + 1024\binom{2q+7}{3} + 4096\binom{2q+7}{5} + 16384\binom{2q+7}{7}.$$

For $q \ge 21$,

$$P(q+1) < 2P(q),$$

because

$$2P(q) - P(q+1) = \frac{256}{315}(512q^7 + 3584q^6 - 11200q^5 - 196000q^4 - 855232q^3 - 1876504q^2 - 2193990q - 1124865) > 0.$$

Indeed the polynomial in parentheses equals 1136887862985 at $q = 21$, its derivative equals 375846246474 at $q = 21$, and its second derivative is

$$112(192q^5 + 960q^4 - 2000q^3 - 21000q^2 - 45816q - 33509),$$

which is positive for $q \ge 21$: the inner polynomial equals 942070507 at $q = 21$, its derivative equals 218690184 at $q = 21$, and its second derivative is $240(2q+5)(8q^2+4q-35) > 0$ for $q \ge 21$. On the other hand,

$$\frac{\binom{2q+9}{q+1}}{\binom{2q+7}{q}} = \frac{(2q+9)(2q+8)}{(q+1)(q+8)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 7q + 48$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 21$ proves the inequality. □

Proposition 4.703 (Residual C seventh nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+7} \le \delta_{2q+7}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta^{C_b}_{2q+7}|$ is the number of column-even Pieri two-strip paths of length $2q+7$ with terminal height at least $2q$. By lemma 4.701,

$$|\delta^{C_b}_{2q+7}| \le \left(256(2q+7) + 1024\binom{2q+7}{3} + 4096\binom{2q+7}{5} + 16384\binom{2q+7}{7}\right)2^{2q}s_{q+7}.$$

Since $q \ge 21$, lemma 4.702 gives

$$|\delta^{C_b}_{2q+7}| \le 2^{2q}\binom{2q+7}{q}s_{q+7}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+7}{q}s_{q+7} \le s_{2q+7}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta^{C_b}_{2q+7}| \le s_{2q+7}/2$, which implies the claimed lower bound. ☐

Lemma 4.704 (*C* eighth-nonboundary height-bad paths have an eight-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+8)}$$

with

$$\ell(\lambda^{(2q+8)}) \ge 2q$$

is at most

$$\left(256 + 1024\binom{2q+8}{2} + 4096\binom{2q+8}{4} + 16384\binom{2q+8}{6} + 65536\binom{2q+8}{8}\right)2^{2q}s_{q+8}.$$

Proof. The terminal height is even and at most $2q+8$. Thus it is

$$2q+8, \qquad 2q+6, \qquad 2q+4, \qquad 2q+2, \qquad \text{or} \qquad 2q.$$

These cases have respectively $0, 2, 4, 6, 8$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+8+p$. Therefore the five cases contribute at most

$$2^{2q+8}s_{q+4}, \quad \binom{2q+8}{2}2^{2q+10}s_{q+5}, \quad \binom{2q+8}{4}2^{2q+12}s_{q+6},$$
$$\binom{2q+8}{6}2^{2q+14}s_{q+7}, \quad \binom{2q+8}{8}2^{2q+16}s_{q+8},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+4}, \dots, s_{q+7}$ by s_{q+8}, the sum is the displayed bound. ☐

Lemma 4.705 (The central binomial factor absorbs the eight-pause loss)**.** *For every $q \ge 21$,*

$$256 + 1024\binom{2q+8}{2} + 4096\binom{2q+8}{4} + 16384\binom{2q+8}{6} + 65536\binom{2q+8}{8} \le \binom{2q+8}{q}.$$

Proof. At $q = 21$,

$$256 + 1024\binom{50}{2} + 4096\binom{50}{4} + 16384\binom{50}{6} + 65536\binom{50}{8} = 35446176998656 < 67327446062800 = \binom{50}{21}.$$

Let

$$P(q) = 256 + 1024\binom{2q+8}{2} + 4096\binom{2q+8}{4} + 16384\binom{2q+8}{6} + 65536\binom{2q+8}{8}.$$

For $q \geq 21$, $P(q+1) < 2P(q)$. Indeed, with $x = q - 21$,

$$\begin{aligned}315(2P(q) - P(q+1)) = {}& 131072x^8 + 23330816x^7 + 1809317888x^6 \\ &+ 79798304768x^5 + 2187606622208x^4 \\ &+ 38136956125184x^3 + 412400070438912x^2 \\ &+ 2525280659192832x + 6690395111258880,\end{aligned}$$

which is positive for $x \geq 0$. On the other hand,

$$\frac{\binom{2q+10}{q+1}}{\binom{2q+8}{q}} = \frac{(2q+10)(2q+9)}{(q+1)(q+9)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 8q + 63$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 21$ proves the inequality. □

Proposition 4.706 (Residual C eighth nonboundary lower correction)**.** *For every residual row C_b with $20 \leq b \leq 217$, put $q = b + 1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+8} \leq \delta_{2q+8}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+8}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q + 8$ with terminal height at least $2q$. By lemma 4.704,

$$|\delta_{2q+8}^{C_b}| \leq \left(256 + 1024\binom{2q+8}{2} + 4096\binom{2q+8}{4} + 16384\binom{2q+8}{6} + 65536\binom{2q+8}{8}\right)2^{2q}s_{q+8}.$$

Since $q \geq 21$, lemma 4.705 gives

$$|\delta_{2q+8}^{C_b}| \leq 2^{2q}\binom{2q+8}{q}s_{q+8}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+8}{q}s_{q+8} \leq s_{2q+8}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+8}^{C_b}| \leq s_{2q+8}/2$, which implies the claimed lower bound. □

Lemma 4.707 (C ninth-nonboundary height-bad paths have a nine-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+9)}$$

with

$$\ell(\lambda^{(2q+9)}) \geq 2q$$

is at most

$$\left(1024(2q+9) + 4096\binom{2q+9}{3} + 16384\binom{2q+9}{5} + 65536\binom{2q+9}{7} + 262144\binom{2q+9}{9}\right)2^{2q}s_{q+9}.$$

Proof. The terminal height is even and at most $2q+9$. Thus it is

$$2q+8, \qquad 2q+6, \qquad 2q+4, \qquad 2q+2, \qquad \text{or} \qquad 2q.$$

These cases have respectively $1, 3, 5, 7, 9$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+9+p$. Therefore the five cases contribute at most

$$(2q+9)2^{2q+10}s_{q+5}, \quad \binom{2q+9}{3}2^{2q+12}s_{q+6}, \quad \binom{2q+9}{5}2^{2q+14}s_{q+7},$$

$$\binom{2q+9}{7}2^{2q+16}s_{q+8}, \quad \binom{2q+9}{9}2^{2q+18}s_{q+9},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+5}, \ldots, s_{q+8}$ by s_{q+9}, the sum is the displayed bound. ∎

Lemma 4.708 (The central binomial factor absorbs the nine-pause loss)**.** *For every $q \geq 23$,*

$$1024(2q+9) + 4096\binom{2q+9}{3} + 16384\binom{2q+9}{5} + 65536\binom{2q+9}{7} + 262144\binom{2q+9}{9} \leq \binom{2q+9}{q}.$$

Proof. At $q = 23$,

$$1024 \cdot 55 + 4096\binom{55}{3} + 16384\binom{55}{5} + 65536\binom{55}{7} + 262144\binom{55}{9} = 1680173121915904 < 1866442158555975 = \binom{55}{23}.$$

Let

$$P(q) = 1024(2q+9) + 4096\binom{2q+9}{3} + 16384\binom{2q+9}{5} + 65536\binom{2q+9}{7} + 262144\binom{2q+9}{9}.$$

For $q \geq 23$, $P(q+1) < 2P(q)$. Indeed, with $x = q - 23$,

$$\begin{aligned} 2835(2P(q) - P(q+1)) = {} & 1048576x^9 + 231211008x^8 + 22580035584x^7 \\ & + 1281321861120x^6 + 46534796378112x^5 \\ & + 1120976919724032x^4 + 17896390057066496x^3 \\ & + 182409816397455360x^2 + 1075667566906386432x \\ & + 2791290177480268800, \end{aligned}$$

which is positive for $x \geq 0$. On the other hand,

$$\frac{\binom{2q+11}{q+1}}{\binom{2q+9}{q}} = \frac{(2q+11)(2q+10)}{(q+1)(q+10)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 9q + 80$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 23$ proves the inequality. ∎

Lemma 4.709 (Exact first two ninth C pause-arithmetic checks)**.** *For $q = 21, 22$, put*

$$P(q) = 1024(2q+9) + 4096\binom{2q+9}{3} + 16384\binom{2q+9}{5} + 65536\binom{2q+9}{7} + 262144\binom{2q+9}{9}.$$

Then

$$2^{2q+1}P(q)s_{q+9} \le s_{2q+9}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_ninth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $q = 21, 22$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`3459b8f1c779c878c9a1666f18b27a03`
`548202729f4ddaa23399b6f2c11c242c`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+9} - 2^{2q+1}P(q)s_{q+9}$
21	805149937982464	95209191753417477281958974480308259512574357282473262219794837952
22	1171869936268288	257420773307602153259394842419604777034268451060763250279932333172864.

This proves the claim. ∎

Proposition 4.710 (Residual C ninth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+9} \le \delta_{2q+9}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+9}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+9$ with terminal height at least $2q$. By lemma 4.707, with $P(q)$ denoting the polynomial from lemma 4.709,

$$|\delta_{2q+9}^{C_b}| \le P(q)\,2^{2q}s_{q+9},$$

If $q = 21, 22$, lemma 4.709 gives

$$|\delta_{2q+9}^{C_b}| \le s_{2q+9}/2.$$

If $q \ge 23$, lemma 4.708 gives

$$|\delta_{2q+9}^{C_b}| \le 2^{2q}\binom{2q+9}{q}s_{q+9}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+9}{q}s_{q+9} \le s_{2q+9}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+9}^{C_b}| \le s_{2q+9}/2$ also for $q \ge 23$, which implies the claimed lower bound. ∎

Lemma 4.711 (*C* tenth-nonboundary height-bad paths have a ten-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+10)}$$

with

$$\ell(\lambda^{(2q+10)}) \geq 2q$$

is at most

$$\left(1024 + 4096\binom{2q+10}{2} + 16384\binom{2q+10}{4} + 65536\binom{2q+10}{6} + 262144\binom{2q+10}{8} + 1048576\binom{2q+10}{10}\right)2^{2q}s_{q+10}.$$

Proof. The terminal height is even and at most $2q+10$. Thus it is

$$2q+10, \quad 2q+8, \quad 2q+6, \quad 2q+4, \quad 2q+2, \quad \text{or} \quad 2q.$$

These cases have respectively $0, 2, 4, 6, 8, 10$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+10+p$. Therefore the six cases contribute at most

$$2^{2q+10}s_{q+5}, \quad \binom{2q+10}{2}2^{2q+12}s_{q+6}, \quad \binom{2q+10}{4}2^{2q+14}s_{q+7},$$
$$\binom{2q+10}{6}2^{2q+16}s_{q+8}, \quad \binom{2q+10}{8}2^{2q+18}s_{q+9}, \quad \binom{2q+10}{10}2^{2q+20}s_{q+10},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+5}, \ldots, s_{q+9}$ by s_{q+10}, the sum is the displayed bound. ☐

Lemma 4.712 (The central binomial factor absorbs the ten-pause loss)**.** *For every $q \geq 26$,*

$$1024 + 4096\binom{2q+10}{2} + 16384\binom{2q+10}{4} + 65536\binom{2q+10}{6} + 262144\binom{2q+10}{8} + 1048576\binom{2q+10}{10} \leq \binom{2q+10}{q}.$$

Proof. At $q = 26$,

$$1024 + 4096\binom{62}{2} + 16384\binom{62}{4} + 65536\binom{62}{6} + 262144\binom{62}{8} + 1048576\binom{62}{10} = 113632146094453760 < 209769429934732479 = \binom{62}{26}.$$

Let

$$P(q) = 1024 + 4096\binom{2q+10}{2} + 16384\binom{2q+10}{4} + 65536\binom{2q+10}{6} + 262144\binom{2q+10}{8} + 1048576\binom{2q+10}{10}.$$

For $q \geq 26$, $P(q+1) < 2P(q)$. Indeed, with $x = q - 26$,

$$\begin{aligned} 14175(2P(q) - P(q+1)) = {} & 4194304x^{10} + 1163919360x^9 + 144947281920x^8 \\ & + 10664395407360x^7 + 513170922012672x^6 \\ & + 16868613697044480x^5 + 383409676620922880x^4 \\ & + 5946352734974115840x^3 + 60177808034062516224x^2 \\ & + 358500237510556200960x + 953522452127742336000, \end{aligned}$$

which is positive for $x \geq 0$. On the other hand,

$$\frac{\binom{2q+12}{q+1}}{\binom{2q+10}{q}} = \frac{(2q+12)(2q+11)}{(q+1)(q+11)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 10q + 99$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 26$ proves the inequality. ☐

Lemma 4.713 (Exact first five tenth C pause-arithmetic checks)**.** *For $q = 21, 22, 23, 24, 25$, put*

$$P(q) = 1024 + 4096\binom{2q+10}{2} + 16384\binom{2q+10}{4} + 65536\binom{2q+10}{6} + 262144\binom{2q+10}{8} + 1048576\binom{2q+10}{10}.$$

Then

$$2^{2q+1}P(q)s_{q+10} \le s_{2q+10}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_tenth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $q = 21, 22, 23, 24, 25$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`abf7ac5edc1f162794cea23d37343d9c`
`7dda9415a30f8eab1a622884ebe7b49c`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+10} - 2^{2q+1}P(q)s_{q+10}$
21	16787109734261760	4903171355778686064964164515745269611213904591723677230177156061824
22	25367621153403904	13771958891177334015216861406002714565114571278870369402339849495106432
23	37711207486899200	416565853854389526348706968118043382164513331065273316670258707547514899 20
24	55219289355244544	1353312636651279198190474101281186781557827945299452052793392523423074655906 56
25	79730383933629440	47105248502538765745603226556988880715061459417076097215082305436524109899185 6896.

This proves the claim. ∎

Proposition 4.714 (Residual C tenth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+10} \le \delta_{2q+10}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+10}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+10$ with terminal height at least $2q$. By lemma 4.711, with $P(q)$ denoting the polynomial from lemma 4.713,

$$|\delta_{2q+10}^{C_b}| \le P(q)\, 2^{2q} s_{q+10}.$$

If $q = 21, 22, 23, 24, 25$, lemma 4.713 gives

$$|\delta_{2q+10}^{C_b}| \le s_{2q+10}/2.$$

If $q \ge 26$, lemma 4.712 gives

$$|\delta_{2q+10}^{C_b}| \le 2^{2q}\binom{2q+10}{q}s_{q+10}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+10}{q}s_{q+10} \le s_{2q+10}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+10}^{C_b}| \le s_{2q+10}/2$ also for $q \ge 26$, which implies the claimed lower bound. ∎

Lemma 4.715 (*C* eleventh-nonboundary height-bad paths have an eleven-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+11)}$$

with

$$\ell(\lambda^{(2q+11)}) \geq 2q$$

is at most

$$\left(4096\binom{2q+11}{1} + 16384\binom{2q+11}{3} + 65536\binom{2q+11}{5} + 262144\binom{2q+11}{7} + 1048576\binom{2q+11}{9} + 4194304\binom{2q+11}{11}\right)2^{2q}s_{q+11}.$$

Proof. The terminal height is even and at most $2q + 11$. Thus it is

$$2q + 10, \quad 2q + 8, \quad 2q + 6, \quad 2q + 4, \quad 2q + 2, \quad \text{or} \quad 2q.$$

These cases have respectively $1, 3, 5, 7, 9, 11$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q + 11 + p$. Therefore the six cases contribute at most

$$\binom{2q+11}{1}2^{2q+12}s_{q+6}, \quad \binom{2q+11}{3}2^{2q+14}s_{q+7}, \quad \binom{2q+11}{5}2^{2q+16}s_{q+8},$$

$$\binom{2q+11}{7}2^{2q+18}s_{q+9}, \quad \binom{2q+11}{9}2^{2q+20}s_{q+10}, \quad \binom{2q+11}{11}2^{2q+22}s_{q+11},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+6}, \ldots, s_{q+10}$ by s_{q+11}, the sum is the displayed bound. ☐

Lemma 4.716 (The central binomial factor absorbs the eleven-pause loss)**.** *For every $q \geq 28$,*

$$4096\binom{2q+11}{1} + 16384\binom{2q+11}{3} + 65536\binom{2q+11}{5} + 262144\binom{2q+11}{7} + 1048576\binom{2q+11}{9} + 4194304\binom{2q+11}{11} \leq \binom{2q+11}{q}.$$

Proof. At $q = 28$,

$$4096\binom{67}{1} + 16384\binom{67}{3} + 65536\binom{67}{5} + 262144\binom{67}{7} + 1048576\binom{67}{9} + 4194304\binom{67}{11} = 5435009637461815296 < 5864393356544251760 = \binom{67}{28}.$$

Let

$$P(q) = 4096\binom{2q+11}{1} + 16384\binom{2q+11}{3} + 65536\binom{2q+11}{5} + 262144\binom{2q+11}{7} + 1048576\binom{2q+11}{9} + 4194304\binom{2q+11}{11}.$$

For $q \geq 28$, $P(q+1) < 2P(q)$. Indeed, with $x = q - 28$,

$$\begin{aligned} 155925(2P(q) - P(q+1)) = {} & 33554432x^{11} + 11072962560x^{10} + 1657022709760x^{9} \\ & + 148393961717760x^{8} + 8834179154313216x^{7} \\ & + 366968144391045120x^{6} + 10849539239009320960x^{5} \\ & + 228200591272315453440x^{4} + 3344494768984406884352x^{3} \\ & + 32506342800225239531520x^{2} + 188410100820385114644480x \\ & + 492831435774247034572800, \end{aligned}$$

which is positive for $x \geq 0$. On the other hand,

$$\frac{\binom{2q+13}{q+1}}{\binom{2q+11}{q}} = \frac{(2q+13)(2q+12)}{(q+1)(q+12)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 11q + 120$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 28$ proves the inequality. ☐

Lemma 4.717 (Exact first seven eleventh C pause-arithmetic checks)**.** *For $21 \le q \le 27$, put*

$$P(q) = 4096\binom{2q+11}{1} + 16384\binom{2q+11}{3} + 65536\binom{2q+11}{5} + 262144\binom{2q+11}{7} + 1048576\binom{2q+11}{9} + 4194304\binom{2q+11}{11}.$$

Then

$$2^{2q+1}P(q)s_{q+11} \le s_{2q+11}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_eleventh_frontier_gmp_certificate.log`

computes the displayed integer inequality for $21 \le q \le 27$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`de69669b8df7784d87b33232131fee8f`
`a6eb22c830fdcc8eeb066d851a409c0c`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+11} - 2^{2q+1}P(q)s_{q+11}$
21	324393072099643392	257362697029215877963949523297153447302766944466517748161416488855680
22	508584122346582016	750559067883129880349273491327444158124656900055892284235763406948255360
23	783391012291121152	2353589660314073337128089738870306304406389791628228318393371452705364754944
24	1187133100402733056	7916862822025459545211278821831496164228888938237813980332978702989743176972672
25	1771912890992054272	28498624463510700078311199299089653133614164147508330272882543415215643039488592640
26	2607774406749368320	109541225766556320001716025709278150174781456225226335841962103028640068955141950940416
27	3787894083819409408	448652128227809777832975850852768721615133669475553568921414615289569671480179369963642880.

This proves the claim. ∎

Proposition 4.718 (Residual C eleventh nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+11} \le \delta_{2q+11}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+11}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+11$ with terminal height at least $2q$. By lemma 4.715, with $P(q)$ denoting the polynomial from lemma 4.717,

$$|\delta_{2q+11}^{C_b}| \le P(q)\, 2^{2q} s_{q+11}.$$

If $21 \le q \le 27$, lemma 4.717 gives

$$|\delta_{2q+11}^{C_b}| \le s_{2q+11}/2.$$

If $q \ge 28$, lemma 4.716 gives

$$|\delta_{2q+11}^{C_b}| \le 2^{2q}\binom{2q+11}{q} s_{q+11}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+11}{q} s_{q+11} \le s_{2q+11}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+11}^{C_b}| \le s_{2q+11}/2$ also for $q \ge 28$, which implies the claimed lower bound. ∎

Lemma 4.719 (*C* twelfth-nonboundary height-bad paths have a twelve-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+12)}$$

with

$$\ell(\lambda^{(2q+12)}) \geq 2q$$

is at most

$$\left(4096 + 16384\binom{2q+12}{2} + 65536\binom{2q+12}{4} + 262144\binom{2q+12}{6} + 1048576\binom{2q+12}{8} + 4194304\binom{2q+12}{10} + 16777216\binom{2q+12}{12}\right)2^{2q}s_{q+12}.$$

Proof. The terminal height is even and at most $2q+12$. Thus it is

$$2q+12, \quad 2q+10, \quad 2q+8, \quad 2q+6, \quad 2q+4, \quad 2q+2, \quad \text{or} \quad 2q.$$

These cases have respectively $0, 2, 4, 6, 8, 10, 12$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+12+p$. Therefore the seven cases contribute at most

$$2^{2q+12}s_{q+6}, \quad \binom{2q+12}{2}2^{2q+14}s_{q+7}, \quad \binom{2q+12}{4}2^{2q+16}s_{q+8},$$
$$\binom{2q+12}{6}2^{2q+18}s_{q+9}, \quad \binom{2q+12}{8}2^{2q+20}s_{q+10}, \quad \binom{2q+12}{10}2^{2q+22}s_{q+11},$$
$$\binom{2q+12}{12}2^{2q+24}s_{q+12},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+6}, \ldots, s_{q+11}$ by s_{q+12}, the sum is the displayed bound. ∎

Lemma 4.720 (The central binomial factor absorbs the twelve-pause loss)**.** *For every $q \geq 31$,*

$$4096 + 16384\binom{2q+12}{2} + 65536\binom{2q+12}{4} + 262144\binom{2q+12}{6} + 1048576\binom{2q+12}{8} + 4194304\binom{2q+12}{10} + 16777216\binom{2q+12}{12} \leq \binom{2q+12}{q}.$$

Proof. At $q = 31$,

$$4096 + 16384\binom{74}{2} + 65536\binom{74}{4} + 262144\binom{74}{6} + 1048576\binom{74}{8} + 4194304\binom{74}{10} + 16777216\binom{74}{12} = 371189423168378720256 < 665858611887899825472 = \binom{74}{31}.$$

Let

$$P(q) = 4096 + 16384\binom{2q+12}{2} + 65536\binom{2q+12}{4} + 262144\binom{2q+12}{6} + 1048576\binom{2q+12}{8} + 4194304\binom{2q+12}{10} + 16777216\binom{2q+12}{12}.$$

For $q \geq 31$, $P(q+1) < 2P(q)$. Indeed, with $x = q - 31$,

$$\begin{aligned} 467775(2P(q) - P(q+1)) = {} & 67108864x^{12} + 26776436736x^{11} + 4887236575232x^{10} \\ & + 539476120043520x^{9} + 40103480490196992x^{8} \\ & + 2114600225942274048x^{7} + 81074772196129243136x^{6} \\ & + 2276668272958085529600x^{5} + 46454921071694256799744x^{4} \\ & + 671432360820605377314816x^{3} + 6521508020414066359468032x^{2} \\ & + 38194452157657256601108480x + 101923801617413045181542400, \end{aligned}$$

which is positive for $x \geq 0$. On the other hand,

$$\frac{\binom{2q+14}{q+1}}{\binom{2q+12}{q}} = \frac{(2q+14)(2q+13)}{(q+1)(q+13)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 12q + 143$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 31$ proves the inequality. ∎

Lemma 4.721 (Exact first ten twelfth C pause-arithmetic checks)**.** *For $21 \le q \le 30$, put*

$$P(q) = 4096 + 16384\binom{2q+12}{2} + 65536\binom{2q+12}{4} + 262144\binom{2q+12}{6} + 1048576\binom{2q+12}{8} + 4194304\binom{2q+12}{10} + 16777216\binom{2q+12}{12}.$$

Then

$$2^{2q+1}P(q)s_{q+12} \le s_{2q+12}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twelfth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $21 \le q \le 30$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`027740d3f1fc0d56fcb27fa4e8d1f32a`
`35bf7f4d0938ae970eeaadef8c4ba875`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+12} - 2^{2q+1}P(q)s_{q+12}$
21	5856171146995879936	13737887743093491029874945629543498680225743342441692732099355335971712
22	9518962187314073600	41649163935462476049958505836798783090818697479057364159395837163257243520
23	15182710965852655616	135329630065808428579424292893519281225757899062377328264131211583751558920064
24	23796267139390607360	471052121438170656227510842657968048509244598133358450149784571617082575423565056
25	36695884124388970496	1752662358687885188987478205394018051759346782906849304190277599933097187266142260992
26	55739965040872656896	6955857126069033829258591712286204229109865339121174753159805133951078138602493998424320
27	83483175402122792960	29386673295598252119315606146972896461970012686091855147040625434158518678060782984083104000
28	123399400507042336768	131909085288222166096090487444534010029062356675207583449019800324247261789364850799927165421056
29	180164856350448209920	627984678371490556154911685166126439591413733396445863377606242170773209583438050710786209392034304
30	260014766495085432832	3165506839235673119694504216537225971338421328159148455404020879282993502672554389276865198890883013120.

This proves the claim. ∎

Proposition 4.722 (Residual C twelfth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+12} \le \delta_{2q+12}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+12}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+12$ with terminal height at least $2q$. By lemma 4.719, with $P(q)$ denoting the polynomial from lemma 4.721,

$$|\delta_{2q+12}^{C_b}| \le P(q)\,2^{2q}s_{q+12}.$$

If $21 \le q \le 30$, lemma 4.721 gives

$$|\delta_{2q+12}^{C_b}| \le s_{2q+12}/2.$$

If $q \ge 31$, lemma 4.720 gives

$$|\delta_{2q+12}^{C_b}| \le 2^{2q}\binom{2q+12}{q}s_{q+12}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+12}{q}s_{q+12} \le s_{2q+12}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+12}^{C_b}| \le s_{2q+12}/2$ also for $q \ge 31$, which implies the claimed lower bound. ∎

Lemma 4.723 (C thirteenth-nonboundary height-bad paths have a thirteen-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+13)}$$

with

$$\ell(\lambda^{(2q+13)}) \geq 2q$$

is at most

$$\left(16384\binom{2q+13}{1} + 65536\binom{2q+13}{3} + 262144\binom{2q+13}{5} + 1048576\binom{2q+13}{7} + 4194304\binom{2q+13}{9} + 16777216\binom{2q+13}{11} + 67108864\binom{2q+13}{13}\right)2^{2q}s_{q+13}.$$

Proof. The terminal height is even and at most $2q+13$. Thus it is

$$2q+12, \quad 2q+10, \quad 2q+8, \quad 2q+6, \quad 2q+4, \quad 2q+2, \quad \text{or} \quad 2q.$$

These cases have respectively $1, 3, 5, 7, 9, 11, 13$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+13+p$. Therefore the seven cases contribute at most

$$\binom{2q+13}{1}2^{2q+14}s_{q+7}, \quad \binom{2q+13}{3}2^{2q+16}s_{q+8}, \quad \binom{2q+13}{5}2^{2q+18}s_{q+9},$$
$$\binom{2q+13}{7}2^{2q+20}s_{q+10}, \quad \binom{2q+13}{9}2^{2q+22}s_{q+11}, \quad \binom{2q+13}{11}2^{2q+24}s_{q+12},$$
$$\binom{2q+13}{13}2^{2q+26}s_{q+13},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+7}, \dots, s_{q+12}$ by s_{q+13}, the sum is the displayed bound. $\square$

Lemma 4.724 (The central binomial factor absorbs the thirteen-pause loss)**.** *For every $q \geq 33$,*

$$16384\binom{2q+13}{1} + 65536\binom{2q+13}{3} + 262144\binom{2q+13}{5} + 1048576\binom{2q+13}{7} + 4194304\binom{2q+13}{9} + 16777216\binom{2q+13}{11} + 67108864\binom{2q+13}{13} \leq \binom{2q+13}{q}.$$

Proof. At $q = 33$,

$$16384\binom{79}{1} + 65536\binom{79}{3} + 262144\binom{79}{5} + 1048576\binom{79}{7} + 4194304\binom{79}{9} + 16777216\binom{79}{11} + 67108864\binom{79}{13} = 17864298942435109027840 < 18723298265888530356770 = \binom{79}{33}.$$

Let

$$P(q) = 16384\binom{2q+13}{1} + 65536\binom{2q+13}{3} + 262144\binom{2q+13}{5} + 1048576\binom{2q+13}{7} + 4194304\binom{2q+13}{9} + 16777216\binom{2q+13}{11} + 67108864\binom{2q+13}{13}.$$

For $q \geq 33$, $P(q+1) < 2P(q)$. Indeed, with $x = q - 33$,

$$\begin{aligned}
6081075(2P(q) - P(q+1)) = {} & 536870912x^{13} + 247765925888x^{12} + 52683411030016x^{11} \\
& + 6833111170023424x^{10} + 603052064423018496x^{9} \\
& + 38235592030949474304x^{8} + 1791431987680749027328x^{7} \\
& + 62782980715340357435392x^{6} + 1645392427080320194445312x^{5} \\
& + 31837793611414160915038208x^{4} + 441932813329921918349869056x^{3} \\
& + 4165225135993797091904323584x^{2} \\
& + 23877555284696150243189882880x + 62833122000287032433950310400,
\end{aligned}$$

which is positive for $x \geq 0$. On the other hand,

$$\frac{\binom{2q+15}{q+1}}{\binom{2q+13}{q}} = \frac{(2q+15)(2q+14)}{(q+1)(q+14)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 13q + 168$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 33$ proves the inequality. $\square$

Lemma 4.725 (Exact first twelve thirteenth C pause-arithmetic checks)**.** *For $21 \le q \le 32$, put*

$$P(q) = 16384\binom{2q+13}{1} + 65536\binom{2q+13}{3} + 262144\binom{2q+13}{5} + 1048576\binom{2q+13}{7} + 4194304\binom{2q+13}{9} + 16777216\binom{2q+13}{11} + 67108864\binom{2q+13}{13}.$$

Then

$$2^{2q+1}P(q)s_{q+13} \le s_{2q+13}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_thirteenth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $21 \le q \le 32$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`6f6ffcfb724ca85fa51310916cd2451e`
`4b38d08bcf0dd224719ce3d8b809e3e9`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+13} - 2^{2q+1}P(q)s_{q+13}$
21	994215784681631088864	731183253555922246274145306849776724283935343412240521338981033999058560
22	167435930009130385408	2349086344497982960380567450134420546006293822947033888650532606552947540480
23	276363361539293691904	7915807283761492701741559988361827399990308007320009827870256865677681228151168
24	447739369047974821888	28498374779999066839524516000230618733300521031838837186021836373553965642135948032
25	712955835787213717504	109541166127051607981301979498068684490946462361460615532644080461796654207629385278720
26	1117151165606205079552	448652113835933266852742777764745511772674732526011424622707823204933417946550301817528320
27	1724410263482076872704	1954211161170261374923664714989792376243604267813600314045649818913786405050060276086835708160
28	2624645313339026522112	9035761316222856032195612149457571806209086085064215683959163378490951478285117127173212392044032
29	3942614982711209148416	44272870337027816746858558646361549656504231610775669646028860443921659475011087072106629992943662592
30	5849642632546285928448	229499010251281522668378046504096718321829854693072240153475734580282679151017260965635067349105902389248
31	8578715686145537064960	1256676780468853148224502722041231129903054470193866885549402798554126364708883036010445696292421349143986688
32	12443791119168545046528	7258238000562607250224480367208885061023620028393530822223165594504067877769292050894174201678791838370956768256.

This proves the claim. ∎

Proposition 4.726 (Residual C thirteenth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+13} \le \delta_{2q+13}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+13}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+13$ with terminal height at least $2q$. By lemma 4.723, with $P(q)$ denoting the polynomial from lemma 4.725,

$$|\delta_{2q+13}^{C_b}| \le P(q)\,2^{2q}s_{q+13}.$$

If $21 \le q \le 32$, lemma 4.725 gives

$$|\delta_{2q+13}^{C_b}| \le s_{2q+13}/2.$$

If $q \ge 33$, lemma 4.724 gives

$$|\delta_{2q+13}^{C_b}| \le 2^{2q}\binom{2q+13}{q}s_{q+13}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+13}{q}s_{q+13} \le s_{2q+13}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+13}^{C_b}| \le s_{2q+13}/2$ also for $q \ge 33$, which implies the claimed lower bound. ∎

Lemma 4.727 (*C* fourteenth-nonboundary height-bad paths have a fourteen-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+14)}$$

with

$$\ell(\lambda^{(2q+14)}) \geq 2q$$

is at most

$$\Bigg(16384\binom{2q+14}{0} + 65536\binom{2q+14}{2} + 262144\binom{2q+14}{4} + 1048576\binom{2q+14}{6} + 4194304\binom{2q+14}{8} + 16777216\binom{2q+14}{10} + 67108864\binom{2q+14}{12} + 268435456\binom{2q+14}{14}\Bigg)2^{2q}s_{q+14}.$$

Proof. The terminal height is even and at most $2q+14$. Thus it is

$$2q+14, \quad 2q+12, \quad 2q+10, \quad 2q+8, \quad 2q+6, \quad 2q+4, \quad 2q+2, \quad \text{or} \quad 2q.$$

These cases have respectively $0, 2, 4, 6, 8, 10, 12, 14$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+14+p$. Therefore the eight cases contribute at most

$$\binom{2q+14}{0}2^{2q+14}s_{q+7}, \quad \binom{2q+14}{2}2^{2q+16}s_{q+8}, \quad \binom{2q+14}{4}2^{2q+18}s_{q+9},$$
$$\binom{2q+14}{6}2^{2q+20}s_{q+10}, \quad \binom{2q+14}{8}2^{2q+22}s_{q+11}, \quad \binom{2q+14}{10}2^{2q+24}s_{q+12},$$
$$\binom{2q+14}{12}2^{2q+26}s_{q+13}, \quad \binom{2q+14}{14}2^{2q+28}s_{q+14},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+7}, \ldots, s_{q+13}$ by s_{q+14}, the sum is the displayed bound. ☐

Lemma 4.728 (The central binomial factor absorbs the fourteen-pause loss)**.** *For every $q \geq 36$,*

$$16384\binom{2q+14}{0} + 65536\binom{2q+14}{2} + 262144\binom{2q+14}{4} + 1048576\binom{2q+14}{6} + 4194304\binom{2q+14}{8} + 16777216\binom{2q+14}{10} + 67108864\binom{2q+14}{12} + 268435456\binom{2q+14}{14} \leq \binom{2q+14}{q}.$$

Proof. At $q = 36$, the left side is

$$1228572339217567202885632 < 2141368094445751179086598 = \binom{86}{36}.$$

Let $P(q)$ denote the displayed left side. For $0 \leq p \leq 14$,

$$\frac{\binom{2q+16}{p}}{\binom{2q+14}{p}} \leq \frac{(2q+16)(2q+15)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 36),$$

because $4q^2 - 50q - 236 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+16}{q+1}}{\binom{2q+14}{q}} = \frac{(2q+16)(2q+15)}{(q+1)(q+15)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 14q + 195$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 36$ proves the inequality. ☐

Lemma 4.729 (Exact first fifteen fourteenth C pause-arithmetic checks)**.** *For $21 \le q \le 35$, put*

$$P(q) = 16384\binom{2q+14}{0} + 65536\binom{2q+14}{2} + 262144\binom{2q+14}{4} + 1048576\binom{2q+14}{6} + 4194304\binom{2q+14}{8} + 16777216\binom{2q+14}{10} + 67108864\binom{2q+14}{12} + 268435456\binom{2q+14}{14}.$$

Then

$$2^{2q+1} P(q) s_{q+14} \le s_{2q+14}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_fourteenth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $21 \le q \le 35$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`315dcc3f51bd178dba5b186d4cb4f3e0`
`0c636284f3b43d8428915d46a90af82b`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+14} - 2^{2q+1}P(q)s_{q+14}$
21	1596271720409684787200	30917529858094002525675633417861850565638275718143447666636062145035675520
22	2783455765485702692864	1326722045731361459505006408226455512169208420360096695426849301832089609153 28
23	4751439282346409738240	47038978333693918541277537384201679152883202544940272438570869155306294350134 8096
24	7952613960499958595584	17524960645367438223002643107891947202410010290614948200389653722792158025466 12195072
25	13069126500807444807680	69558150386219031731922483004113319002274680942450337819791079733843152737035 13778708736
26	21114496385003122933760	29386662551758018082154847284853106262575512858314384278735038682345181630868 645711657918720
27	33574047686425773228032	13190908252057244194598164507571618674047768968753504093590540250901850359002 1353816558633190912
28	52596819785025308016640	62798467765173204402418550603424162925173500850873894971197450548175566718120 37592719811819711 37024
29	81255101945107809845248	31655068390466403082763746648305704856035699565809917596278019294222665779009 04929727336361298654190080
30	123892006741556730347520	16868160644315292137234793151061476357142219647945787848677390455912307942796 862950762802857240886238943744
31	186582701460026967343104	94879010830031532349875873741874771299670970548527039813930028602579942534927 063585645793842574676669159095296
32	277741219368227086221312	56251297361757894973871505141360237043829681482279076836928061320443771798450 4076997473538041680135009142325285888
33	408912361098124892585984	35104972877347779883521543952142040248106635448386585774112480045049469860952 11218898758792608835147398481712965860352
34	595797281870381039828992	23031461318491612561361483118389880070206975243636714799442705876023076415533 79974090893869250277731023689891426608091 8528
35	859572181508965409112064	15865776319387448332653513932405926192909507302965849366664309811554099372856 9364080099703614686993911702096986997787803625472.

This proves the claim. ∎

Proposition 4.730 (Residual C fourteenth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2} s_{2q+14} \le \delta_{2q+14}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+14}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+14$ with terminal height at least $2q$. By lemma 4.727, with $P(q)$ denoting the polynomial from lemma 4.729,

$$|\delta_{2q+14}^{C_b}| \le P(q)\, 2^{2q} s_{q+14}.$$

If $21 \le q \le 35$, lemma 4.729 gives

$$|\delta_{2q+14}^{C_b}| \le s_{2q+14}/2.$$

If $q \ge 36$, lemma 4.728 gives

$$|\delta_{2q+14}^{C_b}| \le 2^{2q}\binom{2q+14}{q} s_{q+14}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+14}{q} s_{q+14} \le s_{2q+14}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta^{C_b}_{2q+14}| \leq s_{2q+14}/2$ also for $q \geq 36$, which implies the claimed lower bound. □

Lemma 4.731 (Even-column standard tableaux have a fixed-point-free involution count)**.** *For every $n \geq 1$, the number of standard Young tableaux of size $2n$ whose columns all have even length is*

$$(2n-1)!! = \frac{(2n)!}{2^n n!}.$$

Proof. Under the Robinson–Schensted correspondence, involutions on $\{1,\dots,2n\}$ are in bijection with standard Young tableaux of size $2n$. Rains [5, Lemma 3.3] records the Schensted fact that the number of fixed points of the involution is the number of odd columns of the tableau. Thus the tableaux with only even columns are exactly the fixed-point-free involutions. These are perfect matchings of $2n$ labelled points, counted by

$$(2n-1)(2n-3)\cdots 3\cdot 1 = \frac{(2n)!}{2^n n!}.$$

□

Lemma 4.732 (C fifteenth-nonboundary height-bad paths have a fifteen-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+15)}$$

with

$$\ell(\lambda^{(2q+15)}) \geq 2q$$

is at most

$$\begin{aligned}\Bigg(&65536\binom{2q+15}{1} + 262144\binom{2q+15}{3} + 1048576\binom{2q+15}{5} + 4194304\binom{2q+15}{7}\\ &+ 16777216\binom{2q+15}{9} + 67108864\binom{2q+15}{11} + 268435456\binom{2q+15}{13} + 1073741824\binom{2q+15}{15}\Bigg)2^{2q}s_{q+15}.\end{aligned}$$

Proof. The terminal height is even and at most $2q+15$. Thus it is

$$2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $1,3,5,7,9,11,13,15$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+15+p$. Therefore the eight cases contribute at most

$$\binom{2q+15}{1}2^{2q+16}s_{q+8},\quad \binom{2q+15}{3}2^{2q+18}s_{q+9},\quad \binom{2q+15}{5}2^{2q+20}s_{q+10},$$

$$\binom{2q+15}{7}2^{2q+22}s_{q+11},\quad \binom{2q+15}{9}2^{2q+24}s_{q+12},\quad \binom{2q+15}{11}2^{2q+26}s_{q+13},$$

$$\binom{2q+15}{13}2^{2q+28}s_{q+14},\quad \binom{2q+15}{15}2^{2q+30}s_{q+15},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+8},\dots,s_{q+14}$ by s_{q+15}, the sum is the displayed bound. □

Lemma 4.733 (The first fifteenth C pause case is absorbed by the fixed-point-free count)**.** *For $q = 21$, the number of column-even Pieri two-strip paths of length $2q + 15$ with terminal height at least $2q$ is at most*

$$52920726139837306981793811162459494689272852275711217409665468750 \leq \frac{1}{2} s_{57}.$$

Proof. Use the pause-set injection from lemma 4.732, but count the target even-column standard tableaux exactly by lemma 4.731. For $q = 21$, this gives the upper bound

$$\sum_{\substack{1 \leq p \leq 15 \\ p \text{ odd}}} \binom{57}{p} (56 + p)!! = 52920726139837306981793811162459494689272852275711217409665468750.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_fifteenth_frontier_gmp_certificate.log`

checks the displayed integer sum and verifies

$$\begin{aligned} &s_{57} - 2 \cdot 52920726139837306981793811162459494689272852275711217409665468750 \\ &\quad = 23535921044447447581362090995492172207851717396924002454173990495180668066 28 > 0. \end{aligned}$$

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`5f829fb541bf8d84f097ff14aa6be272`
`508554407503bd74b37532e62c1484b9`.

□

Lemma 4.734 (The central binomial factor absorbs the fifteen-pause loss)**.** *For every $q \geq 38$,*

$$\begin{aligned} &65536\binom{2q+15}{1} + 262144\binom{2q+15}{3} + 1048576\binom{2q+15}{5} + 4194304\binom{2q+15}{7} \\ &\quad + 16777216\binom{2q+15}{9} + 67108864\binom{2q+15}{11} + 268435456\binom{2q+15}{13} + 1073741824\binom{2q+15}{15} \leq \binom{2q+15}{q}. \end{aligned}$$

Proof. At $q = 38$, the left side is

$$5939573054020994286583808 0 < 60468953239450827809517780 = \binom{91}{38}.$$

Let $P(q)$ denote the displayed left side. For $1 \leq p \leq 15$,

$$\frac{\binom{2q+17}{p}}{\binom{2q+15}{p}} \leq \frac{(2q+17)(2q+16)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 38),$$

because $4q^2 - 54q - 268 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+17}{q+1}}{\binom{2q+15}{q}} = \frac{(2q+17)(2q+16)}{(q+1)(q+16)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 15q + 224$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 38$ proves the inequality. □

Lemma 4.735 (Exact fifteenth C pause-arithmetic checks after the first row)**.** *For $22 \leq q \leq 37$, put*

$$\begin{aligned} P(q) = {}&65536\binom{2q+15}{1} + 262144\binom{2q+15}{3} + 1048576\binom{2q+15}{5} + 4194304\binom{2q+15}{7} \\ &+ 16777216\binom{2q+15}{9} + 67108864\binom{2q+15}{11} + 268435456\binom{2q+15}{13} + 1073741824\binom{2q+15}{15}. \end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+15} \leq s_{2q+15}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_fifteenth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $22 \leq q \leq 37$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`5f829fb541bf8d84f097ff14aa6be272`
`508554407503bd74b37532e62c1484b9`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+15} - 2^{2q+1}P(q)s_{q+15}$
22	43942992213014570008576	6384638072633941694505837874695804644380836826145144184602242817435341577156992
23	77532692687157148844032	28093110657666714909057277466061899267359556118402886615164731950107952805050108672
24	133989774466389220458496	109433303742710598263276391321287752551411989136851033518696417662108760558860099134720
25	227135493100253359439872	448623222915029924737440591476055129159320823405207288109005870309482339727342404079149056
26	378177045530579061637120	1954203368170796853773594264062320589909216147679262173635093966623002616542398755564634177792
27	619178862806272017498112	9035759198108574868114204314427314125944410433316935740711085947218950678406318049415253887202816
28	997959096073050057277440	44272869756652934516466706008181558607533958678344603437118091479604466403261918234480788648856860160
29	1584918071910533189664768	229499010090896465645712757794233230267780943277870916185560831482062042822298289774311766288716060723200
30	2482459843722246457589760	1256676780424136439434570598877690300477266610257380710030351114922366397990844287348484853177097779871728128
31	3837862004424133603295232	7258238000550025020641273629755535532744340586317480271914940646430973318818221399020564801617450002227297973248
32	5860691101463253572321280	44157232836689126022129312225382056021822413875402709397622545669276564133063527278884565304776200276824534556351488
33	8846161207169290968563712	282594820649085148751501837552075377224088586036858394452218557433550485230437025354535128980184244547955074857087049728
34	13206202986736261774376960	1900094241916969821015326783899098957068686376895632009991267385626884091921948782430230615100558163036290402493373876509696
35	19510463367286790841761792	13406572350450220246623533421950480016032212094880317356032197960348111543033308580701452562064817221218366308625135077652318208
36	28540007123376295459880960	99151536294250092631965221915300023892901248805730607655964371666677718184985273915999787809330644818264446243230358332146284193792
37	41357159172102519624630272	767803834154796949332039766767729572150243961827925415210051698803158660857002361916356753846249297137223116592514977515242497715314688.

This proves the claim. ∎

Proposition 4.736 (Residual C fifteenth nonboundary lower correction)**.** *For every residual row C_b with $20 \leq b \leq 217$, put $q = b + 1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+15} \leq \delta_{2q+15}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+15}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q + 15$ with terminal height at least $2q$. If $q = 21$, lemma 4.733 gives

$$|\delta_{2q+15}^{C_b}| \leq s_{2q+15}/2.$$

For $q \geq 22$, lemma 4.732, with $P(q)$ denoting the polynomial from lemma 4.735, gives

$$|\delta_{2q+15}^{C_b}| \leq P(q)\, 2^{2q} s_{q+15}.$$

If $22 \leq q \leq 37$, lemma 4.735 gives

$$|\delta_{2q+15}^{C_b}| \leq s_{2q+15}/2.$$

If $q \geq 38$, lemma 4.734 gives

$$|\delta_{2q+15}^{C_b}| \leq 2^{2q}\binom{2q+15}{q}s_{q+15}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+15}{q}s_{q+15} \leq s_{2q+15}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+15}^{C_b}| \leq s_{2q+15}/2$ also for $q \geq 38$, which implies the claimed lower bound. ∎

Lemma 4.737 (*C* sixteenth-nonboundary height-bad paths have a sixteen-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+16)}$$

with

$$\ell(\lambda^{(2q+16)}) \geq 2q$$

is at most

$$\begin{aligned}\Bigg(65536\binom{2q+16}{0} &+ 262144\binom{2q+16}{2} + 1048576\binom{2q+16}{4} + 4194304\binom{2q+16}{6} \\ &+ 16777216\binom{2q+16}{8} + 67108864\binom{2q+16}{10} + 268435456\binom{2q+16}{12} \\ &+ 1073741824\binom{2q+16}{14} + 4294967296\binom{2q+16}{16}\Bigg)2^{2q}s_{q+16}.\end{aligned}$$

Proof. The terminal height is even and at most $2q+16$. Thus it is

$$2q+16, \quad 2q+14, \quad 2q+12, \quad 2q+10, \quad 2q+8, \quad 2q+6, \quad 2q+4, \quad 2q+2, \quad \text{or} \quad 2q.$$

These cases have respectively $0, 2, 4, 6, 8, 10, 12, 14, 16$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+16+p$. Therefore the nine cases contribute at most

$$\begin{gathered}\binom{2q+16}{0}2^{2q+16}s_{q+8}, \quad \binom{2q+16}{2}2^{2q+18}s_{q+9}, \quad \binom{2q+16}{4}2^{2q+20}s_{q+10}, \\ \binom{2q+16}{6}2^{2q+22}s_{q+11}, \quad \binom{2q+16}{8}2^{2q+24}s_{q+12}, \quad \binom{2q+16}{10}2^{2q+26}s_{q+13}, \\ \binom{2q+16}{12}2^{2q+28}s_{q+14}, \quad \binom{2q+16}{14}2^{2q+30}s_{q+15}, \quad \binom{2q+16}{16}2^{2q+32}s_{q+16},\end{gathered}$$

by lemmas 4.681 and 4.682. After replacing $s_{q+8}, \ldots, s_{q+15}$ by s_{q+16}, the sum is the displayed bound. ☐

Lemma 4.738 (The first two sixteenth *C* pause cases are absorbed by the fixed-point-free count)**.** *For $q = 21, 22$, the number of column-even Pieri two-strip paths of length $2q+16$ with terminal height at least $2q$ is at most $s_{2q+16}/2$.*

Proof. Use the pause-set injection from lemma 4.737, but count the target even-column standard tableaux exactly by lemma 4.731. For $q = 21, 22$, this gives the upper bound

$$\sum_{\substack{0 \leq p \leq 16 \\ p \text{ even}}} \binom{2q+16}{p}(2q+15+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_sixteenth_frontier_gmp_certificate.log`

checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j - 2\mathrm{FPF}(q,j)$
21	58	14006591587695753581293514913909039719080789693690052832937036640625	135331264326267413349868887869322821868469953370931674269172602356060215383198
22	60	1965135565412826288161569628583208390623978721644886751970133067109375	471052485158774046188987143394085149976302513690976318522627253040049506175567106.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`9276304d5e5aff2105d4267604664e9f`
`14dbcc42df60e0a643c335b028250009`.

☐

Lemma 4.739 (The central binomial factor absorbs the sixteen-pause loss)**.** *For every* $q \geq 41$*,*

$$\begin{aligned}65536\binom{2q+16}{0} &+ 262144\binom{2q+16}{2} + 1048576\binom{2q+16}{4} + 4194304\binom{2q+16}{6} \\ &+ 16777216\binom{2q+16}{8} + 67108864\binom{2q+16}{10} + 268435456\binom{2q+16}{12} \\ &+ 1073741824\binom{2q+16}{14} + 4294967296\binom{2q+16}{16} \leq \binom{2q+16}{q}.\end{aligned}$$

Proof. At $q = 41$, the left side is

$$41060740887827289382052823 04 < 6953379637389094006997330256 = \binom{98}{41}.$$

Let $P(q)$ denote the displayed left side. For $0 \leq p \leq 16$,

$$\frac{\binom{2q+18}{p}}{\binom{2q+16}{p}} \leq \frac{(2q+18)(2q+17)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 41),$$

because $4q^2 - 58q - 302 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+18}{q+1}}{\binom{2q+16}{q}} = \frac{(2q+18)(2q+17)}{(q+1)(q+17)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 16q + 255$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 41$ proves the inequality. $\square$

Lemma 4.740 (Exact sixteenth C pause-arithmetic checks after the first two rows)**.** *For* $23 \leq q \leq 40$*, put*

$$\begin{aligned}P(q) = 65536\binom{2q+16}{0} &+ 262144\binom{2q+16}{2} + 1048576\binom{2q+16}{4} + 4194304\binom{2q+16}{6} \\ &+ 16777216\binom{2q+16}{8} + 67108864\binom{2q+16}{10} + 268435456\binom{2q+16}{12} \\ &+ 1073741824\binom{2q+16}{14} + 4294967296\binom{2q+16}{16}.\end{aligned}$$

Then

$$2^{2q+1} P(q) s_{q+16} \leq s_{2q+16}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_sixteenth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $23 \leq q \leq 40$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`9276304d5e5aff2105d4267604664e9f`
`14dbcc42df60e0a643c335b028250009`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+16} - 2^{2q+1}P(q)s_{q+16}$
23	1205807351611961202507776	1509858095356523969157575750967910253082932258610380162720969783022376907880344178432
24	2150483723975075628646400	6887439374866769116850907359416051398468346401926423260145098621747260172624887220429056
25	3758461644764183387242496	29367302157090546319937862855313938422330258031617816279038033437555273144503561727521153280
26	6446046066848765913399296	131903570306498698303191973572111336293213641721840023392678650626740716482555395230962178557440
27	10862292206316129681735680	627983098510752268297381811601656593400968420701540151719427015916832474624724607008670529749451264
28	18004415858340125211688960	3165506383593940812468228722041173064141891020166206247078653265267460916650382989359944786840604957184
29	29383678802502665771483136	16868160512001818037962791490485900780051661543048128204573475148033250265173410976979575043152094991129088
30	47261085444415729746640896	94879010791298187280337279767813612901594464804195239499568072174104637185659235242519350402043059657655106560
31	74978658256448367095513088	562512973606149046916957038460615249681855162365167296181745038752772515224273173045181013557820531175807000417280
32	117420327558262762294542336	3510497287731376952825487778351155018445004309094586720427210166473009943659609766816499788004630733426037417146432512
33	181647019438047091761676288	23031461318490591842248538642177456855384557959341538185195457548228614660390539828570591712060418628420937657600481737728
34	277763886822011089319428096	158657763193874174278210820082600662167588545781561877476243192916863780523554044405574573232077161704573954546870411113637888
35	420094438856161969507729408	1146261223216104381673805361166568220687909854353528630247868414286977064356805899615664456043856543189889053552940393928314664960
36	628757338415691257940606976	8675754396856524847250323812657370220257430919244929339951771850070937018943755082721469025674495507686966397278040733024039425030144
37	931767749965878824904491008	68718405964968875833321335891201862224993158588072239675646218556428209629218997623557804536218766867319872635571876223450378612674213888
38	1367817381467253149340860416	569039354428064809581655104841603985107343828899994063684659330770025354114924528949837165102483062905369652375645316754042484144682497888256
39	1989927006546544129956708352	4921474373856215123551282404435396424010032555189249540385631351130693692793295114168251678989705870335566866580160241189491462836436345101971456
40	2870213714624525227454955520	44415035728604135032970217655684809370220791946857006628208622398005414380915524485038453845820483617257238753156086416783366815115957157569424199680.

This proves the claim. □

Proposition 4.741 (Residual C sixteenth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+16} \le \delta_{2q+16}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+16}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+16$ with terminal height at least $2q$. If $q = 21, 22$, lemma 4.738 gives

$$|\delta_{2q+16}^{C_b}| \le s_{2q+16}/2.$$

For $q \ge 23$, lemma 4.737, with $P(q)$ denoting the polynomial from lemma 4.740, gives

$$|\delta_{2q+16}^{C_b}| \le P(q)\, 2^{2q} s_{q+16}.$$

If $23 \le q \le 40$, lemma 4.740 gives

$$|\delta_{2q+16}^{C_b}| \le s_{2q+16}/2.$$

If $q \ge 41$, lemma 4.739 gives

$$|\delta_{2q+16}^{C_b}| \le 2^{2q}\binom{2q+16}{q}s_{q+16}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+16}{q}s_{q+16} \le s_{2q+16}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

$$\begin{array}{c}\texttt{124068c5daa1123088a3cd79c3f6b839}\\ \texttt{fee45a1aaef637ed82161c465d86bf45}\end{array}.$$

Thus $|\delta_{2q+16}^{C_b}| \le s_{2q+16}/2$ also for $q \ge 41$, which implies the claimed lower bound. □

Lemma 4.742 (C seventeenth-nonboundary height-bad paths have a seventeen-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+17)}$$

with

$$\ell(\lambda^{(2q+17)}) \ge 2q$$

is at most

$$\left(262144\binom{2q+17}{1}+1048576\binom{2q+17}{3}+4194304\binom{2q+17}{5}+16777216\binom{2q+17}{7}\right.$$
$$+67108864\binom{2q+17}{9}+268435456\binom{2q+17}{11}+1073741824\binom{2q+17}{13}$$
$$\left.+4294967296\binom{2q+17}{15}+17179869184\binom{2q+17}{17}\right)2^{2q}s_{q+17}.$$

Proof. The terminal height is even and at most $2q+17$. Thus it is

$$2q+16,\quad 2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $1, 3, 5, 7, 9, 11, 13, 15, 17$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+17+p$. Therefore the nine cases contribute at most

$$\binom{2q+17}{1}2^{2q+18}s_{q+9},\quad \binom{2q+17}{3}2^{2q+20}s_{q+10},\quad \binom{2q+17}{5}2^{2q+22}s_{q+11},$$
$$\binom{2q+17}{7}2^{2q+24}s_{q+12},\quad \binom{2q+17}{9}2^{2q+26}s_{q+13},\quad \binom{2q+17}{11}2^{2q+28}s_{q+14},$$
$$\binom{2q+17}{13}2^{2q+30}s_{q+15},\quad \binom{2q+17}{15}2^{2q+32}s_{q+16},\quad \binom{2q+17}{17}2^{2q+34}s_{q+17},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+9}, \ldots, s_{q+16}$ by s_{q+17}, the sum is the displayed bound. ☐

Lemma 4.743 (The first three seventeenth C pause cases are absorbed by the fixed-point-free count)**.** *For $q = 21, 22, 23$, the number of column-even Pieri two-strip paths of length $2q+17$ with terminal height at least $2q$ is at most $s_{2q+17}/2$.*

Proof. Use the pause-set injection from lemma 4.742, but count the target even-column standard tableaux exactly by lemma 4.731. For $q = 21, 22, 23$, this gives the upper bound

$$\sum_{\substack{1\le p\le 17\\ p \text{ odd}}}\binom{2q+17}{p}(2q+16+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_seventeenth_frontier_gmp_certificate.log`

checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j - 2\mathrm{FPF}(q,j)$
21	59	3646487365485267435348823185481307926680796826335393924646909705390625	7916863325871225525484448344511608155646694346196510796830518427492569348859198
22	61	5430426900387764628193497756837352347403291543287143448347952320139843 75	2849862457075894901864880292629187263673479415275418945729557298637815550920153653 0
23	63	809365302235703922330140036816089188023448622426115176115895952166149218 75	1095412257896723028712113547119374915400532847825471786628492462945348573214299882761 86.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`a1241f73bdcbce52caf77e4ec67c6ef4`
`44661548ce303dfae09cf932a1a100a1`.

☐

Lemma 4.744 (The central binomial factor absorbs the seventeen-pause loss)**.** *For every $q \geq 44$,*

$$\begin{aligned}262144\binom{2q+17}{1}+1048576\binom{2q+17}{3}+4194304\binom{2q+17}{5}+16777216\binom{2q+17}{7}\\+67108864\binom{2q+17}{9}+268435456\binom{2q+17}{11}+1073741824\binom{2q+17}{13}\\+4294967296\binom{2q+17}{15}+17179869184\binom{2q+17}{17}\leq\binom{2q+17}{q}.\end{aligned}$$

Proof. At $q=44$, the left side is

$$284005981622630843449868550144 < 801458485881381465643043319600 = \binom{105}{44}.$$

Let $P(q)$ denote the displayed left side. For $1 \leq p \leq 17$,

$$\frac{\binom{2q+19}{p}}{\binom{2q+17}{p}} \leq \frac{(2q+19)(2q+18)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 44),$$

because $4q^2-62q-338>0$. Hence $P(q+1)<2P(q)$. On the other hand,

$$\frac{\binom{2q+19}{q+1}}{\binom{2q+17}{q}} = \frac{(2q+19)(2q+18)}{(q+1)(q+18)} > 3,$$

since the numerator minus three times the denominator is $q^2+17q+288$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q=44$ proves the inequality. ☐

Lemma 4.745 (Exact seventeenth C pause-arithmetic checks after the first three rows)**.** *For $24 \leq q \leq 43$, put*

$$\begin{aligned}P(q)=262144\binom{2q+17}{1}+1048576\binom{2q+17}{3}+4194304\binom{2q+17}{5}+16777216\binom{2q+17}{7}\\+67108864\binom{2q+17}{9}+268435456\binom{2q+17}{11}+1073741824\binom{2q+17}{13}\\+4294967296\binom{2q+17}{15}+17179869184\binom{2q+17}{17}.\end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+17} \leq s_{2q+17}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_seventeenth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $24 \leq q \leq 43$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`a1241f73bdcbce52caf77e4ec67c6ef4`
`44661548ce303dfae09cf932a1a100a1`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+17} - 2^{2q+1}P(q)s_{q+17}$
24	32998770011970420546469888	406133143944118295207614240726718448502652325289719017791476084366503925692809970595356672
25	59432295280479833893371904	1941499182848369406719839490518623819407961920803338186507903767856284985075833970054084624640
26	104949831086780696214372352	9031945222395551182913192167976209468851868232302089682305445821776409870033578901180701443798528
27	181941306932409381388288000	44271719229880890189547872064109949525701403361559606499910776314872736408892370485018870394678883840
28	310009288261961582165360640	229498661129310833547231936224305438372558721366368822440852078362829082699391190525256472750860834473984
29	519720029531414377302589440	1256676673946322185324179381725643201647556271171114628526512576061321133074616189635261271593410393514951168
30	858089355587184271954280448	7258237967849954884931042357231583835023780403411800254704455350097305906761294820608346704844363133786011923456
31	1396512829567985071065661440	44157232826577367463066310769097404935543041447060462616239168375501541075669014522999300413448063226800971749792768
32	2242104392410675112264859648	282594820645935588180606434340648656948538070201354651900555743018447857962611806536203980023318890837733402268871467008
33	3553741968600343190861250560	1900094241915981371041842174158949860514688227184176473508776365234209320123587584120550457574615370757135590943323242863616
34	5564553815388651717026643968	13406572350449907596228493749260500982499361952017590125882132323276823171683987748495924295842451134934399752657927118730708992
35	8613141912361358819620290560	99151536294249992938880297472579366309800773550293850709355141394036929696778156090420365629994503489283327867410263638212840220672
36	13186560505712868155787051008	767803834154796917279602379262274067098099810859656697933211098066247212563468397527059996962310916734459925523186258372175378763849728
37	19978987958686258441028894720	6219012485561720158730296415209176480235169093635443707357945941299397925258623737956824977251408429175755006309554416848552204030717450240
38	29971195321780924549902368768	52636114508684281019658038026429125714487520644936565969511426459184031113379933896133095832852464595560948122472557133352339093727835296108953 6
39	44537382011599040538580090880	465079114888095472746257509756462428220107222726481931184292362018946686954802126127939516408279551958947220911963307970936286875122253708863459328
40	65587785241707120173479362560	4286049101830469787585145826064000317264620522713040807151224853474572863140823397090827911938405137037628615146236916602459850403706257452358927187968
41	95757755839909279244180258816	41162110219623827986713952512453953258690562246881118283934277469268456583091093250575747832004523306959225351920567914455615250061826130952059064530898944
42	138656824216398390088785199104	411610493361212659403148765054814801562910207570797881088266812485092456976604116455115892512067925145910330139553773678408593345734093956515364107748362182656
43	199194769184163601401791643648	4282289551634348924637847924815314882271058168493559037093783432851176072866241963481233049813442652660078840873862478967639526137222901228961502751701895318822912.

This proves the claim. ∎

Proposition 4.746 (Residual C seventeenth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+17} \le \delta_{2q+17}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+17}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+17$ with terminal height at least $2q$. If $q = 21, 22, 23$, lemma 4.743 gives

$$|\delta_{2q+17}^{C_b}| \le s_{2q+17}/2.$$

For $q \ge 24$, lemma 4.742, with $P(q)$ denoting the polynomial from lemma 4.745, gives

$$|\delta_{2q+17}^{C_b}| \le P(q)\, 2^{2q} s_{q+17}.$$

If $24 \le q \le 43$, lemma 4.745 gives

$$|\delta_{2q+17}^{C_b}| \le s_{2q+17}/2.$$

If $q \ge 44$, lemma 4.744 gives

$$|\delta_{2q+17}^{C_b}| \le 2^{2q}\binom{2q+17}{q} s_{q+17}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+17}{q} s_{q+17} \le s_{2q+17}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+17}^{C_b}| \le s_{2q+17}/2$ also for $q \ge 44$, which implies the claimed lower bound. ∎

Lemma 4.747 (C eighteenth-nonboundary height-bad paths have an eighteen-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+18)}$$

with

$$\ell(\lambda^{(2q+18)}) \ge 2q$$

is at most

$$\Bigg(262144\binom{2q+18}{0}+1048576\binom{2q+18}{2}+4194304\binom{2q+18}{4}+16777216\binom{2q+18}{6}$$
$$+67108864\binom{2q+18}{8}+268435456\binom{2q+18}{10}+1073741824\binom{2q+18}{12}$$
$$+4294967296\binom{2q+18}{14}+17179869184\binom{2q+18}{16}+68719476736\binom{2q+18}{18}\Bigg)2^{2q}s_{q+18}.$$

Proof. The terminal height is even and at most $2q+18$. Thus it is

$$2q+18,\quad 2q+16,\quad 2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $0,2,4,6,8,10,12,14,16,18$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+18+p$. Therefore the ten cases contribute at most

$$\binom{2q+18}{0}2^{2q+18}s_{q+9},\quad \binom{2q+18}{2}2^{2q+20}s_{q+10},\quad \binom{2q+18}{4}2^{2q+22}s_{q+11},$$
$$\binom{2q+18}{6}2^{2q+24}s_{q+12},\quad \binom{2q+18}{8}2^{2q+26}s_{q+13},\quad \binom{2q+18}{10}2^{2q+28}s_{q+14},$$
$$\binom{2q+18}{12}2^{2q+30}s_{q+15},\quad \binom{2q+18}{14}2^{2q+32}s_{q+16},\quad \binom{2q+18}{16}2^{2q+34}s_{q+17},$$
$$\binom{2q+18}{18}2^{2q+36}s_{q+18},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+9},\dots,s_{q+17}$ by s_{q+18}, the sum is the displayed bound. □

Lemma 4.748 (The first three eighteenth C pause cases are absorbed by the fixed-point-free count)**.** *For* $q=21,22,23$*, the number of column-even Pieri two-strip paths of length* $2q+18$ *with terminal height at least* $2q$ *is at most* $s_{2q+18}/2$*.*

Proof. Use the pause-set injection from lemma 4.747, but count the target even-column standard tableaux exactly by lemma 4.731. For $q=21,22,23$, this gives the upper bound

$$\sum_{\substack{0\le p\le 18\\ p \text{ even}}}\binom{2q+18}{p}(2q+17+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_eighteenth_frontier_gmp_certificate.log`

checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j-2\mathrm{FPF}(q,j)$
21	60	936103804833627766619558658829368839169497669297772127543300376719453125	471052483290496707652557262731290971574731252133228937370372771457389018870879606
22	62	147792316180051306675315989515099722426677110364389754267875360964914531250	1752662440282196942849401054887494894303585264196585842685687141835826067117925676956
23	64	23313182866657367477677417211957455715304495181498386549564779747042798203125	6955857144680513605018656639438248482599989467501880284347279981375714830577588469923606.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`2c1768d73f6e84f175648fe543b44e67`
`f791c9221e749ff42253c8a69a6ed7a0`.

□

Lemma 4.749 (The central binomial factor absorbs the eighteen-pause loss)**.** *For every $q \geq 46$,*

$$262144\binom{2q+18}{0} + 1048576\binom{2q+18}{2} + 4194304\binom{2q+18}{4} + 16777216\binom{2q+18}{6} + 67108864\binom{2q+18}{8} + 268435456\binom{2q+18}{10} + 1073741824\binom{2q+18}{12} + 4294967296\binom{2q+18}{14} + 17179869184\binom{2q+18}{16} + 68719476736\binom{2q+18}{18} \leq \binom{2q+18}{q}.$$

Proof. At $q = 46$, the left side is

$$1382611867563723018654817845 2480 < 2274736282411066517941618538 3175 = \binom{110}{46}.$$

Let $P(q)$ denote the displayed left side. For $0 \leq p \leq 18$,

$$\frac{\binom{2q+20}{p}}{\binom{2q+18}{p}} \leq \frac{(2q+20)(2q+19)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 46),$$

because $4q^2 - 66q - 376 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+20}{q+1}}{\binom{2q+18}{q}} = \frac{(2q+20)(2q+19)}{(q+1)(q+19)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 18q + 323$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 46$ proves the inequality. $\square$

Lemma 4.750 (Exact eighteenth C pause-arithmetic checks after the first three rows)**.** *For $24 \leq q \leq 45$, put*

$$P(q) = 262144\binom{2q+18}{0} + 1048576\binom{2q+18}{2} + 4194304\binom{2q+18}{4} + 16777216\binom{2q+18}{6} + 67108864\binom{2q+18}{8} + 268435456\binom{2q+18}{10} + 1073741824\binom{2q+18}{12} + 4294967296\binom{2q+18}{14} + 17179869184\binom{2q+18}{16} + 68719476736\binom{2q+18}{18}.$$

Then

$$2^{2q+1}P(q)s_{q+18} \leq s_{2q+18}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_eighteenth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $24 \leq q \leq 45$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`2c1768d73f6e84f175648fe543b44e67`
`f791c9221e749ff42253c8a69a6ed7a0`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+18} - 2^{2q+1}P(q)s_{q+18}$
24	485690524492741371143389184	3415569985153575367274533197969017388038334138152550172056441941023544842155192750098702592
25	901013500420353108094484480	123718617515897586083287220196259693056833787006661964394946443203878730757386403376819652756992
26	1637475412045018423195074560	625394681032748308818491724462077828049815837548051194740929889237728311134709854075006194072093184
27	2919209192203235399393607680	3164684960453723419033988171777179335593364112200150921402526606711948345746203707148891262097133197824
28	5111212844565486053329469440	16867898744030385967552373953954359844277505317807397078701856006530416894479655755013361259277853426358784
29	8798834219976758419406454784	94878926971285181538362280432357436386352886938210148090373554441858993703952358346277164182311821723573861376
30	14907371697563580919094444032	562512946623016005557246402126450655392812786058689979945775339139007064297727803540268437148796723348746175943680
31	24879815802004287614091526144	3510497278994553285352124342714737035974598674202364154782135377114916374403314125803840200958448825657422499966467072
32	40937925700357484315497725952	23031461315644103996802830544002057424805173049182115216330189013059475845295474945514034743505845226678652754150313999360
33	66461509138151476374200909824	158657763192940671359587645926155393584322449555853777732149024169128998067201988327914212410814814936932576676042471785998336
34	106533717472108512681303212032	1146261223215796127281153047067350611449763178692448923941319717385463465830104573061717280272938022298371243734483970747397126144
35	168717341749300791726141865984	8675754396856422326985058396204698935972320285039604537325007431446181745610465588672873452556769296343249163706982015629959205238784
36	264149708380352676845388562432	68718405964968841483731398906052289974250895191315055265268798505297825076375599885681674624931303886893466708197283025059823822024030208
37	409073328050388683175147864064	569039354428064797984913670253094170681299410754373761692490833218806022921157367406202814323741402252155125718796725562049124835847063359488
38	626957812521160029184913047552	4921474373856215119605507272183515287971251329422263637050499515179821579622753202390863736087838752303061601368975952103575187622073710273327104
39	951418036509207647414246440960	44415035728604135031616963184031838155041986870000616121704484654814288749672989373715732481361386194193420854811113890872159621802739070075973079040
40	1430196899008872205162802053120	417889612984311197972413715741748323772344422906441677583908278778108905721753151975371538494906635256729786947618421020890410173026737895363840681381888
41	2130561760307622776104959606784	4095628359235226170344268509952848370247001712060145269184776415125987077056785411510818545864925263991570640882173727642204791341532558420075830159180062720
42	3146565859795394230612091207680	41778449637156136111589853722521634686017301903581847378885849827856525293907642812020576950191088240405075026665981979218121837877137192564493707361175613218816
43	4608754783361488019049362489344	443216814209338571157347337580637569052962560356503201840060461059507360951427126728892198626465957510509130212037216979092424360355571471425307335054956552396906496
44	6697059394790832303211325685760	4886351313102937424311841329203021597323852241640700645228150343195504189732562797678949425782069485468123700774333533211215477761493812931761484720621722531497970933760
45	9657817815894010546150315393024	5594257935357431929992845141961271365482965620541010924040136972424288636070390049761950825626661116297743699395165386419324954141421672360147943644470193623896920549659361280.

This proves the claim. ∎

Proposition 4.751 (Residual C eighteenth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b + 1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+18} \le \delta_{2q+18}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+18}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q + 18$ with terminal height at least $2q$. If $q = 21, 22, 23$, lemma 4.748 gives

$$|\delta_{2q+18}^{C_b}| \le s_{2q+18}/2.$$

For $q \ge 24$, lemma 4.747, with $P(q)$ denoting the polynomial from lemma 4.750, gives

$$|\delta_{2q+18}^{C_b}| \le P(q)\, 2^{2q} s_{q+18}.$$

If $24 \le q \le 45$, lemma 4.750 gives

$$|\delta_{2q+18}^{C_b}| \le s_{2q+18}/2.$$

If $q \ge 46$, lemma 4.749 gives

$$|\delta_{2q+18}^{C_b}| \le 2^{2q}\binom{2q+18}{q} s_{q+18}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+18}{q} s_{q+18} \le s_{2q+18}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+18}^{C_b}| \le s_{2q+18}/2$ also for $q \ge 46$, which implies the claimed lower bound. ∎

Lemma 4.752 (C nineteenth-nonboundary height-bad paths have a nineteen-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+19)}$$

with

$$\ell(\lambda^{(2q+19)}) \ge 2q$$

is at most

$$\begin{aligned}\Bigg(&1048576\binom{2q+19}{1}+4194304\binom{2q+19}{3}+16777216\binom{2q+19}{5}\\&+67108864\binom{2q+19}{7}+268435456\binom{2q+19}{9}+1073741824\binom{2q+19}{11}\\&+4294967296\binom{2q+19}{13}+17179869184\binom{2q+19}{15}\\&+68719476736\binom{2q+19}{17}+274877906944\binom{2q+19}{19}\Bigg)2^{2q}s_{q+19}.\end{aligned}$$

Proof. The terminal height is even and at most $2q+19$. Thus it is

$$2q+18,\quad 2q+16,\quad 2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $1,3,5,7,9,11,13,15,17,19$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+19+p$. Therefore the ten cases contribute at most

$$\begin{array}{lll}\binom{2q+19}{1}2^{2q+20}s_{q+10}, & \binom{2q+19}{3}2^{2q+22}s_{q+11}, & \binom{2q+19}{5}2^{2q+24}s_{q+12},\\ \binom{2q+19}{7}2^{2q+26}s_{q+13}, & \binom{2q+19}{9}2^{2q+28}s_{q+14}, & \binom{2q+19}{11}2^{2q+30}s_{q+15},\\ \binom{2q+19}{13}2^{2q+32}s_{q+16}, & \binom{2q+19}{15}2^{2q+34}s_{q+17}, & \binom{2q+19}{17}2^{2q+36}s_{q+18},\\ & \binom{2q+19}{19}2^{2q+38}s_{q+19}, & \end{array}$$

by lemmas 4.681 and 4.682. After replacing $s_{q+10},\dots,s_{q+18}$ by s_{q+19}, the sum is the displayed bound. $\square$

Lemma 4.753 (The first four nineteenth C pause cases are absorbed by the fixed-point-free count)**.** *For $q=21,22,23,24$, the number of column-even Pieri two-strip paths of length $2q+19$ with terminal height at least $2q$ is at most $s_{2q+19}/2$.*

Proof. Use the pause-set injection from lemma 4.752, but count the target even-column standard tableaux exactly by lemma 4.731. For $q=21,22,23,24$, this gives the upper bound

$$\sum_{\substack{1\le p\le 19\\ p\ \text{odd}}}\binom{2q+19}{p}(2q+18+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_nineteenth_frontier_gmp_certificate.log`

checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j-2\mathrm{FPF}(q,j)$
21	61	237470108332329713171082643084030364430708345727294881364488502819416328125	28498624096904817734066929509765286020041535760818156311363238947070740334396849030
22	63	39700601747085018754484911634048710728947977468719606083510193151637269218750	109541225710432972437488457987433696279319081162255913449895257162737650208588679682436
23	65	6620723654392057477395380854479066940610645630038027813080726841442956808203125	448652128219640090423345837323001612054965492564703062730327497930184002811740199033637142
24	67	1102869093764874549303216415018746428293136265004334004265436419435902759984375000	1954211164679237519738993223359785151237160761354975954454229478971669293616074699472392992592.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`519d2680d84e75ce37567e5b89e5b1d7`
`3642f0ddf74f02f115aeb728df4168dd`.

$\square$

Lemma 4.754 (The central binomial factor absorbs the nineteen-pause loss)**.** *For every $q \geq 49$,*

$$\begin{aligned} &1048576\binom{2q+19}{1} + 4194304\binom{2q+19}{3} + 16777216\binom{2q+19}{5} \\ &\quad + 67108864\binom{2q+19}{7} + 268435456\binom{2q+19}{9} + 1073741824\binom{2q+19}{11} \\ &\quad + 4294967296\binom{2q+19}{13} + 17179869184\binom{2q+19}{15} \\ &\quad + 68719476736\binom{2q+19}{17} + 274877906944\binom{2q+19}{19} \leq \binom{2q+19}{q}. \end{aligned}$$

Proof. At $q = 49$, the left side is

$$959886034421822120920097112457216 < 26316059485030149793559591064174750 = \binom{117}{49}.$$

Let $P(q)$ denote the displayed left side. For $1 \leq p \leq 19$,

$$\frac{\binom{2q+21}{p}}{\binom{2q+19}{p}} \leq \frac{(2q+21)(2q+20)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 49),$$

because $4q^2 - 70q - 416 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+21}{q+1}}{\binom{2q+19}{q}} = \frac{(2q+21)(2q+20)}{(q+1)(q+20)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 19q + 360$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 49$ proves the inequality. ☐

Lemma 4.755 (Exact nineteenth C pause-arithmetic checks after the first four rows)**.** *For $25 \leq q \leq 48$, put*

$$\begin{aligned} P(q) = {} &1048576\binom{2q+19}{1} + 4194304\binom{2q+19}{3} + 16777216\binom{2q+19}{5} \\ &+ 67108864\binom{2q+19}{7} + 268435456\binom{2q+19}{9} + 1073741824\binom{2q+19}{11} \\ &+ 4294967296\binom{2q+19}{13} + 17179869184\binom{2q+19}{15} \\ &+ 68719476736\binom{2q+19}{17} + 274877906944\binom{2q+19}{19}. \end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+19} \leq s_{2q+19}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_nineteenth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $25 \leq q \leq 48$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`519d2680d84e75ce37567e5b89e5b1d7`
`3642f0ddf74f02f115aeb728df4168dd`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+19} - 2^{2q+1}P(q)s_{q+19}$
25	1313365675899009227252511539 2	3842380120851551944142473147494552077681333347048494952440197066852371374503365116937097548576256
26	24554262757961748518655754240	42544608825317433825017194072171563221014792045929523136375250871080764354516346378152381902243182080
27	44996875384669081611592007680	228922596340449498597709272052905627396620123708835017028698459928221956938741047037001684394505925330944
28	80926429529001584723581992960	1256483959754054882893658262044304324534475077506191506224914323532222922862475479568718579330232663657660928
29	143001582816313022041992200192	72581732630316133704403563723406132773925515511213779088022197615164762322830481261360182799492647803683800097536
30	248531126707426721043479265280	44157211009580330347799852365610389867646375399980869260837407956655471123510405335703346449862803360041499264293888
31	425225447850373260819160891392	282594813254761709829596426296813127388409677795124834347777328446962215948880445410153649680606686689313802356801970176
32	716855183174662333546904944640	1900094239398941214011041594489397140198908647419788401645661251705045828808782696643946980715692190935359115928224915495936
33	1191687384782268653486732214272	13406572349587921939227856292443348922331560285939150037786070042479321733339567833058205414911318838003001756814446538540075008
34	1954924263512918327464731607040	991515362939530332755774025272139703012928189836927031804322921222987963134509850699234881707810248042075756206025854469457276702 72
35	3166852726909542920159941885952	767803834154693969111441063073248733253484113702637951830009933315836735283213525532997041978901467597701890997890436243798336028909568
36	5069065425860958444430457569280	6219012485561684234815308755293489245290611952063363904592571478250765036838029158742019756523969181319495597983229484671359842652228147200
37	8021988242925087774775178166272	52636114508684268398522185144146687794616579709169747400742525957491221275634629893719559369982111069191703146698594889510674349422317297250304
38	12558111617945873215456044646400	465079114888095468280900911393862494883254664227088421555258592053187518620789596123454586082158443011484736251024310918198276707854260417171017728
39	19456857797833949746152734720000	4286049101830469785993894980348880248457611926002771098965364284788320044849264995689920858827118083458305537023652897763538610343833892233575482851328
40	29849028426037613464679481344000	41162110219623827986142713682581476779460775136967730397659528658517488946880722663220898289704131309362223206341041876417959103036743621919134021636722688
41	45361398415060047205612274057216	411610493361212659402942152779612426035671413503991194745996214371682855617877400942854625714083379577612360765418708044785145376427765560092741084947008479232
42	68315416105960826809088440532992	4282289551634348924637772622467391571052467599720276217270259836464965846273656436276081095407377104121802416263129745110015125003409433024359161898911655254777856
43	101998338065906113339054674149376	46316141414884575423512758181052631642077578418249613878781970668380438640450664053709209657765008846809126867873962396578394707199903601592712393830542561227664687104
44	151030717518270678269547382833152	520396248607814162823456888678083574419340659794434686095011665287540075473832731821460177752393256807330106483766578985757206447421175601487977177258999265323742719582208
45	221861281805804846522903362011136	6069768027148561641469987055484821179900569544002511222699863695645693775147353430401562009777811585923752404130006219706714588038484040173569056915379457859318318461889757184
46	323429246142932688920017619648512	73442633136598658548124378675009530522796332345920669968955863270601680097171534430050096731843267205526088096581132111831528144658814838049197342622924998895013179392542669258752
47	468045466676533938884971133403136	921245524013205551953790796064948432422207505327530462415741404473788266278303523601102448802363164678225300392139893987795028748170393657917251102898282026062915325631567166665785344
48	672558077360155695305232772759552	11972270293412322391994921932842237116039497959401574743677868601647816015977507065502280690490490564115364142207230527762613568271988347220584619482063464562880063648323635618825263759360.

This proves the claim. ☐

Proposition 4.756 (Residual C nineteenth nonboundary lower correction). *For every residual row C_b with $20 \le b \le 217$, put $q = b + 1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+19} \le \delta_{2q+19}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+19}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+19$ with terminal height at least $2q$. If $q = 21, 22, 23, 24$, lemma 4.753 gives

$$|\delta_{2q+19}^{C_b}| \le s_{2q+19}/2.$$

For $q \ge 25$, lemma 4.752, with $P(q)$ denoting the polynomial from lemma 4.755, gives

$$|\delta_{2q+19}^{C_b}| \le P(q)\, 2^{2q} s_{q+19}.$$

If $25 \le q \le 48$, lemma 4.755 gives

$$|\delta_{2q+19}^{C_b}| \le s_{2q+19}/2.$$

If $q \ge 49$, lemma 4.754 gives

$$|\delta_{2q+19}^{C_b}| \le 2^{2q}\binom{2q+19}{q}s_{q+19}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+19}{q}s_{q+19} \le s_{2q+19}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+19}^{C_b}| \le s_{2q+19}/2$ also for $q \ge 49$, which implies the claimed lower bound. ☐

Lemma 4.757 (C twentieth-nonboundary height-bad paths have a twenty-pause code). *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+20)}$$

with

$$\ell(\lambda^{(2q+20)}) \ge 2q$$

is at most

$$\Bigg(1048576\binom{2q+20}{0}+4194304\binom{2q+20}{2}+16777216\binom{2q+20}{4}$$
$$+67108864\binom{2q+20}{6}+268435456\binom{2q+20}{8}+1073741824\binom{2q+20}{10}$$
$$+4294967296\binom{2q+20}{12}+17179869184\binom{2q+20}{14}$$
$$+68719476736\binom{2q+20}{16}+274877906944\binom{2q+20}{18}+1099511627776\binom{2q+20}{20}\Bigg)2^{2q}s_{q+20}.$$

Proof. The terminal height is even and at most $2q+20$. Thus it is

$$2q+20,\quad 2q+18,\quad 2q+16,\quad 2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $0,2,4,6,8,10,12,14,16,18,20$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+20+p$. Therefore the eleven cases contribute at most

$$\binom{2q+20}{0}2^{2q+20}s_{q+10},\quad \binom{2q+20}{2}2^{2q+22}s_{q+11},\quad \binom{2q+20}{4}2^{2q+24}s_{q+12},$$
$$\binom{2q+20}{6}2^{2q+26}s_{q+13},\quad \binom{2q+20}{8}2^{2q+28}s_{q+14},\quad \binom{2q+20}{10}2^{2q+30}s_{q+15},$$
$$\binom{2q+20}{12}2^{2q+32}s_{q+16},\quad \binom{2q+20}{14}2^{2q+34}s_{q+17},$$
$$\binom{2q+20}{16}2^{2q+36}s_{q+18},\quad \binom{2q+20}{18}2^{2q+38}s_{q+19},\quad \binom{2q+20}{20}2^{2q+40}s_{q+20},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+10},\ldots,s_{q+19}$ by s_{q+20}, the sum is the displayed bound. □

Lemma 4.758 (The first five twentieth C pause cases are absorbed by the fixed-point-free count)**.** *For $q=21,22,23,24,25$, the number of column-even Pieri two-strip paths of length $2q+20$ with terminal height at least $2q$ is at most $s_{2q+20}/2$.*

Proof. Use the pause-set injection from lemma 4.757, but count the target even-column standard tableaux exactly by lemma 4.731. For $q=21,22,23,24,25$, this gives the upper bound

$$\sum_{\substack{0\le p\le 20\\ p\text{ even}}}\binom{2q+20}{p}(2q+19+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentieth_frontier_gmp_certificate.log`

checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j-2\mathrm{FPF}(q,j)$
21	62	59640178498959305514120228939240938313534246071836194292902921336161643046875	1752662321297424577290892639997668994851908081981447919742078064565734116724468645706
22	64	10546296264465058052491781427837459696489984515461788143252233884332368579453125	6955857123634547441821855269410040461348984985952520243786700467970376621406936907423606
23	66	1857401551044838044392852676786302182793147532481339984781861148229496918789062500	29386673296186905862659799277973863836030358812579483861024063304433700545101686805296669240
24	68	326274573471185200824831553398486097764741139539400805519386410474790347556680468750	131909085288574662554847144278149265223016674231510355370233483779079385857606782407378241090532
25	70	57237304288628801488251204476412623107980093291037502641817121101346066389375667187500	627984678371613795768240992643835579421151419539030578383154290148591005899825266194087500660974504.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`431f9748488a0ab0453777cc498a2260`
`996f1ddcf8b393545d30eaed4c291232`.

□

Lemma 4.759 (The central binomial factor absorbs the twenty-pause loss)**.** *For every $q \geq 51$,*

$$\begin{aligned}1048576&\binom{2q+20}{0} + 4194304\binom{2q+20}{2} + 16777216\binom{2q+20}{4}\\&+ 67108864\binom{2q+20}{6} + 268435456\binom{2q+20}{8} + 1073741824\binom{2q+20}{10}\\&+ 4294967296\binom{2q+20}{12} + 17179869184\binom{2q+20}{14}\\&+ 68719476736\binom{2q+20}{16} + 274877906944\binom{2q+20}{18} + 1099511627776\binom{2q+20}{20} \leq \binom{2q+20}{q}.\end{aligned}$$

Proof. At $q = 51$, the left side is

$$468327889553295627017064178264309 76 < 748566892409484031970576013523462 32 = \binom{122}{51}.$$

Let $P(q)$ denote the displayed left side. For $1 \leq p \leq 20$,

$$\frac{\binom{2q+22}{p}}{\binom{2q+20}{p}} \leq \frac{(2q+22)(2q+21)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 51),$$

because $4q^2 - 74q - 458 > 0$. The $p = 0$ summand has ratio 1, so $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+22}{q+1}}{\binom{2q+20}{q}} = \frac{(2q+22)(2q+21)}{(q+1)(q+21)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 20q + 399$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 51$ proves the inequality. ☐

Lemma 4.760 (Exact twentieth C pause-arithmetic checks after the first five rows)**.** *For $26 \leq q \leq 50$, put*

$$\begin{aligned}P(q) = 1048576&\binom{2q+20}{0} + 4194304\binom{2q+20}{2} + 16777216\binom{2q+20}{4}\\&+ 67108864\binom{2q+20}{6} + 268435456\binom{2q+20}{8} + 1073741824\binom{2q+20}{10}\\&+ 4294967296\binom{2q+20}{12} + 17179869184\binom{2q+20}{14}\\&+ 68719476736\binom{2q+20}{16} + 274877906944\binom{2q+20}{18} + 1099511627776\binom{2q+20}{20}.\end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+20} \leq s_{2q+20}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentieth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $26 \leq q \leq 50$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`431f9748488a0ab0453777cc498a2260`
`996f1ddcf8b393545d30eaed4c291232`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+20}-2^{2q+1}P(q)s_{q+20}$
26	354778401204155339898251902976	2029321248636848132156229336256516289542273932570394327907502732535074464896495180780860172670218852864
27	668046057206256226400690241536	16470227858291556398868251953084445249011371801453753871162632928765387742332661304825544355924844558064128
28	1233678087925221127336074674176	94739387771535650729044632822516217732154802554740302342213832584467959027972591565288745136376182913113701376
29	2236909402936097168961508999168	562463859776645134721777408681391508043653821256254339964732794455721571638853258638459822769907471937211924648960
30	3986640469434493642603832016896	3510479956626479945064065093422685541292954668941382938283717075048742433937775444835212124123584113313494890413145088
31	6990366999405411127331654729728	230314551798925740607823095788981813959032222640563653328774210532276095875378614529615852789872015683095394932605117316096
32	12070201653596990046658930671616	158657761010391709928635933825879720014712852712073244608307670442668130031117226778978817473473585747790460038708177250920448
33	20540316585644715652565573828608	1146261222435806504222778451299208601166705583012514909657009562955583359670235855046916129914704307012836690340197544826148829184
34	34475171214194326069382575292416	8675754396576257580484489516247180010264966731523629612851197647094670914521681322218341646884687895051098503694139034331675589883904
35	57110515561481856763043840524288	68718405964867662561904265761192598509676134512816715127234748064104426081861403054347005524062565333813465621844609512483362290630735872
36	93436643634283881488961284079616	569039354428028048305125528256416769925482817745425010937525872867883145721910344150248856572253383726847704527847292505155439949686697775104
37	151067376328878364756360343584768	4921474373856201691215779968823141681781070217450565831646060308813907493185792261003037411457430418525374388607043173375760716448729015439552512
38	241500944272106790013050091995136	44415035728604130094111381673734028437943733224721232844576881782370385048998090673231228820108791056631562677240757154011985894251611947057038168064
39	381933118070631065623644026699776	417889612984311196145159757609244052940786047117277660521046813847855539611165828196138705286819557264874768639678794385676266980195285401264201818640384
40	597842195094790018058475044601856	4095628359235226169663528226641720275352265015692941967199710360533629510006236451097708164910355021586777313423004997897464440572239901608627322159762530304
41	926644394250348431285693883351040	41778449637156136111334507680682619050342599001186727105782802836361858860912943936296248029821397585233723366523096658006234188789438324382011452093718432587776
42	1422822669341308738068607948292096	4432168142093385711572508864814069794749057469419801237577233797645273474038533093760437188916935117335714336722481961238742438206595770680086111816259655729215873024
43	2165069297334773836476255541657600	4886351313102937424311804637063514269142801222360408920750974420230475310776237230430790833270890535720334871326791916987350075999262451330911831063433982028819318087680
44	3266162087164285945781253393350656	55942579353574319299284500136393250019253802996105767741717827327750023959806505303037207821440900962673973606705673647141080068618186498072870306737127232347646415078277120
45	4886527256556420549226075243151360	664639403789427225561471728776951853213131700423439820015699549221259574648825594145736675540347744091130636990556912333149491891418212169359254452149986608480755678679176003584
46	7252743326889686377601445345099776	8188851318177335552057955122474484220669788144328372380064385876691784910041920651064776549594771008032396196789683388361323422691106803592343436618548320690259347370517086087921664
47	10682627600811694024363297152696320	1045613394299578699768683778153873349669549304900031960275195719026724096964883666059081832549517994817715486780369400386533553702764124718896091215301537878502754751974900060488129101824
48	15619041865727584281493799321993216	1382796873366955667733698787057210090554400524921852847079418204016369853723477421040920207577688871959694107167306467123879148341547199311427242878603156125689114580950476133409875551748096
49	22675183846767937430683325589094400	18928751602053607333207431922902761406058558552419382632691632458421281957173923901607181230033775141234959749975745287719800608602511100883284529879312119407326828567600736159251147259345338368
50	32694928538565825284208708373446656	268045196852917110920858864292769236947110205860353984103918140593860870508962594303438435038934331300021389007635651385858812594502889092383930755683908658212795237567567873604653202227239233486848.

This proves the claim. ☐

Proposition 4.761 (Residual C twentieth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+20} \le \delta_{2q+20}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+20}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+20$ with terminal height at least $2q$. If $q = 21, 22, 23, 24, 25$, lemma 4.758 gives

$$|\delta_{2q+20}^{C_b}| \le s_{2q+20}/2.$$

For $q \ge 26$, lemma 4.757, with $P(q)$ denoting the polynomial from lemma 4.760, gives

$$|\delta_{2q+20}^{C_b}| \le P(q)\, 2^{2q} s_{q+20}.$$

If $26 \le q \le 50$, lemma 4.760 gives

$$|\delta_{2q+20}^{C_b}| \le s_{2q+20}/2.$$

If $q \ge 51$, lemma 4.759 gives

$$|\delta_{2q+20}^{C_b}| \le 2^{2q}\binom{2q+20}{q} s_{q+20}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+20}{q} s_{q+20} \le s_{2q+20}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+20}^{C_b}| \le s_{2q+20}/2$ also for $q \ge 51$, which implies the claimed lower bound. ☐

Lemma 4.762 (C twenty-first-nonboundary height-bad paths have a twenty-one-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+21)}$$

with

$$\ell(\lambda^{(2q+21)}) \ge 2q$$

is at most

$$\begin{aligned}\Bigg(&4194304\binom{2q+21}{1}+16777216\binom{2q+21}{3}+67108864\binom{2q+21}{5}\\&+268435456\binom{2q+21}{7}+1073741824\binom{2q+21}{9}+4294967296\binom{2q+21}{11}\\&+17179869184\binom{2q+21}{13}+68719476736\binom{2q+21}{15}\\&+274877906944\binom{2q+21}{17}+1099511627776\binom{2q+21}{19}\\&+4398046511104\binom{2q+21}{21}\Bigg)2^{2q}s_{q+21}.\end{aligned}$$

Proof. The terminal height is even and at most $2q+20$. Thus it is

$$2q+20,\quad 2q+18,\quad 2q+16,\quad 2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $1,3,5,7,9,11,13,15,17,19,21$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+21+p$. Therefore the eleven cases contribute at most

$$\binom{2q+21}{1}2^{2q+22}s_{q+11},\quad \binom{2q+21}{3}2^{2q+24}s_{q+12},\quad \binom{2q+21}{5}2^{2q+26}s_{q+13},$$

$$\binom{2q+21}{7}2^{2q+28}s_{q+14},\quad \binom{2q+21}{9}2^{2q+30}s_{q+15},\quad \binom{2q+21}{11}2^{2q+32}s_{q+16},$$

$$\binom{2q+21}{13}2^{2q+34}s_{q+17},\quad \binom{2q+21}{15}2^{2q+36}s_{q+18},$$

$$\binom{2q+21}{17}2^{2q+38}s_{q+19},\quad \binom{2q+21}{19}2^{2q+40}s_{q+20},\quad \binom{2q+21}{21}2^{2q+42}s_{q+21},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+11},\dots,s_{q+20}$ by s_{q+21}, the sum is the displayed bound. □

Lemma 4.763 (The first six twenty-first C pause cases are absorbed by the fixed-point-free count)**.** *For $21\le q\le 26$, the number of column-even Pieri two-strip paths of length $2q+21$ with terminal height at least $2q$ is at most $s_{2q+21}/2$.*

Proof. Use the pause-set injection from lemma 4.762, but count the target even-column standard tableaux exactly by lemma 4.731. For $21\le q\le 26$, this gives the upper bound

$$\sum_{\substack{1\le p\le 21\\ p\ \text{odd}}}\binom{2q+21}{p}(2q+20+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyfirst_frontier_gmp_certificate.log`

checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j-2\mathrm{FPF}(q,j)$
21	63	14853304761259119316408268236115782519914732648875198936203636640975622669609375	10954119608322465341341986267986704731585146279068657063693659692248475456061787890118 6
22	65	27751674629796855710346775074595219183825205081085428328503305745954393825894531 25	44865212268254661177275881020843735884487978968088333777331788785568430730374734747113 7142
23	67	51564329917723877250808400677962379741844913183677027485697101029809821239670312 5000	19542111636501566595720454274422235705079506593746639633106975972662581458587500801989 55492592
24	69	95425704873602417500267836363618545745961310075692241308086946707711680690633486 718750	90357613168977610344917864204674837828027447967133109270433813167283483788122704614231 73781686212
25	71	17612308918214414042573132661483881464445143576594538854748006859427149621063875 057812500	44272870337208473463377756399122746412754928148091432553108900919798710002070327734342 3035235198028 40
26	73	32457158626647630408691036016048614980892858475666839900321678617394028482574143 71356250000	22949901025132948234184891382396649107612692476996658479854369983161705730538368533911 24764939767242037 44.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`a5593e2a97e9cadcf10f028940d40366`
`500494e1c24910ccb6d39bf66334659c`.

∎

Lemma 4.764 (The central binomial factor absorbs the twenty-one-pause loss)**.** *For every $q \geq 54$,*

$$\begin{aligned}&4194304\binom{2q+21}{1}+16777216\binom{2q+21}{3}+67108864\binom{2q+21}{5}\\&\quad+268435456\binom{2q+21}{7}+1073741824\binom{2q+21}{9}+4294967296\binom{2q+21}{11}\\&\quad+17179869184\binom{2q+21}{13}+68719476736\binom{2q+21}{15}\\&\quad+274877906944\binom{2q+21}{17}+1099511627776\binom{2q+21}{19}+4398046511104\binom{2q+21}{21}\leq\binom{2q+21}{q}.\end{aligned}$$

Proof. At $q=54$, the left side is

$$326121930128066934702816983397695488 0 < 868600466232156172648388929658129376 0 = \binom{129}{54}.$$

Let $P(q)$ denote the displayed left side. For $1 \leq p \leq 21$,

$$\frac{\binom{2q+23}{p}}{\binom{2q+21}{p}} \leq \frac{(2q+23)(2q+22)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 54),$$

because $4q^2-82q-502>0$. Hence $P(q+1)<2P(q)$. On the other hand,

$$\frac{\binom{2q+23}{q+1}}{\binom{2q+21}{q}} = \frac{(2q+23)(2q+22)}{(q+1)(q+22)} > 3,$$

since the numerator minus three times the denominator is $q^2+21q+440$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q=54$ proves the inequality. ∎

Lemma 4.765 (Exact twenty-first C pause-arithmetic checks after the first six rows)**.** *For $27 \leq q \leq 53$, put*

$$\begin{aligned}P(q)=&\,4194304\binom{2q+21}{1}+16777216\binom{2q+21}{3}+67108864\binom{2q+21}{5}\\&+268435456\binom{2q+21}{7}+1073741824\binom{2q+21}{9}+4294967296\binom{2q+21}{11}\\&+17179869184\binom{2q+21}{13}+68719476736\binom{2q+21}{15}\\&+274877906944\binom{2q+21}{17}+1099511627776\binom{2q+21}{19}+4398046511104\binom{2q+21}{21}.\end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+21} \leq s_{2q+21}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyfirst_frontier_gmp_certificate.log` computes the displayed integer inequality for $27 \le q \le 53$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

$$\texttt{a5593e2a97e9cadcf10f028940d40366}\\\texttt{500494e1c24910ccb6d39bf66334659c}.$$

It verifies the exact positive margins

q	$P(q)$	$s_{2q+21} - 2^{2q+1}P(q)s_{q+21}$
27	9575151515658781798713933168640	985756847345365397668549487566928101202076121316509614773458892525721909334173324823800152168696519104905728
28	18149753346596525461825218674688	7158613480929762268448578947015233616259200760118484622452008356339934295745671832573210225646317695966635815936
29	33757003245024294460845415464960	44120544910882978721202463250197610593423144328107849325051457714354837298459106119318176500002942147786488526937088
30	61673628973181416064980403355648	282581280956550544245902388664810830472849239087675220981796874701933208175009681903504692235627848615039916457727139840
31	110792957802818732910527066931200	1900089231343788999457655862281293726410243695825185147085100958512450452135588410019209568532250630002737425851101313473536
32	195884888100800055947048033189888	13406570490101813922018918189286451995570875976146293397721077830650280881295367612048770932348077151989216336599429795824683008
33	341140422629441482711784908390400	99151535600921383911565449474296514088484211000218384693673187906296230264541841980058648203681457949706172485338995008571746772992
34	585663004126789398027409805017088	767803833895313764326235266814275193563047625851401244038831046389056682051747169658886346826304346253313066959885699612377886736359424
35	991877484988091565409201459036160	6219012485464160250675640067306720643382853351710774079025192427642045817909714769622082087040159484335381141565790416914558656285666170880
36	1658265587248587282210004410564608	52636114508647419356166723022649218158505616690899857107851748355726480996916376230376601818458115970203040338570196147029654984636923862102016
37	2738452785992812795864484308582400	465079114888081471940289531648985472257412313538510479133327670261543342836070296766671018659891071562531351537598771608274494691753346478791446528
38	4469530554282533144521588330201088	4286049101830464440406495420109395031352184405316405765232512254118723890648838508922916455277402916881167657458652528105893626761155031379100672000000
39	7213685774082238473572296788279296	41162110219623825932734233843601208328244031510942473643645999673220062414159622962599623642949315785137013910206388827775437555030430641183869871915929600
40	11518838880023956802605225443590144	411610493361212658609438075514938007634401958271237047730983098090314390859983980123617643307938811174731755930873962223083674536089370308191059126797667164160
41	18206211659385783497600980125483008	4282289551634348924329239706730221927448619999726851143512536260039107673812641791340906021596074618433248591945285593607784491053429855502204991712173465248161792
42	28495746356420880173204428663291904	46316141414884575423392030248557324373347325062095570344235381917801863515870883225873510751105223475181076803224145416339184981913216938954396074847474257008629284864
43	44184327326044573051589598210162688	520396248607814162823409340377204289423884505013387801432379105605759316422427722633448328522511783624410942238068866455533107801385148766155276289836858986466794937466 88
44	67897132543066529765526867231637504	6069768027148561641469968204121477225392587834421204200319935034455051444628527600774726608600034576509190781323760816421657347083690230906723463358250976247744896370331303936
45	103439570086625016614969015384670208	73442633136598658548124371150402878780418925317719173754823781157768800231555088625820275366047569203753463140761080174217107996317324264158042584075646414540008862401898571612160
46	156286648763455584613060296231092224	921245524013205551953790793040788704469227867486519229414132271679590770819733900881763224757941771074895208163079441738001066902956981888865641346434818510776139017905718779864547328
47	234258942103124593785807135318212608	11972270293412322391994921931618335297324123090001871438912085148481002653952144852197350460078416907811449979858628414106791753929314838209815288040538475845166144128257893060512461209600
48	348450347791182776912873805530005504	161095796530604867861834758162350912768004867949482092683233794758296912819818344505858193204479649301413475234827580628029280764588690022696949547732290599270861392001864705314215920490545152
49	514493642014577122423618856443445248	2243056546216084944751807745638494487795511108402461697692038010895203957275122305700540804864135054948937866971399581912136423639155355415999523093805870238736018467778793229609840817849089032192
50	754276652865419534823025158519783424	32299439121457320747514179580679347285477123267840979907021820508147995918657351519455895498157028869409313165113431597572787105302564960829477825501250590157674270183153800297502833245176389900042240
51	1098256307260626870214220222983831552	480736573437627632444442336999710451279209290573111695742968139549979963787136574785586233584619561334575411074277790549878241774693597957150767990068780604458283993443317895087741144498332202068961492992
52	1588561791218806248281669312079462400	7391682389000089802339052812312253432891207688078572676092652130829134490225275535427940381802945635126455768704119046862514125047041133491380667739344626287931386744851356466887954376470515971521003007967232
53	2283134001974655790967430957034897408	117348456612109982890333740601748666125536761977221951143034700801029267699557908273501065501788081486464104857575988336392926173121182550684645021410313364752493141763532200784298388022736314368175040141366460416.

This proves the claim. ∎

Proposition 4.766 (Residual C twenty-first nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+21} \le \delta_{2q+21}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+21}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+21$ with terminal height at least $2q$. If $21 \le q \le 26$, lemma 4.763 gives

$$|\delta_{2q+21}^{C_b}| \le s_{2q+21}/2.$$

For $q \ge 27$, lemma 4.762, with $P(q)$ denoting the polynomial from lemma 4.765, gives

$$|\delta_{2q+21}^{C_b}| \le P(q)\, 2^{2q} s_{q+21}.$$

If $27 \le q \le 53$, lemma 4.765 gives

$$|\delta_{2q+21}^{C_b}| \le s_{2q+21}/2.$$

If $q \ge 54$, lemma 4.764 gives

$$|\delta_{2q+21}^{C_b}| \le 2^{2q}\binom{2q+21}{q}s_{q+21}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+21}{q}s_{q+21} \le s_{2q+21}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

$$\texttt{124068c5daa1123088a3cd79c3f6b839}\\\texttt{fee45a1aaef637ed82161c465d86bf45}.$$

Thus $|\delta_{2q+21}^{C_b}| \le s_{2q+21}/2$ also for $q \ge 54$, which implies the claimed lower bound. ∎

Lemma 4.767 (*C* twenty-second-nonboundary height-bad paths have a twenty-two-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+22)}$$

with

$$\ell(\lambda^{(2q+22)}) \geq 2q$$

is at most

$$\begin{aligned}\Bigg(4194304 &+ 16777216\binom{2q+22}{2} + 67108864\binom{2q+22}{4} \\ &+ 268435456\binom{2q+22}{6} + 1073741824\binom{2q+22}{8} \\ &+ 4294967296\binom{2q+22}{10} + 17179869184\binom{2q+22}{12} \\ &+ 68719476736\binom{2q+22}{14} + 274877906944\binom{2q+22}{16} \\ &+ 1099511627776\binom{2q+22}{18} + 4398046511104\binom{2q+22}{20} \\ &+ 17592186044416\binom{2q+22}{22}\Bigg)2^{2q}s_{q+22}.\end{aligned}$$

Proof. The terminal height is even and at most $2q + 22$. Thus it is

$$2q+22,\quad 2q+20,\quad 2q+18,\quad 2q+16,\quad 2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q + 22 + p$. Therefore the twelve cases contribute at most

$$\binom{2q+22}{0}2^{2q+22}s_{q+11},\quad \binom{2q+22}{2}2^{2q+24}s_{q+12},\quad \binom{2q+22}{4}2^{2q+26}s_{q+13},$$

$$\binom{2q+22}{6}2^{2q+28}s_{q+14},\quad \binom{2q+22}{8}2^{2q+30}s_{q+15},\quad \binom{2q+22}{10}2^{2q+32}s_{q+16},$$

$$\binom{2q+22}{12}2^{2q+34}s_{q+17},\quad \binom{2q+22}{14}2^{2q+36}s_{q+18},$$

$$\binom{2q+22}{16}2^{2q+38}s_{q+19},\quad \binom{2q+22}{18}2^{2q+40}s_{q+20},$$

$$\binom{2q+22}{20}2^{2q+42}s_{q+21},\quad \binom{2q+22}{22}2^{2q+44}s_{q+22},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+11}, \ldots, s_{q+21}$ by s_{q+22}, the sum is the displayed bound. □

Lemma 4.768 (The first seven twenty-second *C* pause cases are absorbed by the fixed-point-free count)**.** *For $21 \leq q \leq 27$, the number of column-even Pieri two-strip paths of length $2q + 22$ with terminal height at least $2q$ is at most $s_{2q+22}/2$.*

Proof. Use the pause-set injection from lemma 4.767, but count the target even-column standard tableaux exactly by lemma 4.731. For $21 \le q \le 27$, this gives the upper bound

$$\sum_{\substack{0\le p\le 22\\ p \text{ even}}} \binom{2q+22}{p}(2q+21+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentysecond_frontier_gmp_certificate.log`

checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j - 2\mathrm{FPF}(q,j)$
21	64	3673546597525622629835972748936048001604247648717791222500228955528081485766953125	6955849797633944919506711702448105444927901170437191839127831754016933333908702532423606
22	66	724448100514946445082843487720702486409131621639371094099504071412022564047695312500	29386671851005507934856585201072593748197990359902535647244555074989280017515552547484169240
23	68	141871669249161368402940347157334384670147654103190649625545483370186803889439492968750	131909085005483873203466777873918234015144877086744529442930986139027191938182755323612616090532
24	70	27634119982394172339974879118216636692043911713376867907724295133823961896204868831250000	627984678316460030412029905566862323593670971401158715142982629338426049874380034534456514332849504
25	72	5361259500752364359897462463162369710276578512029857877855952385609760942353222041075000000	3165506839225007550037522624909651084680151266649452263508220962981924683953410568391326644961152767104
26	74	1037279652297170971042023343593013599978416878321007651590281460717021843855347766405181250000	16868160644363383382648743540649283907739197828775273879545440525301805222446792965089236201138755714508384
27	76	200357862346821966866479160795239368809935057938118084156631161999245725604543961093357734375000	94879010830044579516248919504777610596192027957411393212496424058368337060965458872883294753762971846510206800.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`c2db04c00a2fcf719ada1acdb7dadc65`
`e468bbaa1c87e3dc9a5b9acd34d653a4`.

∎

Lemma 4.769 (The central binomial factor absorbs the twenty-two-pause loss). *For every $q \ge 56$,*

$$\begin{aligned} 4194304 &+ 16777216\binom{2q+22}{2} + 67108864\binom{2q+22}{4} \\ &+ 268435456\binom{2q+22}{6} + 1073741824\binom{2q+22}{8} + 4294967296\binom{2q+22}{10} \\ &+ 17179869184\binom{2q+22}{12} + 68719476736\binom{2q+22}{14} \\ &+ 274877906944\binom{2q+22}{16} + 1099511627776\binom{2q+22}{18} \\ &+ 4398046511104\binom{2q+22}{20} + 17592186044416\binom{2q+22}{22} \le \binom{2q+22}{q}. \end{aligned}$$

Proof. At $q = 56$, the left side is

$$159401924255242975364410651900970532864 < 247522931562325802835549014143162387440 = \binom{134}{56}.$$

The margin in this base comparison is

$$88121007307082827471138362242191854576.$$

Let $P(q)$ denote the displayed left side. For $1 \le p \le 22$,

$$\frac{\binom{2q+24}{p}}{\binom{2q+22}{p}} \le \frac{(2q+24)(2q+23)}{(2q+2)(2q+1)} < 2 \qquad (q \ge 56),$$

because $4q^2 - 82q - 548 > 0$. The $p = 0$ summand is constant, so $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+24}{q+1}}{\binom{2q+22}{q}} = \frac{(2q+24)(2q+23)}{(q+1)(q+23)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 22q + 483$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 56$ proves the inequality. ∎

Lemma 4.770 (Exact twenty-second C pause-arithmetic checks after the first seven rows)**.** *For $28 \le q \le 55$, put*

$$\begin{aligned}P(q) = {} & 4194304 + 16777216\binom{2q+22}{2} + 67108864\binom{2q+22}{4} \\ & + 268435456\binom{2q+22}{6} + 1073741824\binom{2q+22}{8} + 4294967296\binom{2q+22}{10} \\ & + 17179869184\binom{2q+22}{12} + 68719476736\binom{2q+22}{14} \\ & + 274877906944\binom{2q+22}{16} + 1099511627776\binom{2q+22}{18} \\ & + 4398046511104\binom{2q+22}{20} + 17592186044416\binom{2q+22}{22}.\end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+22} \le s_{2q+22}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentysecond_frontier_gmp_certificate.log`

computes the displayed integer inequality for $28 \le q \le 55$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`c2db04c00a2fcf719ada1acdb7dadc65`
`e468bbaa1c87e3dc9a5b9acd34d653a4`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+22} - 2^{2q+1}P(q)s_{q+22}$
28	258232049829169720156803397844992	492349767652154619284693481786288236237095661454957232098550035795270121549384242662073768384508577547526308795392
29	492497525342386521140182577577984	3483466832503280899440696727311462501859329870562046674826263147602027156248074999507174034923285076608985660313680896
30	922100309616885951378364492152832	23021035901630077745090780235767623748628262348446900102731034684383355912289547211532393139652769256132053981105193794560
31	1696598100048949236625274551926784	158653734981690342641968115177593341474947050157045800007738266761342885201012596839444267358036127005508076604328738254071808
32	3070553978397797843455115926700032	1146259663079073097306649009879055070333578319532153953047054468587448893082794094141322410050692192209058271041018414794310498304
33	5471032294063014254708346012565504	8675753790859187255712452925703518902648658778434713496189514486069017491610845330889282049022569281121595780823032473306550480418816
34	9604733587650220221817503248023552	68718405728791263253484986922521028203919796920417966346542649695134802135107818487443516702903452389799510920930377584476819313273944064
35	16625985218571243036882413471924224	569039354335669849819008724579438469567142552046944983975232083374057359886892591671233129940426017607192272678504658484351539179176662028288
36	28397123618411399190739068211167232	4921474373819918838683817296764244584777652081219805826664610970133397418874362450249976850500456182072442333371476474691242392451199903922614272
37	47887667885091847721553177182470144	44415035728589812279713961893426735092172720379532192786694968595671509511430731568027943561845742821062085339556719266251982264347818846018985201664
38	79779897210998404237517082352156672	417889612984305518965438070040311560961682595139748428620620496601259413473948915178980011868502142614972383370048260382306062008566079544888620030562304
39	131378453514526214976117152231194624	4095628359235223907171693595308447469685570290256003419980443287620150570215116701754057883771055467620817324098051565097422202099071276405893028422027862016
40	213963689433201229108023355066286080	41778449637156135204880743980620049571726476214323010731457798253478559865629334122986823597490076283409222181556211534045305735822695321102779051188247054524416
41	344787072400281922188260093936533504	443216814209338570792073820551216354419973112758402700338092272454352348596724497056836886578504665717212600329057875964593464077596315238810447825914438752603643904
42	549987880542188023764797796070195200	4886351313102937424163844168436993005369033802655085360800939373298610015956158588367731328155910975754084907719311852731338450735790070239123350159963463854245030830080
43	868821390393188226793894493084975104	55942579353574319299224195792255582674764724022074526287177488596497496454085998525810084385577411281816539885431999477349922235188568687166823504138484803217270484688027648
44	1359739857343091889785609376644464640	664639403789427225561447001257947088689596883194726062726764916014453948769926032949660602233088806325443799349840701811653413945303865214746461812035408280020595571965659398144
45	2109071981931189293917744295831404544	8188851318177335552057944920067218736961260629180059375073073831568531584640283434479098811517956149189419564211439790004871185452306550851967206677127407892155854900611086429962240
46	3243321278808598988780603540158545920	104561339429957869976868373579255172011486290623368195306618721369753278347677539018842384040878915322483438369998714641325618451290557286305762968418028605245083618700982482883560587264
47	4946470774020608634677247868261105664	1382796873366955667733698785286980074157267178875875009827842356900495714534498979670691906462326499095594377512415918035149174228171124814684607860324735857888746044981734609236485822119936
48	7484168886498429501476337523349258240	18928751602053607333207431922158155511380168384351702050049851409072414021915098555592542513334707916076451323223062626090117125759494679195067731890424030322737468780917517426791668431713435648
49	11237315081066759481974355165471309824	268045196852917110920858864292453955721628047290841296537909620843269638828018884291466095874231486883221822347063069993588841213358044098104087940256347085193274303754733181433547765556279492182016
50	16748409447373062556011217099654430720	3924381011990010004171741527608164978530497770328942545279534321141635845996580399288371378142292912158949545428989308238780575834481797389619120737978038693740120789100928933597441050984972807914356736
51	24785135282172804514497676198302187520	59370954705574675743610932769466315206466503907905141886507701961765487474125475516394128490557956737635659706359250187603604941972386279628758590982808875081565464849590043216124284499591297379482818379776
52	36427080369369564745217083061571682304	927655959519066445608888540023467196375398960443697367332179503419659131760171301641400034247109897251509193238322104155202394251583784209240902686384555013003831791878095058868402456183887015322265348665704448
53	53183361546847072514389722488322916352	14961925445802499547156199139453750585162613576092731610569356185714633640126882589729405206798510690213251303787180327029784781597086269309420092302838918018746668809333889373083060424970909729714224614870077210624
54	77151311285441864814811229339500150784	248977571795961719585950638301180814987846650628626904010669237221170333286101822758486894115008628284986898588281356442069723383176481067975883126730727507661999142058502771176147292000203655218840591177770868374831104
55	111229454697260018616582403493170511872	42726404062304128946162314258743042343412284576381376255075629677900438499311048883915472404656047215926768659250609166161501882550473575298039440765850867027979580999584752754517841396553135819466820130913129644173222608896.

This proves the claim. ☐

Proposition 4.771 (Residual C twenty-second nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b + 1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+22} \le \delta_{2q+22}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+22}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+22$ with terminal height at least $2q$. If $21 \le q \le 27$, lemma 4.768 gives

$$|\delta_{2q+22}^{C_b}| \le s_{2q+22}/2.$$

For $q \ge 28$, lemma 4.767, with $P(q)$ denoting the polynomial from lemma 4.770, gives

$$|\delta_{2q+22}^{C_b}| \le P(q)\,2^{2q}s_{q+22}.$$

If $28 \le q \le 55$, lemma 4.770 gives

$$|\delta_{2q+22}^{C_b}| \le s_{2q+22}/2.$$

If $q \geq 56$, lemma 4.769 gives

$$|\delta^{C_b}_{2q+22}| \leq 2^{2q}\binom{2q+22}{q} s_{q+22}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+22}{q} s_{q+22} \leq s_{2q+22}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

$$\texttt{124068c5daa1123088a3cd79c3f6b839}$$
$$\texttt{fee45a1aaef637ed82161c465d86bf45}.$$

Thus $|\delta^{C_b}_{2q+22}| \leq s_{2q+22}/2$ also for $q \geq 56$, which implies the claimed lower bound. □

Lemma 4.772 (*C* twenty-third-nonboundary height-bad paths have a twenty-three-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+23)}$$

with

$$\ell(\lambda^{(2q+23)}) \geq 2q$$

is at most

$$\begin{aligned}\Bigg(&16777216\binom{2q+23}{1} + 67108864\binom{2q+23}{3} + 268435456\binom{2q+23}{5} \\ &+ 1073741824\binom{2q+23}{7} + 4294967296\binom{2q+23}{9} \\ &+ 17179869184\binom{2q+23}{11} + 68719476736\binom{2q+23}{13} \\ &+ 274877906944\binom{2q+23}{15} + 1099511627776\binom{2q+23}{17} \\ &+ 4398046511104\binom{2q+23}{19} + 17592186044416\binom{2q+23}{21} \\ &+ 70368744177664\binom{2q+23}{23}\Bigg) 2^{2q} s_{q+23}.\end{aligned}$$

Proof. The terminal height is even and at most $2q + 22$. Thus it is

$$2q+22,\quad 2q+20,\quad 2q+18,\quad 2q+16,\quad 2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q + 23 + p$. Therefore the twelve cases contribute at most

$$\binom{2q+23}{1}2^{2q+24}s_{q+12},\quad \binom{2q+23}{3}2^{2q+26}s_{q+13},\quad \binom{2q+23}{5}2^{2q+28}s_{q+14},$$
$$\binom{2q+23}{7}2^{2q+30}s_{q+15},\quad \binom{2q+23}{9}2^{2q+32}s_{q+16},\quad \binom{2q+23}{11}2^{2q+34}s_{q+17},$$
$$\binom{2q+23}{13}2^{2q+36}s_{q+18},\quad \binom{2q+23}{15}2^{2q+38}s_{q+19},$$

$$\binom{2q+23}{17}2^{2q+40}s_{q+20},\quad \binom{2q+23}{19}2^{2q+42}s_{q+21},$$
$$\binom{2q+23}{21}2^{2q+44}s_{q+22},\quad \binom{2q+23}{23}2^{2q+46}s_{q+23},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+12},\dots,s_{q+22}$ by s_{q+23}, the sum is the displayed bound. $\square$

Lemma 4.773 (The first eight twenty-third C pause cases are absorbed by the fixed-point-free count)**.** *For $21 \le q \le 28$, the number of column-even Pieri two-strip paths of length $2q+23$ with terminal height at least $2q$ is at most $s_{2q+23}/2$.*

Proof. Use the pause-set injection from lemma 4.772, but count the target even-column standard tableaux exactly by lemma 4.731. For $21 \le q \le 28$, this gives the upper bound

$$\sum_{\substack{1\le p\le 23\\ p\ \mathrm{odd}}}\binom{2q+23}{p}(2q+22+p)!!.$$

The exact C++/GMP/OpenMP replay

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checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j - 2\mathrm{FPF}(q,j)$
21	65	903395458925233862459538367562195635591789894032637801584533085396134624499776953125	448650321441963687264405033201057249372652442866136288714800384490174664225377113096137142
22	67	187855136252296598892955089953165371907925245701375187830773864045211057311947484375000	19542107889711707533333251865482116777364544383610708242338624854324720760328318810973929925 92
23	69	38737318451337022431677803596238118856006496549152483169175211416709091377624641299218750	9035761239613975541564946392112412263053744176192240448889799460994099438809511067555158156686212
24	71	7934575627332601543808973311549601252211573686508191850366035064164098120241572629940625000	4427287032137454682654898213958994605497869340659717546724570629677613588746098579310128601375417784 0
25	73	161675515429151038819799520046789660254358162219720144708546721008760401878712779457057500000 0	229499010248102463464991222573652238882394341287875600125844430305426187220931956107234735733578286703744
26	75	32813662001863269013681826769923089985447027288228611046332046859237180803800533089144765625000 0	12566767804682107696072798190572380273871732818025422920871295193273421377553607593062983596762882719282495 68
27	77	66413080832039621078408360898635664063082364466853853287713048631560118881554333641485725859375000	725823800056247801159015960551177078120923798834344051569679013039515421616506965069874533679096852464950503956 8
28	79	13417863213593947455210578777994636773263119312159516026950171056539369741242161635850168708281250000	441572328366926739893373273868865608373153462228331660494372240970374845172793154650623637550877936267470299880966 08.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`e095f8a87dbf4718ab5ac066d438ca1b`
`f6b32452d617172d90acd926094f717d`.

$\square$

Lemma 4.774 (The central binomial factor absorbs the twenty-three-pause loss)**.** *For every $q \ge 59$,*

$$\begin{aligned}&16777216\binom{2q+23}{1}+67108864\binom{2q+23}{3}+268435456\binom{2q+23}{5}\\&+1073741824\binom{2q+23}{7}+4294967296\binom{2q+23}{9}+17179869184\binom{2q+23}{11}\\&+68719476736\binom{2q+23}{13}+274877906944\binom{2q+23}{15}\\&+1099511627776\binom{2q+23}{17}+4398046511104\binom{2q+23}{19}\\&+17592186044416\binom{2q+23}{21}+70368744177664\binom{2q+23}{23}\le\binom{2q+23}{q}.\end{aligned}$$

Proof. At $q = 59$, the left side is

$$111277250433536009526517645580629232844 80 < 2879246904417615947316182815401169018440 0 = \binom{141}{59}.$$

The margin in this base comparison is

$$1766474400082255852051006359594876689992 0.$$

Let $P(q)$ denote the displayed left side. For $1 \le p \le 23$,

$$\frac{\binom{2q+25}{p}}{\binom{2q+23}{p}} \le \frac{(2q+25)(2q+24)}{(2q+2)(2q+1)} < 2 \qquad (q \ge 59),$$

because $4q^2 - 86q - 596 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+25}{q+1}}{\binom{2q+23}{q}} = \frac{(2q+25)(2q+24)}{(q+1)(q+24)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 23q + 528$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 59$ proves the inequality. $\square$

Lemma 4.775 (Exact twenty-third C pause-arithmetic checks after the first eight rows)**.** *For* $29 \le q \le 58$, *put*

$$\begin{aligned} P(q) = 16777216\binom{2q+23}{1} + 67108864\binom{2q+23}{3} + 268435456\binom{2q+23}{5} \\ + 1073741824\binom{2q+23}{7} + 4294967296\binom{2q+23}{9} + 17179869184\binom{2q+23}{11} \\ + 68719476736\binom{2q+23}{13} + 274877906944\binom{2q+23}{15} \\ + 1099511627776\binom{2q+23}{17} + 4398046511104\binom{2q+23}{19} \\ + 17592186044416\binom{2q+23}{21} + 70368744177664\binom{2q+23}{23}. \end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+23} \le s_{2q+23}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentythird_frontier_gmp_certificate.log`

computes the displayed integer inequality for $29 \le q \le 58$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`e095f8a87dbf4718ab5ac066d438ca1b`
`f6b32452d617172d90acd926094f717d`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+23} - 2^{2q+1}P(q)s_{q+23}$
29	6959847218612122581155773302177792	262922528006441399142918828714267913476240941026880200453666150140515503922968654849511691381689387860747668371565223936
30	13349871631976307535143002785710080	1892170132553195255385066987251681003257253233938376946794730614650933439139381536060337297887916915077038793292293500804096
31	25149969107522801701576766130225152	13403377699060053823326297425495888171169641624155236894853080179073898100276887289882919721260803741044277077572913129163737088
32	46580242725092383554064925012787200	99150246432412329359110239867329923011459192488748269341318123080059828273442363353162494250882430439994970226462670807207920494592
33	84890254315525433868343134579064832	767803312298115206380124590668612460791274271862995854727843082867091063826624240691954764532928491414937207520650052984921343060713472
34	152357514950698565274444603682979840	6219012273889054177034092486988637274704739875148842753030412471980785838598511677571292654154841980142843945402545248304669930662407690240
35	269495356939459044522956688113795072	52636114422569219004636129831169799132744683673390939573364040719629053243787870968559597926987305851214436239395378243551911491927636317351936
36	470140096490772841533509163234099200	465079114852941847109337451532761244551657005922639408120056162898473524839335483180893314956745356991369976299307356592016696596429156918280572928
37	809426194995622618572102831756541952	4286049101816065352548816412291165840736929919932053033562139352115348522378338693864707924050357567766085932781402410495866424166034454015541524365312
38	1376152883815089322159894077290577920	41162110219617901459938343912727016995032964405718622103152182328431309860201648692830782045089087510021594915880343847359751470524721781452447226109759488
39	2311761539545983226391674546527666176	411610493361210210278036077216449246138717985604567682817755037715590356945504900132723968061308007186773211343651116355590493055514696891731003676472012726272
40	3839169632931209750414177779960512512	4282289551634347907816754155702309537420341739733932749455332650336185199996541562922057933792687213828764334222267524154107507646881537891794193660483552093888512
41	6306166825583292591958642849445052416	46316141414884574999278514354751190723683581587119411306516820118331046817057041749061927161393316662469528250007192849882506291454443207541753731276688588647892680704
42	10250137928218062699305754359393943552	520396248607814162645551522457218886000865448647499950977610131637324894849314023735577938113626570881665247771377589166129637712336954223196431663672309527955139077545984
43	16493759814146855344831476732711141376	6069768027148561641394983873006235848758633131329060018188780965648660991304548118709174862174951542140599406612359545027161788828471247303697820683932754176760780541456760832
44	26285324552202525986867921870160658432	73442633136598658548092583997206106244129174619756220499063477959640329719278812421882687208268535534747162966382810056474726357803539373668862114529391698886709427292551794900992
45	41502867286266040072370371990657171456	921245524013205551953777241599390509538033643838635537858724440216766205151030737106199475416558020952643195672431491630645201243760244807930963838362248003732892803572924638528667648
46	64948851136519542628806235572228063232	11972270293412322391994916120839570364588466494434545579204000751779310045645561126319238554558813958457046659734240577640501538048460257507585892239217486398860293974401087078667404820480
47	100772474255034418205651304353225506816	161095796530604867861834755655929337116832809482232977447217696241572720858675336379385221376004734206837014945705637570677799870025510126092123324762454928914784168526611307414727296326598656
48	155070619782362929786856023651539484672	224305654621608494475180774455082982143708344357883747565088811653403115196389599913280925727581331323280352358389887252881610986210833932846662174401863431525322250557906298994635856291922608128
49	236737242158663771683872963272214839296	32299439121457320747514179580204447478536294093536632919278349022770251264859372565817296877455521722053168509855264311989953107779487328574724529339707177671473609272332813331451421920588725507719168
50	358656091291801559574794297533133750272	480736573437627632444442336999501802805755996801583315801384527688198597503928749544131647440241477723182700554271834130032342822099517110352662739530519037412643531988839667104360900947678056046412267520
51	539365072633013965767348183551809945600	7391682389000089802339052812312161182018208476891540798328492101603557949501945206358782661892505978036863459971499597026992005569884320157167074154459232558304870357220376082129956691372000203058185383903232
52	805364726928469583756578139051112005632	117348456612109982890333740601748625076815645934981681377185477259115506924646139921757826444013957158394337167279004355581966235040023322792971525788818976101815076207701796976685813626706130492474124166910771200
53	1194301475271651671216217023645136977920	1922607071870099047776047162593793145592773628626072961895351827009711892130090998431432884637000474239943340163933837619543574148599943214272620603215396738506279148253608052534802540106562918916934564589422187118592
54	1759332456734695880182256141017506906112	32491567507955082463681564380506379353319468845032910194418789643177304843769025445998468879835810297202419594889959121319796925928702813330102984652722155794792947323374944189599248658270279058688799122755945889615904768
55	2575078096059693015444602118006783344640	566124760447716687947008385494882991678864446076761149394571930976456960410609033369704293260812043563934665210799271341102690487997456224691017721311838704364558034715785658390916552648533894078965897569040137338076385378304
56	3745697405479725315540839646260669448192	10165191425994752876422337440716385533946880872422015491545245926867342861225075660298553306122004331610801138552774695576281167402950706831210383230666808778113818700821527377373669236113759299023385134796437454437422956969066496
57	5415787503093167348264225623949610844160	188012781990683033386658460821740358242778748176665131804483924314196340966432997952525302602383176767503528742780793537648809060014067002204485854529494320580445704666997291627966385065197699421166550718325734760740888836933251760128
58	7785022976611373546066487491729788239872	3580467372446220814152304752927059603961612402007422412364145090431399938181342628210340495226086617922725443098177687766978775250286104160884924615340097997872838051027943034756480292897594729869171895988990327969980597424148683188994048.

This proves the claim. $\square$

Proposition 4.776 (Residual C twenty-third nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b+1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+23} \le \delta_{2q+23}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+23}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+23$ with terminal height at least $2q$. If $21 \le q \le 28$, lemma 4.773 gives

$$|\delta_{2q+23}^{C_b}| \le s_{2q+23}/2.$$

For $q \ge 29$, lemma 4.772, with $P(q)$ denoting the polynomial from lemma 4.775, gives

$$|\delta_{2q+23}^{C_b}| \le P(q)\, 2^{2q} s_{q+23}.$$

If $29 \le q \le 58$, lemma 4.775 gives

$$|\delta_{2q+23}^{C_b}| \le s_{2q+23}/2.$$

If $q \ge 59$, lemma 4.774 gives

$$|\delta_{2q+23}^{C_b}| \le 2^{2q}\binom{2q+23}{q} s_{q+23}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+23}{q} s_{q+23} \le s_{2q+23}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+23}^{C_b}| \le s_{2q+23}/2$ also for $q \ge 59$, which implies the claimed lower bound. ☐

Lemma 4.777 (C twenty-fourth-nonboundary height-bad paths have a twenty-four-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+24)}$$

with

$$\ell(\lambda^{(2q+24)}) \ge 2q$$

is at most

$$\begin{aligned}\Bigg(&16777216\binom{2q+24}{0} + 67108864\binom{2q+24}{2} + 268435456\binom{2q+24}{4} \\ &+ 1073741824\binom{2q+24}{6} + 4294967296\binom{2q+24}{8} \\ &+ 17179869184\binom{2q+24}{10} + 68719476736\binom{2q+24}{12} \\ &+ 274877906944\binom{2q+24}{14} + 1099511627776\binom{2q+24}{16} \\ &+ 4398046511104\binom{2q+24}{18} + 17592186044416\binom{2q+24}{20} \\ &+ 70368744177664\binom{2q+24}{22} + 281474976710656\binom{2q+24}{24}\Bigg)2^{2q} s_{q+24}.\end{aligned}$$

Proof. The terminal height is even and at most $2q+24$. Thus it is

$$2q+24,\quad 2q+22,\quad 2q+20,\quad 2q+18,\quad 2q+16,\quad 2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+24+p$. Therefore the thirteen cases contribute at most

$$\binom{2q+24}{0}2^{2q+24}s_{q+12},\quad \binom{2q+24}{2}2^{2q+26}s_{q+13},\quad \binom{2q+24}{4}2^{2q+28}s_{q+14},$$

$$\binom{2q+24}{6}2^{2q+30}s_{q+15},\quad \binom{2q+24}{8}2^{2q+32}s_{q+16},\quad \binom{2q+24}{10}2^{2q+34}s_{q+17},$$

$$\binom{2q+24}{12}2^{2q+36}s_{q+18},\quad \binom{2q+24}{14}2^{2q+38}s_{q+19},$$

$$\binom{2q+24}{16}2^{2q+40}s_{q+20},\quad \binom{2q+24}{18}2^{2q+42}s_{q+21},$$

$$\binom{2q+24}{20}2^{2q+44}s_{q+22},\quad \binom{2q+24}{22}2^{2q+46}s_{q+23},\quad \binom{2q+24}{24}2^{2q+48}s_{q+24},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+12},\dots,s_{q+23}$ by s_{q+24}, the sum is the displayed bound. $\square$

Lemma 4.778 (The first nine twenty-fourth C pause cases are absorbed by the fixed-point-free count)**.** *For $21 \le q \le 29$, the number of column-even Pieri two-strip paths of length $2q+24$ with terminal height at least $2q$ is at most $s_{2q+24}/2$.*

Proof. Use the pause-set injection from lemma 4.777, but count the target even-column standard tableaux exactly by lemma 4.731. For $21 \le q \le 29$, this gives the upper bound

$$\sum_{\substack{0\le p\le 24\\ p\ \text{even}}} \binom{2q+24}{p}(2q+23+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyfourth_frontier_gmp_certificate.log`

checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j - 2\mathrm{FPF}(q,j)$
21	66	221151264435901656238544121628617366307975586291767907173573858313983738121479335937500	29386230997372837161436998278516311954870347226993195390172396126280794874084437684202919240
22	68	48444237602328436386124860047000459772495816702949179309459451070871911425056949258593750	131908988400752007044916742430078834328894101435406431750953666471216016934733512988593084840532
23	70	10509247527460596851185631453984174845850214095312956407091681807621698204689554487971875000	627984657353233215455624547875549173861754553084818347943823550970511024898631548947757276051599504
24	72	2261695805248852696343321513446275820373208213247535091658572763717440987569198217182676562500	3165506834712338458541321960942802982713924365323588994037210495420491061289748115137636654677949642104
25	74	483611060862391083361785820148019768484334416474890938047383312006436030727572497310988189062500	16868160643398235820228555715867796314130344319006561880352300664510219161355354947321801902049589698883384
26	76	102883760855512811712602813999666930338888613813951595055864607578377497600957196005762736652343750 0	94879010829839212710263356914459287274520279829371240806333268393419358171721907412315359921569903828932081800
27	78	21802604109980250245943707808188627322588633384955492241589185041273962279804658011997949891132812500	562512973617538987809161181648947151950133441437868744026688123560159091712555975513054430372461930273937912609368
28	80	4607287018232315624867407379250784363152357786636938322685439776958548701986208857622629213478867187500	3510497287734769771463259888233565327785269489229364491602610280576683474421257133947258007110692430889480343347613096
29	82	971787874854834355775051826541159058553705103847742341196288915488789687821798939394191225757695546875000	230314613184916108939794293491550616259765254448828464926843108356296268304635316003474778963050309688202964770108943 24496.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`ccd56620d91d6a3c2c9e23673211e25d`
`cb5e1ae5d1a4ddc71e699a7bf80fab4a`.

$\square$

Lemma 4.779 (The central binomial factor absorbs the twenty-four-pause loss)**.** *For every $q \geq 61$,*

$$\begin{aligned}
&16777216\binom{2q+24}{0} + 67108864\binom{2q+24}{2} + 268435456\binom{2q+24}{4} \\
&\quad + 1073741824\binom{2q+24}{6} + 4294967296\binom{2q+24}{8} + 17179869184\binom{2q+24}{10} \\
&\quad + 68719476736\binom{2q+24}{12} + 274877906944\binom{2q+24}{14} \\
&\quad + 1099511627776\binom{2q+24}{16} + 4398046511104\binom{2q+24}{18} \\
&\quad + 17592186044416\binom{2q+24}{20} + 70368744177664\binom{2q+24}{22} + 281474976710656\binom{2q+24}{24} \leq \binom{2q+24}{q}.
\end{aligned}$$

Proof. At $q = 61$, the left side is

$$544718913148137508039167628581708985532416 < 821730205038148901357321269744201244708160 = \binom{146}{61}.$$

The margin in this base comparison is

$$277011291890011393318153641162492259175744.$$

Let $P(q)$ denote the displayed left side. For $0 \leq p \leq 24$,

$$\frac{\binom{2q+26}{p}}{\binom{2q+24}{p}} \leq \frac{(2q+26)(2q+25)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 61),$$

because $4q^2 - 90q - 646 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+26}{q+1}}{\binom{2q+24}{q}} = \frac{(2q+26)(2q+25)}{(q+1)(q+25)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 24q + 575$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 61$ proves the inequality. □

Lemma 4.780 (Exact twenty-fourth C pause-arithmetic checks after the first nine rows)**.** *For $30 \leq q \leq 60$, put*

$$\begin{aligned}
P(q) = {}&16777216\binom{2q+24}{0} + 67108864\binom{2q+24}{2} + 268435456\binom{2q+24}{4} \\
&+ 1073741824\binom{2q+24}{6} + 4294967296\binom{2q+24}{8} + 17179869184\binom{2q+24}{10} \\
&+ 68719476736\binom{2q+24}{12} + 274877906944\binom{2q+24}{14} \\
&+ 1099511627776\binom{2q+24}{16} + 4398046511104\binom{2q+24}{18} \\
&+ 17592186044416\binom{2q+24}{20} + 70368744177664\binom{2q+24}{22} + 281474976710656\binom{2q+24}{24}.
\end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+24} \leq s_{2q+24}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyfourth_frontier_gmp_certificate.log` computes the displayed integer inequality for $30 \le q \le 60$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

$$\texttt{ccd56620d91d6a3c2c9e23673211e25d}$$
$$\texttt{cb5e1ae5d1a4ddc71e699a7bf80fab4a}.$$

It verifies the exact positive margins

q	$P(q)$	$s_{2q+24} - 2^{2q+1}P(q)s_{q+24}$
30	18748018961090625339551362145 9501056	152704139097871013142492293033505400204450787722883583472906050548331311475764735005352067097574849785076655327088181524932608
31	36153437287030547922212335767 0473728	1143758394960371245824775272138777286197289961127712644006871745955823899716077881608915871881023064108931577623916442620700735488
32	685047451016679782159360902945570816	8674701576976002221340231670277008106211846902476029819141182616454829383827423296141980454215545348175084055401841452803912290547712
33	1276632968835875025030772443377041408	68717962553457316450163404628290001830418790062539983147014073651171664844139435380666010990910886657581546429970698206330588303294543872
34	2341817351039060023677249220872503296	569039167350865492708009571033614789086355217943909095763027482581285546713478937410490743215664810430298854029513202517521731680933802037248
35	4231764443057592491270784110232076288	4921474294751144034097965894068037494532894778122830609219036972770117815348456633246982381248669855194609966701734773780274142113579384696696832
36	7538531237089192836487032324726194176	44415035695065473896178353901038559626357468942020052875814250827407796446093372793764425908144752760286941584525536875302579274571065508312100835328
37	13247738110892704221519004143859007488	417889612970048052584838800345578251023938118249645145308732633218462357627139197413789731047380341548699659967893576284569528541661734550608515016101888
38	22980482826037444330997883444549124096	4095628359229139677790038545688124387171460094377123147086100355618200738154337207887266653473368634147100419996259154525445667751484644340129805782617186304
39	39372519886172890702513405381071339520	41778449637153529114988857676920442216434294820583551155303705617283849502220728999700911080583796811511858074713729772019113902426963035230344049853311574614016
40	66662457918691349040098453939694862336	4432168142093374500127577436853366579368467924164521114694532925407719028703210886488664489914993361670611702690660893193079319969217922510918757316999783678645018624
41	111595199364953179020755435232270745600	48863513131029369400894315183290400712659896321652638172506273520799298034820697252829333376625720911925638885182168931019584440059515329326767117987606443519060849 86880
42	184796496702022672234964705370299170816	55942579353574319089200121285804183052364702460701355050031430265656095048183347996058690570092644767757844738449620019786837271458656742323612799747115143354520763894710272
43	302845432968906503364914207342628700160	664639403789427225469891417386216893082563736850625297308327739656035774053930804706232181854421856374729538204123701396077875271442978374414386388001637320109565305493531869184
44	491372191669037241023287832277602533376	8188851318177335552017835681331553521625823830781908333652171621113540647458560250228121805877747120829475283379658046339080133859498482993158249283469910028404252050618882778972160
45	789649978248266532899566899469557432320	1045613394299578699768507121921750818571474214764847803890616437039115190850568072618542511189381880131711996938410096388365507652986033322289916194570072231216888865548132983818834919424
46	1257347646879485290109527127404845203456	13827968733669556677336909672868432899877560544468167665763061422259778518022836760180582071814789951583539451636456171696127758820599452882773163555606057458636272131807416829792162204516352
47	1984383908535430897599901596338471567360	18928751602053607333207428442632680590630278194344844191327581739462380415346384804482035375812513526928445438587250165859941025626381579194999341590383000375782799509489626024548721085820141568
48	3105202164610359463871128385177199312896	268045196852917110920858862735216711572063370817254142509937383503318988311628113834360690049627929249456804003219236246555964396658406311356616801508849616168364547390102893840621428747416912232448
49	4819303060582601485630522408707377070080	3924381011990010004171741526907272586495603870423361575392707824762601471585680924553350870861239559700866686574877569056358603034039575504451563512173433948640478037121021242996500490315935122277236736
50	7420577239759044961332697315377976180736	59370954705574675743610932769149025465094289810529648106620169776007660610556910037344232180976623135577124841747554271684563774016932445259164116138560932914791515937422523018030042940298179964487048429568
51	11338935656308391815239519964619620745216	927655959519066445608888540023322715448747914219128683282635294869591395692061194841427738089327973005580665179266955557076837192895161539607023283101987835212100901642493531445922514576581493767038266907754496
52	17199020185826235449368671748043369873408	14961925445802499547156199139453684401279518482264563401424189166222589424479010156630396068268404871891208739432982833721131117716272311608051343551231908351519506336160508468874054838159713449019081760279254597632
53	25902498203249895658868172377391299035136	248977571795961719585950638301180784486799423456352646420822065655939772346125968183993343870566658365093321610647117879421404109833500428096376864853831449495281315669040256649352151116701808525065523336902784363954176 0
54	38742736876560392863289241836099351871488	4272640406230412894616231425874304220198669125066632079275818659721857857874177360526533028844979016924453049306754765019679374880365495383738540194548780751821493553638049054888081589560303686059241204283801258202895876096
55	57563690369881776688980740324262890438656	75577643333990134698069877278328635352321569315538900459211690836676740954423723133993408884547070454384064113995262120758493235433877324039820733695675135744315812568214010134836725143227766104671995958113024160958350002814976 0
56	84978838825635691607476887063932717498368	1377383225901130709058774297082485196193647821316210892873165874812255280770769610403573420585038148052677808687218856863625869339304535321929707807768019097471330223083123948450179889247172169638062092234766147823974979332032036864
57	124671275785201319272217273168479362482176	25851753711347463363316510715681015703504022257923548656702277033750767963944257680535643073473161572427713087490666596424649025118435583782164772815108315930482094312588863771235523325040744136755959383234056418447936408023796331184128
58	181802910737646130354949964751042351464448	499475127943829349601459914348599634357544242939569285100818023472633461738484986402944050685573262176246738556897543676399772756881237235189485775302590445531590212646520909031594569174238005396547000445248699515668215518024066853056020480
59	263569690794744095879283258861947239530496	9929937435148106573450272984529735153079341429560554937441580616711155559194984648276011040902352293824575853749226582600472588912408161956314562915518505560698302644754055982883008822570634014071799606774763499537795436032697053430196575587532 8
60	379951322839051563495119197660121481084928	2030547549274188134742472959750111053042948717656593143817610971325555222916107888913160703630461085345328588829923486244851097272234333650098879971618439265800584939081041676930588219918859907630933542133237893137584833307801748113963972123099136 00.

This proves the claim. ∎

Proposition 4.781 (Residual C twenty-fourth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b + 1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+24} \le \delta_{2q+24}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+24}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q + 24$ with terminal height at least $2q$. If $21 \le q \le 29$, lemma 4.778 gives

$$|\delta_{2q+24}^{C_b}| \le s_{2q+24}/2.$$

For $q \ge 30$, lemma 4.777, with $P(q)$ denoting the polynomial from lemma 4.780, gives

$$|\delta_{2q+24}^{C_b}| \le P(q)\, 2^{2q} s_{q+24}.$$

If $30 \le q \le 60$, lemma 4.780 gives

$$|\delta_{2q+24}^{C_b}| \le s_{2q+24}/2.$$

If $q \ge 61$, lemma 4.779 gives

$$|\delta_{2q+24}^{C_b}| \le 2^{2q}\binom{2q+24}{q} s_{q+24}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+24}{q} s_{q+24} \le s_{2q+24}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

$$\texttt{124068c5daa1123088a3cd79c3f6b839}$$
$$\texttt{fee45a1aaef637ed82161c465d86bf45}.$$

Thus $|\delta_{2q+24}^{C_b}| \le s_{2q+24}/2$ also for $q \ge 61$, which implies the claimed lower bound. ∎

Lemma 4.782 (*C* twenty-fifth-nonboundary height-bad paths have a twenty-five-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+25)}$$

with

$$\ell(\lambda^{(2q+25)}) \geq 2q$$

is at most

$$\begin{aligned}\Bigg(&67108864\binom{2q+25}{1} + 268435456\binom{2q+25}{3} + 1073741824\binom{2q+25}{5}\\ &+ 4294967296\binom{2q+25}{7} + 17179869184\binom{2q+25}{9}\\ &+ 68719476736\binom{2q+25}{11} + 274877906944\binom{2q+25}{13}\\ &+ 1099511627776\binom{2q+25}{15} + 4398046511104\binom{2q+25}{17}\\ &+ 17592186044416\binom{2q+25}{19} + 70368744177664\binom{2q+25}{21}\\ &+ 281474976710656\binom{2q+25}{23} + 1125899906842624\binom{2q+25}{25}\Bigg)2^{2q}s_{q+25}.\end{aligned}$$

Proof. The terminal height is even and at most $2q+24$. Thus it is

$$2q+24,\quad 2q+22,\quad 2q+20,\quad 2q+18,\quad 2q+16,\quad 2q+14,\quad 2q+12,\quad 2q+10,\quad 2q+8,\quad 2q+6,\quad 2q+4,\quad 2q+2,\quad \text{or}\quad 2q.$$

These cases have respectively $1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+25+p$. Therefore the thirteen cases contribute at most

$$\binom{2q+25}{1}2^{2q+26}s_{q+13},\quad \binom{2q+25}{3}2^{2q+28}s_{q+14},\quad \binom{2q+25}{5}2^{2q+30}s_{q+15},$$

$$\binom{2q+25}{7}2^{2q+32}s_{q+16},\quad \binom{2q+25}{9}2^{2q+34}s_{q+17},\quad \binom{2q+25}{11}2^{2q+36}s_{q+18},$$

$$\binom{2q+25}{13}2^{2q+38}s_{q+19},\quad \binom{2q+25}{15}2^{2q+40}s_{q+20},$$

$$\binom{2q+25}{17}2^{2q+42}s_{q+21},\quad \binom{2q+25}{19}2^{2q+44}s_{q+22},$$

$$\binom{2q+25}{21}2^{2q+46}s_{q+23},\quad \binom{2q+25}{23}2^{2q+48}s_{q+24},\quad \binom{2q+25}{25}2^{2q+50}s_{q+25},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+13}, \ldots, s_{q+24}$ by s_{q+25}, the sum is the displayed bound. □

Lemma 4.783 (The first ten twenty-fifth *C* pause cases are absorbed by the fixed-point-free count)**.** *For $21 \leq q \leq 30$, the number of column-even Pieri two-strip paths of length $2q+25$ with terminal height at least $2q$ is at most $s_{2q+25}/2$.*

Proof. Use the pause-set injection from lemma 4.782, but count the target even-column standard tableaux exactly by lemma 4.731. For $21 \le q \le 30$, this gives the upper bound

$$\sum_{\substack{1 \le p \le 25 \\ p \text{ odd}}} \binom{2q+25}{p}(2q+24+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyfifth_frontier_gmp_certificate.log`

checks the displayed integer sums and verifies

q	j	$\mathrm{FPF}(q,j)$	$s_j - 2\mathrm{FPF}(q,j)$
21	67	5394554710412640738332229160295504374471155221065232256689297937672360104745837600000000	1954103273587235005111756327875185673979708831107140922339104361021446719252851588240361742592
22	69	12436988606154125333980540152151971543403099228006940852375342097616651767138901047939843750	9035736443111400135988323294387715151586895082006777533313061048660327038924159545002344875436212
23	71	283587387296495168429636711721038512608308588922167475152219777193339256882588729987408125000	442728646654959521513108166344736582573076437448485443969125869531126621490040443818098315254729278 40
24	73	64070095189198344841918302340092495036831233784468170386313569685171548064857470421847780156250 0	22949900896993406998960734651168218248148023375633808768087542547332572793767620284765958288576383357 8744
25	75	1436525496394165246701877259186139771005082021530635554380138505406709174465019987456326009765625 00	1256676780181561943568484035097136212085343789401234828326766980672241077611203668029370372846805965287624568
26	77	3200916608382644491919041781372233949942838761908408597945188948157355429561813813295128517996093750 0	7258238000498592505584170794915546762303590580672709905187555666142825863299185662345272169192348925741301914568
27	79	709714187874766352154239395808437663972667476530516880646253245041301124816759975256832374534207031250 0	44157232836678506541306259247738386470556733458827259226145238078456613352720602521305511039905928679593762409971608
28	81	156756758502348119249090792792631622755946691244061390740657276329752202536541499285047454190344761718750 0	2825948206490830388198838372235905135284284493812923377863965436125759879784868655889635148660241828422065771521517984 40
29	83	345252126968540163371614107551115155897950374319690614908709560770110350594373567232369892938377255703125000	190009424191696942742695530305860859700569615680785405122348734942271607946719530144068864723796310392923806996744399961 3936
30	85	758930358844355318010655070873796097979490337547648545594115960030106763760304941569529345840073763992187500 00	134065723504502201816908579208459943645561248150606674425220823843369222202752803229391985539292658567348646406961644433159545 92.

The replay exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`1bd9d79c07c867c4d2d40c446cc351f3`
`a2fbd100861b423ff9e32c0bfc41ed91`.

∎

Lemma 4.784 (The central binomial factor absorbs the twenty-five-pause loss)**.** *For every $q \ge 64$,*

$$\begin{aligned}
&67108864\binom{2q+25}{1} + 268435456\binom{2q+25}{3} + 1073741824\binom{2q+25}{5} \\
&\quad + 4294967296\binom{2q+25}{7} + 17179869184\binom{2q+25}{9} + 68719476736\binom{2q+25}{11} \\
&\quad + 274877906944\binom{2q+25}{13} + 1099511627776\binom{2q+25}{15} \\
&\quad + 4398046511104\binom{2q+25}{17} + 17592186044416\binom{2q+25}{19} \\
&\quad + 70368744177664\binom{2q+25}{21} + 281474976710656\binom{2q+25}{23} + 1125899906842624\binom{2q+25}{25} \le \binom{2q+25}{q}.
\end{aligned}$$

Proof. At $q = 64$, the left side is

$$38106144974042111076149000047527760958062592 < 95784491736452493294473263363363867621614750 = \binom{153}{64}.$$

The margin in this base comparison is

$$57678346762410382218324263315836106663552158.$$

Let $P(q)$ denote the displayed left side. For $1 \le p \le 25$,

$$\frac{\binom{2q+27}{p}}{\binom{2q+25}{p}} \le \frac{(2q+27)(2q+26)}{(2q+2)(2q+1)} < 2 \qquad (q \ge 64),$$

because $4q^2 - 94q - 698 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+27}{q+1}}{\binom{2q+25}{q}} = \frac{(2q+27)(2q+26)}{(q+1)(q+26)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 25q + 624$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 64$ proves the inequality. ∎

Lemma 4.785 (Exact twenty-fifth C pause-arithmetic checks after the first ten rows). *For* $31 \le q \le 63$, *put*

$$\begin{aligned} P(q) = 67108864\binom{2q+25}{1} &+ 268435456\binom{2q+25}{3} + 1073741824\binom{2q+25}{5} \\ &+ 4294967296\binom{2q+25}{7} + 17179869184\binom{2q+25}{9} + 68719476736\binom{2q+25}{11} \\ &+ 274877906944\binom{2q+25}{13} + 1099511627776\binom{2q+25}{15} \\ &+ 4398046511104\binom{2q+25}{17} + 17592186044416\binom{2q+25}{19} \\ &+ 70368744177664\binom{2q+25}{21} + 281474976710656\binom{2q+25}{23} + 1125899906842624\binom{2q+25}{25}. \end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+25} \le s_{2q+25}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyfifth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $31 \le q \le 63$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`1bd9d79c07c867c4d2d40c446cc351f3`
`a2fbd100861b423ff9e32c0bfc41ed91`.

It verifies the exact positive margins

q	$P(q)$	$s_{2q+25} - 2^{2q+1}P(q)s_{q+25}$
31	5047910995737832421064635984025485312	97212057122575785915759525641356783823015024808426402433408095662647193929202815446110778772212386010016690567990609326507159939072
32	9783006720269792541474484407385456640	766954353939393667569022992733765659627320447644051088035886803401787878320182866061951295169165970091790231820563514577699107988938752
33	18637826401908385799827173801136750592	6218640262642298623471717586378681599886035423752070670239226479836892062882930723067076544314900919400470699573351001037903803081716652032
34	34934644971009688944854053909189099520	52635951248913380363096554670940018998687904181126439744860647599390334254131045206609871659195754850200902899681575443089135964970022965653504
35	64476834804439084162653019781224988672	465079043174301178184341342065280398037363358103801609164006057017319303807352772327207631445866712638835105197486933485620701727521928112464187392
36	117262843157300261390013131767288954880	4286049070267780244518691917077741591716599511447976648288701620850991830045484746836775910594577952463729420787301739274381220880340765557565978902528
37	210293928924511933265011519610056343552	41162110205699485727215339801851569360003072364737654866917945016614751335571302428572672472798228228711291271476228604848252895925404917830532364590518272
38	372119345391770058758436281929800089600	411610493355052818279014621341250612631165594694676111232575854973800296983548758460568261542246356358629772292125569504547950917834663202207777683562435993600
39	650111319848417452756276732233928146944	4282289551631615494114688941497730836944521750231747800052976634031574713602713948849735091626026989751657091606628419354322738950073704573212257614722887140204544
40	1121984269071048889070437519272118845440	46316141414883358323830991089558890165390225909696775048318027260804124004205204517583514728370573190721470757390521348826419219367353011835469893188271706081902886912
41	1913847194781341639765142715539163971584	520396248607813618888056039970573737576416291340395281490594120292579772502192842603237098490209560027663514896065142965844760721482200834085989460537478851693417009979392
42	3228216961546034014455968288751254241280	6069768027148561397417163478880042011925471821600363021004117538289020366099964328475453167457782340779278129204724950767844356450563316479059783686633218731847336200460255232
43	5387080268270936355857978319768905580544	73442633136598658438163607192844136475707651745106740958765256070499076775127961249335956052117258511331063033472811270436778910496225407164003459074427953452926495052654312144896
44	8897492608787993868254394228067155312640	921245524013205551904028317863614222246461577870892453852060285820248778078754660373931401545439176523407938139700609748110410402480627055246962113183739202592674854077332786853576704
45	14550646558193869490293003471973100027904	11972270293412322391972298372881474972085005273394744721007237567823874685518333821840780991436392594167382261878408470787919634600269140872721519305117362453639004378093553889982574346240
46	23570246115045439129108320722056212643840	161095796530604867861824423617067152648031497712093719386881574559529007608349199329781763181124243339030954893722313330215338715067674829545443756428457008831494197652298776309589290820599808
47	37832957795248335618617112623112377073664	2243056546216084944751803001437049750979070591224284350737505927179198261499391693157746798390083114365256346365526907849146018844682726368164164309735661165889189340361632518946794316479695585280
48	60193445195613404452996443326105339822080	32299439121457320747514177391710501466402092013167029819537121997675012086905136846153307770441919376514822370157341648520337832282033997695320641249503788354839009498550571069608190232062149340037120
49	94960073805735595991098655183408918953984	480736573437627632444442335984447992758354598717036525772465499694966097182063008643230425156751454414918321002551296174079102396480233844441394778693326607787941698384630667798698320081033742881625571328
50	148586194263139130728563524945902588395520	7391682389000089802339052811838849786799161367624831879592272371152075216279428155128810413595237593926056365691267517347572277772116461285241814936498613248305165130191867065090020008937044521827778026995712
51	230667838351478041798202113726030595227648	117348456612109982890333740601526720425634039231943514917811519681247869748671876191084801320015732361202214980943820531875682865413198777596986053270162399161544278752412159590327995033056968977269327412359135232
52	355374164207232412053046237485559762124800	1922607071870099047776047162593688531479358066518748369246194954730854888092945685178853701385041334706893164259079373685120146923773528088807053189989346528763513830541244285549829051141276279812942327357524297121792
53	5434853262228885150584532144126367962759168	32491567507955082463681564380506329756099361850786123605078230688004342363121789624083883531168901810822933010391881362621541071865472893154598898264224694491620874781379870349527441819467624069729627135545497384388722688
54	825277901933613159407929980558882667233280	566124760447716687947008385494882968030422223724738218851281121371358940877288842482231544590966245332113470500812557495446042670764535058543187273532245544870224535589925949609113656707493662860365386231723749130006894608384
55	1244586175355712284672662269264877257228288	10165191425994752876422337440716385522605723819592550729513028500215155175192921427722956972046854885385937400190163185442477018643508316492391326576879283861269740390446388176969310416712669107041559568470212003681397813032648704
56	1864485727473122212483823931436168286044160	188012781990683033386658460821740358237307957685411817590725441491092560213796608265641197240689771422190045382237578537049242206596189477928506009016149968348824263444879476718798924817215488957217930139312987883480563982282019831808
57	2775203333705242089089334506800809412395008	3580467372446220814152304752927059603958957759728714717147664963961223152885589177654770551741669607637811670657243819612277939584357826341648955599856718236096642572078066287964077616899528630241087526389498003046604030327927160032067584
58	4105066251980629155191870424941471612272640	70176245780667169039660641024950160871814512823772855044674054024142339441971273403262985812659257151710402437723248175993895202994953619792695850924242539208302196882729896265159600504407155498256600381672385080263056788469466295044219863040
59	6035580177644924306983708262908042755964928	1415015897186040195814710359626221483641510356304576359437403210437788218086343920080774700404849706913793543228190221865190813197916310392245894008023846536238144237555869876458939223054380406225148700756855569502945946175081253496396124536963072
60	8822088353065283435060801970710521598443520	29341408362910814023208684901539056904326733923073805664012069255904450899146645447130641572453654341904998696268843792579400652808960655321886463893138720432761065009541439744228660820847402449458619482872522491780259203009878747259150551038025531392
61	12821939899535888663003786654326644694581248	625433971413586789951272620870545357688048891354032157322278449762852737797980863532905728627516721165489285111497510465167355125870679556099090045558938890224274076497116789371673717536597241881722814919070900043582052109922419638514253462917872432709632
62	18532715539394626734029159870149216226508800	13699346116150342953286001332419072291155123064090619006771840983382965352684994833634945501050462429269621670438555819957805858762949285838961068087306789612807502765254114172662733184099825479528595127580714149719364531177577768150764748951569985734363840512
63	26643864127563813718862612702497080520736768	308231792878096742189917625001089406902867307567147701783229431502508469269082006751052137576273510382282813560168170544798567917743462138864167361895574640290273721012328235634333610674470073931773144502044554253748726884868826347977693548123023697889900683067392.

This proves the claim. ∎

Proposition 4.786 (Residual C twenty-fifth nonboundary lower correction). *For every residual row* C_b *with* $20 \le b \le 217$, *put* $q = b+1$ *and write* $s_j = m_j^{\mathrm{st}}$ *and* $m_j^{C_b} = s_j + \delta_j^{C_b}$. *Then*

$$-\frac{1}{2}s_{2q+25} \le \delta_{2q+25}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+25}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+25$ with terminal height at least $2q$. If $21 \le q \le 30$, lemma 4.783 gives

$$|\delta_{2q+25}^{C_b}| \le s_{2q+25}/2.$$

For $q \ge 31$, lemma 4.782, with $P(q)$ denoting the polynomial from lemma 4.785, gives

$$|\delta_{2q+25}^{C_b}| \le P(q)\, 2^{2q} s_{q+25}.$$

If $31 \le q \le 63$, lemma 4.785 gives

$$|\delta_{2q+25}^{C_b}| \le s_{2q+25}/2.$$

If $q \geq 64$, lemma 4.784 gives

$$|\delta_{2q+25}^{C_b}| \leq 2^{2q} \binom{2q+25}{q} s_{q+25}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1} \binom{2q+25}{q} s_{q+25} \leq s_{2q+25}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+25}^{C_b}| \leq s_{2q+25}/2$ also for $q \geq 64$, which implies the claimed lower bound. ☐

Lemma 4.787 (C twenty-sixth-nonboundary height-bad paths have a twenty-six-pause code)**.** *For $q \geq 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+26)}$$

with

$$\ell(\lambda^{(2q+26)}) \geq 2q$$

is at most

$$\begin{aligned}\Bigg(&67108864\binom{2q+26}{0} + 268435456\binom{2q+26}{2} + 1073741824\binom{2q+26}{4} \\ &+ 4294967296\binom{2q+26}{6} + 17179869184\binom{2q+26}{8} \\ &+ 68719476736\binom{2q+26}{10} + 274877906944\binom{2q+26}{12} \\ &+ 1099511627776\binom{2q+26}{14} + 4398046511104\binom{2q+26}{16} \\ &+ 17592186044416\binom{2q+26}{18} + 70368744177664\binom{2q+26}{20} \\ &+ 281474976710656\binom{2q+26}{22} + 1125899906842624\binom{2q+26}{24} \\ &+ 4503599627370496\binom{2q+26}{26}\Bigg) 2^{2q} s_{q+26}.\end{aligned}$$

Proof. The terminal height is even and at most $2q + 26$. Thus it is

$$2q+26, \quad 2q+24, \quad 2q+22, \quad 2q+20, \quad 2q+18, \quad 2q+16, \quad 2q+14, \quad 2q+12, \quad 2q+10, \quad 2q+8, \quad 2q+6, \quad 2q+4, \quad 2q+2, \quad \text{or} \quad 2q.$$

These cases have respectively $0, 2, 4, \ldots, 26$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q + 26 + p$. Therefore the fourteen cases contribute at most

$$\binom{2q+26}{0} 2^{2q+26} s_{q+13}, \quad \binom{2q+26}{2} 2^{2q+28} s_{q+14}, \quad \binom{2q+26}{4} 2^{2q+30} s_{q+15},$$
$$\binom{2q+26}{6} 2^{2q+32} s_{q+16}, \quad \binom{2q+26}{8} 2^{2q+34} s_{q+17}, \quad \binom{2q+26}{10} 2^{2q+36} s_{q+18},$$

$$\binom{2q+26}{12}2^{2q+38}s_{q+19},\quad \binom{2q+26}{14}2^{2q+40}s_{q+20},$$
$$\binom{2q+26}{16}2^{2q+42}s_{q+21},\quad \binom{2q+26}{18}2^{2q+44}s_{q+22},$$
$$\binom{2q+26}{20}2^{2q+46}s_{q+23},\quad \binom{2q+26}{22}2^{2q+48}s_{q+24},$$
$$\binom{2q+26}{24}2^{2q+50}s_{q+25},\quad \binom{2q+26}{26}2^{2q+52}s_{q+26},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+13},\dots,s_{q+25}$ by s_{q+26}, the sum is the displayed bound. ☐

Lemma 4.788 (The first eleven twenty-sixth C pause cases are absorbed by the fixed-point-free count)**.** *For $21 \le q \le 31$, the number of column-even Pieri two-strip paths of length $2q+26$ with terminal height at least $2q$ is at most $s_{2q+26}/2$.*

Proof. Use the pause-set injection from lemma 4.787, but count the target even-column standard tableaux exactly by lemma 4.731. For $21 \le q \le 31$, this gives the upper bound

$$\sum_{\substack{0\le p\le 26\\ p\ \text{even}}} \binom{2q+26}{p}(2q+25+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentysixth_frontier_gmp_certificate.log`

checks the displayed integer sums, exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`6a107c7c20c107e6a71db9aa65cf2884`
`df0e1e94db82c20d7d3f1ae3acc4c07d`.

Its transcript contains exactly eleven `C_FPF_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+26} - 2\,\mathrm{FPF}(q, 2q+26).$$

This proves the half-stable bound for these first eleven cases. ☐

Lemma 4.789 (The central binomial factor absorbs the twenty-six-pause loss)**.** *For every $q \ge 66$,*

$$\begin{aligned}
&67108864\binom{2q+26}{0} + 268435456\binom{2q+26}{2} + 1073741824\binom{2q+26}{4}\\
&+ 4294967296\binom{2q+26}{6} + 17179869184\binom{2q+26}{8} + 68719476736\binom{2q+26}{10}\\
&+ 274877906944\binom{2q+26}{12} + 1099511627776\binom{2q+26}{14}\\
&+ 4398046511104\binom{2q+26}{16} + 17592186044416\binom{2q+26}{18}\\
&+ 70368744177664\binom{2q+26}{20} + 281474976710656\binom{2q+26}{22}\\
&+ 1125899906842624\binom{2q+26}{24} + 4503599627370496\binom{2q+26}{26} \le \binom{2q+26}{q}.
\end{aligned}$$

Proof. At $q = 66$, the left side is

$$186772573187334875256801623442931510752260 9152 < 2737158422982075697084497097760673619727797850 = \binom{158}{66}.$$

The margin in this base comparison is

$$869432691108726944516480863331358512205188698.$$

Let $P(q)$ denote the displayed left side. For $0 \le p \le 26$,

$$\frac{\binom{2q+28}{p}}{\binom{2q+26}{p}} \le \frac{(2q+28)(2q+27)}{(2q+2)(2q+1)} < 2 \qquad (q \ge 66),$$

where the $p = 0$ term has ratio 1, and the displayed strict inequality follows from $4q^2 - 98q - 752 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+28}{q+1}}{\binom{2q+26}{q}} = \frac{(2q+28)(2q+27)}{(q+1)(q+27)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 26q + 675$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 66$ proves the inequality. $\square$

Lemma 4.790 (Exact twenty-sixth C pause-arithmetic checks after the first eleven rows)**.** *For $32 \le q \le 65$, put*

$$\begin{aligned} P(q) = {} & 67108864\binom{2q+26}{0} + 268435456\binom{2q+26}{2} + 1073741824\binom{2q+26}{4} \\ & + 4294967296\binom{2q+26}{6} + 17179869184\binom{2q+26}{8} + 68719476736\binom{2q+26}{10} \\ & + 274877906944\binom{2q+26}{12} + 1099511627776\binom{2q+26}{14} \\ & + 4398046511104\binom{2q+26}{16} + 17592186044416\binom{2q+26}{18} \\ & + 70368744177664\binom{2q+26}{20} + 281474976710656\binom{2q+26}{22} \\ & + 1125899906842624\binom{2q+26}{24} + 4503599627370496\binom{2q+26}{26}. \end{aligned}$$

Then

$$2^{2q+1} P(q) s_{q+26} \le s_{2q+26}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentysixth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $32 \le q \le 65$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`6a107c7c20c107e6a71db9aa65cf2884`
`df0e1e94db82c20d7d3f1ae3acc4c07d`.

Its transcript contains exactly thirty-four `C_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+26} - 2^{2q+1} P(q) s_{q+26}.$$

This proves the claim. $\square$

Proposition 4.791 (Residual C twenty-sixth nonboundary lower correction)**.** *For every residual row C_b with $20 \le b \le 217$, put $q = b + 1$ and write $s_j = m_j^{\mathrm{st}}$ and $m_j^{C_b} = s_j + \delta_j^{C_b}$. Then*

$$-\frac{1}{2}s_{2q+26} \le \delta_{2q+26}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+26}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q + 26$ with terminal height at least $2q$. If $21 \le q \le 31$, lemma 4.788 gives

$$|\delta_{2q+26}^{C_b}| \le s_{2q+26}/2.$$

For $q \ge 32$, lemma 4.787, with $P(q)$ denoting the polynomial from lemma 4.790, gives

$$|\delta_{2q+26}^{C_b}| \le P(q)\, 2^{2q} s_{q+26}.$$

If $32 \le q \le 65$, lemma 4.790 gives

$$|\delta_{2q+26}^{C_b}| \le s_{2q+26}/2.$$

If $q \ge 66$, lemma 4.789 gives

$$|\delta_{2q+26}^{C_b}| \le 2^{2q}\binom{2q+26}{q} s_{q+26}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+26}{q} s_{q+26} \le s_{2q+26}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta_{2q+26}^{C_b}| \le s_{2q+26}/2$ also for $q \ge 66$, which implies the claimed lower bound. ∎

Lemma 4.792 (C twenty-seventh-nonboundary height-bad paths have a twenty-seven-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+27)}$$

with

$$\ell(\lambda^{(2q+27)}) \ge 2q$$

is at most

$$\begin{aligned}\Bigg(268435456\binom{2q+27}{1} &+ 1073741824\binom{2q+27}{3} + 4294967296\binom{2q+27}{5}\\ &+ 17179869184\binom{2q+27}{7} + 68719476736\binom{2q+27}{9}\\ &+ 274877906944\binom{2q+27}{11} + 1099511627776\binom{2q+27}{13}\\ &+ 4398046511104\binom{2q+27}{15} + 17592186044416\binom{2q+27}{17}\\ &+ 70368744177664\binom{2q+27}{19} + 281474976710656\binom{2q+27}{21}\\ &+ 1125899906842624\binom{2q+27}{23} + 4503599627370496\binom{2q+27}{25}\\ &+ 18014398509481984\binom{2q+27}{27}\Bigg)2^{2q}s_{q+27}.\end{aligned}$$

Proof. The terminal height is even and at most $2q+27$. Thus it is

$2q+26$, $2q+24$, $2q+22$, $2q+20$, $2q+18$, $2q+16$, $2q+14$, $2q+12$, $2q+10$, $2q+8$, $2q+6$, $2q+4$, $2q+2$, or $2q$.

These cases have respectively $1,3,5,\ldots,27$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+27+p$. Therefore the fourteen cases contribute at most

$$\binom{2q+27}{1}2^{2q+28}s_{q+14},\quad \binom{2q+27}{3}2^{2q+30}s_{q+15},\quad \binom{2q+27}{5}2^{2q+32}s_{q+16},$$
$$\binom{2q+27}{7}2^{2q+34}s_{q+17},\quad \binom{2q+27}{9}2^{2q+36}s_{q+18},\quad \binom{2q+27}{11}2^{2q+38}s_{q+19},$$
$$\binom{2q+27}{13}2^{2q+40}s_{q+20},\quad \binom{2q+27}{15}2^{2q+42}s_{q+21},$$
$$\binom{2q+27}{17}2^{2q+44}s_{q+22},\quad \binom{2q+27}{19}2^{2q+46}s_{q+23},$$
$$\binom{2q+27}{21}2^{2q+48}s_{q+24},\quad \binom{2q+27}{23}2^{2q+50}s_{q+25},$$
$$\binom{2q+27}{25}2^{2q+52}s_{q+26},\quad \binom{2q+27}{27}2^{2q+54}s_{q+27},$$

by lemmas 4.681 and 4.682. After replacing $s_{q+14},\ldots,s_{q+26}$ by s_{q+27}, the sum is the displayed bound. ☐

Lemma 4.793 (The first twelve twenty-seventh C pause cases are absorbed by the fixed-point-free count)**.** *For $21 \le q \le 32$, the number of column-even Pieri two-strip paths of length $2q+27$ with terminal height at least $2q$ is at most $s_{2q+27}/2$.*

Proof. Use the pause-set injection from lemma 4.792, but count the target even-column standard tableaux exactly by lemma 4.731. For $21 \le q \le 32$, this gives the upper bound

$$\sum_{\substack{1\le p\le 27\\ p \text{ odd}}}\binom{2q+27}{p}(2q+26+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyseventh_frontier_gmp_certificate.log`

checks the displayed integer sums, exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`67df773882805ff4321cd3659e16b1b9`
`985db56a76a94d7136448011ef30f80d`.

Its transcript contains exactly twelve `C_FPF_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+27} - 2\,\mathrm{FPF}(q, 2q+27).$$

This proves the half-stable bound for these first twelve cases. $\square$

Lemma 4.794 (The central binomial factor absorbs the twenty-seven-pause loss). *For every $q \geq 69$,*

$$\begin{aligned}
&268435456\binom{2q+27}{1} + 1073741824\binom{2q+27}{3} + 4294967296\binom{2q+27}{5} \\
&\quad + 17179869184\binom{2q+27}{7} + 68719476736\binom{2q+27}{9} + 274877906944\binom{2q+27}{11} \\
&\quad + 1099511627776\binom{2q+27}{13} + 4398046511104\binom{2q+27}{15} \\
&\quad + 17592186044416\binom{2q+27}{17} + 70368744177664\binom{2q+27}{19} \\
&\quad + 281474976710656\binom{2q+27}{21} + 1125899906842624\binom{2q+27}{23} \\
&\quad + 4503599627370496\binom{2q+27}{25} + 18014398509481984\binom{2q+27}{27} \leq \binom{2q+27}{q}.
\end{aligned}$$

Proof. At $q = 69$, the left side is

$$130891286985753943793524421574917285603272818688 < 319620688063970456986413452647207180184213345250 = \binom{165}{69}.$$

The margin in this base comparison is

$$188729401078216513192889031072289894580940526562.$$

Let $P(q)$ denote the displayed left side. For $1 \leq p \leq 27$,

$$\frac{\binom{2q+29}{p}}{\binom{2q+27}{p}} \leq \frac{(2q+29)(2q+28)}{(2q+2)(2q+1)} < 2 \qquad (q \geq 69),$$

and the displayed strict inequality follows from $4q^2 - 102q - 808 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+29}{q+1}}{\binom{2q+27}{q}} = \frac{(2q+29)(2q+28)}{(q+1)(q+28)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 27q + 728$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 69$ proves the inequality. $\square$

Lemma 4.795 (Exact twenty-seventh C pause-arithmetic checks after the first twelve rows)**.** *For* $33 \le q \le 68$*, put*

$$\begin{aligned}P(q) = 268435456\binom{2q+27}{1} &+ 1073741824\binom{2q+27}{3} + 4294967296\binom{2q+27}{5}\\ &+ 17179869184\binom{2q+27}{7} + 68719476736\binom{2q+27}{9} + 274877906944\binom{2q+27}{11}\\ &+ 1099511627776\binom{2q+27}{13} + 4398046511104\binom{2q+27}{15}\\ &+ 17592186044416\binom{2q+27}{17} + 70368744177664\binom{2q+27}{19}\\ &+ 281474976710656\binom{2q+27}{21} + 1125899906842624\binom{2q+27}{23}\\ &+ 4503599627370496\binom{2q+27}{25} + 18014398509481984\binom{2q+27}{27}.\end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+27} \le s_{2q+27}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyseventh_frontier_gmp_certificate.log`

computes the displayed integer inequality for $33 \le q \le 68$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`67df773882805ff4321cd3659e16b1b9`
`985db56a76a94d7136448011ef30f80d`.

Its transcript contains exactly thirty-six `C_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+27} - 2^{2q+1}P(q)s_{q+27}.$$

This proves the claim. □

Proposition 4.796 (Residual C twenty-seventh nonboundary lower correction)**.** *For every residual row* C_b *with* $20 \le b \le 217$*, put* $q = b+1$ *and write* $s_j = m_j^{\mathrm{st}}$ *and* $m_j^{C_b} = s_j + \delta_j^{C_b}$*. Then*

$$-\frac{1}{2}s_{2q+27} \le \delta_{2q+27}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+27}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+27$ with terminal height at least $2q$. If $21 \le q \le 32$, lemma 4.793 gives

$$|\delta_{2q+27}^{C_b}| \le s_{2q+27}/2.$$

For $q \ge 33$, lemma 4.792, with $P(q)$ denoting the polynomial from lemma 4.795, gives

$$|\delta_{2q+27}^{C_b}| \le P(q)\,2^{2q}s_{q+27}.$$

If $33 \le q \le 68$, lemma 4.795 gives

$$|\delta_{2q+27}^{C_b}| \le s_{2q+27}/2.$$

If $q \ge 69$, lemma 4.794 gives

$$|\delta_{2q+27}^{C_b}| \le 2^{2q}\binom{2q+27}{q}s_{q+27}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+27}{q}s_{q+27} \le s_{2q+27}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta^{C_b}_{2q+27}| \le s_{2q+27}/2$ also for $q \ge 69$, which implies the claimed lower bound. □

Lemma 4.797 (*C* twenty-eighth-nonboundary height-bad paths have a twenty-eight-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+28)}$$

with

$$\ell(\lambda^{(2q+28)}) \ge 2q$$

is at most

$$\begin{aligned}\Bigg(&268435456\binom{2q+28}{0} + 1073741824\binom{2q+28}{2} + 4294967296\binom{2q+28}{4}\\ &+ 17179869184\binom{2q+28}{6} + 68719476736\binom{2q+28}{8}\\ &+ 274877906944\binom{2q+28}{10} + 1099511627776\binom{2q+28}{12}\\ &+ 4398046511104\binom{2q+28}{14} + 17592186044416\binom{2q+28}{16}\\ &+ 70368744177664\binom{2q+28}{18} + 281474976710656\binom{2q+28}{20}\\ &+ 1125899906842624\binom{2q+28}{22} + 4503599627370496\binom{2q+28}{24}\\ &+ 18014398509481984\binom{2q+28}{26} + 72057594037927936\binom{2q+28}{28}\Bigg)2^{2q}s_{q+28}.\end{aligned}$$

Proof. The terminal height is even and at most $2q + 28$. Thus it is one of

$2q+28,\ 2q+26,\ 2q+24,\ 2q+22,\ 2q+20,\ 2q+18,\ 2q+16,\ 2q+14,\ 2q+12,\ 2q+10,\ 2q+8,\ 2q+6,\ 2q+4,\ 2q+2,\ 2q.$

These cases have respectively $0, 2, 4, \ldots, 28$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q + 28 + p$. Hence the p-pause contribution is at most

$$\binom{2q+28}{p}2^{2q+28+p}s_{q+14+p/2} \qquad (p = 0, 2, 4, \ldots, 28)$$

by lemmas 4.681 and 4.682. After replacing $s_{q+14}, s_{q+15}, \ldots, s_{q+27}$ by s_{q+28}, the sum is the displayed bound. □

Lemma 4.798 (The first thirteen twenty-eighth *C* pause cases are absorbed by the fixed-point-free count)**.** *For $21 \le q \le 33$, the number of column-even Pieri two-strip paths of length $2q + 28$ with terminal height at least $2q$ is at most $s_{2q+28}/2$.*

Proof. Use the pause-set injection from lemma 4.797, but count the target even-column standard tableaux exactly by lemma 4.731. For $21 \le q \le 33$, this gives the upper bound

$$\sum_{\substack{0 \le p \le 28 \\ p \text{ even}}} \binom{2q+28}{p} (2q+27+p)!!.$$

The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyeighth_frontier_gmp_certificate.log`

checks the displayed integer sums, exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`400e6b4c50b78e1c490ce93daca73864`
`a2ebfc3380e93c50ac7d502205299b0e`.

Its transcript contains exactly thirteen `C_FPF_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+28} - 2\,\mathrm{FPF}(q, 2q+28).$$

This proves the half-stable bound for these first thirteen cases. □

Lemma 4.799 (The central binomial factor absorbs the twenty-eight-pause loss)**.** *For every $q \ge 71$,*

$$\begin{aligned}
&268435456\binom{2q+28}{0} + 1073741824\binom{2q+28}{2} + 4294967296\binom{2q+28}{4} \\
&\quad + 17179869184\binom{2q+28}{6} + 68719476736\binom{2q+28}{8} + 274877906944\binom{2q+28}{10} \\
&\quad + 1099511627776\binom{2q+28}{12} + 4398046511104\binom{2q+28}{14} \\
&\quad + 17592186044416\binom{2q+28}{16} + 70368744177664\binom{2q+28}{18} \\
&\quad + 281474976710656\binom{2q+28}{20} + 1125899906842624\binom{2q+28}{22} \\
&\quad + 4503599627370496\binom{2q+28}{24} + 18014398509481984\binom{2q+28}{26} + 72057594037927936\binom{2q+28}{28} \le \binom{2q+28}{q}.
\end{aligned}$$

Proof. At $q = 71$, the left side is

$$642246738998335834440489586521956587876845722009 6 < 91435523498595804295645744071617910834198059340 00 = \binom{170}{71}.$$

The margin in this base comparison is

$$272108495987622208515967854194222520465134871390 4.$$

Let $P(q)$ denote the displayed left side. For $0 \le p \le 28$,

$$\frac{\binom{2q+30}{p}}{\binom{2q+28}{p}} \le \frac{(2q+30)(2q+29)}{(2q+2)(2q+1)} < 2 \qquad (q \ge 71),$$

and the displayed strict inequality follows from $4q^2 - 106q - 866 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+30}{q+1}}{\binom{2q+28}{q}} = \frac{(2q+30)(2q+29)}{(q+1)(q+29)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 28q + 783$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 71$ proves the inequality. □

Lemma 4.800 (Exact twenty-eighth C pause-arithmetic checks after the first thirteen rows)**.** *For* $34 \le q \le 70$*, put*

$$\begin{aligned}P(q) = 268435456&\binom{2q+28}{0} + 1073741824\binom{2q+28}{2} + 4294967296\binom{2q+28}{4}\\&+ 17179869184\binom{2q+28}{6} + 68719476736\binom{2q+28}{8} + 274877906944\binom{2q+28}{10}\\&+ 1099511627776\binom{2q+28}{12} + 4398046511104\binom{2q+28}{14}\\&+ 17592186044416\binom{2q+28}{16} + 70368744177664\binom{2q+28}{18}\\&+ 281474976710656\binom{2q+28}{20} + 1125899906842624\binom{2q+28}{22}\\&+ 4503599627370496\binom{2q+28}{24} + 18014398509481984\binom{2q+28}{26} + 72057594037927936\binom{2q+28}{28}.\end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+28} \le s_{2q+28}.$$

Proof. The exact C++/GMP/OpenMP replay

`certificates/post_m29/bc_twentyeighth_frontier_gmp_certificate.log`

computes the displayed integer inequality for $34 \le q \le 70$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`400e6b4c50b78e1c490ce93daca73864`
`a2ebfc3380e93c50ac7d502205299b0e`.

Its transcript contains exactly thirty-seven `C_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+28} - 2^{2q+1}P(q)s_{q+28}.$$

This proves the claim. □

Proposition 4.801 (Residual C twenty-eighth nonboundary lower correction)**.** *For every residual row* C_b *with* $20 \le b \le 217$*, put* $q = b+1$ *and write* $s_j = m_j^{\mathrm{st}}$ *and* $m_j^{C_b} = s_j + \delta_j^{C_b}$*. Then*

$$-\frac{1}{2}s_{2q+28} \le \delta_{2q+28}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+28}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+28$ with terminal height at least $2q$. If $21 \le q \le 33$, lemma 4.798 gives

$$|\delta_{2q+28}^{C_b}| \le s_{2q+28}/2.$$

For $q \ge 34$, lemma 4.797, with $P(q)$ denoting the polynomial from lemma 4.800, gives

$$|\delta_{2q+28}^{C_b}| \le P(q)\,2^{2q}s_{q+28}.$$

If $34 \le q \le 70$, lemma 4.800 gives

$$|\delta_{2q+28}^{C_b}| \le s_{2q+28}/2.$$

If $q \ge 71$, lemma 4.799 gives

$$|\delta_{2q+28}^{C_b}| \le 2^{2q}\binom{2q+28}{q}s_{q+28}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+28}{q}s_{q+28} \le s_{2q+28}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta^{C_b}_{2q+28}| \le s_{2q+28}/2$ also for $q \ge 71$, which implies the claimed lower bound. □

Lemma 4.802 (*C* twenty-ninth-nonboundary height-bad paths have a twenty-nine-pause code)**.** *For $q \ge 1$, the number of column-even Pieri two-strip paths*

$$\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(2q+29)}$$

with

$$\ell(\lambda^{(2q+29)}) \ge 2q$$

is at most

$$\begin{aligned}\Bigg(&1073741824\binom{2q+29}{1} + 4294967296\binom{2q+29}{3} + 17179869184\binom{2q+29}{5}\\ &+ 68719476736\binom{2q+29}{7} + 274877906944\binom{2q+29}{9}\\ &+ 1099511627776\binom{2q+29}{11} + 4398046511104\binom{2q+29}{13}\\ &+ 17592186044416\binom{2q+29}{15} + 70368744177664\binom{2q+29}{17}\\ &+ 281474976710656\binom{2q+29}{19} + 1125899906842624\binom{2q+29}{21}\\ &+ 4503599627370496\binom{2q+29}{23} + 18014398509481984\binom{2q+29}{25}\\ &+ 72057594037927936\binom{2q+29}{27} + 288230376151711744\binom{2q+29}{29}\Bigg)2^{2q}s_{q+29}.\end{aligned}$$

Proof. The terminal height is even and at most $2q+29$. Thus it is one of

$$2q+28,\ 2q+26,\ 2q+24,\ 2q+22,\ 2q+20,\ 2q+18,\ 2q+16,\ 2q+14,\ 2q+12,\ 2q+10,\ 2q+8,\ 2q+6,\ 2q+4,\ 2q+2,\ 2q.$$

These cases have respectively $1, 3, 5, \dots, 29$ steps which fail to create a new row. Deleting the first column, shifting left, and standardizing the repeated pause labels injects the paths with p pauses into the choice of the p-element pause set and an even-column standard tableau of size $2q+29+p$. Hence the p-pause contribution is at most

$$\binom{2q+29}{p}2^{2q+29+p}s_{q+14+(p+1)/2} \qquad (p = 1, 3, 5, \dots, 29)$$

by lemmas 4.681 and 4.682. After replacing $s_{q+15}, s_{q+16}, \dots, s_{q+28}$ by s_{q+29}, the sum is the displayed bound. □

Lemma 4.803 (The first fourteen twenty-ninth *C* pause cases are absorbed by the fixed-point-free count)**.** *For $21 \le q \le 34$, the number of column-even Pieri two-strip paths of length $2q+29$ with terminal height at least $2q$ is at most $s_{2q+29}/2$.*

Proof. Use the pause-set injection from lemma 4.802, but count the target even-column standard tableaux exactly by lemma 4.731. For $21 \le q \le 34$, this gives the upper bound

$$\sum_{\substack{1 \le p \le 29 \\ p \text{ odd}}} \binom{2q+29}{p} (2q+28+p)!!.$$

The exact C++/GMP replay

`certificates/post_m29/bc_twentyninth_c_frontier_gmp_certificate.log`

checks the displayed integer sums, exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`deaa91314bb4c09ba9e41a2c5b3586dc`
`b434b76f15943a8870c4acb5b8cdf72f`.

Its transcript contains exactly fourteen `C_FPF_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+29} - 2\,\mathrm{FPF}(q, 2q+29).$$

This proves the half-stable bound for these first fourteen cases. □

Lemma 4.804 (The central binomial factor absorbs the twenty-nine-pause loss)**.** *For every $q \ge 74$,*

$$\begin{aligned}
&1073741824\binom{2q+29}{1} + 4294967296\binom{2q+29}{3} + 17179869184\binom{2q+29}{5} \\
&\quad + 68719476736\binom{2q+29}{7} + 274877906944\binom{2q+29}{9} + 1099511627776\binom{2q+29}{11} \\
&\quad + 4398046511104\binom{2q+29}{13} + 17592186044416\binom{2q+29}{15} \\
&\quad + 70368744177664\binom{2q+29}{17} + 281474976710656\binom{2q+29}{19} \\
&\quad + 1125899906842624\binom{2q+29}{21} + 4503599627370496\binom{2q+29}{23} \\
&\quad + 18014398509481984\binom{2q+29}{25} + 72057594037927936\binom{2q+29}{27} + 288230376151711744\binom{2q+29}{29} \le \binom{2q+29}{q}.
\end{aligned}$$

Proof. At $q = 74$, the left side is

$$450783285331425602968260277664824699027879842807808 < 1069333076921382630709408439483450169458590305354000 = \binom{177}{74}.$$

The margin in this base comparison is

$$618549791589957027741148161818625470430710462546192.$$

Let $P(q)$ denote the displayed left side. For $1 \le p \le 29$,

$$\frac{\binom{2q+31}{p}}{\binom{2q+29}{p}} \le \frac{(2q+31)(2q+30)}{(2q+2)(2q+1)} < 2 \qquad (q \ge 74),$$

and the displayed strict inequality follows from $4q^2 - 110q - 926 > 0$. Hence $P(q+1) < 2P(q)$. On the other hand,

$$\frac{\binom{2q+31}{q+1}}{\binom{2q+29}{q}} = \frac{(2q+31)(2q+30)}{(q+1)(q+30)} > 3,$$

since the numerator minus three times the denominator is $q^2 + 29q + 840$. Thus the binomial side grows by a larger factor than $P(q)$, and induction from $q = 74$ proves the inequality. □

Lemma 4.805 (Exact twenty-ninth C pause-arithmetic checks after the first fourteen rows)**.** *For* $35 \le q \le 73$, *put*

$$\begin{aligned}P(q) = 1073741824\binom{2q+29}{1} &+ 4294967296\binom{2q+29}{3} + 17179869184\binom{2q+29}{5}\\ &+ 68719476736\binom{2q+29}{7} + 274877906944\binom{2q+29}{9} + 1099511627776\binom{2q+29}{11}\\ &+ 4398046511104\binom{2q+29}{13} + 17592186044416\binom{2q+29}{15}\\ &+ 70368744177664\binom{2q+29}{17} + 281474976710656\binom{2q+29}{19}\\ &+ 1125899906842624\binom{2q+29}{21} + 4503599627370496\binom{2q+29}{23}\\ &+ 18014398509481984\binom{2q+29}{25} + 72057594037927936\binom{2q+29}{27} + 288230376151711744\binom{2q+29}{29}.\end{aligned}$$

Then

$$2^{2q+1}P(q)s_{q+29} \le s_{2q+29}.$$

Proof. The exact C++/GMP replay

`certificates/post_m29/bc_twentyninth_c_frontier_gmp_certificate.log`

computes the displayed integer inequality for $35 \le q \le 73$. It exits with status 0, reports `SUMMARY failures=0`, and has SHA-256

`deaa91314bb4c09ba9e41a2c5b3586dc`
`b434b76f15943a8870c4acb5b8cdf72f`.

Its transcript contains exactly thirty-nine `C_MARGIN` records, one for each displayed q, and each record is the exact positive integer

$$s_{2q+29} - 2^{2q+1}P(q)s_{q+29}.$$

This proves the claim. ∎

Proposition 4.806 (Residual C twenty-ninth nonboundary lower correction)**.** *For every residual row* C_b *with* $20 \le b \le 217$, *put* $q = b+1$ *and write* $s_j = m_j^{\mathrm{st}}$ *and* $m_j^{C_b} = s_j + \delta_j^{C_b}$. *Then*

$$-\frac{1}{2}s_{2q+29} \le \delta_{2q+29}^{C_b}.$$

Proof. As in the proof of proposition 4.685, $|\delta_{2q+29}^{C_b}|$ is the number of column-even Pieri two-strip paths of length $2q+29$ with terminal height at least $2q$. If $21 \le q \le 34$, lemma 4.803 gives

$$|\delta_{2q+29}^{C_b}| \le s_{2q+29}/2.$$

For $q \ge 35$, lemma 4.802, with $P(q)$ denoting the polynomial from lemma 4.805, gives

$$|\delta_{2q+29}^{C_b}| \le P(q)\, 2^{2q} s_{q+29}.$$

If $35 \le q \le 73$, lemma 4.805 gives

$$|\delta_{2q+29}^{C_b}| \le s_{2q+29}/2.$$

If $q \ge 74$, lemma 4.804 gives

$$|\delta_{2q+29}^{C_b}| \le 2^{2q}\binom{2q+29}{q}s_{q+29}.$$

The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks the inequality

$$2^{2q+1}\binom{2q+29}{q}s_{q+29} \le s_{2q+29}$$

on this residual row and moment as part of its all-row residual-window audit; it reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

Thus $|\delta^{C_b}_{2q+29}| \le s_{2q+29}/2$ also for $q \ge 74$, which implies the claimed lower bound. □

Development archive: superseded residual-tail routes.

Remark 4.807 (Status of the following development archive)**.** The results from this point through the return marker below record conditional compression suppliers, obstructions, and superseded decoder constructions. They are retained to document why those routes were abandoned. No result in this block is an incoming node of propositions 4.840 and 4.841 and theorem 4.842. In particular, a proposition in this block whose statement begins with "suppose" remains only a conditional implication and is not an accepted proof premise. The active proof resumes at the explicitly marked half-stable bridge.

Lemma 4.808 (Rank-free Pieri compression fails in low rank)**.** *Let s_n be the stable Littlewood moment sequence of lemma 4.65. In content (2^7), the two bad-shape sums*

$$A^w = \sum_{\substack{\lambda'\colon \mathrm{even}\\ \lambda_1\ge 6}} a_7(\lambda), \qquad A^h = \sum_{\substack{\lambda'\colon \mathrm{even}\\ \ell(\lambda)\ge 6}} a_7(\lambda)$$

both equal 211, *whereas*

$$\binom{7}{3}s_4 = 210.$$

Equivalently, by lemma 4.94,

$$|\delta^{B_2}_7| = |\delta^{C_2}_7| = 211.$$

Consequently the rank-plus-one Pieri-compression estimate

$$\sum_{\substack{\lambda'\colon \mathrm{even}\\ \lambda_1\ge 2q}} a_j(\lambda) \le \binom{j}{q}s_{j-q}, \qquad \sum_{\substack{\lambda'\colon \mathrm{even}\\ \ell(\lambda)\ge 2q}} a_j(\lambda) \le \binom{j}{q}s_{j-q},$$

cannot hold as a rank-free theorem for all q, j.

Proof. By lemma 4.97, $a_7(\lambda)$ is the number of seven-step horizontal-two-strip Pieri paths from $\varnothing$ to λ. Applying this finite recursion gives the following nonzero column-even terminal contributions with $\lambda_1 \ge 6$:

λ	$a_7(\lambda)$
$(7,7)$	36
$(6,6,1,1)$	175.

Thus $A^w = 36 + 175 = 211$. The same recursion gives the following nonzero column-even terminal contributions with $\ell(\lambda) \ge 6$:

λ	$a_7(\lambda)$
$(5,5,1,1,1,1)$	49
$(4,4,2,2,1,1)$	126
$(3,3,2,2,2,2)$	15
$(3,3,3,3,1,1)$	21.

Hence $A^h = 49 + 126 + 15 + 21 = 211$.

Finally, lemma 4.67 gives

$$s_0 = 1, \quad s_1 = 0, \quad s_2 = 1, \quad s_3 = 1, \quad s_4 = 6,$$

so $\binom{7}{3}s_4 = 35 \cdot 6 = 210$. Taking $q = 3$ and $j = 7$ contradicts both displayed rank-free estimates. $\square$

Proposition 4.809 (Rank-plus-one Pieri compression would imply the residual B/C tail)**.** *For the rows*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Suppose that, for every such row and every j in the residual window of remark 4.839, the correction satisfies

$$|\delta_j^G| \le \binom{j}{q} s_{j-q}.$$

Then remark 4.839 holds.

Proof. The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_compression_gmp_certificate.log`

checks, over all 402 rows of the original B/C range and all corresponding moments, the integer inequalities

$$2\binom{j}{q} s_{j-q} \le s_j.$$

It checks 3,904,626 inequalities, reports `failures=0`, and has SHA-256

$$\texttt{af7d338f58854db62c6f17ca23975e17}$$
$$\texttt{5aee5e2206ef7c2386ead79875007558}.$$

The replay reports its largest ratio diagnostic at B_{14}, moment 30: $\log_{10}(2\binom{j}{q} s_{j-q}/s_j) \approx -11.6664415829297177$. The exact checked integer inequality, rather than this rounded diagnostic, shows that the displayed compression hypothesis gives $|\delta_j^G| \le s_j/2$ throughout the residual windows. Since proposition 4.62 gives $\delta_j^G \le 0$, this is the lower bound $\delta_j^G \ge -s_j/2$ required in remark 4.839. $\square$

Proposition 4.810 (q-bit-loss Pieri compression would imply the residual B/C tail)**.** *For the rows*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Suppose that, for every such row and every j in the residual window of remark 4.839, the correction satisfies

$$|\delta_j^G| \le 2^q \binom{j}{q} s_{j-q}.$$

Then remark 4.839 holds.

Proof. The exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_compression_pow2_gmp_certificate.log`

checks, over all 402 rows of the original B/C range and all corresponding moments, the integer inequalities

$$2^{q+1}\binom{j}{q} s_{j-q} \le s_j.$$

It checks 3,904,626 inequalities, reports `failures=0`, and has SHA-256

$$\texttt{a98540c3b38d65a7c414898302214411}$$
$$\texttt{2663801ab59fe2cb8deec1b34f6c5f7e}.$$

The replay again reports its largest ratio diagnostic at B_{14}, moment 30: $\log_{10}(2^{q+1}\binom{j}{q}\, s_{j-q}/s_j) \approx -7.15099164796999887$. The exact checked integer inequality, rather than this rounded diagnostic, shows that the displayed controlled-loss compression hypothesis gives $|\delta_j^G| \le s_j/2$ throughout the residual windows. Combining this with proposition 4.62 gives the lower bound $\delta_j^G \ge -s_j/2$, which is exactly remark 4.839. $\square$

Proposition 4.811 (Residual B/C bad-shape reduction)**.** *For the rows*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b+1$ for $G = B_b$ or $G = C_b$. Suppose that, for every such row and every j in the residual window of remark 4.839, the corresponding bad-shape estimate

$$\sum_{\substack{\lambda' :\, \text{even} \\ \lambda_1 \ge 2q}} a_j(\lambda) \le 2^{2q} \binom{j}{q} s_{j-q}$$

holds on the B_b side, and

$$\sum_{\substack{\lambda' :\, \text{even} \\ \ell(\lambda) \ge 2q}} a_j(\lambda) \le 2^{2q} \binom{j}{q} s_{j-q}$$

holds on the C_b side. Then remark 4.839 holds.

Proof. By lemma 4.94, the displayed bad-shape estimates give

$$|\delta_j^G| \le 2^{2q} \binom{j}{q} s_{j-q}$$

for the relevant residual row and moment. The controlled-loss exact GMP/OpenMP replay

`certificates/post_m29/bc_pieri_powerloss_2q_gmp_certificate.log`

checks, over all residual rows and windows, the integer inequalities

$$2^{2q+1} \binom{j}{q} s_{j-q} \le s_j.$$

It reports `failures=0` and has SHA-256

`124068c5daa1123088a3cd79c3f6b839`
`fee45a1aaef637ed82161c465d86bf45`.

The replay reports its largest ratio diagnostic at B_{14}, moment 30, as the approximation $\log_{10}(2^{2q+1}\binom{j}{q}\, s_{j-q}/s_j) \approx -2.63554171301028006$. The conclusion uses the exact checked integer inequality, not this rounded display, and therefore gives $|\delta_j^G| \le s_j/2$. Together with the sign $\delta_j^G \le 0$ from proposition 4.62, this is exactly $-s_j/2 \le \delta_j^G$ on the residual window. $\square$

Corollary 4.812 (Pieri crossing estimates imply the residual B/C tail)**.** *For the rows*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b+1$ for $G = B_b$ or $G = C_b$. Suppose that, for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following direct Pieri-crossing estimates hold:

$$\#\{\lambda^\bullet : \lambda^\bullet \textit{ is a column-even Pieri two-strip path of length } j,\ C_q^w(\lambda^\bullet) = S,\ \lambda_1^{(j)} \ge 2q\} \le 2^{2q} s_{j-q}$$

for the B_b rows, and

$$\#\{\lambda^\bullet : \lambda^\bullet \textit{ is a column-even Pieri two-strip path of length } j,\ C_q^h(\lambda^\bullet) = S\} \le 2^{2q} s_{j-q}$$

for the C_b rows. Then remark 4.839 holds.

Proof. Fix a residual B_b row and an admissible j. By corollary 4.114, the width fixed-witness fiber with witness S is equinumerous with the first displayed Pieri-crossing set. Summing the assumed bound over all q-element subsets $S \subset \{1, \ldots, j\}$ gives

$$\sum_{\substack{\lambda' : \text{even} \\ \lambda_1 \geq 2q}} a_j(\lambda) \leq 2^{2q} \binom{j}{q} s_{j-q}.$$

Thus the B_b bad-shape estimate required in proposition 4.811 holds.

The C_b case is the same, using the height version of corollary 4.114. For each witness set S, the height fixed-witness fiber is equinumerous with the second displayed Pieri-crossing set, so summing over S gives

$$\sum_{\substack{\lambda' : \text{even} \\ \ell(\lambda) \geq 2q}} a_j(\lambda) \leq 2^{2q} \binom{j}{q} s_{j-q}.$$

Hence the C_b bad-shape estimate required in proposition 4.811 also holds. Since the residual row and moment were arbitrary, that proposition gives remark 4.839. □

Proposition 4.813 (Replacement-zone supplier implies the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \ldots, j\}$, the following replacement-zone repair data exist for the width fixed-witness fiber when $G = B_b$, and for the height fixed-witness fiber when $G = C_b$:

1. *deterministic replacement zones Z_α satisfying the hypotheses of proposition 4.252;*
2. *a repaired path $P^*(T) \in \mathcal{P}^{\text{even}}_{j-q}$ for each tableau T in the fiber; and*
3. *branch words*

$$\beta_\alpha(T) \in \{0, 1\}^{2|Z_\alpha \cap I(S)|}, \qquad I(S) = \{j - s + 1 : s \in S\},$$

for which the local inverses required in proposition 4.252 recover the original replacement-zone subpaths.

Then remark 4.839 holds.

Proof. Fix a B_b residual row, an admissible j, and one q-element witness set S. Applying proposition 4.541 to the width fiber gives

$$\#\{T : \lambda(T)' \text{ even},\ \lambda_1(T) \geq 2q,\ W_q(T) = S\} \leq 2^{2q} s_{j-q},$$

and summing over S as in proposition 4.563 gives the B_b bad-shape estimate.

The same argument for a C_b residual row applies proposition 4.541 to the height fiber:

$$\#\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\} \leq 2^{2q} s_{j-q}.$$

Summing over S gives the C_b bad-shape estimate. Since the row and moment were arbitrary in the residual lists and residual windows, proposition 4.811 applies and yields remark 4.839. □

Proposition 4.814 (Canonical halo repairs imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b+1$ for $G = B_b$ or $G = C_b$. Suppose that for every such row, every j in the residual window of remark 4.839, every q-element subset $S \subset \{1, \ldots, j\}$, the canonical halo zones of lemma 4.254 satisfy the following repair requirements.

1. *For the width fixed-witness fiber when $G = B_b$, they admit the parser, target path, replacement windows, branch words, and local inverses listed in proposition 4.540.*

(2) *For the height fixed-witness fiber when $G = C_b$, either the same local repair data exist, or the canonical halo zones can be ordered as $Z_1, \ldots, Z_L$ with $Z_L = \{a_L, \ldots, j\}$ and satisfy the right-boundary auxiliary hypotheses of proposition 4.542.*

Then remark 4.839 holds.

Proof. Fix one residual row, one admissible j, and one witness set S. In the B_b case, apply proposition 4.540 to the width fixed-witness fiber. It gives

$$\#\{T : \lambda(T)' \text{ even},\ \lambda_1(T) \geq 2q,\ W_q(T) = S\} \leq 2^{2q} s_{j-q}.$$

In the C_b case, apply proposition 4.540 if the local repair data of item (2) are available, and otherwise apply proposition 4.542 to the ordered canonical halo zones supplied by item (2). In both alternatives one obtains

$$\#\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\} \leq 2^{2q} s_{j-q}.$$

Since these estimates hold for every residual row, every admissible j, and every S, proposition 4.563 gives the corresponding bad-shape estimates. Then proposition 4.811 yields remark 4.839. ☐

Proposition 4.815 (Split target-window canonical halo repairs imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b+1$ for $G = B_b$ or $G = C_b$. Suppose that for every such row, every j in the residual window of remark 4.839, every q-element subset $S \subset \{1, \ldots, j\}$, the canonical halo zones of lemma 4.254 satisfy the following repair requirements.

(1) *For the width fixed-witness fiber when $G = B_b$, they admit the parser, target path, replacement windows, branch words, and local inverses listed in proposition 4.540.*

(2) *For the height fixed-witness fiber when $G = C_b$, either the same local repair data exist, or the canonical halo zones can be ordered as $Z_1, \ldots, Z_L$ with $Z_L = \{a_L, \ldots, j\}$ so that the hypotheses of proposition 4.253 hold for the parser, target path, endpoint pairs, zones strictly left of Z_L, and their local inverses. For the final zone, write*

$$Z_L = \{j - 2m + 1, \ldots, j\}.$$

Assume it is in the non-full tight-height setting of corollary 4.320, and that the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, and the labelled continuation boxes. If $m \geq 29$, let

$$\Omega_{\text{tw}} = \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

with the lexicographic order of definition 4.354; when $\Omega_{\text{tw}} \neq \varnothing$, let $(\gamma; e_1, \ldots, e_{m-1})$ be the associated fixed-even datum of definition 4.379. In the $m \geq 29$ cases not already covered by the condition

$$\Omega_{\text{tw}} = \varnothing \quad \textit{or} \quad \big(\Omega_{\text{tw}} \neq \varnothing \quad \textit{and} \quad \big(m = 29 \quad \textit{or} \quad m = 30 \quad \textit{or} \quad m = 31 \quad \textit{or} \quad (\gamma, e_{m-1}) \notin \mathcal{P}_m^{\text{env}} \quad \textit{or} \quad (\gamma, e_{m-1}, e_{m-2}) \notin \mathcal{P}_m^{\text{env},2} \quad \textit{or} \quad \exists d,\ 3 \leq d \leq m-1,\ (\gamma, e_{m-1}, e_{m-2}, \ldots, e_{m-d}) \notin \mathcal{P}_m^{\text{env},d}\big)\big),$$

assume in addition that the same data determine either the commuting normal-form signature κ of the true fixed-even standard tableau, or a canonically ordered finite set

$$\Sigma_{\text{tw}}$$

of commuting normal-form signature words such that the true fixed-even standard tableau has signature in Σ_{tw} and

$$|\Sigma_{\text{tw}}| \leq 2^m.$$

Then remark 4.839 holds.

Proof. We check the two alternatives in proposition 4.814. The B_b width hypothesis is exactly item (1) of that proposition.

For a C_b height fiber, the first alternative in item (2) is again exactly the local-repair alternative of proposition 4.814. In the right-boundary alternative, proposition 4.505 applies to the displayed final-zone target-window data. It supplies the fixed inverse for the final right-boundary zone with

$$A(P) = \mathcal{D}_{\text{out}}(P).$$

Together with the assumed parser, endpoint pairs, and local inverses for $Z_1, \ldots, Z_{L-1}$, this gives the right-boundary auxiliary hypotheses required in item (2) of proposition 4.814. Applying that proposition yields remark 4.839. ☐

Proposition 4.816 (Terminal-source target-window canonical halo repairs imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b+1$ for $G = B_b$ or $G = C_b$. Suppose that for every such row, every j in the residual window of remark 4.839, every q-element subset $S \subset \{1, \ldots, j\}$, the canonical halo zones of lemma 4.254 satisfy the following repair requirements.

1. *For the width fixed-witness fiber when $G = B_b$, they admit the parser, target path, replacement windows, branch words, and local inverses listed in proposition 4.540.*
2. *For the height fixed-witness fiber when $G = C_b$, either the same local repair data exist, or the canonical halo zones can be ordered as $Z_1, \ldots, Z_L$ with $Z_L = \{a_L, \ldots, j\}$ so that the hypotheses of proposition 4.253 hold for the parser, target path, endpoint pairs, zones strictly left of Z_L, and their local inverses. For the final zone, write*

$$Z_L = \{j - 2m + 1, \ldots, j\}.$$

Assume it is in the non-full tight-height setting of corollary 4.320. If $m \le 28$, assume the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, and the labelled continuation boxes. If $m \ge 29$ and $\lambda^{(2m)}$ is column-even, assume the final-zone replacement window and branch word are chosen by the decoder of proposition 4.494. If $m \ge 29$ and $\lambda^{(2m)}$ is not column-even, assume that the target-window data

$$S, \quad Z_L, \quad V_L^*, \quad \mathcal{D}_{\text{out}}(P), \quad \rho_{2a_L-2}, \rho_{2j}$$

determine $\lambda^{(2m)}$, the retained even-label strips $E_1, \ldots, E_m$, and the labelled continuation boxes. Let

$$\Omega_{\text{tw}} = \Omega(\lambda^{(2m)}; E_1, \ldots, E_m)$$

with the lexicographic order of definition 4.354; when $\Omega_{\text{tw}} \ne \varnothing$, let $(\gamma; e_1, \ldots, e_{m-1})$ be the associated fixed-even datum of definition 4.379. In the $m \ge 29$ cases not already covered by the condition

$$\Omega_{\text{tw}} = \varnothing \quad \textit{or} \quad \big(\Omega_{\text{tw}} \ne \varnothing \quad \textit{and} \quad \big(m = 29 \quad \textit{or} \quad m = 30 \quad \textit{or} \quad m = 31 \quad \textit{or} \quad (\gamma, e_{m-1}) \notin \mathcal{P}_m^{\text{env}} \quad \textit{or} \quad (\gamma, e_{m-1}, e_{m-2}) \notin \mathcal{P}_m^{\text{env},2} \quad \textit{or} \quad \exists d,\ 3 \le d \le m-1,\ (\gamma, e_{m-1}, e_{m-2}, \ldots, e_{m-d}) \notin \mathcal{P}_m^{\text{env},d}\big)\big),$$

form from the target-window data the degree-constrained ambient set and residual profile

$$r_1 = \#\{1 \le a \le 2m : d_a = 1\}, \qquad r_2 = \#\{1 \le a \le 2m : d_a = 2\}$$

of definition 4.440. Also put

$$r_0 = \#\{1 \le a \le 2m : d_a = 0\}, \qquad D = \sum_{a=1}^{2m} d_a, \qquad C = \sum_{a=1}^{2m} \sum_{t=2m+1}^{j} u_{\{a,t\}},$$

and let $A(m)$ and $B(m)$ be the frontier functions of proposition 4.452. If

$$\mathsf{N}(r_1, r_2) > 2^{2m}, \qquad r_0 \le 2m - A(m), \qquad C \le 4m - B(m),$$

assume in addition that one of the following six source hypotheses holds. First, the same target-window data determine a set

$$U_{\mathrm{tw}} \subseteq \{1 \le a \le 2m : d_a > 0\}$$

and determine the true Step 3 entries on every low-low cell incident to U_{tw}, with

$$D(U_{\mathrm{tw}}) = \sum_{a \in U_{\mathrm{tw}}} d_a,$$

and with one of the following positive-residual threshold tests:

$$|U_{\mathrm{tw}}| > r_1 + r_2 - A(m) \qquad \text{or} \qquad D(U_{\mathrm{tw}}) > D - B(m).$$

Second, the same target-window data determine a partial low-low readout

$$\mathcal{E}_{\mathrm{tw}}, \qquad p_{\{a,b\}} \quad (\{a,b\} \in \mathcal{E}_{\mathrm{tw}})$$

as in proposition 4.447, agreeing with the true entries on $\mathcal{E}_{\mathrm{tw}}$, with residual profile satisfying

$$\mathsf{N}(r_1^{\mathcal{E}}, r_2^{\mathcal{E}}) \le 2^{2m},$$

or, with

$$P_{\mathcal{E}} = \sum_{\{a,b\} \in \mathcal{E}_{\mathrm{tw}}} p_{\{a,b\}},$$

satisfying

$$2P_{\mathcal{E}} > D - B(m).$$

Third, the same target-window data determine a canonically ordered set

$$\Omega_{\mathrm{src}} \subseteq \Omega(\lambda^{(2m)}; E_1, \dots, E_m)$$

containing the true deleted odd-packet assignment and satisfying

$$|\Omega_{\mathrm{src}}| \le 2^{2m}.$$

Fourth, the same target-window data determine a canonically ordered finite set

$$\Sigma_{\mathrm{tw}}$$

of commuting normal-form signature words such that the true fixed-even standard tableau has signature in Σ_{tw} and

$$|\Sigma_{\mathrm{tw}}| \le 2^m.$$

Fifth, the same target-window data determine a canonically ordered finite set

$$\mathcal{L}_{\mathrm{tw}}$$

of possible Step 3 entry assignments on L_{2m}, the true assignment belongs to this set, and

$$|\mathcal{L}_{\mathrm{tw}}| \le 2^{2m}.$$

Sixth, the same target-window data determine one of the following signature-route readouts: the commuting normal-form signature κ of the true fixed-even standard tableau; or the even standard-time endpoint chain

$$\sigma^{(0)}, \sigma^{(2)}, \sigma^{(4)}, \dots, \sigma^{(4m)}$$

of the doubled standardization; or, for each semistandard label $1 \le a \le 2m$, the two boxes carrying label a.

Then remark 4.839 holds.

Proof. As in the proof of proposition 4.815, it is enough to verify the right-boundary alternative in item (2) of proposition 4.814; all other cases are exactly the local-repair hypotheses of that proposition.

In the right-boundary alternative with $m \leq 28$, proposition 4.505 applies to the displayed base target-window data. No commuting-signature hypothesis is needed in this range, by the small-compatibility and full-branch broad-Ω cases built into that proposition.

In the right-boundary alternative with $m \geq 29$ and column-even prefix endpoint, proposition 4.494 applies to the displayed final-zone replacement data. In the remaining right-boundary alternative, where $m \geq 29$ and the prefix endpoint is not column-even, the displayed target-window data determine the unrefined compatibility set. If this set is empty, there is no current fiber. If $m = 29$, then proposition 4.487 supplies the fixed right-boundary inverse. If $m = 30$, then proposition 4.488 supplies the fixed right-boundary inverse. If $m = 31$, then proposition 4.489 supplies the fixed right-boundary inverse. If $m \geq 32$ and the associated fixed-even top pair $(\gamma, e_{m-1}) \notin \mathcal{P}_m^{\mathrm{env}}$, then proposition 4.484 supplies the fixed right-boundary inverse. Hence it remains only to handle the top-fixed envelope-frontier cases. If $m \geq 32$ and $(\gamma, e_{m-1}, e_{m-2}) \notin \mathcal{P}_m^{\mathrm{env},2}$, then proposition 4.485 supplies the fixed right-boundary inverse. Hence it remains only to handle the top-two fixed envelope-frontier cases. If $m \geq 32$ and there is a d with $3 \leq d \leq m-1$ such that

$$(\gamma, e_{m-1}, e_{m-2}, \dots, e_{m-d}) \notin \mathcal{P}_m^{\mathrm{env},d},$$

then proposition 4.486 supplies the fixed right-boundary inverse. Hence it remains only to handle the cases that stay in every fixed-depth envelope frontier:

$$m \geq 32, \qquad (\gamma, e_{m-1}, e_{m-2}) \in \mathcal{P}_m^{\mathrm{env},2}, \qquad (\gamma, e_{m-1}, e_{m-2}, \dots, e_{m-d}) \in \mathcal{P}_m^{\mathrm{env},d} \quad (3 \leq d \leq m-1).$$

Taking $d = m-1$ and applying corollary 4.397, this remaining branch has

$$|\Omega_{\mathrm{tw}}| > 2^{2m}.$$

By corollary 4.431, it also realizes more than 2^m commuting normal-form signatures. Thus the unrefined broad odd-packet class itself is genuinely over budget; the following source alternatives must refine it, find a bounded signature list, or replace it by an equivalent low-low candidate source. In those cases form the degree-constrained ambient set and residual degree-one and degree-two counts r_1, r_2, together with r_0 and C, from definition 4.440. If

$$\mathsf{N}(r_1, r_2) \leq 2^{2m},$$

proposition 4.446 supplies the fixed right-boundary inverse. If instead $\mathsf{N}(r_1, r_2) > 2^{2m}$ but either

$$r_0 > 2m - A(m) \qquad \text{or} \qquad C > 4m - B(m),$$

then corollary 4.453 supplies the same fixed right-boundary inverse. If the exact profile and continuation-frontier closures both fail, then we are in the displayed live branch. By corollaries 4.456 and 4.457, this two-threshold positive-residual form is equivalent to the earlier four-test incidence-cover form. Under the incidence-cover source hypothesis, corollary 4.450 supplies the fixed right-boundary inverse in the positive-label-count threshold subcase, while corollary 4.451 supplies it in the positive residual-degree threshold subcase. Under the partial residual-profile source hypothesis, proposition 4.447 supplies the fixed right-boundary inverse in the exact residual-profile subcase, while corollary 4.449 supplies it in the read-edge-mass subcase. Under the full odd-packet-class source hypothesis, proposition 4.480 applies to Ω_{src}; equivalently, corollary 4.464 converts the same source into a full-budget low-low candidate list. Under the bounded signature-list source hypothesis, proposition 4.481 applies to the displayed signature list, and corollary 4.465 gives the same candidate-list form. Under the full candidate-list source hypothesis, proposition 4.478 applies to the displayed candidate-list hypothesis. Under the signature source hypothesis, there are three subcases. If the source is κ, proposition 4.468 applies to the already recovered base target-window datum. If the source is the even standard-time endpoint chain, then proposition 4.469 applies, using the labelled continuation boxes already recovered in the base target-window datum. If the source is the low-label box readout, then proposition 4.470 applies, again using the recovered labelled continuation boxes. In all three subcases we obtain the fixed right-boundary inverse with $A(P) = \mathcal{D}_{\mathrm{out}}(P)$. The earlier carrier and Hall formulations are not additional alternatives in the present statement. The direct low-low carrier route is reduced by corollary 4.502; in particular, corollary 4.185 turns any parsed carrier containing L_{2m} into a terminal-boundary source after the dual-RSK' backward sweep. The internal-Q-cut and direct Hall-data routes are reduced by corollaries 4.499 and 4.500, and the continuation-segment Hall route feeds the same hypotheses through proposition 4.497 and corollary 4.219. The incidence-cover carrier relaxation is the partial-carrier version of the same reduction: proposition 4.493 and corollary 4.216 feed the positive-residual two-threshold incidence source into the full-budget low-low candidate-list source, after the general four-test form has been projected by corollaries 4.456 and 4.457, and corollary 4.501 records the direct Hall-data shared form. A full-budget terminal-boundary candidate list is not a separate terminal artifact: by corollary 4.475 it induces a full-budget low-low candidate list. The graph-side counterparts are corollaries 4.202 and 4.203. Full-budget odd-packet classes feed both sides through proposition 4.480 and corollary 4.204, and they feed the low-low candidate-list source through corollary 4.464. Bounded signature lists feed both sides through proposition 4.481 and corollary 4.205 and induce the same full-budget low-low candidate-list source by corollary 4.465. Finally, corollary 4.466 turns a recovered signature fiber into a bounded low-low candidate class, while propositions 4.469 and 4.470 record concrete parser-facing ways to recover the same signature-route inverse. The matching graph-side statement is corollary 4.471. Thus the live source obligation is an incidence-cover readout below the certified positive-label-count or residual-degree frontier, a partial low-low readout below the residual profile frontier, a full-budget odd-packet class, a bounded signature list, bounded low-low candidate recovery, or one of these signature-route readouts, in the exact degree-profile dense branch where $\mathsf{N}(r_1, r_2) > 2^{2m}$, inside every fixed-depth recursive-envelope frontier with $m \geq 32$; by corollaries 4.449 and 4.451, the first two alternatives include the covered-degree and read-edge-mass tests stated there, with covered-label tests normalized by corollaries 4.456 and 4.457. By proposition 4.452, this branch has at least $A(m)$ positive residual labels and total residual degree at least $B(m)$. By corollary 4.453, the same branch has at most $2m - A(m)$ fully continued low labels and total low-continuation incidence at most $4m - B(m)$. Exact terminal-boundary recovery is a singleton terminal-boundary list, hence a special case of this low-low candidate-list input. We obtain the fixed inverse for the final right-boundary zone with

$$A(P) = \mathcal{D}_{\mathrm{out}}(P).$$

Together with the assumed parser, endpoint pairs, and local inverses for $Z_1, \ldots, Z_{L-1}$, this gives the right-boundary auxiliary hypotheses required in item (2) of proposition 4.814. Applying that proposition yields remark 4.839. □

Proposition 4.817 (Below-wall packet-splice carrier repairs imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b+1$ for $G = B_b$ or $G = C_b$. Suppose that for every such row, every j in the residual window of remark 4.839, every q-element subset $S \subset \{1, \ldots, j\}$, the canonical halo zones of lemma 4.254 satisfy the following repair requirements.

1. *For the width fixed-witness fiber when $G = B_b$, they satisfy the hypotheses of proposition 4.538.*
2. *For the height fixed-witness fiber when $G = C_b$, either they satisfy the hypotheses of proposition 4.538, or they satisfy the right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. By corollary 4.539, the below-wall packet-splice hypotheses in item (1) supply the local-repair hypotheses required in item (1) of proposition 4.816. The same corollary supplies the first height alternative in item (2) of that proposition. The second height alternative is exactly assumed in item (2) above. Therefore the hypotheses of proposition 4.816 hold, and that proposition gives remark 4.839. ☐

Proposition 4.818 (Step–3 carrier packet readouts plus the terminal target imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.560.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the canonical halo zones satisfy the hypotheses of proposition 4.560.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*
(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the Step–3 carrier packet-readout hypotheses of proposition 4.559, using the parser, window, branch word, and endpoint data supplied by this right-boundary package. The final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. For a B_b row, item (1) and proposition 4.560 give

$$\#\{T : \lambda(T)' \text{ even},\ \lambda_1(T) \geq 2q,\ W_q(T) = S\} \leq 2^{2q} s_{j-q}.$$

For a C_b row with no zero-surplus height halo zone, item (2) gives the corresponding height fixed-fiber estimate by the same carrier local-repair criterion. If the zero-surplus zone is full, item (3) and corollary 4.110 give the height estimate, as in proposition 4.819.

It remains to treat the non-full zero-surplus case. By corollary 4.324, the zero-surplus zone is terminal and every zone strictly to its left has positive surplus. For each left zone, item (4) and proposition 4.559 supply the missing local inverse in the right-boundary auxiliary package. The final-zone target-window/right-boundary alternative supplies the fixed inverse for Z_L. Therefore proposition 4.542 gives

$$\#\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\} \leq 2^{2q} s_{j-q}$$

also in this case.

Thus the required width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S. Applying proposition 4.563 and then proposition 4.811 gives remark 4.839. ☐

Proposition 4.819 (Terminal height-zone split implies the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*

(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the canonical halo zones satisfy the hypotheses of proposition 4.538.*

(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \ldots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \ldots, j\}$, the zones strictly left of Z_L are supplied by below-wall packet-splice local inverses, and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. Fix a residual row, an admissible j, and a witness set S.

For a B_b row, item (1) and proposition 4.538 give the width fixed-fiber estimate

$$\#\{T : \lambda(T)' \text{ even},\ \lambda_1(T) \geq 2q,\ W_q(T) = S\} \leq 2^{2q} s_{j-q}.$$

Now consider a C_b row. By corollary 4.324, there is at most one zero-surplus height halo zone. If no such zone exists, item (2) and proposition 4.538 give the height fixed-fiber estimate.

If the zero-surplus zone is full, then corollary 4.320 writes it as $Z_\gamma = \{j - 2m + 1, \ldots, j\}$. Since the full-zone hypothesis says $Z_\gamma = \{1, \ldots, j\}$, one has $j = 2m$. The full zone contains every deleted block, so $m = |I(S)| = q$. The same corollary then gives

$$j = 2q, \qquad S = \{1, 3, \ldots, 2q-1\}.$$

Since residual rows have $q \geq 15$, corollary 4.110 applies and gives the same height fixed-fiber estimate.

It remains to treat the non-full zero-surplus case. By corollary 4.324, the zero-surplus zone is the terminal right-boundary zone and every earlier zone has positive surplus. Item (4), together with corollary 4.539, gives the right-boundary auxiliary hypotheses used in the proof of proposition 4.816; equivalently, proposition 4.542 applies to the height fixed-witness fiber. Hence

$$\#\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\} \leq 2^{2q} s_{j-q}$$

also holds in this case.

Thus the required width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S. Applying proposition 4.563 and then proposition 4.811 gives remark 4.839. ☐

Proposition 4.820 (Anchor-pair positive height zones imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \ldots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*

(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion, and every zone satisfies the anchor-pair hypotheses of proposition 4.532.*

(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \ldots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \ldots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the anchor-pair hypotheses of proposition 4.532, and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. Fix a residual row, an admissible j, and a witness set S.

The B_b width estimate is exactly as in proposition 4.819: item (1) and proposition 4.538 give

$$\#\{T : \lambda(T)' \text{ even},\ \lambda_1(T) \geq 2q,\ W_q(T) = S\} \leq 2^{2q} s_{j-q}.$$

Now let $G = C_b$. By corollary 4.324, there is at most one zero-surplus height halo zone.

If no zero-surplus height halo zone exists, then every height halo zone has positive surplus. For each zone, proposition 4.532 supplies the missing local inverse in proposition 4.540. Therefore that canonical local-repair criterion applies to the height fixed-witness fiber and gives

$$\#\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\} \leq 2^{2q} s_{j-q}.$$

If the zero-surplus zone is full, then the same argument as in proposition 4.819 applies corollary 4.110 and gives the same height fixed-fiber estimate.

It remains to treat the non-full zero-surplus case. By corollary 4.324, the zero-surplus zone is terminal and every zone strictly to its left has positive surplus. Applying proposition 4.532 to each such left zone supplies the local inverses required in proposition 4.253. The final-zone target-window/right-boundary alternative supplies the fixed inverse for Z_L with the auxiliary datum specified there. Hence the hypotheses of proposition 4.542 hold for the height fixed-witness fiber, and this again gives

$$\#\{T : \lambda(T)' \text{ even},\ \ell(\lambda(T)) \geq 2q,\ H_q(T) = S\} \leq 2^{2q} s_{j-q}.$$

Thus the required width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S. Applying proposition 4.563 and then proposition 4.811 gives remark 4.839. ☐

Proposition 4.821 (Two-label carrier positive height zones imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \ldots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*

(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into the unchanged surplus-anchor blocks and one merged window M_t for each deleted singleton u_t, as in proposition 4.532. Moreover, with $s_t = j - u_t + 1$, with E_{s_t, s_t+1} the two-label Step–3 carrier, and with e_t its unique cell entry, the data*

$$M_t, \quad S, \quad Z_\gamma, \quad t, \quad \rho_{2a_\gamma - 2}, \rho_{2b_\gamma}$$

determine the left/bottom boundary of E_{s_t,s_t+1} *and the corresponding two-bit word satisfies* $\beta_t = \kappa(e_t)$.

(3) *If a height zero-surplus zone exists and is the full alternating zone* $j = 2q$ *with* $S = \{1, 3, \ldots, 2q-1\}$, *use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written* $Z_L = \{a_L, \ldots, j\}$, *the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of* Z_L. *Each zone strictly left of* Z_L *satisfies the two-label carrier hypotheses stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The B_b width estimate is supplied by item (1), exactly as in proposition 4.820.

Now consider a C_b row. If no zero-surplus height halo zone exists, then every height halo zone has positive surplus by corollary 4.318. For each such zone, item (2) gives the parser decomposition, left/bottom two-label carrier boundary, and cell-entry branch encoding required by proposition 4.272. Hence that proposition supplies the missing local inverse for every zone in proposition 4.540, and the height fixed-fiber estimate follows from that criterion.

If the zero-surplus zone is full, item (3) and corollary 4.110 give the height fixed-fiber estimate as in proposition 4.820.

It remains to treat the non-full zero-surplus case. By corollary 4.324, the zero-surplus zone is terminal and every zone strictly to its left has positive surplus. Item (4) and proposition 4.272 supply the local inverses for those left zones, while the final-zone target-window/right-boundary alternative supplies the fixed inverse for Z_L. Thus the hypotheses of proposition 4.542 hold for the height fixed-witness fiber, giving the same height fixed-fiber estimate.

Therefore the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S. Applying proposition 4.563 and then proposition 4.811 gives remark 4.839. □

Proposition 4.822 (Coupled collar carriers imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b + 1$ *for* $G = B_b$ *or* $G = C_b$. *Fix once and for all an injection*

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j *in the residual window of remark 4.839, and every* q*-element subset* $S \subset \{1, \ldots, j\}$, *the following repair data are available.*

(1) *For the width fixed-witness fiber when* $G = B_b$, *the canonical halo zones satisfy the hypotheses of proposition 4.538.*

(2) *For the height fixed-witness fiber when* $G = C_b$, *order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone* Z_γ, *the parser decomposes* V_γ^* *into unchanged surplus-anchor blocks and merged windows* $M_1, \ldots, M_m$ *for the increasing deleted singleton list*

$$Z_\gamma \cap I = \{u_1 < \cdots < u_m\}.$$

These merged windows satisfy the coupled collar hypotheses of proposition 4.317: each M_t *determines the left-collar block* $B_{u_t-1}(P)$, *the parsed zone data determine the left/bottom boundary of* E_{s_t,s_t+1} *after the parser-local right collars are recovered, and the two-bit word satisfies* $\beta_t = \kappa(e_t)$ *for the unique carrier entry* e_t.

(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the coupled collar hypotheses stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The B_b width estimate is supplied by item (1), exactly as in proposition 4.821.

Now consider a C_b row. If no zero-surplus height halo zone exists, then every height halo zone has positive surplus by corollary 4.318. For each such zone, item (2) supplies the hypotheses of proposition 4.317. That proposition gives the missing local inverse for every zone in proposition 4.540, and the height fixed-fiber estimate follows.

If the zero-surplus zone is full, item (3) and corollary 4.110 give the height fixed-fiber estimate as in proposition 4.821.

It remains to treat the non-full zero-surplus case. By corollary 4.324, the zero-surplus zone is terminal and every zone strictly to its left has positive surplus. Item (4) and proposition 4.317 supply the local inverses for those left zones, while the final-zone target-window/right-boundary alternative supplies the fixed inverse for Z_L. Therefore the hypotheses of proposition 4.542 hold for the height fixed-witness fiber, giving the same height fixed-fiber estimate.

The width and height fixed-fiber estimates now hold for every residual row, admissible j, and witness set S. Applying proposition 4.563 and then proposition 4.811 gives remark 4.839. □

Proposition 4.823 (Padded carrier valleys imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b+1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0,1,2\} \hookrightarrow \{0,1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*

(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into unchanged surplus-anchor blocks and merged windows M_t for the increasing deleted singleton list. Each merged window is the padded carrier valley of proposition 4.276: for $s_t = j - u_t + 1$, if*

$$\begin{matrix} \nu_t & \lambda_t \\ \rho_t & \mu_t \end{matrix}$$

are the corner labels of E_{s_t, s_t+1} and e_t is its cell entry, then

$$M_t = \mathrm{Pad}_{e_t}(\nu_t, \rho_t, \mu_t),$$

and the two-bit word satisfies $\beta_t = \kappa(e_t)$.

(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the padded-carrier hypotheses stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is the same reduction as proposition 4.822, with proposition 4.276 replacing the coupled-collar local inverse. Item (1) supplies the B_b width estimate.

For a C_b row with no zero-surplus height halo zone, every height halo zone has positive surplus by corollary 4.318. Item (2) and proposition 4.276 supply the missing local inverse for every zone in proposition 4.540, giving the height fixed-fiber estimate.

If the zero-surplus zone is full, item (3) and corollary 4.110 give the height fixed-fiber estimate. In the remaining non-full zero-surplus case, corollary 4.324 makes that zero-surplus zone terminal and every zone strictly to its left positive. Item (4) supplies the padded-carrier inverses for those left zones and the final-zone target-window/right-boundary inverse for Z_L, so proposition 4.542 gives the height fixed-fiber estimate.

Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S. Applying proposition 4.563 and then proposition 4.811 gives remark 4.839. ☐

Proposition 4.824 (Tagged collective padded splices imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* as a tagged collective padded splice satisfying proposition 4.279, with the same fixed injection κ.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q - 1\}$, use the fixed full-zone decoder of corollary 4.110.*
(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the tagged collective padded-splice hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The reduction is the same as proposition 4.823, with proposition 4.279 as the positive-height local inverse. Item (1) supplies the B_b width estimate.

For a C_b row with no zero-surplus height halo zone, every height halo zone has positive surplus by corollary 4.318. Item (2) and proposition 4.279 supply the missing local inverse for every zone in proposition 4.540, giving the height fixed-fiber estimate.

If the zero-surplus zone is full, item (3) and corollary 4.110 give the height fixed-fiber estimate. In the remaining non-full zero-surplus case, corollary 4.324 makes that zero-surplus zone terminal and every zone strictly to its left positive. Item (4) supplies the tagged collective padded-splice inverses for those left zones and the final-zone target-window/right-boundary inverse for Z_L, so proposition 4.542 gives the height fixed-fiber estimate.

Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S. Applying proposition 4.563 and then proposition 4.811 gives remark 4.839. ☐

Proposition 4.825 (Grouped collective carrier splices imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \ldots, j\}$, the following repair data are available.

1. *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
2. *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* as a grouped collective carrier splice satisfying proposition 4.281, with the same fixed injection κ.*
3. *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \ldots, 2q - 1\}$, use the fixed full-zone decoder of corollary 4.110.*
4. *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \ldots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the grouped collective carrier-splice hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The reduction is the same as proposition 4.824, with proposition 4.281 as the positive-height local inverse. The B_b width estimate is supplied by item (1).

For a C_b row with no zero-surplus height halo zone, every height halo zone has positive surplus by corollary 4.318. Item (2) and proposition 4.281 supply the missing local inverse for every zone in proposition 4.540, giving the height fixed-fiber estimate.

If the zero-surplus zone is full, item (3) and corollary 4.110 give the height fixed-fiber estimate. In the remaining non-full zero-surplus case, corollary 4.324 makes that zero-surplus zone terminal and every zone strictly to its left positive. Item (4) supplies the grouped collective carrier-splice inverses for those left zones and the final-zone target-window/right-boundary inverse for Z_L, so proposition 4.542 gives the height fixed-fiber estimate.

Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S. Applying proposition 4.563 and then proposition 4.811 gives remark 4.839. ☐

Proposition 4.826 (Step–3 boundary grouped splices imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V^*_γ into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.282, with the same fixed injection κ.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q - 1\}$, use the fixed full-zone decoder of corollary 4.110.*
(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the Step–3 boundary grouped-splice hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The reduction is the same as proposition 4.825, with proposition 4.282 providing the positive-height local inverse. Item (1) supplies the B_b width estimate.

For a C_b row with no zero-surplus height halo zone, every height halo zone has positive surplus by corollary 4.318. Item (2) and proposition 4.282 supply the missing local inverse for every zone in proposition 4.540, giving the height fixed-fiber estimate.

If the zero-surplus zone is full, item (3) and corollary 4.110 give the height fixed-fiber estimate. In the remaining non-full zero-surplus case, corollary 4.324 makes that zero-surplus zone terminal and every zone strictly to its left positive. Item (4) supplies the Step–3 boundary grouped-splice inverses for those left zones and the final-zone target-window/right-boundary inverse for Z_L, so proposition 4.542 gives the height fixed-fiber estimate.

Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S. Applying proposition 4.563 and then proposition 4.811 gives remark 4.839. □

Proposition 4.827 (One-cell corner grouped splices imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*

(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.283, with the same fixed injection κ.*

(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \ldots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \ldots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the one-cell corner grouped-splice hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.826, with proposition 4.283 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone one-cell corner inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. ☐

Proposition 4.828 (Grouped padded-valley readouts imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b+1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \ldots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*

(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.284, with the same fixed injection κ.*

(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \ldots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \ldots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the grouped padded-valley hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.827, with proposition 4.284 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone grouped padded-valley inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. □

Proposition 4.829 (Grouped midpoint-defect readouts imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.294, with the same fixed injection κ.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q - 1\}$, use the fixed full-zone decoder of corollary 4.110.*
(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the grouped midpoint-defect hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.828, with proposition 4.294 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone grouped midpoint-defect inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. □

Proposition 4.830 (Grouped singleton-defect row readouts imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.295, with the same fixed injection κ.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*
(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the grouped singleton-defect-row hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.829, with proposition 4.295 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone grouped singleton-defect-row inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. ☐

Proposition 4.831 (Grouped northeast-box row readouts imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.296, with the same fixed injection κ.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*
(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the grouped northeast-box-row hypothesis stated in item (2),*

and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.830, with proposition 4.296 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone grouped northeast-box-row inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. □

Proposition 4.832 (Visible-gated northeast-box row readouts imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

1. *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
2. *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.299, with the same fixed injection κ.*
3. *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q - 1\}$, use the fixed full-zone decoder of corollary 4.110.*
4. *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the visible-gated northeast-box row hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.831, with proposition 4.299 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone visible-gated northeast-box row inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. □

Proposition 4.833 (Grouped removable-row readouts imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.297, with the same fixed injection κ.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*
(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the grouped removable-row hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.831, with proposition 4.297 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone grouped removable-row inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. □

Proposition 4.834 (Visible-gated grouped removable-row readouts imply the residual B/C tail)**.** *For each row*

$$B_{14}, \dots, B_{61}, \qquad C_{20}, \dots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \dots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.301, with the same fixed injection κ.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \dots, 2q-1\}$, use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \ldots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the visible-gated grouped removable-row hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.833, with proposition 4.301 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone visible-gated removable-row inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. ☐

Proposition 4.835 (Visible-gated midpoint readouts imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \ldots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*

(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of proposition 4.302, with the same fixed injection κ.*

(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \ldots, 2q - 1\}$, use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \ldots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the visible-gated midpoint grouped-splice hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.834, with proposition 4.302 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone visible-gated midpoint inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. ☐

Proposition 4.836 (Visible-gated Step–3 boundary readouts imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \ldots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. For every such height zone Z_γ, the parser decomposes V_γ^* into grouped subwindows and parser-visible tag sets satisfying the hypotheses of corollary 4.308, with the same fixed injection κ.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \ldots, 2q - 1\}$, use the fixed full-zone decoder of corollary 4.110.*
(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \ldots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the visible-gated Step–3 boundary hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. For every positive height zone appearing in items (2) and (4), corollary 4.308 converts the stated visible-gated Step–3 boundary readout into the hypotheses of proposition 4.302. Thus the hypotheses of proposition 4.835 are satisfied with the same width data, the same full-zone decoder, and the same terminal right-boundary alternative. Applying that proposition gives remark 4.839. □

Proposition 4.837 (Visible-gated target blocks imply the residual B/C tail)**.** *For each row*

$$B_{14}, \ldots, B_{61}, \qquad C_{20}, \ldots, C_{61},$$

put $q = b + 1$ for $G = B_b$ or $G = C_b$. Fix once and for all an injection

$$\kappa : \{0, 1, 2\} \hookrightarrow \{0, 1\}^2.$$

Suppose that for every such row, every j in the residual window of remark 4.839, and every q-element subset $S \subset \{1, \ldots, j\}$, the following repair data are available.

(1) *For the width fixed-witness fiber when $G = B_b$, the canonical halo zones satisfy the hypotheses of proposition 4.538.*
(2) *For the height fixed-witness fiber when $G = C_b$, order the canonical halo zones increasingly. If no height halo zone has zero surplus, then the global parser, target path, replacement windows, branch words, and endpoint pairs satisfy the hypotheses of proposition 4.540 except for the local inverse assertion. Every such height zone Z_γ satisfies the visible-gated target-block hypotheses of proposition 4.303, with the same fixed injection κ.*
(3) *If a height zero-surplus zone exists and is the full alternating zone $j = 2q$ with $S = \{1, 3, \ldots, 2q - 1\}$, use the fixed full-zone decoder of corollary 4.110.*

(4) *If a height zero-surplus zone exists and is not full, then, with this terminal zone written $Z_L = \{a_L, \dots, j\}$, the global parser, target path, replacement windows, branch words, endpoint pairs, and unchanged complement blocks satisfy the right-boundary auxiliary hypotheses of proposition 4.253 except for the local inverse assertions on the zones strictly left of Z_L. Each zone strictly left of Z_L satisfies the visible-gated target-block hypothesis stated in item (2), and the final zone satisfies the final-zone target-window/right-boundary alternative listed in item (2) of proposition 4.816.*

Then remark 4.839 holds.

Proof. The proof is identical to proposition 4.835, with proposition 4.303 as the positive-height local inverse. Item (1) gives the B_b width estimate. In the no-zero-surplus C_b case, item (2) supplies the local inverse for every positive height zone in proposition 4.540. In the full zero-surplus case, item (3) supplies the fixed full-zone decoder. In the remaining terminal zero-surplus case, item (4) supplies the left-zone visible-gated target-block inverses and the final-zone target-window/right-boundary inverse, so proposition 4.542 applies. Thus the width and height fixed-fiber estimates hold for every residual row, admissible j, and witness set S, and propositions 4.563 and 4.811 give remark 4.839. □

Return to the active proof: the half-stable bridge.

Proposition 4.838 (Post-$m = 29$ B/C half-stable bridge reduction)**.** *For the rows*

$$B_{14}, \dots, B_{123}, \qquad C_{20}, \dots, C_{217},$$

let r_G be the rank of G and let $N^{BC}_{\mathrm{dir}}(G)$ be the direct-tail onset of proposition 4.59. Put $s_j = m^{\mathrm{st}}_j$ and $m^G_j = s_j + \delta^G_j$. Suppose that

$$-\frac{1}{2}s_j \le \delta^{B_b}_j$$

for every $14 \le b \le 41$ and every integer j with

$$2b + 31 \le j \le N^{BC}_{\mathrm{dir}}(B_b) + 2,$$

and that

$$-\frac{1}{2}s_j \le \delta^{B_b}_j$$

for every $42 \le b \le 123$ and every integer j with

$$2b + 32 \le j \le N^{BC}_{\mathrm{dir}}(B_b) + 2,$$

and that

$$-\frac{1}{2}s_j \le \delta^{C_b}_j$$

for every $20 \le b \le 217$ and every integer j with

$$2b + 32 \le j \le N^{BC}_{\mathrm{dir}}(C_b) + 2.$$

Then, for every integer m with

$$N^{(29)}_{\mathrm{hit}}(G) \le 2m + 1 < N^{BC}_{\mathrm{dir}}(G),$$

one has

$$D_G(2m + 1) \ge 0.$$

More locally, for a fixed displayed row and a fixed such m, the same conclusion follows if

$$-\frac{1}{2}s_j \le \delta^G_j \le 0 \qquad (2r_G + 2 \le j \le 2m + 5)$$

without any hypothesis on later correction variables.

Proof. First combine the hypotheses with propositions 4.567, 4.570, 4.573, 4.576, 4.579, 4.582, 4.585, 4.589, 4.593, 4.597, 4.601, 4.605, 4.609, 4.613, 4.617, 4.621, 4.625, 4.629, 4.633, 4.637, 4.680, 4.685, 4.688, 4.691, 4.694, 4.697, 4.700, 4.703, 4.706, 4.710, 4.714, 4.718, 4.722, 4.726, 4.730, 4.736, 4.741, 4.746, 4.751 and 4.756. Also use the twentieth, twenty-first, twenty-second, twenty-third, twenty-fourth, twenty-fifth, twenty-sixth, twenty-seventh, twenty-eighth, B-side high-row twenty-ninth, and C-side twenty-ninth nonboundary theorems propositions 4.641, 4.645, 4.649, 4.653, 4.657, 4.661, 4.665, 4.669, 4.674, 4.678, 4.761, 4.766, 4.771, 4.776, 4.781, 4.786, 4.791, 4.796, 4.801 and 4.806. For B_b rows, these propositions supply $j = 2b + 2$, $j = 2b + 3$, $j = 2b + 4$, $j = 2b + 5$, $j = 2b + 6$, $j = 2b+7$, $j = 2b+8$, and $j = 2b+9$, $j = 2b+10$, $j = 2b+11$, $j = 2b+12$, $j = 2b+13$, $j = 2b+14$, $j = 2b + 15$, $j = 2b + 16$, $j = 2b + 17$, $j = 2b + 18$, and $j = 2b + 19$, $j = 2b + 20$, $j = 2b + 21$, $j = 2b+22$, $j = 2b+23$, $j = 2b+24$, $j = 2b+25$, $j = 2b+26$, $j = 2b+27$, $j = 2b+28$, $j = 2b+29$, and $j = 2b + 30$. For $42 \leq b \leq 123$, proposition 4.678 supplies $j = 2b + 31$, while the hypothesis supplies every value $j \geq 2b + 32$. For $14 \leq b \leq 41$, the hypothesis supplies every value $j \geq 2b + 31$. For C_b rows, propositions 4.680, 4.685, 4.688, 4.691, 4.694, 4.697, 4.700, 4.703, 4.706, 4.710, 4.714, 4.718, 4.722, 4.726, 4.730, 4.736, 4.741, 4.746, 4.751 and 4.756, together with propositions 4.761, 4.766, 4.771, 4.776, 4.781, 4.786, 4.791, 4.796, 4.801 and 4.806, supplies $j = 2b + 2$, $j = 2b + 3$, $j = 2b + 4$, $j = 2b + 5$, $j = 2b + 6$, and $j = 2b + 7$, $j = 2b + 8$, $j = 2b + 9$, $j = 2b + 10$, $j = 2b + 11$, $j = 2b+12$, $j = 2b+13$, $j = 2b+14$, $j = 2b+15$, $j = 2b+16$, $j = 2b+17$, $j = 2b+18$, $j = 2b+19$, $j = 2b+20$, $j = 2b+21$, $j = 2b+22$, $j = 2b+23$, $j = 2b+24$, $j = 2b+25$, $j = 2b+26$, $j = 2b+27$, $j = 2b + 28$, $j = 2b + 29$, $j = 2b + 30$, and $j = 2b + 31$, while the hypothesis supplies every value $j \geq 2b + 32$. Thus every displayed row has

$$-\frac{1}{2}s_j \leq \delta_j^G$$

for every correction variable in the full interval $2r_G + 2 \leq j \leq N_{\mathrm{dir}}^{BC}(G) + 2$.

Expand the Chain difference in the stable moments and correction variables as in the proof of proposition 4.44. By proposition 4.62, the B/C corrections are nonpositive. Terms with negative linear coefficient are therefore nonnegative and may be discarded, while a positive linear coefficient $L_{m,j}$ contributes at worst $-\frac{1}{2}L_{m,j}s_j$. Likewise positive quadratic terms may be discarded and a negative quadratic coefficient $Q_{m,i,j}$ contributes at worst $\frac{1}{4}Q_{m,i,j}s_is_j$.

For each Chain index m, the verifier takes the smallest active boundary $2\,\mathrm{rank}(G) + 2$ among rows whose finite bridge still includes $D_G(2m + 1)$. Using this frontier only adds possible negative terms, so it gives a common lower bound for every active row. The exact GMP/OpenMP replay

`certificates/post_m29/bc_lambda_half_bridge_gmp_certificate.log`

computes these lower bounds for all $m = 31, \ldots, 15447$ with $\lambda = 1/2$. It exits with status 0, reports `failures=0`, and has SHA-256

`426fc6ccf924e70d27fdfd90f0e4314f`
`02e2bc87fe09a6fe84cfdce92f2dbec7`.

Its worst positive lower bound is attained at $m = 15447$, odd exponent 30897, with frontier rank 217 and $\log_{10}$ lower bound $125316.825861440928 > 0$. Hence every active bridge Chain difference is nonnegative under the displayed assumptions and the cited boundary and initial nonboundary theorems. For a fixed m, the two moment-formula diagonals in $D_G(2m+1)$ involve moment indices only through $2m + 5$. The same coefficient calculation may therefore be truncated at that index, proving the local clause. □

Remark 4.839 (Superseded half-stable route)**.** Before proposition 4.43, the only unresolved route used the following two-row condition. It is recorded solely for provenance and is not an assumption of any theorem in this paper. For $b = 14, 15$, let $N_{\mathrm{dir}}^{BC}(B_b)$ be the direct-tail onset of proposition 4.59.

With $s_j = m_j^{\mathrm{st}}$ and $m_j^{B_b} = s_j + \delta_j^{B_b}$, the former condition was

$$-\frac{1}{2}s_j \le \delta_j^{B_b} \qquad \left(2b+31 \le j \le N_{\mathrm{dir}}^{BC}(B_b)+2\right).$$

The earlier values are theorems: $j = 2b+2$ for B_b rows by proposition 4.567, $j = 2b+3$ for B_b rows by proposition 4.570, $j = 2b+4$ for B_b rows by proposition 4.573, $j = 2b+5$ for B_b rows by proposition 4.576, $j = 2b+6$ for B_b rows by proposition 4.579, $j = 2b+7$ for B_b rows by proposition 4.582, and $j = 2b+8$ for B_b rows by proposition 4.585, and $j = 2b+9$ for B_b rows by proposition 4.589, and $j = 2b+10$ for B_b rows by proposition 4.593, and $j = 2b+11$ for B_b rows by proposition 4.597, and $j = 2b+12$ for B_b rows by proposition 4.601, and $j = 2b+13$ for B_b rows by proposition 4.605, and $j = 2b+14$ for B_b rows by proposition 4.609, and $j = 2b+15$ for B_b rows by proposition 4.613, and $j = 2b+16$ for B_b rows by proposition 4.617, and $j = 2b+17$ for B_b rows by proposition 4.621, and $j = 2b+18$ for B_b rows by proposition 4.625, and $j = 2b+19$ for B_b rows by proposition 4.629, and $j = 2b+20$ for B_b rows by proposition 4.633, and $j = 2b+21$ for B_b rows by proposition 4.637, and $j = 2b+22$ for B_b rows by proposition 4.641, and $j = 2b+23$ for B_b rows by proposition 4.645, and $j = 2b+24$ for B_b rows by proposition 4.649, and $j = 2b+25$ for B_b rows by proposition 4.653, and $j = 2b+26$ for B_b rows by proposition 4.657, and $j = 2b+27$ for B_b rows by proposition 4.661, and $j = 2b+28$ for B_b rows by proposition 4.665, and $j = 2b+29$ for B_b rows by proposition 4.669, and $j = 2b+30$ for B_b rows by proposition 4.674.

Proposition 4.840 (Active residual B/C correction-prefix contract)**.** *Put $q = b+1$, write $s_j = m_j^{\mathrm{st}}$, and write $m_j^G = s_j + \delta_j^G$. The following bounds hold:*

$$-\frac{1}{2}s_{2q+r} \le \delta_{2q+r}^{B_b} \le 0 \quad (14 \le b \le 123, 0 \le r \le 28), \tag{4.840.1}$$

and

$$-\frac{1}{2}s_{2q+r} \le \delta_{2q+r}^{C_b} \le 0 \quad (20 \le b \le 217, 0 \le r \le 28). \tag{4.840.2}$$

The lower bound at each offset is supplied by the exact result in the same row of the following table; the common upper bound is proposition 4.62.

r	*type B lower-bound result*	*type C lower-bound result*
0	*proposition 4.567*	*proposition 4.680*
1	*proposition 4.570*	*proposition 4.685*
2	*proposition 4.573*	*proposition 4.688*
3	*proposition 4.576*	*proposition 4.691*
4	*proposition 4.579*	*proposition 4.694*
5	*proposition 4.582*	*proposition 4.697*
6	*proposition 4.585*	*proposition 4.700*
7	*proposition 4.589*	*proposition 4.703*
8	*proposition 4.593*	*proposition 4.706*
9	*proposition 4.597*	*proposition 4.710*
10	*proposition 4.601*	*proposition 4.714*
11	*proposition 4.605*	*proposition 4.718*
12	*proposition 4.609*	*proposition 4.722*
13	*proposition 4.613*	*proposition 4.726*
14	*proposition 4.617*	*proposition 4.730*
15	*proposition 4.621*	*proposition 4.736*
16	*proposition 4.625*	*proposition 4.741*
17	*proposition 4.629*	*proposition 4.746*
18	*proposition 4.633*	*proposition 4.751*
19	*proposition 4.637*	*proposition 4.756*
20	*proposition 4.641*	*proposition 4.761*
21	*proposition 4.645*	*proposition 4.766*
22	*proposition 4.649*	*proposition 4.771*
23	*proposition 4.653*	*proposition 4.776*
24	*proposition 4.657*	*proposition 4.781*

r	*type B lower-bound result*	*type C lower-bound result*
25	*proposition 4.661*	*proposition 4.786*
26	*proposition 4.665*	*proposition 4.791*
27	*proposition 4.669*	*proposition 4.796*
28	*proposition 4.674*	*proposition 4.801*

*The additional offset-*29 *results in the detailed supplement are not needed by the main-theorem propagation below.*

Proof. For each table row, the stated lower-bound result has exactly the family, rank interval, and moment index printed in (4.840).1 or (4.840).2. Those results and their proofs occur earlier in this detailed build. The Pieri omitted-summand formula in proposition 4.62 gives $\delta_j^G \le 0$ for both families. Combining the two sides proves the contract. □

Proposition 4.841 (Residual B/C post-$m = 29$ closure)**.** *For*

$$G = B_b \quad (14 \le b \le 123), \qquad or \qquad G = C_b \quad (20 \le b \le 217),$$

one has

$$Q_3^G(n) \ge 0$$

for every odd integer $n \ge N_{\text{hit}}^{(29)}(G)$.

Proof. Write $m_j^G = s_j + \delta_j^G$, where s_j is the stable moment. The complete active correction import is proposition 4.840; in particular, its table replaces the former prose reference to a suite of unnamed contracts.

We record why this prefix is sufficient. The analytic suppliers used below start no later than the odd exponent

$$N_{\text{tail}}(b) = 2b + 29.$$

To propagate to an odd target $t < N_{\text{tail}}(b)$, the last Chain difference used is $D_G(t-2)$. By lemma 2.3 and definition 2.10, that difference involves moments only through m_{t+2}^G. Since $t \le 2b+27$, its largest possible correction index is

$$t + 2 \le 2b + 29 = 2q + 27.$$

Thus only offsets $0 \le r \le 27$ of proposition 4.840 are consumed by the main theorem. Offset 28 is retained as a checked overlap; the offset-29 results and the longer conditional development routes printed in the source archive are not premises of this proposition.

Substitution of these bounds into the local half-stable bridge leaves an integer lower bound in stable moments. The exact GMP/OpenMP verifier checks it for $m = 31, \ldots, 15447$, reports no failures, and has replay SHA-256

`426fc6ccf924e70d27fdfd90f0e4314f`
`02e2bc87fe09a6fe84cfdce92f2dbec7`.

Together with the first-hit proposition and Chain propagation, this reaches the applicable analytic supplier. The layered-MGF proposition supplies $B_{14}, \ldots, B_{61}$ and $C_{20}, \ldots, C_{61}$; the tilted-MGF and direct interval propositions supply $B_{62}, \ldots, B_{123}$ and $C_{62}, \ldots, C_{217}$. For $B_{14}, \ldots, B_{17}$ the layered-MGF onset already precedes $N_{\text{hit}}^{(29)}(G)$. These ranges are disjoint and exhaust the displayed rows, proving every stated odd exponent. No diagnostic or superseded route in the expanded archive is used. □

Theorem 4.842 (Classical B, C, D closure)**.** *For every compact connected simple group whose Lie algebra has type B_b, C_b, or D_b, and for every $n \ge 0$,*

$$Q_3^G(n) \ge 0.$$

Proof. Even n is lemma 2.9. For odd n, the moment formula and lemma 2.6 give the strict base value $Q_3^G(1) = 2m_3^G > 0$ in each adjoint classical row. The stable edge lemma 4.3 and the row-gated bridge proposition 4.16 supply the needed nonnegative Chain differences before the post-bridge hit, so lemma 2.11 gives the pre-tail odd exponents. The high-rank rows are covered by proposition 4.14. The post-bridge finite rows listed in proposition 4.17 are covered by the trace-tail criterion. The low-rank post-$m = 29$ rows in proposition 4.18 are covered by their direct pushforward-tail certificates and proposition 4.54, with D_{19} also supplied by proposition 4.50. The rows $D_{25}, \dots, D_{52}$ are covered unconditionally by proposition 4.51. Every row $D_{53}, \dots, D_{295}$ is covered unconditionally by proposition 4.49; D_{297} is covered by proposition 4.17, and all other higher type-D rows are covered by proposition 4.14. The high-edge rows B_b, C_b for $218 \le b \le 275$ or $b = 277$ are covered by proposition 4.19. The rows $B_{124}, \dots, B_{217}$ are covered by proposition 4.21. For the next 308 rows, the first post-bridge hit is covered by proposition 4.44, the boundary and initial nonboundary theorems propositions 4.567, 4.570, 4.573, 4.576, 4.579, 4.582, 4.585, 4.589, 4.593, 4.597, 4.601, 4.605, 4.609, 4.613, 4.617, 4.621, 4.625, 4.629, 4.633, 4.637, 4.680, 4.685, 4.688, 4.691, 4.694, 4.697, 4.700, 4.703, 4.706, 4.710, 4.714, 4.718, 4.722, 4.726, 4.730, 4.736, 4.741, 4.746, 4.751 and 4.756, the twentieth, twenty-first, twenty-second, twenty-third, twenty-fourth, twenty-fifth, twenty-sixth, twenty-seventh, twenty-eighth, B-side high-row twenty-ninth, and C-side twenty-ninth nonboundary theorems propositions 4.641, 4.645, 4.649, 4.653, 4.657, 4.661, 4.665, 4.669, 4.674, 4.678, 4.761, 4.766, 4.771, 4.776, 4.781, 4.786, 4.791, 4.796, 4.801 and 4.806. For $B_{18}, \dots, B_{61}$ and $C_{20}, \dots, C_{61}$, the applicable theorems in these lists, together with proposition 4.62, give the half-stable correction bound through the prefix used by the local clause of proposition 4.838. That clause and lemma 2.11 cover every intervening odd exponent below $2b + 29$, and proposition 4.43 supplies $2b + 29$ and every later one. The four additional rows $B_{14}, B_{15}, B_{16}, B_{17}$ are supplied directly from odd exponents $57, 59, 61, 63$, respectively, by the same proposition. For $B_{62}, \dots, B_{123}$, the B-side theorems in these two lists together with proposition 4.62 give $-s_j/2 \le \delta_j^{B_b} \le 0$ through $j = 2b+31$. The prefix-local clause of proposition 4.838 and lemma 2.11 therefore reach the smaller of $2b+29$ and the logged direct-tail onset. From there, either proposition 4.59 already applies or proposition 4.29 supplies every later odd exponent. For the rows $C_{62}, \dots, C_{217}$, the C-side theorems in the two preceding lists together with proposition 4.62 give $-s_j/2 \le \delta_j^{C_b} \le 0$ through $j = 2b + 31$. The prefix-local clause of proposition 4.838 and lemma 2.11 therefore reach the smaller of $2b + 29$ and the logged direct-tail onset. From there, either proposition 4.59 already applies or proposition 4.33 supplies every later odd exponent. These alternatives exhaust every non-A classical row and every odd exponent. □

5. Exceptional Lie Groups

The exceptional proof uses the exact Chain difference of definition 2.10. For every exceptional adjoint family, the moment formula and lemma 2.6 give $Q_3^G(1) = 2m_3^G > 0$. Nonnegative Chain differences therefore propagate positivity by lemma 2.11.

Group	exact prefix	tail/bridge closure
G_2	Chain $m = 0, \dots, 97$	direct rectangular tail from odd $n \ge 21$
F_4	Chain $m = 0, \dots, 107$	direct rectangular tail from odd $n \ge 65$
E_6	Chain $m = 0, \dots, 37$	direct Chain tail from odd $n \ge 75$
E_7	Chain $m = 0, \dots, 32$	direct Chain tail from odd $n \ge 65$
E_8	Chain $m = 0, \dots, 32$	tail from odd $n \ge 133$ plus bridges to $n = 77$

Table 2. Exceptional exact prefixes and tail certificates.

Lemma 5.1 (Exceptional adjoint endpoint and Chain negative region)**.** *Let G be a compact exceptional simple group of rank r, and put $C_G = r$. Then*

$$\chi_{\mathrm{ad}}(g) \geq -C_G \qquad (g \in G).$$

For odd n, with the pushforward measure ψ_G of lemma 4.5,

$$D_G(n) = \int_{\mathbb{R}} u^n(u^2-4)\,d\psi_G(u).$$

Consequently, if $L(n)$ is any lower bound for a positive part of this integral, then

$$D_G(n) \geq L(n) - U_G(n), \qquad U_G(n) = 2\big((2C_G)^2+4\big)(2C_G)^n. \tag{5.1.1}$$

Proof. Garibaldi–Guralnick–Rains [13, Theorem 2.1] prove $\operatorname{Tr}\operatorname{Ad}(g) \geq -\operatorname{rank} G$ for every element of every compact Lie group. The displayed identity follows from definition 2.10 and lemma 4.5. For odd n, its integrand is negative only on

$$[-2C_G, -2] \cup [0, 2].$$

On this union its absolute value is at most $((2C_G)^2+4)(2C_G)^n$. Since $\psi_G(\mathbb{R}) = 2$ by lemma 4.5, integration gives (5.1).1. □

Lemma 5.2 (Normalized exceptional Weyl–rectangle coefficient)**.** *Normalize long roots to have squared length two. Let $Q^\vee \subset P^\vee$ be the coroot and coweight lattices, put*

$$z_R = \prod_{\alpha \in R_+} \frac{|\alpha|^2}{2}, \qquad \nu_R = \frac{|P^\vee/Q^\vee|^2}{\operatorname{covol}(Q^\vee)^2},$$

and let d_i be the invariant degrees. For the exceptional root systems,

R	G_2	F_4	E_6	E_7	E_8
z_R	1/27	1/4096	1	1	1
$\operatorname{covol}(Q^\vee)^2$	3	4	3	2	1
$\lvert P^\vee/Q^\vee \rvert$	1	1	3	2	1
ν_R	1/3	1/4	3	2	1
$\min_{0\neq\lambda\in P^\vee} \lvert\lambda\rvert^2$	2	2	4/3	3/2	2

(5.2.1)

Let $a = d/2$, where $d = \dim G$, and let

$$A_R = \nu_R z_R^2 \frac{8ad^2(d/\kappa)^d \prod_i ((d_i-1)!)^2}{(2\pi)^r}. \tag{5.2.2}$$

For a retained radial cell $[s_0, s_1] \times [t_0, t_1]$ and its transpose, suppose the local character bounds are g_R, h_R and the product of the certified Weyl-sine and optional radial exponential losses is $\eta_R(n) > 0$. Then their combined positive contribution to $D_G(n)$ is at least

$$\frac{A_R(2d)^n}{n^{d+2}} \frac{g_R(n)^2\eta_R(n)}{a\Gamma(a)^2} \left(\int_{s_0}^{s_1} s^{a-1}\,ds\right)\left(\int_{t_0}^{t_1} t^{a-1}\,dt\right) h_R(n)^n\big((2dh_R(n))^2-4\big). \tag{5.2.3}$$

All exceptional caps used below lie in an injective exponential chart of the adjoint maximal torus.

Proof. For the adjoint global form the exponential kernel is $2\pi P^\vee$. Thus normalized torus Haar measure in Euclidean exponential coordinates is

$$\frac{dv}{(2\pi)^r\operatorname{covol}(P^\vee)}, \qquad \operatorname{covol}(P^\vee) = \frac{\operatorname{covol}(Q^\vee)}{|P^\vee/Q^\vee|}. \tag{5.2.4}$$

Apply the Weyl integration formula [1, Chapter IV, (1.11)] in both variables. If $\Delta_R(v) = \prod_{\alpha>0} \alpha(v)$ and Δ_{std} uses the squared-length-two normal to every reflecting hyperplane, then $\Delta_R^2 = z_R\Delta_{\mathrm{std}}^2$. The Macdonald–Mehta identity [12, §2.1 and Theorem 3.1(i)], scaled by $(d/\kappa)^{1/2}$, gives

$$\int e^{-\kappa|v|^2/(2d)}\Delta_R(v)^2\,dv = z_R(2\pi)^{r/2}(d/\kappa)^a \prod_i d_i!.$$

The square of this identity, the two factors in (5.2).4, and $|W| = \prod_i d_i$ give

$$\nu_R z_R^2 \frac{(d/\kappa)^d \prod_i((d_i-1)!)^2}{(2\pi)^r}.$$

In the scaled radial variable $s = n\kappa|v|^2/(2d)$, homogeneity of Δ_R^2 gives the radial density $s^{a-1}/\Gamma(a)$. The character-gap square contributes $(2d/n)^2 g_R^2$; adjoining the transposed cell contributes the factor two. Writing $8d^2$ as $(8ad^2)/a$ yields exactly (5.2).3. The retained cells have $s_0 \ge t_1$, so the cells and their transposes are disjoint up to null boundaries.

The Cartan and simple-coroot Gram determinants give the middle three rows of (5.2).1; the short-root products give the first row. For E_6, E_7, inversion of the Cartan matrix and exact enumeration of the simple-root pairings in $\{-1, 0, 1\}$ give the displayed coweight minima. This enumeration is global: if one pairing has absolute value at least two, Cauchy's inequality and $|\alpha_i|^2 = 2$ give $|\lambda|^2 \ge 2$. The same calculation gives the G_2, F_4 minima; for E_8, determinant one and the even integral Cartan form give minimum two. These determinant and minimum calculations are repeated in the exact C++ checkers.

Finally, a cap point satisfies $|v|^2 < \delta^2/2$. For G_2, F_4 one has $\delta \le \pi$; for E_6, E_7, E_8 one has $\delta \le 14/3$. The last row of (5.2).1 and the rational inequality $\pi > 333/106$ show that the diameter of every cap is strictly smaller than $2\pi \min_{0\ne\lambda\in P^\vee} |\lambda|$. Hence no two cap points differ by a nonzero exponential-kernel vector. □

Lemma 5.3 (Rectangular Chain certificate template)**.** *Let G be exceptional, let $d = \dim G$, and write $D_G(n) = Q_3^G(n+2) - 4Q_3^G(n)$. Suppose a finite union of rational rectangles $R = [s_0, s_1]\times[t_0, t_1]$ in radial variables satisfies, for all odd $n \ge n_0$,*

$$\chi(y) - \chi(x) \ge \frac{2d}{n} g_R, \qquad \frac{\chi(x)+\chi(y)}{2d} \ge h_R(n),$$

with $g_R > 0$, $h_R(n) > 0$, and with every rectangle contained in the specified root-angle cap $|\alpha\cdot v| \le \delta$ on which the stated scalar trigonometric minorants have been proved. Let $L(n) > 0$ be the resulting sum of integrated rectangular lower bounds, and suppose an explicit negative-region estimate $U(n) > 0$ satisfies, for every odd $n \ge n_0$,

$$D_G(n) \ge L(n) - U(n), \qquad L(n_0) \ge U(n_0), \qquad L(n+2)U(n) \ge L(n)U(n+2) \tag{5.3.1}$$

Then $D_G(n) \ge 0$ for every such n.

Proof. The two local bounds supply the squared gap and positive-power factors; the cap makes the stated scalar bounds uniform. The middle comparison gives $L(n_0)/U(n_0) \ge 1$, and the last makes this ratio nondecreasing along odd n. Hence $L(n) \ge U(n)$ and $D_G(n) \ge 0$. □

Proposition 5.4 (Direct rectangular tails for E_6 and E_7)**.** *The following exact rectangular certificates hold:*

G	n_0	*grid*	*retained cells*	$\log_{10}(L(n_0)/U(n_0))$	*odd-step lower bound*
E_6	75	1/10	23088	8.904	$> 10^{0.573}$
E_7	65	1/20	66925	> 0.42	$> 101/100$

Consequently $D_{E_6}(n) \ge 0$ for every odd $n \ge 75$, and $D_{E_7}(n) \ge 0$ for every odd $n \ge 65$.

Proof. Put $\delta = 14/3$. The root constants are

G	d	r	$\lvert R_+\rvert$	$\kappa : \sum_{\alpha>0}(\alpha\cdot u)^2 = \kappa\lvert u\rvert^2$	$q : \sum_{\alpha>0}(\alpha\cdot u)^4 = Q(u)^2/q$
E_6	78	6	36	12	16
E_7	133	7	63	18	27.

The checker generates the 36 and 63 positive roots from the two Cartan matrices by exact simple reflections, expands both degree-four polynomials coefficientwise in the simple-root coordinates, and verifies $16\sum(\alpha\cdot u)^4 = Q(u)^2$ and $27\sum(\alpha\cdot u)^4 = Q(u)^2$, respectively.

On $|z| \le \delta$, exact Bernstein coefficients applied to alternating Taylor minorants prove

$$\begin{aligned} 2(1-\cos z) &\ge a_1 z^2 + a_2 z^4 + a_3 z^6, \\ 2\cos z &\ge 2 - z^2 + cz^4, \\ \frac{\sin(z/2)}{z/2} &\ge \exp\{-z^2/24 - z^4/2880 - z^6/103800\}, \end{aligned} \tag{5.4.1}$$

where

$$(a_1,a_2,a_3) = \left(\frac{99119225}{10^8}, -\frac{7802858}{10^8}, \frac{169078}{10^8}\right), \qquad c = \frac{83}{2000}.$$

Since $\sum z^6 \ge (\sum z^4)^2/\sum z^2$ and $\sum z^6 \le \delta^2 \sum z^4$, these scalar inequalities give, on a cell $R = [s_0,s_1]\times[t_0,t_1]$,

$$\begin{aligned} g_R(n) &= a_1 s_0 - t_1 + a_2\frac{2d}{q}\frac{s_1^2}{n} + a_3\frac{(2d)^2}{q^2}\frac{s_0^3}{n^2} + c\frac{2d}{q}\frac{t_0^2}{n}, \\ h_R(n) &= 1 - \frac{s_1+t_1}{n} + c\frac{2d}{q}\frac{s_0^2+t_0^2}{n^2}. \end{aligned} \tag{5.4.2}$$

The sine-density loss is bounded using

$$2\frac{2d}{24n}(s_1+t_1) + 2\left(\frac{1}{2880} + \frac{\delta^2}{103800}\right)\frac{(2d)^2}{qn^2}(s_1^2+t_1^2). \tag{5.4.3}$$

For E_6 the checker partitions $[10,30]\times[5,20]$ into 1/10-cells. The cap value at $n = 75$ is $2450/39 > 30$. It retains exactly 23088 cells. It uses lemma 5.1 with $C_{E_6} = 6$, hence $U_{E_6}(n) = 296\cdot 12^n$.

For E_7, let

$$P(S) = \frac{T_{26}((2S-273)/259)}{T_{26}(-301/259)}, \qquad I = \int S^{14}(S^2+4)P(S)^2\, d\psi_{E_7}(S).$$

Exact Bernstein coefficients on $[-14,-2]$ and $[0,2]$ prove at $n = 65$

$$|S^{65}(S^2-4)| \le \frac{24}{25}14^{51}S^{14}(S^2+4)P(S)^2. \tag{5.4.4}$$

The integral I is an exact linear combination of $Q_3^{E_7}(0),\dots,Q_3^{E_7}(68)$ and therefore uses only the moments $m_0,\dots,m_{70}$ regenerated by lemmas 2.4 and 2.5. Thus

$$U_{E_7}(n) = \frac{24}{25}14^{n-14}I$$

is a valid negative-region bound for every later odd n: multiplication by 14^2 preserves (5.4).4 because $|S| \le 14$ on both negative-sign intervals. The checker partitions $[15,35]\times[8,28]$ into 1/20-cells; the cap is $910/19 > 35$, and exactly 66925 cells survive.

For every retained cell the checker verifies the two inequalities in lemma 5.3, positivity of the direct factor, and the cleared monotonicity inequalities for $g_R(n)$ and $h_R(n)$. It also verifies cellwise

$$\left(\frac{d}{C_G}h_R(n_0)\right)^2\left(\frac{n_0}{n_0+2}\right)^\alpha > 1$$

(and $> 101/100$ for E_7), while the cap and sine factors improve. The positive sums use the normalized Weyl factors $\nu_{E_6} = 3$ and $\nu_{E_7} = 2$ from lemma 5.2; the checker verifies the Cartan determinants, coweight minima, and disjointness from every transposed cell before summing. OpenMP parallelizes the cell sum, but every retained-cell value and both final comparisons use GMP rationals. The resulting base-ratio logarithms are those in the table, so lemma 5.3 proves both tails. □

Proposition 5.5 (E_8 rectangular bridge and tail ledger)**.** *For E_8, the following exact rectangular certificates prove $D_{E_8}(n) \geq 0$ at the listed odd exponents:*

n	*retained cells*	$\log_{10}$(*base ratio*)
77	57817	0.19
79	68095	0.79
81	79060	0.27
83	91138	0.28
85	103495	0.29
87	116526	0.44
89	130740	0.47
91	69502	0.13
93	53822	0.04
95	29784	0.70
97	35869	0.27
99	42662	1.70
101	49478	0.68
103	56387	0.68
105	63660	0.19
107	70562	0.08
109	77454	1.15
111	84702	0.39
113	91581	1.89
115	98455	0.52
117	111427	0.05
119	15000	0.66
121	74053	0.10
123	28121	1.10
125	8497	1.11
127	10528	1.04
129	12719	0.35
131	17797	1.59

The analytic tail from $n \geq 133$ uses 13558 *retained cells, base-ratio logarithm* 0.02*, and odd-step logarithm* 0.85*.*

Proof. The E_8 constants are $d = 248$, $|R_+| = 120$, rank 8, and $C_G = 8$. For the analytic tail the negative-region estimate is $U(n) = 2((2C_G)^2+4)(2C_G)^n = 520 \cdot 16^n$. The certificates use the quartic Weyl identity

$$\sum_{\alpha>0} (\alpha \cdot u)^4 = \frac{1}{50} \left(\sum_{\alpha>0} (\alpha \cdot u)^2 \right)^2 .$$

On each retained rational cell the checker proves the two local character bounds of lemma 5.3; all retained cells are inside the root-angle cap. It also verifies the factor-one normalized Weyl coefficient and transposed-cell disjointness required by lemma 5.2. For the tail certificate at $n = 133$, for instance, the cell formula is

$$g_R = s_0 - t_1 - \frac{248}{300n} s_1^2, \qquad h_R(n) = 1 - \frac{s_1 + t_1}{n} + \frac{1}{18}\frac{496}{50}\frac{s_0^2 + t_0^2}{n^2},$$

and the retained cells have

$$\min h_R(133) = \frac{176378749}{318402000}, \qquad \min(496^2 h_R(133)^2 - 4) > 0.$$

Before the table is consumed, `verify_e8_negative_bounds_gmp.cpp` checks its exact 29-row schema. For $n = 77, 79, \ldots, 113$, it reads the locally verified and ancillary moment chain through m_{100}, reconstructs the rational square-majorant from the pinned factor witness, proves both pointwise inequalities by exact Bernstein coefficients, and integrates the witness with GMP. Only $n = 81$ and $n = 85$ require subdivision, of depths three and five respectively. For $n = 115, 117, \ldots, 133$, the audit recomputes $520 \cdot 16^n$ from lemma 5.1. Every resulting rational must equal the corresponding table entry; no historical Python row module is active.

Only then does the OpenMP/GMP replay `direct_chain_rect_e8_parallel_certificate.cpp` consume that table. It checks every retained-cell count, the $n = 133$ tail row, and the two exact rational comparisons for each case. The base comparisons and tail odd-step comparison are exact rational checks. For completeness, on the fixed $n = 133$ tail cells one has $s_1 < 40$, $t_1 \leq 31$, and $s_1+t_1 > 63$. With $A_R = s_1+t_1$ and $B_R = (124/225)(s_0^2+t_0^2)$, one has $133A_R > 8379 > 2824 > 2B_R$, so $h_R(n) = 1-A_R/n+B_R/n^2$ increases thereafter. The gap, cap, sine-density, and direct factors also improve. Hence every cell has odd-step quotient at least $((248/8)\min_R h_R(133))^2(133/135)^{250} > 1$, the logged exact value. Thus (5.3).1 holds throughout the tail, proving the listed differences and every odd $n \geq 133$. ∎

Proposition 5.6 (G_2 and F_4)**.** *For $G = G_2$ or $G = F_4$ and every $n \geq 0$,*

$$Q_3^G(n) \geq 0.$$

Proof. The base value $Q_3^G(1) > 0$ is lemmas 2.3 and 2.6. Put $Q(u) = \sum_{\alpha>0}(\alpha \cdot u)^2$. Exact expansion of the positive roots gives

G	d	r	$\lvert R_+\rvert$	$\kappa : Q = \kappa\lvert u\rvert^2$	$q : \sum_{\alpha>0}(\alpha \cdot u)^4 = Q(u)^2/q$	(d_i)
G_2	14	2	6	4	16/5	$(2, 6)$
F_4	52	4	24	9	54/5	$(2, 6, 8, 12)$.

The Macdonald–Mehta identity [12, Theorem 3.1(i)], whose root normals have squared length 2, gives the positive-saddle normalization

$$A_0 = \nu_G z_G^2 \frac{8ad^2(d/\kappa)^d \prod_i((d_i - 1)!)^2}{(2\pi)^r}, \qquad a = d/2.$$

Here the actual short roots change the squared discriminant by $z_{G_2} = 1/27$ and $z_{F_4} = 1/4096$; these are respectively the products of $|\alpha|^2/2$ over the three and twelve positive short roots. The normalized-torus factors from lemma 5.2 are $\nu_{G_2} = 1/3$ and $\nu_{F_4} = 1/4$.

Apply lemma 5.3 with $C = 2, 4$ and with

G	n_0	$[s_0, s_1] \times [t_0, t_1]$	g	$h(n_0)$
G_2	21	$[19/5, 21/5] \times [7/5, 9/5]$	111/80	1661/2268
F_4	65	$[51/4, 53/4] \times [31/4, 33/4]$	3023/1296	44063/63180.

Writing q for the preceding quartic denominator, the exact local bounds are

$$g = s_0 - t_1 - \frac{ds_1^2}{6qn_0}, \qquad h(n) = 1 - \frac{s_1 + t_1}{n} + \frac{2d(s_0^2 + t_0^2)}{18qn^2}.$$

The normalized rectangular radial mass is bounded below by

$$g^2\,3^{-\lceil s_1+t_1\rceil}\frac{(s_1^a-s_0^a)(t_1^a-t_0^a)}{a^3((a-1)!)^2}.$$

Using $\pi<355/113$, the Weyl-sine lower factors are 3^{-1} and 3^{-4}. The resulting exact base comparisons with $U(n)=2((2C)^2+4)(2C)^n$ exceed 1 and 10^{13}, respectively. Moreover the odd-step quotients are bounded below by

$$\left(\frac{d}{C}h(n_0)\right)^2\left(\frac{n_0}{n_0+2}\right)^{d+2}>6,10,$$

respectively. The exact checks also give $n_0(s_1+t_1)>2d(s_0^2+t_0^2)/(9q)$, so $h(n)$ increases; the gap and cap improve with n. They also verify the coroot Gram and Cartan determinants, the coweight minima, and transposed-cell disjointness required by lemma 5.2. Thus the rectangular Chain tails begin at odd $n=21,65$. The exact prefixes reach odd $n=197,217$ and overlap both tails. Lemma 2.11 closes all odd exponents; even exponents are lemma 2.9. □

Proposition 5.7 (E_6 and E_7)**.** *For $G=E_6$ or $G=E_7$ and every $n\geq 0$,*

$$Q_3^G(n)\geq 0.$$

Proof. The base value $Q_3^G(1)>0$ is lemmas 2.3 and 2.6. The exact prefixes verify the Chain differences through $m=37$ for E_6 and $m=32$ for E_7, reaching the odd exponents 75 and 65, respectively. The direct rectangular certificates of proposition 5.4 prove the corresponding Chain tails from those same odd exponents onward. The prefix endpoint and the tail start overlap exactly, so lemma 2.11 gives $Q_3^G(n)\geq 0$ for every odd n. Even n is lemma 2.9. □

Proposition 5.8 (E_8 finite prefix and bridge)**.** *For E_8, the finite prefix Chain differences*

$$D_{E_8}(n)\quad(n=1,3,\ldots,65)$$

are positive. The finite bridge values

$$Q_3^{E_8}(n)\quad(n=69,71,73,75,77)$$

are positive, and the Chain differences

$$D_{E_8}(n)\quad(n=67,69,71,73,75,77,79,81,83,85,87,89,91,93,95)$$

are nonnegative.

Proof. The exact source-pairing replay applies lemmas 2.4 and 2.5 through $\mathrm{ad}^{\otimes 50}$ and independently regenerates $m_0,\ldots,m_{100}$ from the E_8 Cartan datum. It checks every value against the archived local prefix and the Bourjaily–Plesser–Vergu comparison table, with zero mismatches. For odd $n\leq 98$, lemma 2.3 computes $Q_3^{E_8}(n)$ using only these moments. Exact integer evaluation gives

$$\begin{aligned}
&\min D_{E_8}(1,3,\ldots,65) && =20,\\
&\log_{10}\min Q_3^{E_8}(69,71,73,75,77) && \approx 100.94,\\
&\log_{10}\min D_{E_8}(67,69,\ldots,77) && \approx 100.94,\\
&\log_{10}\min D_{E_8}(67,69,\ldots,95) && \approx 100.94.
\end{aligned}$$

The replay flags are

```
moment_log_0_100_arxiv_extension    OK,
prefix_chain_diff_1_65_positive     OK,
bridge_q3_69_77_positive            OK,
chain_diff_67_77_positive           OK,
extended_chain_diff_67_95_positive  OK.
```

□

Proposition 5.9 (E_8 tail)**.** *For E_8, the Chain difference is nonnegative for every odd $n \geq 77$.*

Proof. The direct rectangular certificates in proposition 5.5 prove the Chain tail for odd $n \geq 133$ and the successive bridge steps

$$n = 131, 129, 127, 125, 123, 121, 119, 117, 115, 113, 111, 109, 107, 105, 103, 101,$$

$$n = 99, 97, 95, 93, 91, 89, 87, 85, 83, 81, 79, 77.$$

Together with proposition 5.8, these bridge steps connect the exact prefix to the $n = 133$ tail. Therefore $D_{E_8}(n) \geq 0$ for every odd $n \geq 77$. □

Theorem 5.10 (Exceptional closure)**.** *For every $G \in \{G_2, F_4, E_6, E_7, E_8\}$ and every $n \geq 0$,*

$$Q_3^G(n) \geq 0.$$

Proof. Use proposition 5.6 and proposition 5.7 for G_2, F_4, E_6, E_7. For E_8, proposition 5.8 supplies the prefix Chain differences through $n = 65$, the bridge values at $69, 71, 73, 75, 77$, and the bridge Chain differences through 95; then proposition 5.9 supplies every later Chain difference. Applying lemma 2.11 from the strict base value $Q_3^{E_8}(1) > 0$ of lemmas 2.3 and 2.6 gives every odd exponent. Even exponents are covered by lemma 2.9. □

6. Assembly of the Classification

Proof of theorem 2.2. By lemma 2.8, $Q_3^G(n)$ depends only on the compact simple Lie algebra. In the irreducible-root-system classification, $B_1 = C_1 = A_1$ and $D_3 = A_3$ are already in the type-A branch, D_2 is not simple, and the remaining classical types have B_b, C_b with $b \geq 2$ and D_b with $b \geq 4$. The classification therefore leaves type A, the non-A classical types B, C, D, and the exceptional types G_2, F_4, E_6, E_7, E_8. Type A is theorem 3.16; types B, C, D are theorem 4.842; the exceptional types are theorem 5.10. These cases exhaust all compact connected simple Lie groups. □

7. Conclusion and limitations

The proof establishes one uniform sign inequality for the adjoint tensor category of every compact simple type. Its common structure is the passage from the moment identity in lemma 2.3 to a stable prefix, a finite-rank correction interval, and an analytic tail. The classification is needed because the mechanism controlling the middle interval differs between type A, the three other classical families, and the exceptional groups.

The theorem does not compare unequal characters and does not prove Ginibre's condition on the cone of all central positive-definite functions. It also does not assert that the exactly-two-minus argument extends directly to four or more minus factors. Part III addresses the latter hierarchy only on the closed multiplicative convex cone generated by χ_{ad}, using the present theorem as an input. These scope boundaries separate the mathematical conclusion from possible applications to larger character cones or to model-specific correlation inequalities.

Remark 7.1 (Final-facing dependency map)**.** The proof graph is the following; it deliberately omits superseded diagnostic routes retained in the expanded archive. All incoming nodes are proved in the manuscript or discharged by the authenticated exact replays.

closure	load-bearing incoming results
theorem 3.16	theorems 3.1, 3.5, 3.7 and 3.14, lemma 3.9, and corollary 3.15
theorem 4.842	lemma 4.3 and propositions 4.14, 4.16 to 4.19, 4.21, 4.49, 4.840 and 4.841
theorem 5.10	propositions 5.6 to 5.9
theorem 2.2	lemma 2.8 and theorems 3.16, 4.842 and 5.10

Inside proposition 4.841, the only live residual route is the proved correction-prefix bridge followed by the layered/tilted MGF or direct interval supplier. Local fixed-witness lemmas used to prove the boundary-through-twenty-ninth correction contract are represented in the accepted correction-prefix suite. The expanded archive supplies a line-by-line audit of those contracts. Global all-row fixed-witness/compression interfaces explicitly marked superseded have no incoming edge from this map.

Appendix A. Exact Certificate Protocol

All finite certificates used above are exact. Moment computations use either the hook-length partition formula in type A or the exact Racah–Speiser iteration of lemma 2.4 together with the pairing identities of lemma 2.5. Signs are decided only after reducing the target inequality to an integer or rational comparison. The replay outputs quoted in the paper are therefore certificate outputs, not floating-point tests.

The common checker contract is the following.

Input	root datum, highest weights, exact moment or delta logs, and row list,
Moment step	compute $m_k = \mathrm{mult}(\mathbf{1}, \mathrm{ad}^{\otimes k})$ by integer arithmetic,
Target step	form $Q_3(n)$ or $D_G(n)$ from lemma 2.3 and definition 2.10,
Reduction step	rewrite every tail or bridge claim as an integer/rational inequality,
Decision step	accept only if every required integer/rational margin is > 0.

For every classical or exceptional source table passed to an active root-datum source-replay stage, the checker reconstructs the relevant Cartan matrix and its full root set, verifies the adjoint root multiplicities and rank-dimensional zero-weight space, rejects negative output multiplicities and coordinate overflow, regenerates the moments through half degree, and compares every claimed moment or full correction. In particular, the E_8 bridge regenerates $m_0, \dots, m_{100}$ from $\mathrm{ad}^{\otimes 0}, \dots, \mathrm{ad}^{\otimes 50}$. For each classical source check, the checker independently regenerates the stable sequence from lemma 4.67, verifies every archived OEIS value through the largest required degree, and uses the regenerated value in the finite correction. Thus the OEIS and arXiv tables are independent comparisons rather than accepted source axioms.

The finite row-gated bridge is now source-generated. The standard command runs the Pieri/CRT verifier of proposition 4.15 for all 870 B/C steps and the exact-GMP interval mode of proposition 4.48 for the remaining 36 high-rank type-D steps. Together with the active D_4–D_{52} replays, these computations replace the historical row-gated ledger completely. Its old raw logs remain authenticated provenance, but no theorem consumes their classifications, statuses, or printed margins.

For type D, a bare historical line `D_b Delta_j = B_j` with $j > 2b$ records only the nonnegative determinant component B_j, not the full correction $A_j + B_j$. Both the source checker and the downstream interval checker reject such a line as an exact moment. Bare lines with $j \le 2b$, where $A_j = 0$, and explicit full or exact-Weyl moment lines are regenerated and compared exactly.

Certificate	Output used in the proof
Type A finite-overlap replay	The endpoint constants, rows $N = 6, \dots, 18$, and the final finite-overlap band are checked; the final line is `All SU(N) repair certificates verified.`
Classical Rains-square cutoff replay	The cutoff inequalities are checked for $C_G = 296, \dots, 10000$, the pre-cutoff bad range is $248, \dots, 295$, and the replay exits with status 0.

Certificate	Output used in the proof
Classical first-hit trace replay	The replay tests 881 finite rows, closes exactly 57 first-hit rows, and records the minimum margin at D_{281}.
Source-generated classical row-gated bridge	The common Pieri traversal regenerates 832 exact B/C corrections and checks all 870 row-gated inequalities: 416 are direct exact evaluations and 454 use proved one-sided correction intervals. It reports zero failures and minimum lower margin 1046. The separate type-D mode checks all 36 nonvacuous D_{53}–D_{63} steps with zero failures. Both programs are rebuilt and executed in the isolated replay; the historical raw bridge archive is provenance only.
Classical post-$m = 29$ first-hit replay	The replay checks the first post-$m = 29$ Chain step in 796 rows, generates stable moments through m_{557}, and exits with status 0. The five bucket row counts are $14, 26, 490, 25, 241$. The replay is used only for the first-hit claim of proposition 4.44; the all-later B/C closure is the separate correction-prefix and analytic-supplier argument of proposition 4.841. Its input delta-log manifest and artifact classification are recorded in `certificates/post_m29`. The input, accepted, diagnostic, and classification manifest hashes are, respectively, `f8ca9e149f896c28b22ad5c53ba4bb8d` `e877aca153cc56d6f04f21c04c25c95e`, `ce34ba6dd902f8058923219f5e3b9b29` `faf66a541e875af4782b854cc9fb0540`, `01f053f2d50173900e3b7d2d773629b6` `ec6b3a46bad5fd1eed9cdb4fb50f640f`, and `31e37c889291d5ab2504472163f89d0e` `2ada74f621eea2315291350f09733455`.
Classical B_{11} post-$m = 29$ bridge	The replay checks 28 copied exact B_{11} boundary-delta logs $\Delta_{24}, \dots, \Delta_{51}$, verifies that each records `__EXIT_STATUS=0`, recomputes Chain margins for $m = 31, \dots, 36$, and exits with status 0. The minimum margin is attained at $m = 31$ and is positive. The delta-log manifest hash is `c689442193378fd04a2e3db6cbfa9b315` `3839d88b37f06055ca13e71ac9aa129`.

Certificate	Output used in the proof
Classical B_{12} post-$m = 29$ finite tail	The replay checks one existing exact B_{12} log for $\Delta_{26}, \ldots, \Delta_{39}$ and 14 copied exact B_{12} boundary-delta logs $\Delta_{40}, \ldots, \Delta_{53}$, verifies that each records `__EXIT_STATUS=0`, recomputes Chain margins for $m = 31, \ldots, 38$, and checks the direct Chebyshev tail from odd exponent 81 onward. The minimum bridge margin is attained at $m = 31$ and is positive. The delta-log manifest hash is `480b77eb57b215cded54e4c195a15e` `61c119894aae01f3dc04c58eb2d22f2e09`.
Classical B_{13} post-$m = 29$ finite tail	The replay checks one existing exact B_{13} log for $\Delta_{28}, \ldots, \Delta_{39}$ and 18 copied exact B_{13} boundary-delta logs $\Delta_{40}, \ldots, \Delta_{57}$, verifies that each records `__EXIT_STATUS=0`, recomputes Chain margins for $m = 31, \ldots, 43$, checks stable-envelope bounds for the missing negative quadratic pairs at $m = 42, 43$, and checks the direct Chebyshev tail from odd exponent 91 onward. The delta-log manifest and replay log hashes are `0fe4e21a1533ba290a7bfad3c07609bf` `ff8d70a0d7ad6d46976faf60d7ed3b37` and `08b9666bc771897e9702c0132620f4ca` `e9215b30c92c160ed52795fe6db096ee`.
Classical C_{13} post-$m = 29$ finite tail	The replay checks one existing exact C_{13} log for $\Delta_{28}, \ldots, \Delta_{39}$ and four copied exact C_{13} boundary-delta logs $\Delta_{40}, \ldots, \Delta_{43}$, verifies that each records `__EXIT_STATUS=0`, recomputes Chain margins for $m = 31, 32, 33$, and checks the direct Chebyshev tail from odd exponent 71 onward. The minimum bridge margin is attained at $m = 31$ and is positive. The delta-log manifest hash is `e2c407483c5f0894d4ceee5b887c2e22` `aa6e5604b85bde037ace25923c2f0320`.

Certificate	Output used in the proof
Classical C_{14} post-$m = 29$ finite tail	The replay checks one existing exact C_{14} log for $\Delta_{30}, \dots, \Delta_{39}$ and 12 copied exact C_{14} boundary-delta logs $\Delta_{40}, \dots, \Delta_{51}$, verifies that each records `__EXIT_STATUS=0`, recomputes Chain margins for $m = 31, \dots, 39$, and checks the direct Chebyshev tail from odd exponent 83 onward. The minimum bridge margin is attained at $m = 31$ and is positive. The delta-log manifest hash is `aa13f907722ad5e1bd384c243ac042ce` `1fddd429782c641c246c146acabf4756`; the replay log has SHA-256 `6dc9a3f519c64568b378b8689dd78867` `e5311abdf1257268f37ba79408feb1c3`.
Classical C_{15} post-$m = 29$ direct tail	The replay checks the existing exact C_{15} log for $\Delta_{32}, \dots, \Delta_{39}$, verifies that it records `__EXIT_STATUS=0`, and checks the direct Chebyshev tail from odd exponent 63 onward. The replay log has SHA-256 `68fd43b912006354f77fb2c051475767` `f8c6d85b3355b2a3ba3c438cd7a2ad68`.
Classical C_{16} post-$m = 29$ direct tail	The replay checks the existing exact C_{16} log for $\Delta_{34}, \dots, \Delta_{39}$, verifies that it records `__EXIT_STATUS=0`, and checks the direct Chebyshev tail from odd exponent 63 onward. The replay log has SHA-256 `9e15444c6e789d3aba9f83feaea03a22` `f5dd423e9c2a138b68f959bdb22c719f`.
Classical C_{17} post-$m = 29$ finite tail	The replay checks exact C_{17} boundary-delta logs $\Delta_{36}, \dots, \Delta_{45}$, verifies that each records `__EXIT_STATUS=0`, recomputes Chain margins for $m = 31, \dots, 34$, and checks the direct Chebyshev tail from odd exponent 73 onward. The delta-log manifest and replay log hashes are `47b4939674358a15b1c66ecb8d3c33a8` `f684a33a53a0c3ef25c1e7004749ccff` and `dcd5eb8e782be93d926a6486f18d8a5e` `c445d33f8784ba251f60dd0203d8eb32`.
Classical C_{18} post-$m = 29$ finite tail	The replay checks exact C_{18} boundary-delta logs $\Delta_{38}, \dots, \Delta_{50}$, verifies that each records `__EXIT_STATUS=0`, recomputes Chain margins for $m = 31, \dots, 40$, and checks the direct Chebyshev tail from odd exponent 85 onward. The delta-log manifest and replay log hashes are `5e61b65a8c745b131b16839fab178eb31` `73b29c90b3dcd922942ea64fa26a001` and `e25bfaef2b3b6289631431fd26717704` `dc82c78009cc02cf9c8e8dc8861b5541`.

Certificate	Output used in the proof
Classical C_{19} post-$m = 29$ finite tail	The replay checks exact C_{19} boundary-delta logs $\Delta_{40}, \ldots, \Delta_{58}$, verifies that each records `__EXIT_STATUS=0`, recomputes Chain margins for $m = 31, \ldots, 47$, and checks the direct Chebyshev tail from odd exponent 99 onward. The delta-log manifest and replay log hashes are `5567ba76b394f5579832d7fce8991e71` `70d98cef1f3cf4cbb43a57bd06450448` and `034d55810933481b342966686cf249a33` `649fac5333ad335df9b94e8f95b8273`.
Superseded determinant-only D_{12}–D_{97} tail routes	These historical replays treated the determinant-isotypic contribution as the full finite-rank correction after the omitted-even-row term became nonzero. They are diagnostic artifacts and no numbered result consumes them. The accepted replacements are the full-signed-moment D_{12}–D_{24} onset-63 certificate, the two-sided D_{25}–D_{52} bridge and exact/mixed MGF certificate, and the A-free D_{53}–D_{295} frontier.
Classical high-edge no-TV Chernoff trace	The directed-rounding C++/MPFR replay checks the one-trace source range, the Chernoff admissibility window, and the trace-pushforward inequality for B_b, C_b with $218 \leq b \leq 275$ or $b = 277$, using the three checked B/C parameter sets. It exits with status 0 and has SHA-256 `ca3a938c9d45d65eb892f6ba637ed8c3` `a7b1d32a571a5be52e2f5efb80477176`.
Classical middle type-B unitary square trace	The 384-bit directed-rounding MPFR replay checks the four strict logarithmic comparisons in proposition 4.21 for all 94 rows $B_{124}, \ldots, B_{217}$. It reports `rows_checked=94`, `failures=0`, and minimum log margin greater than 126. Its SHA-256 is `7780bfe3253bf9e030a9a544dcd7327f` `cd905aed8a2890ded191dc38efae8230`.

Certificate	Output used in the proof
Classical middle type-B/C exponentially tilted MGFs	The 384-bit directed-rounding MPFR replay checks the CJL source domains, positive tilted interval masses, and strict comparisons in propositions 4.29 and 4.33 for all 62 rows $B_{62}, \ldots, B_{123}$ and all 156 rows $C_{62}, \ldots, C_{217}$. The type-C part additionally checks both orthogonal-component source domains from the Rains square decomposition. It reports `rows_checked=218`, `failures=0`, and has SHA-256 `ca6860204466e4a6b38d9c5597511cd1` `1cfebb4b1b65cab88db6ee45f9477111`.
Classical residual layered $B/C/D$ MGFs	The 384-bit directed-rounding MPFR replay checks exact low-rank Toeplitz–Hankel determinants, Bessel-series remainder bounds, positive interval-Cholesky pivots, the extended CJL trace-norm domains, eight nested tilted-mass bounds, exact stable push moments, one-sided polynomial expectations, and the final strict comparison in propositions 4.43 and 4.49. It covers 48 type-B, 42 type-C, and 243 type-D rows, reports `failures=0`, and has minimum natural-log margin $0.0078804615513483267\ldots$ at D_{220}. Its SHA-256 is `3e4900e5e2de2ec1a7fb04830ac59dea` `7369f8eebb1932ec08ac9ef73ba11654`.
Classical D_{25}–D_{52} exact/mixed finite-rank MGFs	The 384-bit directed-rounding verifier checks 28 exact/mixed MGF onsets, including exact normalized SO-even defining determinants at ranks 25–28 and exact Rains square-component determinants at every rank. It reports `failures=0`, minimum final log margin $0.013751090874926124\ldots$, and has SHA-256 `f87ab4bce0b189878ef4c3528ce34687` `1721bff05cb00ccb553a6c4e84e1f049`.
Classical D_{25}–D_{52} correction-aware bridge	The exact C++/GMP/OpenMP verifier checks all 707 finite Chain steps using the proved two-sided correction boxes. It reports `failures=0`, minimum exact lower bound 4695986280539928492451072, and has SHA-256 `e47196c8097fb2a087b4673d47a5e34c` `aae1cd1a316058759e17b13708e2d888`.
Classical D_{53}–D_{295} A-free frontier	The exact GMP/OpenMP verifier checks every Chain step $m = 30, \ldots, 163$, requires every direct onset to satisfy $N_D(b) + 2 \leq 2b$, reports minimum A-free slack 1 and `failures=0`, and has SHA-256 `876d3f752d03b8f35183f0b28af7c15` `e9e28d104396fc65c215e5d854d8ae31f`.

Certificate	Output used in the proof
Type-D legacy-envelope counterexample	The exact machine-C C++/GMP replay computes $B_{18}(4) - A_{18}(4) = -80781084484224$ and has SHA-256 `463e3c9632bb92936620fa65c03d3ff5` `abdb86940624f1621ffa976156d7d02f`.
Classical B/C interval direct tail	The exact GMP negative-tail verifier checks the degree-8, scale-12000, cutoff-80 Chebyshev input for all 402 residual B/C rows and reports `failures=0`. The directed-rounding MPFR verifier then checks the direct pushforward-tail inequality at the logged row-dependent onset for each residual row and reports `failures=0`. The two replay SHA-256 hashes are `80fce36378ec2192dca2c0b616c083b46168cf` `5134b66e73775156b995ea303b` and `53530cfe3d4b1947a77bf45c5c5aaa56` `a052ec1cde93c0704b68e004f22f2cb5`.
Classical B/C correction-prefix bridge and superseded tail routes	The exact GMP/OpenMP verifier checks the residual B/C finite bridge under the Pieri sign theorem and the lower bound $\delta_j \geq -s_j/2$. The expanded correction propositions prove the prefix hypotheses used in proposition 4.841. It covers $m = 31, \ldots, 15447$, reports `failures=0`, and its worst positive lower bound occurs at B_{217}, odd exponent 30897. The replay SHA-256 hash is `426fc6ccf924e70d27fdfd90f0e4314f` `02e2bc87fe09a6fe84cfdce92f2dbec7`. The Pieri and fixed-witness arithmetic verifiers above are conditional arithmetic reductions: they do not discharge their combinatorial fixed-witness or compression premises. Their detailed ledger is retained in the source archive but excluded from the publication proof ledger because those routes are not used after proposition 4.43.
E_8 finite bridge replay	The replay checks the $m_0, \ldots, m_{100}$ source table, verifies the prefix Chain differences from 1 through 65, verifies $Q_3^{E_8}(69), \ldots, Q_3^{E_8}(77) > 0$, and verifies the extended Chain differences from 67 through 95.
E_6, E_7 rectangular tail replays	The combined C++/OpenMP/GMP replay verifies the rank-corrected negative endpoints, exact root identities, scalar and Chebyshev Bernstein predicates, retained-cell counts, base-ratio lower bounds, and uniform odd-step bounds for both tails, exiting with `RESULT: ALL PASS`.

Certificate	Output used in the proof
E_8 rectangular OpenMP replay	The exact source audit first reconstructs all 29 negative-bound rows from their pointwise majorants, moment sources, and moment integrals and reports `E8_NEGATIVE_BOUND_AUDIT_GMP: ALL PASS`. The C++/OpenMP replay then consumes the authenticated table, verifies every expected retained-cell count, prints all base-ratio lower bounds and the $n = 133$ odd-step ratio, and exits with `RESULT: ALL PASS` and status 0.

The following table is an identity ledger, not the active-build manifest. For historical source certificates, the first hash is the recorded generator identity and need not identify a current source snapshot; the accepted output is the immutable output boundary stated in the replay contract. Current sources rebuilt by the standard command are instead pinned by `replay_sources.sha256`. The repaired type-A and first-hit trace rows below give both the current source hash and a newly archived accepted output hash. This table authenticates identities; it does not override the accepted/diagnostic/superseded classification, and a superseded recorded output is not proof evidence.

Replay	Generation/checker identity	Recorded output identity
Type A finite-overlap	`3a0cacb5c1f402dfc954e07513da4335` `219d96cd9335cc15ade6356d54d784a0`	`7339140167ea9e12101a7e0a316abc9d` `9be1e52c021a44c23e77f46ab776ff05`
Classical Rains-square cutoff	`ea02f81ac518eaaf849039112fd06781f` `908c4094619d54f0685987a7b2424c3`	`ca3a938c9d45d65eb892f6ba637ed8c3` `a7b1d32a571a5be52e2f5efb80477176`
Classical first-hit trace	`ea02f81ac518eaaf849039112fd06781f` `908c4094619d54f0685987a7b2424c3`	`ca3a938c9d45d65eb892f6ba637ed8c3` `a7b1d32a571a5be52e2f5efb80477176`
Classical B/C source-generated row-gated bridge	`fbd5565215a7b976b67a4529ad21daa2` `f235f2988da463c502370e3dab3ad797`	`d019c183f7dedff1049199b31a50d681` `3e0c17a0c565df1e845119e2a9fba054`
Classical D_{53}–D_{63} source-generated row-gated bridge	`1e574e0c53e624df30e9c69bc24280b` `8e6c56a099923428f0a27036b74a7454b`	`a84df851316bf376f00933ff958b7ad0d` `ac444acfa8f45d05d4221f17ddadd12`
Classical post-$m = 29$ first-hit	`4499d62b37fc54d700fbe80973d07dca` `4f3e91a6ee17ed7b397595aadf84863f`	`21642c87b90affa8e618bb9f97bf6ad3` `46289380832690546488c6dcc7a7300e`

Classical B/C interval direct tail	`cba8addb7581d99cbd7d92d33c11f01a`
	`2c78267b0dd8d17d39f7c938bf565876`
		`80fce36378ec2192dca2c0b616c083b461`
		`68cf5134b66e73775156b995ea303b`
		`53530cfe3d4b1947a77bf45c5c5aaa56`
		`a052ec1cde93c0704b68e004f22f2cb5`

Replay	Generation/checker identity	Recorded output identity
Classical middle type-B unitary square trace	`d0d503ffc4bd6cb0485a97a4cef2178a` `930f3f00c38ffd4486cb66aa01652aa7`	`7780bfe3253bf9e030a9a544dcd7327f` `cd905aed8a2890ded191dc38efae8230`
Classical middle type-B/C exponentially tilted MGFs	`0f5ad1037bcf738ab4ddd9c5b12f151a` `adc1bff2b5c4726c60622f608ef9e0e7`	`ca6860204466e4a6b38d9c5597511cd1` `1cfebb4b1b65cab88db6ee45f9477111`
Classical residual layered $B/C/D$ MGFs	`71cc2293c4ad5ba50d216c8f939cd1ec` `604ce7b86d6a19e9be1fcb27adc421fe`	`3e4900e5e2de2ec1a7fb04830ac59dea` `7369f8eebb1932ec08ac9ef73ba11654`
Classical D_{25}–D_{52} exact/mixed finite-rank MGFs	`71cc2293c4ad5ba50d216c8f939cd1ec` `604ce7b86d6a19e9be1fcb27adc421fe`	`f87ab4bce0b189878ef4c3528ce34687` `1721bff05cb00ccb553a6c4e84e1f049`
Classical D_{25}–D_{52} correction-aware bridge	`9a6cf4da5b866d3b8fc2ce6d1d1a442f` `d3b419702fda2296da83d2fb788379e4`	`e47196c8097fb2a087b4673d47a5e34c` `aae1cd1a316058759e17b13708e2d888`
Classical D_{53}–D_{295} A-free frontier	`444cf4dbe59d5000a3315f31054ef07e` `c1cf710edf91a7c685f973644d65ad74`	`876d3f752d03b8f35183f0b28af7c15` `e9e28d104396fc65c215e5d854d8ae31f`
Type-D legacy-envelope counterexample	`bc22efcafafe2d40b5f5430cb8f9cb046` `bc6b9b188da684e61dc6fdfaa27077e`	`463e3c9632bb92936620fa65c03d3ff5` `abdb86940624f1621ffa976156d7d02f`
Classical B_{11} post-$m = 29$ bridge	`6ae0d3bf506737b32480c336a6213ef6` `820e6561497899977311199daa9177f56`	`14f8c9813d3fda536e02a5c5809cae13` `6e6f531669d9f20806352415702db9f3`
Classical B_{12} post-$m = 29$ finite tail	`1afb4bb14311864103dd5ebd381842bd` `238f09f90c2064257f0b05e34e705443`	`a8e6a7c58eccf3565f578c8ed9d9187a` `507000b497efdca6593c40013c4f6854`
Classical B_{13} post-$m = 29$ finite tail	`d6e6f57279872ce94c81d31ed45e5e7b` `c0948993ef335a45d9717bc64f7a9031`	`08b9666bc771897e9702c0132620f4ca` `e9215b30c92c160ed52795fe6db096ee`
Classical C_{13} post-$m = 29$ finite tail	`1511406f2e1015184027f9d90ae380f8` `9789c5d2ac5b8f49a599608d62ca1f1f`	`1f66ffebfe9e722f443073031ec74401` `06b87116a1d1db279d154967a7e45049`
Classical C_{14} post-$m = 29$ finite tail	`000e286d0dd04595a66c9efeeafaae9b` `a38318cab396f08f282540b6713d696f`	`6dc9a3f519c64568b378b8689dd78867` `e5311abdf1257268f37ba79408feb1c3`

Replay	Generation/checker identity	Recorded output identity
Classical C_{15} post-$m = 29$ direct tail	d7ec8b228ee53f304d4de0060c371121 b9e21c3c456f8263346ad862b20fa4c1	68fd43b912006354f77fb2c051475767 f8c6d85b3355b2a3ba3c438cd7a2ad68
Classical C_{16} post-$m = 29$ direct tail	23af44fbdc597cf0332c422d716cb2c 483a47bafd1756f15b48486062b77a1b9	9e15444c6e789d3aba9f83feaea03a22 f5dd423e9c2a138b68f959bdb22c719f
Classical C_{17} post-$m = 29$ finite tail	710f53d2736eec18956752a744f2239bc 1461627736ddf8fb1c17842c9bb7ff5	dcd5eb8e782be93d926a6486f18d8a5e c445d33f8784ba251f60dd0203d8eb32
Classical C_{18} post-$m = 29$ finite tail	43f9efa815f1cf4ec325932889e383a26 cfd8e829b2848043feb2efec41fea68	e25bfaef2b3b6289631431fd26717704 dc82c78009cc02cf9c8e8dc8861b5541
Classical C_{19} post-$m = 29$ finite tail	282f6e3ab2c952155473cfa46c4f82e4 cf331462f4de406d76baab4302c1f307	034d55810933481b342966686cf249a33 649fac5333ad335df9b94e8f95b8273
Classical D_{12} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	f9b0a67a24912120f30b23f7781c88d5 a2126144a88e341d5f9b7a0df1988c2a
Classical D_{13} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	6f6ee3ffe7f92d0127c89ac1ce429e30 bb59a3a83cd113b54a60faa8c37b088a
Classical D_{14} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	aa4ad190afbe3a76383ad6999501f547 4098036724a8bd48144620bc4fa045dc
Classical D_{15} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	d7977837cc415e9ab1550123a5829832 e07f9cc9c3163b54ceb980fab46d8108
Classical D_{16} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	2a037f4f94b10af7f9da01425c6c22be eee1f492e4dda55e26d6962ee37a0ca8
Classical D_{17} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	dc513cfa308133f6e0229107cd44ee ebdaa8093ec31491febbde2b223641a3a4
Classical D_{18} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	7241b564ab367ced09f6faed98e3876 a5cf65d5b4de6e06a09d445b4e5bee455

Replay	Generation/checker identity	Recorded output identity
Classical D_{19} exact tail and bridge	1f296ce3a32756fbee27bdc44eb9749d 03dc88e74107eb7f15d91f8fda3a15e5 edec7e651f112a342909230068651c9a 59dc44a58bffcb5fa9e0420bf9f7ad8b	42e987f476bacc0e7422adb3825562227 aa0511a58a6984f61adcf4c08b3e4b5 a0a46e0274a2fa4ded548edfbec2f2d8 3cc7d5cecc13be155aa37d77fe9e11a0
Classical D_{20} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	d114e6a029422bde33cf360dab227cce 8703fcf746c8fe928fcb24529e09736b
Classical D_{21} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	87cb3ddbec928e5b22ae74f2d4a4a0 e70cc5a6001774558db105a0fdc1c7c235
Classical D_{22} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	8b321bacfdb915016fea96fd04b3672d 33a1758b0163ebe06d68daab36034c64
Classical D_{23} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	366460e47742ef94d5ac2371169d70742 d96e17426b89a1216bdcc6a8346e593
Classical D_{24} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	32e4c416211b54709ebb88d9458a0bc6 3a5e99cd144680e0d9d8fac05c9ba761
Classical D_{25} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	49cfdb89f74c85d5cd7b0fd90dc9d3bf e81cc219872d39bd4751e6327efa0d77
Classical D_{26} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	3becbf73915fac1362361aa8181e60af5 7a3456497b1751d8b22f42e266c5b1b
Classical D_{27} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	6c2405c8fa6304bd3dd215ed05d00eda a9be0cef7bf6fdcae4998ed0de6fd290
Classical D_{28} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	20d8b9bebc9b09cecd90c6e860c30079 ac13cec71ae13d716625d351da18ef7d
Classical D_{29} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	3340f19beacc7fcd2efa4bfdc4776c4f 6a21cbecf97d68de7a83aa95f2d8402e
Classical D_{30} post-$m = 29$ finite tail	6455ea1539e414087cba2335aa75bcfe 176d936edffc15536a7c84321147aa0f	c6afbd1a77b10d874d82e5ff142128b 837d7c09602f0dda5c3c7da92893a8e66

Replay	Generation/checker identity	Recorded output identity
Classical D_{31} post-$m = 29$ finite tail	`6455ea1539e414087cba2335aa75bcfe` `176d936edffc15536a7c84321147aa0f`	`beb8d9008c08783096f4ad91753676bd` `2ffff514189cc42603cbfe5b1faf9f1f`
Classical D_{32} post-$m = 29$ finite tail	`6455ea1539e414087cba2335aa75bcfe` `176d936edffc15536a7c84321147aa0f`	`e245d7ee1124af187c02e53365a1bd3d` `dc4f72893c1b8fa280ccb527a3357df2`
Classical D_{33} post-$m = 29$ finite tail	`930c0b9a705e079599af852d13bd05a5` `228337f36ec2fd981ea899f5c39e34ec`	`b7d813cde42faaf8ca92a803946a20ce` `1b93322f9102f81a095e6771d14d2f9e`
Classical D_{34} post-$m = 29$ finite tail	`3cdcda5f9bd87351035730d42c459d40` `63545cc85a734c87ed29970f6c62f72e`	`6700bcf5bd99f8a4b512a79e98708ae8` `b76ca06a6d1cbbd795f5acb87c96e800`
Classical D_{35} post-$m = 29$ finite tail	`e4b5f031f4adbde095ac9fd2ee463d62` `eadb03849825bef593d3caff6b9053fb`	`1100c50808f2b21a9977602e41b24166` `9da29afcdb2e36da9db445550bcd6bbc`
Classical D_{36} post-$m = 29$ finite tail	`8b9bc37a35ee3d03433876a1cb24f9a1` `0e10977c095d1f08a5fde35416f58e27`	`a9a3b4aa024930a41cdaf0eb2e97d89c` `16dac9f72f58e3016283185d1a8a03fa`
Classical D_{37} post-$m = 29$ finite tail	`66642943003e5bce9d3ae2e182fcd310` `42a3131d237f6c10bd1e932dc16aa225`	`8b7f4c16de0354d97d3a6c6233883ad` `4e0f9640c9ac74c47bbdf1b760009e15b`
Classical D_{38} post-$m = 29$ finite tail	`d143978e45e6ba1819d0dcdc9658cfc5` `dc02ca5df5478c0726023220969edbb6`	`44641f3b21f4d23764e653f2bda16d90` `6bd9d664e13f6928bb5ba961f3930ca3`
Classical D_{39} post-$m = 29$ finite tail	`9985e56e5313cb20f7e1b368bacde6a0` `b3112b31dc2dcc149fd636f5dc5cbc0f`	`6f052631b33f7be04fb0e9e32dc34d6` `3b1b2bf401a8ee3b479870d53ce9240a7`
Classical D_{40} post-$m = 29$ finite tail	`3fd9763dc1afc89af81a27758e0098a1` `0af7d28ddc1b3101e36ed101d01b6220`	`d3e94418ad9f6fd1e0911a89e19c2f1e` `5168d7727d5dc57be9ddd2b491a5e098`
Classical D_{41} post-$m = 29$ finite tail	`8352225e575eb2a427244fb81df383e9` `030498b60b85463707fe2edd5bcf9577`	`475339350b7802bb40f28f76de4f10aa` `9587942ac1e0df86873c0119463e4e51`
Classical D_{42} post-$m = 29$ finite tail	`b903e1adf2c11e389d32bb0bc5f0001f` `ba4c5d089c937c34fadae21f6c749ca9`	`22ff552c35e6800d669e62379cf7e690` `235cf90c96cfe21c0d1205ba161569fa`

Replay	Generation/checker identity	Recorded output identity
Classical D_{43} post-$m = 29$ finite tail	6c448a0b987d603c81037325fb96bba2 0bf2eeff52ed66f9bd713fd45f836f5a	d42e5f1fa39b939d92d21b4b8632b6a5 1c8a646d517a6900bbde580b3ccfdb09
Classical D_{44} post-$m = 29$ finite tail	b6ac9839cedf8c1a9457f0d3f9fe5a37 5da947589749585716dfb3df07d9a3d6	4a6e05c3837db8d3aa30cb7f46c78735 9337b394808c563544e6ace72a4740b8
Classical D_{45} post-$m = 29$ finite tail	964825593c8c7a3314d4f12587c3ca55 b5a5d77fcd928c6f7fa5d98b1f946efa	1264b3c773d580ff2d08a8355b118316 64b1bea28f3595f8ccbef028c41fd93b
Classical D_{46} post-$m = 29$ finite tail	29fc071c7d01d990c17532c1594f20a7 f9a8a3b0f07900a9082818bbb8508c18	ef7655165b98bcd29261a1a3dbd39336 a48e038df122b738bb63f39dc5668f41
Classical D_{47} post-$m = 29$ finite tail	1b663aef0410dfba5ca577df7bacc51e 8dd1cfac397d6730774d02089dd3717c	897adf7d7720993a66ff9ecd2e9ee58b5 a8cfc4211005b3f0b652a0a75eec147
Classical D_{48} post-$m = 29$ finite tail	6e38bb995342e904702ca544397c0ecc 9d31fcda8b043ab5b16525bf3becdbda	d66d077c5538c716da9e6551903db589 67d4a41df3779aaad8fcd3ba544c6cbd
Classical D_{49} post-$m = 29$ finite tail	cf9f373a83a0dce0b398ee8421d7c449 887cb2998f3e5d9c756a884973ce36b8	1c4a2651719271c321e11075425f3904 e080fb06339cc31c2630300ed9fdec3d
Classical D_{50} post-$m = 29$ finite tail	54d814fe2824c0a7a7fd09a2fa8973ea 584e694099e4b6f2ac426a113f5ca14d	1d88e66f53c82abd5d2162a3c746d4e6 dd6c25657ec1ce66b89d8a4cbf256c12
Classical D_{51} post-$m = 29$ finite tail	95f5e39b8ad38e5f197a09f22aa52445 0471a61e53f3ef49011fbbd033327158	828271646ebe60b7b8ad37af84860882 c96526486cad8be2d70a600eaaa1c49f
Classical D_{52} post-$m = 29$ finite tail	d05e95b2414c036e5c6f88077ced3287 4e9af1e153492e4597a4f933c9e47f35	e8e7dc7aae4c08d7c571e41dd2939213 bf17b4ed0cc322adf86734505976803b
Classical D_{53} post-$m = 29$ finite tail	bc9d5a804fcc9e72bf1b49b394de72c1 a8269ad0beac4d1d7365845af2701bab	a3c2559355281edd74889039564bbae1 3dfba5ce10ec92d957fa3a33a6227e48
Classical D_{54} post-$m = 29$ finite tail	43df015ed783f51dfb134d19c778968a c0eee616769fd97d5bf625af4ca087b6	0f54bb8b1e66209fcc79179573d49dc2 93e1e79fdb36a396e081b261a1b5b303

Replay	Generation/checker identity	Recorded output identity
Classical D_{55} post-$m = 29$ finite tail	`54e76fa1b690aee3e2f2c02214850c38` `d2a69e2eac388a90217ea43df2a3eb93`	`6d5a489a0f135317a3305e66cc0cd6fe` `da03a6cfcf37a04337da5872bdcbc127`
Classical D_{56} post-$m = 29$ finite tail	`066731d6ee5d0e150645591e84c2a0cb` `dc1da3bc2a4c86855ffba5f1d9a57817`	`26806ffc35f50a94493c9ee7f0c196c2` `92616ca1cf00a22acdbf1e2a0415336d`
Classical D_{57} post-$m = 29$ finite tail	`bc130849ef80bf0902757e96f3c8cc6` `c84c4ef72db2fa965eca1d4467c968dde`	`c6e69b001e3632bd578195126f266b6e` `01f8bbc80ae9e9451243ee3afbaf5dac`
Classical D_{58} post-$m = 29$ finite tail	`063c93acf978ef1816b1a0c8c927b721` `7add401a80704d6111f3c1fb26de4554`	`6347fb99063973f4f19b3b15354aa73f` `68d9e471f749adafcd8a60d3fac027ab`
Classical D_{59} post-$m = 29$ finite tail	`5cbce0bacd52cfee3d85bec546f79a37` `f3c1e119f15901c073387978e37ea273`	`aeb95cd95e84ef81a5bd97dae7b05839` `2e0acb24a23ad2f3f7e2c4ecee0fd401`
Classical D_{60} post-$m = 29$ finite tail	`9b196b4efba5c85abafe14a3b51c3038` `a3440c6fd590485a922469e1cdb4327d`	`38370d31bad7b33df73ecc854ad4c3` `a97a2bcb8e7e85dda7758aaad252e56a07`
Classical D_{61} post-$m = 29$ finite tail	`9982522dc6db42b69563e313dd51c576` `e230db1d2635f72d1dec277200db968b`	`3920dbf5295facfac7ca14dba3f03792` `9eb9b80a07f650da1300f6e35d81c419`
Classical D_{62} post-$m = 29$ finite tail	`3aaef754cca0cf37fbc475cf88685351` `395a3568c1dd8c7f82e7d3cf4caa2398`	`1656ddcb8569128be98151eed8f8fb6d` `798d249e837b6879aed8afd8d7a5dfa9`
Classical D_{63} post-$m = 29$ finite tail	`247b133c4bcd160998fe093f6bb429ac` `6989e50590642ccd328ddd9c56639f66`	`9df3f216e7718fcf3c33278b98523017` `41f6f98f681e17ed008584085220c9c1`
Classical D_{64} post-$m = 29$ finite tail	`9c4afa3e54ca488f848cb3b9e2043cda` `d9afbab05fb018b3510db7e162e9719b`	`d4dbd28548a01801d0f4fb42fef3aeae` `9eb112e3f2cac518437abbf10ddfc532`
Classical D_{65} post-$m = 29$ finite tail	`350c95515bbe01694ac24071f0483159` `50470f5b31d9ab7a7510a7b55bf896c1`	`022308c3011e70f9743db140e618fa92` `8d8ebaaf808ba7ace61a06133b968e4e`
Classical D_{66} post-$m = 29$ finite tail	`d5ff976201f0e668893e78254dcd9822` `c86af1d4d5505c50049ca0464e8476d9`	`8d8b8df1ee57bd6567f1a3d08e6f39f4` `6ac24447c12248beaadd5ecb1d505f1c`

Replay	Generation/checker identity	Recorded output identity
Classical D_{67} post-$m = 29$ finite tail	807ed743b04aee27029719abdcd350b5 ae03eb8941f2f6daa4346703f4e747da	fb5499210f42feb3db19d0bbccf3589c b8622f6b0aa8edae43f917709e68876a
Classical D_{68} post-$m = 29$ finite tail	e52df93f12a68fd3dfd92ea1a489aee4 be887fc1be7cf545f2fd6e21bd89e15d	e6ca2fce3bde6cf2fbfb79308053aa fe39ad3a891a3cc0d7fe37084a24177055
Classical D_{69} post-$m = 29$ finite tail	e6b77aad80fe398dfe2b5c9770548904 ccdf2b6f4c8e5810ba5aa95694d59910	86bd7a3e527be1c8e91fd24859ccfe66 3918950c79483e0fb7d24a5d1f84691e
Classical D_{70} post-$m = 29$ finite tail	4a42ae7ba0e238fed18f221b44d8f347 5b539524f0ffe32ada8c8bcf8f351572	510fa3957eaf02a26a9e4e69945bb4a1 b34d33e36738a9efda17ab36b10dccf6
Classical D_{71} post-$m = 29$ finite tail	b07c0663d7d4b70129d2cf4d56373458 2afbce31b82b78c98a5d2253aa854150	602b99dd2bd22b16e4a336b26c15b491 1f044367e7c7ff421c4bea53b358fb53
Classical D_{72} post-$m = 29$ finite tail	650d9d3cb069d9f220100577d4c2250 a8ff8a85041b743b45a5a3f227a982c9e	6c1b1070bf528fcbf5e7cfcdbbbe670 443c122ed2837ab8e2c944abcbbb97ddb
Classical D_{73} post-$m = 29$ finite tail	322b1d38392ccdba434f172e51d2f657 c8efabf0a0652296051889e31cf50fb8	b2ee7080c19dae3f253a08a57de0351 2a4d0f4da5c27429fd034c37498f6a12d
Classical D_{74} post-$m = 29$ finite tail	2a9b94175879a4b2456c1bfc6ff0fe96 12e14aee280aff9acdf61d1ff5ac5fe4	eee9ed7239c9671ffdcf480e2b3d0890 7332adef7e930af5930181c826eb46f8
Classical D_{75} post-$m = 29$ finite tail	9901fdae225b08bead8b02665fefc17d 5c476f5172d7403ba3171954ba916c30	c804e87fec00b21fa05056729c74be7a ece3f28f2f69ff4d8c64691c8af212db
Classical D_{76} post-$m = 29$ finite tail	210cb83e64c1ec7b78e3873d8d991d2 681a183be71452c87c261ebdb586ebc86	0f29482007a812ffbc79483f77fa0aa f69c033d2c2f288813568e90ad0955230
Classical D_{77} post-$m = 29$ finite tail	4dcbbcdcfeef7b6b4fb73c48b1920184 a4cba920baac9b89cf832d4515c9e98e	46d8bbe29c15c4dc3315ba9ebf7be9a 7db533ecae0ec87c36d29a184dd481f34
Classical D_{78} post-$m = 29$ finite tail	f74b68a2c3a5746d82acf30e5b10b641 b0ab66416c566b769e5dacded19e21c9	796de51fc5a01c17bbe0f6eafa4c127e 29389b521e38bcc4dda32610f55f6b79

Replay	Generation/checker identity	Recorded output identity
Classical D_{79} post-$m = 29$ finite tail	96061823694f2984e928f7bfa37b0f3 0106b9d212356c39d5add8493617baede	a2475a6f521776dac099fc21ccc46974cc 2abb2cc86dba75fc3aef8af6568fd9
Classical D_{80} post-$m = 29$ finite tail	df1d12680cea80faeda9b017deee3bc4 07cc423e34e447a7fc87f4857f192a38	8c1264fa4292d8f7f67ef978d9109d99 8fa39f759a4db622a58030cefa854851
Classical D_{81} post-$m = 29$ finite tail	457989dcabfe1c5aa97e8dde03da4b5f 2fff9eb48aba4d8b0204d44b36685b96	a295f235c7e29ad7719bd8f1552f0831 c5220c6c1f08a6c4b2445fad822be839
Classical D_{82} post-$m = 29$ finite tail	d114c05a9e22475cb14e643c7e4db545 7aa911ff10656c657d1e9d3216df2db6	1e7af36a394f21b0057c5c845571681 06c7068239b15c22b2ce9480871c14ad8
Classical D_{83} post-$m = 29$ finite tail	f8ae51c0443ca3a14b3529367cc9f641 303180eae84e20b9148212076a922582	b14ea35f764564dd5d7685f89adc9669 48c39e414b9fe46752ef7f21a3b1d43f
Classical D_{84} post-$m = 29$ finite tail	2acea25cd9b982e5010ec51e8d049590 bbe0230c6e099c76265a1f83b05a18a5	92693ae7cb8f56d258f9824442f3804e c8ca7a2e8f15b4da3ceec0b725920e66
Classical D_{85} post-$m = 29$ finite tail	6b109272136ec349ff69c00b1f270660 1e95c1efbbe01dd159c2f5dbd222237a	0d8ec01ede79008e9a2dcffe97fa8ce8 c1c81307bef53038f7a3036ddd4baa5f
Classical D_{86} post-$m = 29$ finite tail	c4288a10881cb4c628091fb91354852f e73d8499158fdf52bb69230e546b7ec3	5ab292178239b92c9e0e4e4445e9e0f9 42b75fb3ebe00e3e5706cdce3e2d71d4
Classical D_{87} post-$m = 29$ finite tail	5f1f823bf1133549db47e3e9220c71299 353f1fdc2a0c41c4b9b39bf8d595cbe	9de45735f34728dc14228c0f95a42f88f 84be059311c8cf9dbb2577ce6d706f3
Classical D_{88} post-$m = 29$ finite tail	adcb516f50700d151d05cf926b484531 a6a9224a32bd2bea197b006f81a49b19	4bb7b5143445fd7b5cc613c14bdf3800 f0e5cd8e302adfdf178ef4db540422f1
Classical D_{89} post-$m = 29$ finite tail	920265217898e51e8825c43a16a47ce0 bf4590ae92a00b676bd3d89018a71561	7cf2e4118d1c3815b907bcb3849a847 0598de3ec0d96f0b066c7385f713b5f5a
Classical D_{90} post-$m = 29$ finite tail	5b579ff50cdac23aa775f319400fd54095 defefe2704410cf69f8255450b7550	ba3e023245c270566c72eba6723612d39 9c979a46e25b1d56ea17b8ea08696b8

Replay	Generation/checker identity	Recorded output identity
Classical D_{91} post-$m = 29$ finite tail	`518e1358dbf4de57ba79592cc90d4117` `d5c99ffe6db660e23f8c75b312f82a8e`	`71c6d6b1b8bc17b21422f55cc2c1dbb4` `001a5d5e7bdf7868ec91ca62c2a22abb`
Classical D_{92} post-$m = 29$ finite tail	`62ae14c767b1b01b3081b6093ada47ba` `0e0f36e3886f9d0e5e29212fc67d9b39`	`d5c20243f621cfb443c7ca6f1619e3c` `ed149dbffbb92ab179320be96699deae3`
Classical D_{93} post-$m = 29$ finite tail	`282efa573810e3ea137038b47a447e56` `f1bb0ba67023d853d9255c146b25f8fe`	`27d61a5fff05d9bf688cfbddfbffbc21` `d5b90ade28e3be86d83e039268cbe7a3`
Classical D_{94} post-$m = 29$ finite tail	`9f63c2508b241a3534130fa35681eebf` `322401c6ddfe3ae08e0ece045655341d`	`690fea30cfb9ad79d887d57653d24291` `1b5dfed68157669ab38b8a351331d545`
Classical D_{95} post-$m = 29$ finite tail	`d1c2aa46419f77c0503dae8ca16cf7c` `0c2e349564dca90a1a7fb0229607a6f2c`	`b37110f3c85f43bd736599b0715f512` `bf8cf48e2ef964939dd64657845c96ceb`
Classical D_{96} post-$m = 29$ finite tail	`23569a4501d98a11e41af838660ddc4c` `c39a0ed580fe53eb632b2fcc295c699a`	`1ac17e2524c6501556f5fd55a61ef95a` `fe744a47ccd469c109492f20467e02b7`
Classical D_{97} post-$m = 29$ finite tail	`171c1586935f6f787cef2bdea17e4771` `3bb13c48f1a0b67dbb374aba58a10a16`	`26be4b2020bf5eb25a1abd7565520a55` `0e67abf90a452893ca7c22132ec90ca9`
Classical high-edge no-TV Chernoff trace	`ea02f81ac518eaaf849039112fd06781f` `908c4094619d54f0685987a7b2424c3`	`ca3a938c9d45d65eb892f6ba637ed8c3` `a7b1d32a571a5be52e2f5efb80477176`
E_8 finite bridge	`46900d26a211d572bd41cbe92dd6e59` `3936e4255af0e1560824a0f29ba299129`	`17cd75bbedc88d6e9e6ac85ca8ae8c4` `f4825018e5e8de5a5cfdf1cf60ed2414c`
E_6/E_7 rank-corrected rectangular tails	`20c89e8befe8a7172455f62e4b5467c6` `19f9f7f01d93e717f78ca60c832b4cb3`	`8ded845d7746e5d46d2cb834b09f0faa` `ac5e2e8cefc0b983cf734627eba5713a`
E_8 rectangular OpenMP source	`2024a73d5ad98c33d80dbc9c12820927` `37d0172a3088e7db527aa7cd7406d764`	`6bc3c5bf1774e5c50017360f9c7dda09` `932454f2f4fe73f7c8cfbba0650ee025`
E_8 negative-bound exact audit	`7758379e442ca9b65844db4de7fbcb9d` `49cd0105b142d958c0ca5300c8678046`	`ffd490645e0fe1858980f7ab9bbda23a` `2e1cc56764d1c1180460271e57ff3789`

The classical no-TV Chernoff high-edge replay is a load-bearing proof replay for proposition 4.19. Its archived log hash is `ca3a938c9d45d65eb892f6ba637ed8c3` `a7b1d32a571a5be52e2f5efb80477176`.

Appendix B. Post-$m = 29$ Classical Replay Transcript

The post-$m = 29$ first-hit proof is the union of the active 530-row B/C replay and the two exact type-D bridge replays. Its composite ledger is:

finite cutoff	296,
rows checked	783,
B/C stable moments generated	$m_0, \ldots, m_{557}$,
scope	one Chain step at $N_{\mathrm{hit}}^{(29)}(G)$,
status	all three accepted replays exit with status 0.

The bucket decomposition is:

bucket	rows	minimum margin row
BC exact $m = 30$ window	14	B_{11}
BC stable-tail $m = 30$	26	B_{19}
BC stable-tail sloped	490	B_{32}
D exact two-sided interval	28	D_{25}
D A-free frontier	225	D_{53}.

The row count identity is

$$14 + 26 + 490 + 28 + 225 = 783.$$

The B/C replay is not a pushforward-tail replay and is not used by itself to propagate any row beyond its first post-$m = 29$ hit. The type-D bridge replays check every intervening Chain step through their independently certified monotone-tail onsets.

Appendix C. E8 Finite Bridge Transcript

The final finite bridge replay for E_8 reports:

prefix Chain odd exponents	$1, 3, \ldots, 65$,
bridge odd exponents	$69, 71, 73, 75, 77$,
Chain odd exponents	$67, 69, 71, 73, 75, 77$,
extended Chain odd exponents	$67, 69, \ldots, 95$,
minimum prefix Chain difference	20,
$\log_{10}$ minimum bridge value	100.94,
$\log_{10}$ minimum Chain difference	100.94,
$\log_{10}$ minimum extended Chain difference	100.94.

The exact replay flags are all `OK`.

Appendix D. Type A Certificate Details

The type A replay is organized around a finite set of endpoint and finite-overlap inequalities. The stable-rank theorem covers $N \geq n+2$. The finite-rank theorem covers $N < n+2$ by separating the following cases:

case	rank range	closure mechanism
small ranks	$N = 2, 3, 4, 5$	finite transition-ratio identities
finite overlap	$6 \leq N \leq 18$	integer moment replay
middle endpoint	$24 \leq N \leq 27$	row endpoint certificate
large endpoint	$N \geq 30$	monotone endpoint comparison
interpolation	all remaining $N \geq 6$	ratio monotonicity

For the small ranks the proof is not a numerical exception. For $N=2$, Weyl integration and the change of variables $\theta=\alpha+\beta$, $\phi=\alpha-\beta$ identify

$$\frac{1}{2}Q_3^{\mathrm{SU}(2)}(n)=\int_{\mathrm{USp}(4)}\mathrm{tr}\big((\Lambda^2V)^{\otimes n}\big)\,dA,$$

which is the multiplicity of the trivial representation in $(\Lambda^2V)^{\otimes n}$ and hence is a nonnegative integer. For $N=3,4,5$, Gessel's determinant gives the following adjoint-moment recurrences:

N	recurrence
3	$(k+3)^2m_{k+1}=k(7k+11)m_k+8k(k+1)m_{k-1}$
4	$(k^3+22k^2+161k+392)m_{k+4}$ $-(16k^3+230k^2+1054k+1524)m_{k+3}$ $+(10k^3-340k-750)m_{k+2}$ $+(72k^3+522k^2+1242k+972)m_{k+1}$ $+(45k^3+270k^2+495k+270)m_k=0$
5	$192(k+1)(k+2)(k+3)(k+6)m_k$ $+16(k+2)(k+3)(22k^2+184k+309)m_{k+1}$ $+(k+3)(129k^3+1245k^2+2570k-1112)m_{k+2}$ $-(k+3)(30k^3+642k^2+4487k+10196)m_{k+3}$ $+(k+8)^2(k+10)^2m_{k+4}=0.$

Substitution into the boundary reduction proves the ratio monotonicity needed for every odd $n\geq 5$ in these rows; the finite initial cases are checked by exact integer moments.

The replay computes adjoint moments by iterating tensor multiplication with the adjoint representation and reducing each tensor product by the Weyl dot-action. For each row it evaluates the exact integer target

$$R_{N,k}:=m_{k+2}^{(N)}m_k^{(N)}-(m_{k+1}^{(N)})^2.$$

The sign check $R_{N,k}\geq 0$ is exactly $\rho_k^{(N)}\leq\rho_{k+1}^{(N)}$. The replay output contains the lines

```
          Propositions 64-65 endpoint constants:  ok,
                        Proposition 66 row N=24:  ok,
                        Proposition 66 row N=25:  ok,
                        Proposition 66 row N=26:  ok,
                        Proposition 66 row N=27:  ok,
                  Proposition 67 band certificate:  ok,
  Proposition 67 tail and endpoint certificates:  ok,
                         Proposition 68 row N=6:  ok,
                         Proposition 68 row N=7:  ok,
                         Proposition 68 row N=8:  ok,
                         Proposition 68 row N=9:  ok,
                        Proposition 68 row N=10:  ok,
                        Proposition 68 row N=11:  ok,
                        Proposition 68 row N=12:  ok,
                        Proposition 68 row N=13:  ok,
                        Proposition 68 row N=14:  ok,
                        Proposition 68 row N=15:  ok,
                        Proposition 68 row N=16:  ok,
                        Proposition 68 row N=17:  ok,
                        Proposition 68 row N=18:  ok.
```

The terminal line is

```
All SU(N) repair certificates verified.
```

Appendix E. Classical Row Ledger

This appendix records the final row coverage used in theorem 4.842. The high-rank theorem covers all rows with $C_G \geq 296$. The finite rows with $C_G < 296$ are divided by the bridge and post-bridge certificates.

High-rank and trace rows.

family	rank condition	certificate
B, C, D	$C_G \geq 296$	Rains-square trace cutoff
B, C	$b = 276, 278, \dots, 295$	first-hit trace
D	$b = 276, 278, 280, \dots, 295, 297$	first-hit trace

Direct post-$m = 29$ low-rank tails.

family	rank condition	certificate
B	$2 \leq b \leq 13$	*proposition* 4.18
C	$2 \leq b \leq 19$	*proposition* 4.18
D	$4 \leq b \leq 24$	*proposition* 4.18

Post-$m = 29$ first-hit rows.

family	rank interval	number of rows
B	$11 \leq b \leq 275,\ b = 277$	266
C	$13 \leq b \leq 275,\ b = 277$	264
D	$25 \leq b \leq 275,\ b = 277, 279$	253
total		783

The bucket decomposition is:

bucket	B ranks	C ranks	D ranks
BC exact $m = 30$ window	$11, \dots, 18$	$13, \dots, 18$	–
BC stable-tail $m = 30$	$19, \dots, 31$	$19, \dots, 31$	–
BC stable-tail sloped	$32, \dots, 275, 277$	$32, \dots, 275, 277$	–
D exact two-sided interval	–	–	$25, \dots, 52$
D A-free frontier	–	–	$53, \dots, 275, 277, 279$

The replay checks that every row in this table has a strictly positive integer lower margin at its first post-$m = 29$ hit. The subsequent odd exponents for B_{11}, B_{12}, B_{13} and for $C_{13}, \dots, C_{19}$ are covered by proposition 4.18; $D_{25}, \dots, D_{52}$ are covered by proposition 4.51, and $D_{53}, \dots, D_{295}$ by proposition 4.49; the B/C high-edge rows with $218 \leq b \leq 275$ or $b = 277$ are covered by proposition 4.19; the D interval rows $D_{98}, \dots, D_{275}, D_{277}, D_{279}$ are covered by proposition 4.58. The remaining B/C rows are exhausted by the explicit split in the proof of theorem 4.842. Namely, proposition 4.43 supplies every odd exponent $n \geq 2b + 29$ for $B_{14}, \dots, B_{61}$ and $C_{20}, \dots, C_{61}$. It starts directly at odd exponents $57, 59, 61, 63$ for $B_{14}, \dots, B_{17}$; for the other rows, the proved boundary and non-boundary correction theorems and the exact half-stable bridge are assembled in proposition 4.841, which supplies the required earlier prefixes. The rows $B_{62}, \dots, B_{123}$ and $C_{62}, \dots, C_{217}$ are closed by the same finite-prefix mechanism followed by propositions 4.29, 4.33 and 4.59; $B_{124}, \dots, B_{217}$ are covered by proposition 4.21. The exact bridge proposition 4.54 and the terminal-tail proposition proposition 4.55 close the remaining $D_4, \dots, D_{24}$ rows.

Appendix F. Exceptional Certificate Ledger

The exceptional proof uses exact prefixes and direct Chain tails. The following table records the exact moment source and the terminal Chain coverage.

G	moments checked	Chain prefix	tail start
G_2	$m_0, \ldots, m_{200}$	$m = 0, \ldots, 97$	inside prefix
F_4	$m_0, \ldots, m_{220}$	$m = 0, \ldots, 107$	inside prefix
E_6	$m_0, \ldots, m_{80}$	$m = 0, \ldots, 37$	$n \geq 75$
E_7	$m_0, \ldots, m_{70}$	$m = 0, \ldots, 32$	$n \geq 65$
E_8	$m_0, \ldots, m_{100}$	$m = 0, \ldots, 32$	$n \geq 133$

For E_8, the bridge chain consists of the odd exponents

$$131, 129, 127, 125, 123, 121, 119, 117, 115, 113, 111, 109, 107, 105, 103, 101,$$

$$99, 97, 95, 93, 91, 89, 87, 85, 83, 81, 79, 77.$$

The finite bridge supplies $Q_3(n) > 0$ at $n = 69, 71, 73, 75, 77$ and nonnegative Chain differences from 67 through 95. Therefore the finite bridge, the explicit bridge chain down to 77, and the tail from 133 form one continuous Chain-propagation interval.

Appendix G. External Inputs and Replay Boundary

The proof uses two kinds of external data. First, it cites published theorems with the specific loci used above: Weyl integration, character orthogonality, classification through irreducible root systems, root-system conventions, and classical compact groups in [1, Chapters II, IV–VI], the finite-Coxeter Macdonald–Mehta integral and squared-length-two root-normal convention in [12, §2.1 and Theorem 3.1(i)], Ginibre's condition (Q3) in [2, §1], the universal compact-group adjoint trace bound in [13, Theorem 2.1], the unitary Schur-expansion moment calculation in [5, §1], Gessel's determinant in [3, §7, Theorem 16 and the formula after (26)], the exact orthogonal and symplectic joint trace moments in [4, Theorems 4 and 6], Rains' orthogonal and symplectic Schur-integral stabilization in [5, §3, Lemmas 3.1–3.2 and Theorem 3.4], Albion's column-even Littlewood identity in [7, Introduction, (1.1c)], Krattenthaler's semistandard growth-diagram conversion in [8, Theorem 4], Krattenthaler's arbitrary-filling growth-diagram local rules in [9, §4.1, Theorem 7], Huh–Kim–Krattenthaler–Okada's marked even-bound vacillating-tableau model in [10, Definitions 8.1, 8.5, 8.6 and Theorem 8.7], the orthogonal and symplectic power-map decomposition of Rains [6, Theorem 1.5, equations (22)–(23), and Theorem 1.7, equation (37)], and the Courteaut–Johansson–Lambert trace estimates [11, Theorems 1.1, 1.4, 4.1 and §4.2]. The ancillary table of Bourjaily–Plesser–Vergu [14, Appendix B and ancillary file `adjoint_tensor_ranks.tex`] is used only as an independent comparison for the E_8 values $m_{71}, \ldots, m_{100}$. The proof input is regenerated from the E_8 Cartan datum by the exact Racah–Speiser iteration and character-pairing identities of lemmas 2.4 and 2.5; the replay requires termwise equality with the local transcription of the published table.

The source notes in this repository are not invoked as additional analytic lemmas. They are part of the computer-assisted certificate boundary: the manuscript states the theorem, reductions, certificate criteria, row ledgers, replay transcripts, and artifact hashes, while the replay implementations and hashed ledgers check the exact integer/rational certificate statements printed above.

The cited primary-source files used in the source audit are vendored in `ginibre_q3/references`. In particular, the local folder contains Ginibre's paper, Rains' 1998 and 2003 papers, Gessel's author PDF, Albion's arXiv PDF, Krattenthaler's arXiv PDF, Krattenthaler's 2006 arXiv PDF on arbitrary-filling growth diagrams, Huh–Kim–Krattenthaler–Okada's author PDF, Courteaut–Johansson–Lambert's arXiv PDF, Bröcker–tom Dieck's GTM text, the Bourjaily–Plesser–Vergu arXiv paper, and the BPV ancillary `adjoint_tensor_ranks.tex` file used for the independent E_8 comparison. The folder also contains the primary Etingof arXiv PDF and the primary Garibaldi–Guralnick–Rains journal PDF; these replace the former Markdown-only mappings for those two citations. The fourteen bibliography keys and fifteen corresponding local source files are mapped to their verified theorem/equation loci in `references/active_paper_sources.tsv`. The complete directory is pinned by `references/references_manifest.sha256`; the clean-room replay rejects a missing, changed, unlisted, or newly added reference file.

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